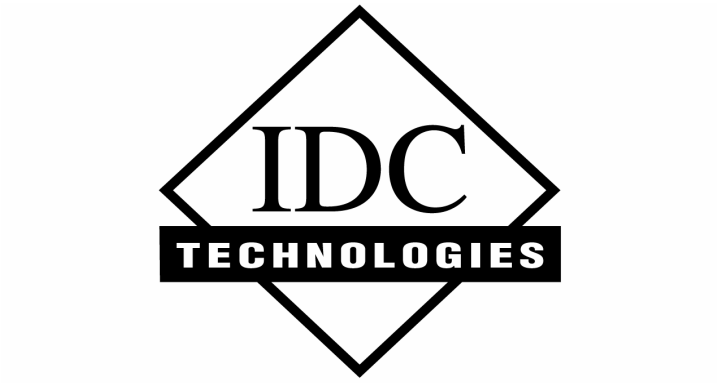

Technology Training that Works

Presents

Practical

Fieldbus, DeviceNet and Ethernet

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Acknowledgements

IDC Technologies expresses its sincere thanks to all those engineers and technicians on our training workshops who freely made available their expertise in preparing this manual.

Who is IDC Technologies?

IDC Technologies is a specialist in the field of industrial communications, telecommunications, automation and control and has been providing high quality training for more than six years on an international basis from offices around the world.

IDC consists of an enthusiastic team of professional engineers and support staff who are committed to providing the highest quality in their consulting and training services.

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The technological world today presents tremendous challenges to engineers, scientists and technicians in keeping up to date and taking advantage of the latest developments in the key technology areas.

- The immediate benefits of attending IDC workshops are:
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All workshops have been carefully structured to ensure that attendees gain maximum benefits. A combination of carefully designed training software, hardware and well written documentation, together with multimedia techniques ensure that the workshops are presented in an interesting, stimulating and logical fashion.

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Technology Training that Works

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IDC is continually developing high quality state of the art workshops aimed at assisting engineers, technicians and scientists. Current workshops include:

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- Practical Alarm Management for Engineers and Technicians
- Practical Programmable Logic Controller's (PLCs) for Automation and Process Control
- Practical Batch Management & Control (Including S88) for Industry
- Practical Boiler Control and Instrumentation for Engineers and Technicians
- Practical Programming for Industrial Control - using (IEC 1131-3 and OPC)
- Practical Distributed Control Systems (DCS) for Engineers & Technicians
- Practical Data Acquisition using Personal Computers and Standalone Systems
- Best Practice in Process, Electrical & Instrumentation Drawings and Documentation
- Practical Troubleshooting of Data Acquisition & SCADA Systems
- Practical Industrial Flow Measurement for Engineers and Technicians
- Practical Hazops, Trips and Alarms
- Practical Hazardous Areas for Engineers and Technicians
- A Practical Mini MBA in Instrumentation and Automation
- Practical Instrumentation for Automation and Process Control
- Practical Intrinsic Safety for Engineers and Technicians
- Practical Tuning of Industrial Control Loops
- Practical Motion Control for Engineers and Technicians
- Practical SCADA and Automation for Managers, Sales and Administration
- Practical Automation, SCADA and Communication Systems: A Primer for Managers
- Practical Fundamentals of OPC (OLE for Process Control)
- Practical Process Control for Engineers and Technicians
- Practical Process Control & Tuning of Industrial Control Loops
- Practical Industrial Programming using 61131-3 for PLCs
- Practical SCADA & Telemetry Systems for Industry
- Practical Shutdown & Turnaround Management for Engineers and Managers
- Practical Safety Instrumentation and Shut-down Systems for Industry
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- Practical Safety Instrumentation & Emergency Shutdown Systems for Process Industries
- Control Valve Sizing, Selection and Maintenance

Communications

- Best Practice in Industrial Data Communications
- Practical Data Communications & Networking for Engineers and Technicians
- Practical DNP3, 60870.5 & Modern SCADA Communication Systems
- Practical Troubleshooting & Problem Solving of Ethernet Networks
- Practical FieldBus and Device Networks for Engineers and Technicians
- Practical Fieldbus, DeviceNet and Ethernet for Industry
- Practical Use and Understanding of Foundation FieldBus for Engineers and Technicians
- Practical Fiber Optics for Engineers and Technicians
- Data Communications, Networking and Protocols for Industry - Back to Basics
- Practical Troubleshooting & Problem Solving of Industrial Data Communications
- Practical Troubleshooting, Design & Selection of Industrial Fibre Optic Systems for Industry
- Practical Industrial Networking for Engineers & Technicians
- Troubleshooting Industrial Ethernet & TCP/IP Networks
- Practical Local Area Networks for Engineers and Technicians
- Practical Routers & Switches (including TCP/IP and Ethernet) for Engineers & Technicians
- Practical TCP/IP and Ethernet Networking for Industry
- Practical Fundamentals of Telecommunications and Wireless Communications
- Practical Radio & Telemetry Systems for Industry
- Practical TCP/IP Troubleshooting & Problem Solving for Industry
- Practical Troubleshooting of TCP/IP Networks
- Practical Fundamentals of Voice over IP (VOIP) for Engineers and Technicians
- Wireless Networking and Radio Telemetry Systems for Industry
- Wireless Networking Technologies for Industry

Electrical

- Practical Maintenance & Troubleshooting of Battery Power Supplies
- Practical Electrical Network Automation & Communication Systems
- Safe Operation & Maintenance of Circuit Breakers and Switchgear
- Troubleshooting, Maintenance & Protection of AC Electrical Motors and Drives
- Practical Troubleshooting of Electrical Equipment and Control Circuits
- Practical Earthing, Bonding, Lightning & Surge Protection
- Practical Distribution & Substation Automation (incl. Communications) for Electrical Power Systems
- Practical Solutions to Harmonics in Power Distribution
- Practical High Voltage Safety Operating Procedures for Engineers and Technicians

- Practical Electrical Wiring Standards - National Rules for Electrical Installations - ET 101:2000
- Lightning, Surge Protection and Earthing of Electrical & Electronic Systems in Industrial Networks
- Practical Power Distribution
- Practical Power Quality: Problems & Solutions
- Practical Power Systems Protection for Engineers and Technicians
- Wind & Solar Power - Renewable Energy Technologies
- Practical Power Transformers: Operation, Maintenance & Testing
- Maintenance and Troubleshooting of UPS Systems and Battery Power Supplies
- Practical Variable Speed Drives for Instrumentation and Control Systems
- Practical Electrical Wiring Standards - IEE BS7671 - 2001 Edition

Project & Financial Management

- Practical Financial Fundamentals and Project Investment Decision Making
- How to Manage Consultants
- Marketing for Engineers and Technical Personnel
- Practical Project Management for Engineers and Technicians
- Practical Specification and Technical Writing for Engineers & Other Technical People

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- Practical Boiler Plant Operation and Management for Engineers and Technicians
- Practical Bulk Materials Handling (Conveyors, Bins, Hoppers & Feeders)
- Practical Pumps and Compressors: Control, Operation, Maintenance & Troubleshooting
- Practical Cleanroom Technology and Facilities for Engineers and Technicians
- Gas Turbines: Troubleshooting, Maintenance & Inspection
- Practical Hydraulic Systems: Operation and Troubleshooting
- Practical Lubrication Engineering for Engineers and Technicians
- Practical Safe Lifting Practice and Maintenance
- Practical Mechanical Drives (Belts, Chains etc) for Engineers & Technicians
- Fundamentals of Mechanical Engineering
- Practical Pneumatics: Operations and Troubleshooting for Engineers & Technicians
- Practical Centrifugal Pumps - Optimising Performance
- Practical Machinery and Automation Safety for Industry
- Practical Machinery Vibration Analysis and Predictive Maintenance

Electronics

- Practical Digital Signal Processing Systems for Engineers and Technicians
- Practical Embedded Controllers: Troubleshooting and Design
- Practical EMC and EMI Control for Engineers and Technicians
- Practical Industrial Electronics for Engineers and Technicians

- Practical Image Processing and Applications
- Power Electronics and Variable Speed Drives: Troubleshooting & Maintenance
- Practical Shielding, EMC/EMI, Noise Reduction, Earthing and Circuit Board Layout

Information Technology

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- Industrial Network Security for SCADA, Automation, Process Control and PLC Systems
- SNMP Network Management: The Essentials
- Practical VisualBasic Programming for Industrial Automation, Process Control & SCADA Systems

Chemical Engineering

- Practical Fundamentals of Chemical Engineering

Civil Engineering

- Hazardous Waste Management and Pollution Prevention
- Structural Design for non-structural Engineers
- Best Practice in Sewage and Effluent Treatment Technologies

Technology Training that Works

Comprehensive Training Materials

All IDC workshops are fully documented with complete reference materials including comprehensive manuals and practical reference guides.

Software

Relevant software is supplied with most workshop. The software consists of demonstration programs which illustrate the basic theory as well as the more difficult concepts of the workshop.

Hands-On Approach to Training

The IDC engineers have developed the workshops based on the practical consulting expertise that has been built up over the years in various specialist areas. The objective of training today is to gain knowledge and experience in the latest developments in technology through cost effective methods. The investment in training made by companies and individuals is growing each year as the need to keep topical and up to date in the industry which they are operating is recognized. As a result, the IDC instructors place particular emphasis on the practical hands-on aspect of the workshops presented.

On-Site Workshops

In addition to the quality of workshops which IDC presents on a world-wide basis, all IDC courses are also available for on-site (in-house) presentation at our clients premises. On-site training is a cost effective method of training for companies with many delegates to train in a particular area. Organizations can save valuable training $$$'s by holding courses on-site, where costs are significantly less. Other benefits are IDC's ability to focus on particular systems and equipment so that attendees obtain only the greatest benefits from the training.

All on-site workshops are tailored to meet with clients training requirements and courses can be presented at beginners, intermediate or advanced levels based on the knowledge and experience of delegates in attendance. Specific areas of interest to the client can also be covered in more detail. Our external workshops are planned well in advance and you should contact us as early as possible if you require on-site/customized training. While we will always endeavor to meet your timetable preferences, two to three month's notice is preferable in order to successfully fulfil your requirements. Please don't hesitate to contact us if you would like to discuss your training needs.

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In addition to standard on-site training, IDC specializes in customized courses to meet client training specifications. IDC has the necessary engineering and training expertise and resources to work closely with clients in preparing and presenting specialized courses.

These courses may comprise a combination of all IDC courses along with additional topics and subjects that are required. The benefits to companies in using training is reflected in the increased efficiency of their operations and equipment.

Training Contracts

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Some of the thousands of Companies worldwide that have supported and benefited from IDC workshops are:

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References from various international companies to whom IDC is contracted to provide on-going technical training are available on request.

Preface

The Fieldbus and DeviceNet standards are becoming a standard at the field and instrumentation level and replacing the traditional approaches in the plant today. Ethernet is also fast becoming the obvious choice for industrial control networking worldwide. While the basic structure of Ethernet has not changed much, the faster technologies such as Fast Ethernet and Gigabit Ethernet have increased the complexity and choices you have available in planning and designing these systems. There has also been a convergence between Fieldbus and DeviceNet standards in that they are also increasingly becoming based on industrial Ethernet for the higher speed data transfer applications.

There is a fair degree of confusion about where Fieldbus, DeviceNet and Ethernet, are applied and the workshop commences with a clear comparison between the different standards and where they are applied. The first day focuses on ASi-bus, DeviceNet, Profibus and Foundation Fieldbus technologies in a simple and understandable manner. A detailed discussion is then held on the application of the technologies in your plant today. There are many misconceptions on the best standard to apply in a given section of the plant. This workshop will promote the theme which is rapidly growing strength in that you should focus on your application and apply the particular fieldbus or DeviceNet to match this application and ensure easy interconnectivity between the different standards. Selecting one standard to match all applications is not really a practical approach.

On the second day, Ethernet is then discussed with a brief outline of the fundamentals of Ethernet and its operation. The method of access is discussed in depth and topics such as full duplex and auto negotiation are explained. The best methods of designing and installing the cabling systems are then explored with the discussion ranging from 10Base-T over twisted pair to Gigabit Ethernet cabling. Methods of optimizing Ethernet to obtain best performance are then defined.

As Ethernet has become more complex, a number of misconceptions have arisen as to how Ethernet functions and how the system should be optimally configured and what exactly industrial Ethernet means. This workshop addresses these issues in a clear and practical manner, thus enabling you to apply the technology quickly and effectively in your next project. There is also a practical discussion on how to connect Fieldbus and DeviceNet with Ethernet.

We would hope that you gain the following benefits from this book. After reading this book and attending the associated workshop you should be able to:

- Compare the Ethernet and Fieldbus/Devicenet standards
- Troubleshoot and fix simple DeviceNet, Profibus and Foundation Fieldbus problems
- Design and install simple Ethernet networks
- Know when to use repeaters, bridges, switches, and routers
- Apply switched Ethernet systems effectively
- Install the cabling and hardware for a typical industrial
- Ethernet Network
- Decide on the best cabling and connectors for your harsh or office environment
- Apply the structured cabling system concepts to your next project
- Perform simple troubleshooting tasks on a Network

Typical people who will find this book useful include:

- IT Managers working with Networks
- Electrical engineers
- Project engineers
- Design engineers
- Electrical and instrumentation technicians
- Maintenance engineers and supervisors
- Systems engineers
- Instrumentation and control system engineers
- Process Control Designers and Systems Engineers
- Instrumentation Technologists and Engineers
- Gain useful practical know-how into how to apply the latest Fieldbus, DeviceNet and Ethernet Technologies to your plant
- Anyone involved in the installation, design and support of industrial communications systems

The structure of the book is as follows.

Chapter 1 Fundamental Principles of Industrial Communications
A brief overview of the key building blocks of data communications in an industrial context.

Chapter 2 RS-232 Fundamentals
A detailed discussion of the important issues with RS-232.

Chapter 3 RS-485 Fundamentals
A detailed discussion of the important issues with the balanced and very popular industrial standard RS-485.

Chapter 4 Modbus Overview
A review of the Modbus protocol representing a Data Link and Application Layer implementatation.

Chapter 5 AS- interface
A discussion of the important and simple AS-i industrial communications interface.

Chapter 6 DeviceNet
A brief review of the key elements of DeviceNet.

Chapter 7 Profibus PA/DP Overview
A review of arguably the most popular Fieldbus standard in the world today - Profibus PA and DP.

Chapter 8 Foundation Fieldbus
A review of arguably and technically the most sophisticated Fieldbus with a very well developed user layer.

Chapter 9 Operation of Ethernet Systems
The fundamentals of the operation of Ethernet.

Chapter 10 Physical Layer implementation of Ethernet Media Systems
The fundamentals of the physical part of Ethernet.

Chapter 11 Ethernet Cabling and connectors
The key issues with Ethernet cabling ranging from coaxial, twisted pair and fiber.

Chapter 12 LAN System components
An overview of hubs, repeaters, switches and routers all of which represent key components of Ethernet networks.

Chapter 13 Structured Cabling
Cabling represents one of the most important and often neglected issues with setting up an Ethernet system and this chapter reviews the key issues here.

Chapter 14 Multi segment Configuration guidelines for half duplex Ethernet Systems
A brief review of multi segment configuration guidelines.

Chapter 15 Industrial Ethernet
A summary of the key underlying features of Industrial Ethernet.

Chapter 16 Troubleshooting Ethernet
Typical strategies in troubleshooting Ethernet.

Chapter 17 Network Protocols – Part one - IP
A discussion of the Internet Protocol (IP).

Chapter 18 Network Protocols – Part two – TCP/UDP
A review of the TCP/IP protocols – the connection oriented TCP and the connectionless UDP.

Chapter 19 Industrial Application Layer Protocols

Chapter 20 Connecting Ethernet, Fieldbus and DeviceNet
A brief description of how to go about connecting the different fieldbus, devicenet, Ethernet.

Chapter 21 Virtual LANs (VLANs) using Ethernet

Appendix A Comparison of the different standards
A tabular comparison of the different Fieldbus, DeviceNet and Ethernet based standards.

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1

Fundamental principles of industrial communications

1.1 Overview

This manual can be divided into three distinct sections:

1.1.1 Introductory material

Chapter 1 covers general topics such as the OSI model, systems engineering concepts, physical (layer 1) connections, protocols, and noise and ingress protection. Chapters 2, 3 and 4 cover the key fundamental standards of RS-232, RS-485 and Modbus

1.1.2 DeviceNet and Fieldbus

Chapters 5 to 8 cover the various DeviceNet and Fieldbus standards such as AS-I bus, DeviceNet, Profibus and Foundation Fieldbus.

1.1.3 Industrial Ethernet

Chapters 9 to 18 focus on Industrial Ethernet commencing with a study of the fundamentals and covering cabling, LAN system components, industrial versus commercial Ethernet, troubleshooting and network protocols such as the TCP/IP suite. Chapter 19 focuses on industrial application protocols. Chapter 20 looks at the tricky but important issue of connecting Ethernet, Fieldbus and DeviceNet together. The book is completed with a discussion of Virtual LANs.

Note: Throughout this manual we will refer to RS-232, RS-422 and RS-485. One is often criticized for using these terms of reference, since in reality they are obsolete. However, if we briefly examine the history of the organization that defined these standards, it is not difficult to see why they are still in use today, and will probably continue as such.

The Electronics Industry Association (EIA) of America defined the common serial interface RS-232. ‘RS’ stands for ‘recommended standard’, and the number (suffix -232)

refers to the interface specification of the physical device. The EIA has since established many standards and amassed a library of white papers on various implementations of them. So to keep track of them all it made sense to change the prefix to EIA. (It is interesting to note that most of the white papers are NOT free).

The Telecommunications Industry Association (TIA) was formed in 1988, by merging the telecommunications arm of the EIA and the United States Telecommunications Suppliers Association. The prefix changed again to EIA/TIA-232, (along with all the other serial implementations of course). So now we have TIA-232, TIA-485 etc.

It should also be noted that the TIA is a member of the Electronics Industries Alliance (EIA). This alliance is made up of several trade organizations (including the CEA, ECA, GEIA...) that represent the interests of manufacturers of electronics-related products. Now when someone refers to 'EIA' they are talking about the Alliance, not the Association!

If we still use the terms EIA-232, EIA-422 etc, then they are just as equally obsolete as the 'RS' equivalents. However, when they are referred to as TIA standards some people might give you a quizzical look and ask you to explain yourself... So to cut a long story short, one says 'RS-xxx' -- and the penny drops. 'RS' has become more or less a *de facto* approach, as a search on the Internet will testify.

Copies of the relevant standards are available from Global Engineering documents, the official suppliers of EIA documents. A brief perusal of their website (http://global.ihs.com) will reveal the name changes over time, since names were not changed retroactively. The latest '232' revision refers to TIA-232, but earlier revisions and other related documents still refer to TIA/EIA-232, EIA-232 and RS-232.

1.2 OSI reference model

Faced with the proliferation of closed network systems, the *International Organization for Standardization* (ISO) defined a 'Reference Model for Communication between Open Systems' in 1978. This has become known as the Open Systems Interconnection Reference model, or simply as the OSI model (ISO7498). The OSI model is essentially a data communications management structure, which breaks data communications down into a manageable hierarchy of seven layers.

Each layer has a defined purpose and interfaces with the layers above it and below it. By laying down standards for each layer, some flexibility is allowed so that the system designers can develop protocols for each layer independent of each other. By conforming to the OSI standards, a system is able to communicate with any other compliant system, anywhere in the world.

At the outset it should be realized that the OSI reference model is not a protocol or set of rules for how a protocol should be written but rather an overall framework in which to define protocols. The OSI model framework specifically and clearly defines the functions or services that have to be provided at each of the seven layers (or levels).

Since there must be at least two sites to communicate, each layer also appears to converse with its peer layer at the other end of the communication channel in a virtual ('logical') communication. These concepts of isolation of the process of each layer, together with standardized interfaces and peer-to-peer virtual communication, are fundamental to the concepts developed in a layered model such as the OSI model. The OSI layering concept is shown in Figure 1.1.

The actual functions within each layer are provided by entities that are abstract devices, such as programs, functions, or protocols that implement the services for a particular layer on a single machine. A layer may have more than one entity – for example a protocol entity and a management entity. Entities in adjacent layers interact through the common upper and lower boundaries by passing physical information through *Service Access Points* (SAPs). A SAP could be compared to a pre-defined 'post-box' where one layer would collect data from the previous layer. The relationship between layers, entities, functions and SAPs are shown in Figure 1.2.

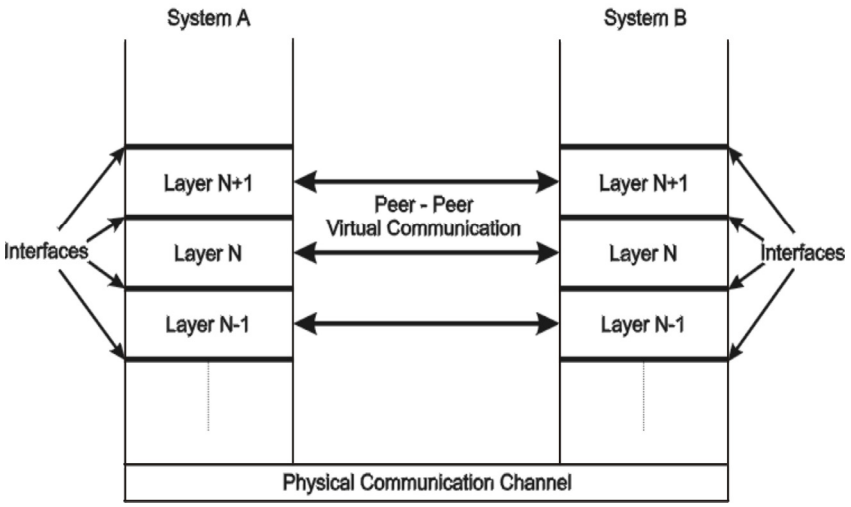


Figure 1.1
OSI layering concept

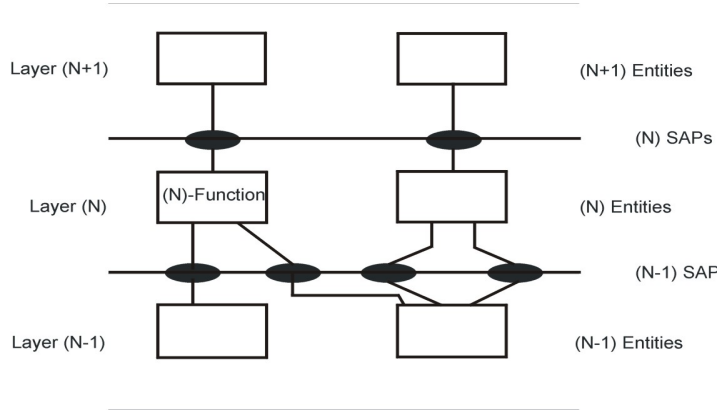


Figure 1.2
Relationship between layers, entities, functions and SAPs

In the OSI model, the entity in the next higher layer is referred to as the N+1 entity and the entity in the next lower layer as N–1. The services available to the higher layers are the result of the services provided by all the lower layers.

The functions and capabilities expected at each layer are specified in the model. However, the model does not prescribe how this functionality should be implemented. The focus in the model is on the 'interconnection' and on the information that can be passed over this connection. The OSI model does not concern itself with the internal operations of the systems involved.

When the OSI model was being developed, a number of principles were used to determine exactly how many layers this communication model should encompass. These principles are:

- A layer should be created where a different level of abstraction is required
- Each layer should perform a well-defined function
- The function of each layer should be chosen with thought given to defining internationally standardized protocols
- The layer boundaries should be chosen to minimize the information flow across the boundaries
- The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy

The use of these principles led to seven layers being defined, each of which has been given a name in accordance with its process purpose. Figure 1.3 shows the seven layers of the OSI model.

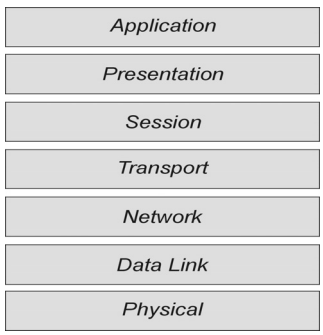


Figure 1.3
The OSI reference model

At the transmitter, the user invokes the system by passing data and control information (physically) to the highest layer of the protocol stack. The system then passes the data physically down through the seven layers, adding headers (and possibly trailers), and invoking functions in accordance with the rules of the protocol. At each level, this combined data and header 'packet' is termed a protocol data unit or PDU. At the receiving site, the opposite occurs with the headers being stripped from the data as it is passed up through the layers. These header and control messages invoke services and a peer-to-peer logical interaction of entities across the sites.

At this stage, it should be quite clear that there is NO connection or direct communication between the peer layers of the network. Rather, all communication is across the physical layer, or the lowest layer of the stack. Communication is down through the protocol stack on the transmitting stack and up through the stack on the receiving stack. Figure 1.4 shows the full architecture of the OSI model, whilst Figure 1.5 shows the effects of the addition of headers (protocol control information) to the respective PDUs at each layer. The net effect of this extra information is to reduce the overall bandwidth of the communications channel, since some of the available bandwidth is used to pass control information.

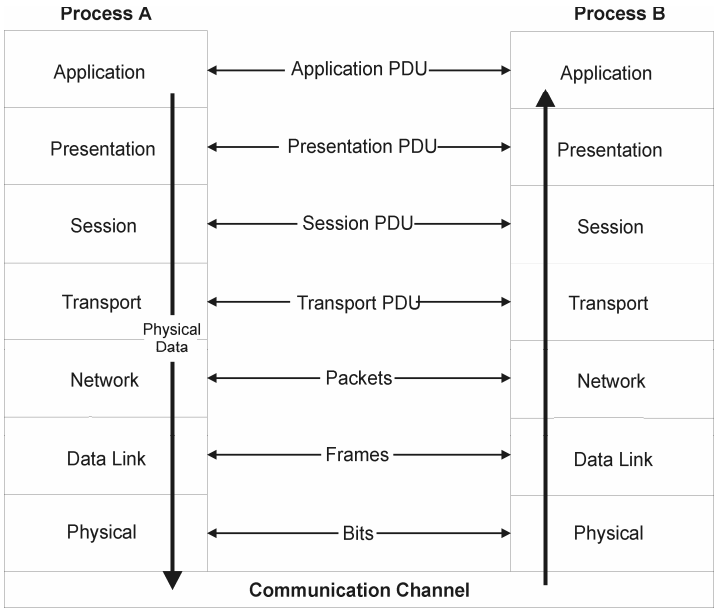


Figure 1.4
Full architecture of OSI model

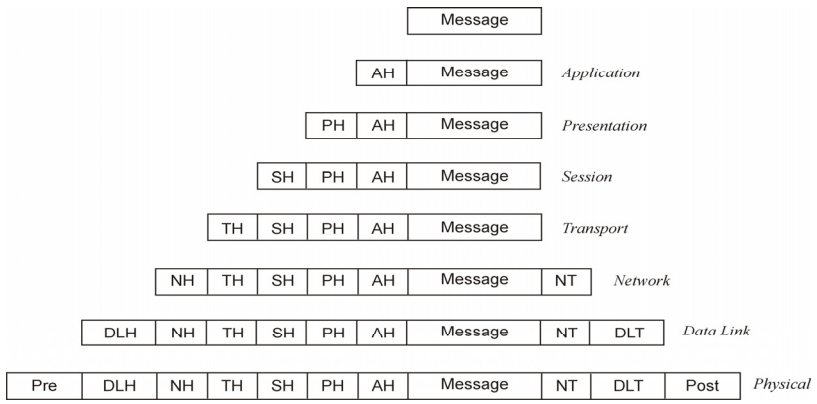


Figure 1.5
OSI message passing

1.2.1 OSI layer services

Briefly, the services provided at each layer of the stack are:

- **Application (layer 7)** – the provision of network services to the user's application programs (clients, servers etc.). Note: the user's actual application programs do NOT reside here
- **Presentation (layer 6)** – maps the data representations into an external data format that will enable correct interpretation of the information on receipt. The mapping can also possibly include encryption and/or compression of data
- **Session (layer 5)** – control of the communications between the users. This includes the grouping together of messages and the coordination of data transfer between grouped layers. It also effects checkpoints for (transparent) recovery of aborted sessions
- **Transport (layer 4**) – the management of the communications between the two end systems
- **Network (layer 3)** – responsible for the control of the communications network. Functions include routing of data, network addressing, fragmentation of large packets, congestion and flow control
- **Data Link (layer 2)** – responsible for sending a frame of data from one system to another. Attempts to ensure that errors in the received bit stream are not passed up into the rest of the protocol stack. Error correction and detection techniques are used here
- **Physical (layer 1)** – Defines the electrical and mechanical connections at the physical level, or the communication channel itself. Functional responsibilities include modulation, multiplexing and signal generation. Note that the Physical layer defines, but does NOT include the medium. This is located below the physical layer and is sometimes referred to as layer 0

A more specific discussion of each layer is now presented.

Application layer

The application layer is the topmost layer in the OSI model. This layer is responsible for giving applications access to the network. Examples of application-layer tasks include file transfer, electronic mail services, and network management. Application-layer services are more varied than the services in lower layers, because the entire range of application and task possibilities is available here. To accomplish its tasks, the application layer passes program requests and data to the presentation layer, which is responsible for encoding the application layer's data in the appropriate form.

Presentation layer

The presentation layer is responsible for presenting information in a manner suitable for the applications or users dealing with the information. Functions, such as data conversion from EBCDIC to ASCII (or vice versa), use of special graphics or character sets, data compression or expansion, and data encryption or decryption are carried out at this layer. The presentation layer provides services for the application layer above it, and uses the session layer below it. In practice, the presentation layer rarely appears in pure form, and is the least well defined of the OSI layers. Application- or session-layer programs will often encompass some or all of the presentation layer functions.

Session layer

The session layer is responsible for synchronizing and sequencing the dialogue and packets in a network connection. This layer is also responsible for making sure that the connection is maintained until the transmission is complete, and ensuring that appropriate security measures are taken during a 'session' (that is, a connection). The session layer is used by the presentation layer above it, and uses the transport layer below it.

Transport layer

In the OSI model the transport layer is responsible for providing data transfer at an agreed-upon level of quality, such as at specified transmission speeds and error rates. To ensure delivery, outgoing packets are assigned numbers in sequence. The numbers are included in the packets that are transmitted by lower layers. The transport layer at the receiving end checks the packet numbers to make sure all have been delivered and to put the packet contents into the proper sequence for the recipient. The transport layer provides services for the session layer above it, and uses the network layer below it to find a route between source and destination. In many ways the transport layer is crucial because it sits between the upper layers (which are strongly application-dependent) and the lower ones (which are network-based).

The layers below the transport layer are collectively known as the subnet layers. Depending on how well (or not) they perform their function, the transport layer has to interfere less (or more) in order to maintain a reliable connection.

Network layer

The network layer is the third lowest layer, or the uppermost subnet layer. It is responsible for the following tasks:

- Determining addresses or translating from hardware to network addresses. These addresses may be on a local network or they may refer to networks located elsewhere on an internetwork. One of the functions of the network layer is, in fact, to provide capabilities needed to communicate on an internetwork
- Finding a route between a source and a destination node or between two intermediate devices
- Establishing and maintaining a logical connection between these two nodes, to establish either a connectionless or a connection-oriented communication. The data is processed and transmitted using the data-link layer below the network layer. Responsibility for guaranteeing proper delivery of the packets lies with the transport layer, which uses network-layer services
- Fragmentation of large packets of data into frames which are small enough to be transmitted by the underlying data link layer. The corresponding network layer at the receiving node undertakes re-assembly of the packet

Data link layer

The data link layer is responsible for creating, transmitting, and receiving data packets. It provides services for the various protocols at the network layer, and uses the physical layer to transmit or receive material. The data link layer creates packets appropriate for the network architecture being used. Requests and data from the network layer are part of the data in these packets (or frames, as they are often called at this layer). These packets are passed down to the physical layer and from there the data is transmitted to the

physical layer on the destination machine. Network architectures (such as Ethernet, ARCnet, token ring, and FDDI) encompass the data-link and physical layers, which is why these architectures support services at the data-link level. These architectures also represent the most common protocols used at the data-link level.

The IEEE802.x networking working groups have refined the data-link layer into two sub-layers: the logical-link control (LLC) sub-layer at the top and the media-access control (MAC) sub-layer at the bottom. The LLC sub-layer must provide an interface for the network layer protocols, and control the logical communication with its peer at the receiving side. The MAC sublayer must provide access to a particular physical encoding and transport scheme.

Physical layer

The physical layer is the lowest layer in the OSI reference model. This layer gets data packets from the data link layer above it, and converts the contents of these packets into a series of electrical signals that represent 0 and 1 values in a digital transmission. These signals are sent across a transmission medium to the physical layer at the receiving end. At the destination, the physical layer converts the electrical signals into a series of bit values. These values are grouped into packets and passed up to the data-link layer.

The mechanical and electrical properties of the transmission medium are defined at this level. These include the following:

- The type of cable and connectors used. A cable may be coaxial, twisted-pair, or fiber optic. The types of connectors depend on the type of cable.
- The pin assignments for the cable and connectors. Pin assignments depend on the type of cable and also on the network architecture being used.
- The format for the electrical signals. The encoding scheme used to signal 0 and 1 values in a digital transmission or particular values in an analog transmission depend on the network architecture being used. Most networks use digital signalling, and most use some form of Manchester encoding for the signal.

1.3 Systems engineering approach

1.3.1 System specifications

Systems engineering, especially in a military context, is a fully-fledged subject and proper treatment thereof will warrant a two-day workshop on its own. However, the basic principles of systems engineering can be applied very advantageously throughout the life cycle of any project, and hence we will briefly look at the concepts. The project, in the context of this workshop, would involve the planning, installation, commissioning and ongoing maintenance of some sort of industrial data communications system.

The question is: what is a system, where does it start and where does it end? The answer is a bit ambiguous – it depends where the designer draws the boundaries. For example; the engine of a motor vehicle, excluding gearbox, radiator, battery and engine mounts, but including fuel injection system, could be seen as a system in its own right. On the other hand, the car in its entirety could be seen as a system, and the engine one of its subsystems. Other subsystems could include the gearbox, drive train, electrical system, etc. In similar fashion a SCADA system integrator could view the entire product as the ‘system’ with, for example, the RTUs as subsystems, whereas for a hardware developer the RTU could be viewed as a ‘system’ in its own right.

The point of departure should be the physical, mechanical and electrical environment in which the system operates. For a car engine this could include the dimensions of the engine compartment, minimum and maximum ambient temperatures and levels of humidity. An engine operating in Alaska in mid-winter faces different problems than its counterpart operating in Saudi Arabia.

In similar fashion an RTU developer or someone contemplating an RTU installation should consider:

- Minimum and maximum temperatures
- Vibration
- Humidity
- Mounting constraints
- IP rating requirements
- Power supply requirements (voltage levels, tolerances, current consumption, power backup and redundancy, etc)

These should all be included in the specifications. Let is return to the engine. There are five attributes necessary to fully describe it, but we will initially look at the first three attributes namely inputs, outputs and functions.

Inputs

What goes 'into' the system? Inputs would include fuel from the fuel pump, air input from the air filter, cold-water input from the radiator and electrical power from the battery. For each input, the mechanical, electrical and other details, as required, must be stated. For example, for the electrical inputs of the engine, the mechanical details of the +12 V and ground terminals must be given, as well as the voltage and current limits.

For an RTU the inputs could include:

- Digital inputs (e.g. contact closures)
- Analog inputs (e.g. 4-20 mA)
- Communication input (RS-232)
- Power (e.g. 12 Vdc at 100 mA)

Specifications should include all relevant electrical and mechanical considerations including connector types, pin allocations, minimum and maximum currents, minimum and maximum voltage levels, maximum operating speeds, and any transient protection.

Stated in general; in the mathematical equation $y = f(x)$, where x would be the input.

Outputs

What comes 'out of' the system? Engine outputs would include torque output to the gearbox, hot water to the radiator and exhaust gases to the exhaust system. For each output, the exact detail (including flange dimensions, bolt sizes, etc) has to be stated. The reason for this is simple. Each output of the engine has to mate exactly with the corresponding input of the associated auxiliary subsystem. Unless the two mating entities are absolutely complementary, dimensionally and otherwise, there will be a problem.

For an RTU the outputs could include:

- Relay outputs
- Open collector transistor outputs

Specifications should include maximum voltages and currents as well as maximum operating speeds, relay contact lifetime and transient protection.

Stated in general; in the mathematical equation y = f (x), y (the output) occurs as a result of x, the input.

Functions

What does the system (viewed as a 'black box') do? The functions are built into the system black box. They convert the input(s) to the output(s) according to some built-in transfer function(s). The system can be seen as having a singular function with several sub-functions, or as simply having several separate functions. The overall function of the engine would be to convert fuel plus air into energy. Its main sub-function would be to convert the fuel plus air into torque to drive the car, another sub-function could be to provide electrical energy to the battery. In the mathematical equation above, this refers to the f ( ) part, in other words it takes 'x' and does something to it in order to produce 'y'.

The three items mentioned so far describes the behavior of the system in terms of 'what' it has to do, but not 'how'. It has, in other words, not described a specific implementation, but just a functional specification. Once this has been documented, reviewed (several times!) and ratified, the solution can be designed.

The full (detailed) specification has to include the 'how'. For this, two additional descriptions are necessary. They are the structure of elements and couplings, and the state transition diagram.

Structure of elements and couplings

It is also referred to as the EC diagram. This refers to all the 'building blocks' of the system and their interrelationship, but does not elucidate the way they operate. In a car engine this would show the engine block, pistons, connecting rods, crankshaft, etc, and the way they are attached to each other.

For an RTU this would include a full electronic circuit diagram as well as a component placement diagram.

1.4 State transition structure

This is also referred to as the ST diagram. This is the 'timing diagram' of the system. It explains, preferably in diagram form (e.g. flowchart), how all the building blocks interact. For the engine, it would show the combustion cycle of the engine, plus the firing sequence of the spark plugs etc.

For an RTU this would be an explanation of the system operation by means of a flow chart. Flowcharts could be drawn for the initial setup, normal system operation (from an operator point of view) and program flow (from a software programmer's point of view) etc.

1.4.1 System life cycle

Our discussion this far has focused on the specification of the system, but not on the implementation thereof. Here is a possible approach. Each phase mentioned here should be terminated with a proper design review. The further a system implementation progresses, the more costly it becomes to rectify mistakes.

1.4.2 Conceptual phase

In this phase, the functional specification is developed. Once it has been agreed upon, one or more possible solutions can be put together and evaluated on paper.

1.4.3 Validation phase

If there are any untested assumptions in the design concept, now is the time to validate it. This could involve setting up a small pilot system or a test network, in order to confirm that the design objectives can be achieved.

1.5 Detailed design

Once the validation has been completed, it is time to do the full, detailed design of the system.

1.5.1 Implementation

This phase involves the procurement of the equipment, the installation, and subsequent commissioning of the system.

1.5.2 Maintenance/troubleshooting

Once the system is operational, these actions will be performed for the duration of its service life. At the end of its useful life the system will be replaced, overhauled or scrapped. In fact often overlooked is the monetary cost of maintaining a system over its useful life, including the cost of parts, maintenance and service infrastructure that could exceed the initial purchase cost be a factor of 5 or more.

1.6 Media

For any communication to take place between two entities there must be some form of medium between them. The OSI model does not include the actual medium (although it may specify it). The medium is sometimes referred to as ‘layer 0’ (being below layer 1) although, in fact, there is no such thing. In the context of Data Communications we can distinguish between two basic groupings namely conductive media and radiated media.

In the case of conductive media there is a physical cable between the two devices. This cable could be either a copper cable or an optic fiber cable.

In copper cable, the signal is conducted as electrical impulses. This type of cable can be in the form of:

- Coaxial cable, for example RG-58
- Twisted pair cable (single or multi-pair), for example EIA/TIA-568 Cat 5, or
- Untwisted (parallel) cable, for example, the flat cables for DeviceNet or AS-i

Twisted pair cable can be unshielded or shielded with foil, woven braid or a combination thereof.

In the case of optic fiber, the signal is conducted as impulses of light. There are two main approaches possible with fiber optic cables, namely:

- Single mode (monomode) cabling, and
- Multimode cabling

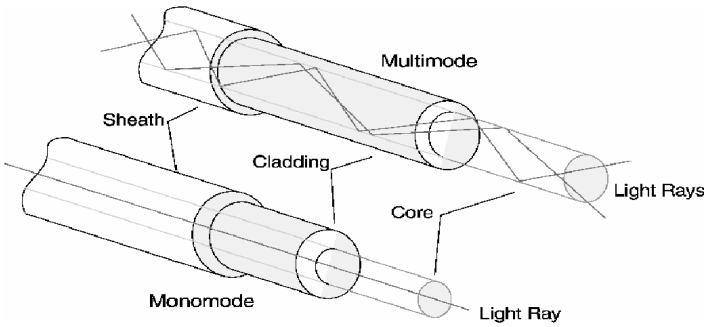


Figure 1.6
Monomode and multimode optic fibers

This is widely used throughout industrial communications systems because of immunity to electrical noise and optical isolation from surges and transients. As a result, fiber is tending to dominate in all new installations that require reasonable levels of traffic.

An alternative to conductive media is radiated media. Here the medium is actually free space, and various techniques are used to transmit the signal. These include infrared transmission as well as VHF transmission (30 MHz–300 MHz) and UHF transmission (300 MHz–3 GHz). A very popular band is the unlicensed 2.4 GHZ ISM (industrial, scientific and medical) band as used in IEEE 802.15 Bluetooth and most wireless LANs e.g. IEEE802.11.

In microwave transmission a differentiation is often made in terms of terrestrial systems (i.e. transmission takes place in a predominantly horizontal plane) and satellite transmission, where transmission takes place in a predominantly vertical plane.

1.7 Physical connections

This refers to layer 1 of the OSI model and deals with the mechanism of placing an actual signal on the conductor for the purpose of transmitting 1s and 0s. Many network standards such as Ethernet and AS-i have their own unique way of doing this. Many others, such as Data Highway Plus and Profibus, use the RS-485 standard.

Here follows a brief summary of RS-485, although it is covered in detail elsewhere. RS-485 is a balanced (differential) system with up to 32 ‘standard’ transmitters and receivers per line, speeds up to 10 Mbps and distances up to 1200 m.

The RS-485 standard is very useful for instrumentation and control systems, where several instruments or controllers may be connected together on the same multipoint network.

A diagram of a typical RS-485 systems is shown in Figure 1.7.

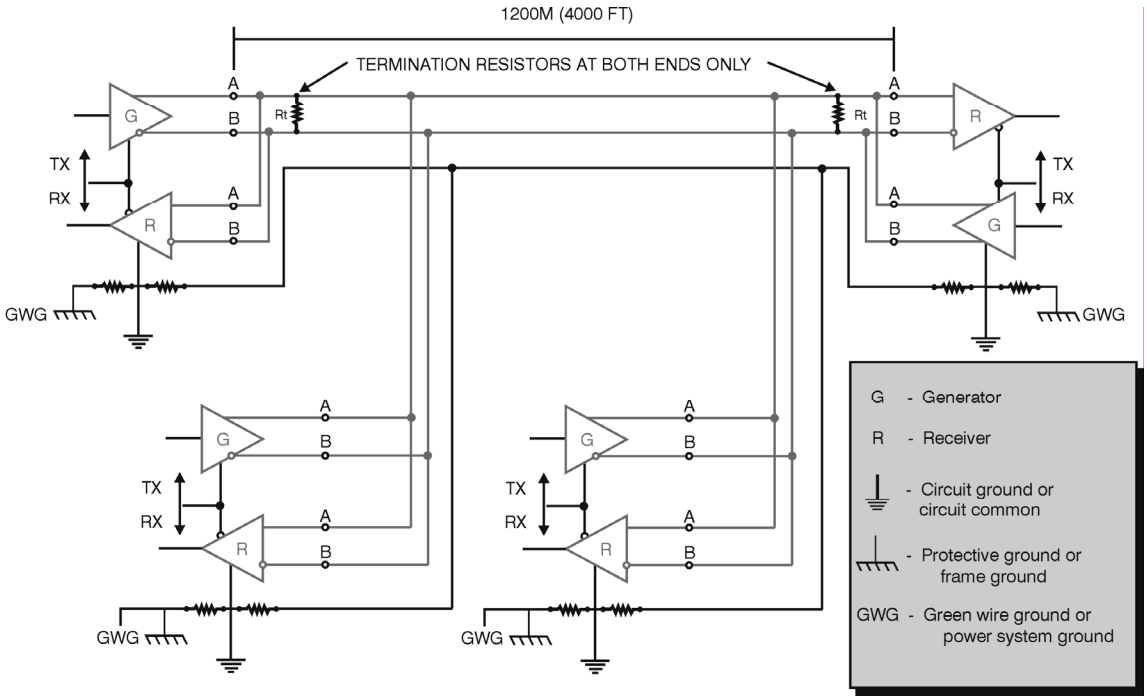


Figure 1.7
Typical two-wire multidrop network for RS-485

1.8 Protocols

It has been shown that there are protocols operating at layers 2 to 7 of the OSI model. Layer 1 is implemented by physical standards such as RS-232 and RS-485, which are mechanisms for 'putting the signal on the wire' and are therefore not protocols. Protocols are the sets of rules by which communication takes place, and are implemented in software.

Protocols vary from the very simple (such as ASCII based protocols) to the very sophisticated (such as TCP and IP), which operate at high speeds transferring megabits of data per second. There is no right or wrong protocol, the choice depends on a particular application.

Examples of protocols include:

- Layer 2: SDLC, HDLC
- Layer 3: IP, IPX
- Layer 4: TCP, UDP, SPX
- Layers 5+6+7: CIP, HTTP, FTP, POP3, NetBIOS

Depending on their functionality and the layer at which they operate, protocols perform one or more of the following functions.

- **Segmentation (fragmentation) and reassembly:** Each protocol has to deal with the limitations of the PDU (protocol data unit) or packet size associated with the protocol below it. For example, the Internet Protocol (IP) (layer 3) can only handle 65536 bytes of data, hence the Transmission Control Protocol (TCP) (layer 4) has to segment the data received from layer 5 into pieces no bigger than that. IP (layer 3), on the other hand, has to be aware that Ethernet (layer 2) cannot accept more than 1500 bytes of data at a time, and has to fragment the data accordingly. The term 'fragmentation' is normally

associated with layer 3 whereas the term 'segmentation' is normally associated with layer 4. The end result of both is the same but the mechanisms differ. Obviously the data stream fragmented by a protocol on the transmitting side has to be re-assembled by its corresponding peer on the receiving side, so each protocol involved in the process of fragmentation has to add appropriate parameters in the form of sequence numbers, offsets and flags to facilitate this.

- **Encapsulation:** Each protocol has to handle the information received from the layer above it 'without prejudice'; i.e. it carries forwards it without regard for its content. For example, the information passed on to IP (layer 3) could contain a TCP header (layer 4) plus an FTP header (layers 5,6,7) plus data from an FTP client (e.g. Cute FTP). IP simply regards this as a package of information to be forwarded, adds its own header with the necessary control information, and passes it down to the next layer (e.g. Ethernet)
- **Connection control:** Some layer 4 protocols such as TCP create logical connections with their peers on the other side. For example, when browsing the Internet, TCP on the client (user) side has to establish a connection with TCP on the server side before a web site can be accessed. Obviously there are mechanisms for terminating the connection as well
- **Ordered delivery:** Large messages have to be cut into smaller fragments, but on a packet switching network the different fragments can theoretically travel via different paths to their destination. This results in fragments arriving at their destination out of sequence, which creates problems in rebuilding the original message. This issue is normally addressed at layer 3 and sometimes at layer 4 (anywhere that fragmentation and segmentation takes pace) and different protocols use different mechanisms, including sequence numbers and fragment offsets
- **Flow control:** The protocol on the receiving side must be able to liaise with its counterpart on the sending side in order not to be overrun by data. In simple protocols this is accomplished by a lock-step mechanism (i.e. each packet sent needs to be acknowledged before the next one can be sent) or XON/XOFF mechanisms where the receiver sends an XOFF message to the sender to pause transmission, then sends an XON message to resume transmission.

  More sophisticated protocols us 'sliding windows'. Here, the sliding window is a number that represents the amount of unacknowledged data that can still be sent. The receiver does not have to acknowledge every message, but can from time to time issue blanket acknowledgements for all data received up to a point. As the sender sends data, the window shrinks and as the receiver acknowledges, the window expands accordingly. When the window becomes zero, the transmitter stops until some acknowledgment is received and the window opens up again
- **Error control:** The sender needs some mechanism by which it can ascertain if the data received is the same as the data sent. This is accomplished by performing some form of checksum on the data to be transmitted, and including the checksum in the header or in a trailer after the data. Types of checksum include vertical and longitudinal parity, *block check count* (BCC) and c*yclic redundancy checking* (CRC)
- **Addressing:** Protocols at various levels need to identify the physical or logical recipient on the other side. This is done by various means. Layer 4

protocols such as TCP and UDP use port numbers. Layer 3 protocols use a protocol address (such as the IP address for the Internet Protocol) and layer 2 protocols use a hardware (or 'media') address such as a station number or MAC address

- **Routing:** In an internetwork, i.e. a larger network consisting of two or more smaller networks interconnected by routers, the routers have to communicate with each other in order to know the best path to a given destination on the network. This is achieved by routing protocols (RIP, OSPF etc) residing on the routers
- **Multiplexing:** Some higher-protocols such as TCP can create several 'logical' channels on one physical channel. The opposite can be done some lower-level protocols such as PPP where one logical stream of data can be sent over several physical (e.g. dial-up) connections. This mechanism is called multiplexing

1.9 Noise

1.9.1 Sources of electrical noise

Typical sources of noise are devices that produce quick changes (or spikes) in voltage or current, such as:

- Large electrical motors being switched on
- Fluorescent lighting tubes
- Lightning strikes
- High voltage surging due to electrical faults
- Welding equipment

From a general point of view, there must be three contributing factors for the existence of an electrical noise problem. They are:

- A source of electrical noise
- A mechanism coupling the source to the affected circuit
- A circuit conveying the sensitive communication signals

1.9.2 Electrical coupling of noise

There are four forms of coupling of electrical noise into the sensitive data communications circuits. They are:

- Impedance coupling (sometimes referred to as conductance coupling)
- Electrostatic coupling
- Magnetic or inductive coupling
- Radio frequency radiation (a combination of electrostatic and magnetic)

Each of these noise forms will be discussed in some detail in the following sections.

1.9.3 Impedance coupling (or common impedance coupling)

For situations where two or more electrical circuits share common conductors, there can be some coupling between the different circuits with harmful effects on the connected circuits. Essentially, this means that the signal current from the one circuit proceeds back

along the common conductor resulting in an error voltage along the return bus that affects all the other signals. The error voltage is due to the impedance of the return wire. This situation is shown in the figure 1.8.
Obviously, the quickest way to reduce the effects of impedance coupling is to minimize the impedance of the return wire. The best solution is to use a separate return for each individual signal.

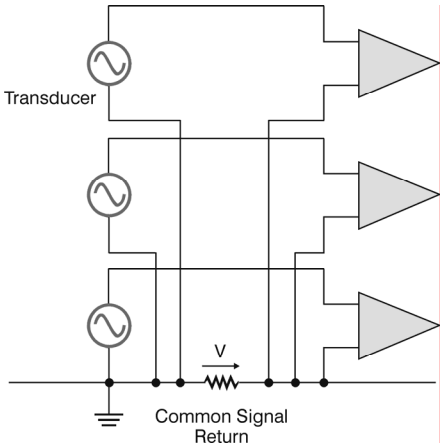


Figure 1.8
Impedance coupling

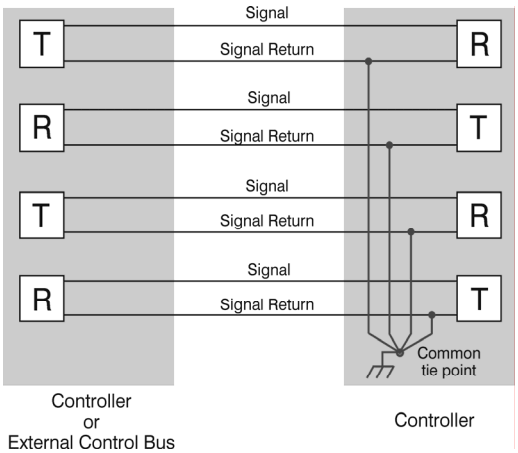


Figure 1.9
Impedance coupling eliminated with separate ground returns

1.9.4 Electrostatic or capacitive coupling

This form of coupling is proportional to the capacitance between the noise source and the signal wires. The magnitude of the interference depends on the rate of change of the noise voltage and the capacitance between the noise circuit and the signal circuit.

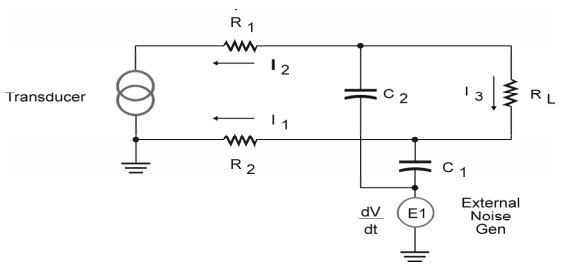


Figure 1.10
Electrostatic coupling

In the figure above, the noise voltage is coupled into the communication signal wires through two capacitors, C_1 and C_2, and a noise voltage is produced across the resistance in the circuit. The size of the noise (or error) voltage in the signal wires is proportional to the:

- Inverse of the distance of noise voltage from each of the signal wires
- Length (and hence impedance) of the signal wires into which the noise is induced
- Amplitude (or strength) of the noise voltage
- Frequency of the noise voltage
- There are four methods for reducing the noise induced by electrostatic coupling

They are:

- Shielding of the signal wires
- Separating from the source of the noise
- Reducing the amplitude of the noise voltage (and possibly the frequency)
- Twisting of the signal wires

The problem can be addressed by installing an electrostatic shield around the signal wires. The currents generated by the noise voltages prefer to flow down the lower impedance path of the shield rather than the signal wires. If one of the signal wires and the shield are tied to the ground at one point, which ensures that the shield and the signal wires are at an identical potential, then reduced signal current flows between the signal wires and the shield.

The shield must be of a low resistance material such as aluminum or copper. For a loosely braided copper shield (85% braid coverage), the screening factor is about 100 times or 20 dB. For a low resistance multi layered screen, this screening factor can be 35 dB or 3000 times.

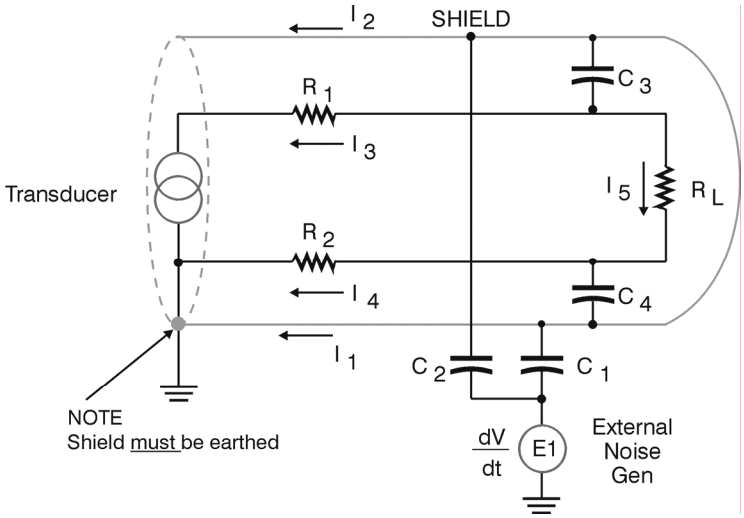


Figure 1.11
Shield to minimize electrostatic coupling

Twisting of the signal wires provides a slight improvement in reducing the induced noise voltage by ensuring that C_1 and C_2 are closer together in value; thus ensuring that any noise voltages induced in the signal wires tend to cancel each other out.

Provision of a shield by a cable manufacturer ensures that the capacitance between the shield and each wire is equal in value, thus eliminating any noise voltages by cancellation.

1.9.5 Magnetic or inductive coupling

This depends on the rate of change of the noise current and the mutual inductance between the noise system and the signal wires. Expressed slightly differently, the degree of noise induced by magnetic coupling will depend on the:

- Magnitude of the noise current
- Frequency of the noise current
- Area enclosed by the signal wires (through which the noise current magnetic flux cuts)
- Inverse of the distance from the disturbing noise source to the signal wires

The effect of magnetic coupling is shown in Figure 1.12.

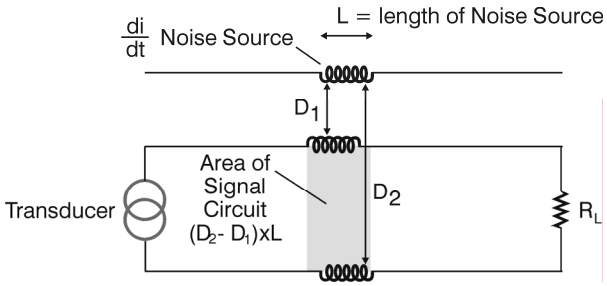


Figure 1.12
Magnetic coupling

The easiest way of reducing the noise voltage caused by magnetic coupling is to twist the signal conductors. This results in lower noise due to the smaller area for each loop. This means less magnetic flux to cut through the loop and consequently, a lower induced noise voltage. In addition, the noise voltage that is induced in each loop tends to cancel out the noise voltages from the next sequential loop. It is assumed that the noise voltage is induced in equal magnitudes in each signal wire due to the twisting of the wires giving a similar separation distance from the noise voltage.

The second approach is to use a magnetic shield around the signal wires. The magnetic flux generated from the noise currents induces small eddy currents in the magnetic shield. These eddy currents then create an opposing magnetic flux ϕ_1 to the original flux ϕ_2. This means a lesser flux ($\phi_2 - \phi_1$) reaches our circuit.

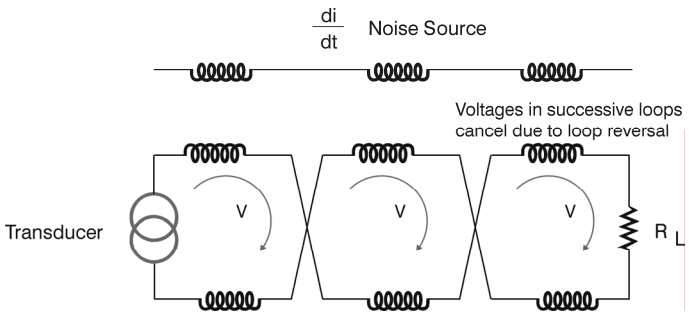


Figure 1.13
Twisting of wires to reduce magnetic coupling

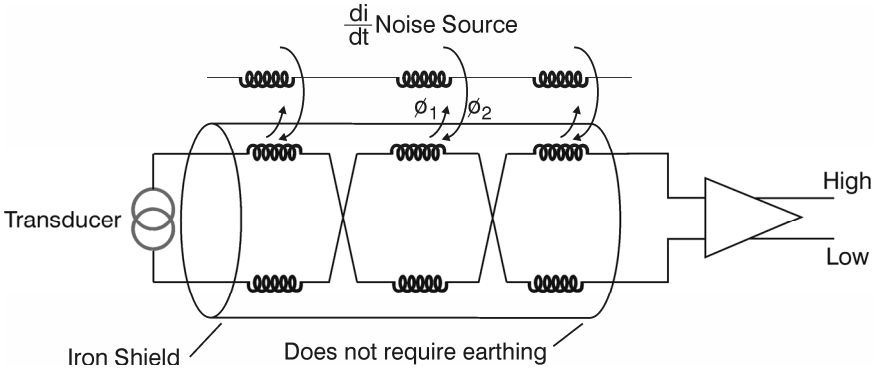


Figure 1.14
Use of magnetic shield to reduce magnetic coupling

Note: The magnetic shield does not require grounding. It works merely by being present. High permeability steel makes best magnetic shields for special applications. However, galvanized steel conduit makes quite an effective shield.

1.9.6 Radio frequency radiation

The noise voltages induced by electrostatic and inductive coupling (discussed above) are manifestations of the near field effect, which is electromagnetic radiation close to the source of the noise. This sort of interference is often difficult to eliminate. It requires close attention to grounding of the adjacent electrical circuit, and the ground connection is only effective for circuits in close proximity to the electromagnetic radiation. The effects of electromagnetic radiation can be neglected unless the field strength exceeds 1 volt/meter. This can be calculated by the formula:

Field strength = $\sqrt{2}$(Power) /Distance

Where:

Field strength	volt/meter
Power	kilowatt
Distance	km

The two most commonly used mechanisms to minimize electromagnetic radiation are:

- Proper shielding (iron)
- Capacitors to shunt the noise voltages to ground

Any incompletely shielded conductors will perform as a receiving aerial for the radio signal and hence care should be taken to ensure good shielding of any exposed wiring.

1.9.7 Shielding

It is important that electrostatic shielding is only grounded at one point. More than one ground point will cause circulating currents. The shield should be insulated to prevent inadvertent contact with multiple ground points, which could result in circulating currents. The shield should never be left floating because that would tend to allow capacitive coupling, rendering the shield useless.
Two useful techniques for isolating one circuit from the other are by the use of opto-isolation as shown in the Figure 1.15, and transformer coupling as shown in Figure 1.16.

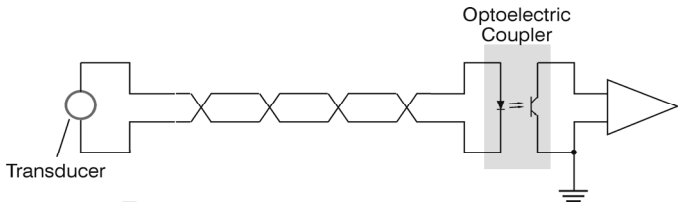


Figure 1.15
Opto-isolation of two circuits

Although opto-isolation does isolate one circuit from the other, it does not prevent noise or interference being transmitted from one circuit to another.

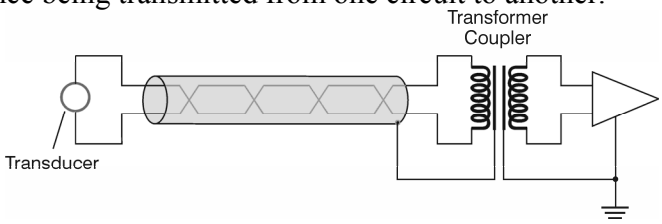


Figure 1.16
Transformer coupling

Transformer coupling can be preferable to optical isolation when there are high-speed transients in one circuit. There is some capacitive coupling between the LED and the base of the transistor, which is in the opto-coupler, can allow these types of transients to penetrate one circuit from another. This is not the case with transformer coupling.

1.9.8 Good shielding performance ratios

The use of some form of low resistance material covering the signal conductors is considered good shielding practice for reducing electrostatic coupling. When comparing shielding with no protection, this reduction can vary from copper braid (85% coverage), which returns a noise reduction ratio of 100:1 to aluminum Mylar tape with drain wire, with a ratio of 6000:1.

Twisting the wires to reduce inductive coupling reduces the noise (in comparison to no twisting) by ratios varying from 14:1 (for four-inch lay) to 141:1 (for one inch lay). In comparison, putting parallel (untwisted) wires into steel conduit only gives a noise reduction of 22:1.

On very sensitive circuits with high levels of magnetic and electrostatic coupling the approach is to use coaxial cables. Double-shielded cable can give good results for very sensitive circuits.

Note: With double shielding, the outer shield could be grounded at multiple points to minimize radio frequency circulating loops. This distance should be set at intervals of less than 1/8 of the wavelength of the radio frequency noise.

1.9.9 Cable ducting or raceways

These are useful in providing a level of attenuation of electric and magnetic fields. These figures are done at 60 Hz for magnetic fields and 100 kHz for electric fields.

Typical screening factors are:

- 5 cm (2 inch) aluminum conduit with 0.154 inch thickness: magnetic fields (at 60 Hz) 1.5:1, electric fields (at 100 kHz) 8000:1
- Galvanized steel conduit 5 cm (2 inch), wall thickness 0.154 inch width: magnetic fields (at 60 Hz) 40:1, electric fields (at 100 kHz) 2000:1

1.10 Cable spacing

In situations where there are a large number of cables varying in voltage and current levels, the IEEE518–1982 standard has developed a useful set of tables indicating separation distances for various classes of cables. There are four classification levels of susceptibility for cables. Susceptibility, in this context, is understood to be an indication of how well the signal circuit can differentiate between the undesirable noise and required signal. It follows a data communication physical standard such as RS-232E that would have a high susceptibility and a 1000 volt, 200 amp ac cable that has a low susceptibility. The four susceptibility levels defined by the IEEE 518 standard are briefly:

- **Level 1: High**
  This is defined as analog signals less than 50 volt and digital signals less than 15 volt. This would include digital logic buses and telephone circuits. Data communication cables fall into this category
- **Level 2: Medium**
  This category includes analog signals greater than 50 volt and switching circuits
- **Level 3: Low**
  This includes switching signals greater than 50 volt and analog signals greater than 50 volt. Currents less than 20 amp are also included in this category
- **Level 4: Power**
  This includes voltages in the range 0–1000 volt and currents in the range 20–800 amps. This applies to both ac and dc circuits

IEEE 518 also provides for three different situations when calculating the separation distance required between the various levels of susceptibilities.

In considering the specific case where one cable is a high susceptibility cable and the other cable has a varying susceptibility, the required separation distance would vary as follows:

- **Both cables contained in a separate tray**
  1. Level 1 to Level 2–30 mm
  2. Level 1 to Level 3–160 mm
  3. Level 1 to Level 4–670 mm
- **One cable contained in a tray and the other in conduit**
  4. Level 1 to Level 2–30 mm
  5. Level 1 to Level 3–110 mm
  6. Level 1 to Level 4–460 mm
- **Both cables contained in separate conduit**
  7. Level 1 to Level 2–30 mm
  8. Level 1 to Level 3–80 mm
  9. Level 1 to Level 4–310 mm

Figures are approximate as the original standard is quoted in inches. A few words need to be said about the construction of the trays and conduits. The trays are to be manufactured from metal and firmly grounded with complete continuity throughout the length of the tray. The trays should also be fully covered preventing the possibility of any area being without shielding.

1.10.1 Grounding requirements

This is a contentious issue and a detailed discussion laying out all the theory and practice is possibly the only way to minimize the areas of disagreement. The picture is further complicated by different national codes, which whilst not actively disagreeing with the basic precepts of other countries, tend to lay down different practical techniques in the implementation of a good grounding system.

A typical design should be based around three separate ground systems. They are:

- The equipment (or instrument) ground
- The chassis (or safety) ground
- The earth ground

The aims of these systems are:

- To minimize the electrical noise in the system
- To reduce the effects of fault or ground loop currents on the instrumentation system
- To minimize the hazardous voltages on equipment due to electrical faults

Ground is defined as a common reference point for all signals in equipment situated at zero potential. Below 10 MHz, a single point grounding system is the optimum solution. Two key concepts to be considered when setting up an effective grounding system are:

- To minimize the effects of impedance coupling between different circuits (i.e. when three different currents, for example, flow through a common impedance)
- To ensure that ground loops are not created (for example, by mistakenly tying the screen of a cable at two points to ground)

There are three types of grounding system possible as shown in Figure 1.17. The series single point is perhaps the more common; while the parallel single point is the preferred approach with a separate ground system for different groups of signals.

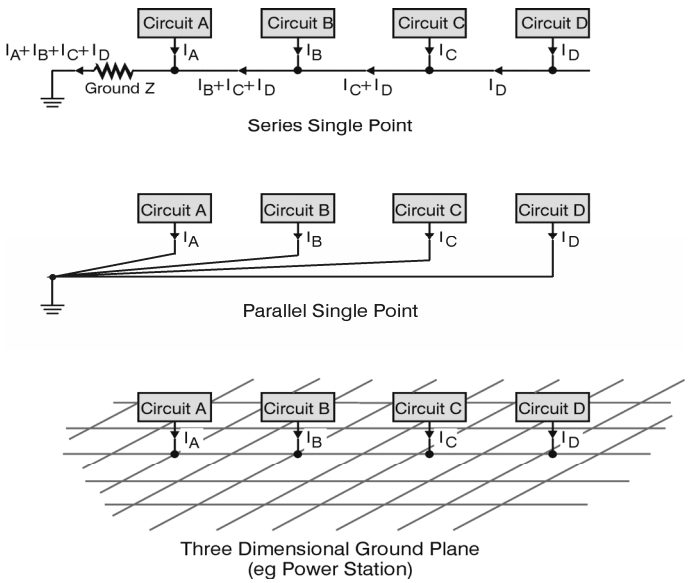


Figure 1.17
Various grounding configurations

1.10.2 Suppression techniques

It is often appropriate to approach the problem of electrical noise proactively by limiting the noise at the source. This requires knowledge of the electrical apparatus that is causing the noise and then attempting to reduce the noise caused here. The two main approaches are shown here.

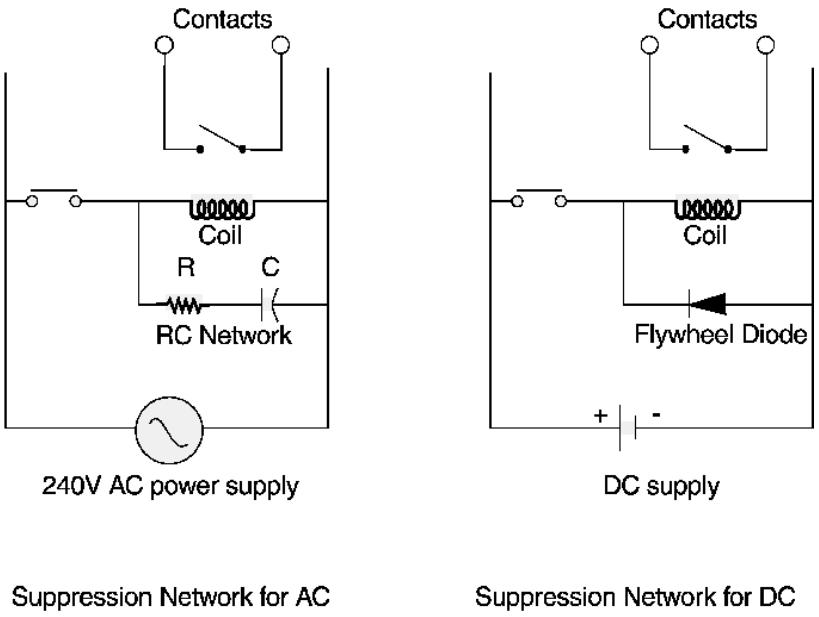


Figure 1.18
Suppression networks (snubbers)

In Figure 1.18, the inductance will generate a back emf across the contacts when the voltage source applied to it is switched off. This RC network then takes this back emf and thus reduces damage to the contacts.

The voltage can be limited by various combinations of devices, depending on whether the circuit is ac or dc.

Circuit designers should be aware that the response time of the coil could be reduced significantly. For example, the dropout time of a coil can be increased by a factor of ten. Hence this should be approached with caution, where quick response is required from regular switched circuits (apart from the obvious negative impact on safety due to slowness of operation).

Silicon controlled rectifiers (SCRs) and triacs generate considerable electrical noise due to the switching of large currents. A possible solution is to place a correctly sized inductor in series with the switching device.

1.10.3 Filtering

Filtering should be done as close to the source of noise as possible. A table below summarizes some typical sources of noise and possible filtering means.

Typical sources of noise	Filtering remedy	Comments
Ac voltage varies	Improved ferroresonant transformer	Conventional ferroresonant transformer fails
Notching of ac waveform form	Improved ferroresonant transformer	Conventional ferroresonant transformer fails
Missing half cycle in ac waveform	Improved ferroresonant transformer	Conventional ferroresonant transformer fails
Notching in dc line	Storage capacitor	For extreme cases active power line filters are required
Random excessively high voltage spikes or transients	Non-linear filters	Also called limiters
High frequency components	Filter capacitors across the line	Called low pass filtering –great care should be taken with high frequency vs performance of 'capacitors' at this frequency
Ringing of filters	Use T filters	From switching transients or high level of harmonics
60 hz or 50Hz interference	Twin-T RC notch filter networks	Sometimes low pass filters can be suitable
Common mode voltages	Avoid filtering (isolation transformers or common-mode filters)	Opto isolation is preferred eleiminates ground loop
Excessive noise	Auto or cross correlation techniques	Extracts the signal spectrum from the closely overlapping noise spectrum

Table 1.1
Typical noise sources and some possible means of filtering

1.11 Ingress protection

The *ingress protection* (IP) rating system is recognized in most countries and is described by several standards, including IEC 60529. It describes the degree of protection offered by an enclosure. This enclosure can be of any description, including a cable, cable assembly, connector body, the casing of a network hub or a large cabinet used to enclose electronic equipment.
Enclosures are rated in the format 'IP xy' or 'IP xyz.'

- The first digit of the IP designation (x) describes the degree of protection against access to hazardous parts and ingress of solid objects
- The second digit (y) designates the degree of protection against water. Refer to the appropriate sections of IEC 60529 for complete information regarding applications, features, and design tests
- The third digit (z) describes the degree of protection against mechanical impacts and is often omitted. It does, for example, apply to metal enclosures but not to cables or cable assemblies

Here follows a list of meanings attributed to the digits of the IP rating.

1st	Protection against foreign objects	2nd	Protection against moisture	3rd	Protection against mechanical impacts
0	Not protected	0	Not protected	0	Not protected
1	Protected against objects greater than 50 mm diameter (e.g. hand contact)	1	Protected against dripping water (falling vertically, e.g. condensation)	1	Impact 0.225 joule
2	Protected against objects greater than 12 mm (e.g. fingers)	2	Protected against dripping water when tilted 15° to either side	2	Impact 0.375 joule
3	Protected against objects greater than 2.5 mm (e.g. tolls, wires)	3	Protected against rain up to 60 degrees from vertical	3	Impact 0.60 joule
4	Protected against objects greater than 1.0 mm (e.g. small tools, small wires)	4	Protected against splashing water, any direction	4	N/a
5	Dust protected – limited ingress permitted (no harmful deposits)	5	Protected against water jets (with nozzles)	5	Impact 2.00 joule
6	Dust tight – totally protected against dust (no deposits at all)	6	Protected against heavy seas	6	N/a
7	N/a	7	Protection against effects of immersion	7	Impact 6.00 joule
8	N/a	8	Protection against submersion	8	N/a
9	N/a	9	N/a	9	Impact 20.00 joule

For example, a marking of IP 68 would indicate a dust tight (first digit = 6) piece of equipment that is protected against submersion in water (second digit = 8).

2

RS-232 fundamentals

Objectives

When you have completed study of this chapter, you will be able to:

- List the main features of the RS-232 standard
- Fix the following problems:
  - Incorrect RS-232 cabling
  - Male/female D-type connector confusion
  - Wrong DTE/DCE configuration
  - Handshaking
  - Incorrect signaling voltages
  - Excessive electrical noise
  - Isolation

2.1 RS-232 Interface standard (CCITT V.24 Interface standard)

The RS-232 interface standard was developed for the single purpose of interfacing *data terminal equipment* (DTE) and data *circuit terminating equipment* (DCE) employing serial binary data interchange. In particular, RS-232 was developed for interfacing data terminals to modems.

The RS-232 interface standard was issued in the USA in 1969 by the engineering department of the RS. Almost immediately, minor revisions were made and RS-232C was issued. RS-232 was originally named RS-232, (Recommended Standard), which is still in popular usage. The prefix 'RS' was superseded by 'EIA/TIA' in 1988. The current revision is EIA/TIA-232E (1991), which brings it into line with the international standards ITU V.24, ITU V.28 and ISO-2110.

Poor interpretation of RS-232 has been responsible for many problems in interfacing equipment from different manufacturers. This had led some users to dispute as to whether it is a 'standard.' It should be emphasized that RS-232 and other related RS standards

define the electrical and mechanical details of the interface (layer 1 of the OSI model) and do not define a protocol.

The RS-232 interface standard specifies the method of connection of two devices – the DTE and DCE. DTE refers to data terminal equipment, for example, a computer or a printer. A DTE device communicates with a DCE device. DCE, on the other hand, refers to data communications equipment such as a modem. DCE equipment is now also called data circuit-terminating equipment in EIA/TIA-232E. A DCE device receives data from the DTE and retransmits to another DCE device via a data communications link such as a telephone link.

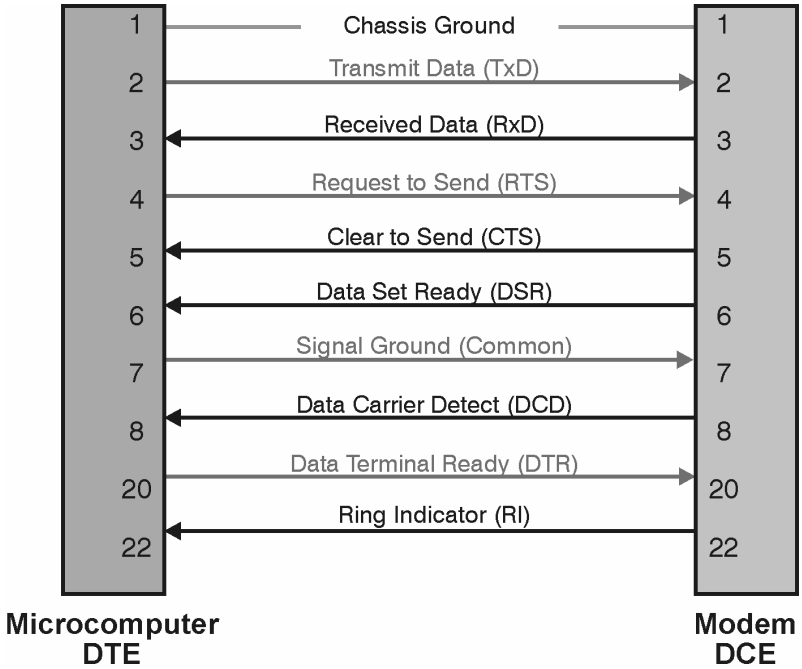


Figure 2.1
Connections between the DTE and the DCE using DB-25 connectors

2.1.1 The major elements of RS-232

The RS-232 standard consists of three major parts, which define:

- Electrical signal characteristics
- Mechanical characteristics of the interface
- Functional description of the interchange circuits

Electrical signal characteristics

RS-232 defines electrical signal characteristics such as the voltage levels and grounding characteristics of the interchange signals and associated circuitry for an unbalanced system.

The RS-232 transmitter is required to produce voltages in the range +/1 5 to +/– 25 V as follows:

- Logic 1: –5 V to –25 V
- Logic 0: +5 V to +25 V
- Undefined logic level: +5 V to –5 V

At the RS-232 receiver, the following voltage levels are defined:

- Logic 1: –3 V to –25 V
- Logic 0: +3 V to +25 V
- Undefined logic level: –3 V to +3 V

Note: The RS-232 transmitter requires a slightly higher voltage to overcome voltage drop along the line.

The voltage levels associated with a microprocessor are typically 0 V to +5 V for *Transistor-Transistor Logic* (TTL). A line driver is required at the transmitting end to adjust the voltage to the correct level for the communications link. Similarly, a line receiver is required at the receiving end to translate the voltage on the communications link to the correct TTL voltages for interfacing to a microprocessor. Despite the bipolar input voltage, TTL compatible RS-232 receivers operate on a single +5 V supply.

Modern PC power supplies usually have a standard +12 V output that could be used for the line driver.

The control or 'handshaking' lines have the same range of voltages as the transmission of logic 0 and logic 1, except that they are of opposite polarity. This means that:

- A control line asserted or made active by the transmitting device has a voltage range of +5 V to +25 V. The receiving device connected to this control line allows a voltage range of +3 V to +25 V
- A control line inhibited or made inactive by the transmitting device has a voltage range of –5 V to –25 V. The receiving device of this control line allows a voltage range of –3 V to –25 V

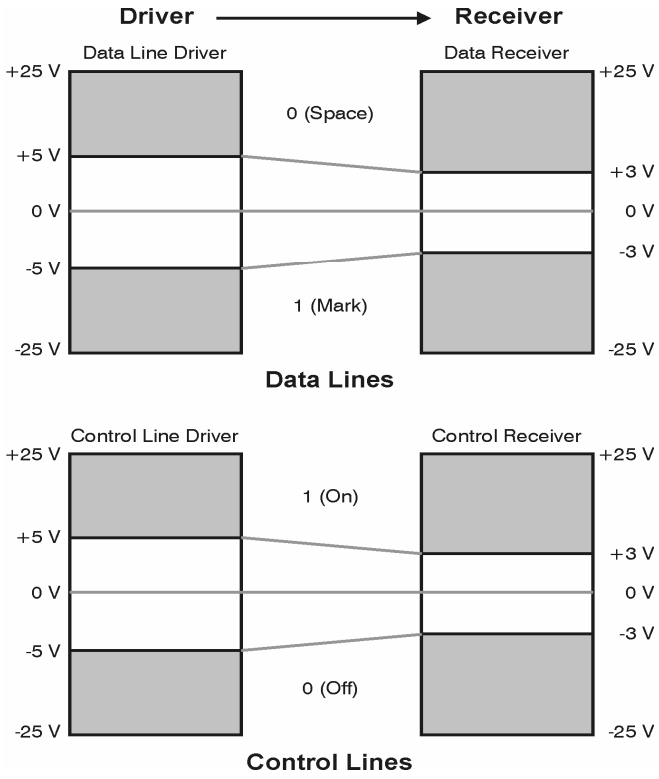


Figure 2.2
Voltage levels for RS-232

At the receiving end, a line receiver is necessary in each data and control line to reduce the voltage level to the 0 V and +5 V logic levels required by the internal electronics.

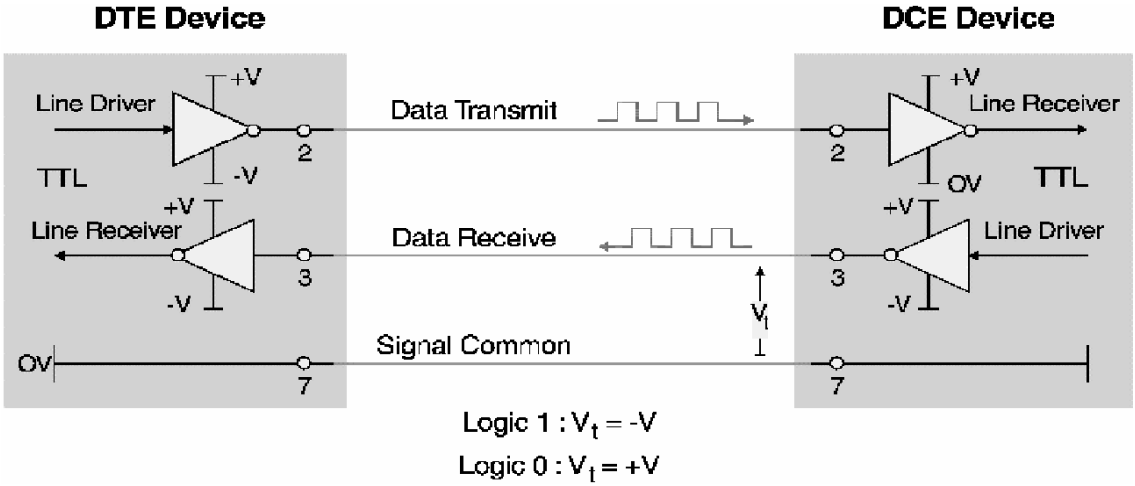


Figure 2.3
RS-232 transmitters and receivers

The RS-232 standard defines 25 electrical connections. The electrical connections are divided into four groups viz:

- Data lines
- Control lines
- Timing lines
- Special secondary functions

Data lines are used for the transfer of data. Data flow is designated from the perspective of the DTE interface. The transmit line, on which the DTE transmits and the DCE receives, is associated with pin 2 at the DTE end and pin 2 at the DCE end for a DB-25 connector. These allocations are reversed for DB-9 connectors. The receive line, on which the DTE receives, and the DCE transmits, is associated with pin 3 at the DTE end and pin 3 at the DCE end. Pin 7 is the common return line for the transmit and receive data lines.

Control lines are used for interactive device control, which is commonly known as hardware handshaking. They regulate the way in which data flows across the interface.

The four most commonly used control lines are:

- RTS: Request to send
- CTS: Clear to send
- DSR: Data set ready (or DCE ready in RS-232D/E)
- DTR: Data terminal ready (or DTE ready in RS-232D/E)

It is important to remember that with the handshaking lines, the enabled state means a positive voltage and the disabled state means a negative voltage.

Hardware handshaking is the cause of most interfacing problems. Manufacturers sometimes omit control lines from their RS-232 equipment or assign unusual applications to them. Consequently, many applications do not use hardware handshaking but, instead, use only the three data lines (transmit, receive and signal common ground) with some form of software handshaking. The control of data flow is then part of the application

program. Most of the systems encountered in data communications for instrumentation and control use some sort of software-based protocol in preference to hardware handshaking.

There is a relationship between the allowable speed of data transmission and the length of the cable connecting the two devices on the RS-232 interface. As the speed of data transmission increases, the quality of the signal transition from one voltage level to another, for example, from –25V to +25 V, becomes increasingly dependent on the capacitance and inductance of the cable.

The rate at which voltage can 'slew' from one logic level to another depends mainly on the cable capacitance and the capacitance increases with cable length. The length of the cable is limited by the number of data errors acceptable during transmission. The RS-232 D&E standard specifies the limit of total cable capacitance to be 2500 pF. With typical cable capacitance having improved from around 160 pF/m to only 50 pF/m in recent years, the maximum cable length has extended from around 15 meters (50 feet) to about 50 meters (166 feet).

The common data transmission rates used with RS-232 are 110, 300, 600, 1200, 2400, 4800, 9600 and 19200 bps. For short distances, however, transmission rates of 38400, 57600 and 115200 can also be used. Based on field tests, table 2.1 shows the practical relationship between selected baud rates and maximum allowable cable length, indicating that much longer cable lengths are possible at lower baud rates. Note that the achievable speed depends on the transmitter voltages, cable capacitance (as discussed above) as well as the noise environment.

Baud Rate	Cable Length (meters)
110	850
300	800
600	700
1200	500
2400	200
4800	100
9600	70
19200	50
115K	20

Table 2.1
Demonstrated maximum cable lengths with RS-232 interface

Mechanical characteristics of the interface

RS-232 defines the mechanical characteristics of the interface between the DTE and the DCE. This dictates that the interface must consist of a plug and socket and that the socket will normally be on the DCE.

Although not specified by RS-232C, the DB-25 connector (25 pin, D-type) is closely associated with RS-232 and is the *de facto* standard with revision D. Revision E formally specifies a new connector in the 26-pin alternative connector (known as the ALT A connector). This connector supports all 25 signals associated with RS-232. ALT A is physically smaller than the DB-25 and satisfies the demand for a smaller connector suitable for modern computers. Pin 26 is not currently used. On some RS-232 compatible equipment, where little or no handshaking is required, the DB-9 connector (9 pin, D-type) is common. This practice originated when IBM decided to make a combined

serial/parallel adapter for the AT&T personal computer. A small connector format was needed to allow both interfaces to fit onto the back of a standard ISA interface card. Subsequently, the DB-9 connector has also became an industry standard to reduce the wastage of pins. The pin allocations commonly used with the DB-9 and DB-25 connectors for the RS-232 interface are shown in table 2.2. The pin allocation for the DB-9 connector is not the same as the DB-25 and often traps the unwary.

The data pins of DB-9 IBM connector are allocated as follows:

- Data transmit pin 3
- Data receive pin 2
- Signal common pin 5

Pin No. DTE	DB-9 Connector IBM Assignment	DB-25 Connector EIA-232 Pin Assignment	DB-25 Connector EIA-530 Pin Assignment
1	Received Line Signal	Shield	Shield
2	Received Data	Transmitted Data	Transmitted Data (A)
3	Transmitted Data	Received Data	Received Data (A)
4	DTE Ready	Request to Send	Request to Send (A)
5	Signal/Common Ground	Clear to Send	Clear to Send (A)
6	DCE Ready	DCE Ready	DCE Ready (A)
7	Request to Send	Signal/Common Ground	Signal/Common Ground
8	Clear to Send	Received Line Signal	Received Line Signal (A)
9	Ring Indicator	+Voltage (testing)	Receiver Signal DCE Element Timing (B)
10		-Voltage (testing)	Received Line (B)
11		Unassigned	Transmitter Signal DTE Element Timing (B)
12		Sec Received Line Signal Detector/Data Signal	Transmitter Signal DCE Element Timing
13		Sec Clear to Send	Clear to Send (B)
14		Sec Transmitted Data	Transmitted Data (B)
15		Transmitter Signal DCE Element Timing	Transmitter Signal DCE Element Timing (A)
16		Sec Received Data	Received Data (B)
17		Receiver Signal DCE Element Timing	Receiver Signal DCE Element Timing (A)
18		Local Loopback	Local Loopback
19		Sec Request to Send	Request to Send (B)
20		DTE Ready	DTE Ready (A)
21		Remote Loopback/Signal Quality Detector	Remote Loopback
22		Ring Indicator	DCE Ready (B)
23		Data Signal Rate	DTE Ready (B)
24		Transmit Signal DTE Element Timing	Transmitter Signal DTE Element Timing (A)
25		Test Mode	Test Mode

Table 2.2
Common DB-9 and DB-25 pin assignments for RS-232 and EIA/TIA-530 (often used for RS-422 and RS-485)

Functional description of the interchange circuits

RS-232 defines the function of the data, timing and control signals used at the interface of the DTE and DCE. However, very few of the definitions are relevant to applications for data communications for instrumentation and control.

The circuit functions are defined with reference to the DTE as follows:

- **Protective ground (shield)**
  The protective ground ensures that the DTE and DCE chassis are at equal potentials (remember that this protective ground could cause problems with circulating earth currents)
- **Transmitted data (TxD)**
  This line carries serial data from the DTE to the corresponding pin on the DCE. The line is held at a negative voltage during periods of line idle
- **Received data (RxD)**
  This line carries serial data from the DCE to the corresponding pin on the DTE
- **Request To Send (RTS)**
  (RTS) is the request to send hardware control line. This line is placed active (+V) when the DTE requests permission to send data. The DCE then activates (+V) the CTS (Clear To Send) for hardware data flow control
- **Clear To Send (CTS)**
  When a half-duplex modem is receiving, the DTE keeps RTS inhibited. When it is the DTE's turn to transmit, it advises the modem by asserting the RTS pin. When the modem asserts the CTS, it informs the DTE that it is now safe to send data
- **DCE ready**
  Formerly called data set ready (DSR). The DTE ready line is an indication from the DCE to the DTE that the modem is ready
- **Signal ground (common)**
  This is the common return line for all the data transmit and receive signals and all other circuits in the interface. The connection between the two ends is always made
- **Data Carrier Detect (DCD)**
  This is also called the received line signal detector. It is asserted by the modem when it receives a remote carrier and remains asserted for the duration of the link
- **DTE ready (data terminal ready)**
  Formerly called data terminal ready (DTR). DTE ready enables but does not cause, the modem to switch onto the line. In originate mode, DTE ready must be asserted in order to auto dial. In answer mode, DTE ready must be asserted to auto answer
- **Ring indicator**
  This pin is asserted during a ring voltage on the line
- **Data Signal Rate Selector (DSRS)**
  When two data rates are possible, the higher is selected by asserting DSRS; however, this line is not used much these days

Pin No.	CCITT No.	Circuit	Description	Circuit Direction
1	-	-	Shield	To DCE
2	103	BA	Transmitted Data	To DCE
3	104	BB	Received Data	From DCE
4	105/133	CA/CJ	Request to Send/Ready for Receiving	To DCE
5	106	CB	Clear to Send	From DCE
6	107	CC	DCE Ready	From DCE
7	102	AB	Signal Common	-
8	109	CF	Received Line Signal Detector	From DCE
9	-	-	Reserved for Testing	-
10	-	-	Reserved for Testing	-
11	126	See Note	Unassigned	-
12	122/112	SCF/CI	Secondary Received Line Signal Detector/ Data Signal Rate Selector	From DCE
13	121	SCB	Secondary Clear to Send	From DCE
14	118	SBA	Secondary Transmitted Data	To DCE
15	114	DB	Transmitter Signal Element Timing (DTE Source)	From DCE
16	119	SBB	Secondary Received Data	From DCE
18	141	LL	Local Loopback	To DCE
19	120	SCA	Secondary Request to Send	To DCE
20	108/112	CD	DTE Ready	To DCE
21	140/110	RL/CG	Remote Loopback/Signal Quality Detector	T/F DCE
22	125	CE	Ring Indicator	From DCE
23	111/112	CH/CI	Data Signal Rate Selector (DTE/DCE Source)	T/F DCE
24	113	DA	Transmit Signal Element Timing (DTE Source)	To DCE
25	142	TM	Test Mode	From DCE
26		None	(Alt A Connector) No Connection at this Time	

Table 2.3
ITU-T V24 pin assignment (ISO 2110)

2.2 Half-duplex operation of the RS-232 interface

The following description of one particular operation of the RS-232 interface is based on half-duplex data interchange. The description encompasses the more generally used full duplex operation.

Figure 2.4 shows the operation with the initiating user terminal, DTE, and its associated modem DCE on the left of the diagram and the remote computer and its modem on the right.

The following sequence of steps occurs when a user sends information over a telephone link to a remote modem and computer:

- The initiating user manually dials the number of the remote computer
- The receiving modem asserts the *Ring Indicator* line (RI) in a pulsed ON/OFF fashion reflecting the ringing tone. The remote computer already has its *Data Terminal Ready* (DTR) line asserted to indicate that it is ready to receive calls. Alternatively, the remote computer may assert the DTR line after a few rings. The remote computer then sets its *Request To Send* (RTS) line to ON

- The receiving modem answers the phone and transmits a carrier signal to the initiating end. It asserts the DCE ready line after a few seconds
- The initiating modem asserts the *data carrier detect* (DCD) line. The initiating
  terminal asserts its DTR, if it is not already high. The modem responds by asserting its DTE ready line
- The receiving modem asserts its *clear to send* (CTS) line, which permits the transfer of data from the remote computer to the initiating side
- Data is transferred from the receiving DTE (transmitted data) to the receiving modem. The receiving remote computer then transmits a short message to indicate to the originating terminal that it can proceed with the data transfer. The originating modem transmits the data to the originating terminal
- The receiving terminal sets its request to send (RTS) line to OFF The receiving modem then sets its *clear to send* (CTS) line to OFF
- The receiving modem switches its carrier signal OFF
- The originating terminal detects that the *data carrier detect* (DCD) signal has been switched OFF on the originating modem and switches its RTS line to the ON state. The originating modem indicates that transmission can proceed by setting its CTS line to ON
- Transmission of data proceeds from the originating terminal to the remote computer
- When the interchange is complete, both carriers are switched OFF and in many cases; the DTR is set to OFF. This means that the CTS, RTS and DCE ready lines are set to OFF

Full duplex operation requires that transmission and reception occur simultaneously. In this case, there is no RTS/CTS interaction at either end. The RTS line and CTS line are left ON with a carrier to the remote computer.

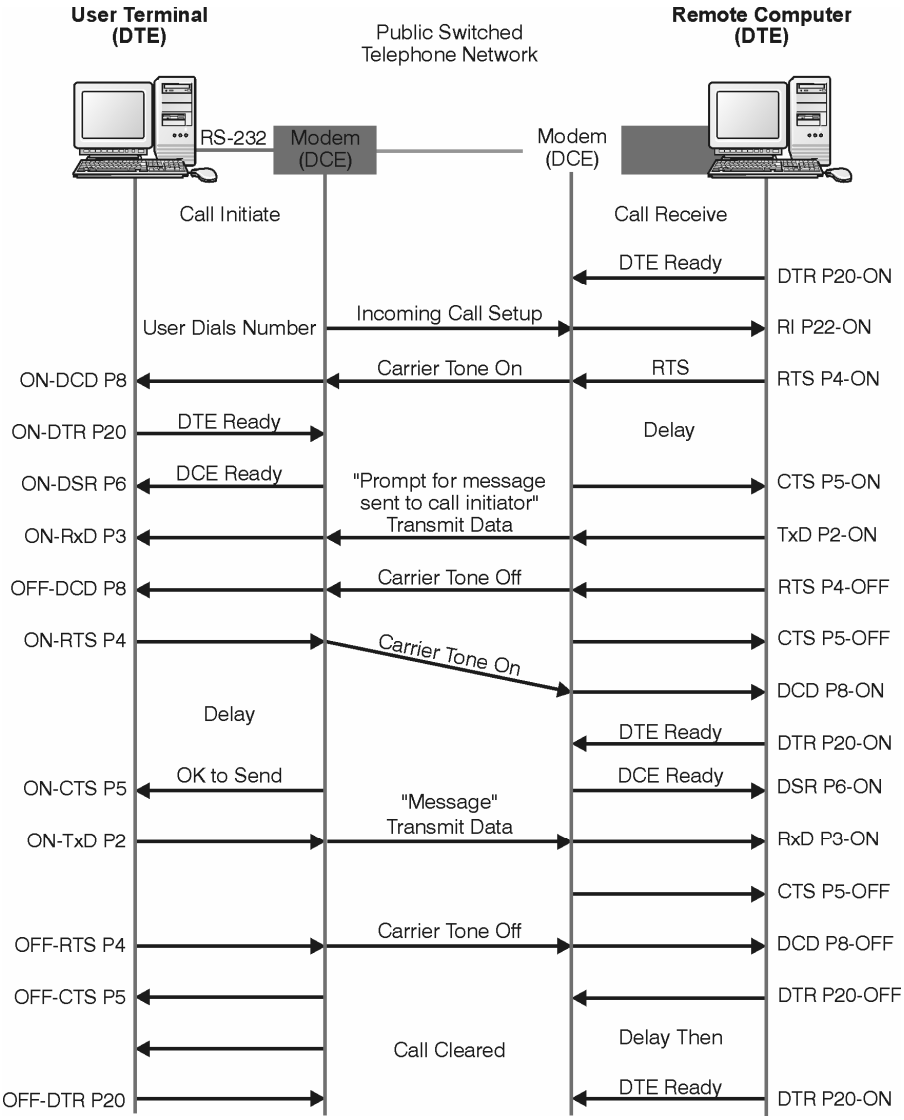


Figure 2.4
Half duplex operational sequence of RS-232

2.3 Summary of EIA/TIA-232 revisions

A summary of the main differences between RS-232 revisions, C, D and E are discussed below.

2.3.1 Revision D – RS-232D

The 25 pin D type connector was formally specified. In revision C, reference was made to the D type connector in the appendices and a disclaimer was included revealing that it was not intended to be part of the standard; however, it was treated as the *de facto* standard.

The voltage ranges for the control and data signals were extended to a maximum limit of 25 V from the previously specified 15 V in revision C.

The 15 meter (50 foot) distance constraint, implicitly imposed to comply with circuit capacitance, was replaced by 'circuit capacitance shall not exceed 2500 pF' (Standard RS-232 cable has a capacitance of 50 pF/ft.).

2.3.2 Revision E – RS-232E

Revision E formally specifies the new 26 pin alternative connector, the ALT A connector. This connector supports all 25 signals associated with RS-232, unlike the 9-pin connector, which has become associated with RS-232 in recent years. Pin 26 is currently not used. The technical changes implemented by RS-232E do not present compatibility problems with equipment confirming to previous versions of RS-232.

This revision brings the RS-232 standard into line with international standards CCITT V.24, V.28 and ISO 2110.

2.4 Limitations

In spite of its popularity and extensive use, it should be remembered that the RS-232 interface standard was originally developed for interfacing data terminals to modems. In the context of modern requirements, RS-232 has several weaknesses. Most have arisen as a result of the increased requirements for interfacing other devices such as PCs, digital instrumentation, digital variable speed drives, power system monitors and other peripheral devices in industrial plants.

The main limitations of RS-232 when used for the communications of instrumentation and control equipment in an industrial environment are:

- The point-to-point restriction, a severe limitation when several 'smart' instruments are used
- The distance limitation of 15 meters (50 feet) end-to-end, too short for most control systems
- The 20 Kbps rate, too slow for many applications
- The –3 to –25 V and +3 to +25 V signal levels, not directly compatible with modern standard power supplies

Consequently, a number of other interface standards have been developed by the RS to overcome some of these limitations. The RS-485 interface standards are increasingly being used for instrumentation and control systems.

2.5 RS-232 troubleshooting

2.5.1 Introduction

Since RS-232 is a point-to-point system, installation is fairly straightforward and simple and all RS-232 devices use either DB-9 or DB-25 connectors. These connectors are used because they are cheap and allow multiple insertions. None of the 232 standards define which device uses a male or female connector, but traditionally the male (pin) connector is used on the DTE and the female type connector (socket) is used on DCE equipment. This is only traditional and may vary on different equipment. It is often asked why a 25-pin connector is used when only 9 pins are needed. This was done because RS-232 was used before the advent of computers. It was therefore used for hardware control (RTS/CTS). It was originally thought that, in the future, more hardware control lines would be needed hence the need for more pins.

When doing an initial installation of an RS-232 connection it is important to note the following:

- Is one device a DTE and the other a DCE?
- What is the sex and size of connectors at each end?
- What is the speed of the communication?
- What is the distance between the equipment?
- Is it a noisy environment?
- Is the software set up correctly?

2.6 Typical approach

When troubleshooting a serial data communications interface, one needs to adopt a logical approach in order to avoid frustration and wasting many hours. A procedure similar to that outlined below is recommended:

- Check the basic parameters. Are the baud rate, stop/start bits and parity set identically for both devices? These are sometimes set on DIP switches in the device. However, the trend is towards using software, configured from a terminal, to set these basic parameters
- Identify which is DTE or DCE. Examine the documentation to establish what actually happens at pins 2 and 3 of each device. On the 25 pin DTE device, pin 2 is used for transmission of data and should have a negative voltage (mark) in idle state, whilst pin 3 is used for the receipt of data (passive) and should be approximately at 0 Volts. Conversely, at the DCE device, pin 3 should have a negative voltage, whilst pin 2 should be around 0 Volts. If no voltage can be detected on either pin 2 or 3, then the device is probably not RS-232 compatible and could be connected according to another interface standard, such as RS-422, RS-485, etc

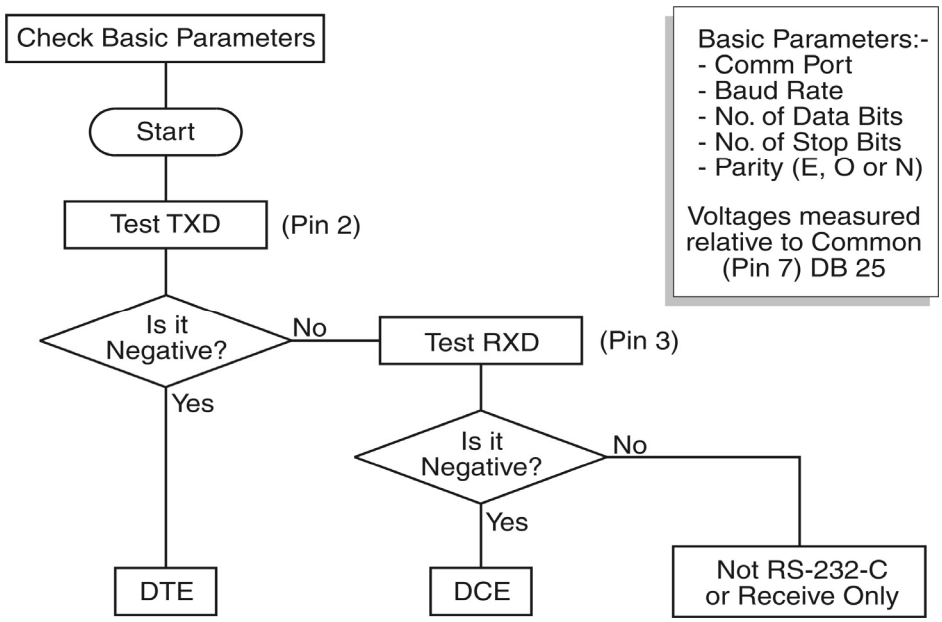


Figure 2.5
Flowchart to identify an RS-232 device as either a DTE or DCE

- Clarify the needs of the hardware handshaking when used. Hardware handshaking can cause the greatest difficulties and the documentation should be carefully studied to yield some clues about the handshaking sequence. Ensure all the required wires are correctly terminated in the cables
- Check the actual protocol used. This is seldom a problem but, when the above three points do not yield an answer, it is possible that there are irregularities in the protocol structure between the DCE and DTE devices
- Alternatively, if software handshaking is utilized, make sure both have compatible application software. In particular, check that the same ASCII character is used for XON and XOFF

2.7 Test equipment

From a testing point of view, the RS-232-E interface standard states that:

'The generator on the interchange circuit shall be designed to withstand an open circuit, a short circuit between the conductor carrying that interchange circuit in the interconnecting cable and any other conductor in that cable including signal ground, without sustaining damage to itself or its associated equipment.'

In other words, any pin may be connected to any other pin, or even earth, without damage and, theoretically, one cannot blow up anything! This does not mean that the RS-232 interface cannot be damaged. The incorrect connection of incompatible external voltages can damage the interface, as can static charges.

If a data communication link is inoperable, the following devices may be useful when analyzing the problem:

- A digital multimeter. Any cable breakage can be detected by measuring the continuity of the cable for each line. The voltages at the pins in active and inactive states can also be ascertained by the multimeter to verify its compatibility to the respective standards.
- An LED. The use of an LED is to determine which are the asserted lines or whether the interface conforms to a particular standard. This is laborious and accurate pin descriptions should be available.
- A breakout box
- PC-based protocol analyzer (including software)
- Dedicated hardware protocol analyzer (e.g. Hewlett Packard)

2.7.1 The breakout box

The breakout box is an inexpensive tool that provides most of the information necessary to identify and fix problems on data communications circuits, such as the serial RS-232, RS-422, RS-423 and RS-485 interfaces and also on parallel interfaces.

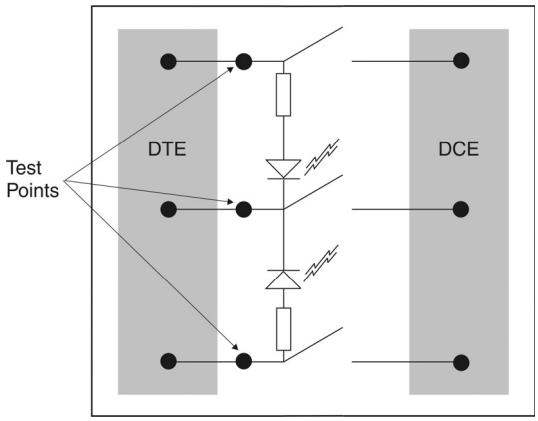


Figure 2.6
Breakout box showing test points

A breakout box is connected to the data cable, to bring out all conductors in the cable to accessible test points. Many versions of this equipment are available on the market, from the 'homemade' using a back-to-back pair of male and female DB-25 sockets to fairly sophisticated test units with built-in LEDs, switches and test points.

Breakout boxes usually have a male and a female socket and by using two standard serial cables, the box can be connected in series with the communication link. The 25 test points can be monitored by LEDs, a simple digital multimeter, an oscilloscope or a protocol analyzer. In addition, a switch in each line can be opened or closed while trying to identify the problem.

The major weakness of the breakout box is that while one can interrupt any of the data lines, it does not help much with the interpretation of the flow of bits on the data communication lines. A protocol analyzer is required for this purpose.

2.7.2 Null modem

Null modems look like DB-25 'through' connectors and are used when interfacing two devices of the same gender (e.g. DTE to DTE, DCE to DCE) or devices from different manufacturers with different handshaking requirements. A null modem has appropriate internal connections between handshaking pins that 'trick' the terminal into believing conditions are correct for passing data. A similar result can be achieved by soldering extra loops inside the DB-25 plug. Null modems generally cause more problems than they cure and should be used with extreme caution and preferably avoided.

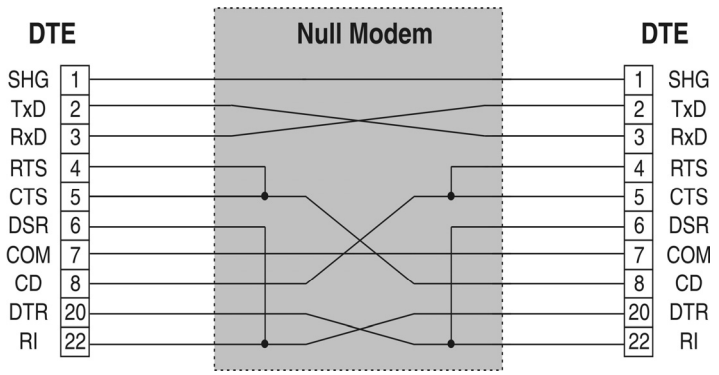


Figure 2.7
Null modem connections

Note that the null modem may inadvertently connect pins 1 together, as in Figure 2.7. This is an undesirable practice and should be avoided.

2.7.3 Loop back plug

This is a hardware plug which loops back the transmit data pin to receive data pin and similarly for the hardware handshaking lines. This is another quick way of verifying the operation of the serial interface without connecting to another system.

2.7.4 Protocol analyzer

A protocol analyzer is used to display the actual bits on the data line, as well as the special control codes, such as STX, DLE, LF, CR, etc. The protocol analyzer can be used to monitor the data bits, as they are sent down the line and compared with what should be on the line. This helps to confirm that the transmitting terminal is sending the correct data and that the receiving device is receiving it. The protocol analyzer is useful in identifying incorrect baud rate, incorrect parity generation method, incorrect number of stop bits, noise, or incorrect wiring and connection. It also makes it possible to analyze the format of the message and look for protocol errors.

When the problem has been shown not to be due to the connections, baud rate, bits or parity, then the content of the message will have to be analyzed for errors or inconsistencies. Protocol analyzers can quickly identify these problems.

Purpose built protocol analyzers are expensive devices and it is often difficult to justify the cost when it is unlikely that the unit will be used very often. Fortunately, software has been developed that enables a normal PC to be used as a protocol analyzer. The use of a PC as a test device for many applications is a growing field, and one way of connecting a PC as a protocol analyzer is shown in Figure 2.8.

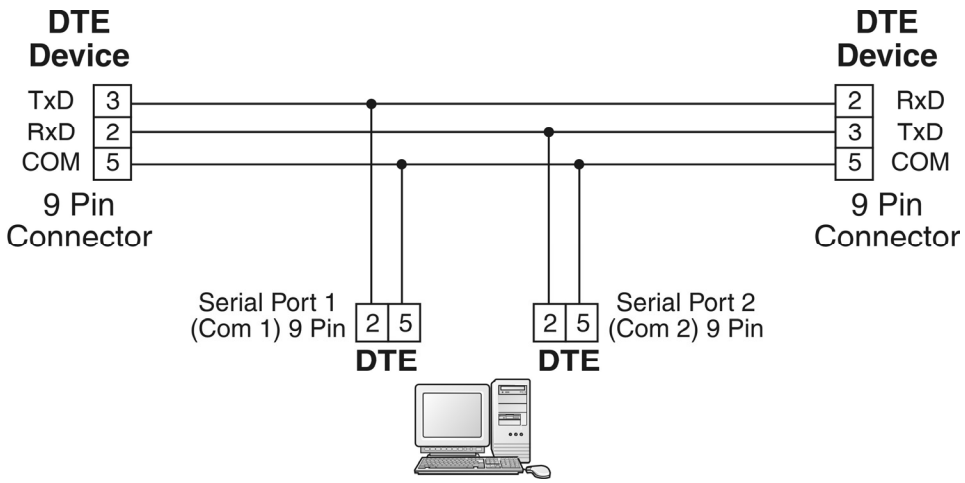


Figure 2.8
Protocol analyzer connection

The above figure has been simplified for clarity and does not show the connections on the control lines (For Example, RTS and CTS).

2.8 Typical RS-232 problems

Below is a list of typical RS-232 problems, which can arise because of inadequate interfacing. These problems could equally apply to two PCs connected to each other or to a PC connected to a printer.

Problem	Probable cause of problem
Garbled or lost data	Baud rates of connecting ports may be different
	Connecting cables could be defective
	Data formats may be inconsistent (Stop bit/ parity/ number of data bits)
	Flow control may be inadequate
	High error rate due to electrical interference
	Buffer size of receiver inadequate
First characters garbled	The receiving port may not be able to respond quickly enough. Precede the first few characters with the ASCII (DEL) code to ensure frame synchronization.
No data communications	Power for both devices may not be on
	Transmit and receive lines of cabling may be incorrect
	Handshaking lines of cabling may be incorrectly connected
	Baud rates mismatch
	Data format may be inconsistent
	Earth loop may have formed for RS-232 line
	Extremely high error rate due to electrical interference for transmitter and receiver
	Protocols may be inconsistent/ Intermittent communications
	Intermittent interference on cable
ASCII data has incorrect spacing	Mismatch between 'LF' and 'CR' characters generated by transmitting device and expected by receiving device.

Table 2.4
A list of typical RS-232 problems

To determine whether the devices are DTE or DCE, connect a breakout box at one end and note the condition of the TX light (pin 2 or 3) on the box. If pin 2 is ON, then the device is probably a DTE. If pin 3 is ON, it is probably a DCE. Another clue could be the sex of the connector, male are typically DTEs and females are typically DCEs, but not always.

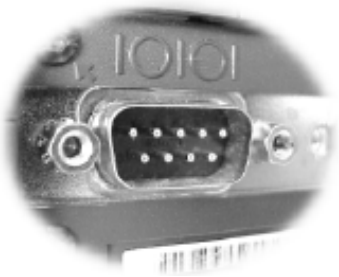


Figure 2.9
A 9 pin RS-232 connector on a DTE

When troubleshooting an RS-232 system, it is important to understand that there are two different approaches. One approach is followed if the system is new and never been run before and the other if the system has been operating and for some reason does not

communicate at present. New systems that have never worked have more potential problems than a system that has been working before and now has stopped. If a system is new it can have three main problems viz. mechanical, setup or noise. A previously working system usually has only one problem, viz. mechanical. This assumes that no one has changed the setup and noise has not been introduced into the system. In all systems, whether having previously worked or not, it is best to check the mechanical parts first.

This is done by:

- Verifying that there is power to the equipment
- Verifying that the connectors are not loose
- Verifying that the wires are correctly connected
- Checking that a part, board or module has not visibly failed

2.8.1 Mechanical problems

Often, mechanical problems develop in RS-232 systems because of incorrect installation of wires in the D-type connector or because strain reliefs were not installed correctly.

The following recommendations should be noted when building or installing RS-232 cables:

- Keep the wires short (20 meters maximum)
- Stranded wire should be used instead of solid wire (solid wire will not flex.)
- Only one wire should be soldered in each pin of the connector.
- Bare wire should not be showing out of the pin of the connector
- The back shell should reliably and properly secure the wire

The speed and distance of the equipment will determine if it is possible to make the connection at all. Most engineers try to stay less than 50 feet or about 16 meters at 115200 bits per second. This is a very subjective measurement and will depend on the cable, voltage of the transmitter and the amount and noise in the environment. The transmitter voltage can be measured at each end when the cable has been installed. A voltage of at least +/– 5 V should be measured at each end on both the TX and RX lines.

An RS-232 breakout box is placed between the DTE and DCE to monitor the voltages placed on the wires by looking at pin 2 on the breakout box. Be careful here because it is possible that the data is being transmitted so fast that the light on the breakout box doesn't have time to change. If possible, lower the speed of the communication at both ends to something like 2 bps.

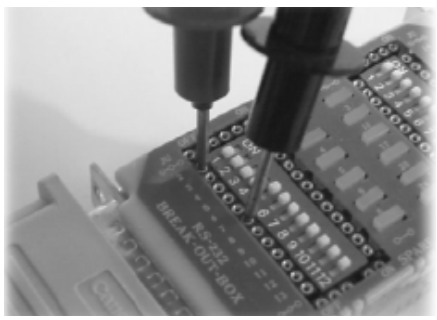


Figure 2.10
Measuring the voltage on RS-232

Once it has been determined that the wires are connected as DTE to DCE and that the distance and speed are not going to be a problem, the cable can be connected at each end. The breakout box can still be left connected with the cable and both pin 2 and 3 lights on the breakout box should now be on.

The color of the light depends on the breakout box. Some breakout boxes use red for a one and others use green for a one. If only one light is on then that may mean that a wire is broken or there is a DTE to DTE connection. A clue to a possible DTE to DTE connection would be that the light on pin 3 would be off and the one on pin 2 would be on. To correct this problem, first check the wires for continuity then turn switches 2 and 3 off on the breakout box and use jumper wires to swap them. If the TX and RX lights come on, a null modem cable or box will need to be built and inserted in-line with the cable.

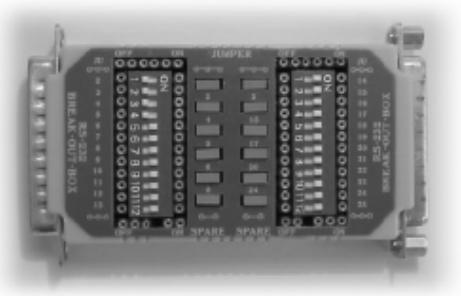


Figure 2.11
A RS-232 breakout box

If the pin 2 and pin 3 lights are on, one end is transmitting and the control is correct, then the only thing left is the protocol or noise. Either a hardware or software protocol analyzer will be needed to troubleshoot the communications between the devices. On new installations, one common problem is mismatched baud rates. The protocol analyzer will tell exactly what the baud rates are for each device. Another thing to look for with the analyzer is the timing. Often, the transmitter waits some time before expecting a proper response from the receiver. If the receiver takes too long to respond or the response is incorrect, the transmitter will 'time out.' This is usually denoted as a ‘communications error or failure.’

2.8.2 Setup problems

Once it is determined that the cable is connected correctly and the proper voltage is being received at each end, it is time to check the setup. The following circumstances need to be checked before trying to communicate:

- Is the software communications setup at both ends for either 8N1, 7E1 or 7O1?
- Is the baud rate the same at both devices? (1200, 4800, 9600, 19200 etc.)
- Is the software setup at both ends for binary, hex or ASCII data transfer?
- Is the software setup for the proper type of control?

Although the 8 data bits, no parity and 1 stop bit is the most common setup for asynchronous communication, often 7 data bits even parity with 1 stop bit is used in industrial equipment. The most common baud rate used in asynchronous communications is 9600. Hex and ASCII are commonly used as communication codes.

If one device is transmitting but the other receiver is not responding, then the next thing to look for is what type of control the devices are using. The equipment manual may define whether hardware or software control is being used. Both ends should be set up either for hardware control, software control or none.

2.8.3 Noise problems

RS-232, being a single ended (unbalanced) type of circuit, lends itself to receiving noise. There are three ways that noise can be induced into an RS-232 circuit.

- Induced noise on the common ground
- Induced noise on the TX or RX lines
- Induced noise on the indicator or control lines

Ground induced noise

Different ground voltage levels on the ground line (pin7) can cause ground loop noise. Also, varying voltage levels induced on the ground at either end by high power equipment can cause intermittent noise. This kind of noise can be very difficult to reduce. Sometimes, changing the location of the ground on either the RS-232 equipment or the high power equipment can help, but this is often not possible. If it is determined that the noise problem is caused by the ground it may be best to replace the RS-232 link with a fiber optic or RS-422 system. Fiber optic or RS-422 to RS-232 adapters are relatively cheap, readily available and easy to install. When the cost of troubleshooting the system is included, replacing the system often is the cheapest option.

Induced noise on the TX or RX lines

Noise from the outside can cause the communication on an RS-232 system to fail, although this voltage must be quite large. Because RS-232 voltages in practice are usually between +/– 7 and +/– 12, the noise voltage value must be quite high in order to induce errors. This type of noise induction is noticeable because the voltage on the TX or RX will be outside of the specifications of RS-232. Noise on the TX line can also be induced on the RX line (or *vice versa*) due to the common ground in the circuit. This type of noise can be detected by comparing the data being transmitted with the received communication at the other end of the wire (assuming no broken wire). The protocol analyzer is plugged into the transmitter at one end and the data monitored. If the data is correct, the protocol analyzer is then plugged into the other end and the received data monitored. If the data is corrupt at the receiving end, then noise on that wire may be the problem. If it is determined that the noise problem is caused by induced noise on the TX or RX lines, it may be best to move the RS-232 line and the offending noise source away from each other. If this doesn't help, it may be necessary to replace the RS-232 link with a fiber optic or RS-485 system.

Induced noise on the indicator or control lines

This type of noise is very similar to the previous TX/RX noise. The difference is that noise on these wires may be harder to find. This is because the data is being received at both ends, but there still is a communication problem. The use of a voltmeter or oscilloscope would help to measure the voltage on the control or indicator lines and therefore locate the possible cause of the problem, although this is not always very accurate. This is because the effect of noise on a system is governed by the ratio of the power levels of the signal and the noise, rather than a ratio of their respective voltage levels. If it is determined that the noise is being induced on one of the indicator or control

lines, it may be best to move the RS-232 line and the offending noise source away from each other. If this doesn't help, it may be necessary to replace the RS-232 link with a fiber optic or RS-485 system.

2.9 Summary of troubleshooting

Installation

- Is one device a DTE and the other a DCE?
- What is the sex and size of the connectors at each end?
- What is the speed of the communications?
- What is the distance between the equipment?
- Is it a noisy environment?
- Is the software set up correctly?

Troubleshooting new and old systems

- Verify that there is power to the equipment
- Verify that the connectors are not loose
- Verify that the wires are correctly connected
- Check that a part, board or module has not visibly failed

Mechanical problems on new systems

- Keep the wires short (20 meters maximum)
- Stranded wire should be used instead of solid wire (stranded wire will flex)
- Only one wire should be soldered in each pin of the connector
- Bare wire should not be showing out of the connector pins
- The back shell should reliably and properly secure the wire

Setup problems on new systems

- Is the software communications set up at both ends for either 8N1, 7E1 or 7O1?
- Is the baud rate the same for both devices? (1200,4800,9600,19200 etc.)
- Is the software set up at both ends for binary, hex or ASCII data transfer?
- Is the software setup for the proper type of control?

Noise problems on new systems

- Noise from the common ground
- Induced noise on the TX or RX lines
- Induced noise on the indicator or control lines

3

RS-485 fundamentals

Objectives

When you have completed study of this chapter, you will be able to:

- Describe the RS-485 standard
- Remedy the following problems:
- Incorrect RS-485 wiring
- Excessive common mode voltage
- Faulty converters
- Isolation
- Idle state problems
- Incorrect or missing terminations
- RTS control via hardware or software

3.1 The RS-485 interface standard

The RS-485-A standard is one of the most versatile of the RS interface standards. It is an extension of RS-422 and allows the same distance and data speed but increases the number of transmitters and receivers permitted on the line. RS-485 permits a 'multidrop' network connection on 2 wires and allows reliable serial data communication for:

- Distances of up to 1200 m (4000 feet, same as RS-422)
- Data rates of up to 10 Mbps (same as RS-422)
- Up to 32 line drivers on the same line
- Up to 32 line receivers on the same line

The maximum bit rate and maximum length can, however, not be achieved at the same time. For 24 AWG twisted pair cable the maximum data rate at 4000 ft (1200 m) is approximately 90 kbps. The maximum cable length at 10 Mbps is less than 20 ft (6m). Better performance will require a higher-grade cable and possibly the use of active (solid state) terminators in the place of the 120-ohm resistors.

According to the RS-485 standard, there can be 32 'standard' transceivers on the network. Some manufacturers supply devices that are equivalent to ½ or ¼ standard

device, in which case this number can be increased to 64 or 128. If more transceivers are required, repeaters have to be used to extend the network.

The two conductors making up the bus are referred to as A in B in the specification. The A conductor is alternatively known as A–, TxA and Tx+. The B conductor, in similar fashion, is called B+, TxB and Tx–. Although this is rather confusing, identifying the A and B wires are not difficult. In the MARK or OFF state (i.e. when the RS-232 TxD pin is LOW (e.g. minus 8 V), the voltage on the A wire is more negative than that on the B wire.

The differential voltages on the A and B outputs of the driver (transmitter) are similar (although not identical) to those for RS-422, namely:

- –1.5V to –6V on the A terminal with respect to the B terminal for a binary 1 (MARK or OFF) state
- +1.5V to +6V on the A terminal with respect to the B terminal for a binary 0 (SPACE or ON state)

As with RS-422, the line driver for the RS-485 interface produces a ±5V differential voltage on two wires.

The major enhancement of RS-485 is that a line driver can operate in three states called tri-state operation:

- Logic 1
- Logic 0
- High-impedance

In the high impedance state, the line driver draws virtually no current and appears not to be present on the line. This is known as the 'disabled' state and can be initiated by a signal on a control pin on the line driver integrated circuit. Tri-state operation allows a multidrop network connection and up to 32 transmitters can be connected on the same line, although only one can be active at any one time. Each terminal in a multidrop system must be allocated a unique address to avoid conflicting with other devices on the system. RS-485 includes current limiting in cases where contention occurs.

The RS-485 interface standard is very useful for systems where several instruments or controllers may be connected on the same line. Special care must be taken with the software to coordinate which devices on the network can become active. In most cases, a master terminal, such as a PC or computer, controls which transmitter/receiver will be active at a given time.

The two-wire data transmission line does not require special termination if the signal transmission time from one end of the line to the other end (at approximately 200 meters per microsecond) is significantly smaller than one quarter of the signal's rise time. This is typical with short lines or low bit rates. At high bit rates or in the case of long lines, proper termination becomes critical. The value of the terminating resistors (one at each end) should be equal to the characteristic impedance of the cable. This is typically 120 Ohms for twisted pair wire.

Figure 3.1 shows a typical two-wire multidrop network. Note that the transmission line is terminated on both ends of the line but not at drop points in the middle of the line.

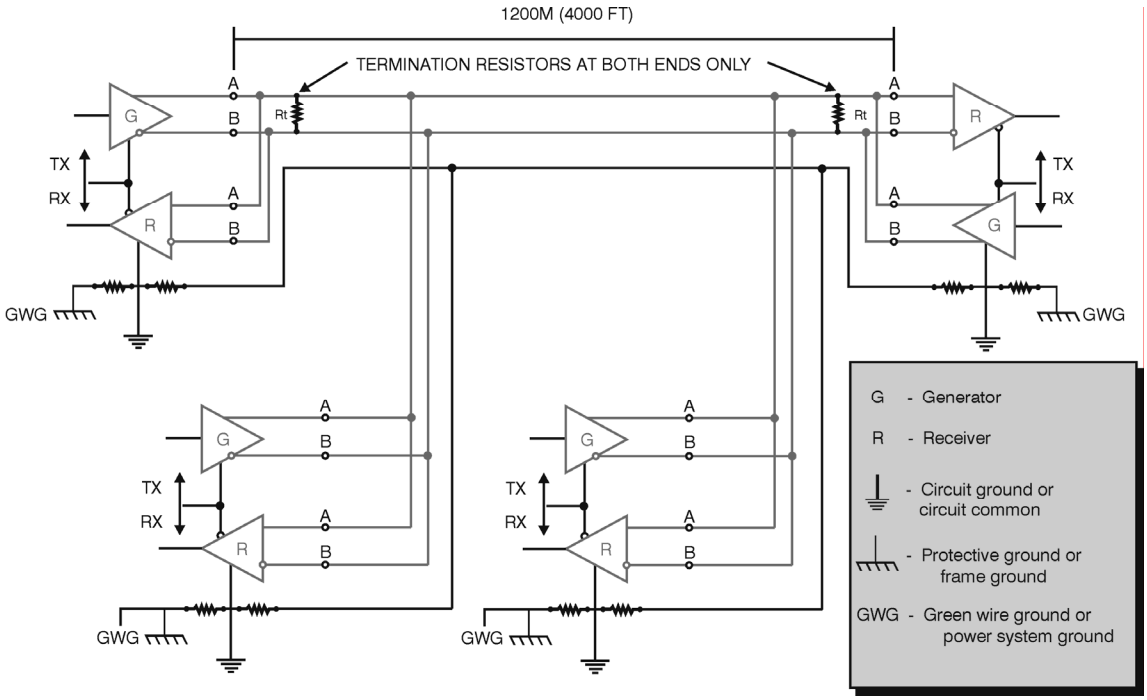


Figure 3.1
Typical two wire multidrop network

An RS-485 network can also be connected in a four wire configuration as shown in figure 3.2. In this type of connection it is necessary that one node is a master node and all others slaves. The master node communicates to all slaves, but a slave node can communicate only to the master. Since the slave nodes never listen to another slave's response to the master, a slave node cannot reply incorrectly to another slave node. This is an advantage in a mixed protocol environment.

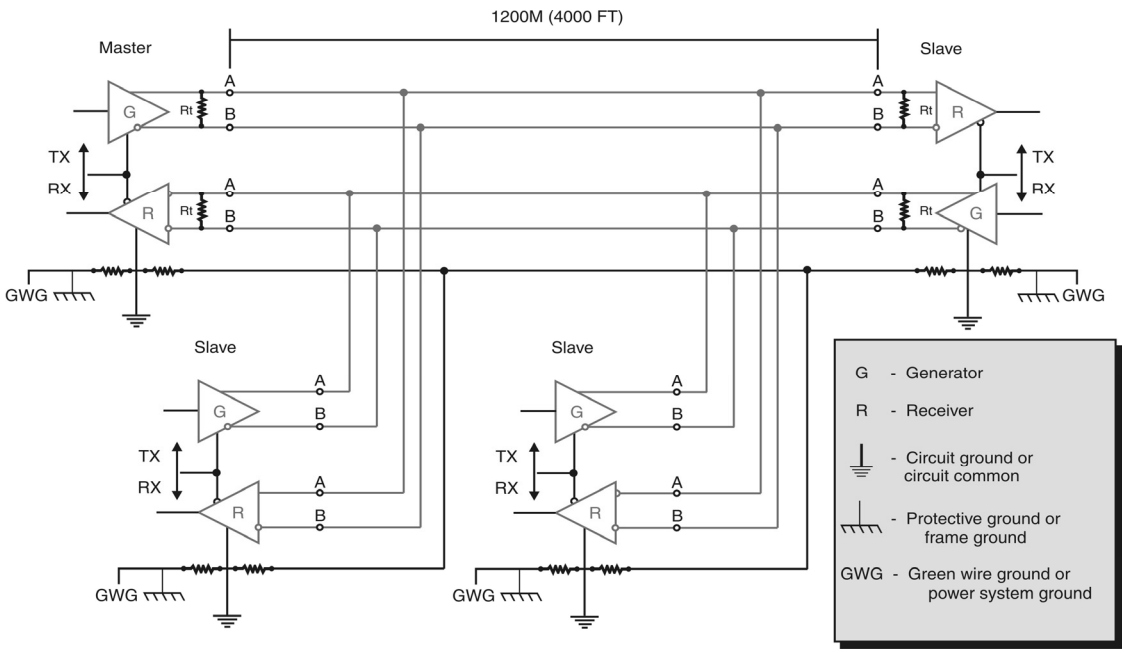


Figure 3.2
Four wire network configuration

During normal operation there are periods when all RS-485 drivers are off, and the communications lines are in the idle, high impedance state. In this condition the lines are susceptible to noise pick up, which can be interpreted as random characters on the communications line. If a specific RS-485 system has this problem, it should incorporate bias resistors, as indicated in figure 3.3. The purpose of the bias resistors is not only to reduce the amount of noise picked up, but to keep the receiver biased in the IDLE state when no input signal is received. For this purpose the voltage drop across the 120 Ohm termination resistor must exceed 200 mV AND the A terminal must be more negative than the B terminal. Keeping in mind that the two 120-Ohm resistors appear in parallel, the bias resistor values can be calculated using Ohm's Law. For a +5V supply and 120-Ohm terminators, a bias resistor value of 560 Ohm is sufficient. This assumes that the bias resistors are only installed on ONE node.

Some commercial systems use higher values for the bias resistors, but then assume that all or several nodes have bias resistors attached. In this case the value of all the bias resistors in parallel must be small enough to ensure 200 mV across the A and B wires.

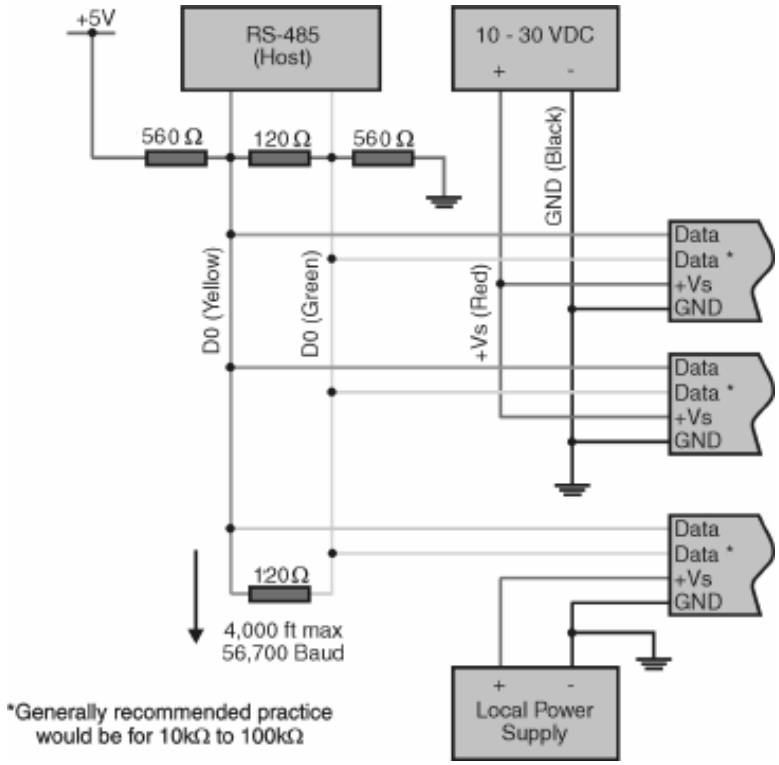


Figure 3.3
Suggested installation of resistors to minimize noise

RS-485 line drivers are designed to handle 32 nodes. This limitation can be overcome by employing an RS-485 repeater connected to the network. When data occurs on either side of the repeater, it is transmitted to the other side. The RS-485 repeater transmits at full voltage levels, consequently another 31 nodes can be connected to the network. A diagram for the use of RS-485 with a bi-directional repeater is given in figure 3.4.

The 'gnd' pin of the RS-485 transceiver should be connected to the logic reference (also known as circuit ground or circuit common), either directly or through a 100-Ohm ½ Watt resistor. The purpose if the resistor is to limit the current flow if there is a significant potential difference between the earth points. This is not shown in figure 3.2. In addition, the logic reference is to be connected to the chassis reference (protective

ground or frame ground) through a 100 Ohm 1/2 Watt resistor. The chassis reference, in turn, is connected directly to the safety reference (green wire ground or power system ground).

If the grounds of the nodes are properly interconnected, then a third wire running in parallel with the A and B wires are technically speaking not necessary. However, this is often not the case and thus a third wire is added as in figure 3.2. If the third wire is added, a 100-ohm ½ W resistor is to be added at each end as shown in figure 3.2.

The 'drops' or 'spurs' that interconnect the intermediate nodes to the bus need to be as short as possible since a long spur creates an impedance mismatch, which leads to unwanted reflections. The amount of reflection that can be tolerated depends on the bit rate. At 50 kbps a spur of, say, 30 meters could be in order, whilst at 10 Mbps the spur might be limited to 30 cm. Generally speaking, spurs on a transmission line are "bad news" because of the impedance mismatch (and hence the reflections) they create, and should be kept as short as possible.

Some systems employ RS-485 in a so-called 'star' configuration. This is not really a star, since a star topology requires a hub device at its center. The 'star' is in fact a very short bus with extremely long spurs, and is prone to reflections. It can therefore only be used at low bit rates.

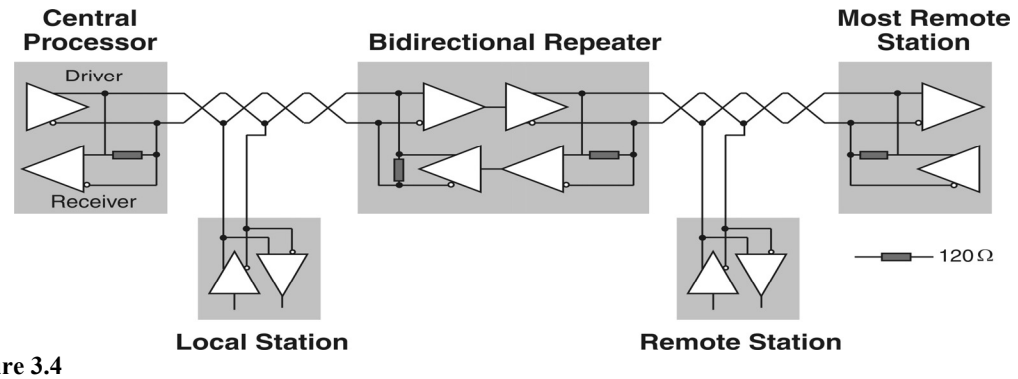


Figure 3.4
RS-485 used with repeaters

The 'decision threshold' of the RS-485 receiver is identical to that of both RS-422 & RS-423 receivers (not discussed as they have been superseded by RS-423) at 400 mV (0.4 V), as indicated in figure 3.5.

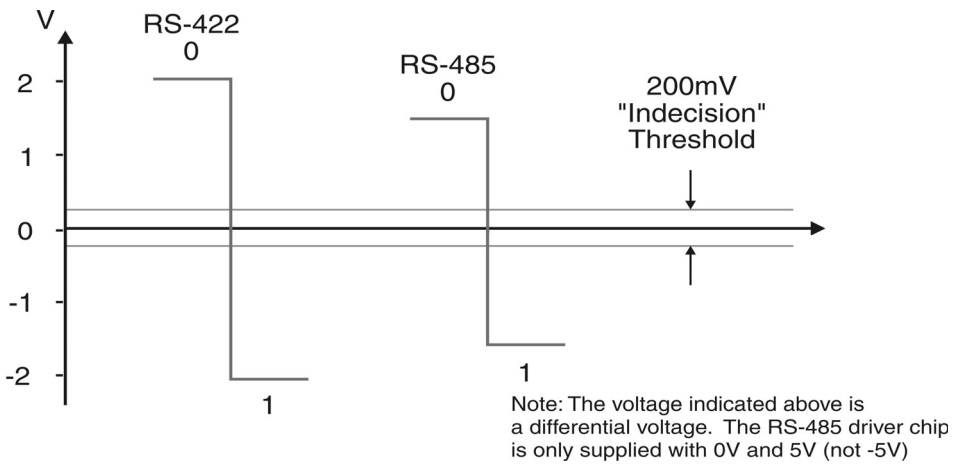


Figure 3.5
RS-485/422 & 423 receiver sensitivities

3.2 RS-485 troubleshooting

3.2.1 Introduction

RS-485 is the most common asynchronous voltage standard in use today for multi-drop communication systems since it is very resistant to noise, can send data at high speeds (up to 10 Mbps), can be run for long distances (5 km at 1200 Bps, 1200m at 90 Kbps), and is easy and cheap to use.

The RS-485 line drivers/receivers are differential chips. This means that the TX and RX wires are referenced to each other. A one is transmitted, for example, when one of the lines is +5 Volts and the other is 0 Volts. A zero is then transmitted when the lines reverse and the line that was + 5 volts is now 0 volts and the line that was 0 Volts is now +5 volts. In working systems the voltages are usually somewhere around +/– 2 volts with reference to each other. The indeterminate voltage levels are +/– 200 mV. Up to 32 devices can be connected on one system without a repeater. Some systems allow the connection of five legs with four repeaters and get 160 devices on one system.

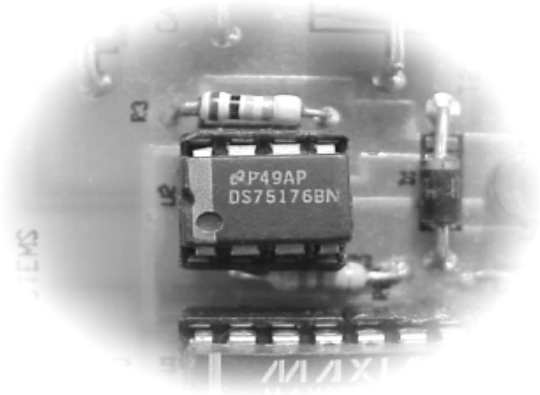


Figure 3.6
RS-485 Chip

Resistors are sometimes used on RS-485 systems to reduce noise, common mode voltages and reflections.

Bias resistors of values from 560 Ohms to 4k Ohms can sometimes be used to reduce noise. These resistors connect the B+ line to + 5 volts and the A- line to ground. Higher voltages should not be used because anything over +12 volts will cause the system to fail. Unfortunately, sometimes these resistors can increase the noise on the system by allowing a better path for noise from the ground. It is best not to use bias resistors unless required by the manufacturer.

Common mode voltage resistors usually have a value between 100k and 200k Ohms. The values will depend on the induced voltages on the lines. They should be equal and as high as possible and placed on both lines and connected to ground. The common mode voltages should be kept less then +7 Volts, measured from each line to ground. Again, sometimes these resistors can increase the noise on the system by allowing a better path for noise from the ground. It is best not to use common mode resistors unless required by the manufacturer or as needed.

The termination resistor value depends on the cable used and is typically 120 Ohms. Values less than 110 Ohms should not be used since the driver chips are designed to drive

a load resistance not less than 54 Ohms, being the value of the two termination resistors in parallel plus any other stray resistance in parallel. These resistors are placed between the lines (at the two furthest ends, not on the stubs) and reduce reflections. If the lines are less than 100 meters long and speeds are 9600 baud or less, the termination resistor usually becomes redundant, but having said that, you should always follow the manufacturers' recommendations.

3.3 RS-485 vs. RS-422

In practice, RS-485 and RS-422 are very similar to each other and manufacturers often use the same chips for both. The main working difference is that RS 485 is used for 2 wire multi-drop half duplex systems and RS-422 is for 4-wire point-to-point full duplex systems. Manufacturers often use a chip like the 75154 with two RS-485 drivers on board as an RS-422 driver. One driver is used as a transmitter and the other is dedicated as a receiver. Because the RS-485 chips have three states, TX, RX and high impedance, the driver that is used as a transmitter can be set to high impedance mode when the driver is not transmitting data. This is often done using the RTS line from the RS-232 port. When the RTS goes high (+ voltage) the transmitter is effectively turned off by being put the transmitter in the high impedance mode. The receiver is left on all the time, so data can be received when it comes in. This method can reduce noise on the line by having a minimum of devices on the line at a time.

3.4 RS-485 installation

Installation rules for RS-485 vary per manufacturer and since there are no standard connectors for RS-485 systems, it is difficult to define a standard installation procedure. Even so, most manufacture procedures are similar. The most common type of connector used on most RS-485 systems is either a one-part or two-part screw connector. The preferred connector is the 2-part screw connector with the sliding box under the screw (phoenix type). Other connectors use a screw on top of a folding tab. Manufacturers sometimes use the DB-9 connector instead of a screw connector to save money. Unfortunately, the DB-9 connector has problems when used for multidrop connections. The problem is that the DB-9 connectors are designed so that only one wire can be inserted per pin. RS-485 multidrop systems require the connection of two wires so that the wire can continue down the line to the next device. This is a simple matter with screw connectors, but it is not so easy with a DB-9 connector. With a screw connector, the two wires are twisted together and inserted in the connector under the screw. The screw is then tightened down and the connection is made. With the DB-9 connector, the two wires must be soldered together with a third wire. The third wire is then soldered to the single pin on the connector.

Note: When using screw connectors, the wires should NOT be soldered together. Either the wires should be just twisted together or a special crimp ferrule should be used to connect the wires before they are inserted in the screw connector.

Figure 3.7
A bad RS-485 connection

Serious problems with RS-485 systems are rare (that is one reason it is used) but having said that, there are some possible problems that can arise in the installation process:

- The wires get reversed. (e.g. black to white and white to black)
- Loose or bad connections due to improper installation
- Excessive electrical or electronic noise in the environment
- Common mode voltage problems
- Reflection of the signal due to missing or incorrect terminators
- Shield not grounded, grounded incorrectly or not connected at each drop
- Starring or tee-ing of devices (i.e. long stubs)

To make sure the wires are not reversed, check that the same color is connected to the same pin on all connectors. Check the manufacturer's manual for proper wire color codes.

Verifying that the installers are informed of the proper installation procedures can reduce loose connections. If the installers are provided with adjustable torque screwdrivers, then the chances of loose or over tightened screw connections can be minimized.

3.5 Noise problems

RS-485, being a differential type of circuit, is resistant to receiving common mode noise. There are five ways that noise can be induced into an RS-485 circuit.

- Induced noise on the A/B lines
- Common mode voltage problems
- Reflections
- Unbalancing the line
- Incorrect shielding

3.5.1 Induced noise

Noise from the outside can cause communication on an RS-485 system to fail. Although the voltages on an RS-485 system are small (+/– 5 volts), because the output of the receiver is the difference of the two lines, the voltage induced on the two lines must be different. This makes RS-485 very tolerant to noise. The communications will also fail if the voltage level of the noise on the either or both lines is outside of the minimum or maximum RS-485 specification. Noise can be detected by comparing the data communication being transmitted out of one end with the received communication at the other (assuming no broken wire.) The protocol analyzer is plugged into the transmitter at one end and the data monitored. If the data is correct, the protocol analyzer is then plugged into the other end and the received data monitored. If the data is corrupt at the received end, then the noise on that wire may be the problem. If it is determined that the noise problem is caused by induced noise on the A or B lines it may be best to move the RS-485 line or the offending noise source away from each other.

Excessive noise is often due to the close proximity of power cables. Another possible noise problem could be caused by an incorrectly installed grounding system for the cable shield. Installation standards should be followed when the RS-485 pairs are installed close to other wires and cables. Some manufacturers suggest biasing resisters to limit noise on the line while others dissuade the use of bias resistors completely. Again, the procedure is to follow the manufacturer's recommendations. Having said that, it is usually found that biasing resisters are of minimal value, and that there are much better methods of reducing noise in an RS-485 system.

3.5.2 Common mode noise

Common mode noise problems are usually caused by a changing ground level. The ground level can change when a high current device is turned on or off. This large current draw causes the ground level as referenced to the A and B lines to rise or decrease. If the voltages on the A or B line are raised or lowered outside of the minimum or maximum as defined by the manufacturer specifications, it can prohibit the line receiver from operating correctly. This can cause a device to float in and out of service. Often, if the common mode voltage gets high enough, it can cause the module or device to be damaged. This voltage can be measured using a differential measurement device like a handheld digital voltmeter. The voltage between A and ground and then B to ground is measured. If the voltage is outside of specifications then resistors of values between 100K ohm and 200K ohm are placed between A and ground and B and ground. It is best to start with the larger value resistor and then verify the common mode voltage. If it is still too high, try a lower resistor value and recheck the voltage. At idle the voltage on the A line should be close to 0 and the B line should be between 2 and 6 volts. It is not uncommon for an RS-485 manufacturer to specify a maximum common voltage value of +12 and –7 volts, but it is best to have a system that is not near these levels. It is important to follow the manufacturer's recommendations for the common mode voltage resistor value or whether they are needed at all.

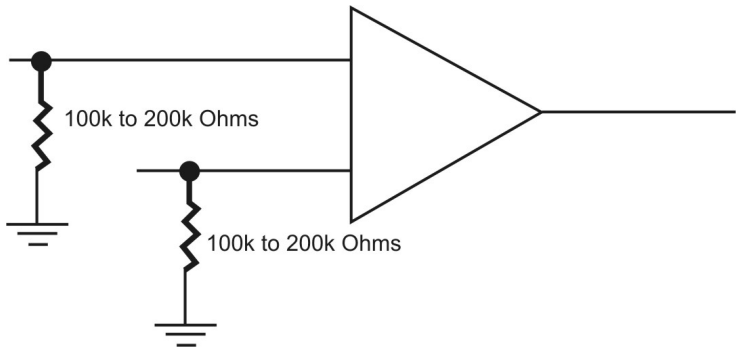


Figure 3.8
Common mode resistors

Note: When using bias resistors, neither the A nor the B line on the RS-485 system should ever be raised higher than +12 volts or lower than - 7 volts. Most RS-485 driver chips will fail if this happens. It is important to follow the manufacturer recommendations for bias resistor values or whether they are needed at all.

3.5.3 Reflections or ringing

Reflections are caused by the signal reflecting off the end of the wire and corrupting the signal. It usually affects the devices near the end of the line. It can be detected by placing a balanced ungrounded oscilloscope across the A and B lines. The signal will show ringing superimposed on the square wave. A termination resistor of typically 120 Ohms is placed at each end of the line to reduce reflections. This is more important at higher speeds and longer distances.

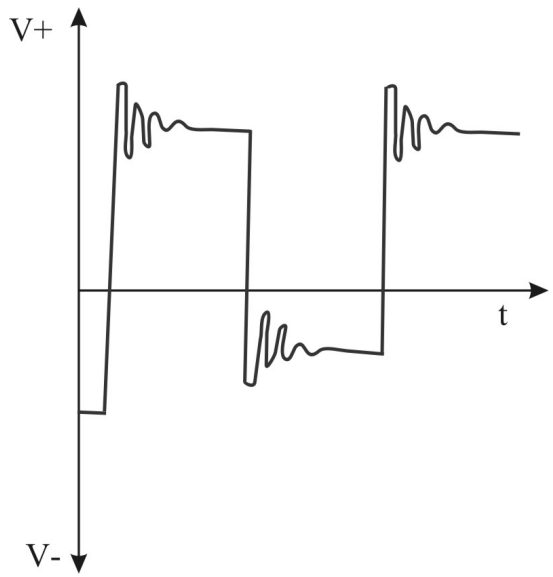


Figure 3.9
Ringing on an RS-485 signal

3.5.4 Unbalancing the line

Unbalancing the line does not actually induce noise, but it does make the lines more susceptible to noise. A line that is balanced will more or less have the same resistance,

capacitance and inductance on both conductors. If this balance is disrupted, the lines then become affected by noise more easily. There are a few ways most RS-485 lines become unbalanced:

- Using a star topology
- Using a 'tee' topology
- Using unbalanced cable
- Damaged transmitter or receiver

There should, ideally, be no stars or tees in the RS-485-bus system. If another device is to be added in the middle, a two-pair cable should be run out and back from the device. The typical RS-485 system would have a topology that would look something like the following:

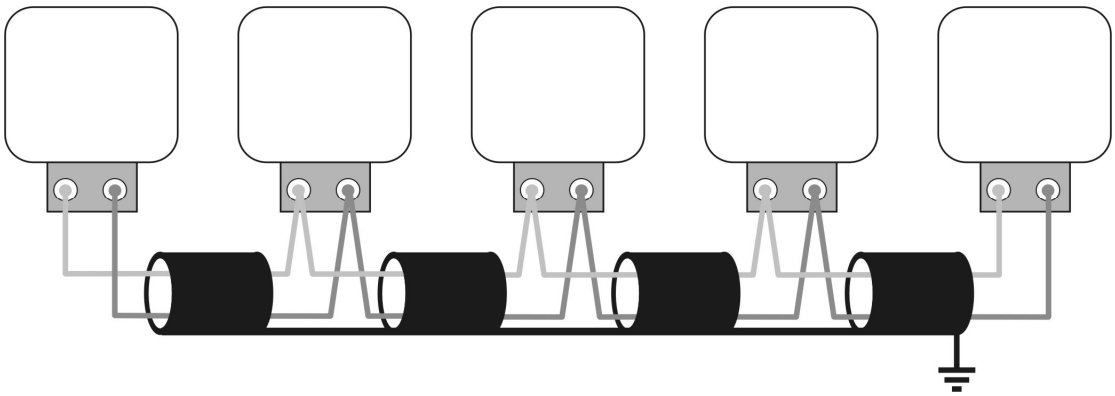


Figure 3.10
A typical RS-485

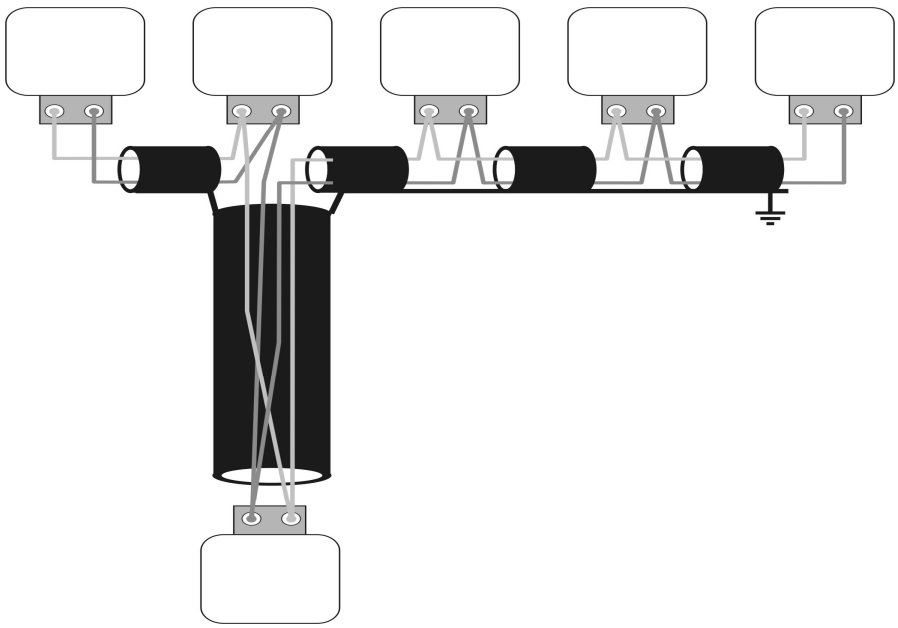


Figure 3.11
Adding a new device to a RS-485 bus

The distance between the end of the shield and the connection in the device should be no more than 10 mm or 1/2 inch. The end of the wires should be stripped only far enough to fit all the way into the connector, with no exposed wire outside the connector. The wire

should be twisted tightly before insertion into the screw connector. Often, installers will strip the shield from the wire and connect the shields together at the bottom of the cabinet. This is incorrect, as there would be from one to two meters of exposed cable from the terminal block at the bottom of the cabinet to the device at the top. This exposed cable will invariably receive noise from other devices in the cabinet. The pair of wires should be brought right up to the device and stripped as mentioned above.

3.5.5 Shielding

The choices of shielding for an RS-485 installation are:

- Braided
- Foil (with drain wire)
- Armored

From a practical point of view, the noise reduction difference between the first two is minimal. Both the braided and the foil will provide the same level of protection against capacitive noise. The third choice, armored cable has the distinction of protecting against magnetic induced noise. Armored cable is much more expensive than the first two and therefore braided and the foil types of cable are more popular. For most installers, it is a matter of personal choice when deciding to use either braided or foil shielded wire.

With the braided shield, it is possible to pick the A and B wires between the braids of the shield without breaking the shield. If this method is not used, then the shields of the two wires should be soldered or crimped together. A separate wire should be run from the shield at the device down to the ground strip in the bottom of the cabinet, but only one per bus, not per cabinet. It is incorrect in most cases to connect the shield to ground in each cabinet, especially if there are long distances between cabinets.

3.6 Test equipment

When testing or troubleshooting an RS-485 system, it is important to use the right test equipment. Unfortunately, there is very little in generic test equipment specifically designed for RS-485 testing. The most commonly used are the multimeter, oscilloscope and the protocol analyzer. It is important to remember that both of these types of test equipment must have floating differential inputs. The standard oscilloscope or multimeter each has their specific uses in troubleshooting an RS-485 system.

3.6.1 Multimeter

The multimeter has three basic functions in troubleshooting or testing an RS-485 system.

- Continuity verification
- Idle voltage measurement
- Common mode voltage measurement

Continuity verification

The multimeter can be used before start-up to check that the lines are not shorted or open. This is done as follows:

- Verify that the power is off
- Verify that the cable is disconnected from the equipment
- Verify that the cable is connected for the complete distance
- Place the multimeter in the continuity check mode

- Measure the continuity between the A and B lines.
- Verify that it is open.
- Short the A and B at the end of the line.
- Verify that the lines are now shorted.
- Un-short the lines when satisfied that the lines are correct.

If the lines are internally shorted before they are manually shorted as above, then check to see if an A line is connected to a B line. In most installations the A line is kept as one color wire and the B is kept as another. This procedure keeps the wires away from accidentally being crossed.

The multimeter is also used to measure the idle and common mode voltages between the lines.

Idle voltage measurement

At idle the master usually puts out a logical "1" and this can be read at any station in the system. It is read between A and B lines and is usually somewhere between –1.5 volts and –5 volts (A with respect to B). If a positive voltage is measured, it is possible that the leads on the multimeter need to be reversed. The procedure for measuring the idle voltage is as follows:

- Verify that the power is on
- Verify that all stations are connected
- Verify that the master is not polling
- Measure the voltage difference between the A– and B+ lines starting at the master
- Verify and record the idle voltage at each station

If the voltage is zero, then disconnect the master from the system and check the output of the master alone. If there is idle voltage at the master, then plug in each station one at a time until the voltage drops to or near zero. The last station probably has a problem.

Common mode voltage measurement

Common mode voltage is measured at each station, including the master. It is measured from each of the A and B lines to ground. The purpose of the measurement is to check if the common mode voltage is getting close to maximum tolerance. It is important therefore to know what the maximum common mode voltage is for the system. In most cases, it is +12 and –7 volts. A procedure for measuring the common mode voltage is:

- Verify that the system is powered up.
- Measure and record the voltage between the A and ground and the B and ground at each station.
- Verify that voltages are within the specified limits as set by the manufacturer.

If the voltages are near or out of tolerance, then either contact the manufacturer or install resistors between each line to ground at the station that has the problem. It is usually best to start with a high value such as 200k Ohms 1/4 watt and then go lower as needed. Both resistors should be of the same value.

3.6.2 Oscilloscope

Oscilloscopes are used for:

- Noise identification
- Ringing
- Data transfer

Noise identification

Although the Oscilloscope is not the best device for noise measurement, it is good for detection of some types of noise. The reason the oscilloscope is not that good at noise detection is that it is a two dimensional voltmeter; whereas the effect of the noise is seen in the ratio of the power of a signal vs. the power of the noise. But having said that, the oscilloscope is useful for determining noise that is constant in frequency. This can be a signal such as 50/60 Hz hum, motor induced noise or relays clicking on and off. The oscilloscope will not show intermittent noise, high frequency radio waves or the power ratio of the noise vs. the signal.

Ringing

Ringing is caused by the reflection of signals at the end of the wires. It happens more often on higher baud rate signals and longer lines. The oscilloscope will show this ringing as a distorted square wave.

As mentioned before, the 'fix' for ringing is a termination resistor at each end of the line. Testing the line for ringing can be done as follows:

- Use a two-channel oscilloscope in differential (A-B) mode
- Connect the probes of the oscilloscope to the A and B lines. Do NOT use a single channel oscilloscope, connecting the ground clip to one of the wires will short that wire to ground and prevent the system from operating
- Setup the oscilloscope for a vertical level of around 2 volts per division
- Setup the oscilloscope for horizontal level that will show one square wave of the signal per division
- Use an RS-485 driver chip with a TTL signal generator at the appropriate baud rate. Data can be generated by allowing the master to poll, but because of the intermittent nature of the signal, the oscilloscope will not be able to trigger. In this case a storage oscilloscope will be useful.
- Check to see if the waveform is distorted

Data transfer

Another use for the oscilloscope is to verify that data is being transferred. This is done using the same method as described for observing ringing, and by getting the master to send data to a slave device. The only difference is the adjustment of the horizontal level. It is adjusted so that the screen shows complete packets. Although this is interesting, it is of limited value unless noise is noted or some other aberration is displayed.

3.6.3 Protocol analyzer

The protocol analyzer is a very useful tool for checking the actual packet information. Protocol analyzers come in two varieties, hardware and software. Hardware protocol analyzers are very versatile and can monitor, log and interpret many types of protocols.

When the analyzer is hooked-up to the RS-485 system, many problems can be displayed such as:

- Wrong baud rates
- Bad data
- The effects of noise
- Incorrect timing
- Protocol problems

The main problem with the hardware protocol analyzer is the cost and the relatively rare use of it. The devices can cost from US$ 5000 to US$ 10000 and are often used only once or twice a year.

The software protocol analyzer, on the other hand, is cheap and has most of the features of the hardware type. It is a program that sits on the normal PC and logs data being transmitted down the serial link. Because it uses existing hardware, (the PC) it can be a much cheaper but useful tool. The software protocol analyzer can see and log most of the same problems a hardware type can.

The following procedure can be used to analyze the data stream:

- Verify that the system is on and the master is polling.
- Set up the protocol analyzer for the correct baud rate and other system parameters.
- Connect the protocol analyzer in parallel with the communication bus.
- Log the data and analyze the problem.

3.7 Summary

Installation

- Are the connections correctly made?
- What is the speed of the communications?
- What is the distance between the equipment?
- Is it a noisy environment?
- Is the software setup correctly?
- Are there any tees or stars in the bus?

Troubleshooting new and old systems

- Verify that there is power to the equipment
- Verify that the connectors are not loose
- Verify that the wires are correctly connected
- Check that a part, board or module has not visibly failed

Mechanical problems on new systems

- Keep the wires short, if possible
- Stranded wire should be used instead of solid wire (stranded wire will flex.)
- Only one wire should be soldered in each pin of the connector
- Bare wire should not be showing out of the pin of the connector
- The back shell should reliably and properly secure the wire

Setup problems on new systems

- Is the software communications setup at both ends for 8N1, 7E1 or 7O1?

- Is the baud rate the same at both devices? (1200,4800,9600,19200 etc.)
- Is the software setup at both ends for binary, hex or ASCII data transfer?
- Is the software setup for the proper type of control?

Noise problems on new systems

- Induced noise on the A or B lines?
- Common mode voltage noise?
- Reflection or ringing?

4

Modbus overview

Objectives

When you have completed study of this chapter, you will be able to:

- List the main Modbus structures and frames used
- Identify and correct problems with:
  - No response to protocol messages
  - Exception reports
  - Noise

4.1 General overview

Modbus ® is a transmission protocol (note: a *protocol* only), developed by Gould Modicon (now Schneider Electric) for process control systems. It is, however, regarded as a 'public' protocol and has become the *de facto* standard in multi–vendor integration. In contrast to other buses and protocols, no physical (OSI layer 1) interface has been defined.

MODBUS is a simple, flexible, publicly published protocol, which allows devices to exchange discrete and analog data. End users are aware that specifying MODBUS as the required interface between subsystems is a way to achieve multi–vendor integration with the most purchasing options and at the lowest cost. Small equipment makers are also aware that they must offer MODBUS with EIA–232 and/or EIA–485 to sell their equipment to system integrators for use in larger projects.

System integrators know that MODBUS is a safe interface to commit to, as they can be sure of finding enough equipment on the market to both realize the required designs and handle the inevitable 'change orders,' which come along. However, Modbus suffers from the limitations imposed by EIA–232/485 serial links, including the following:

- Serial lines are relatively slow – 9600 to 115,000 baud means only 0.010 Mbps to 0.115 Mbps. Compare that to today's common 'control network' speeds of 5 to 16 Mbps – or even the new Ethernet speeds of 100 Mbps, and 1Gbps and 10 Gbps!

- While it is easy to link 2 devices by EIA–232 and 20–30 devices by EIA–485, the only solution to link 500 devices with EIA–485 is a complex hierarchy of masters and slaves in a deeply nested tree structure. Such solutions are never simple or easy to maintain.
- Serial links with Modbus are inherently single–master designs. That means, only one device can talk to a group of slave devices – so only that one device (the master) is aware of all the current real–time data.

Designers share this data with multiple operator workstations, control systems, database systems, customized process optimizing workstations and all the other potential users of the data by ending up with complex, fragile hierarchies of master/slave groups shuffling data, up the ladder. Apart from the complexity involved, lower levels of the hierarchy (even expensive DCS systems) waste valuable time shuffling data solely for the benefit of higher levels of the hierarchy.

Even with all these limitations, Modbus has the advantage of wide acceptance among instrument manufacturers and users with many systems in operation. It can therefore be regarded as a *de facto* industrial standard with proven capabilities.

Certain characteristics of the Modbus protocol are fixed, such as frame format, frame sequences, handling of communications errors and exception conditions and the functions performed. Other characteristics are selectable. These are transmission medium, transmission characteristics and transmission mode, viz. RTU or ASCII. The user characteristics are set at each device and cannot be changed when the system is running.

The two transmission modes in which data is exchanged are:

- ASCII – readable; used, for example, for testing. (ASCII format)
- RTU – compact and faster; used for normal operation. (Hexadecimal format)

The RTU mode (sometimes also referred to as Modbus–B for Modbus Binary) is the preferred Modbus mode. The ASCII transmission mode (sometimes referred to as Modbus-A) has a typical message that is about twice the length of the equivalent RTU message.

Modbus also provides an error check for transmission and communication errors. Communication errors are detected by character framing, a parity check, a redundancy check or a sixteen bit cyclic redundancy check (CRC-16). The latter varies depending on whether the RTU or ASCII transmission mode is being used.

Modbus packets can also be sent over local area and wide area networks by encapsulating the Modbus data in a TCP/IP packet.

4.2 Modbus protocol structure

The following illustrates a typical Modbus message frame format.

Address field	Function field	Data field	Error check field
1 byte	1 byte	Variable	2 bytes

Table 4.1
Format of Modbus message frame

The first field in each message frame is the address field, which consists of a single byte of information. In request frames, this byte identifies the controller to which the request is being directed. The resulting response frame begins with the address of the responding device. Each slave can have an address field between 1 and 247; although practical limitations will limit the maximum number of slaves. A typical Modbus installation will have one master and two or three slaves.

The second field in each message is the function field, which also consist of a single byte of information. In a host request, this byte identifies the function that the target PLC is to perform.

If the target PLC is able to perform the requested function, the function field of its response will echo that of the original request. Otherwise, the function field of the request will be echoed with its most–significant bit set to one, thus signaling an exception response. Table 4.2 summarizes the typical functions used.

The third field in a message frame is the data field, which varies in length according to which function is specified in the function field. In a host request, this field contains information the PLC may need to complete the requested function. In a PLC response, this field contains any data requested by that host.

The last two bytes in a message frame comprise the error–check field. The numeric value of this field is calculated by performing a Cyclic Redundancy Check (CRC–16) on the message frame. This error checking assures that devices do not react to messages that may have been damaged during transmission.

Table 9.2 lists the address range and offsets for these four data types, as well as the function codes that apply to each. The diagram above also gives an easy reference to the Modbus data types.

Data type	Absolute addresses	Relative addresses	Function codes	Description
Coils	00001 to 09999	0 to 9998	01	Read coil status
Coils	00001 to 09999	0 to 9998	05	Force single coil
Coils	00001 to 09999	0 to 9998	15	Force multiple coils
Discrete inputs	10001 to 19999	0 to 9998	02	Read input status
Input registers	30001 to 39999	0 to 9998	04	Read input registers
Holding registers	40001 to 49999	0 to 9998	03	Read holding register
Holding registers	40001 to 49999	0 to 9998	06	Preset single register
Holding registers	40001 to 49999	0 to 9998	16	Preset multiple registers
–	–	–	07	Read exception status
–	–	–	08	Loopback diagnostic test

Table 4.2
Modicon addresses and function codes

4.3 Function codes

Each request frame contains a function code that defines the action expected for the target controller. The meaning of the request data fields is dependent on the function code specified.

The following paragraphs define and illustrate most of the popular function codes supported. In these examples, the contents of the message–frame fields are shown as hexadecimal bytes.

4.3.1 Read coil or digital output status (Function code 01)

This function allows the host to obtain the ON/OFF status of one or more logic coils in the target device.

The data field of the request consists of the relative address of the first coil followed by the number of coils to be read. The data field of the response frame consists of a count of the coil bytes followed by that many bytes of coil data.

The coil data bytes are packed with one bit for the status of each consecutive coil (1=ON, 0=OFF). The least significant bit of the first coil data byte conveys the status of the first coil read. If the number of coils read is not an even multiple of eight, the last data byte will be padded with zeros on the high end. Note that if multiple data bytes are requested, the low order bit of the first data byte in the response of the slave contains the first addressed coil.

In the following example, the host requests the status of coils 000A (decimal 00011) and 000B (decimal 00012). The target device's response indicates both coils are ON.

Request Message

Address	Function Code	Initial Coil Offset		Number of Points		CRC
		Hi	Lo	Hi	Lo	
01	01	00	0A	00	02	9D C9

Response Frame

Address	Function Code	Byte Count	Coil Data	CRC
01	01	01	03	11 89

Figure 4.1
Example of read coil status

4.3.2 Read digital input status (Function code 02)

This function enables the host to read one or more discrete inputs in the target device.

The data field of the request frame consists of the relative address of the first discrete input followed by the number of discrete inputs to be read. The data field of the response frame consists of a count of the discrete input data bytes followed by that many bytes of discrete input data.

The discrete–input data bytes are packed with one bit for the status of each consecutive discrete input (1=ON, 0=OFF). The least significant bit of the first discrete input data byte conveys the status of the first input read. If the number of discrete inputs read is not an even multiple of eight, the last data byte will be padded with zeros on the high end. The low order bit of the first byte of the response from the slave contains the first addressed digital input.

In the following example, the host requests the status of discrete inputs with offsets 0000 and 0001 hex i.e. decimal 10001 and 10002. The target device's response indicates that discrete input 10001 is OFF and 10002 are ON.

Request Message

Address	Function Code	Initial Coil Offset		Number of Points		CRC
		Hi	Lo	Hi	Lo	
01	02	00	00	00	02	F9 CB

Response Frame

Address	Function Code	Byte Count	Input Data	CRC
01	02	01	02	20 49

Figure 4.2
Example of read input status

4.3.3 Read holding registers (Function code 03)

This function allows the host to obtain the contents of one or more holding registers in the target device.

The data field of the request frame consists of the relative address of the first holding register followed by the number of registers to be read. The data field of the response time consists of a count of the register data bytes followed by that many bytes of holding register data.

The contents of each requested register (16 bits) are returned in two consecutive data bytes (most significant byte first).

In the following example, the host requests the contents of holding register hexadecimal offset 0002 or decimal 40003. The controller's response indicates that the numerical value of the register's contents is hexadecimal 07FF or decimal 2047. The first byte of the response register data is the high order byte of the first addressed register.

Request Message

Address	Function Code	Starting Register		Register Count		CRC	
		Hi	Lo	Hi	Lo		
01	03	00	02	00	01	25	CA

Response Frame

Address	Function Code	Byte Count	Register Data		CRC	
			Hi	Lo		
01	03	02	07	FF	FA	34

Figure 4.3
Example of Reading Holding Register

4.3.4 Reading input registers (Function code 04)

This function allows the host to obtain the contents of one or more input registers in the target device.

The data field of the request frame consists of the relative address of the first input register followed by the number of registers to be read. The data field of the response frame consists of a count of the register–data bytes followed by that many bytes of input–register data.

The contents of each requested register are returned in two consecutive register–databytes (most–significant byte first). The range for register variables is 0 to 4095.

In the following example, the host requests the contents of input register hexadecimal offset 000 or decimal 30001. The PLC's response indicates that the numerical value of that register's contents is 03FFH, which would correspond to a data value of 25 percent (if the scaling of 0 to 100 percent is adopted) and a 12 bit D to A converter with a maximum reading of 0FFFH is used.

Request Message

Address	Function Code	Starting Register		Register Count		CRC	
		Hi	Lo	Hi	Lo		
01	04	00	00	00	01	31	CA

Response Frame

Address	Function Code	Byte Count	Register Data		CRC	
			Hi	Lo		
01	04	02	03	FF	F9	80

Figure 4.4
Example of Reading Input Register

4.3.5 Force single coil (Function code 05)

This function allows the host to alter the ON/OFF status of a single logic coil in the target device.

The data field of the request frame consists of the relative address of the coil followed by the desired status for that coil. A hexadecimal status value of FF00 will activate the coil, while a status value of 0000H will deactivate it. Any other status value is illegal.

If the controller is able to force the specified coil to the requested state, the response frame will be identical to the request. Otherwise, an exception response will be returned.

If the address 00 is used to indicate broadcast mode, all attached slaves will modify the specified coil address to the state required.

The following example illustrates a successful attempt to force coil 11 (decimal) OFF.

Request Message

Address	Function Code	Coil Offset		New Coil Status		CRC	
		Hi	Lo	Hi	Lo		
01	05	00	0A	00	00	ED	C8

Response Frame

Address	Function Code	Coil Offset		New Coil Status		CRC	
		Hi	Lo	Hi	Lo		
01	05	00	0A	00	00	ED	C8

Figure 4.5
Example of forcing a single coil

4.3.6 Preset single register (Function code O6)

This function enables the host to alter the contents of a single holding register in the target device.

The data field of the request frame consists of the relative address of the holding register followed by the new value to be written to that register (most–significant byte first).

If the controller is able to write the requested new value to the specified register, the response frame will be identical to the request. Otherwise, an exception response will be returned.

The following example illustrates a successful attempt to change the contents of holding register 40003 to 3072 (0C00 Hex).

When slave address is set to 00 (broadcast mode), all slaves will load the specified register with the value specified.

Request Message

Address	Function Code	Register Offset		Register Value		CRC	
		Hi	Lo	Hi	Lo		
01	06	00	02	0C	00	2D	0A

Response Frame

Address	Function Code	Register Offset		Register Value		CRC	
		Hi	Lo	Hi	Lo		
01	06	00	02	0C	00	2D	0A

Figure 4.6
Example of Presetting a Single Register

4.3.7 Read exception status (Function code 07)

This is a short message requesting the status of eight digital points within the slave device.

This will provide the status of eight predefined digital points in the slave. For example, this could be items such as the status of the battery, whether memory protect has been enables or the status of the remote input/output racks connected to the system.

Request Message

Address	Function Code	CRC	
11	07	..	..

Response Frame

Address	Function Code	Coil Station	CRC	
11	07	02	..	..

Figure 4.7
Read exception status query message

4.3.8 Loopback test (Function code 08)

The objective of this function code is to test the operation of the communications system without affecting the memory tables of the slave device. It is also possible to implement additional diagnostic features in a slave device (should this be considered necessary) such as number of CRC errors, number of exception reports etc.

The most common implementation will only be considered in this section; namely, a simple return of the query messages.

Request Frame

Address	Function Code	Data Diagnostic Code		Data		CRC
		Hi	Lo	Hi	Lo	
11	08	00	00	A5	37	.. ..

Response Frame

Address	Function Code	Data Diagnostic Code		Data		CRC
		Hi	Lo	Hi	Lo	
11	08	00	00	A5	37	.. ..

Figure 4.8
Loopback test message

4.3.9 Force multiple coils or digital outputs (Function code 0F)

This forces a contiguous (or adjacent) group of coils to an ON or OFF state. The following example sets 10 coils starting at address 01 Hex (at slave address 01) to the ON state. If slave address 00 is used in the request frame, broadcast mode will be implemented resulting in all slaves changing their coils at the defined addresses.

Request Frame

Address	Function Code	Address	Byte Count	Data Coil Status	CRC
		Hi Lo		Hi Lo	
01	0F	00 01	0F	FF 03	.. ..

Response Frame

Address	Function Code	Address	Number of Coils	CRC
		Hi Lo	Hi Lo	
01	0F	00 01	00 0A	.. ..

Figure 4.9
Example of forcing multiple coils

4.3.10 Force multiple registers (Function code 10)

This is similar to the preset a single register and the forcing of multiple coils. In the example below, a slave address 01 has 2 registers changed commencing at address 10.

Request Frame

Address	Function Code	Address	Quantity	Byte Count	First Register	Second Register	CRC
		Hi Lo	Hi Lo		Hi Lo	Hi Lo	
01	10	00 0A	00 02	04	00 0A	01 02	.. ..

Response Frame

Address	Function Code	Address	Quality	CRC
		Hi Lo	Hi Lo	
01	10	00 0A	00 02	.. ..

Figure 4.10
Example of Presetting Multiple Registers

Table 4.3 lists the most important exception codes that may be returned.

Code	Name	Description
01	Illegal function	Requested function is not supported.
02	Illegal data address	Requested data address is not supported
03	Illegal data value	Specified data value is not supported
04	Failure in associated device	Slave PLC has failed to respond to a message
05	Acknowledge	Slave PLC is processing the command
06	Busy, rejected message	Slave PLC is busy

Table 4.3
Abbreviated list of exception codes returned

An example of an illegal request and the corresponding exception response is shown below. The request in this example is to READ COIL STATUS of points 514 to 521 (eight coils beginning with an offset 0201H). These points are not supported in this PLC, so an exception report is generated indicating code 02 (illegal address).

Request Message

Address	Function Code	Starting Point	Number of Points	CRC
01	01	02 01	00 08	6D B4

Exception Response Message

Address	Function Code	Exception Code	CRC
01	81	02	C1 91

Figure 4.11
Example of an illegal request

4.4 Common problems and faults

No matter what extremes of care you may have taken, there is hardly ever an installation that boasts of trouble-free setup and configuration. Some of the commonly faced

problems and faults that one comes across in the industry are listed below. This list is broken into two distinct groups:

4.4.1 Hardware related

This includes mis-wired communication cabling and faulty communication interfaces.

4.4.2 Software related

Software related issues arise when the master application tries to access non-existent nodes or use invalid function codes, address non-existent memory locations in the slaves, or specify illegal data format types, which obviously the slave devices do not understand. These issues will be dealt with in detail under the Modbus plus protocol, as these issues are common to both the traditional Modbus protocol and the Modbus plus protocol. They are summarized under software related problems. These issues are also applicable to the latest Modbus/TCP protocol.

4.5 Description of tools used

In order to troubleshoot the problems listed above, one would require the use of a few tools, hardware and software, to try and decipher the errors. The most important tool of all would always be the instruction manuals of the various components involved.

The hardware tools that may be required include RS-232 break-out boxes, RS-232 to RS-485 converters, continuity testers, voltmeters, screw drivers, pliers, crimping tools, spare cabling and other similar tools & tackle. These would generally be used to ensure that the cabling and terminations are proper and have been installed as per the recommended procedures detailed in the instruction manuals.

On the software front, one would need some form of protocol analyzer that is in a position to eavesdrop on the communications link between the master and slave modules. This could be either a dedicated hardware protocol analyzer, that is very expensive, or a software based protocol analyzer that could reside on a computer.

Obviously, this second option is more economical and also requires the relevant hardware component support in order to connect to the network.

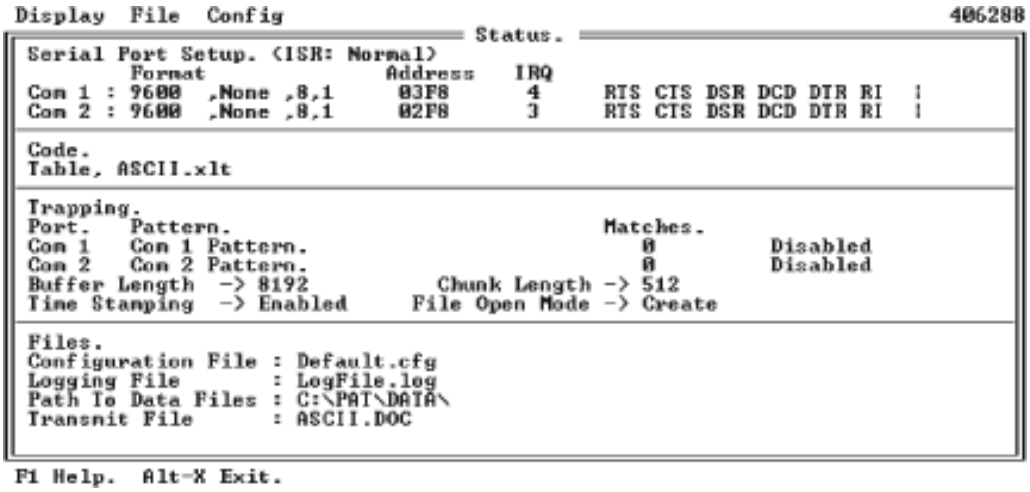


Figure 4.12
Screen shot of the protocol analysis tool

4.6 Detailed troubleshooting

4.6.1 Mis-wired communication cabling

RS-232 wiring for 3-wire point-to-point mode

There are various wiring scenarios such as point-to-point, multi-drop two-wire, multi-drop 4-wire, etc. In a point-to-point configuration, the physical standard used is usually RS-232 on a 25 Pin D connector, which means a minimum of three wires are used. They are: on the DTE (master) side: transmit (TxD – pin 2), receive (RxD – pin 3) and signal ground (common – pin 7); and on the DCE (slave) side: receive (RxD – pin 2), transmit (TxD – pin 3) and signal ground (common – pin 7).

The other pins were primarily used for handshaking between the two devices for data flow control. Nowadays, these pins are rarely used as the flow control is done via software handshaking protocols.

With the advent of VLSI technology, the footprint of the various devices have a tendency to shrink; thereby even the physical connections of communications have been now been standardized on 9-pin D connectors. The pin assignment that has been adopted is the IBM standard in which the pin configurations are: on the DTE (master) side: transmit (TxD – pin 3), receive (RxD – pin 2) and signal ground (common – pin 5); and on the DCE (Slave) side: receive (RxD – pin 3), transmit (TxD – pin 2) and signal ground (common – pin 5).

It follows that the cabling between the two devices must be straight-through pin-for-pin. In this manner, the transmit pin of the master is directly connected to the receive pin of the slave and vice-versa. These cables are standard off-the-shelf products available in standard predetermined lengths. They can also be fabricated to custom lengths, with ease.

Master devices are usually the present day IBM compatible computers, with a Modbus application enabled on it, and therefore have the standard IBM RS-232 port provided. The slave devices usually have a user selectable option to have either an RS-232 or RS 485 port for communication. Unfortunately some manufacturers, in order to force the customers to return to them time-and-again, employ a strategy of modifying these standards to their own advantage.

Illustrated below are a couple of these combinations:

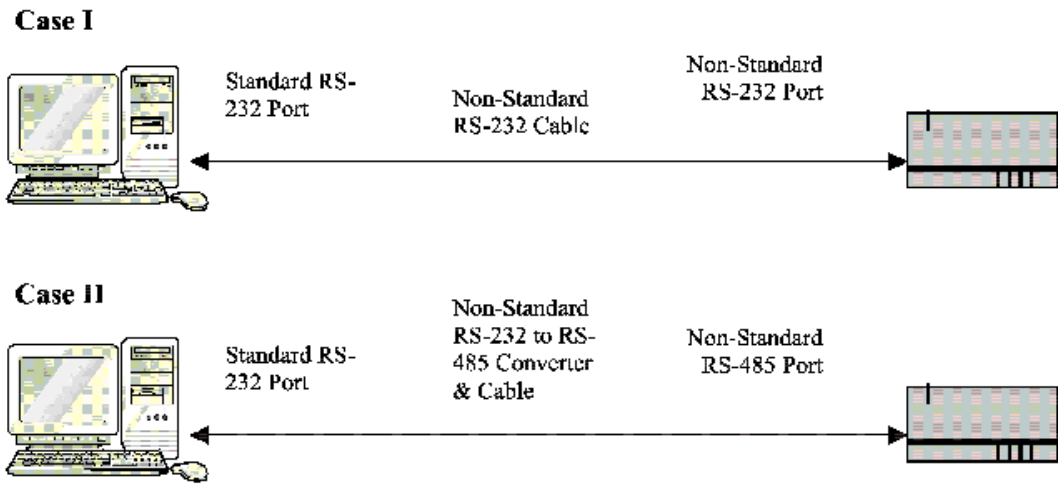


Figure 4.13
Typical customizations in RS-232 cabling

As can be seen in case I, the manufacturer has modified the standard pin-out of the RS-232 connection or has totally replaced the standard IBM RS-232 9-pin DIN connector with his own standard. In situations like these, it is then imperative to possess the manufacturer's manuals of the device being used.

With the aid of the continuity checker, and the use of the RS-232 breakout box, you can determine the non-standard pin-out of the cable, and fabricate a spare cable to the required lengths without having to revert to the supplier.

As in case II, the manufacturer has embedded an RS-232 to RS-485 converter into the cable itself and this would mimic a standard off the shelf RS-232 serial cable. This obviously would not be clearly evident if you were not aware of the modification. With the aid of the voltage tester, you can determine the operating voltages and therefore be in a position to decipher the type of standard being employed.

In other implementations, where multi-dropped RS-485 is used, the installations usually cater for both configurations of two-wire and four-wire communications. In the case of single master configurations, either of the two wiring modes could be used. In the case of multi master configurations, only the two-wire communication can be used.

RS-485/RS-422 wiring for 4-wire, repeat or multi-drop modes

The four-wire configuration is, in actual fact, the RS-422 standard where there is one line driver connected to multiple receivers, or multiple line drivers connected to one line receiver. When using 4-wire RS-422 communications, messages are transmitted on one pair of shielded, twisted wires and received on another (Belden part number 9729, 9829, or equivalent). Both multi-drop and repeat configurations are possible when RS-422 is used.

This is the classic case where the one line driver is the one that belongs to the master and the remaining receivers belong to the multiple slaves. The slaves receive their commands via this link and respond via their line drivers that are all connected to each other and finally connected to the master's receiver.

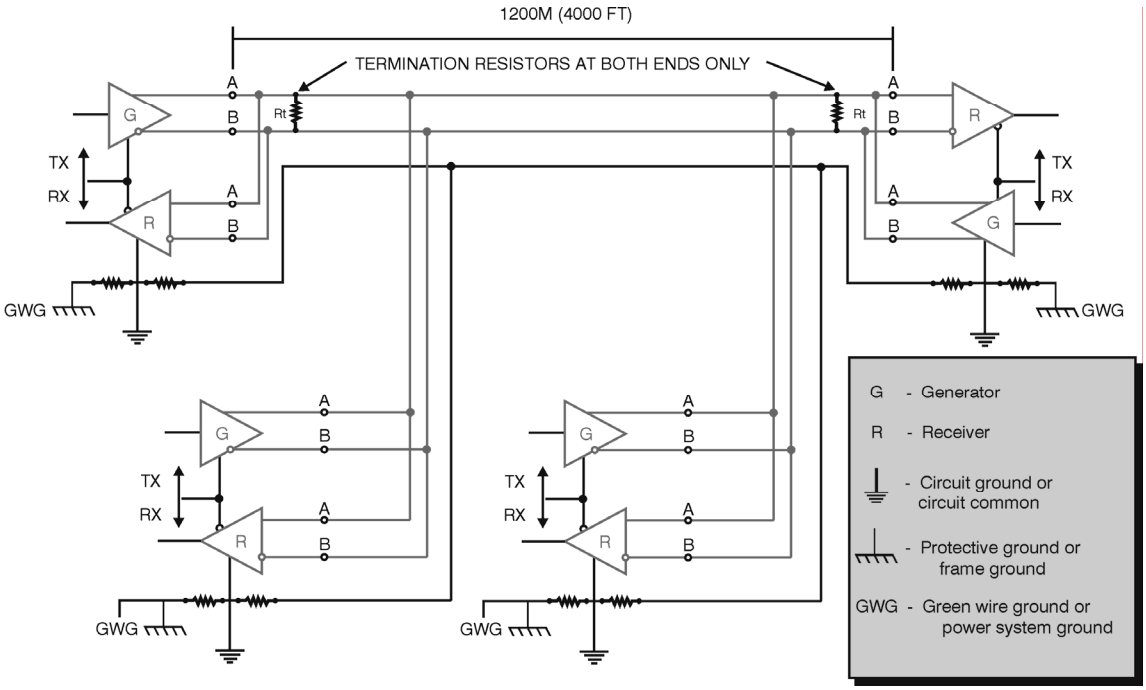


Figure 4.14
Typical two-wire RS-422/485 cabling

RS-485 wiring for 2-wire, multidrop mode only

The two-wire configuration is the actual RS-485 standard where both the line drivers and the receivers are connected to the same pair of communication cables. When using 2-wire RS-485 communications, messages are transmitted and received on the same pair of shielded, twisted wires. Care must be taken to turn the host driver IC on when sending messages and off when receiving messages. This can be accomplished using software or hardware. Only multi-drop wiring configuration is possible when RS-485 is used.

Typically, the maximum number of physical RS-485 nodes on a network is 32. If more physical nodes are placed on the network an RS-485 repeater must be used.

This system works inherently similar to the four-wire system except that the master transmits requests to the various slaves and the slaves respond to the master over the same pair of wires.

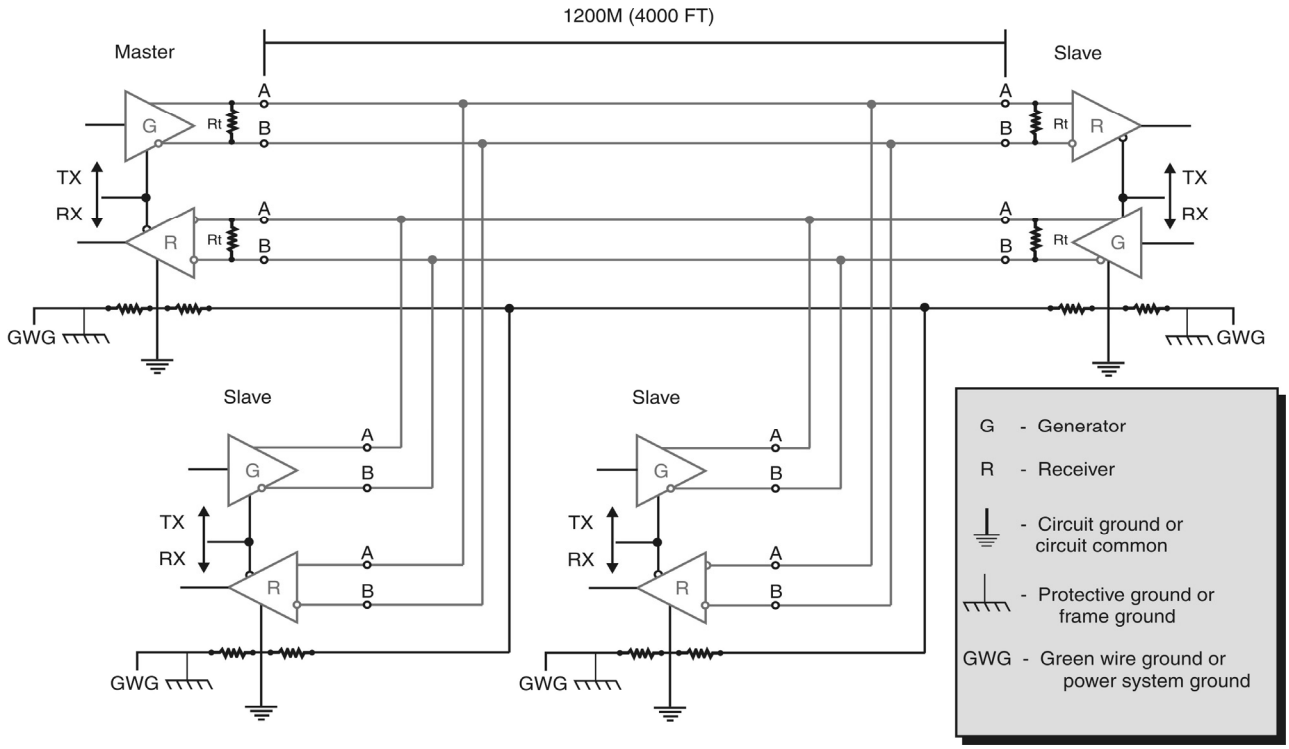


Figure 4.15
Typical four-wire RS-485 multi-drop cabling

Note: The shield grounds should be connected to system ground at one point only. This will eliminate the possibility of ground loops causing erroneous operation.

Grounding

The logic ground of all controllers and any other communication device on the same network must reference the same ground. If one power supply is used for all controllers, then they will be referenced to the same ground. When multiple supplies are used, there are a couple of options to consider:

- Connect the supply output ground to a solid earth ground on all power supplies. If the grounding system is good, all controllers will be referenced to the same ground. (Note that the typical PC follows this practice).

In some older buildings, the earth ground could vary by several volts within the building. In this case, option 2 should be considered

- Use the GND position on each controller to tie all controllers to the same potential. Note that each power supply output must be isolated from the input lines, otherwise irreparable damage may result if two power supplies are powered from different phases of an electrical distribution system. It is not recommended that the shield of a cable be used for this purpose, use a separate conductor
- Where it is not possible to get all controllers referenced to the same ground, use an isolated RS-485 repeater between controllers that are located at different ground potentials. Isolated repeaters are also an excellent way to clean up signals and extend distances in noisy environments

4.6.2 Faulty communication interfaces

RS-232 driver failed

The RS-232 driver may be tested with a good high impedance voltmeter. Place the meter in the DC voltage range. Place the RED probe on the transmit pin (TxD) and the BLACK probe on the signal ground (common). While the node is not transmitting (TX light off), there should be a voltage between –5 VDC and –25 VDC across these pins. This is referred to as the idle state of the RS-232 driver. When the node is transmitting, the voltage will oscillate through between ±5VDC and ±25 VDC. It may be difficult to see the deflection at the higher baud rates and an oscilloscope is suggested for advanced troubleshooting. If the voltage is fixed at anywhere between –25 VDC and +25 VDC and doesn't oscillate as the TX light blinks, then the transmitter is probably damaged and must be returned to the factory for repair or replacement.

RS-232 receiver failed

This requires the use of an RS-232 loop back plug or an RS-232 breakout box with a jumper installed between the transmit and receive pins. After connecting this to the communication port of the node to be tested, the node is made to transmit data, and when the TxD light of the node flashes, the RxD light must also flash, then it can be said that the RS-232 receiver of that node is good. If the RxD light does not flash at the same time as the TxD, then the RS-232 receiver may be bad.

Alternatively, two nodes may be connected to each other with a tested and working communication cable and both their RS-232 drivers working. If one of the nodes is transmitting to the other, the second node indicates a good reception by flashing its RxD light as the first node transmits, then the RS-232 receiver on the second node is good. The same is true when the second node transmits and the first flashes its RxD light.

RS-485 driver failed

The RS-485 driver too may be tested with a good high impedance voltmeter. Place the meter in the DC voltage range. Place the RED probe on the non-inverted transmit terminal (TX+) and the BLACK probe on the inverted transmit terminal (TX–). While the node is not transmitting (TX light off), there should be approximately minus 4 Vdc across these wires. When the node is transmitting, the voltage will oscillate through ±4Vdc. It may be difficult to see the deflection at the higher baud rates and an oscilloscope is suggested for advanced troubleshooting. The minimum voltage obtained during a full 1200 baud to 19200 baud sweep will be around –1.6 VDC. If the voltage is fixed at

around +4, 0, or –4 volts, and doesn't oscillate as the TX light blinks, then the transmitter is probably damaged and must be returned to the factory for repair or replacement.

RS-485 receiver failed

This requires two nodes connected to each other with a tested and working communication cable and both their RS-485 drivers working. If one of the nodes is transmitting to the other, the second node indicates a good reception by flashing its RX light as the first node transmits, then the RS-485 receiver on the second node is good. If the RX light flashes when the first node polls at other baud rates then most likely, the polarity is reversed between the OUT of the first node and the second node. Reverse the TX+ and TX– wires at the first node.

4.6.3 Software related problems

See the suggestions under Modbus Plus. But the important issues to consider are:

No response to message from master to slave

This could mean that either the slave does not exist or there are CRC errors in the transmitted message due to noise (or incorrectly formatted message).

Exception responses

See the list of potential problems reported by the exception responses. This could vary from slave address problems to I/O addresses being illegal.

4.7 Conclusion

We have seen that the master computer sends commands to the various slave units to determine the status of its various process inputs or to change the status of its outputs using the Modbus protocol. The commands are transmitted over a single pair of twisted wires (RS-485) or two pair of twisted wires (RS-422) at speeds of 9600, 19200, 38400, or 57600 baud. The addressed slave decodes the commands and returns the appropriate response. If the master computer is an IBM PC or compatible, inexpensive interface driver software is available. This software dramatically simplifies sending and receiving these messages. If you prefer, you can use one of the off-the-shelf graphics-based data acquisition and control software packages. Many of these packages offer a Modbus compatible driver.

If you were to follow the recommended installation and startup procedure and took extra precaution while setting up the following items, then you would achieve trouble free startup:

- Setting the base address
- Setting the protocol and baud rate
- Serial communications wiring
- Communication wiring termination

5

AS-interface (AS-i) overview

Objectives

When you have completed study of this chapter, you will be able to:

- Describe the main features of AS-i
- Fix problems with:
  - Cabling
  - Connections
  - Gateways to other standards

5.1 Introduction

The actuator sensor interface is an open system network developed by eleven manufacturers. These manufacturers created the AS-i association to develop the AS-i specifications. Some of the more widely known members of the association include Pepperl-Fuchs, Allen-Bradley, Banner Engineering, Datalogic Products, Siemens, Telemecanique, Turck, Omron, Eaton and Festo. The governing body is ATO, the AS-i Trade Organization. The number of ATO members currently exceeds fifty and continues to grow. The ATO also certifies that products under development for the network meet the AS-i specifications. This will assure compatibility between products from different vendors.

AS-i is a bit-oriented communication link designed to connect binary sensors and actuators. Most of these devices do not require multiple bytes to adequately convey the necessary information about the device status, so the AS-i communication interface is designed for bit oriented messages in order to increase message efficiency for these types of devices.

The AS-i interface is just that, an interface for binary sensors and actuators, designed to interface binary sensors and actuators to microprocessor based controllers using bit length ‘messages.’ It was not developed to connect intelligent controllers together since this would be far beyond the limited capability of bit length message streams.

Modular components form the central design of AS-i. Connection to the network is made with unique connecting modules that require minimal, or in some cases no, tools

and provide for rapid, positive device attachment to the AS-i flat cable. Provision is made in the communications system to make 'live' connections, permitting the removal or addition of nodes with minimum network interruption.

Connection to higher level networks (e.g. ProfiBus) is made possible through plug-in PC and PLC cards or serial interface converter modules.

The following sections examine these features of the AS-i network in more detail.

5.2 Layer 1 – The physical layer

AS-i uses a two-wire untwisted, unshielded cable that serves as both communication link and power supply for up to thirty-one slaves. A single master module controls communication over the AS-i network, which can be connected in various configurations such as bus, ring, or tree. The AS-i flat cable has a unique cross-section that permits only properly polarized connections when making field connections to the modules. Alternatively, ordinary 2 wire cable (#16 AWG, 1,5 mm) can be used. A special shielded cable is also available for high noise environments.

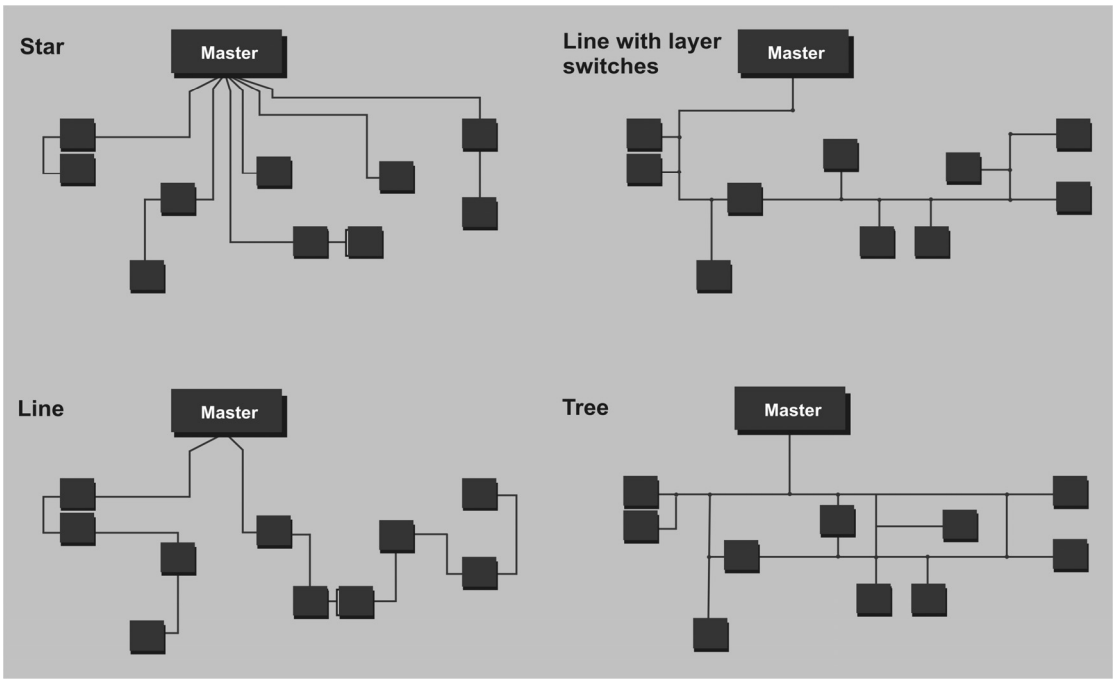


Figure 5.1
Various AS-i configurations

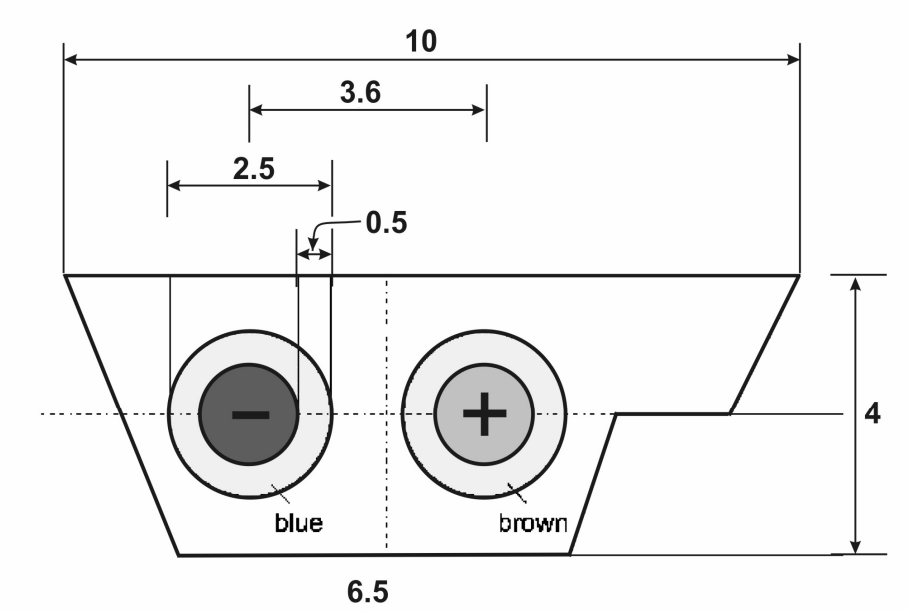


Figure 5.2
Cross section of AS-i cable (mm)

Each slave is permitted to draw a maximum of 65mA from the 30Vdc-power supply. If devices require more than this, separate supplies must be provided for each device. With a total of 31 slaves drawing 65mA, a total limit of 2A has been established to prevent excessive voltage drop over the 100m permitted network length. A 16 AWG cable is specified to insure this condition. If this limitation on power drawn from the (yellow) signal cable is a problem, then a second (black) cable, identical in dimensions to the yellow cable, can be used in parallel for power distribution only.

The slave (or field) modules are available in four configurations:

- Input modules for 2 and 3-wire DC sensors or contact closure
- Output modules for actuators
- Input/output (I/O) modules for dual purpose applications
- Field connection modules for direct connection to AS-i compatible devices.
- 12 bit analog to digital converter

The original AS-i specification (V2) allowed for 31 devices per segment of cable, with a total of 124 digital inputs and 124 digital outputs that is, a total of 248 I/O points. The latest specification, V2.1, allows for 62 devices, resulting in 248 inputs and 186 outputs, a total of 434 I/O points. With the latest specification, even 12 bit A to D converters can be read over 5 cycles.

A unique design allows the field modules to be connected directly into the bus while maintaining network integrity. The field module is composed of an upper and lower section, secured together once the cable is inserted. Specially designed contact points pierce the self-sealing cable providing bus access to the I/O points and/or continuation of the network. True to the modular design concept, two types of lower sections and three types of upper sections are available to permit ‘mix-and-match’ combinations to accommodate various connection schemes and device types. Plug connectors are utilized to interface the I/O devices to the slave (or with the correct choice of modular section screw terminals) and the entire module is sealed from the environment with special seals provided where the cable enters the module. The seals conveniently store away within the module when not in use.

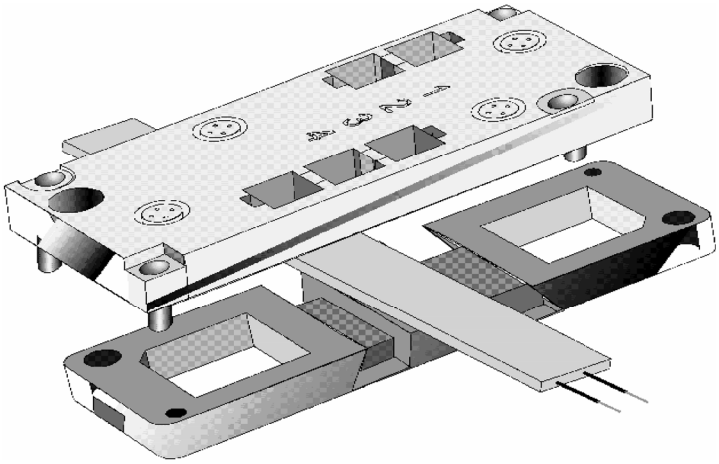


Figure 5.3
Connection to the cable

The AS-i network is capable of a transfer rate of 167 Kbps. Using an access procedure known as 'master-slave access with cyclic polling,' the master continually polls all the slave devices during a given cycle to ensure rapid update times. For example, with 31 slaves and 124 I/O points connected, the AS-i network can ensure a 5mS-cycle time, making the AS-i network one of the fastest available.

A modulation technique called 'alternating pulse modulation' provides this high transfer rate capability as well as high data integrity. This technique will be described in the following section.

5.3 Layer 2 – the data link layer

The data link layer of the AS-i network consists of a master call-up and slave response. The master call-up is exactly fourteen bits in length while the slave response is 7 bits. A pause between each transmission is used for synchronization. Refer to the following figure, for example, call-up and answer frames.

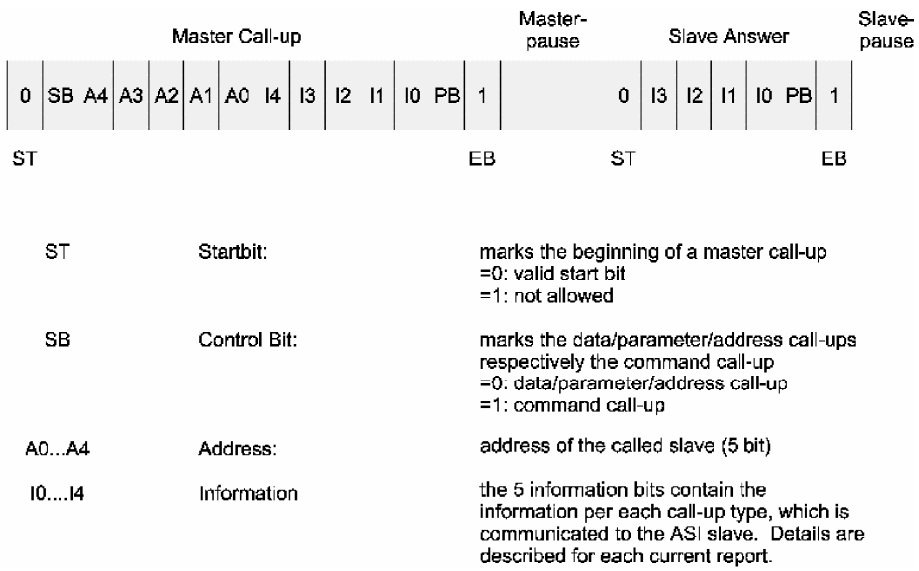


Figure 5.4
Example call up and response frames

Various code combinations are possible in the information portion of the call-up frame and it is precisely these various code combinations that are used to read and write information to the slave devices. Examples of some of the master call-ups are listed in the following figure. A detailed explanation of these call-ups is available from the ATO literature and is only included here to illustrate the basic means of information transfer on the AS-i network.

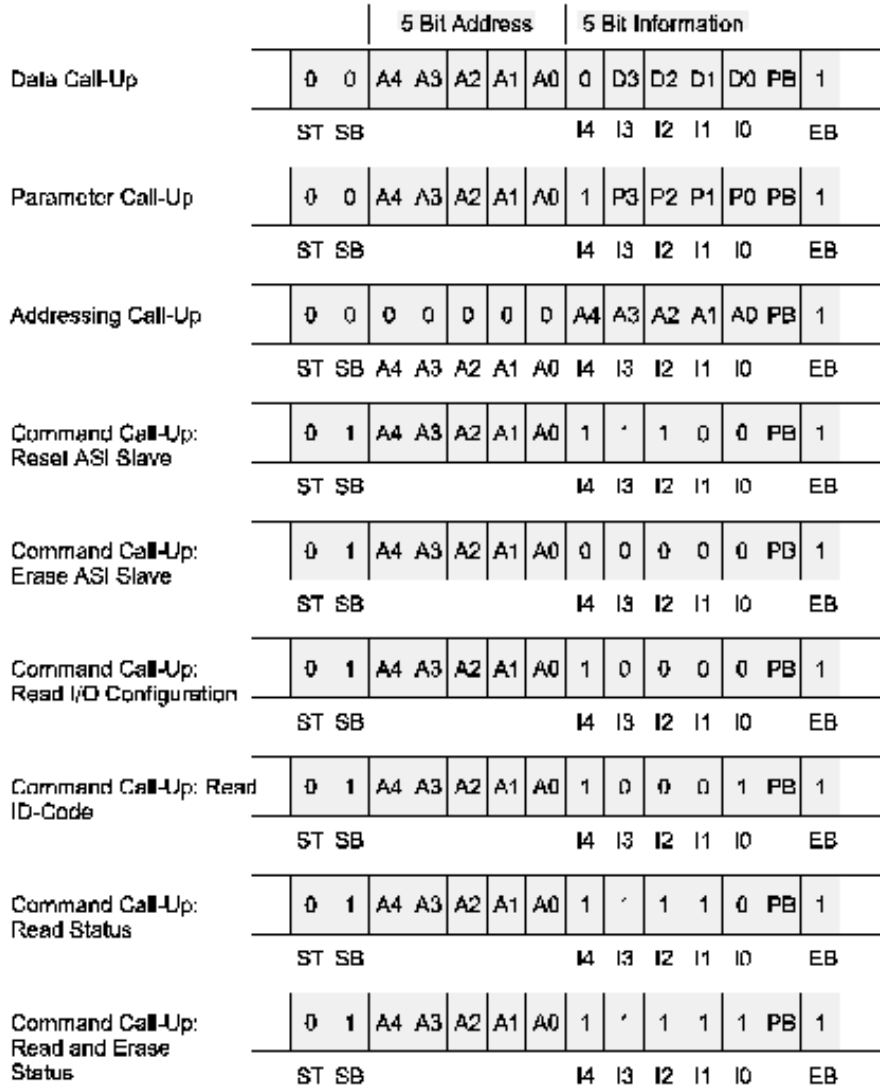


Figure 5.5
Some AS-i call ups

The modulation technique used by AS-i is known as *'Alternating Pulse Modulation'* (APM). Since the information frame is of a limited size, providing conventional error checking was not possible and therefore the AS-i developers chose a different technique to insure high level of data integrity.

Referring to the following figure, the coding of the information is similar to Manchester II coding but utilizing a 'sine squared' waveform for each pulse. This waveform has several unique electrical properties, which reduce the bandwidth required of the transmission medium (permitting faster transfer rates) and reduce the end of line reflections common in networks using square wave pulse techniques. Also, notice that each bit has an associated pulse during the second half of the bit period. This property is

utilized as a bit level of error checking by all AS-i devices. The similarity to Manchester II coding is no accident since this technique has been used for many years to pass synchronizing information to a receiver along with the actual data.

In addition, AS-i developers also established a set of rules for the APM coded signal that is used to further enhance data integrity. For example, the start bit or first bit in the AS-i telegram must be a negative impulse and the stop bit a positive impulse. Two subsequent impulses must be of opposite polarity and the pause between two consecutive impulses should be 3 microseconds. Even parity and a prescribed frame length are also incorporated at the frame level. As a result the 'odd' looking waveform, in combination with the rules of the frame formatting, the set of APM coding rules and parity checking, work together to provide timing information and high level data integrity for the AS-i network.

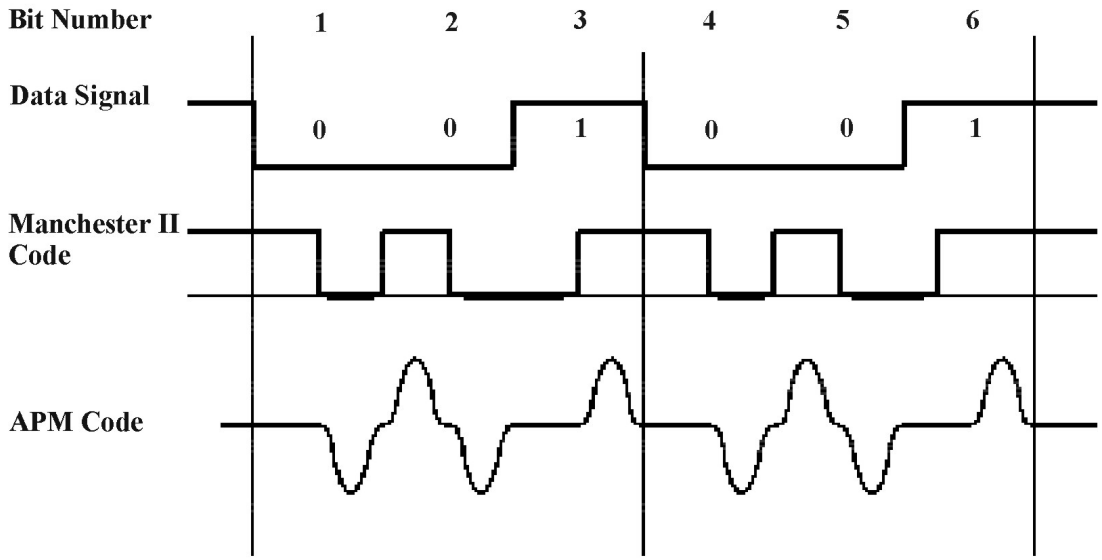


Figure 5.6
Sine squared wave form

5.4 Operating characteristics

AS-i node addresses are stored in nonvolatile memory and can be assigned either by the master or one of the addressing or service units. Should a node fail, AS-i has the ability to automatically reassign the replaced node's address and, in some cases, reprogram the node itself allowing rapid response and repair times.

Since AS-i was designed to be an interface between lower level devices, connection to higher-level systems enables the capability to transfer data and diagnostic information. Plug-in PC cards and PLC cards are currently available. The PLC cards allow direct connection with various Siemens PLCs. Serial communication converters are also available to enable AS-i connection to conventional RS-232, 422, and 485 communication links. Direct connection to a Profibus field network is also possible with the Profibus coupler, enabling several AS-i networks access to a high-level digital network.

Handheld and PC based configuration tools are available, which allow initial start-up programming and also serve as diagnostic tools after the network is commissioned. With these devices, on-line monitoring is possible to aid in determining the health of the network and locating possible error sources.

5.5 Troubleshooting

5.5.1 Introduction

The AS-i system has been designed with a high degree of 'maintenance friendliness' in mind and has a high level of built-in auto-diagnosis. The system is continuously monitoring itself against faults such as:

- Operational slave errors (permanent or intermittent slave failure, faulty configuration data such as addresses, I/O configuration, and ID codes)
- Operational master errors (permanent or intermittent master failure, faulty configuration data such as addresses, I/O configuration, and ID codes)
- Operational cable errors (short circuits, cable breakage, corrupted telegrams due to electrical interference and voltage outside of the permissible range)
- Maintenance related slave errors (false addresses entered, false I/O configuration, false ID codes)
- Maintenance related master errors (faulty projected data such as I/O configuration, ID codes, parameters etc.)
- Maintenance related cable errors (counter poling the AS-i cable)

The fault diagnosis is displayed by means of LEDs on the master.

Where possible, the system will protect itself. During a short-circuit, for example, the power supply to the slaves is interrupted, which causes all actuators to revert to a safe state. Another example is the jabber control on the AS-i chips, whereby a built-in fuse blows if too much current is drawn by a chip, disconnecting it from the bus.

The following tools can be used to assist in faultfinding.

5.6 Tools of the trade

5.6.1 Addressing handheld

Before an AS-i system can operate, all the operating addresses must be assigned to the connected slaves, which store this on their internal nonvolatile memory (EEPROM). Although this can theoretically be done on-line, it requires that a master device with this addressing capability be available.

In the absence of such a master, a specialized battery powered addressing handheld (for example, one manufactured by Pepperl and Fuchs) can be used. The device is capable of reading the current slave address (from 0 to 31) as well as reprogramming the slave to a new address entered via the keyboard.

The slaves are attached to the handheld device, one at a time, by means of a special short cable. They are only powered via the device while the addressing operation takes place (about 1 second) with the result that several hundred slaves can be configured in this way before a battery change is necessary.

5.6.2 Monitor

A monitor is essentially a protocol analyzer, which allows a user to capture and analyze the telegrams on the AS-i bus. A good monitor should have triggering and filtering capabilities, as well as the ability to store, retrieve and analyze captured data. Monitors are usually implemented as PC-based systems.

5.6.3 Service device

An example of such a device is the SIE 93 handheld manufactured by Siemens. It can perform the following:

- Slave addressing, as described above
- Monitoring, i.e. the capturing, analysis and display of telegrams
- Slave simulation, in which case it behaves like a supplementary slave, the user can select its operating address
- Master simulation, in which case the entire cycle of master requests can be issued to test the parameters, configuration and address of a specific slave device (one at a time)

5.6.4 Service book

A 'service book' is a commissioning and servicing tool based on a notebook computer. It is capable of monitoring an operating network, recording telegrams, detecting errors, addressing slaves off-line, testing slaves off-line, maintaining a database of sensor/ actuator data and supplying help functions for user support.

Bus data for use by the software on the notepad is captured, preprocessed and forwarded to the laptop by a specialized network interface, a so-called 'hardware checker.' The hardware checker is based on an 80C535 single chip micro-controller and connects to the notepad via an RS-232 interface.

5.6.5 Slave simulator

Slave simulators are PC based systems used by software developers to evaluate the performance of a slave (under development) in a complete AS-i network. They can simulate the characteristics of up to 32 slaves concurrently and can introduce errors that would be difficult to set up in real situations.

6

DeviceNet overview

Objectives

When you have completed study of this chapter you will be able to:

- List the main features of DeviceNet
- Identify and correct problems with:
- Cable topology
- Power and earthing
- Signal voltage levels
- Common mode voltages
- Terminations
- Cabling
- Noise
- Node communications problems

6.1 Introduction

DeviceNet, developed by Allen-Bradley, is a low-level device oriented network based on the CAN (controller area network) developed by Bosch (GmbH) for the automobile industry. It is designed to interconnect lower level devices (sensors and actuators) with higher level devices (controllers).

The variable, multi-byte format of the CAN message frame is well suited to this task as more information can be communicated per message than with bit type systems. The Open DeviceNet Vendor Association, Inc. (ODVA) has been formed to issue DeviceNet specifications, ensure compliance with the specifications and offer technical assistance for manufacturers wishing to implement DeviceNet. The DeviceNet specification is an open specification and available through the ODVA.

DeviceNet can support up to 64 nodes, which can be removed individually under power and without severing the trunk line. A single, four-conductor cable (round or flat) provides both power and data communications. It supports a bus (trunk line drop line) topology, with branching allowed on the drops. Reverse wiring protection is built into all nodes, protecting them against damage in the case of inadvertent wiring errors.

The data rates supported are 125, 250 and 500K baud, although a specific installation does not have to support all as data rates can be traded for distance.

As DeviceNet was designed to interface lower level devices with higher level controllers, a unique adaptation of the basic CAN protocol was developed. This is similar to the familiar poll/response or master/slave technique but still utilizes the speed benefits of the original CAN.

Figure 6.1 below illustrates the positioning of DeviceNet and CANBUS within in the OSI model. Note that DeviceNet only implements layers 1,2 and 7 of the OSI model. Layers 1 and 2 provide the basic networking infrastructure, whilst layer 7 provides an interface for the application software. Due to the absence of layers 3 and 4, no routing and end-to-end control is possible.

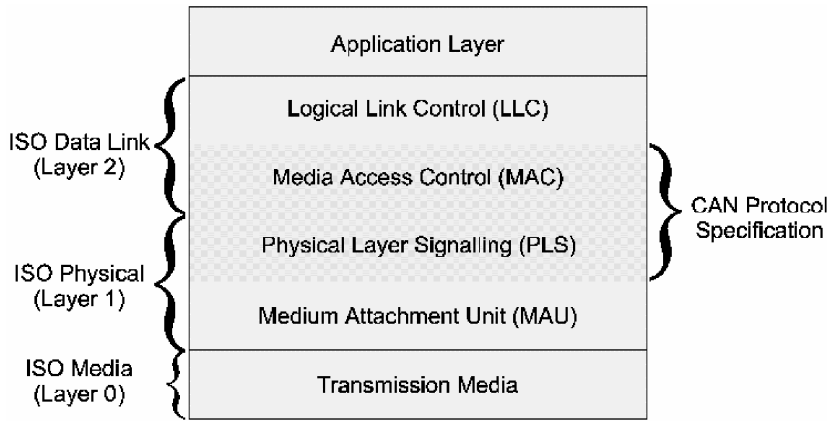


Figure 6.1
DeviceNet vs. the OSI model

6.2 Physical layer

6.2.1 Topology

The DeviceNet media consists of a physical bus topology. The bus or 'trunk' (white and blue wires) is the backbone of the network and must be terminated at either end by a 120 ohm 1/4W resistor.

Drop lines of up to 6 meters (20 feet) in length enable the connection of nodes (devices) to the main trunk line, but care is to be taken not to exceed the total drop line budget for a specific speed. Branching to multiple nodes is allowed only on drop lines.

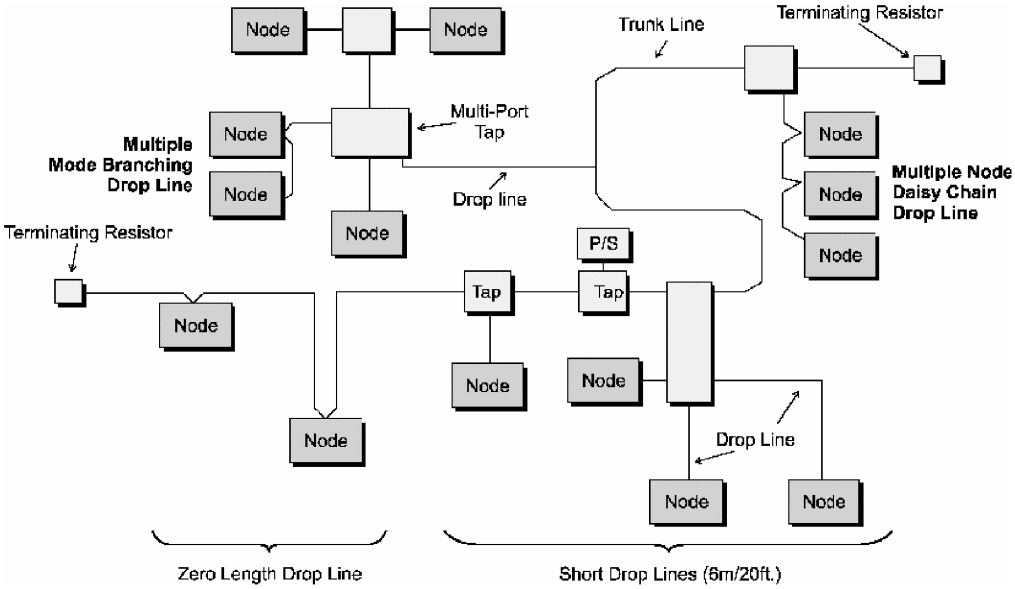


Figure 6.2
DeviceNet topology

Three types of cable are available, all of which can be used as the trunk. They are thick, thin and flat wire.

6.3 Connectors

DeviceNet has adopted a range of open and closed connectors that are considered suitable for connecting equipment onto the bus and drop lines. This range of recommended connectors is described in this section.

DeviceNet users can connect to the system using other proprietary connectors, the only restrictions placed on the user regarding the types of connectors used are as follows:

- All nodes (devices), whether using sealed or unsealed connections, supplying or consuming power, must have male connectors
- Whatever connector is chosen, it must be possible for the related device to connected or disconnected from the DeviceNet bus without compromising the system's operation
- Connectors must be rated to carry high levels (8 amps or more at 24 volts, or 200 VA) of current
- A minimum of 5 isolated connector pins are required, with the possible requirement of a 6th, or metal body shield connection for safety ground use

There are two basic styles of DeviceNet connectors that are used for bus and drop line connections in normal, harsh, and hazardous conditions. These are:

- An open style connector (pluggable or hard wired)
- A closed style connector (mini or micro style)

6.3.1 Pluggable (unsealed) connector

This is a 5 pin, unsealed open connector utilizing direct soldering, crimping, screw terminals, barrier strips or screw type terminations. This type of connector entails removing system power for connection.

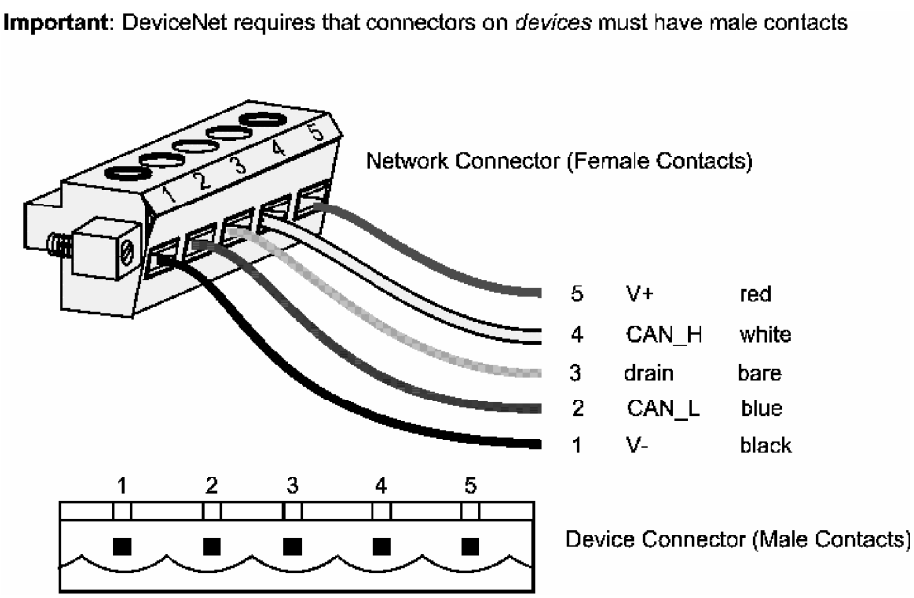


Figure 6.3
Unsealed screw connector

6.3.2 Hard wired (unsealed) connection

Loose wire connections can be used to make direct attachment to a node or a bus tap without the presence of a connector, although this is not a preferred method. It is only a viable option if the node can be removed from the trunk line without severing the trunk.

The ends of the cable are 'live' if the cable has been removed from the node in question and are still connected as part of the bus infrastructure. As such, care MUST be exercised to insulate the exposed ends of the cable.

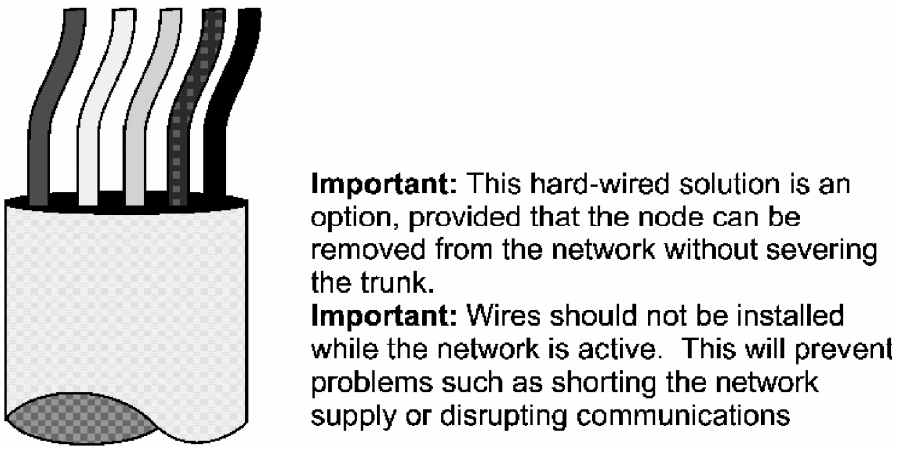


Figure 6.4
Open wire connection

6.3.3 Mini (sealed) connector

This 18mm round connector is recommended for harsh environments (field connections). This connection must meet ANSI/B93.55M-1981. The female connector (attached to the bus cable) must have rotational locking. This connector requires a minimum voltage

rating of 25 volts, and for trunk use a current rating of 8 Amps is required. Additional options can include oil and water resistance.

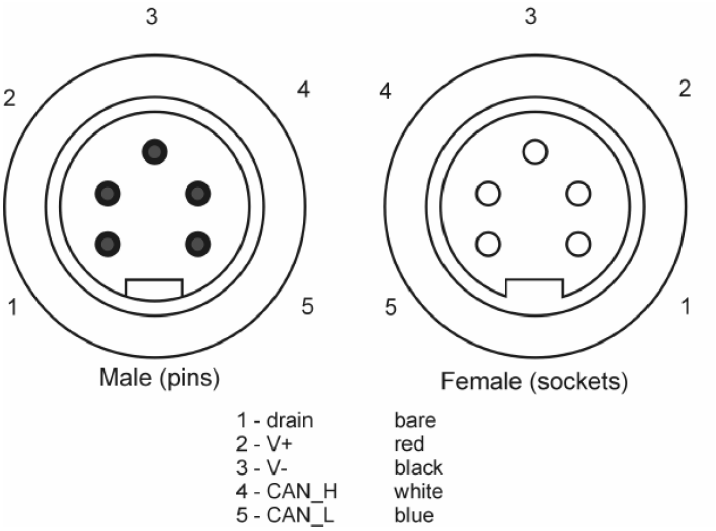


Figure 6.5
Sealed mini-type connector (face views)

6.3.4 Micro (sealed) connector

This connector is effectively a 12mm diameter miniature version of the mini style connector, except its suitability is for thin wire drop connections requiring reduction in both physical and current carrying capacity.

It has 5 pins, 4 in a circular periphery pattern and the fifth pin in the center. This connector should have a minimum voltage rating of 25 volts, and for drop connections a current rating of 3 amps is required. The male component must mate with Lumberg Style RST5-56/xm or equivalent, the female component part must also conform to Lumberg Style RST5-56/xm or equivalent. Additional options can include oil and water resistance.

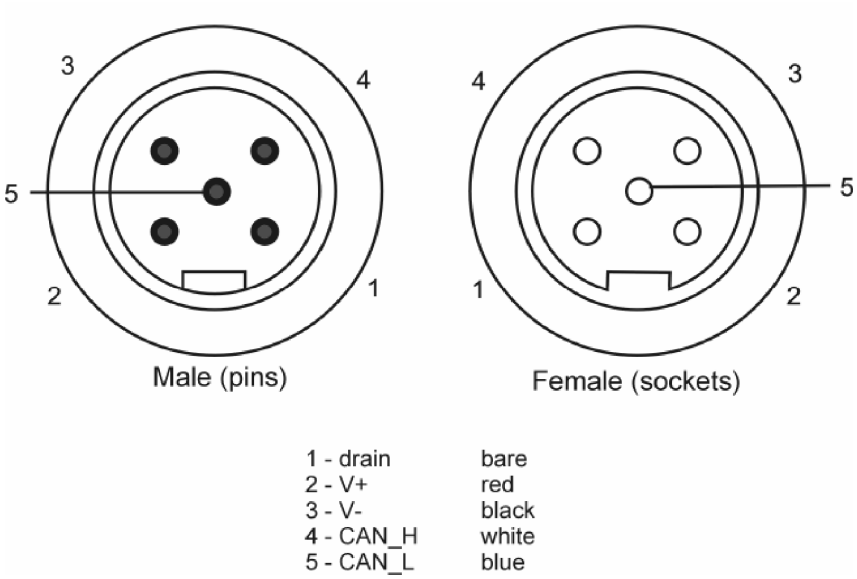


Figure 6.6
Sealed micro-style connector (face views)

6.4 Cable budgets

DeviceNet's transmission media can be constructed of either DeviceNet thick, thin or flat cable or a combination thereof. Thick or flat cable is used for long distances and is stronger and more resilient than the thin cable, which is mainly used as a local drop line connecting nodes to the main trunk line.

The trunk line supports only tap or multi-port taps that connect drop lines into the associated node. Branching structures are allowed only on drop lines and not on the main trunk line.

The following tables show the distance vs. length trade-off for the different types of cable.

DATA RATES	125 k baud	250 k baud	500 k baud
Trunk distance	500 m (1640 ft)	250 m (630 ft)	100 m (328 ft)
Max. drop length	20 ft	20 ft	20 ft
Cumulative drop	512 ft	256 ft	128 ft
Number of nodes	64	64	64

Table 6.1
Constraints: Thick wire

DATA RATES	125 k baud	250 k baud	500 k baud
Trunk distance	100 m (326 ft)	100 m (326 ft)	100 m (328 ft)
Max. drop length	20 ft	20 ft	20 ft
Cumulative drop	512 ft	256 ft	128 ft
Number of nodes	64	64	64

Table 6.2
Constraints: Thin wire

DATA RATES	125 k baud	250 k baud	500 k baud
Trunk distance	420 m (1640 ft)	200 m (630 ft)	75 m (328 ft)
Max. drop length	20 ft	20 ft	20 ft
Cumulative drop	512 ft	256 ft	128 ft
Number of nodes	64	64	64

Table 6.3
Constraints: Flat wire

6.5 Device taps

6.5.1 Sealed taps

Sealed taps are available in single port (T type) and multi-port configurations. Regardless of whether the connectors are mini or micro style, DeviceNet requires that male connectors must have external threads while female connectors must have internal threads. In either case, the direction of rotation is optional.

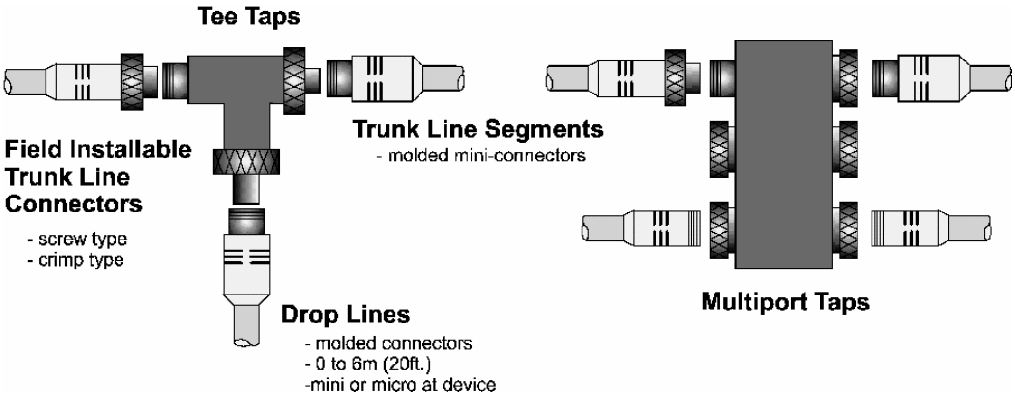


Figure 6.7
Sealed taps

6.5.2 IDC taps

IDCs (insulation displacement connectors) are used for KwikLink flat cable. They are modular, relatively inexpensive and compact. They are compatible with existing media and require little installation effort. The enclosure conforms to NEMA 6P and 13, and IP 67.

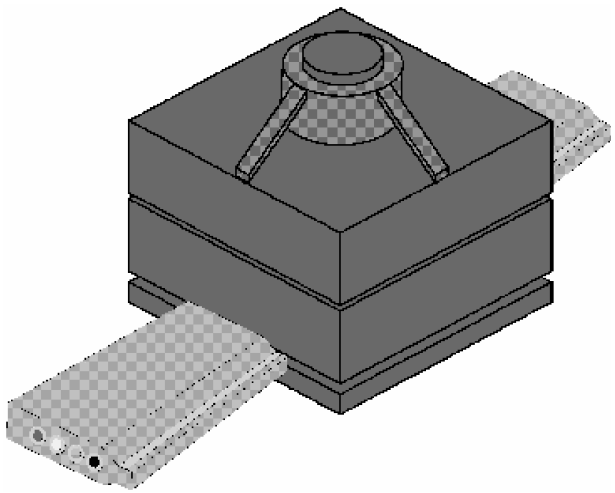


Figure 6.8
Insulation displacement connector

6.5.3 Open style taps

DeviceNet has three basic forms of open taps. They are:

- Zero length drop line, suitable for daisy-chain applications
- Open tap, able to connect a 6 meter (20 foot) drop line onto the trunk
- An open style connector, supporting ‘temporary’ attachment of a node to a drop line

The temporary connector is suitable for connection both to and from the system when the system is powered. It is of similar construction to a standard telephone wall plug, being of molded construction and equipped with finger grips to assist removal, and is

styled as a male pin-out. The side cheeks are polarized to prevent reversed insertion into the drop line open tap connector.

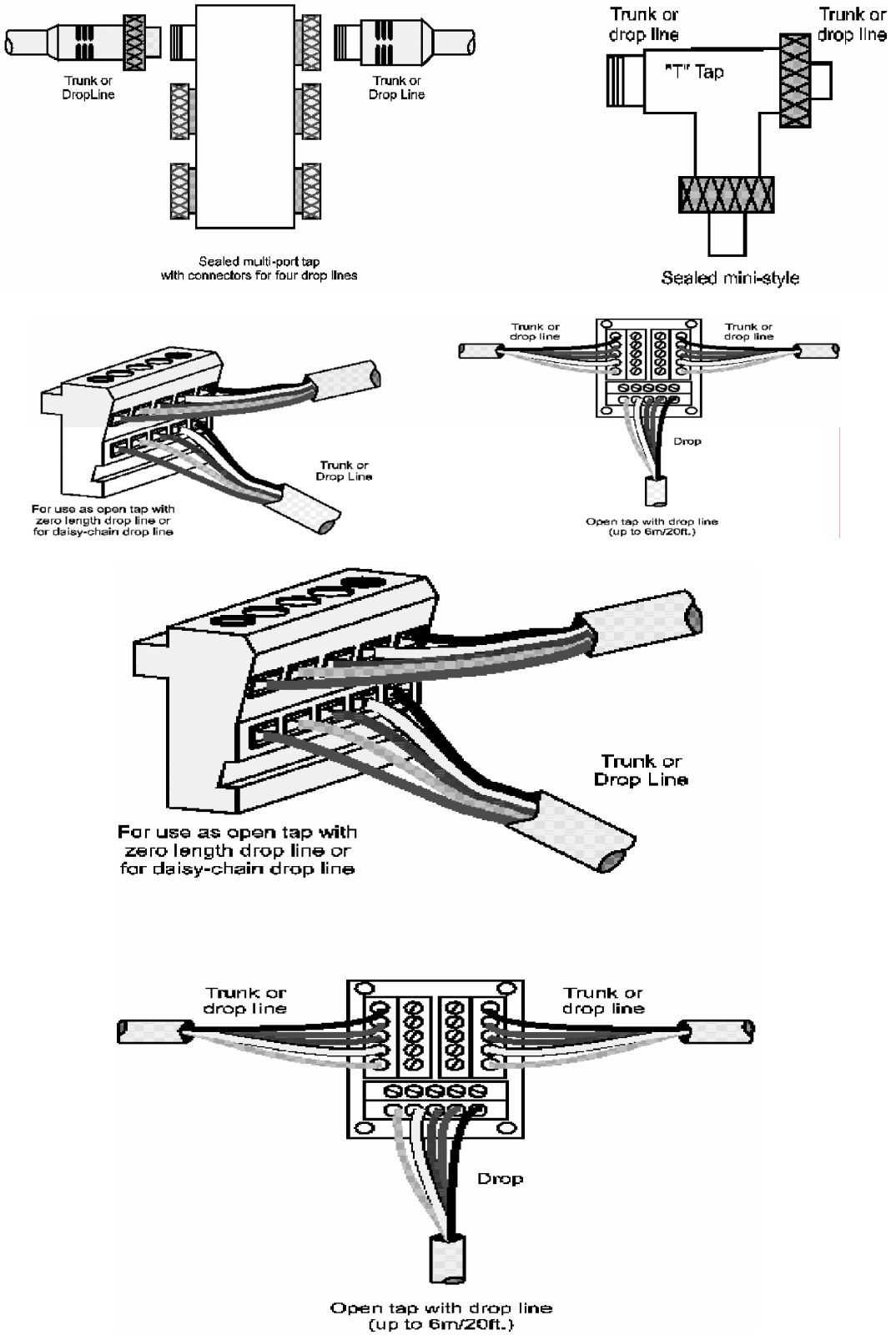


Figure 6.9
Open and temporary DeviceNet taps

6.5.4 Multiport open taps

If a number of nodes or devices are located within a close proximity of each other, e.g. within a control cabinet or similar enclosure, an open tap can be used.

Alternatively, devices can be wired into a DeviceBox multiport tap. The drops from the individual devices are not attached to the box via sealed connectors but are fed in via cable grips and connected to a terminal strip.

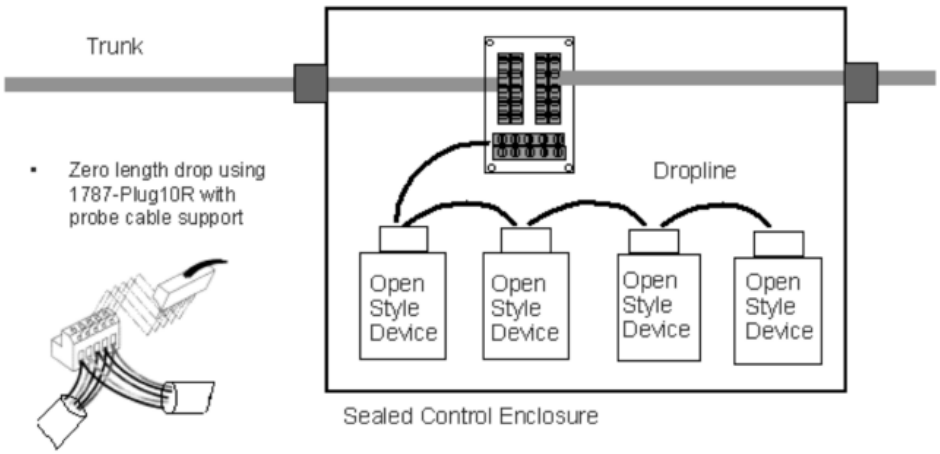


Figure 6.10
Multiport open taps

6.5.5 Power taps

Power taps differ from device taps in that they have to perform four essential functions that are not specifically required by the device taps. These include:

- Two protection devices in the V+ supply
- Connection from the positive output of the supply to the V+ bus line via a Schottky diode
- Provision of a continuous connection for the signaling pair, drain and negative wires through the tap
- Provision of current limiting in both directions from the tap

The following figure illustrates the criteria listed above.

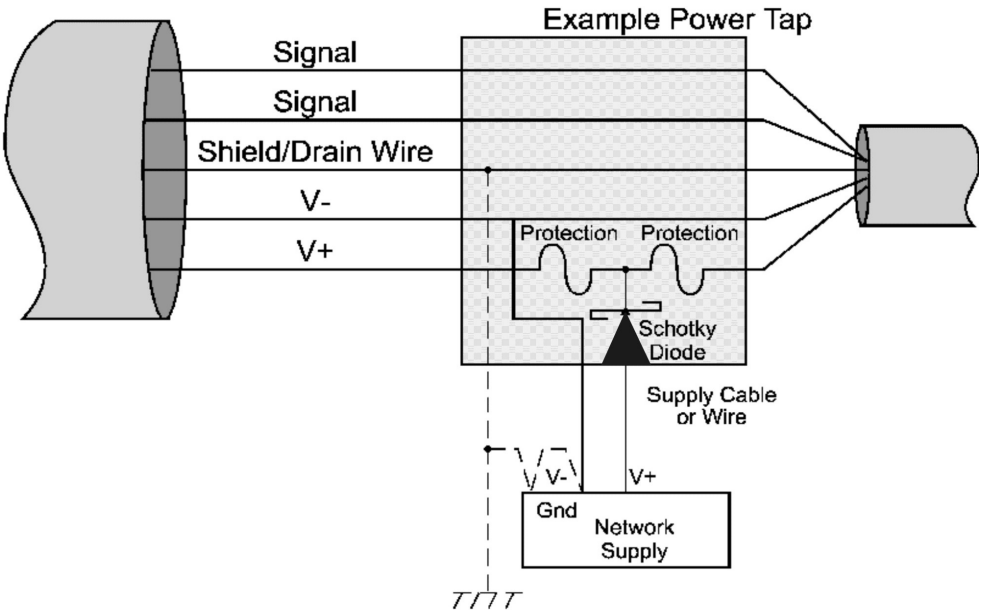


Figure 6.11
Principle of a DeviceNet power tap

6.6 Cable description

The 'original' (round) DeviceNet cable has two shielded twisted pairs. These are twisted on a common axis with a drain wire in the center, and are equipped with an overall braid.

6.6.1 Thick cable

This cable is used as the trunk line when length is important. Overall diameter is 0.480 inches (10.8mm) and it comprises of:

- A signal pair, consisting of one twisted pair (3 twists per foot) coded blue/white with a wire size of #18 (19 x 30 AWG) copper and individually tinned; the impedance is 120 ohms ± 10% at 1 MHz, the capacitance between conductors is 12 pF /foot and the propagation delay is 1.36 nS /foot maximum
- A power pair, consisting of one twisted pair (3 twists per foot) coded black/red with a wire size of #15 (19 x 28 AWG) copper and individually tinned

This is completed by separate aluminized Mylar shields around each pair and an overall foil/braided shield with an #18 (19 × 30 AWG) bare drain wire. The power pair has an 8 amp power capacity and is PVC/nylon insulated. It is also flame resistant and UL oil resistant II.

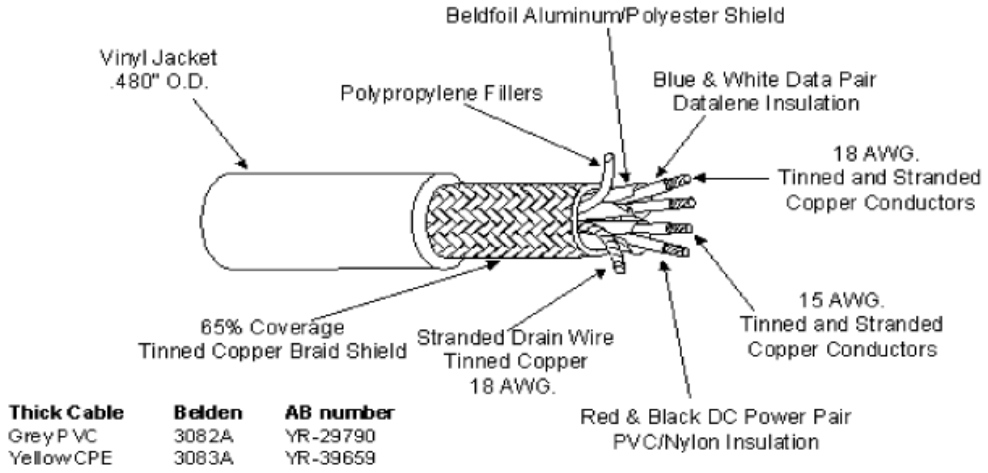


Figure 6.12
DeviceNet thick cable

6.6.2 Thin cable specification

This cable is used for both drop lines as well as short trunk lines. Its overall diameter is 0.27 inches (6.13mm) and it comprises of:

- A signal pair, consisting of a twisted pair (4.8 twists per foot) coded blue/white with a wire size of #24 (19 x 36 AWG) copper and individually tinned; the impedance is 120 ohms ± 10% at 1 MHz, the capacitance between conductors is 12 pF /Foot and the propagation delay is 1.36 nS /foot maximum
- A power pair consisting of one twisted pair (4.8 twists per foot) coded black/red with a wire size of #22 (19 x 34 AWG) copper and individually tinned

This is completed by separate aluminized Mylar shields around each pair and an overall foil/braided shield with an #22 (19 x 34 AWG) bare drain wire. The power pair has a 3 amp power capacity and is PVC insulated.

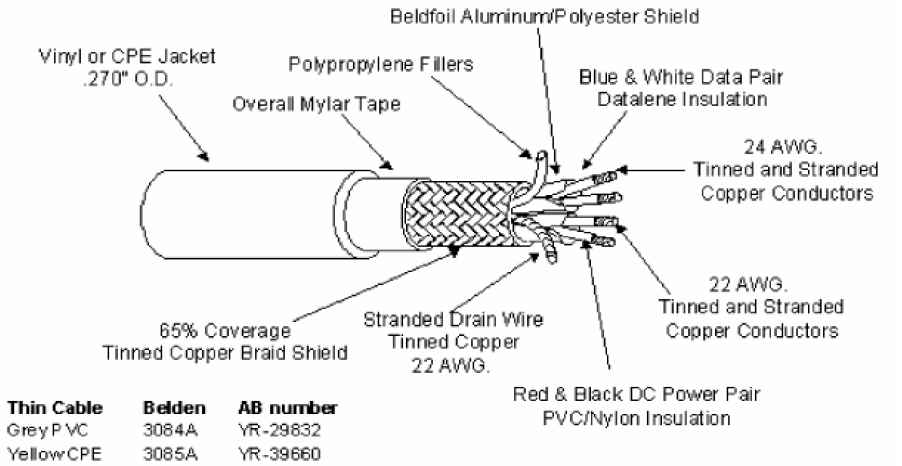


Figure 6.13
DeviceNet thin cable

6.6.3 Flat cable

DeviceNet flat cable is a highly flexible cable that works with existing devices. It has the following specifications:

- 600V 8A rating
- A physical key
- Fitting into 1 inch (25 mm) conduit
- The jacket made of TPE/ Santoprene

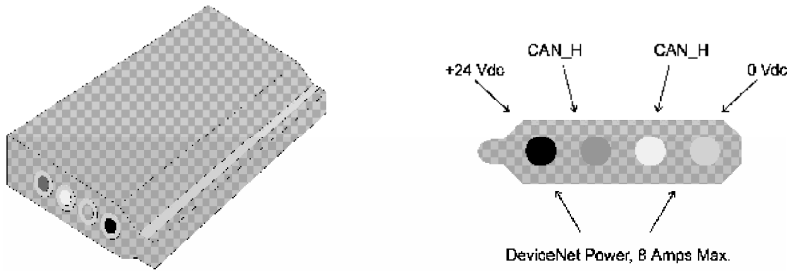


Figure 6.14
DeviceNet thin cable

6.7 Network power

6.7.1 General approach

One or more 24V power supplies can be used to power the devices on the DeviceNet network, provided that the 8A current limit on thick/flat wire and the 3A limit on thin wire is not exceeded. The power supplies used should be dedicated to the DeviceNet cable power ONLY!

Although, technically speaking, any suitable power supply can be used, supplies such as the Rockwell automation 1787-DNPS 5.25A supply are certified specifically for DeviceNet.

The power calculations can be done by hand, but it is easier to use a design spreadsheet such as the Rockwell automation/Allen Bradley DeviceNet Power Supply Configuration toolkit running under Microsoft Excel.

The network can be constructed of both thick and thin cable as long as only one type of cable is used per section of network, comprising a section between power taps or between a power tap and the end of the network.

Using the steps illustrated below, a quick initial evaluation can be achieved as to the power requirements needed for a particular network.

Sum the total current requirements of all network devices, then evaluate the total permissible network length (be conservative here) using the following table:

Thick cable network current distribution and allowable current loading												
Network length in meters	0	25	50	100	150	200	250	300	350	400	450	500
Network length in feet	0	83.3	167	333	500	666	833	999	1166	1332	1499	1665
Maximum current in Amps	8	8	5.42	2.93	2.01	1.53	1.23	1.03	0.89	0.78	0.69	0.63

Thin cable network current distribution and allowable current loading											
Network length in meters	0	10	20	30	40	50	60	70	80	90	100
Network length in feet	0	33	66	99	132	165	198	231	264	297	330
Maximum current in Amps	3.0	3.0	3.0	2.06	1.57	1.26	1.06	0.91	0.80	0.71	0.64

Table 6.4
Thick and thin cable length and power capacity

Depending on the final power requirement cost and network complexity, either a single supply end or center connected can be used.

6.7.2 Single supply – end connected

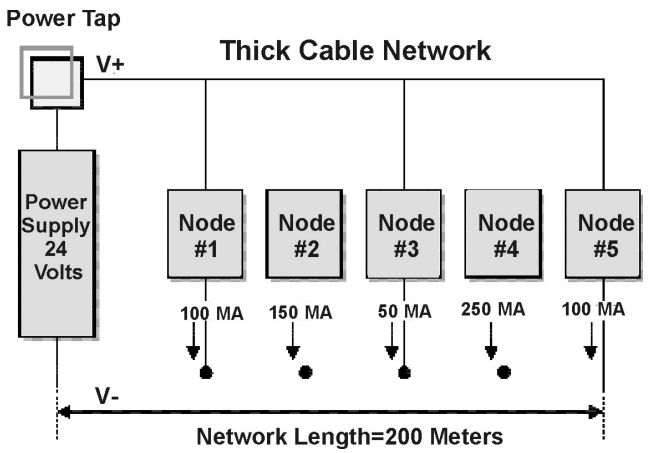


Figure 6.15
Single supply – end connected

Total network length = 200 meters (656 feet)
Total current = Sum of node 1, 2, 3, 4 and 5 currents = 0.65 Amps
Referring to table 14.1 the current limit for 200 meters = 1.53 Amps.
Configurations are correct as long as THICK cable is used.

6.7.3 Single supply – center connected

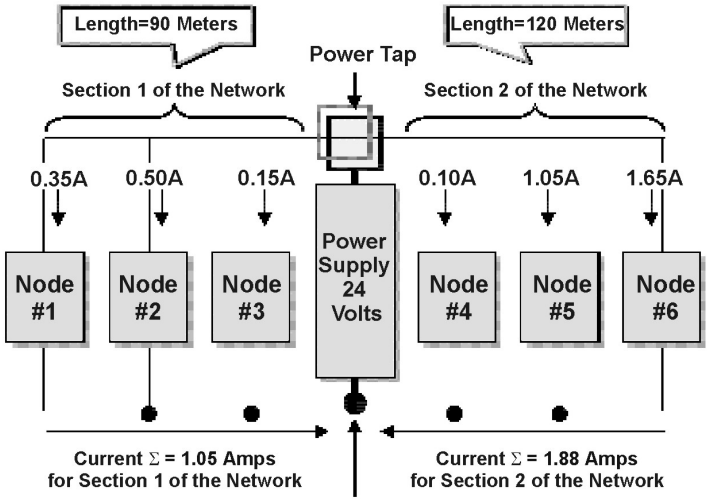


Figure 6.16
Single supply – center connected

Current in section 1 = 1.05 amps over a length of 90 meters (300 feet)
Current in section 2 = 1.88 amps over a length of 120 meters (400 feet)
Current limits for a distance of 90 meters is 3.3 Amps and for 120 meters is 2.63 Amps
Power for both sections is correct, and a 3 Amp (minimum) power supply is required.
The following table indicates parameters that control load limits and allowable tolerances as related to DeviceNet power.

System power load limits	
Max. voltage drop on both the – Ve and +Ve power lines	5.0 volts on each line
Maximum thick cable trunk line current	8.0 Amps in any section
Maximum thin cable trunk line current	3.0 Amps in any section
Maximum drop line current	0.75 to 3.0 Amps
Voltage range at each node	11.0 to 25.0 Volts
Operating current on each product	Specified by the product manufacturer

Table 6.5
System power load limits

Maximum drop line currents	
Current limits are calculated by the following equations, where I = allowable drop line current and L = distance. In Meters: I = 4.57/L In Feet I = 15/L	
Drop length	**Maximum allowable current**
1.00	3.00
3.00	3.00
5.00	3.00
7.50	2.00
10.00	1.50
15.00	1.00
20.00	0.75

Table 6.6
Maximum drop line currents

6.7.4 Suggestions for avoiding errors and power supply options

The following steps can be used to minimize errors when configuring power on a network.

- Ensure the calculations made for current and distances are correct (be conservative)
- Conduct a network survey to verify correct voltages, remembering that a minimum of 11 volts at a node is required and that a maximum voltage drop of 10 volts across each node is allowed
- Allow for a good margin to have reserves of power to correct problems if needed
- If using multiple supplies, it is essential that they be all turned on simultaneously to prevent both power supply and cable overloading occurring
- Power supplies MUST be capable of supporting linear and switching regulators
- Supply MUST be isolated from both the AC supply and chassis

6.8 System grounding

Grounding of the system must be done at one point only, preferably as close to the physical center of the network as possible. This connection should be done at a power tap where a terminal exists for this purpose. A main ground connection should be made from this point to a good earth or building ground via a copper braid or at least a 8 AWG copper conductor not more than 3 meters (10 feet) in length.

At this point of connection, the following conductors and circuits should connected together in the form of a ‘star’ configuration.

- The drain wire of the main trunk cable

- The shield of the main trunk cable
- The negative power conductor
- The main ground connection as described above

If the network is already connected to ground at some other point, do NOT connect the ground terminal of a power tap to a second ground connection. This can result in unwanted ground loop currents occurring in the system. It is essential that a single ground connection is established for the network bus and failure to ground the bus –ve supply at ONE POINT only will result in a low signal to noise ratio appearing in the system.

Care must be exercised when connecting the drain/shield of the bus or drop line cable at nodes, which are already grounded. This can happen when the case or enclosure of the equipment, comprising the node, is connected to ground for electrical safety and/or a signaling connection to other self-powered equipment. In cases where this condition exists, the drain/shield should be connected to the node ground through a 0.01uF/ 500-volt capacitor wired in parallel with a 1 megohm 1/4 watt resistor. If the node has no facility for grounding, the drain and shield must be left UNCONNECTED.

6.9 Signaling

DeviceNet is a two wire differential network. Communication is achieved by switching the CAN–H wire (white) and the CAN–L wire (blue) relative to the V– wire (black). CAN–H swings between 2.5VDC (recessive state) and 4.0VDC (dominant state) while CAN–L swings between 2.5VDC (recessive state) and 1.5VDC (dominant state).

With no network master connected, the CAN–H and CAN–L lines should be in the recessive state and should read (with a voltmeter set to DC mode) between 2.5V and 3.0V relative to V– at the point where the power supply is connected to the network. With a network master connected AND polling the network, the CAN–H to V– voltage will be around 3.2VDC and the CAN–L to V– voltage will be around 2.4 VDC. This is because the signals are switching, which affects the DC value read by the meter.

The voltage values given here assume that no common mode voltages are present. Should they be present, voltages measured closer to the power supply will be higher than those measured furthest from the power supply. However, the differential voltages (CAN–H minus CAN–L) will not be affected.

DeviceNet uses a differential signaling system. A logical ‘1’ is represented by CAN–H being Low (recessive) and CAN–L being High (recessive). Conversely, a logical ‘0’ is represented by CAN–H being High (dominant) and CAN–L being Low (dominant). Figure 14.22 depicts this graphically.

The nodes are all attached to the bus in parallel, resulting in a wired–AND configuration. This means that as long as ANY one node imposes a Low signal (logical 0) on the bus, the resulting signal on the bus will also be low. Only when ALL nodes output a high signal (logical 1), will the signal on the bus be high as well.

6.10 Data link layer

6.10.1 Frame format

The format of a DeviceNet frame is shown here. Note that the data field is rather small (8 bytes) and that any messages larger than this need to be fragmented.

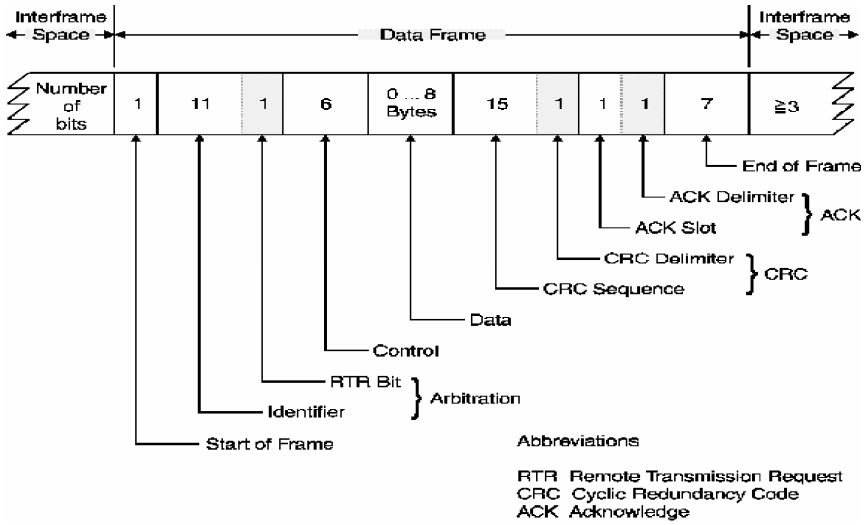


Figure 6.17
DeviceNet frame

This frame will be placed on the bus as sequential 1s and 0s, by changing the levels of the CAN–H and CAN–L in a differential fashion.

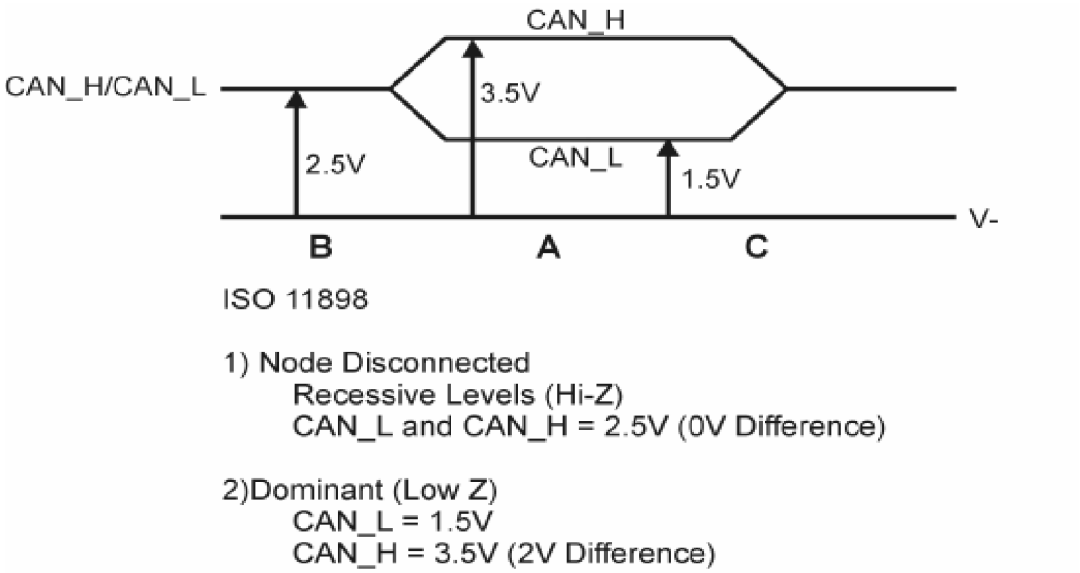


Figure 6.18
DeviceNet transmission

In this figure, A represents the CAN–H signal in its dominant (high) state (+3.5VDC to +4.0VDC), C represents the CAN–L signal in its dominant (low) state (+1.5VDC to 2.5VDC) and B represents both the CAN–H signal Recessive (low) and the CAN–L signal recessive (high) states of +2.5V–3.0VDC.

6.10.2 Medium access

The medium access control method could be described as ‘carrier sense multiple access with bit-wise arbitration,’ where the arbitration takes place on a bit-by-bit basis on the first field in the frame (the 11 bit identifier field). If a node wishes to transmit, it has to

defer to any existing transmission. Once that transmission has ended, the node wishing to transmit has to wait for 3 bit times before transmitting. This is called the interframe space.

Despite this precaution, it is possible for two nodes to start transmitting concurrently. In the following example nodes 1 and 2 start transmitting concurrently, with both nodes monitoring their own transmissions. All goes well for the first few bits since the bit sequences are the same. Then the situation arises where the bits are different. Since the '0' state is dominant, the output of node 2 overrides that of node 1. Node 1 loses the arbitration and stops transmitting. It does, however, still ACK the message by means of the ACK field in the frame.

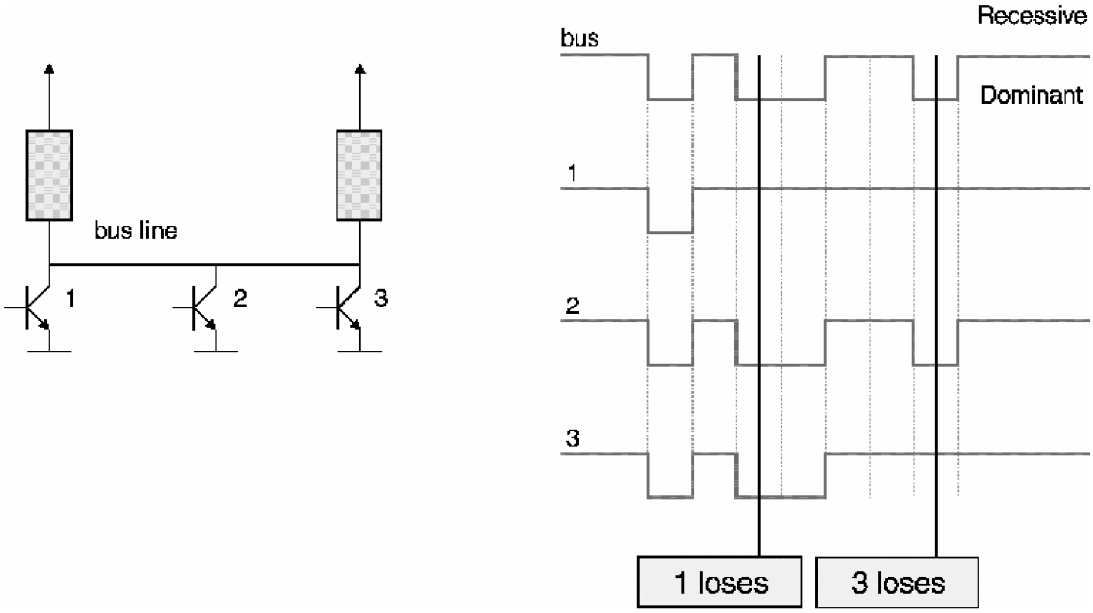


Figure 6.19
DeviceNet arbitration

Because of this method of arbitration, the node with the lowest number (i.e. the most significant '0' s in its identifier field) will win the arbitration.

6.10.3 Fragmentation

Any device that needs more than 8 bytes of data sent in any direction will cause fragmentation to occur. This happens since a frame can only contain 8 bytes of data. When fragmentation occurs, only 7 bytes of data can be sent at a time since the first byte is used to facilitate the reassembly of fragments. It is used as follows:

First byte	**Significance**
00	First fragment (number 0)
41–7F	Intermediate fragment (lower 6 bits of the byte is the fragment number)
80–FF	Last fragment (lower 6 bits of the byte is the fragment number)

Example:

Data packet	**Description**
00 12 34 56 78 90 12 34	First fragment, number 0
41 56 78 90 12 34 56 78	Intermediate fragment number 1
42 90 12 34 56 78 90 12	Intermediate fragment number 2
83 34 56	Last fragment number 3

6.11 The application layer

The CAN specification does not dictate how information within the CAN message frame fields are to be interpreted – that was left up to the developers of the DeviceNet application software.

Through the use of special identifier codes (bit patterns) in the identifier field, master is differentiated from slave. Also, sections of this field tell the slaves how to respond to the master's message. For example, slaves can be requested to respond with information simultaneously in which case the CAN bus arbitration scheme assures the timeliest consecutive response from all slaves in decreasing order of priority. Or, slaves can be polled individually, all through the selection of different identifier field codes. This technique allows the system implementers more flexibility when establishing node priorities and device addresses.

6.12 Troubleshooting

6.12.1 Introduction

Networks, in general, exhibit the following types of problems from time to time.

The first type of problem is of an electronic nature, where a specific node (e.g. a network interface card) malfunctions. This can be due to a component failure or to an incorrect configuration of the device.

The second type is related to the medium that interconnects the nodes. Here, the problems are more often of an electromechanical nature and include open and short circuits, electrical noise, signal distortion and attenuation. Open and short circuits in the signal path are caused by faulty connectors or cables. Electrical interference (noise) is caused by incorrect grounding, broken shields or external sources of electro-magnetic or radio frequency interference. Signal distortion and attenuation can be caused by incorrect termination, failure to adhere to topology guidelines (e.g. drop cables too long), or faulty connectors.

Whereas these are general network-related problems, the following ones are very specific to DeviceNet:

- Missing terminators
- Excessive common mode voltage, caused by faulty connectors or excessive cable length
- Low power supply voltage caused by faulty connectors or excessive cable length
- Excessive signal propagation delays caused by excessive cable length

These problems will be discussed in more detail.

6.13 Tools of the trade

The following list is by no means complete, but is intended to give an overview of the types of tools available for commissioning and troubleshooting DeviceNet networks. Whereas some tools are sophisticated and expensive, many DeviceNet problems can be sorted out with common sense and a multimeter.

6.13.1 Multimeter

A multimeter capable of measuring DC volts, resistance, and current is an indispensable troubleshooting tool. On the current scale, it should be capable of measuring several amperes.

6.13.2 Oscilloscope

An inexpensive 20 MHz dual-trace oscilloscope comes in quite handy. It can be used for all the voltage tests as well as observing noise on the lines, but caution should be exercised when interpreting traces.

Firstly, signal lines must be observed in differential mode (with probes connected to CAN_H and CAN_L. If they are observed one at a time with reference to ground, they may seem unacceptable due to the common mode noise (which is not a problem since it is rejected by the differential mode receivers on the nodes).

6.13.3 Handheld analyzers

Handheld DeviceNet analyzers such as the NetAlert NetMeter or DeviceNet Detective can be used for several purposes. Depending on the capabilities of the specific device, they can configure node addresses and baud rates, monitor power and signal levels, log errors and network events of periods ranging from a few minutes to several days, indicate short circuits and poorly wired connections, and obtain configuration states as well as firmware versions and serial numbers from devices.

Figure 6.20
NetAlert NetMeter

6.13.4 Intelligent wiring components

Examples of these are the NetAlert traffic monitor and NetAlert power monitor. These are ‘intelligent’ tee-pieces that are wired into the system. The first device monitors and displays network traffic by means of LEDs and gives a visual warning if traffic levels exceed 90%. The second device monitors voltages and visually indicates whether they are OK, too high, too low, or totally out of range.

The NetMeter can be attached to the above-mentioned tees for more detailed diagnostics.

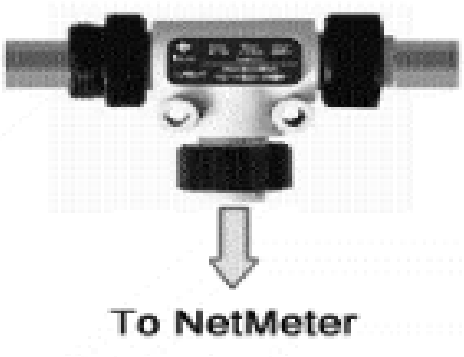


Figure 6.21
Power Monitor tee

6.13.5 Controller software

Many DeviceNet controllers have associated software, running under various operating systems such as Windows 2000 and NT4 that can display sophisticated views of the network for diagnostic purposes. The example given here is one of many generated by the ApplicomIO software and displays data obtained from a device, down to bit level (in hexadecimal).

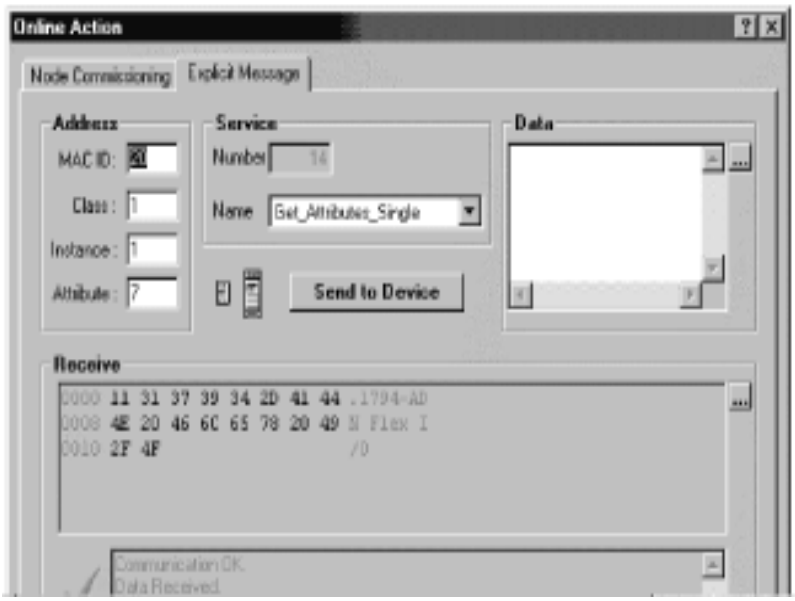


Figure 6.22
ApplicomIO display

6.13.6 CAN bus monitor

Since DeviceNet is based on the controller area network (CAN), CAN protocol analysis software can be used on DeviceNet networks to capture and analyze packets (frames). An example is Synergetic's CAN Explorer for Windows 95/98 and NT, running on a PC. The same vendor also supplies the ISA, PC/104, PCI and parallel port network interfaces for connecting the PC to the DeviceNet network.

A PC with this software can not only function as a protocol analyzer, but also as a data logger.

6.14 Fault finding procedures

In general, the system should not be operating i.e. there should be no communication on the bus, and all devices should be installed.

A low-tech approach to troubleshooting could involve disconnecting parts of the network, and observing the effect on the problem. This does, unfortunately, not work well for problems such as excessive common mode voltage and ground loops since disconnecting part of the network often solves the problem.

6.14.1 Incorrect cable lengths

If the network exhibits problems during the commissioning phase or after modifications/additions have been made to the network, the cable lengths should be double-checked against the DeviceNet topology restrictions. The maximum cable lengths are as follows:

	125 Kbaud	**250 Kbaud**	**500 Kbaud**
Thick trunk length	500m	250m	100m
Thin trunk length	100m	100m	100m
Single drop	6m	6m	6m
Cumulative all drops	156m	78m	39m

For simplicity, only the metric sizes are given here.

They following symptoms are indicative of a topology problem.

- If drop lines too long, i.e. the total amount of drops exceeds the permitted length, variations on CAN signal amplitude will occur throughout the network
- If a trunk line is too long it will cause ‘transmission line’ effects in which reflections in the network cause faulty reception of messages; this will result in CAN frame errors

6.14.2 Power and earthing problems

Shielding

Shielding is checked in the following way.

Connect a 16A DC ammeter from DC common to shield at the end of the network furthest from the power supply. If the power supply is in the middle, then this test must be performed at both ends. In either case, there should be significant current flow. If practical, this test can also be performed at the end of each drop.

If there is no current flowing, then the shield is broken or the network is improperly grounded.

Grounding

In general, the following rules should be observed:

- Physically connect the DC power supply ground wire and the shield together to earth ground at the location of the power supply
- In the case of multiple power supplies, connect this ground only at the power supply closest to the middle of the network
- Ensure that all nodes on the network connect to the shield, the signal and power lines

Note: CAN frame errors are a symptom of grounding problems. CAN error messages can be monitored with a handheld DeviceNet analyzer or CAN bus analyzer.

Break the shield at a few points in the network, and insert a DC ammeter in the shield.

If there is a current flow, then the shield is connected to DC common or ground in more than one place, and a ground loop exists.

Power

This can be measured with a voltmeter or handheld DeviceNet analyzer and is measured between V+ (red) and V– (black).

Measure the network voltage at various points across the network, especially at the ends and at each device. The measured voltage should ideally be 24V, but no more than 25V and not less than 11V DC.

If devices draw a lot of current, then voltages on the bus can fluctuate hence bus voltages should be monitored over time.

If the voltages are not within specification, then:

- Check for faulty or loose connectors
- Check the power system design calculations by measuring the current flow in each section of cable

On some DeviceNet analyzers, one can set a supply alarm voltage below which a warning should be generated. Plug the analyzer in at locations far from the power supply and leave it running over time. If the network voltage falls below this level at any time, this low voltage event will be logged by the analyzer.

Note that ‘THIN’ cable, which has a higher DC resistance, will have greater voltage drop across distance.

6.14.3 Incorrect signal voltage levels

The following signal levels should be observed with a voltmeter, oscilloscope or DeviceNet analyzer. Readings that differ by more than 0.5V with the following values are most likely indicative of a problem.

CAN_H can NEVER be lower than CAN_L and if this is observed, it means that the two wires have probably been transposed.

- If bus communications are OFF (idle) the following values can be observed with any measuring device.
    - CAN_H (white) 2.5 VDC
    - CAN_L (blue) 2.5 VDC
- If bus communications are ON, the following can be observed with a voltmeter:
    - CAN_H (white) 3.0 VDC
    - CAN_L (blue) 2.0 VDC

Alternatively, the voltages can be observed with an oscilloscope or DeviceNet analyzer, in which case both minimum and maximum values can be observed.

These are:

- CAN_H (white) 2.5V min, 4.0V max
- CAN_L (blue) 1.0V min, 2.5V max

6.14.4 Common mode voltage problems

This test assumes that the shield had already been checked for continuity and current flow and can be done with a voltmeter or an oscilloscope:

- Turn all network power supplies on
- Configure all nodes to draw the maximum amount of power from the network
- Turn on all outputs that draw current

Now measure the DC voltage between V– and the shield. The difference should, technically speaking, be less than 4.65 V. For a reasonable safety margin, this value should be kept below 3 V.

These measurements should be taken at the two furthest ends (terminator position), at the DeviceNet master(s) and at each power supply. Should a problem be observed here, a solution could be to relocate the power supply to the middle of the network or to add additional power supplies.

In general, one can design a network using any number of power supplies, providing that:

- The voltage drop in the cable between a power supply and each station it supplies does not exceed 5VDC
- The current does not exceed the cable/connector ratings
- The power supply common ground voltage level does not vary by more than 5V between any two points in the network.

6.14.5 Incorrect termination

These tests can be performed with a MultiMate. They must be done with all bus communications off (bus off) and the meter set to measure resistance.

Check the resistance from CAN_H to CAN_L at each device. If the values are larger than 60 ohms (120 Ohms in parallel with 120 Ohms) there could be a break in one of the signal wires or there could be a missing terminator or terminators somewhere. If, on the other hand, the measured values are less than 50 ohms, this could indicate a short between the network wires, (an) extra terminating resistor(s), one or more faulty transceivers or un-powered nodes.

6.14.6 Noise

Noise can be observed with a loudspeaker or with an oscilloscope. However, more important than the noise itself, is the way in which the noise affects the actual transmissions taking place on the medium. The most common effect of EMI/RFI problems are CAN frame errors, which can be monitored with a CAN analyzer or DeviceNet analyzer.

The occurrence of frame errors must be related to specific nodes and to the occurrence of specific events e.g. a state change on a nearby variable frequency drive.

6.14.7 Node communication problems

Node presence

One method to isolate defective nodes is to use the master configuration software or a DeviceNet analyzer to create a 'live list' to see which nodes are active on the network, and to compare this with the list of nodes that are supposed to be on the network.

Excessive traffic

The master configuration software or a DeviceNet analyzer can measure the percentage traffic on the network. A figure of 30–70% is normal and anything over 90% is excessive.

Loads over 90% indicate problems. High bus loads can indicate any of the following:

- Some nodes could be having difficulty making connections with other nodes and have to retransmit repeatedly to get messages through. Check termination, bus length, topology, physical connections and grounding
- Defective nodes can 'chatter' and put garbage on the network
- Nodes supplied with corrupt or noisy power may chatter
- Change Of State (COS) devices may be excessively busy with rapidly changing data and cause high percentage bus load
- Large quantities of explicit messages (configuration and diagnostic data) being sent can cause high percentage bus load
- Diagnostic instruments such as DeviceNet analyzers add traffic of their own; if this appears to be excessive, the settings on the device can be altered to reduce the additional traffic

MACID/baud rate settings

A network status LED is built into many devices. This LED should always be flashing GREEN. A solid RED indicates a communication fault, possibly an incorrect baud rate or a duplicate station address (MACID).

Network configuration software can be used to perform a 'network who' to verify that all stations are connected and communicating correctly.

In the absence of indicator LEDs, a DeviceNet analyzer will only be able to indicate that one or more devices have wrong baud rates. The devices will have to be found by inspection, and the baud rate settings corrected. First disconnect the device with the wrong baud rate, correct the setting, and then reconnect the device.

In the absence of an indicator LED, there is no explicit way of checking duplicate MACIDs either. If two nodes have the same address, one will just passively remain off-line. One solution is to look for nodes that should appear in the live list, but do not.

7

Profibus PA/DP/FMS overview

Objectives

When you have completed study of this chapter, you will be able to:

- List the main features of Profibus PA/DP/FMS
- Fix problems with:
  - Cabling
  - Fiber
  - Shielding
  - Grounding/earthing
  - Segmentation
  - Color coding
  - Addressing
  - Token bus operation
  - Unsolicited messages
  - Fine tuning of impedance terminations
  - Drop-line lengths
  - GSD files usage
  - Intrinsic safety concerns

7.1 Introduction

ProfiBus (PROcess FIeld BUS) is a widely accepted international networking standard, commonly found in process control and in large assembly and material handling machines. It supports single-cable wiring of multi-input sensor blocks, pneumatic valves, complex intelligent devices, smaller sub-networks (such as AS-i), and operator interfaces.

ProfiBus is nearly universal in Europe and also popular in North America, South America, and parts of Africa and Asia. It is an open, vendor independent standard. It adheres to the OSI model and ensures that devices from a variety of different vendors can communicate together easily and effectively. It has been standardized under the German

National standard as DIN 19 245 Parts 1 and 2 and, in addition, has also been ratified under the European national standard EN 50170 Volume 2.

The development of ProfiBus was initiated by the BMFT (German Federal Ministry of Research and Technology) in cooperation with several automation manufacturers in 1989. The bus interfacing hardware is implemented on ASIC (application specific integrated circuit) chips produced by multiple vendors, and is based on the RS-485 standard as well as the European EN50170 Electrical specification. The standard is supported by the ProfiBus Trade Organization, whose web site can be found at *www.profibus.com*.

ProfiBus uses 9-Pin D-type connectors (impedance terminated) or 12mm quick-disconnect connectors. The number of nodes is limited to 127. The distance supported is up to 24 Km (with repeaters and fiber optic transmission), with speeds varying from 9600 bps to 12 Mbps. The message size can be up to 244 bytes of data per node per message (12 bytes of overhead for a maximum message length of 256 bytes), while the medium access control mechanisms are polling and token passing.

ProfiBus supports two main types of devices, namely, masters and slaves:

- Master devices control the bus and when they have the right to access the bus, they may transfer messages without any remote request. These are referred to as active stations
- Slave devices are typically peripheral devices i.e. transmitters/sensors and actuators. They may only acknowledge received messages or, at the request of a master, transmit messages to that master. These are also referred to as passive stations

There are several versions of the standard, namely, ProfiBus DP (master/slave), ProfiBus FMS (multi-master/peer to peer), and ProfiBus PA (intrinsically safe).

- ProfiBus DP (distributed peripheral) allows the use of multiple master devices, in which case each slave device is assigned to one master. This means that multiple masters can read inputs from the device but only one master can write outputs to that device. ProfiBus-DP is designed for high speed data transfer at the sensor/actuator level (as opposed to ProfiBus-FMS which tends to focus on the higher automation levels) and is based around DIN 19 245 parts 1 and 2 since 1993. It is suitable as a replacement for the costly wiring of 24V and 4–20 mA measurement signals.
  The data exchange for ProfiBus-DP is generally cyclic in nature. The central controller, which acts as the master, reads the input data from the slave and sends the output data back to the slave. The bus cycle time is much shorter than the program cycle time of the controller (less than 10 mS)
- ProfiBus FMS (Fieldbus message specification) is a peer to peer messaging format, which allows masters to communicate with one another. Just as in ProfiBus DP, up to 126 nodes are available and all can be masters if desired. FMS messages consume more overhead than DP messages
- ‘COMBI mode’ is when FMS and DP are used simultaneously in the same network, and some devices (such as Synergetic's DP/FMS masters) support this. This is most commonly used in situations where a PLC is being used in conjunction with a PC, and the primary master communicates with the secondary master via FMS. DP messages are sent via the same network to I/O devices

- The ProfiBus PA protocol is the same as the latest ProfiBus DP with V1 diagnostic extensions, except that voltage and current levels are reduced to meet the requirements of intrinsic safety (class I division II) for the process industry. Many DP/FMS master cards support ProfiBus PA, but barriers are required to convert between DP and PA. PA devices are normally powered by the network at intrinsically safe voltage and current levels, utilizing the transmission technique specified in IEC 61158-2. (which Foundation Fieldbus H1 uses as well)

7.2 ProfiBus protocol stack

The architecture of the ProfiBus protocol stack is summarized in the figure below. Note the addition of an eighth layer, the so-called 'user' layer, on top of the 7-layer OSI model.

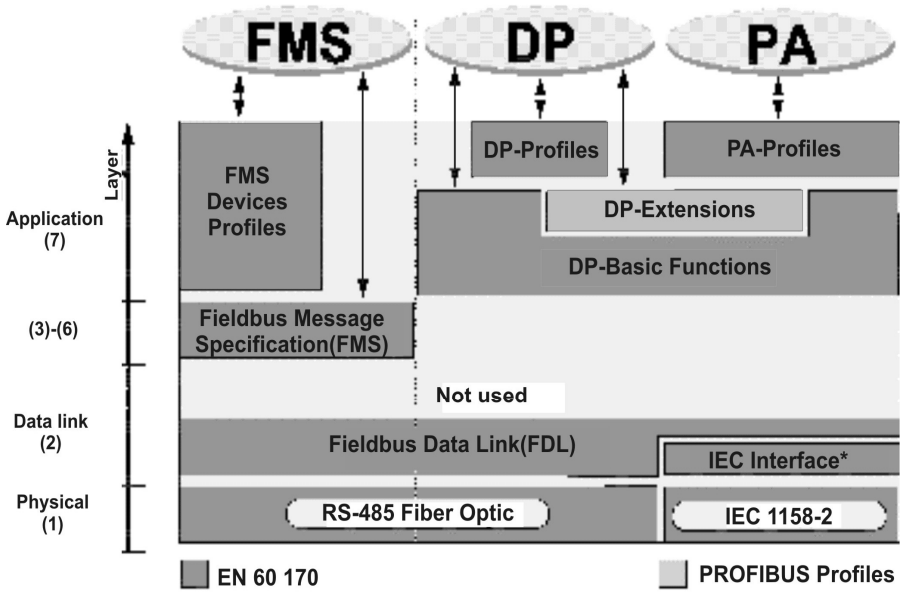


Figure 7.1
ProfiBus protocol stack

All three ProfiBus variations namely FMS, DP and PA use the same data link layer protocol (layer 2). The DP and PA versions us the same physical layer (layer 1) implementation, namely RS-485, while PA uses a variation thereof (as per IEC 61158-2) in order to accommodate intrinsic safety requirements.

7.2.1 Physical layer (layer 1)

The physical layer of the ProfiBus DP standard is based on RS-485 and has the following features:

- The network topology is a linear bus, terminated at both ends
- Stubs are possible
- The medium is a twisted pair cable, with shielding conditionally omitted depending on the application. Type A cable is preferred for transmission speeds greater than 500 kbaud. Type B should only be used for low baud rates and short distances. These are very specific cable types of which the details are given below

- The data rate can vary between 9.6 kbps and 12 Mbps, depending on the cable length. The values are:

9.6 kbps	1200m
19.2 kbps	1200m
93.75 kbps	1200m
187.5 kbps	600m
500 kbps	200m
1.5 Mbps	200m
12 Mbps	100m

The specifications for the two types of cable are as follows:

Type A cable

Impedance:	135 up to 165 Ohm (for frequency of 3 to 20 MHz)
Cable capacity:	<30 pF per meter
Core diameter:	>0.34 mm2 (AWG 22)
Cable type:	twisted pair cable. 1×2 or 2×2 or 1×4.
Resistance:	<110 Ohm per km
Signal attenuation:	max. 9dB over total length of line section
Shielding:	Cu shielding braid or shielding braid and shielding foil

Type B cable

Impedance:	135 up to 165 Ohm (for frequency >100 kHz)
Cable capacity:	<60 pF per meter
Core diameter:	>0.22 mm^2 (AWG 24)
Cable type:	twisted pair cable. 1x2 or 2x2 or 1x4.
Resistance:	<110 Ohm per km
Signal attenuation:	Max. 9dB over total length of line section
Shielding:	Cu shielding braid or shielding braid and shielding foil

For a more detailed discussion of RS-485, refer to the chapter on RS-485 installation and troubleshooting.

7.2.2 Data link layer (layer 2)

The second layer of the OSI model implements the functions of medium access control as well as that of the logical link control i.e. the transmission and reception of the actual frames. The latter includes the data integrity function i.e. the generation and checking of checksums.

The medium access control determines when a station may transmit on the bus and ProfiBus supports two mechanisms, namely, token passing and polling.

Token passing is used for communication between multiple masters on the bus. It involves the passing of software token between masters, in a sequence of ascending addresses. Thus, a logical ring is formed (despite the physical topology being a bus). The polling method (or master-slave method), on the other hand, is used by a master that

currently has the token to communicate with its associated slave devices (passive stations).

ProfiBus can be set up either as a pure master-master system (token passing), or as a polling system (master-slave), or as a hybrid system using both techniques.

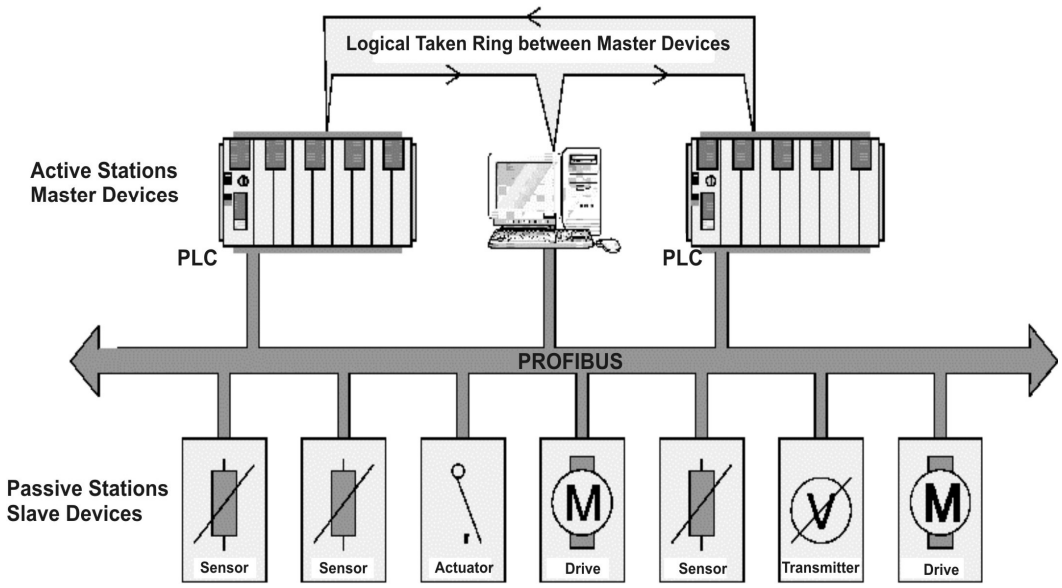


Figure 7.2
Hybrid medium access control

The following is a more detailed description of the token-passing mechanism.

- The token is passed from master station to master station in ascending order
- When a master station receives the token from a previous station, it may then transfer messages to slave devices as well as to other masters.
- If the token transmitter does not recognize any bus activity within the slot time, it repeats the token and waits for another slot time. It retires if it recognizes bus activity. If there is no bus activity, it will repeat the token frame for the last time. If there is still no activity, it will try to pass the token to the next but one master station. It continues repeating the procedure until it identifies a station that is alive
- Each master station is responsible for the addition or removal of stations in the address range from its own station address to the next station. Whenever a station receives the token, it examines one address in the address range between itself and its current successor. It does this maintenance whenever its currently queued message cycles have been completed. Whenever a station replies saying that it is ready to enter the token ring it is then passed the token. The current token holder also updates its new successor
- After a power up and after a master station has waited a predefined period, it claims the token if it does not see any bus activity. The master station with the lowest station address commences initialization. It transmits two token frames addressed to itself. This then informs the other master stations that it is now the only station on the logical token ring. It then transmits a 'request field data link status' to each station in an increasing address order The first master station that responds is then passed the token. The slave

stations and 'master not ready' stations are recorded in an address list called the GAP List

- When the token is lost, it is not necessary to re-initialize the system. The lowest address master station creates a new token after its token timer has timed out. It then proceeds with its own messages and then passes the token onto its successor
- The real token rotation time is calculated by each master station on each cycle of the token. The system reaction time is the maximum time interval between two consecutive high priority message cycles of a master station at maximum bus load. From this, a target token rotation time is defined. The real token rotation time must be less than the target token rotation time for low priority messages to be sent out
- There are two priorities that can be selected by the application layer, namely 'low' and 'High.' The high priority messages are always dispatched first. Independent of the token rotation time, a master station can always transmit one high priority message. The system's target token rotation time depends on the number of stations, the number of high priority messages and the duration of each of these messages. Hence it is important only to set very important and critical messages to high priority. The predefined target token rotation time should contain sufficient time for low priority message cycles with some safety margin built in for retries and loss of messages

Basically the ProfiBus layer 2 operates in a connectionless fashion, i.e. it transmits frames without prior checking as to whether the intended recipient is able or willing to receive the frame. In most cases, the frames are 'unicast,' i.e. they are intended for a specific device, but broadcast and multicast communication is also possible. Broadcast communication means that an active station sends an unconfirmed message to all other stations (masters and slaves). Multicast communication means that a device sends out an unconfirmed message to a group of stations (masters or slaves).

Layer 2 provides data transmission services to layer 7. These services are as defined in DIN 19241-2, IEC 955, ISO 8802-2 and ISO/IEC JTC 1/SC 6N 4960 (LLC Type 1 and LLC Type 3) and comprise three acyclic data services as well as one cyclic data service.

The following data transmission services are defined:

- Send-data-with-acknowledge (SDA) – acyclic.
- Send-data-with-no-acknowledge (SDN) – acyclic.
- Send-and-request-data-with-reply (SRD) – acyclic.
- Cyclic-send-and-request-data-with-reply (CSRD) – cyclic.

All layer 2 services are accessed by layer 7 in software through so-called service access points or SAPs. On both active and passive stations, multiple SAPs (service access points) are allowed simultaneously:

- 32 Stations are allowed without repeaters, but with repeaters this number may be increased to 127
- The maximum bus length is 1200 meters. This may be increased to 4800m with repeaters
- Transmission is half-duplex, using NRZ (non-return to zero) coding.
- The data rate can vary between 9.6 kbps and 12 Mbps, with values of 9.6, 19.2, 93.75, 187.5, 500, 1500 kbps or 12 Mbps

- The frame formats are according to IEC-870-5-1, and are constructed with a hamming distance of 4. This means that despite up to 4 consecutive faulty bits in a frame (and despite a correct checksum), a corrupted message will still be detected
- There are two levels of message priority

7.2.3 Application layer

Layer 7 of the OSI model provides the application services to the user. These services make an efficient and open (as well as vendor independent) data transfer possible between the application programs and layer 2.

The ProfiBus application layer is specified in DIN 19 245 part 2 and consists of:

- The Fieldbus message specification (FMS)
- The lower layer interface (LLI)
- The FieldBus management services layer 7 (FMA 7)

7.2.4 Fieldbus message specification (FMS)

From the viewpoint of an application process (at layer 8), the communication system is a service provider offering communication services, known as the FMS services. These are basically classified as either confirmed or unconfirmed services.

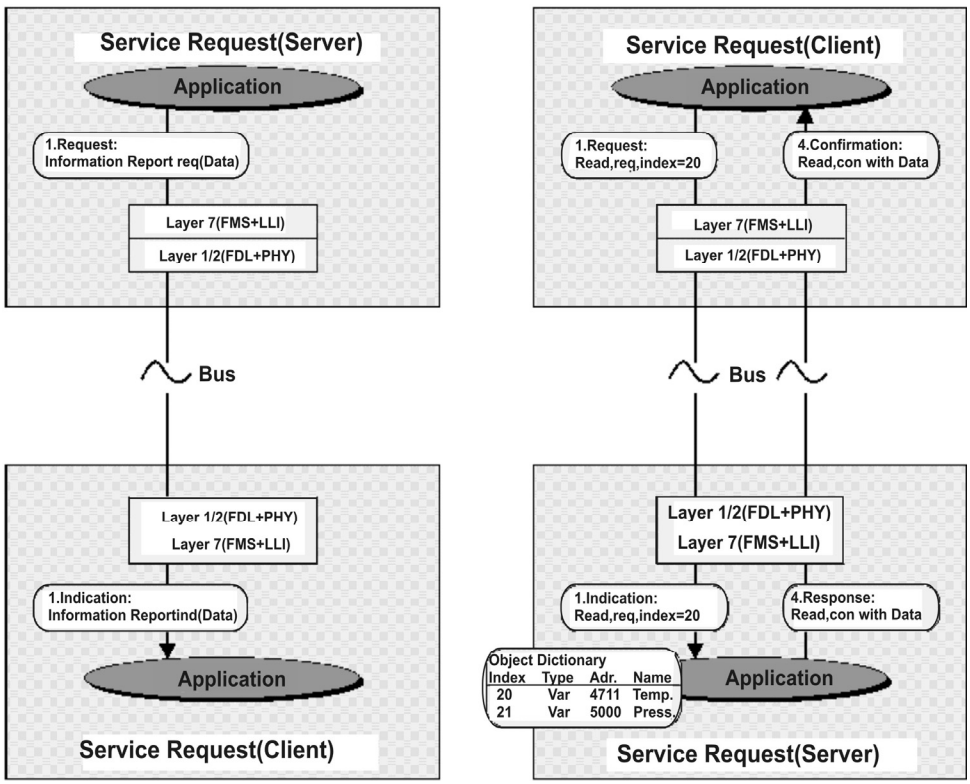


Figure 7.3
Execution of confirmed and unconfirmed services

Confirmed services are only permitted on connection-oriented communication relationships while unconfirmed services may also be used on connectionless

relationships. Unconfirmed services may be transferred with either a high or a low priority.

In the ProfiBus standard, the interaction between requester and responder, as implemented by the appropriate service is described by a service primitive.

The ProfiBus FMS services can be divided into the following groups:

- Context management services allow establishment and release of logical connections, as well as the rejection of inadmissible services
- Variable access services permit access (read and write) to simple variables, records, arrays and variable lists
- The domain management services enable the transmission (upload or download) of contiguous memory blocks. The application process splits the data into smaller segments (fragments) for transmission purposes
- The program invocation services allow the control (start, stop etc.) of program execution
- The event management services are unconfirmed services, which make the transmission of alarm messages possible. They may be used with high or low priority, and messages may be transmitted on broadcast or multicast communication relationships
- The VFD support messages permit device identification and status reports. These reports may be initiated at the discretion of individual devices, and transmitted on broadcast or multicast communication relationships
- The OD management services permit object dictionaries to be read and written. Process objects must be listed as communication objects in an object dictionary (OD). The application process on the device must make its objects visible and available before these can be addressed and processed by the communication services

As can be seen, there are large amounts of ProfiBus-FMS application services to satisfy the various requirements of field devices. Only a few of these (5, in fact) are mandatory for implementation in all ProfiBus devices. The selection of further services depends on the specific application and is specified in the so-called profiles.

7.2.5 Lower layer interface (LLI)

Layer 7 needs a special adaptation to layer 2. This is implemented by the LLI in the ProfiBus protocol. The LLI conducts the data flow control and connection monitoring as well as the mapping of the FMS services onto layer 2, with due consideration of the various types of devices (master or slave).

Communications relationships between application processes with the specific purpose of transferring data must be defined before a data transfer is started. These definitions are listed in layer 7 in the communications relationship list (CRL).

The main tasks of the LLI are:

- Mapping of FMS services onto the data link layer services
- Connection establishment and release
- Supervision of the connection
- Flow control

The following types of communication relationships are supported by:

- Connectionless communication, which can be either
- Broadcast, or
- Multicast, and
- Connection oriented communication, which can be either
- Master/master (cyclic or acyclic), or
- Master/slave – with or without slave initiative – (cyclic or acyclic)

Connection oriented communication relationships represent a logical peer-to-peer connection between two application processes. Before any data can be sent over this connection, it has to be established with an initiate service, one of the context management services. This comprises the connection establishment phase. After successful establishment, the connection is protected against third party access and can then be used for data communication between the two parties involved. This comprises the data transfer phase. In this phase, both confirmed and unconfirmed services can be used. When the connection is no longer needed, it can be released with yet another context management service, the Abort service. This comprises the connection release phase.

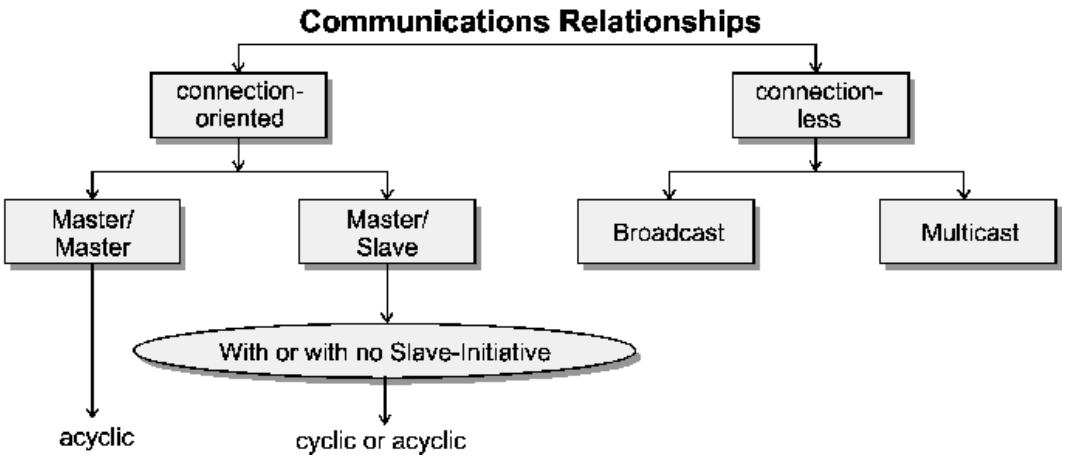


Figure 7.4
Supported communication relationships

7.2.6 Fieldbus management layer (FMA 7)

This describes object and management services. The objects are manipulated locally or remotely using management services. There are three groups here:

- **Context management**

  This provides a service for opening and closing a management connection

- **Configuration management**

  This provides services for the identification of communication components of a station, for loading and reading the communication relationship list (CRL) and for accessing variables, counters and the parameters of the lower layers

- **Fault Management**

  This provides services for recognizing and eliminating errors

7.3 The ProfiBus communication model

From a communication point of view, an application process includes all programs, resources and tasks that are not assigned to one of the communication layers. The ProfiBus communication model permits the combination of distributed application processes into a common process, using communications relationships. This acts to unify distributed application processes to a common process. That part of an application process in a field device that is reachable for communication is called a *Virtual Field Device* (VFD).

All objects of a real device that can be communicated with (such as variables, programs, data ranges) are called communication objects. The VFD contains the communication objects that may be manipulated by the services of the application layer via ProfiBus.

7.4 Relationship between application process and communication

Between two application processes, one or more communication relationships may exist; each one having a unique communication end point as shown in the following diagram:

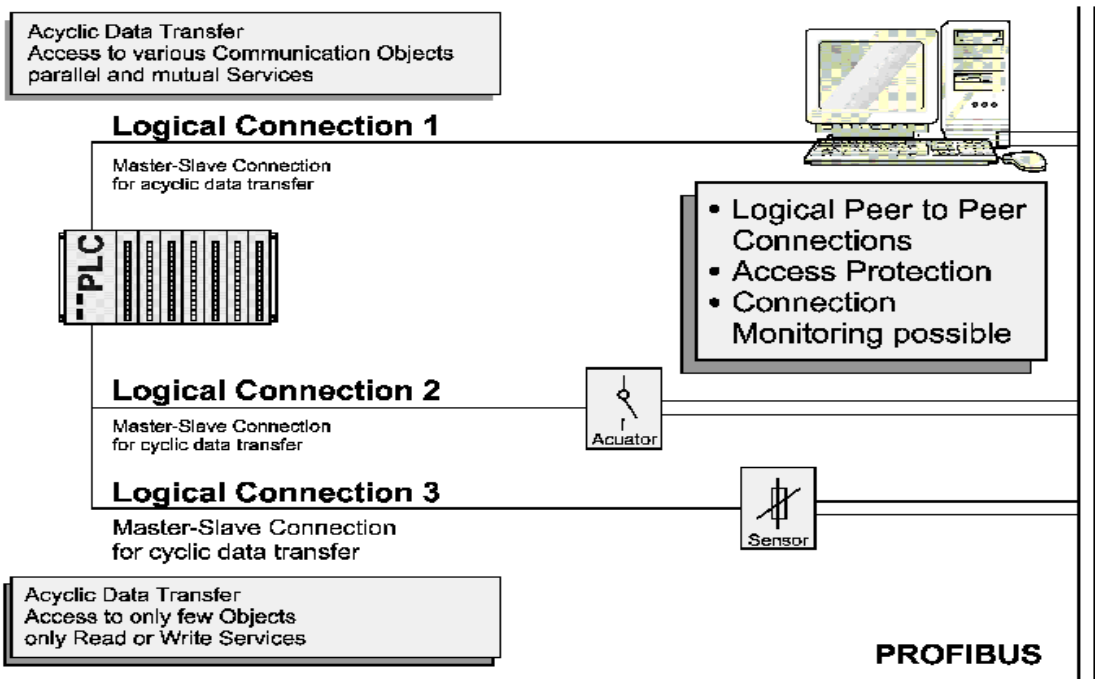


Figure 7.5
Assignment of communication relationships to application process

Mapping of the functions of the VFD onto the real device is provided by the application layer interface. The diagram below shows the relationship between the real field device and the virtual field device.

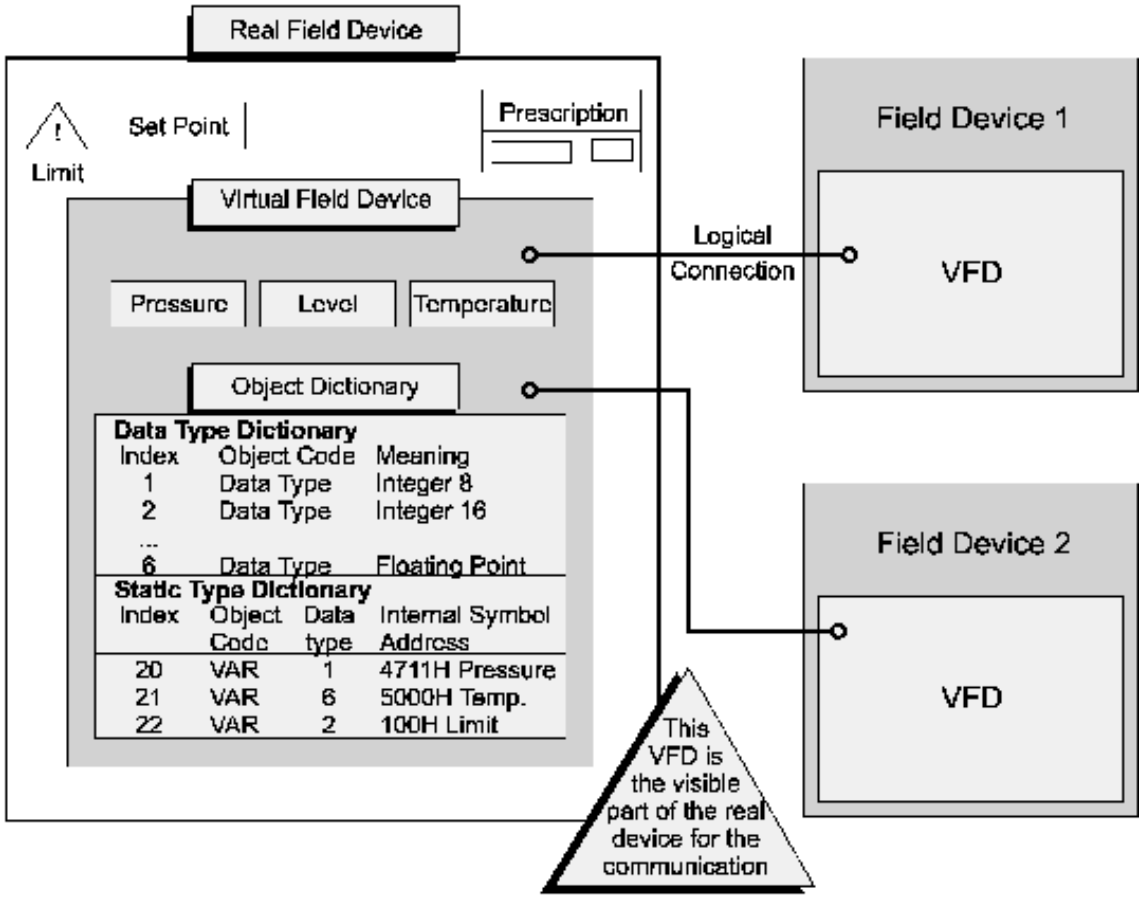


Figure 7.6
Virtual Field Device (VFD) With Object Dictionary (OD)

In this example, only the variables pressure, fill level and temperature may be read or written via two communication relationships.

7.5 Communication objects

All communication objects of a ProfiBus station are entered into a local object dictionary. This object dictionary may be predefined at simple devices; however on more complex devices it is configured and locally or remotely downloaded into the device.

The object dictionary (OD) structure contains:

- A header, which contains information about the structure of the OD
- A static list of types, containing the list of the supported data types and data structures
- A static object dictionary, containing a list of static communication objects
- A dynamic list of variable lists, containing the actual list of the known variable lists, and a dynamic list of program invocations, which contains a list of the known programs

Defined static communication objects include simple variable, array (a sequence of simple variables of the same type), record (a list of simple variables not necessarily of the same type), domain (a data range) and event.

Dynamic communication objects are entered into the dynamic part of the OD. They include program invocation and variable list (a sequence of simple variables, arrays or records). These can be predefined at configuration time, dynamically defined, deleted or changed with the application services in the operational phase.

Logical addressing is the preferred way of addressing of communication objects. They are normally accessed with a short address called an index (unsigned 16 bit). This makes for efficient messaging and keeps the protocol overhead down.

There are, however, two other optional addressing methods:

- Addressing by name, where the symbolic name of the communication objects is transferred via the bus.
- Physical addressing. Any physical memory location in the field device may be accessed with the services PhysRead and PhysWrite.

It is possible to implement password protection on certain objects and also to make them read-access only, for example.

7.6 Performance

A short reaction time is one of the main advantages of ProfiBus-DP. The figures are typical.

512 Inputs and outputs distributed over 32 stations can be accessed:

- In 6 mS at 1.5 Mbps and
- In 2 mS at 12 Mbps.

The chart below gives a visual indication of ProfiBus performance.

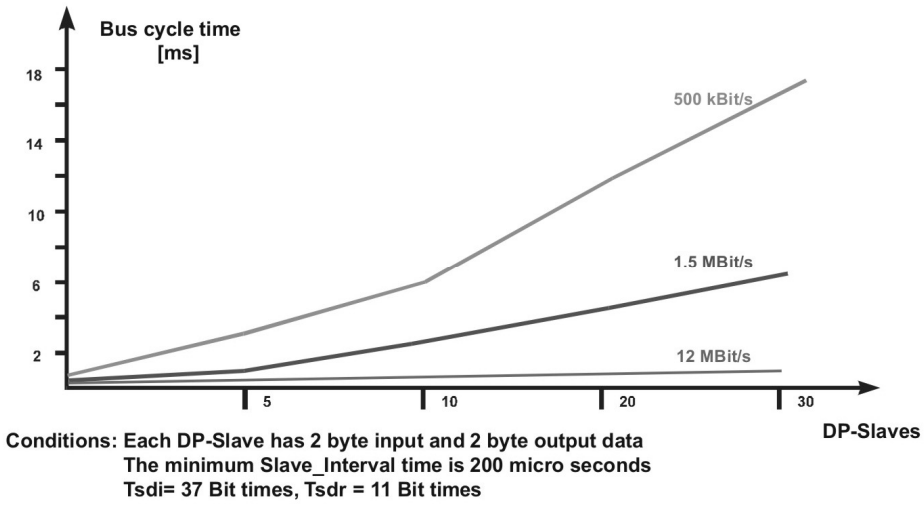


Figure 7.7
Bus cycle time of a ProfiBus DP mono-master system

The main service used to achieve these results is the send and receive data service of layer 2. This allows for the transmission of the input and output data in a single message cycle. Obviously, the other reason for increased performance is the higher transmission speed of 12 Mbps.

7.7 System operation

7.7.1 Configuration

The choice is up to user as to whether the system should be a mono-master or multi-master system. Up to 126 stations (masters or slaves) can be accommodated.

There are different device types:

- DP-master class 1 (DPM1). This is typically a PLC (programmable logical controller)
- DP-master class 2 (DPM2). These devices are used for programming, configuration or diagnostics
- DP-slave A. This is typically a sensor or actuator. The amount of I/O data is limited to 246 bytes

The two configurations possible are shown in the diagrams below.

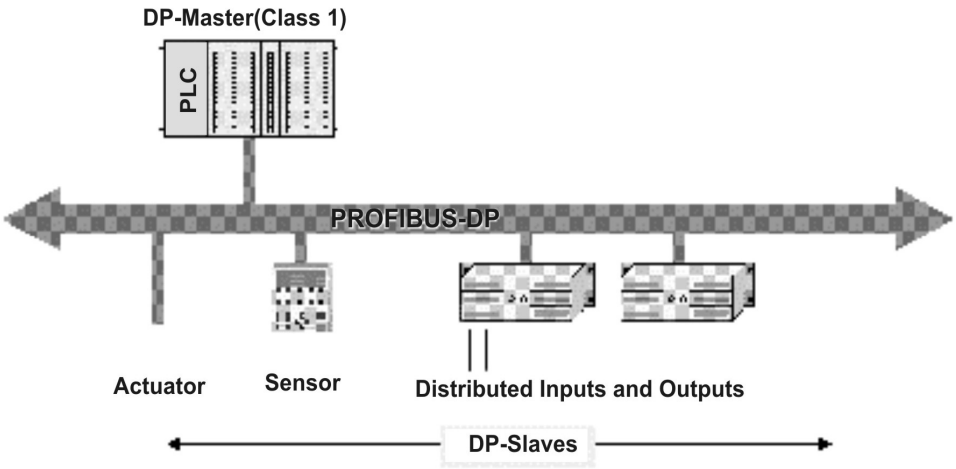


Figure 7.8
ProfiBus DP mono-master system

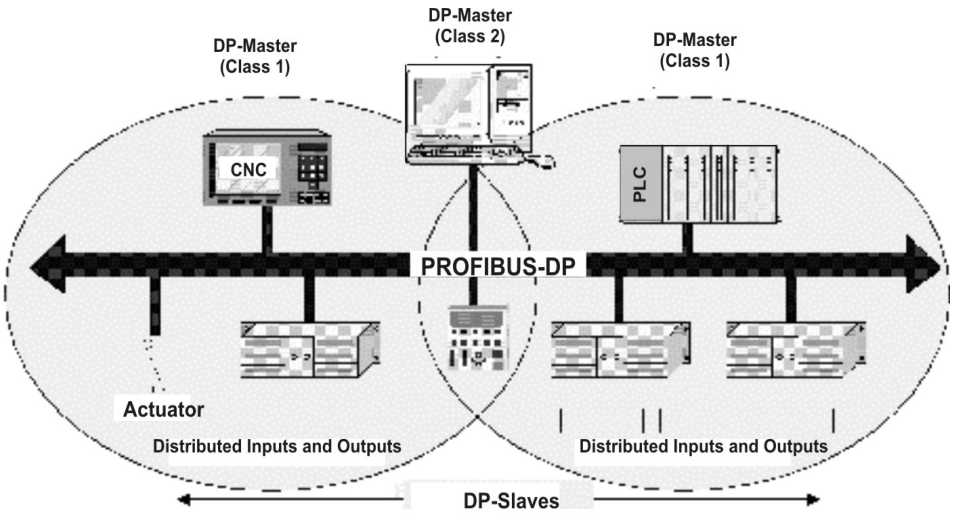


Figure 7.9
ProfiBus DP multi-master system

The following states can occur with DPM1:

- Stop. In this state, no data transfer occurs between the DPM1 and DP-slaves
- Clear. The DPM1 puts the outputs into a fail-safe mode and reads the input data from the DP-slaves
- Operate. The DPM1 is in the data transfer state with a cyclic message sequence where input data is read and output data is written down to the slave

7.7.2 Data transfer between DPM1 and the DP-slaves

During configuration of the system, the user defines the assignment of a DP slave to a DPM1 and which of the DP-slaves are included in the message cycle. In the so-called parameterization and configuration phases, each slave device compares its real configuration with that received from the DPM1.This configuration information has to be identical. This safeguards the user from any configuration faults. Once this has been successfully checked, the slave device will enter into the data transfer phase as indicated in the figure below.

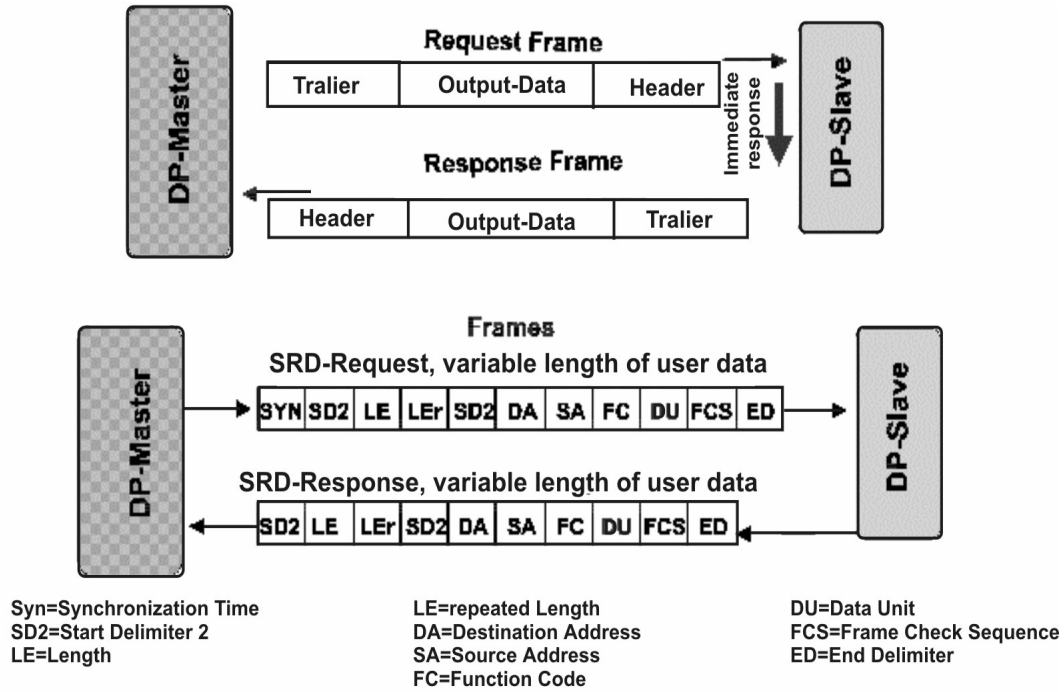


Figure 7.10
User data exchange for ProfiBus-DP

7.7.3 Synchronization and freeze modes

In addition to the standard cyclic data transfer mechanisms automatically executed by the DPM1, it is possible to send control commands from a master to an individual or group of slaves.

If the 'sync' command is transmitted to the appropriate slaves, they enter this state and freeze the outputs. They then store the output data during the next cyclic data exchange. When they receive the next 'sync' command, the stored output data is issued to the field.

If a ‘freeze’ command is transmitted to the appropriate slaves, the inputs are frozen in the present state. The input data is only updated on receiving the next ‘freeze’ command.

7.7.4 Safety and protection of stations

At the DPM1 station, the user data transfer to each slave is monitored with a watchdog timer. If this timer expires indicating that no successful transfer has taken place, the user is informed and the DPM1 leaves the OPERATE state and switches the outputs of all the assigned slave devices to the fail-safe state. The master changes to the CLEAR state. Note that the master ignores the timer if the automatic error reaction has been enabled (Auto_Clear = True).

At the slave devices, the watchdog timer is again used to monitor any failures of the master device or the bus. The slave switches its outputs autonomously to the fail-safe state if it detects a failure.

7.7.5 Mixed operation of FMS and DP stations

Where lower reaction times are acceptable, it is possible to operate FMS and DP devices together on the same bus. It is also possible to use a composite device, which supports both FMS and DP protocols simultaneously. This can make sense if the configuration is done using FMS and the higher speed cyclic operations are done for user data transfer. The only difference between the FMS and DP protocols are of course the application layers.

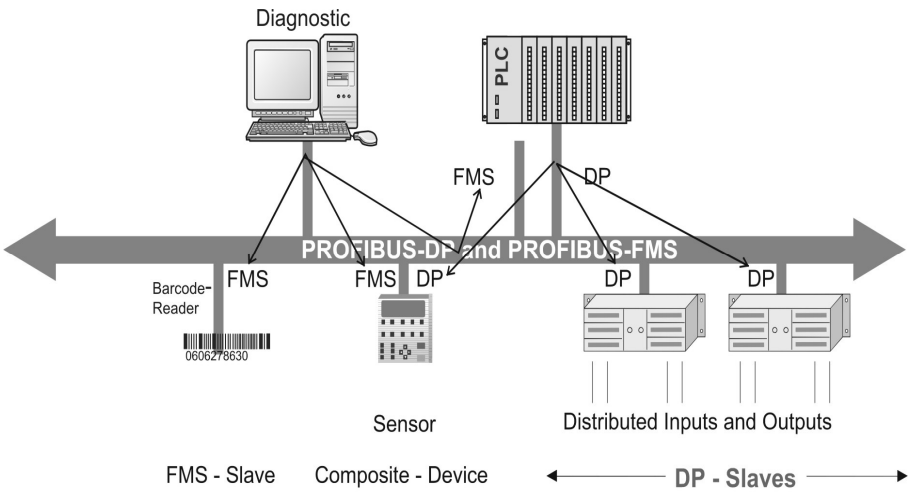


Figure 7.11
Mixed operation of profibus FMS and DP

7.8 Troubleshooting

7.8.1 Introduction

ProfiBus DP and FMS use RS-485 at the physical layer (layer 1) and therefore all the RS-485 installation and troubleshooting guidelines apply. Refer to the appropriate chapter in this manual. Profibus PA uses the same physical layers as the 61158-2 standard (which is

the same as the Foundation Fieldbus H1 standard). This section will discuss some additional specialized tools.

7.9 Troubleshooting tools

7.9.1 Handheld testing device

These are similar to the ones available for DeviceNet, and can be used to check the copper infrastructure before connecting any devices to the cable. A typical example is the unit made by Synergetic.

They can indicate:

- A switch (i.e. reversal) of the A and B lines
- Wire breaks in the A and B lines as well as in the shield
- Short circuits between the A and B lines and the shield
- Incorrect or missing terminations

The error is indicated via text shown in the display of the device.

These devices can also be used to check the RS-485 interfaces of ProfiBus devices, after they have been connected to the network. Typical functions include:

- Creating a list with the addresses of all stations connected to the bus (useful for identifying missing devices)
- Testing individual stations (e.g. identifying duplicate addresses)
- Measuring distance (checking whether the installed segment lengths comply with the Profibus requirements)
- Measuring reflections (e.g. locating an interruption of the bus line)

7.9.2 D-type connectors with built-in terminators

For further location of cable break errors reported by a handheld tester, 9-pin D connectors with integrated terminations are very helpful. When the termination is switched to ‘on’ at the connector, the cable leading out of the connector is disconnected. This feature can be used to identify the location of the error, as follows:

If, for example, the handheld is connected at the beginning of the network and a wire break of the A line is reported, plug the D connector somewhere in the middle of the network and switch the termination to ‘on.’ If the problem is still reported by the tester, it means that the introduced termination is still not ‘seen’ by the tester and thus the cable break must be between the beginning of the network and the D connector.

7.9.3 Configuration utilities

Each ProfiBus network must be configured and various products are commercially available to perform this task. Examples include the ProfiBus DP configuration tool by SST, the Allen Bradley plug & play software and the Siemens COM package. In many cases, the decision on the tool to be used for configuration is made automatically by choosing the controlling device for the bus. The choice of configuration tool should not be treated lightly because the easier the tool is to use, the less likely a configuration error will be made.

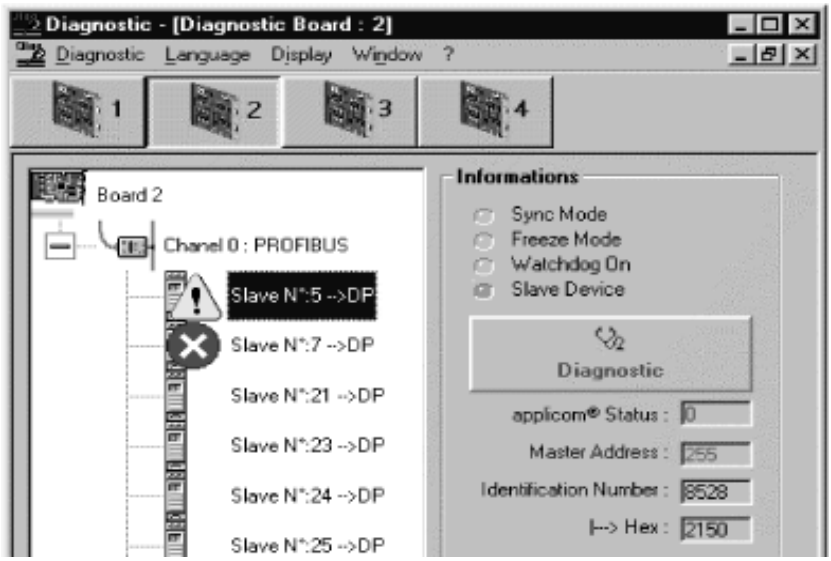


Figure 7.12
Applicom configuration tool

With ProfiBus, all parameters of a device (including text to provide a good explanation of the parameters and of the possible choices, values and ranges) are specified in a so-called GSD file, which is the electronics data sheet of the device. Therefore the configuration software has all the information it needs to create a friendly user interface with no need for the interpretation of hexadecimal values.

7.9.4 Built-in diagnostics

Several diagnostic functions were designed into ProfiBus. The standard defines different timers and counters used by the physical layer to maintain operation and monitor the quality of the network. One counter, for example, counts the number of invalid start delimiters received as an indication of installation problems or an interface not working properly. These timers and counters can be used by the ProfiBus device or by its configuration tool to identify a problem and to indicate it to the user.

For ProfiBus DP, a special diagnostic message is defined, which can be indicated by a ProfiBus DP slave or requested by a ProfiBus DP master. The first 6 bytes are implemented for all ProfiBus DP devices.

This information is used to indicate various problems to the user and could include:

- Configuration of the specific device incorrect
- Required features not supported on the device or
- Device does not answer

The user normally gets access to all this information through the configuration tool. The user selects a device, the tool reads the diagnostic information from the device, and provides high-level text information.

During operation, a DP device automatically reports problems to the ProfiBus DP master. The master stores the diagnostic information and provides it to the user. This can, for example, be done by a PC-based system that utilizes diagnostic flow charts to evaluate the information and then make it available to the operator.

The definition of additional diagnostics enables each manufacturer to simplify matters for end-users. The additional information differentiates between a device-related, an

identifier-related, and a channel-related part. The device-related part provides the opportunity to encode manufacturer specific details. This can be used to report that the module place in slot #4 is not the same as the one configured. Module-related diagnostics provide an overview of the status of all modules; it identifies whether a module supports diagnostics or not. The channel-related part offers the possibility to report problems down to the bit level. This means a DP slave can indicate that channel #3 of the module in slot #5 has a short-circuit to ground.

With the additional diagnostics, a ProfiBus DP device can send very detailed error reports to the controlling device. As a result, the master device is able to provide details to the user such as 'ERROR oven control: lower temperature limit exceeded' or 'station address 23 (conveyor control): wire break at module 2, channel 5.' This feature provides not only the flexibility to report any kind of error at a device but also often how to correct it. Because the protocol for ProfiBus PA is identical to that for DP, the diagnostic mechanism is the same.

7.9.5 Bus monitors

A bus monitor (protocol analyzer) is an additional tool for troubleshooting a ProfiBus network, enabling the user to perform packet content and timing verification. It is capable of monitoring and capturing all ProfiBus network activity (messages, ACKs etc.) on the bus, and then saving the captured data to disk. Each captured message is time-stamped with sub-millisecond resolution A monitor does not have a ProfiBus station address nor does it affect the speed or efficiency of the network.

A monitor provides a wide range of trigger and filter functions that allow capturing messages between two stations only or triggering on a special event like diagnostic requests. Such a tool can be used for the indication of problems with individual devices (e.g., wrong configuration) and also to visualize physical problems.

Bus monitors are typically PCs with a special ProfiBus interface card and the appropriate data capturing software. An example is the ProfiBus DP captures utility by SST. Bus monitors are sophisticated tools and are recommending only for people with a reasonable knowledge of ProfiBus and its protocol.

7.10 Tips

ProfiBus (especially DP) is straightforward from a user's point of view, as the messaging format is fairly simple.

However, the following notes will be helpful in identifying common problems:

- ProfiBus has a relatively high (12 Mbps) maximum data rate but it can also be operated at speeds as low as 9600 baud. If ProfiBus is to be used at high speeds, it might be necessary to use a scope or analyzer to check and fine-tune impedance terminations and drop-line lengths. Such problems are magnified at higher speeds. Users who initially intend to run their network at maximum speed often find that a lower speed setting performs just as well and is easier to get working
- One of the most common problems encountered in configuring a ProfiBus network is selecting the wrong GSD (device description) file for a particular node. As GSD files reside in a separate disk and are not embedded in the product itself, files are sometimes paired with the wrong devices
- When installing a new network, follow the ProfiBus installation guidelines:

- Use connectors suitable for an industrial environment and according to the defined standard
- Use only the specified (blue) cable
- Make sure the cable has no wire break and none of the wires causes a short circuit condition
- Do not crisscross the wires; always use the green wire for A and the red wire as B throughout the whole network
- Make sure the segment length is according to the chosen transmission rate (use repeaters to extend the network)
- Make sure the number of devices/RS-485 drivers per segment does not exceed 32 (use repeaters where necessary)
- Check proper termination of all copper segments (an RS-485 segment must be terminated at both ends)
- If so-called 'activated terminations' are used, they must be powered at all times
- Avoid drop lines or make sure the overall length does not exceed the specified maximum. In case T-drops are needed, use repeaters or active bus terminals
- In case the network connects buildings or runs in a hazardous environment, consider the use of fiber optics
- Check whether the station addresses are set to the correct value
- Check if the network configurations match the physical setup
- For RS-485 implementations (ProfiBus FMS and ProfiBus DP), type A cable is preferred for transmission speeds greater than 500 kbaud; type B should only be used for low baud rates and short distances

The specifications for the two types of cable are as follows:

Type A Cable	
Impedance:	135 up to 165 Ohm (for frequency of 3 to 20 MHz)
Cable Capacity:	<30 pF per Meter
Core Diameter:	>0.34 mm2 (AWG 22)
Cable Type:	twisted pair cable. 1 x2 or 2x2 or 1x4.
Resistance:	<110 Ohm per km
Signal Attenuation:	Max. 9dB over total length of line section
Shielding:	Cu shielding braid or shielding braid and shielding foil

Type B Cable	
Impedance:	135 up to 165 Ohm (for frequency > 100 kHz)
Cable Capacity:	<60 pF per Meter
Core Diameter:	>0.22 mm2 (AWG 24)
Cable Type:	twisted pair cable. 1 x2 or 2x2 or 1x4.
Resistance:	<110 Ohm per km
Signal Attenuation:	Max. 9dB over total length of line section
Shielding:	Cu shielding braid or shielding braid and shielding foil

- The connection between shield and protective ground is made via the metal cases and screw tops of the D-type connectors. Should this not be possible, then the connection should be made via pin 1 of the connectors. This is not an optimum solution and it is probably better to bare the cable shield at the appropriate point and to ground it with a cable as short as possible to the metallic structure of the cabinet

8

Foundation Fieldbus overview

Objectives

When you have completed study of this chapter, you will be able to:

- Describe how Foundation Fieldbus operates
- Remedy problems with:
- Wiring
- Earths/grounds
- Shielding
- Wiring polarity
- Power
- Terminations
- Intrinsic safety
- Voltage drop
- Power conditioning
- Surge protection
- Configuration

8.1 Introduction to Foundation Fieldbus

Foundation Fieldbus (FF) takes full advantage of the emerging ‘smart’ field devices and modern digital communications technology allowing end user benefits such as:

- Reduced wiring
- Communications of multiple process variables from a single instrument
- Advanced diagnostics
- Interoperability between devices of different manufacturers
- Enhanced field level control
- Reduced start-up time
- Simpler integration

The concept behind Foundation Fieldbus is to preserve the desirable features of the present 4–20 mA standard (such as a standardized interface to the communications link, bus power derived from the link and intrinsic safety options) while taking advantage of the new digital technologies.

This will provide the features noted above because of:

- Reduced wiring due to the multi-drop capability
- Flexibility of supplier choices due to interoperability
- Reduced control room equipment due to distribution of control functions to the device level
- Increased data integrity and reliability due to the application of digital communications

Foundation Fieldbus consists of four layers. Three of them correspond to OSI layers 1, 2 and 7. The fourth is the so-called 'user layer' that sits in top of layer 7 and is often said to represent OSI 'layer 8,' although the OSI model does not include such a layer. The user layer provides a standardized interface between the application software and the actual field devices.

8.2 The physical layer and wiring rules

The physical layer standard has been approved and is detailed in the IEC 61158-2 and the ISA standard S50.02-1992. It supports communication rates of 31.25 kbps and uses the Manchester Bi-phase L encoding scheme with four encoding states as shown in figure 16.2. Devices can be optionally powered from the bus under certain conditions. The 31.25 kbps (or H1, or low-speed bus) can support from 2 to 32 devices that are not bus powered, two to twelve devices that are bus powered or two to six devices that are bus powered in an intrinsically safe area. Repeaters are allowed and will increase the length and number of devices that can be put on the bus. The H2 or high speed bus options was not implemented as originally planned, but was superseded by the *High Speed Ethernet* (HSE) standard. This is discussed later in this section.

The low speed (H1) bus is intended to utilize existing plant wiring and uses #22 AWG type B wiring (shielded twisted pair) for segments up to 1200 m (3936 feet) and #18 AWG type A wiring (shielded twisted pair) up to 1900 meters (6232 feet). Two additional types of cabling are specified and are referred to as type C (multi-pair twisted without shield) and type D (multi-core, no shield).

Type C using #26 AWG cable is limited to 400 meters (1312 feet) per segment and type D with #16 AWG is restricted to segments less than 200 meters (660 feet).

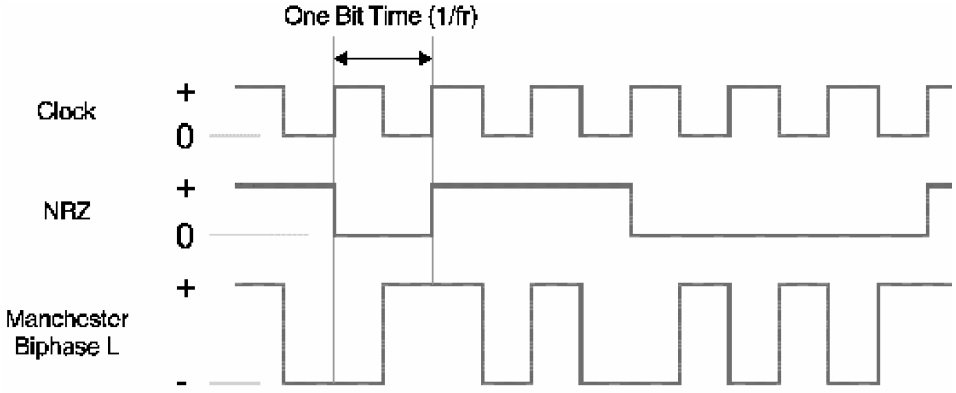
- Type A #18 AWG 1900 m (6232 feet)
- Type B #22 AWG 1200 m (3936 feet)
- Type C #26 AWG 400 m (1312 feet)
- Type D #16 AWG multi-core 200 m (660 feet)

The Foundation Fieldbus wiring is floating/ balanced and equipped with a termination resistor (RC combination) connected across each end of the transmission line. Neither of the wires should ever be connected to ground. The terminator consists of a 100 ohm quarter Watt resistor and a capacitor sized to pass 31.25 kHz. As an option, one of the terminators can be center-tapped and grounded to prevent voltage buildup on the bus. Power supplies must be impedance matched. Off-the-shelf power supplies must be conditioned by fitting a series inductor. If a 'normal power supply' is placed across the

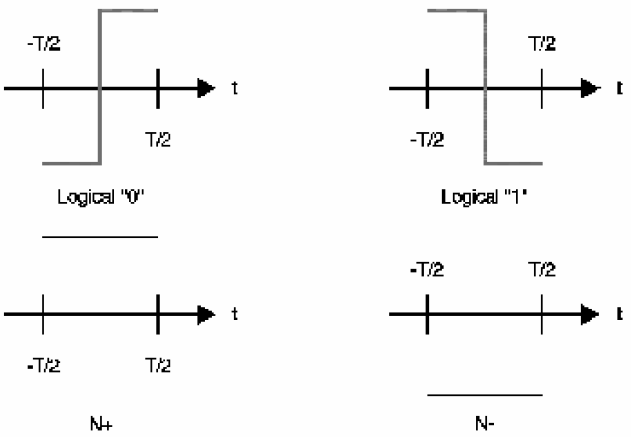
line, it will load down the line due to its low impedance. This will cause the transmitters to stop transmitting.

Fast response times for the bus are one of the FF goals. For example, at 31.25 kbps on the H1 bus, response times as low as 32 microseconds are possible. This will vary, based on the loading of the system, but will average between 32 microseconds and 2.2 ms with an average of approximately 1 ms.

Spurs can be connected to the 'home run.' The length of the spurs depends on the type of wire used and the number of spurs connected. The maximum length is the total length of the spurs and the home run.



ENCODING AND DECODING



ENCODING RULES

Symbols		Encoding
1	(ONE)	Hi-Lo transition (mid-bit)
0	(ZERO)	Lo-Hi transition (mid-bit)
N+	(NON-DATA PLUS)	Hi (No transition)
N-	(NON-DATA MINUS)	Lo (No transition)

ENCODING RULES

Figure 8.1
Foundation Fieldbus physical layer

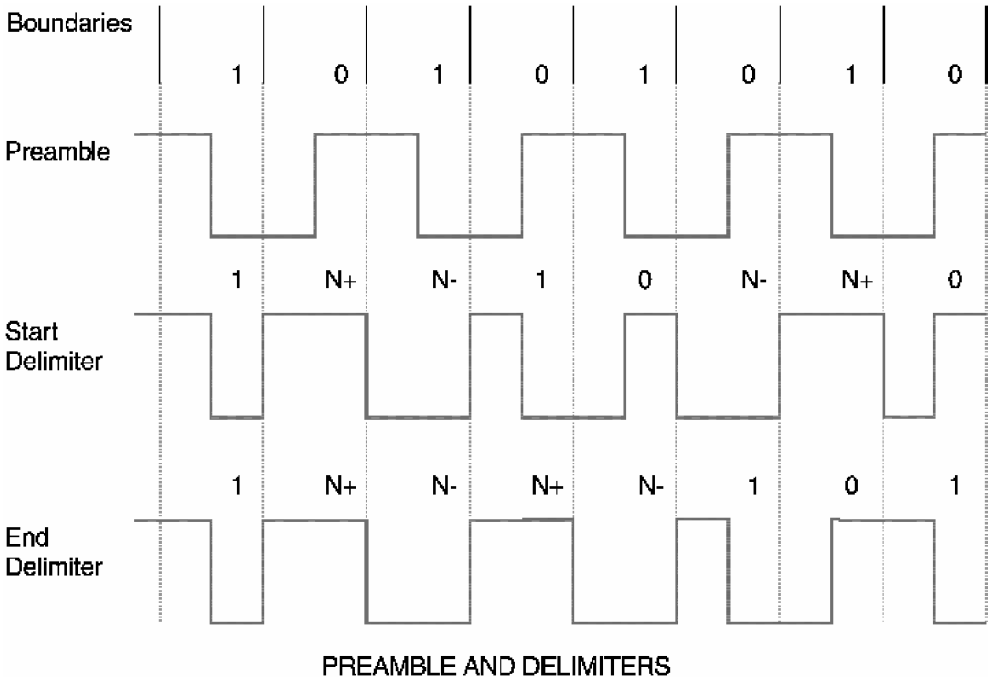


Figure 8.2
Use of N+ and N- encoding states

The physical layer standard has been out for some time. Most of the recent work has been focused on these upper layers and are defined by the FF as the 'communications stack' and the 'user layer.' The following sections will explore these upper layers:

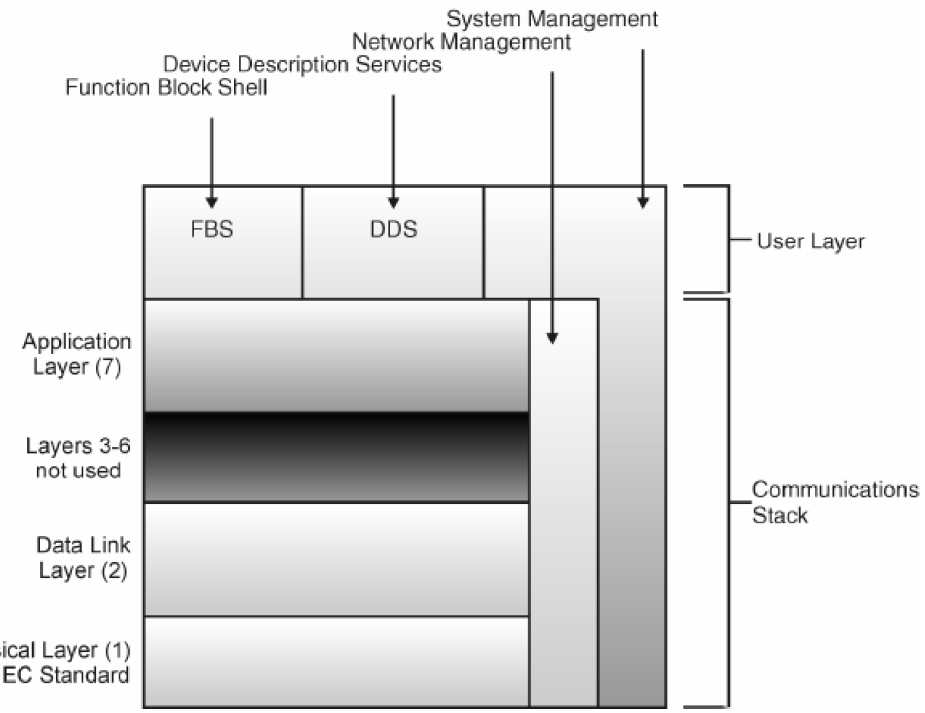


Figure 8.3
The OSI model of the FF protocol stack

8.3 The data link layer

The communications stack as defined by the FF corresponds to OSI layers two and seven, the data link and applications layers. The DLL (data link layer) controls access to the bus through a centralized bus scheduler called the *Link Active Scheduler* (LAS). The DLL packet format is shown below:

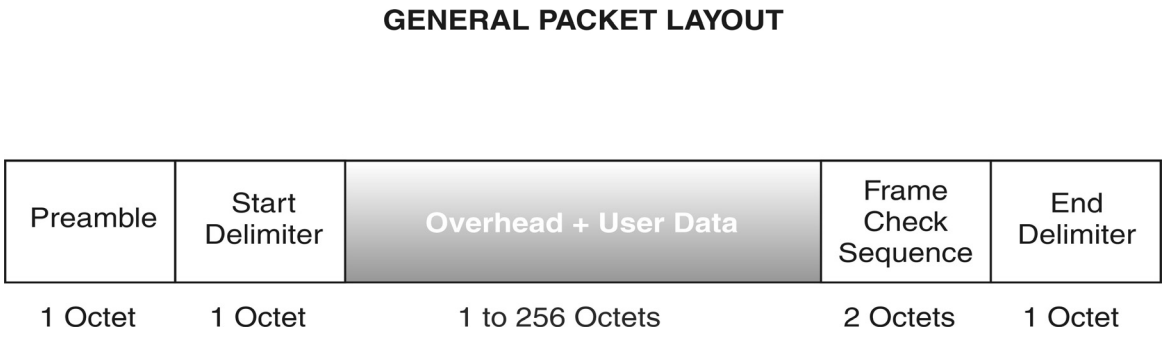


Figure 8.4
Data link layer packet format

The *Link Active Scheduler* (LAS) controls access to the bus by granting permission to each device according to predefined 'schedules.' No device may access the bus without LAS permission. There are two types of schedules implemented: cyclic (scheduled) and acyclic (unscheduled). It may seem odd that one could have an unscheduled 'schedule,' but these terms actually refer to messages that have a periodic or non-periodic routine, or 'schedule.'

The cyclic messages are used for information (process and control variables) that requires regular, periodic updating between devices on the bus. The technique used for information transfer on the bus is known as the publisher-subscriber method. Based on the user predefined (programmed) schedule, the LAS grants permission for each device in turn access to the bus. Once the device receives permission to access the bus, it 'publishes' its available information. All other devices can then listen to the 'published' information and read it into memory (subscribe) if it requires it for its own use. Devices not requiring specific data simply ignore the 'published' information.

The acyclic messages are used for special cases that may not occur on a regular basis. These may be alarm acknowledgment or special commands such as retrieving diagnostic information from a specific device on the bus. The LAS detects time slots available between cyclic messages and uses these to send the acyclic messages.

8.4 The application layer

The application layer in the FF specification is divided into two sub-layers: the Foundation *Fieldbus Access Sublayer* (FAS) and the Foundation *Fieldbus Messaging Specification* (FMS).

The capability to pre-program the 'schedule' in the LAS provides a powerful configuration tool for the end user since the time of rotation between devices can be established and critical devices can be 'scheduled' more frequently to provide a form of prioritization of specific I/O points. This is the responsibility and capability of the FAS. Programming the schedule via the FAS allows the option of implementing (actually, simulating) various 'services' between the LAS and the devices on the bus.

Three such 'services' are readily apparent such as:

- Client/server with a dedicated client (the LAS) and several servers (the bus devices)
- Publisher/subscriber as described above
- Event distribution with devices reporting only in response to a 'trigger' event, or by exception, or other predefined criteria

These variations, of course, depend on the actual application and one scheme need not necessarily be 'right' for all applications, but the flexibility of the Foundation Fieldbus is easily understood from this example.

The second sub-layer, the Foundation *Fieldbus Messaging Specification* (FMS), contains an 'object dictionary' that is a type of database that allows access to Foundation Fieldbus data by tag name or an index number. The object dictionary contains complete listings of all data types, data type descriptions, and communication objects used by the application. The services allow the object dictionary (application database) to be accessed and manipulated.

Information can be read from or written to the object dictionary allowing manipulation of the application and the services provided.

8.5 The user layer

The FF specifies an eighth layer called the user layer that resides 'above' the application layer of the OSI model, this layer is usually referred to as layer 8. In the Foundation Fieldbus, this layer is responsible for three main tasks viz. network management, system management and function block/device description services. Figure 16.5 illustrates how all the layer's information packets are passed to the physical layer.

The network management service provides access to the other layers for performance monitoring and managing communications between the layers and between remote objects (objects on the bus). The system management takes care of device address assignment, application clock synchronization, and function block scheduling. This is essentially the time coordination between devices and the software and ensures correct time stamping of events throughout the bus.

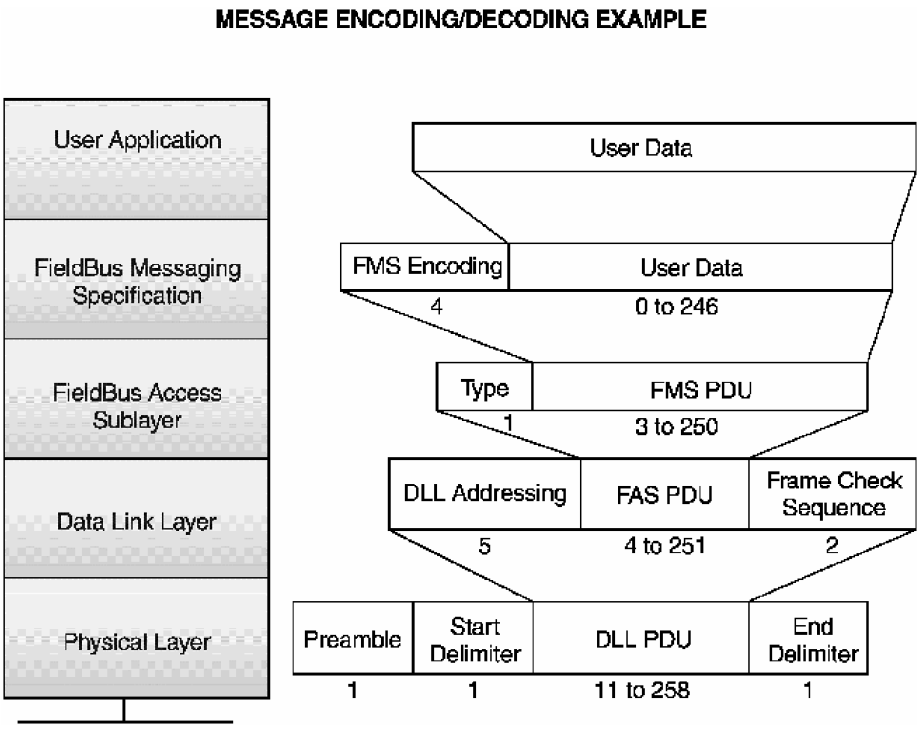


Figure 8.5
The passage of information packets to the physical layer

Function blocks and Device description services provide pre-programmed 'blocks,' which can be used by the end user to eliminate redundant and time-consuming configuration. The block concept allows selection of generic functions, algorithms, and even generic devices from a library of objects during system configuration and programming. This process can dramatically reduce configuration time since large 'blocks' are already configured and simply need to be selected. The goal is to provide an open system that supports interoperability and a *Device Description Language* (DDL), which will enable multiple vendors and devices to be described as 'blocks' or 'symbols.' The user would select generic devices then refine this selection by selecting a DDL object to specify a specific vendor's product. Entering a control loop 'block' with the appropriate parameters would nearly complete the initial configuration for the loop. Advanced control functions and mathematics 'blocks' are also available for more advanced control applications.

8.6 Error detection and diagnostics

FF has been developed as a purely digital communications bus for the process industry and incorporates error detection and diagnostic information. It uses multiple vendors' components and has extensive diagnostics across the stack from the physical link up through the network and system management layers by design.

The signaling method used by the physical layer timing and synchronization is monitored constantly as part of the communications. Repeated messages and the reason for the repetition can be logged and displayed for interpretation.

In the upper layer, network and system management is an integral feature of the diagnostic routines. This allows the system manager to analyze the network 'on-line' and maintain traffic loading information. As devices are added and removed, optimization of the *Link Active Scheduler* (LAS) routine allows communications optimization

dynamically without requiring a complete network shutdown. This ensures optimal timing and device reporting, giving more time to higher priority devices and removing, or minimizing, redundant or low priority messaging.

With the *Device Description* (DD) library for each device stored in the host controller (a requirement for true interoperability between vendors), all the diagnostic capability of each vendors' products can be accurately reported and logged and / or alarmed to provide continuous monitoring of each device.

8.7 High Speed Ethernet (HSE)

High Speed Ethernet (HSE) is the Fieldbus Foundation's backbone network running at 100 Mbits/second. HSE field devices are connected to the backbone via HSE linking devices. A HSE linking device is a device used to interconnect H1 Fieldbus segments to HSE to create a larger network. A HSE switch is an Ethernet device used to interconnect multiple HSE devices such as HSE linking devices and HSE field devices to form an even larger HSE network. HSE hosts are used to configure and monitor the linking devices and H1 devices. Each H1 segment has its own link active scheduler (LAS) located in a linking device. This feature enables the H1 segments to continue operating even if the hosts are disconnected from the HSE backbone. Multiple H1 (31.25 kbps) Fieldbus segments can be connected to the HSE backbone via linking devices.

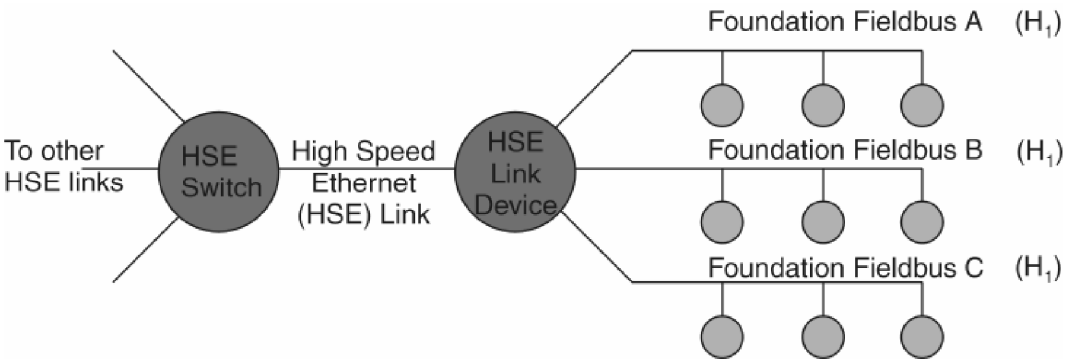


Figure 8.6
High speed Ethernet and Foundation Fieldbus

8.8 Good wiring and installation practice

8.8.1 Termination preparation

If care is taken in the preparation of the wiring, there will be fewer problems later and minimal maintenance required.

A few points to be noted here are:

- Strip 50 mm of the cable sheathing from the cable and remove the cable foil
- Strip 6 mm of the insulation from the ends. Watch out to avoid wire nicking or cutting off strands of wire. Use a decent cable-stripping tool
- Crimp a ferrule on the wire ends and on the end of the shield wire. Crimp ferrules are preferable as they provide a gas tight connection between the wire and the ferrule that is corrosion resistant. It is the same metal as the terminal in the wiring block
- An alternative strategy is to twist the wires together and to tin them with solder. Wires can be put directly into the wire terminal but make sure all

strands are in and they are not touching each other. Make sure that the strands are not stretched to breaking point
- Do not attach shield wires together in a field junction box. This can be the cause of ground loops
- Do not ground the shield in more than one place (inadvertently)
- Use good wire strippers to avoid damaging the wire

8.8.2 Installation of the complete system

Other system components can be installed soon after the cable is installed. This includes the terminators, power supply, power conditioners, spurs and in some cases, the intrinsic safety barriers. Some devices already have terminator built-in. In that case, be careful that you are not doubling up with terminators.

Check whether the grounding is correct. There should only be one point for the shield ground point. Once these checks have been performed; switch on the power supply and check the wiring system.

The Fieldbus tester (or an alternative simpler device) can be used to indicate:

- Polarities are correct
- Power carrying capability of the wire system is ok
- The attenuation and distortion parameters are within specification

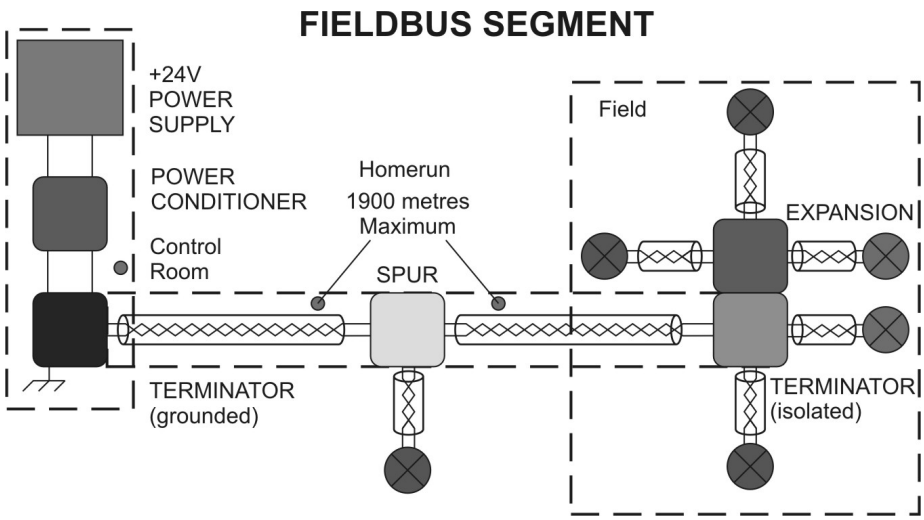


Figure 8.7
Overall diagram of Fieldbus wiring configuration (courtesy Relcom Inc.)

A few additional wiring tips and suggestions with reference to the diagram below:

- It is not possible to run two homerun cables in parallel for redundancy under the H1 standard. H1 Fieldbus is a balanced transmission line that must be terminated at each end. In some cases, it is a good idea to run a parallel cable for future use. In case of physical damage, you need to disconnect the damaged cable and put in the undamaged one. Ensure that, if this is the philosophy, you do not route both cables in the same cable tray
- Do not ground the shield of the cable at each Fieldbus device. The shield of the cable at the transmitter (for example) should be trimmed and covered

with insulating tape or heat shrinkable tubing. The only 'ground' that occurs on the segment is usually at the control room Fieldbus power conditioner

- Note that the ground that is connected to the isolated terminator at the far end of the segment does not connect the shields of the Fieldbus. It only allows for a high frequency path for ac currents
- There has been no provision made for lightning strikes. However, you should specify a terminator that has some type of spark gap arrestor, which will clamp the shields to about 75 V in such a high voltage surge
- A quick way to check that the grounding is correct before powering up is, doing a resistance measurement from the ground bolt on the power conditioner to the earth ground connection point. This measurement should be of the order of megohms. You can then connect the earth protective ground to the power conditioner bolt. Once this has been done, measure the resistance from the cable shields on the isolated terminator at the far end of the segment to a nearby earth ground point. A very low value of resistance should be seen
- A standard power supply cannot be used to power a Fieldbus segment. A standard power supply absorbs most of the Fieldbus signals due its low internal impedance. It is possible for a standard power supply to provide power to a Fieldbus power conditioning device as long as it has sufficient current is a floating power supply and has very low ripple and noise
- Use wiring blocks that hold the wiring securely and will not vibrate loose

Regular testing of an operational Fieldbus network

A Fieldbus tester can be used to get a view on the operation of the network. It is generally connected as follows:

- Red terminal to the (+) wire
- Black terminal to the (–) wire
- Green terminal to the shield

When the network is operating, the tester will build up a record of operational devices and then builds up a record of their signal characteristics. During later routine network maintenance, the results will be compared. If there is deterioration, this will be indicated and could be wiring problems, additional noise or a device, whose transmitter is starting to fail.

8.9 Troubleshooting

8.9.1 Introduction

Estimates are that 70% of network downtime is caused by physical problems. Foundation Fieldbus is more complicated to troubleshoot the most networks because it can and often does use the communication bus to power devices. The troubleshooter needs to know not only whether the communication is working but also whether there is enough power for the devices. Below is a diagram of a typical system. Notice that the power supply on the left is supplying power to the devices in the system.

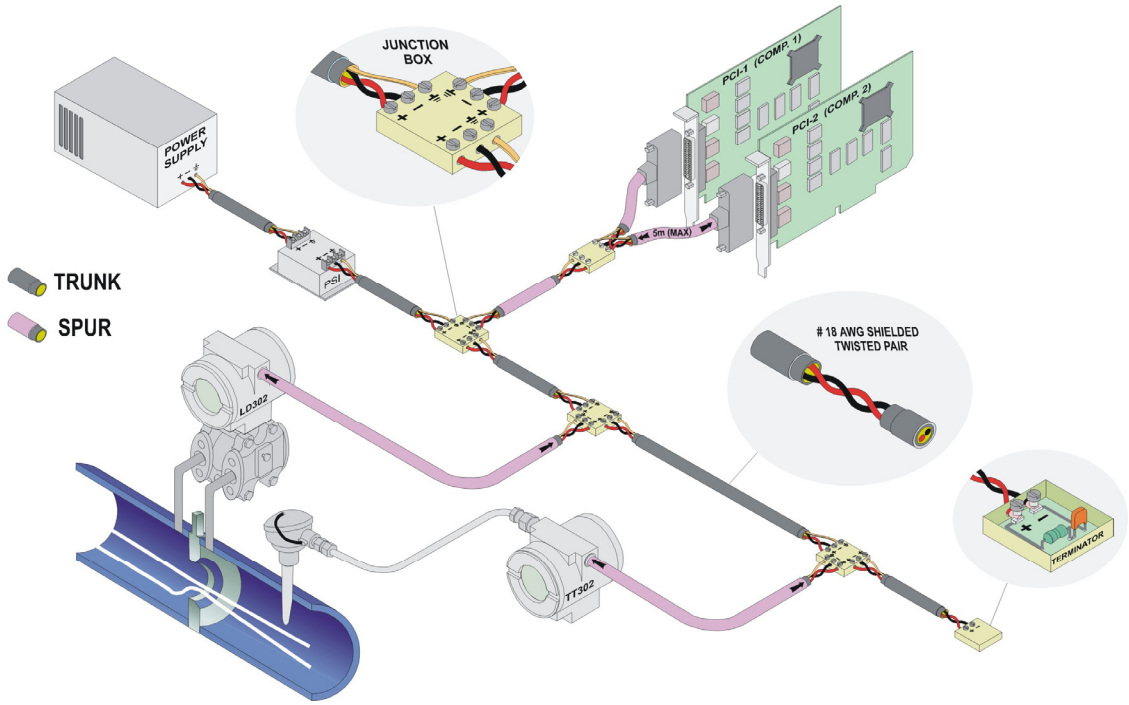


Figure 8.8
A typical Foundation Fieldbus system

When troubleshooting a Foundation Fieldbus system, it is necessary to first determine whether the problem is a power problem or a communications problem. In new systems, it may be found that the problem is both. In working systems it is usually one or the other.

8.10 Power problems

Power problems in a FF system can be divided into two types. One where the system is new and never worked and the other where the system has been up and running for a while. When new devices are added to an existing system and the communications immediately fails, it is easy to realize that the new device had something to do with the problem. If the system has never worked, then the problem could be anywhere and could be caused by multiple devices. The problem could also be with the design itself.

The following need to be known when troubleshooting the power system of a FF system.

- What is the layout of the system? Does each device have at least 9 volts dc?
- What is the supply current?
- What is the supply voltage?
- What is the current draw of each device?
- What is the resistance of each cable leg?

The easiest way of determining a power problem, is to do the following:

- Check each device to see if the power light is on
- Measure the voltage at each device
- Check the connections for opens, corrosion or loose connections
- Measure the current draw of each device to see if it conforms to the manufacturer's specifications

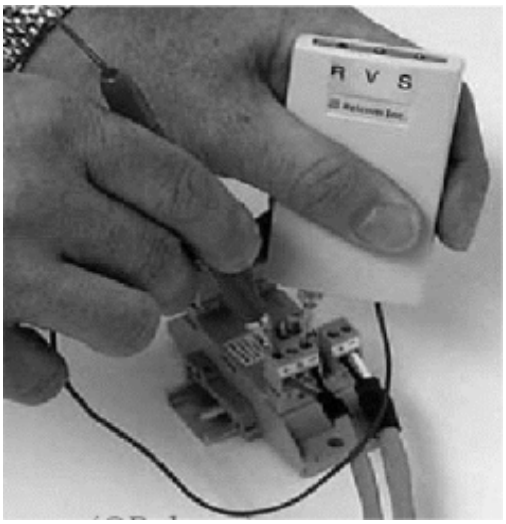


Figure 8.9
Testing the system (courtesy Relcom Inc.)

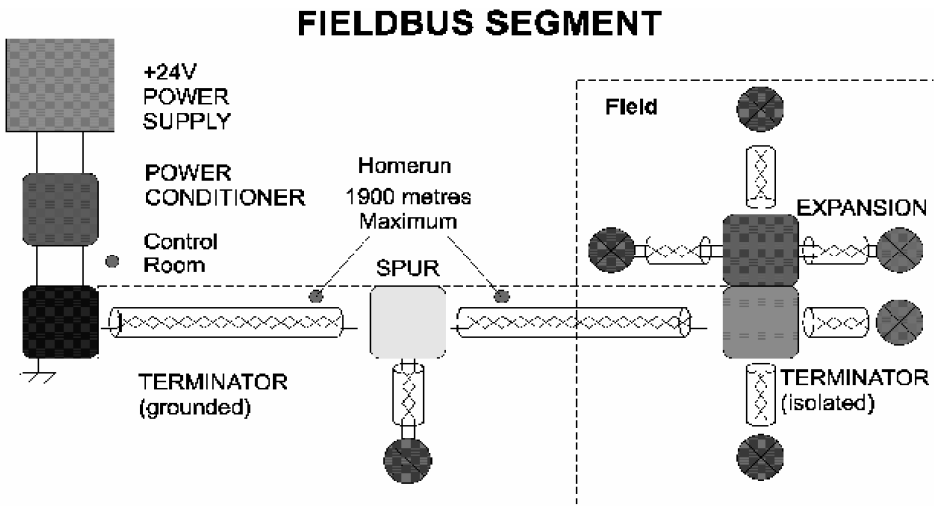


Figure 8.10
Layout of a system

Notice that the home run in the last drawing connects the control room equipment with the devices via common terminal blocks (chickenfoot or crowsfoot). The signal cable also provides the power to the devices. There is a terminator at each end of the cable. Power supplies require power conditioners.

8.10.1 Power example

Here is an example of the power requirements for a system:

- The power supply output is 20 volts
- The two wires are 1 km long with 22 Ohms per wire (44 Ohms total)
- Each device draws 20 mA
- Minimum voltage at each device 9 Volts (20 – 9 = 11 Volts)
- 11 Volts/44 Ohms = 250 mA

Therefore

- 250 mA/20mA = 12 devices on the system

8.11 Communication problems

Once the power is ruled out as a problem, it can be assumed that the communication system is at fault. Initially, it is important to check the following items:

- Are the wires connected correctly?
- Is the shield continuous throughout the system?
- Is the shield grounded at only one place?

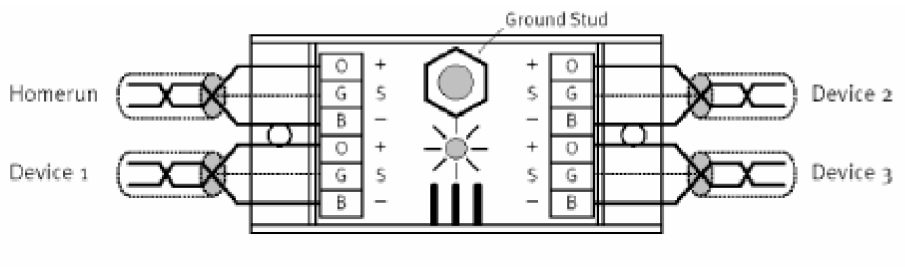


Figure 8.11
Schematic of a terminal block (courtesy Relcom Inc.)

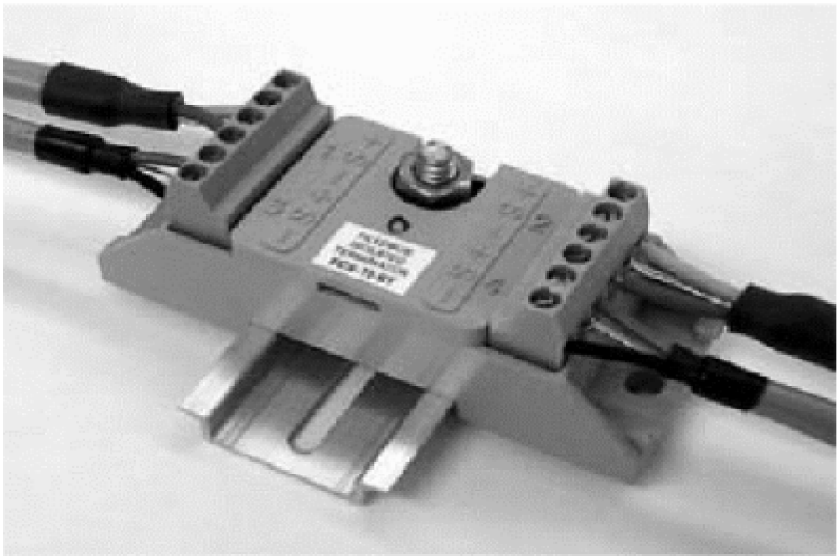


Figure 8.12
Terminal block (courtesy Relcom Inc.)

Once these basics are verified, the next step is to check to make sure that the cables are not too long. To measure the losses through the cable, a FF transmitter device is placed at one end and a receiver test device at the other. The maximum loss is usually around –14 dB. The typical characteristics of a twisted pair cable is:

- Impedance: 100 Ohms
- Wire size: 18 GA (0.8mm^2)
- Shield: 90% coverage
- Capacitive unbalance: 2 nF/km
- Attenuation: 3 dB/km

Using an ungrounded oscilloscope, it is possible to look at the signal. A good transmitter signal might look like this:

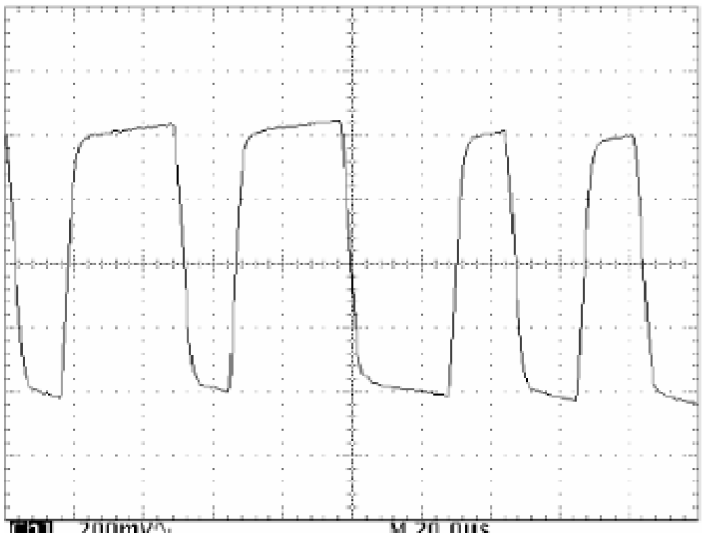


Figure 8.13
A good transmitted signal (courtesy Relcom Inc.)

When it is received it might look like this:

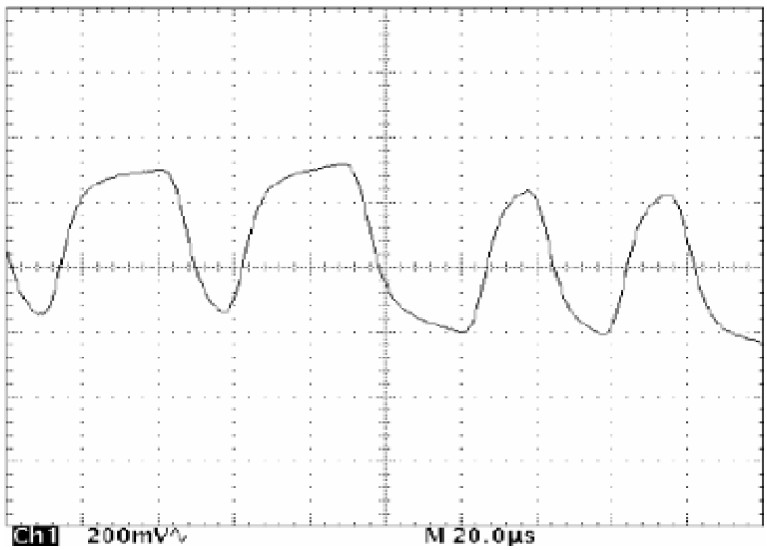


Figure 8.14
Received FF signal (courtesy Relcom Inc.)

Notice that the waveform is a bit distorted and lower in amplitude but still good. The next drawing is what the whole packet might look like.

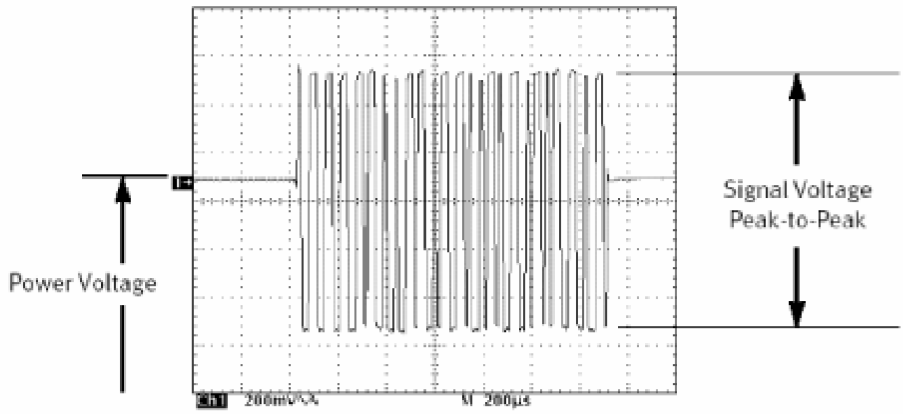


Figure 8.15
Bipolar FF signal (courtesy Relcom Inc.)

8.12 Foundation Fieldbus test equipment

There are a few manufacturers that have brought out test equipment specifically designed for testing FF systems. Some of the test equipment can be used while the system is working and others are used when the system is offline.

Some of the things the test equipment can check for are:

- DC voltage levels
- Link active scheduler probe node frame voltage
- Number of devices on the network
- If devices have been added or removed
- The lowest voltage level transmitted by a device
- Noise level between frames
- Device response noise level

One of the best troubleshooting tools are the LEDs provided on the devices. These LEDs show many different conditions of the system. If the troubleshooter becomes familiar with them, then LEDs can often indicate what is wrong with the system.

9

Operation of Ethernet systems

Objectives

When you have completed study of the chapter, you will have:

- Familiarized yourself with the 802 series of IEEE standards for networking
- Studied the details of the makeup of the data frames under DIX standard, IEEE 802.3, LLC, IEEE 802.1p, and IEEE 802.1Q
- Understood in depth how CSMA/CD operates for Half-Duplex transmissions
- Understood how multiplexing, Ethernet flow control, and PAUSE operations work
- Understood how full-duplex transmissions and auto-negotiation are carried out

9.1 Introduction

The OSI reference model was dealt with in chapter one wherein it was seen that the data link layer establishes communication between stations. It creates, transmits and receives data frames, recognizes link addresses, etc. It provides services for the various protocols at the network layer above it and uses the physical layer to transmit and receive messages. The data link layer creates packets appropriate for the network architecture being used. Network architectures (such as Ethernet, ARCnet, Token Ring, and FDDI) encompass the data link and physical layers

Ethernet standards include a number of systems that all fit within data link and physical layers. The IEEE has facilitated organizing of these systems by dividing these two OSI layers into sub layers, namely, the logical link control (LLC) and media access control (MAC) sub layers.

9.1.1 Logical link control (LLC) sublayer

The IEEE has defined the LLC layer in the IEEE 802.2 standard. This sub-layer provides a choice of three services viz. an unreliable datagram service, an acknowledgement datagram service, and a reliable connection oriented service. For acknowledged datagram or connection-oriented services, the data frames contain a source address, a destination address, a sequence number, an acknowledgement number, and a few miscellaneous bits. For unreliable datagram service, the sequence number and acknowledge number are omitted.

9.1.2 MAC sublayer

MAC layer protocols are a set of rules used to arbitrate access to the communication channel shared by several stations. The method of determining which station can send a message is critical and affects the efficiency of the LAN. Carrier sense multiple access/collision detection (CSMA/CD), the method used for Ethernet, will be discussed in this chapter.

It must be understood that medium access controls are only necessary where more than two nodes need to share the communication path, and separate paths for transmitting and receiving do not exist. Where only two nodes can communicate across separate paths, it is referred to as full-duplex operation and the need for arbitration does not exist.

To understand MAC protocols thoroughly, this chapter will look at the design of data frames, the rules for transmission of these frames, and how these affect design of Ethernet LANs. It was mentioned in chapter one that the original DIX standard and IEEE 802.3 standards differ only slightly. Both these standards will be compared in all aspects so as to make concepts clear and to prevent any confusion. We shall also see how backward compatibility is maintained in the case of faster versions of Ethernet.

9.2 IEEE/ISO standards

The IEEE has been given the task of developing standards for LAN under the auspices of the IEEE 802 committee. Once a draft standard has been agreed and completed, it is passed to the International Standards Organization ISO for ratification. The corresponding ISO standard, which is generally accepted internationally, is given the same number as the IEEE committee, with the addition of an extra ‘8’ in front of the number i.e. IEEE 802 is equivalent to ISO 8802.

These IEEE committees, consisting of various technical, study and working groups, provide recommendations for various features within the networking field. Each committee is given a specific area of interest, and a separate sub-number to distinguish it. The main committees and the related standards are described below.

9.2.1 Internetworking

These standards are responsible for establishing the overall LAN architecture, including internetworking and management. They are a series of sub standards, which include:

802.1B – LAN management

802.1D – Local bridging

802.1p – This is part of 802.1D and provides support for traffic-class expediting (for traffic prioritization in a switching hub) and dynamic multicast filtering)to identify which ports to use when forwarding multicast packages)

802.1Q – This is a VLAN standard providing for a vendor-independent way of implementing VLANs

802.1E – System load protocol
802.1F – Guidelines for layer management standards
802.1G – Remote MAC bridges
802.1I – MAC bridges (FDDI supplement)

802.2 Logical link control

This is the interface between the network layer and the specific network environments at the physical layer. The IEEE has divided the data link layer in the OSI model into two sublayers – the media access MAC sublayer, and the logical link layer LLC. The logical link control protocol (LLC) is common for all IEEE 802 standard network types. This provides a common interface to the network layer of the protocol stack.

The protocol used at this sub layer is based on IBM's SDLC protocol, and can be used in three modes, or types:

- **Type 1** – Unacknowledged connectionless link service
- **Type 2** – Connection-oriented link service
- **Type 3** – Acknowledged connectionless link service, used in real time applications such as manufacturing control

802.3 CSMA/CD local area networks

The carrier-sense, multiple access with collision detection type LAN is commonly, but erroneously, known as an Ethernet LAN. The CSMA/CD relates to the shared medium access method, as does each of the next two standards.

The committee was originally constrained to only investigate LANs with a transmission speed not exceeding 20 Mbps. However, that constraint has now been removed and the committee has published standards for a 100 Mbps 'Fast Ethernet' system and a 1000 Mbps 'Gigabit Ethernet', and, a 10000 Mbps '10 Gigabit Ethernet' system.

802.3ad Link aggregation

This standard is complete and has been implemented. It provides a vendor-neutral way to operate Ethernet links in parallel.

802.3x Full-duplex

This is a full-duplex supplement providing for explicit flow control by optional MAC control and PAUSE frame mechanisms.

802.4 Token bus LANs

The other major access method for a shared medium is the use of a token. This is a type of data frame that a station must possess before it can transmit messages. The stations are connected to a passive bus, although the token logically passes around in a cyclic manner.

802.5 Token ring LANs

As in 802.4, data transmission can only occur when a station holds a token. The logical structure of the network wiring is in the form of a ring, and each message must cycle through each station connected to the ring.

802.6 Metropolitan area networks

This committee is responsible for defining the standards for MANs. It has recommended that a system known as Distributed Queue Data Bus DQDB be utilized as a MAN standard. The explanation of this standard is outside the scope of this manual. The committee is also investigating cable television interconnection to support data transfer.

802.7 Broadband LANs Technical Advisory Group (TAG)

This committee is charged with ensuring that broadband signaling as applied to the 802.3, 802.4 and 802.5 medium access control specifications remains consistent. Note that there is a discrepancy between IEEE 802.7 and ISO 8802.7. The latter is responsible for slotted ring LAN standardization.

802.8 Fiber optic LANs TAG

This is the fiber optic equivalent of the 802.7 broadband TAG. The committee is attempting to standardize physical compatibility with FDDI and synchronous optical networks (SONET). It is also investigating single mode fiber and multimode fiber architectures.

802.9 Integrated voice and data LANs

This committee has recently released a specification for Isochronous Ethernet as IEEE 802.9a. It provides a 6.144 Mbps voice service (96 channels at 64 kbps) multiplexed with 10 Mbps data on a single cable. It is designed for multimedia applications.

802.10 Secure LANs

Proposals for this standard included two methods to address the lack of security in the original specifications. These are:

A secure data exchange (SDE) sublayer that sits between the LLC and the MAC sublayers. There will be different SDEs for different systems, e.g. military and medical.

A secure interoperable LAN System (SILS) architecture. This will define system standards for secure LAN communications.

The standard has now been approved. The approved standards listed below provide IEEE 802 environments with:

- Security association management
- Key management (manual, KDC, and certificate based)
- Security labeling
- Security services (data confidentiality, connectionless integrity, data origin authentication and access control)

The Key Management Protocol (KMP) defined in Clause 3 of IEEE 802.10c is applicable to the secure data exchange (SDE) protocol contained in the standards and other security protocols.

Approved standards (available at http://standards.ieee.org/getieee802/) include:

- IEEE Standard for Interoperable LAN/MAN Security (SILS), IEEE Std 802.10-1998
- Key Management (Clause 3), IEEE Std 802.10c-1998 (supplement)
- Security Architecture Framework (Clause 1), IEEE Std 802.10a-1999 (supplement)

802.11 Wireless LANs

Some of the techniques being investigated by this group include spread spectrum signaling, indoor radio communications, and infrared links.

Several Task Groups are working on this standard. Details and corresponding status of the task groups are as follows:

- The MAC group has completed developing one common MAC for WLAN applications, in conjunction with PHY group. Their work is now part of the standard

- PHY group has developed three PHYs for WLAN applications using Infrared (IR), 2.4 GHz Frequency Hopping Spread Spectrum (FHSS), and 2.4 GHz Direct Sequence Spread Spectrum (DSSS). Work is complete and has been part of the standard since 1997
- The Tga group developed a PHY to operate in the newly allocated UNII band. Work is complete, and is published as 802.11a-1999
- The Tgb group developed a standard for a higher rate PHY in 2.4 GHz band. Work is complete and issued as a part of 802.11b-1999

802.12 Fast LANs

Two new 100 Mbps LAN standards were ratified by the IEEE in July 1995, IEEE 802.3u and IEEE 802.12. These new 100 Mbps standards were designed to provide an upgrade path for the many 10s of millions of 10BaseT and token ring users worldwide. Both of the new standards support installed customer premises cabling, existing LAN management and application software.

IEEE 802.3 and IEEE 802.12 have both initiated projects to develop gigabit per second LANs initially as higher speed backbones for the 100 Mbps systems.

Demand priority summary

Demand priority is a protocol that was developed for IEEE 802.12. It combines the best characteristics of Ethernet (simple, fast access) and token ring (strong control, collision avoidance, and deterministic delay). Control of a demand priority network is centered in the repeaters and is based on a request/grant handshake between the repeaters and their associated end nodes. Access to the network is granted by the repeater to requesting nodes in a cyclic round robin sequence. The round robin protocol has two levels of priority, normal and high priority. Within each priority level, selection of the next node to transmit is determined by its sequential location in the network rather than the time of its request. The demand priority protocol has been shown to be fair and deterministic. IEEE 802.12 transports IEEE 802.3, Ethernet and IEEE 802.5 frames.

Scalability of demand priority: burst mode

The demand priority MAC is not speed sensitive. As the data rate is increased, the network topologies remain unchanged. However, since Ethernet or token ring frame formats are used the efficiency of the network would decrease with increased data rate.

To counteract this a packet burst mode has been introduced into the demand priority protocol. Burst mode allows an end node to transmit multiple packets for a single grant.

The new packet burst mode may also be implemented at 100 megabits per second, as it is backwards compatible with the original MAC. Analysis indicates that minimum efficiencies of 80% are possible at 1062.5 MBaud.

Higher speed IEEE 802.12 physical layers

Higher speed IEEE 802.12 will leverage some of the physical layers and control signaling developed for a fiber channel. The fiber channel baud rates leveraged by IEEE 802.3 are 531 MBaud and 1062.5 MBaud.

Multimode fiber and short wavelength laser transceivers will be used to connect repeaters or switches separated by less than 500 m within buildings. Single mode fiber and laser transceivers operating at a wavelength of 1300 nm will be used for campus backbone links. It has been shown that FC-0 and FC-1 can be used to support an IEEE 802.12 MAC or switch port.

A new physical layer under development in IEEE 802.12 will support desktop connections at 531 MBaud over 100 m of UTP category 5. The physical layer will utilize

all pairs of a four pair cable. The proposed physical layer incorporates a new 8B3N code and provides a continuously available reverse channel for control. There is no requirement for echo cancellation, which simplifies the implementation. Potential for class B radiated emissions compliance has been demonstrated.

9.3 Ethernet frames

The term Ethernet originally referred to a LAN implementation standardized by Xerox, Digital, and Intel; the original DIX standard. The IEEE 802.3 group standardized operation of a CSMA/CD network that was functionally equivalent to the DIX II or 'Bluebook' Ethernet.

Data transmission speeds of 100 Mbps (the IEEE 802.3u standard, also called Fast Ethernet) and 1000 Mbps (the 802.3z standard, also called Gigabit Ethernet) have been achieved, and, these faster versions are also included in term 'Ethernet'.

When the IEEE was asked to develop standards for Ethernet, Token Ring, and other networking technologies, DIX Ethernet was already in use. The objective of the 802 committee was to develop standards and rules that would be generic to all types of LANs so that data could move from one type of network, say Ethernet, to other type, say token ring. This had potential for conflict with the existing DIX Ethernet implementations. The '802' committee was therefore careful to separate rules for the old and the new since it was recognized there would be a coexistence between DIX frames and IEEE 802.3 frames on the same LAN.

These are the reasons why there is a difference between DIX Ethernet and IEEE 802.3 frames. Despite the two types of frames, we generally refer to both as 'Ethernet' frames in the following text.

9.3.1 DIX and IEEE 802.3 frames

A frame is a packet comprising data bits. The packet format is fixed and the bits are arranged in sequence of groups of bytes (fields). The purpose of each field, its size, and its position in the sequence are all meaningful and predetermined. The fields are Preamble, Destination Address, Source Address, Type or Length, DATA/LLC, and Frame Check Sequence, in that order.

DIX and IEEE 802.3 frames are identical in terms of the number and length of fields. The only difference is in the contents of the fields and their interpretation by the stations which send and receive them. Ethernet interfaces therefore can send either of these frames.

Figure 9.1 schematically shows the DIX as well as IEEE 802.3 frame structures.

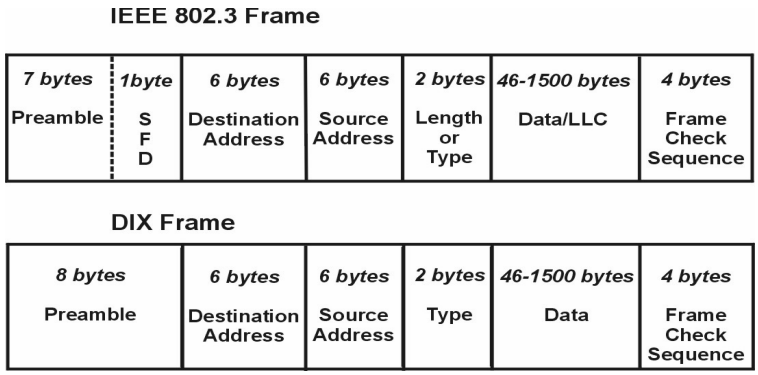


Figure 9.1
IEEE 802.3 and DIX frames

9.3.2 Preamble

The ‘preamble’ of the frame is like the introductory remarks of a speaker. If one misses a few words from a preamble being delivered by a speaker, one does not lose the substance of the speech. Similarly, the preamble in this case is used for synchronization and to protect the rest of the frame even if some start-up losses occur to the signal.

Fast and Gigabit Ethernet have other mechanisms for avoiding signal start-up losses, but in their frames a preamble is retained for purposes of backward compatibility.

Because of the synchronous communication method used, the preamble is necessary to enable all stations to synchronize their clocks. The Manchester encoding method used for 10 Mbps Ethernet is self-clocking since each list contains a signal transition in the module.

Preamble in DIX

Here the preamble consists of eight bytes of alternating ones and zeros, which appear as a square wave with Manchester encoding. The last two bits of the last byte are ‘1,1’. These ‘1,1’ bits signify to the receiving interface that the end of the preamble has occurred and that actual meaningful bits are about to start.

Preamble in IEEE 802.3

Here the preamble is divided in two parts, first one of seven bytes, and another of one byte. This one byte segment is called the start frame delimiter or SFD for short. Here, again, the last two bits of the SFD are ‘1,1’ and with the same purpose as in the DIX standard. There is no practical difference between the preambles of DIX and IEEE – the difference being only semantic.

9.3.3 Ethernet MAC addresses

These addresses are also called hardware addresses or media addresses. Each Ethernet interface needs a unique MAC address, and this is usually allocated at the time of manufacture. The first 24 bits of the MAC address consist of an organizationally unique identifier (OUI), in other words a ‘manufacturer ID’, assigned to a vendor by the IEEE. This is why they are also called vendor codes. The Ethernet vendor combines their 24-bit OUI with a unique 24-bit value that they generate to create a unique 48-bit address for each Ethernet interface they build. The latter 24-bit value is normally issued sequentally.

Organizationally unique identifier (OUI)/‘company_id’

An OUI/‘company_id’ is a 24 bit globally unique assigned number referenced by various standards. OUI is used in the family of the IEEEE802 LAN standards, e.g., Ethernet, Token Ring, etc.

Standards involved with OUI

The OUI defined in the IEEE 802 standard can be used to generate 48-bit universal LAN MAC addresses to identify LAN and MAN stations uniquely, and protocol identifiers to identify public and private protocols. These are used in local and metropolitan area network applications.

The relevant standards include CSMA/CD (IEEE 802.3), Token Bus (IEEE 802.4), Token Ring (IEEE 802.5), IEEE 802.6, and FDDI (ISO 9314-2).

Structure of the MAC addresses

A MAC address is a sequence of six octets. The first three take the values of the three octets of the OUI in order. The last three octets are administered by the vendor. For

example, the OUI AC - DE - 48 could be used to generate the address AC-DE-48-00-00-80

Address administration

An OUI assignment allows the assignee to generate approximately 16 million addresses, by varying the last three octets. The IEEE normally does not assign another OUI to the assignee until the latter has consumed more than 90% of this block of potential addresses. It is incumbent upon the assignee to ensure that large portions of the address block are not left unused in manufacturing facilities.

9.3.4 Destination address

The destination address field of 48 bits follows the preamble. Each Ethernet interface has a unique 48-bit MAC address. This is a physical or hardware address of the interface that corresponds to the address in the destination address field. The field may contain a multicast address or a standard broadcast address.

Each Ethernet interface on the network reads each frame at least up to the end of the destination address field. If the address in the field does not match its own address, then the frame is not read further and is discarded by the interface.

A destination address of all ‘1’s (FF-FF-FF-FF-FF-FF) means that it is a broadcast and that the frame is to be read by all interfaces.

Destination address in DIX standard

The first bit of the address is used to distinguish unicast from multicast/broadcast addresses. If the first bit in the field is zero, then the address is taken to be a physical (unicast) address. If first bit is one, then the address is taken to be multicast address, meaning that the frame is being sent to several (but not all) interfaces.

Destination address in IEEE 802.3

Here, apart from the first bit being significant as in the DIX standard, the second bit is also significant. If the first bit is zero and the second bit is set to zero as well, then the address in the field is a globally administered physical address (assigned by manufacturer of the interface). If first bit is zero but second bit is set to one, then the address is locally administered (by the systems designer/administrator). The latter option is very rarely used.

9.3.5 Source address

This is the next field in the frame after destination address field. It contains the physical address of the interface of transmitting station. This field is not interpreted in any way by the Ethernet MAC protocol, but is provided for use of higher protocols.

Source address field in DIX standard

The DIX standard allows for changing the source address, but the physical address is commonly used.

Source address in IEEE 802.3 standard

IEEE 802.3 does not provide specifically for overriding 48-bit physical addresses assigned by manufacturers, but all interfaces allow for overriding if required by the network administrator.

9.3.6 Type/length field

This field refers to the data field of the frame.

Type field in DIX standard

In the DIX standard, this field describes the type of high-level PDU information that is carried in the data field of the Ethernet frame. For example, the value of 0x0800 (0800 Hex) indicates that the frame is used to carry an IP packet.

Length/type field in IEEE 802.3 standard

When the IEEE 802.3 standard was first introduced, this field was called the length field, indicating length (in bytes) of data to follow. Later on, (in 1997) the standard was revised to include either a type specification or a length specification.

The length field indicates how many bytes are present in data field, from a minimum of zero to a maximum of 1500.

The most important reason for having a minimum length frame is to ensure that all transmitting stations can detect collisions (by comparing what they transmit with what they hear on the network) while they are still transmitting. To ensure this all frames must transmit for more than twice the time it takes a frame to reach the other end.

The data field must contain a minimum 46 bytes or a maximum of 1500 bytes of actual data. The network protocol itself is expected to provide at least 46 bytes of data. If data is less than 46 bytes, padding by dummy data is done to bring the field size to 46 bytes. Before the data in the frame is read, the receiving station must know which of the bytes constitute real data and which part is padding. Upon reception of the frame, the length field is used to determine the length of valid data in the data field, and the pad data is discarded.

If the value in the length/type field is numerically less than or equal to 1500, then the field is being used as a length field, in which case the number in this field represents the number of data bytes in the data field.

If the value in the field is numerically equal to or greater than 1536 (0x0600), then the field is being used as type field, in which case the hexadecimal identifier in the field is used to indicate the type of protocol data being carried in the data field.

9.3.7 Use of type field for protocol identification

The type field number, issued by the IEEE Type Field Registrar, provides a context for interpretation of the data field of the frame. Well-known protocols have defined type numbers.

The IEEE 802.3, Length/type field, originally known as EtherType, is a two-octet field, which takes one of two meanings depending on its numeric value. For numeric evaluation, the first octet is the most significant octet of this field.

When the value of this field is greater than or equal to 1536 decimal, (0x0600) the type field indicates the nature of the MAC client protocol (type interpretation). The length and type interpretations of this field are mutually exclusive.

The type field is very small and therefore its assignment is limited. It is incumbent upon the assignee to ensure that requests for type fields be very limited and only on an as needed basis. Requests for multiple type fields by the same applicant are not granted unless the applicant certifies that they are for unrelated purposes. In particular, only one new type field is necessary to limit reception of a new protocol or protocol family to the intended class of devices. New protocols and protocol families should have provision for a sub type field within their specification to handle different aspects of the application (e.g., control vs. data) and future upgrades.

9.3.8 Data field

Data field in the DIX standard

In the DIX standard, the data field must contain a minimum of 46 bytes and a maximum of 1500 bytes of data. The network protocol software provides at least 46 bytes of data if needed.

Data field in the IEEE 802.3 standard

Here the minimum and the maximum lengths are same as for the DIX standard. The LLC protocol as per IEEE 802.2 may occupy some space in the data field for identifying the type of protocol data being carried by the frame if the type/length field is used for length information. The LLC PDU is carried in the first 10 bytes in the data field.

If the number of LLC octets is less than the minimum number required for the data filed, then pad data octets are automatically added. On receipt of the frame, the length of meaningful data is determined using the length field.

9.3.9 Frame check sequence (FCS) field

This last field in a frame, the same in both DIX and IEEE 802.3 standards, is used to check the integrity of bits in various fields, excluding, of course, the preamble and FCS fields. The CRC (cyclic redundancy check) method is used to compute and check this integrity.

9.4 LLC frames and multiplexing

9.4.1 LLC frames

Since some networks use the IEEE 802.2 LLC standard, it will be useful to examine LLC frames.

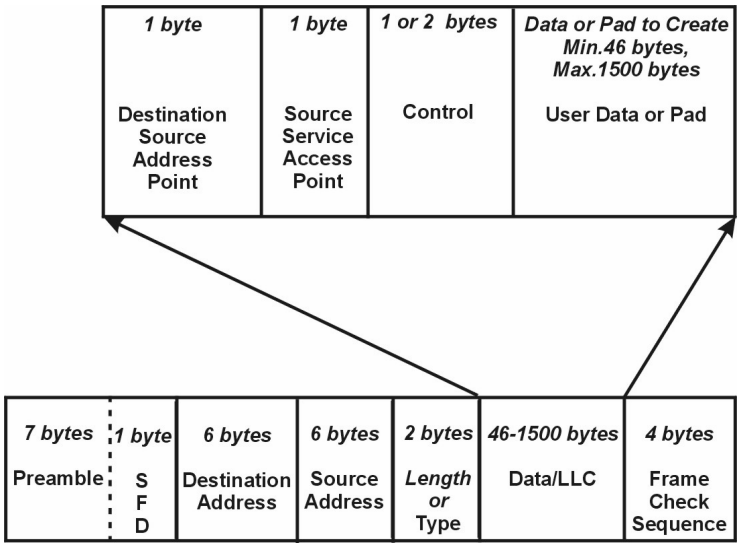


Figure 9.2
LLC protocol data as part of Ethernet frame

9.4.2 LLC and multiplexing

It previous paragraphs it was seen shown that the value of the identifier in the length/type field determines how this field is used. When used as a length field, the IEEE 802.2 LLC identifies the type of high-level protocol being carried in the data field of the frame. The IEEE 802.2 PDU contains a destination service access point or DSAP, (this identifies the high level protocol that the data in the frame is intended for), a source service access point, or SSAP (this identifies the high level protocol from which the data in the frame originated), some control data, and actual user data. Multiplexing and de-multiplexing work in the same way that they do for a frame with a type field. The only difference is that identification of the type of high-level protocol data is shifted to the SSAP, which is located in the LLC PDU. In frames carrying LLC fields, the actual amount of data that can be carried is 3-4 bytes less than in frames that use a type field because of the size of the LLC header.

The reason why the IEEE defined the IEEE 802.2 LLC protocol to provide multiplexing, when the type field does the job equally well, was its objective of standardizing a whole set of LAN technologies and not just IEEE 802.3 Ethernet systems.

802.1p/Q VLAN standards and frames

The IEEE 802.1D standard lays down norms for bridges and switches. The IEEE 802.1p standard, which is a part of 802.1D, provides support for traffic-class expediting and dynamic multicast filtering. It also provides a generic attribute registration protocol (GARP) that can be used by switches and stations to exchange information about various capabilities or attributes that may apply to a switch or port.

The IEEE 802.1Q standard provides a vendor-independent way of implementing VLANs. A VLAN is a group of switch ports that behave as if they are independent switching hub. Provision of VLAN capabilities on a switching hub enables a network manager to allocate a particular set of switch ports to different VLANs.

VLANs can now be based on the content of frames instead of just ports on the hub. There are proprietary frame tagging mechanisms for identifying or tagging frames in terms of which VLAN they belong to. The IEEE 802.1Q provides a vendor independent way of tagging Ethernet frames and thereby implementing VLANs.

Here, 4 bytes of new information containing identification of the protocol and priority information are added after the source address field and before the length/type field. The VLAN tag header is added to the IEEE 802.3 frame as shown below in Figure 9.3:

The tag header consists of two parts. The first part is the tag protocol identifier (TPID), which is a 2-byte field that identifies the frame as a tagged frame. For Ethernet, the value of this field is 0x8100.

The second part is the tag control information (TCI), which is a 2-byte field. The first three bits of this field are used to carry priority information based on the values defined in the IEEE 802.1p standard. The last eleven bits carry VLAN identifier (VID) that uniquely identifies the VLAN to which the frame belongs.

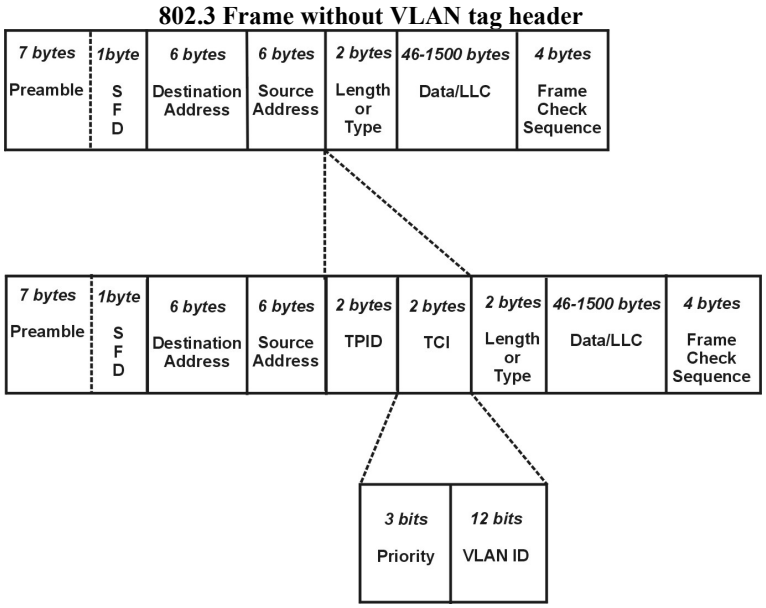


Figure 9.3
IEEE 802.1p/Q VLAN Tag header added to IEEE 802.3 frame

802.1Q thus extends the priority-handling aspects of the IEEE 802.1p standard by providing space in the VLAN tag to indicate traffic priorities.

Addition of the VLAN tag increases the maximum frame size to 1522 bytes.

9.5 Media access control for half-duplex LANs (CSMA/CD)

The MAC protocol for half-duplex Ethernets (DIX as well as IEEE 802.3), decides how to share a single channel for communication between all stations, the transmission being in both directions on the same channel, but not simultaneously.

There is no central controller on the LAN to decide about these matters. Each interface of each station has this protocol and plays by the same MAC rules so that there is a fair sharing of the communication on the channel.

The MAC protocol for half-duplex Ethernet LANs is called CSMA/CD, or carrier sense multiple access/collision detection, after the manner in which the protocol manages the communication traffic.

A detailed look will now be taken at this mechanism.

9.5.1 Terminology of CSMA/CD

Before one gets into a discussion on CSMA/CD, it is necessary to understand the terminology used to describe various features and occurrences of signal transmission in half-duplex Ethernet:

When a message is in the process of being transmitted, the condition is called **carrier**. There is no real carrier signal involved as Ethernet uses a baseband mechanism for transmitting information.

When there is no carrier, the channel is said to be **idle.**

If the channel is not idle, a station wanting to transmit waits for the channel to become idle. This waiting is called **deferring.**

When the channel becomes idle, a station wanting to transmit waits for a predetermined period of time called the **interframe gap.**

When two (or more) signals traveling in opposite directions meet, they **collide** and obstruct each other.

When a transmitting station comes to know of such a collision, the station stops transmission and reschedules it. This is called **collision-detect.**

When collision-detect has taken place, the transmitting station will still transmit 32 bits of data. This data is called a **collision enforcement jam signal** or just **jam signal.** A jam signal is a signal composed of alternating ones and zeroes

After sending a jam signal the transmitting station waits for a random time period before it attempts to transmit again. This waiting is called **back-off.**

The maximum round trip time (RTT) for signal transmission on a LAN (time to go to the farthest station and come back) is called the **slot time.**

9.5.2 CSMA/CD access mechanism

The way the CSMA/CD access mechanism operates is described below:

- A station wishing to transmit listens for the absence of carrier in order to determine if the channel is idle
- If the channel is idle, once the period of inactivity has equaled or exceeded the interframe gap (IFG), the station starts transmitting a frame immediately. If multiple frames are to be transmitted, the station waits for a period of IFG between each successive frame

The IFG is meant to provide recovery time for the interfaces and is equal to the time for the transmission of 96 bits. This is equal to 9.6 microseconds (1 microsecond = 10^{-6} second) for a 10 Mbps network, and 960 nanoseconds and 96 nanoseconds for 100 Mbps and Gigabit networks respectively. 1 nanosecond equals 10^{-9} second.

- If there is a carrier, the station continuously defers till the channel becomes free
- If, after starting transmission, a collision is detected, the station will send another 32 bits of a jam signal. If collision is detected very early or just after the start of transmission, the station will transmit a preamble plus jam signal
- As soon as a collision is detected, two processes are invoked. These are a collision counter and a back-off time calculation algorithm (which is based on random number generation)
- After sending the jam signal, the station stops transmitting and waits for a period equal to the back-off time (which is calculated by the aforementioned

algorithm). On expiry of the back-off time, the station starts transmitting all over again

- If a collision is detected once again, the whole process of backing off and re-transmitting is repeated, but the algorithm, which is given the collision count by the counter, increases the back-off time. This can go on till there are no collisions detected, or upto a maximum of 16 consecutive attempts.
- If the station has managed to send the preamble plus 512 bits, the station has 'acquired' the channel, and will continue to transmit until there is nothing more to transmit. If the network has been designed as per rules, there should be no collisions after acquiring the channel
- The 512-bit slot time mentioned above is for 10 Mbps and 100 Mbps networks. For gigabit networks, this time is increased to 512 bytes (4096 bits).
- On acquiring the channel the collision counter and back-off time calculation algorithm are turned off
- All stations strictly follow the above rules. It is of course assumed that the network is designed as per rules so that all these timing rules provide the intended results

9.5.3 Slot time, minimum frame length, and network diameter

Slot time is based on the maximum round trip signal traveling time of a network. It includes time for a frame to go through all cable segments and devices such as cables, transceivers and repeaters along the longest route.

Slot time has two constituents:

- The time to travel across a maximum sized system from end to end and return
- The maximum time for collision detection, and sending of the jam signal

These two times plus a few bits for safety amount to the slot time of 512-bits for 10 Mbps and 100 Mbps systems. Thus, even when transmitting the smallest legitimate frame, the transmitting station will always get enough time to know about a collision even if collision occurs at the farthest end of the longest route.

The minimum frame size of 512 bits includes 12 bytes for addresses, 2 bytes for length/type field, 46 bytes of data and 4 bytes of FCS. Preamble is not included in this calculation.

The slot time and network size are closely related. This is a trade-off between the smallest frame size and the maximum cable length along the longest route. The signal speed is dependent on the medium – around two-thirds the speed of light on copper, regardless of the bit rate. The transmission speed, for instance 10 Mbps, determines the LENGTH of the frame (in time). Therefore, when going from 10 Mbps to 100 Mbps (factor of 10), the maximum frame size in microseconds will 'shrink' by a factor of 10, and hence the permissible collision domain will reduce from 2500 m. to 250 m. The slot time will reduce from 512 microseconds to 5.12 microseconds. For Gigabit Ethernet the same argument would produce a collision domain of 25 m. and a slot time of 0.512 microseconds, which is ridiculously low. Therefore, the minimum frame size for the latter has been increased to 512 bytes (4096 bits), which gives a physical collision domain of around 200 m. The slot time is used as a basic unit of time for the back-off time calculation algorithm.

In practice many fast Ethernet systems and most Gigabit Ethernet systems operate in full-duplex mode, i.e. with the collision detection circuitry disabled. However, all these

systems have to conform to the collision detection requirements for backwards compatibility with the original IEEE 802.3 standard.

Since a valid collision can occur only within the 512-bit slot time, the length of a frame destroyed in a collision, a 'fragment', will always be smaller than 512 bits. This helps interfaces in detecting fragments and discarding them.

9.5.4 Collisions

Collisions are a normal occurrence in CSMA/CD systems. Collisions are not errors, and they are managed efficiently by the protocol. In a properly designed network, collisions, if they occur, will happen in the first 512 bits of transmission. Any frame encountering a collision is destroyed, its fragments are not bigger than 512 bits. Such a frame is automatically retransmitted without fuss. The number of collisions will increase with transmission traffic.

Collision detection mechanisms

The mechanism for detection of collision depends on the medium of transmission.

On a coaxial cable the transceiver detects collisions by monitoring the average DC signal voltage level which reaches a particular level, triggering the collision detect mechanism circuit.

On link segments, such as twisted-pair or fiber optic media, (which have independent transmit and receive data paths) collisions are detected in a link segment receiver by simultaneous occurrence of activity on both transmit and receive data paths.

Late collisions

A collision in a smoothly functioning network has to occur within the first 512-bit time. If a late collision occurs it signifies a serious error. There is no automatic retransmission of a frame in case of a late collision occurring, and the fault has to be detected by higher-level protocols. The sending interface must wait for acknowledge timers to time-out before resending the frame. This slows down the network.

Late collisions are caused by network segments that exceed the stipulated maximum sizes, or by a mismatch between duplex configurations at each end of a link segment. One end of a link segment may be configured for half-duplex transmission while the other end maybe configured for full-duplex transmission (full-duplex does not use CSMA/CD for obvious reasons).

Excessive signal crosstalk on a twisted pair segment can also result in late collisions.

Collision domains

A collision domain is a single half-duplex Ethernet system of cables; repeaters, station interfaces, and other network hardware where all are a part of the same signal-timing and slot-timing domain. A single collision domain may encompass several segments as long as they are linked together by repeaters. (A repeater enforces any collisions on any segment attached to it, for example, a collision on segment x is enforced by the repeater onto segment y by sending a jam signal.) A repeater makes multiple network segments function like a single cable.

9.6 MAC (CSMA-CD) for gigabit half-duplex networks

Most Gigabit Ethernet systems use the full-duplex method of transmission, so the question of collisions does not arise. However, the IEEE has specified a half-duplex CSMA/CD mode for gigabit networks to insure that gigabit networks get included in the IEEE 802.3 standard. This matter is narrated in brief below for the sake of completeness.

If the same norms (same minimum frame length and slot times) are applied to gigabit networks then the effective network diameter becomes very small at about 20 meters. This is too small a value to be of any practical use. A network diameter of say 200 meters would be more useful.

To solve this problem the slot time is increased while keeping the same minimum frame length. The only way this can be done is by increasing the 'minimum frame length'. Just specifying a longer length would make the system incompatible with systems using 512 bits as the minimum frame length. Appending or suffixing non-data signals, called extension bits, at the end of an FCS field, has overcome this problem. This is called 'carrier extension'.

With carrier extension, the minimum frame size is increased from 512 bits (64 bytes) to 4096 bits (512 bytes). This now increases the slot time proportionately. All this, of course, increases overhead i.e. it decreases the proportion of actually useful data (original data) to total traffic, thereby decreasing efficiency of the network.

All this is somewhat academic since most gigabit Ethernets use full-duplex methods so that CSMA/CD is not at all needed.

9.7 Multiplexing and higher level protocols

Several computers using different high-level protocols can use the same Ethernet network. Identifying which protocol is being used and carried in each data frame is called multiplexing, which allows placing of multiple sources of information on a single system.

The type field was originally used for multiplexing, For example, a higher-level protocol creates a packet of data, and software inserts an appropriate hexadecimal value in the type field of the Ethernet frame. The receiving station uses this value in the type field to de-multiplex the received frame.

The most widely used high-level protocol today is TCP/IP, which can use both type and length fields in the Ethernet frame. Newer high-level protocols developed after the creation of the IEEE 802.2 LLC use the length field and LLC mechanism for multiplexing and de-multiplexing of frames.

9.8 Full-duplex transmissions

The full-duplex mode of transmission allows simultaneous two-way communication between two stations, which must use point-to-point links with media such as twisted-pair or fiber optic cables to provide independent transmit and receive data paths. Because of the absence of CSMA/CD there can only be two nodes (for example an NIC and switch port) in a collision domain.

Full-duplex mode doubles the bandwidth of media as compared with that of half-duplex mode. The maximum segment length limitation imposed by timing requirements of half-duplex mode does not apply to full-duplex mode.

The IEEE specifies full-duplex mode in its 802.3x supplement (100 Mbps Ethernet) along with optional mechanisms for flow control, namely MAC Control and PAUSE.

9.8.1 Features of full-duplex operation

For full-duplex operation, certain requirements must be fulfilled. These include independent data paths for transmit and receive mode in cabling media, a point-to-point link between stations, and the capability and configuration of interfaces of both stations for simultaneous transmission and receipt of data frames.

In full-duplex mode a station wishing to transmit ignores carrier sense and does not have to defer to traffic. However, the station still waits for an interframe gap period between its own frame transmissions, as in half-duplex mode, so that interfaces at each end of the link can keep up with the full frame rate.

Since there are no collisions, the CSMA/CD mechanism is deactivated at both ends.

Although full-duplex mode doubles bandwidth, this usually does not result in a doubling of performance because most network protocols are designed to send data and then wait for an acknowledgment. This could lead to heavy traffic in one direction and negligible traffic in the return direction. However, a network backbone system using full-duplex links between switching hubs will typically carry multiple conversations between many computers and the aggregate traffic on a backbone system will therefore tend to be more symmetrical.

It is essential to configure both ends of a communication link for full-duplex operation, or else serious data errors may result. Auto-negotiation for automatic configuration is recommended wherever possible. Since support for auto-negotiation is optional for most media systems, a vendor may not have provided for it. In such case careful manual configuration of BOTH ends of the links is necessary. One end obeying full duplex while the other is still on half-duplex will definitely result in loss of frames.

9.8.2 Ethernet flow control

Network backbone switches connected by full-duplex links can be subject to heavy traffic, sometimes overloading internal switching bandwidth and packet buffers, which are apportioned to switching ports. To prevent overloading of these limited resources, a variety of flow control mechanisms (for example use of a short burst of carrier signal sent by a switching hub to cause stations to stop sending data if buffers are full) are offered by hub vendors for use on half-duplex segments. These are not, however, useful on full-duplex segments.

The IEEE has provided for optional MAC Control and PAUSE specifications in its 802.3x Full-duplex supplement.

9.8.3 MAC control protocol

The MAC control system provides a way for the station to receive a MAC control frame and act upon it. MAC control frames are identified with a type value of 0x8808. A station equipped with optional MAC control receives all frames using normal Ethernet MAC functions, and then passes the frames to the MAC control software for interpretation. If the frame contains hex value 0x8808 in the type field, then the software reads the frame, reads MAC control operation codes carried in the data field and takes action accordingly.

MAC control frames contain operational codes (opcodes) in the first two bytes of the data field.

9.8.4 PAUSE operation

The PAUSE system of flow control uses MAC control frames to carry PAUSE commands. The opcode for the PAUSE command is 0x0001.

When a station issues a PAUSE command, it sends a PAUSE frame to 48-bit MAC address of 01-80-C2-00-00-01. This special multicast address is reserved for use in PAUSE frames, simplifying the flow control process, because a frame with this address will not be forwarded to any other port of the hub, but will be interpreted and acted upon.

The particular multicast address used is selected from a range of addresses that have been reserved by the IEEE 802.1D standard.

A PAUSE frame includes the PAUSE opcode as well as the period of pause time (in the form of a two byte integer) being requested by the sending station. This time is the length of time for which the receiving station will stop transmitting data. Pause time is measured in units of 'quanta' where one quanta is equal to 512 bit times.

Figure 9.4 shows a PAUSE frame wherein the pause time requested is 3 quantas or 1536 bit times.

A PAUSE request provides real-time flow control between switching hubs, or even between a switching hub and a server, provided of course they are equipped with optional MAC control software and are connected by a full-duplex link.

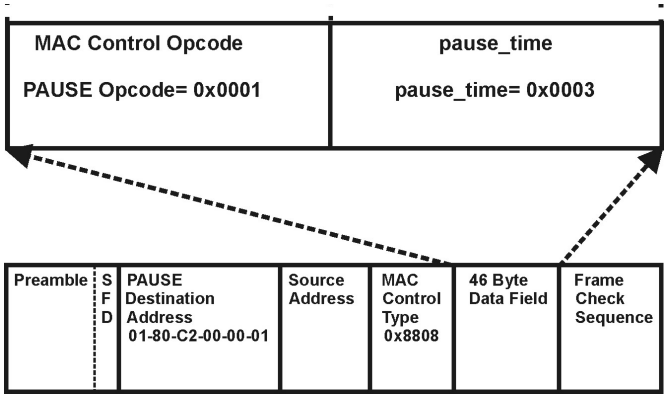


Figure 9.4
PAUSE frame with MAC control opcode and pause time

9.9 Auto-negotiation

9.9.1 Introduction to auto-negotiation

Auto-negotiation involves the exchange of information between stations about their capabilities over a link segment and performing of automatic configuration to achieve the best possible mode of operation over a link. Auto-negotiation enables an Ethernet equipped computer to communicate at the highest speed offered by a multi-speed switching hub port.

A switching hub capable of supporting full-duplex operation on some or all of its ports can announce the fact using auto-negotiation. Stations supporting full-duplex operation and connected to the hub can then automatically configure themselves to use the full-duplex mode when interacting with the hub.

Automatic configuration using auto-negotiation makes it possible to have twisted-pair Ethernet interfaces that can support several speeds. Twisted-pair ports and interfaces can configure themselves to operate at 10 Mbps, 100 Mbps, or 1000 Mbps.

The auto-negotiation system has the following features:

- It is designed to work over link segments only. A link segment can have only two devices connected to it, one at each end
- Auto-negotiation takes place during link initialization. When a device is turned on, or an Ethernet cable is connected, the link is initialized by the devices at each end of the link. This initialization and auto-negotiation happens only once, before transmission of any data over the link

- Auto-negotiation uses its own signaling system. This signaling system is designed for twisted-pair cabling

9.9.2 Signaling in auto-negotiation

Auto-negotiation uses fast link pulse (FLP) signals to carry information. These signals are a modified version of normal link pulse (NLP) signals used for verifying link integrity on 10BaseT system.

FLPs are specified for following twisted-pair media systems:

- 10BaseT
- 100BaseTX (using unshielded twisted-pair)
- 100BaseT4
- 100BaseT2
- 1000BaseT

100BaseTX with shielded twisted-pair cable and 9-pin connectors will not support auto-negotiation. There is also no IEEE auto-negotiation standard for fiber optic Ethernet systems except for fiber optic gigabit Ethernet systems.

9.9.3 FLP details

Fast link pulses are sent in bursts of pulses in 33 pulse positions where each pulse position may contain a pulse. Each pulse is 100 nanoseconds long and the time between each successive burst the same as that between NLPs. This fact deludes a 10BaseT device that is receiving an NLP; thus providing backward compatibility with older 10BaseT equipment that does not support auto-negotiation.

Of the 33 pulse positions, the 17 odd-numbered positions each holds a link pulse that represents clock information. The 16 even-numbered pulse positions carry data. The presence or absence of a pulse in an even numbered position represents logic 1 and logic 0 respectively. This coding in even numbered positions is for the transmission of 16-bit link code words that contain auto-negotiation information.

Each burst of 33 pulse positions contains a 16-bit message, and a device can send as many bursts as are needed. Sometimes the negotiation task may get completed in the first message in the first burst itself. The first message is called the base page. Mapping of the base page message is shown in Figure 9.5.

The 16 bits are labeled D0 through D15. D0 to D4 are used as a selector field for identifying the type of LAN technology in use, allowing the protocol to be extended to other LANs in future. For Ethernet, the S0 bit is set to 1 and S2 to S4 are all set to zero.

The 8-bit field from D5 to D12 is called the technology ability field. Positions A0 to A7 are allotted to the presence or absence of support for various technologies as shown in Figure 9.5. If a device supports one or more technologies, the corresponding bits are set to 1; else they are set to zero as the case may be. Two reserved bit positions, A6 and A7, are for future use.

Bit D13 is called the remote fault indicator. This bit can be set to 1 if a fault has been detected at a remote end.

Bit D14 is set to 1 to acknowledge receipt of the 16-bit message. Negotiation messages are sent repeatedly until the link partner acknowledges them, thus completing the auto-negotiation process.

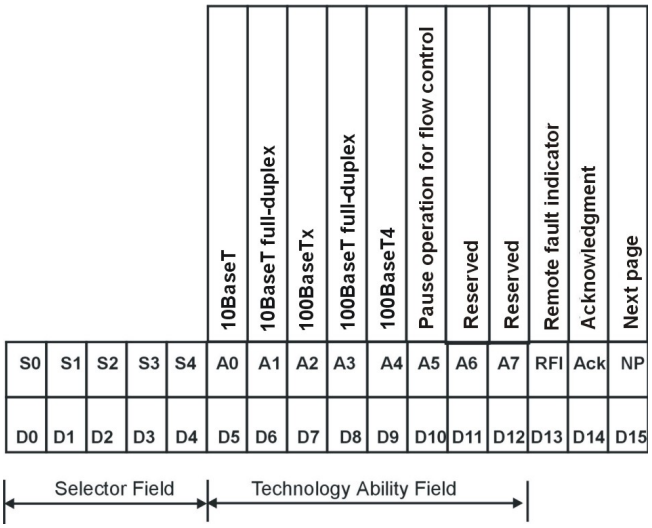


Figure 9.5
Auto-negotiation base page mapping

The link partner sends an acknowledgement only after three consecutive identical messages are received.

Bit D15 is used to indicate if there is a next page, that is, if more information on capabilities is to follow. Next page capability has been provided for sending vendor-specific information or any new configuration commands that may be required in future developments. 1000BaseT gigabit systems use this method for their configuration.

Once auto-negotiation is complete, further bursts of pulses will not be sent unless a link has been down due to any reason and reconnection takes place.

9.9.4 Matching of best capabilities

Once devices connected to two ends of a link have informed each other of their capabilities, the auto-negotiation protocol finds the best match for transmission or the highest common denominator between the capabilities of the two devices. This is based on priorities specified by the standard. Priority is decided by type of technology and not by the order of bits in the technology ability field of base page.

Priorities, ranking from highest to lowest, are as follows:

1	1000BaseT	full duplex
2	1000BaseT	
3	100BaseT2	full duplex
4	100BaseTX	full duplex
5	100BaseT2	
6	100BaseT4	
7	100BaseTX	
8	10BaseT	full duplex
9	10BaseT	

Thus if both devices support say, 100BaseTX as well as 10BaseT, then 100BaseTX will be selected for transmission. Note that other things being the same, full-duplex has higher priority than half-duplex.

If both devices support the PAUSE protocol and the link is configured for full-duplex, then PAUSE control will be selected and used. The priority list above is based on data

rates while PAUSE control has nothing to do with data rates. Therefore, PAUSE is not part of the priority list.

If auto-negotiation does not find any common support on devices at both ends, then a connection will not be made at all and the port will be left in the 'off' position.

9.9.5 Parallel detection

Auto-negotiation is optional for most media systems (except for 1000BaseT systems) because many of the media systems were developed before auto-negotiation was developed. Therefore, the auto-negotiation system has been made compatible with those devices that do not have it. If Auto-negotiation exists only at one end of a link, then the protocol is designed to detect this condition and respond by using 'parallel detection'.

Parallel detection can detect the media system supported at the other end and can hence set the port for that media system. It, however, cannot detect whether the other end supports full-duplex or not. Even if the other end supports full-duplex, parallel detection will set the port for half-duplex mode. Parallel detection is not without problems, and it is preferable for network managers to go for auto-negotiation on all their devices.

9.10 Deterministic Ethernet

Deployment of Ethernet in the industrial environment is increasing but a debate goes on as to whether Ethernet can deliver where mission critical applications are concerned. The issue here is the requirement of a deterministic system when industrial process controls are the application environment

Many people look down upon Ethernet because the CSMA/CD systems are not deterministic. A deterministic system in this context means a system that will deliver the transmission in very short time – as specified and required by the process control parameter. The CSMA/CD style of operation of Ethernet is definitely not deterministic, but probabilistic. However, this lack of determinism is being steadily chiseled away.

The features that are making Ethernet deterministic are as follows:

- Full-duplex operation is deterministic, because CSMA/CD is irrelevant here
- VLANs and prioritized frames (IEEE 802.1p/Q) have also moved Ethernet towards full determinism

More on this subject and on industrial Ethernet will follow later in this manual.

10

Physical layer implementations of Ethernet media systems

Objectives

When you have completed study of this chapter you should be able to:

- Describe essential medium-independent Ethernet hardware
- Explain methods of connection of 10Base5, 10Base2 and 10BaseT networks
- Understand design rules for 10 Mbps Ethernet
- List basic methods used to achieve higher speeds
- Explain various 100 Mbps media systems and their design rules
- Understand 1000 Mbps media systems and their design rules
- Be familiar with proposed 10 Gigabit Ethernet technology standard

10.1 Introduction

Physical layer implementations of media systems for Ethernet networks are dealt with in this chapter. Sending data from one station to another requires a media system based on a set of standard components. Some of these are hardware components specific to the type of media being used while some are common to all media. This chapter will deal with various media systems used, starting from some basic components common to all media, then 10 Mbps Ethernet, 100 Mbps Ethernet, Gigabit Ethernet, and finally 10 Gigabit Ethernet.

10.2 Components common to all media

Hardware components common to all media or medium-independent hardware components include:

- The attachment unit interface (AUI) for 10 Mbps systems
- The medium-independent interface (MII) for 10 and 100 Mbps systems

- The Gigabit medium-independent interface (GMII) for Gigabit systems
- Internal and external transceivers
- Transceiver cables
- Network interface cards

10.2.1 Attachment unit interface (AUI)

The AUI is a medium independent attachment for 10 Mbps systems and can be connected to several 10 Mbps media systems. It connects an Ethernet interface to an external transceiver though a 15-pin male AUI connector, transceiver cable, and a female 15-pin AUI connector. The whole set carries 3 data signals, (transmit data from interface to transceiver, receive data from transceiver to interface, and collision presence signal), and 12-volt power from Ethernet to transceiver. Figure 10.1 shows the mapping of pins of the AUI connector.

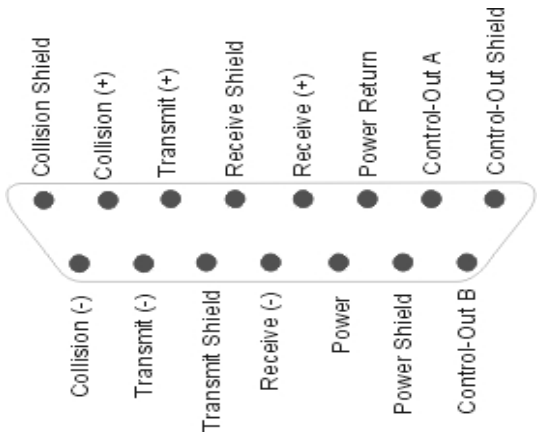


Figure 10.1
AUI 15 Pin connector mapping

The transceiver, also known as a medium attachment unit (MAU), transmits and receives signals to and from the physical medium. Signals that transceivers send to physical media and receive from it are different depending on the media. But signals that travel between transceiver and interface are of the same type, irrespective of the media being used on the other side of the transceiver. A transceiver is specific to each 10 Mbps media system and is not part of the AUI.

The AUI IEEE standard cable has no minimum length specified, and can be as long as 50 m. Office grade AUI cable is thinner and more flexible than IEEE standard cable but suffers from higher signal losses. Office grade cable therefore should not be more than 12.5 m. in length. Signals between the transceiver and Ethernet interface are low voltage (+0.7 volts to –0.7 volts) differential signals, there being two wires for each signal, one for the +ve and the other for the –ve part of the signal

Some external transceivers are small enough to fit directly onto the 15-pin AUI connector of the Ethernet interface, eliminating the need for a cable.

Among the many technical innovations of the 10 Gigabit Ethernet Task Force is an interface called the XAUI (pronounced ‘Zowie’). The ‘AUI’ portion is borrowed from the Ethernet attachment unit interface. The ‘X’ represents the Roman numeral for ten and implies ten gigabits per second. The XAUI is designed as an interface extender, and the interface, which it extends, is the XGMII – the 10 Gigabit media independent interface.

The XAUI is a low pin count, self-clocked serial bus that is directly evolved from the Gigabit Ethernet 1000BaseX PHY. The XAUI interface speed is 2.5 times that of

1000BaseX. By arranging four serial lanes, the 4-bit XAUI interface supports the ten-times data throughput required by 10 Gigabit Ethernet. More information on XAUI can be found in last section of this chapter.

10.2.2 Medium-independent interface (MII)

The MII is an updated version of the original AUI. It supports 10 Mbps as well as 100 Mbps systems. It is designed to make signaling differences among the various media segments transparent to the Ethernet interface.

It does this by converting signals received from various media segments by the PHY (the transceiver of MII is called physical layer device (PHY), and not MAU as in the case of an AUI transceiver) into a standardized digital format and submitting the signals to the networked device over a 4-bit wide data path.

The 4-bit wide data path is clocked at 25 MHz to provide a 100 Mbps transfer speed, or at 2.5 MHz, to provide transfer speed at 10 Mbps. The MII provides a set of control signals for interacting with external transceivers to set and detect various modes of operation.

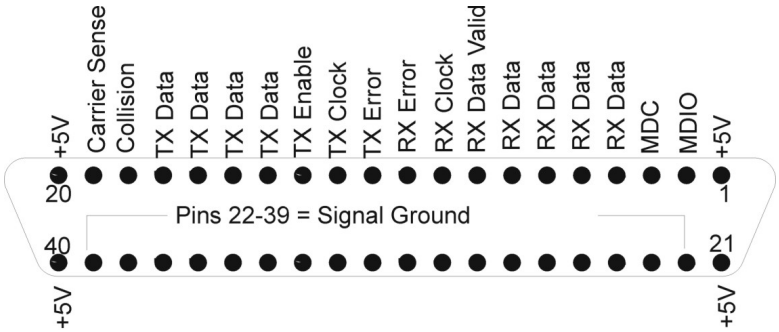


Figure 10.2
MII connector pins mapping

The MII connector has 40 pins whose functions are listed below:

- +5 volts**:** pins 1, 20, 21, and 40 are used to carry +5 volts at a maximum current of 750 milliamps
- Signal ground**:** pins 22 to 39 carry the signal ground wires
- I/O: pin 2 is for a management data input/output signal representing control and status information. This enables functions like setting and resetting of various modes, and testing
- Management data clock**:** this Pin is for clocking purposes for use as a timing reference for serial data sent on pin 2
- RX data**:** pins 4, 5, 6, 7 provide a 4-bit receive data path
- RX data valid: a receive data valid signal is carried by pin 8
- RX clock: pin 9 serves the receive clock running at 25 MHz or 2.5 MHz for 100 Mbps/10 Mbps systems respectively
- RX error**:** the error signal is carried by pin 10
- TX error**:** pin 11 is for the signal used by a repeater to force propagation of received errors. This signal may be used by a repeater but not by a station
- TX clock**:** pin 12 carries the transmit clock running at speeds equal to RX clock
- TX enable**:** pin 13 does the job of sending the transmit enable signal from the DTE to which data is being sent

- TX data**:** pins 14 to 17 provide a 4-bit wide data path from interface to transceiver
- Collision**:** collision signals in the case of half-duplex mode are carried by pin 18. In case of full-duplex mode, this signal is undefined and the collision light may glow erratically and is to be ignored
- Carrier sense**:** pin 19 carries this signal indicating activity on the network segment from interface to transceiver

The MII cable consists of 20 twisted pairs with 40 wires and a maximum length of 0.50 meters. Since the majority of external transceivers sit directly on the MII connector, the MII cable is not needed.

10.2.3 Gigabit medium-independent interface (GMII)

GMII supports 1000 Mbps Gigabit Ethernet systems. High speeds here make it difficult to engineer an externally exposed interface. Unlike AUI and MII, GMII only provides a standard way of interconnecting integrated circuits on circuit boards. Since there is no exposed GMII, an external transceiver cannot be connected to a Gigabit Ethernet system.

Unlike MII, which provides a 4-bit wide data path, GMII provides an 8-bit wide data path. Other features are similar.

GMII supports only 1000 Mbps operation. Transceiver chips that implement both MII and GMII circuits on a given Ethernet port are available, providing support for 10/100/1000 Mbps over twisted-pair cabling with automatic configuration using auto-negotiation.

Devices do not require media independence provided by GMII that only support the 1000BaseX media family, because the 1000BaseX system is based on signaling originally developed for the ANSI fiber channel standard. If only 1000BaseX support is needed, then an interface called ten-bit interface (TBI) is used. The TBI data path is 10 bits wide to accommodate 8B/10B signal encoding.

10.3 10 Mbps media systems

The IEEE 802.3 standard defines a range of cable types that can be used. They include coaxial cable, twisted pair cable and fiber optic cable. In addition, there are different signaling standards and transmission speeds that can be utilized. These include both base band and broadband signaling, and speeds of 1 Mbps and 10 Mbps.

The IEEE 802.3 standard documents (ISO8802.3) support various cable media and transmission rates up to 10 Mbps as follows:

- 10Base2**:** thin wire coaxial cable (0.25 inch diameter), 10 Mbps, single cable bus
- 10Base5**:** thick wire coaxial cable (0.5 inch diameter), 10 Mbps, single cable bus
- 10BaseT**:** unscreened twisted pair cable (0.4 to 0.6 mm conductor diameter), 10 Mbps, with hub topology
- 10BaseF**:** optical fiber cables, 10 Mbps, twin fiber, and used for point–to-point transmissions

10.3.1 10Base5 systems

'Thick Ethernet' or 'Thicknet' was the first Ethernet media system specified in the original DIX standard of 1980. It is limited to speeds of 10 Mbps. The medium is a coaxial cable, of 50-ohm characteristic impedance, and yellow or orange in color. The naming convention 10Base5 means 10 Mbps, baseband signaling on a cable that will support 500-meter (1640 feet) segment lengths.

This system is not of much use as a backbone network due to incompatibility with higher speed systems. If you need to link LANs together at higher speeds, this system will have to be replaced with twisted-pair or fiber optic cables. Virtually all new installations are therefore based on twisted-pair cabling and fiber optic backbone cables. 10Base5 cable has a large bending radius so cannot normally be taken to the node directly. Instead, it is laid in a cabling tray etc and the transceiver (medium attachment unit or MAU) is installed directly on the cable. From there an intermediate cable, known as an attachment unit interface (AUI) cable is used to connect to the NIC. This cable can be a maximum of 50 meters (164 feet) long, compensating for the lack of flexibility of placement of the segment cable. The AUI cable consists of five individually shielded pairs – two each (control and data) for both transmitting and receiving, plus one for power.

The connection to the coaxial cable is made by either cutting the cable or fitting N-connectors and a coaxial T or by using a 'bee sting' or 'vampire' tap. This mechanical connection clamps directly over the cable. The electrical connection is made via a probe that connects to the center conductor and sharp teeth physically puncture the cable sheath to connect to the braid. These hardware components are shown in Figure 10.3.

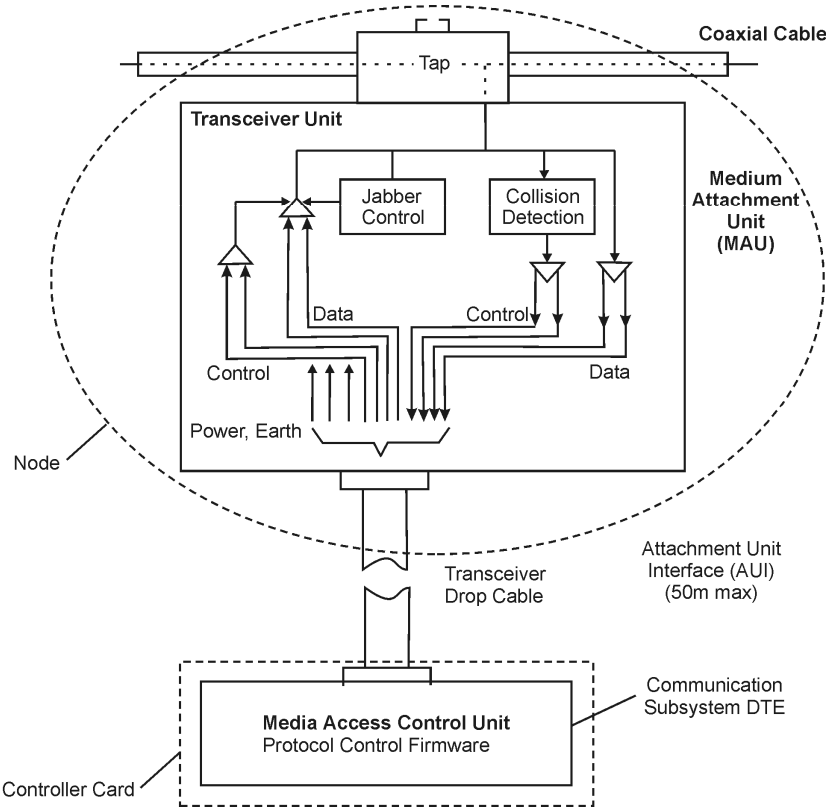


Figure 10.3
10Base5 hardware components

The location of the connection is important to avoid multiple electrical reflections on the cable, and the Thicknet cable is marked every 2.5 meters with a black or brown ring to indicate where a tap should be placed. Fan-out boxes can be used if there are a number of nodes for connection, allowing a single tap to feed each node as though it was individually connected. The connection at either end of the AUI cable is made through a 25 pin D-connector with a slide latch, called a DIX connector.

There are certain requirements if this cable architecture is used in a network.

These include:

- Segments must be less than 500 meters in length to avoid signal attenuation problems
- No more than 100 taps on each segment i.e. not every potential connection point can support a tap
- Taps must be placed at integer multiples of 2.5 meters
- The cable must be terminated with an N type 50-ohm terminator at each end
- It must not be bent at a radius less than 25.4 cm or 10 inches
- One end of the cable screen must be earthed

The physical layout of a 10Base5 Ethernet segment is shown in Figure 10.4.

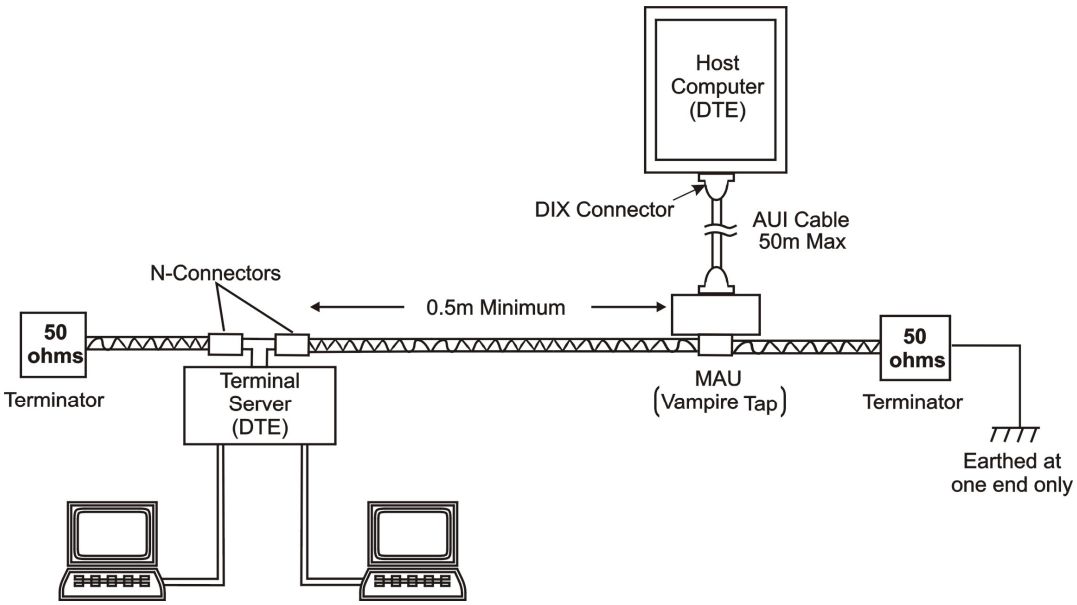


Figure 10.4
10Base5 Ethernet segment

The Thicknet cable was extensively used as a backbone cable, but now use of twisted pair and fiber is more popular. Note that when an MAU (tap) and AUI cable is used, the on-board transceiver on the NIC is not used. Instead, there is a transceiver in the MAU and this is fed with power from the NIC via the AUI cable.

Since the transceiver is remote from the NIC, the node needs to be aware that the transceiver can detect collisions if they occur. A signal quality error (SQE), or heartbeat, test function in the MAU performs this confirmation. The SQE signal is sent from the MAU to the node on detecting a collision on the bus. However, on completion of every frame transmission by the MAU, the SQE signal is asserted to ensure that the circuitry remains active, and that collisions can be detected.

Not all components (e.g. 10Base5 repeaters) support the SQE test and mixing those that do with those that don't could cause problems. Specifically, if the 10Base5 repeater was to receive an SQE signal after a frame had been sent, and it was not expecting it, it could think it was seeing a collision. In turn, the 10Base5 repeater will then transmit a jam signal every time.

Encoding is done using Manchester encoding and only half-duplex mode is possible.

10.3.2 10Base2 systems

The other type of coaxial cable Ethernet network is 10Base2, often referred to as 'Thin net' or sometimes 'thin wire Ethernet'. It uses type RG-58 A/U or C/U with 50-ohm characteristic impedance and a 5 mm diameter. The cable is normally connected to the NICs in the nodes by means of a BNC T-piece connector.

Connectivity requirements include:

- It must be terminated at each end with a 50-ohm terminator
- The maximum length of a cable segment is 185 meters and NOT 200 meters
- No more than 30 transceivers can be connected to any one segment
- There must be a minimum spacing of 0.5 meters between nodes
- It may not be used as a link segment between two 'Thicknet' segments
- The minimum bend radius is 5 cm

The physical layout of a 10Base2 Ethernet segment is shown in Figure 10.5.

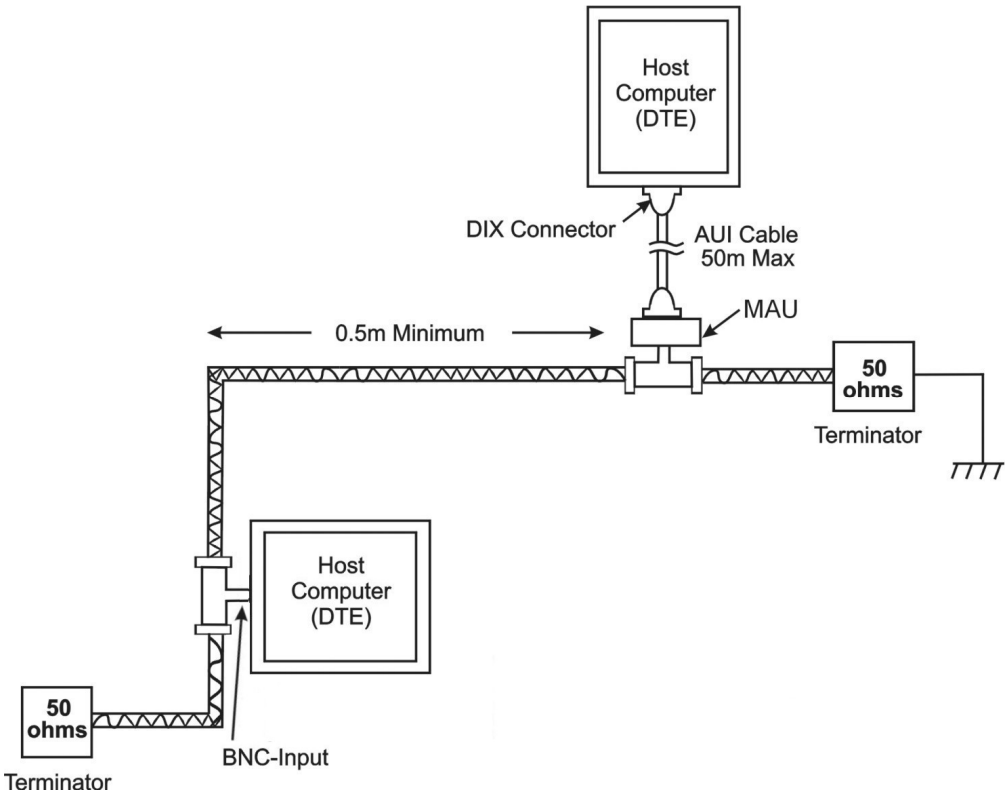


Figure 10.5
10 Base2 Ethernet segment

There is no need for an externally attached transceiver and transceiver cable. However, there are disadvantages with this approach. A cable fault can bring the whole system

down very quickly. To avoid such a problem, the cable is often taken to wall connectors with a make-break connector incorporated. There is also provision for remote MAUs in this system, with AUI cables making the node connection, in a similar manner to the Thicknet connection, but to do this one has to remove the vampire tap from the MAU and replace it with a BNC T-piece.

As with 10Base5, 10Base2 can work at speeds of 10 Mbps only. This system can be useful for small groups of computers, or for setting up temporary set-ups in a computer lab. As is the case with 10Base5, 10Base2 components are no longer readily available.

10.3.3 10BaseT

10BaseT was developed in the early 1990s and soon became very popular. The 10BaseT standard for Ethernet networks uses AWG24 unshielded twisted pair (UTP) cable for connection to the node. The physical topology of the standard is a star, with nodes connected to a hub. The hubs can be connected to a backbone cable that may be coax or fiber optic. They can alternatively be daisy-chained with UTP cables, or interconnected with special interconnecting cables via their backplanes. The node cable should be at least category 5 cable. The node cable has a maximum length of 100 meters, consists of two pairs for receiving and transmitting, and is connected via RJ45 plugs. The wiring hub can be considered a local bus internally, and so the logical topology is still considered a bus topology. Figure 10.6 schematically shows how the hub interconnects the nodes.

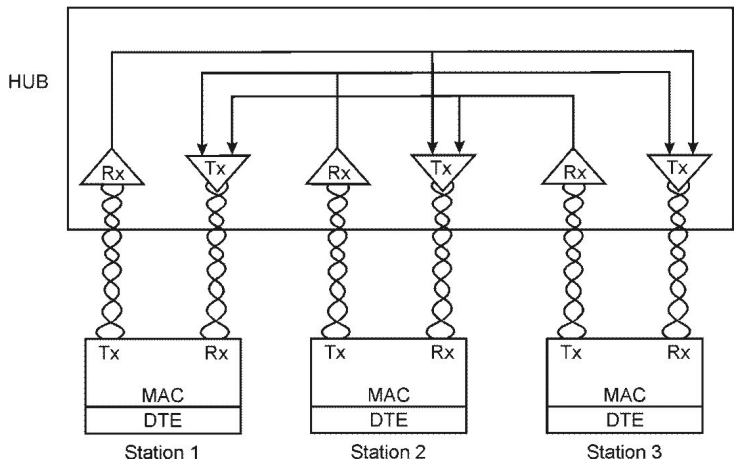


Figure 10.6
10BaseT system

Collisions are detected by the NIC and so the hub must retransmit an input signal on all outputs. The electronics in the hub must ensure that the stronger retransmitted signal does not interfere with the weaker input signal. The effect is known as 'far end cross talk' (FEXT), and is handled by special adaptive cross talk echo cancellation circuits.

The standard has disadvantages that should be noted.

- The UTP cable is not very resistant to electrostatic electrical noise, and is not be suitable for some industrial environments. In this case screened twisted pair should be used. Whilst the cable is relatively inexpensive, there is the additional cost of the associated wiring hubs to be considered.
- The node cable is limited to 100 m.

Advantages of the system include:

- Ordinary shared hubs can be replaced with switching hubs, which pass frames only to the intended destination node. This not only improves the security of the network but also increases the available bandwidth.
- Flood wiring could be installed in a new building, providing many more wiring points than are initially needed, but giving great flexibility for future expansion. When this is done, patch panels – or punch down blocks – are often installed for even greater flexibility.

10.3.4 10BaseF

10BaseF uses fiber optic media and light pulses to send signals. Fiber optic link segments can carry signals for much longer distances as compared to copper media; even two kilometer distances are possible. This fiber optic media can also carry signals at much higher speeds, so fiber optic media installed today for 10 Mbps speed can be used for Fast or Gigabit Ethernet systems in the future.

A major advantage of fiber optic media is its immunity to electrical noise, making it useful on factory floors.

There are two 10 Mbps link segment types in use, the original fiber optic inter-repeater link (FOIRL) segment, and 10BaseFL segment. The FOIRL specification describes a link segment up to 1000 meters between repeaters only. The cost of repeaters has since been coming down and the capacities of repeater hubs have increased. It now makes sense to link individual computers to fiber optic ports on a repeater hub.

A new standard 10BaseF was developed to specify a set of fiber optic media including link segments to allow direct attachments between repeater ports and computers. Three types of fiber optic segments have been specified:

10BaseFL

This fiber link segment standard is a 2 km upgrade to the existing fiber optic inter repeater link (FOIRL) standard. The original FOIRL as specified in the IEEE 802.3 standard was limited to a 1 km fiber link between two repeaters.

Note that this is a link between two repeaters in a network, and cannot have any nodes connected to it.

If older FOIRL equipment is mixed with 10BaseFL equipment then the maximum segment length may only be 1 km.

Figures 10.7 and 10.8 show a 10BaseFL transceiver and a typical station connected to a 10BaseFL system.

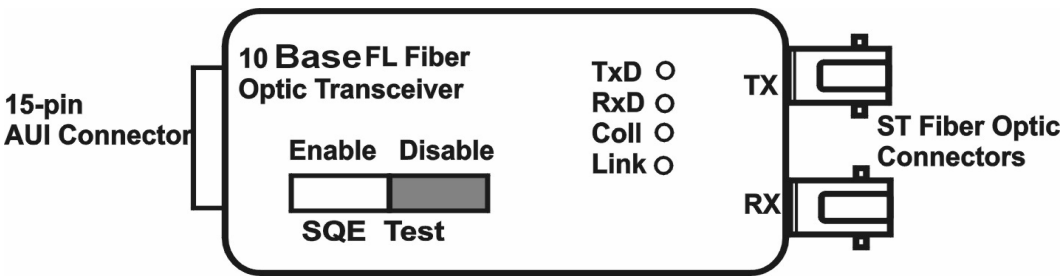


Figure 10.7
10BaseFL transceiver

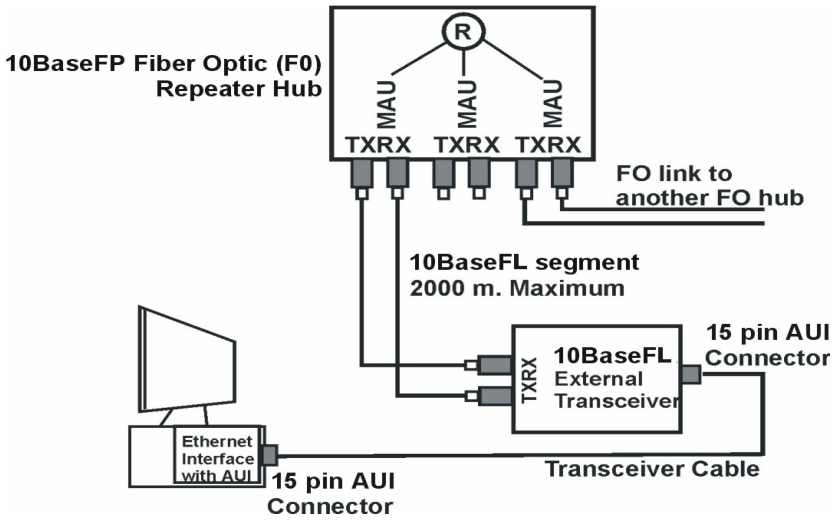


Figure 10.8
Connecting a station to a 10BaseFL system

10BaseFP

FP here means 'fiber passive'. This is a set of specifications for a 'passive fiber optic mixing segment', and, is based on a non-powered device acting as a fiber optic signal coupler, linking multiple computers. A star topology network based on the use of a passive fiber optic star coupler can be 500 m long and up to 33 ports are available per star. The passive hub is completely immune to external noise and is an excellent choice for extremely noisy industrial environments, inhospitable to electrical repeaters.

Passive fiber systems can be implemented using standard fiber optic components (splitters and combiners). However these are hard-wired networks and LAN equipment has not become commercially available to readily implement this method.

This variation has, however, not become commercially available.

10BaseFB

This is a fiber backbone link segment in which data is transmitted synchronously. It was designed only for connecting repeaters, and for repeaters to use this standard, they must include a built in transceiver. This reduces the time taken to transfer a frame across the repeater hub. The maximum link length is 2 km, although up to 15 repeaters can be cascaded, giving greater flexibility in network design. This has been made technologically obsolete by single mode fiber cable where 100 km is possible with no repeaters!

This variation has also not been commercially available.

10.3.5 Obsolete systems

10Broad36

This architecture, whilst included in the IEEE 802.3 standard, is now extinct. This was a broadband version of Ethernet, and used a 75-ohm coaxial cable for transmission. Each transceiver transmitted on one frequency and received on a separate one. The Tx/Rx streams each required 14 MHz bandwidth and an additional 4 MHz was required for collision detection and reporting. The total bandwidth requirement was thus 36 MHz. The cable was limited to 1800 meters because each signal had to traverse the cable twice, so the worst-case distance was 3600 m. This figure gave the system its nomenclature.

1Base5

This architecture, whilst included in the IEEE 802.3 standard, is also extinct. It was hub-based and used UTP as a transmission medium over a maximum length of 500 meters. However, signaling took place at 1 Mbps, and this meant special provision had to be made if it was to be incorporated in a 10 Mbps network. It has been superseded by 10BaseT.

10.3.6 10 Mbps design rules

The following design rules on length of cable segment, node placement and hardware usage should be strictly observed.

Length of the cable segment

It is important to maintain the overall Ethernet requirements as far as length of the cable is concerned. Each segment has a particular maximum allowable length. For example, 10Base2 allows 185 m maximum segment lengths. The recommended maximum length is 80% of this figure. Some manufacturers advise that you can disregard this limit with their equipment. This can be a risky strategy and should be carefully considered.

Cable segments need not be made from a single homogenous length of cable, and may comprise multiple lengths joined by coaxial connectors (two male plugs and a connector barrel). Although ThickNet (10Base5) and Thin Net (10Base2) cables have the same nominal 50-ohm impedance they can only be mixed within the same 10Base2 cable segment to achieve greater segment length.

System	Maximum	Recommended
10Base5	500 m	400 m
10Base2	185 m	150 m
10BaseT	100 m	80 m

To achieve maximum performance on 10Base5 cable segments, it is preferable that the total segment be made from one length of cable or from sections off the same drum of cable. If multiple sections of cable from different manufacturers are used, then these should be standard lengths of 23.4 m, 70.2 m or 117 m (± 0.5 m), which are odd multiples of 23.4 m (half wavelength in the cable at 5 MHz). These lengths ensure that reflections from the cable-to-cable impedance discontinuities are unlikely to add in phase. Using these lengths, exclusively a mix of cable sections should be able to makeup the full 500 m segment length.

If the cable is from different manufacturers and potential mismatch problems are suspected, then check that signal reflections due to impedance mismatches do not exceed 7% of the incident wave.

Maximum transceiver cable length

In 10Base5 systems, the maximum length of transceiver cables is 50 m but it should be noted that this only applies to specified IEEE 802.3 compliant cables. Other AUI cables using ribbon or office grade cables can only be used for short distances (less than 12.5 m) so check the manufacturer's specifications for these.

Node placement rules

Connection of the transceiver media access units (MAU) to the cable causes signal reflections due to their bridging impedance. Placement of the MAUs must therefore be controlled to ensure that reflections from them do not significantly add in phase.

In 10Base5, systems the MAUs are spaced at 2.5 m multiples, coinciding with the cable markings. In 10Base2 systems, the minimum node spacing is 0.5 m.

Maximum transmission path

The total number of segments can be made up of a maximum of five segments in series, with up to four repeaters, no more than three 'mixing segments' and this is known as the 5-4-3-2 rule.

Note that the maximum sized network of four repeaters supported by IEEE 802.3 can be susceptible to timing problems. The maximum configuration is limited by propagation delay.

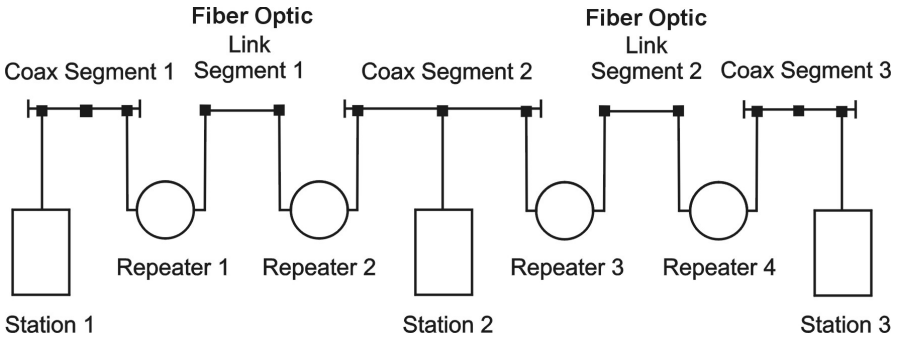


Figure 10.9
Maximum transmission path

Maximum network size

This refers to the maximum possible distance between two nodes.
10Base5 = 2800 m node to node (5 × 500 m segments + 8 repeater cables + 2 AUI cables)
10Base2 = 925 m node to node (5 × 185 m segments)
10BaseT = 100 m node to hub
While determining the maximum network size, collision domain distance and number of segments are both to be considered simultaneously. For example three 10Base2 systems and two 10BaseFL (which can run up to 2 km) together can be used to the 2500 m limit.

Repeater rules

Repeaters are connected to the cable via transceivers that count as one node on the segments.

Special transceivers are used to connect repeaters and these do not implement the signal quality error test (SQE).

Fiber optic repeaters are available giving up to 3000 m links at 10 Mbps. Check the vendor's specifications for adherence with IEEE 802.3 and 10BaseFL requirements.

Cable system grounding

Grounding has safety and noise implications. IEEE 802.3 states that the shield conductor of each coaxial cable shall make electrical contact with an effective earth reference at one point only.

The single point earth reference for an Ethernet system is usually located at one of the terminators. Most terminators for Ethernet have a screw terminal to which a ground lug can be attached using a braided cable (preferably) to ensure good earthing.

Ensure that all other splices taps or terminators are jacketed so that no contact can be made with any metal objects. Insulating boots or sleeves should be used on all in-line coaxial connectors to avoid unintended earth contacts.

10.4 100 Mbps media systems

10.4.1 Introduction

Although 10 Mbps Ethernet, with over 500 million installed nodes worldwide, was a very popular method of linking computers on a network, its speed is too slow for data intensive or some real-time applications.

From a philosophical point of view, there are several ways to increase speed on a network. The easiest, conceptually, is to increase the bandwidth and allow faster changes of the data signal. This requires a high bandwidth medium and generates a considerable amount of high frequency electrical noise on copper cables, which is difficult to suppress.

The second approach is to move away from the serial transmission of data on one circuit to a parallel method of transmitting over multiple circuits at each instant. A third approach is to use data compression techniques to enable more than one bit to transfer for each electrical transition. A fourth approach is to operate circuits full-duplex, enabling simultaneous transmission in both directions.

Three of these approaches are used to achieve 100 Mbps Fast Ethernet and 1000 Mbps Gigabit Ethernet transmission on both fiber optic and copper cables using current high-speed LAN technologies.

Cabling limitations

Typically most LAN systems use either coaxial cable, shielded (STP) or Unshielded Twisted Pair (UTP) or fiber optic cables. The choice of media depends on collision domain distance limitations and whether the operation is to full-duplex or not.

The unshielded twisted pair is obviously popular because of ease of installation and low cost. This is the basis of the 10BaseT Ethernet standard. The category 3 cable allows only 10 Mbps over 100 m while category 5 cable supports 100 Mbps data rates. The four pairs in the standard cable allow several parallel data streams to be handled.

10.4.2 100BaseT (100BaseTX, T4, FX, T2) systems

100 Mbps Ethernet uses the existing Ethernet MAC layer with various enhanced physical media dependent (PMD) layers to improve the speed. These are described in the IEEE 802.3u and 802.3y standards as follows:

IEEE 802.3u defines three different versions based on the physical media:

- 100BaseTX, which uses two pairs of category 5 UTP or STP
- 100BaseT4 which uses four pairs of wires of category 3,4 or 5 UTP
- 100BaseFX, which uses multimode or single-mode fiber optic cable

IEEE 802.3y defines 100BaseT2 which uses two pairs of wires of category 3,4 or 5 UTP.

This approach is possible because the original IEEE 802.3 specifications defined the MAC layer independently of the various physical PMD layers it supports. The MAC layer defines the format of the Ethernet frame and defines the operation of the CSMA/CD access mechanism. The time dependent parameters are defined in the IEEE 802.3 specifications in terms of bit-time intervals and so it is speed independent. The 10 Mbps Ethernet interframe gap is actually defined as an absolute time interval of 9.60 microseconds, equivalent to 96 bit times while the 100 Mbps system reduces this by ten times to 960 nanoseconds.

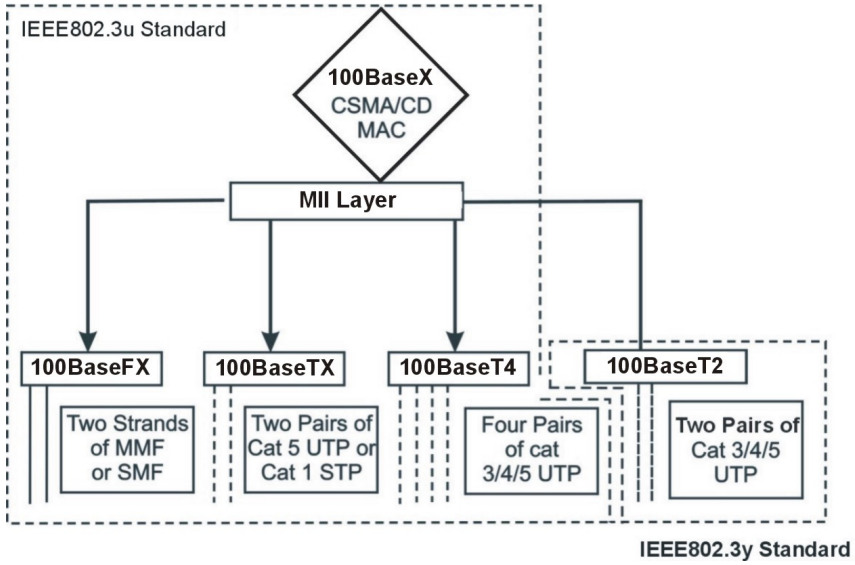


Figure 10.10
Summary of 100BaseX standards

One of the limitations of the 100BaseT system is the size of the collision domain if operating in CSMA/CD mode. This is the maximum sized network in which collisions can be detected; being one tenth of the size of the maximum 10 Mbps network. This limits the distance between a workstation and hub to 100 m, the same as for 10BaseT, but the number of hubs allowed in a collision domain will depend on the type of hub. This means that networks larger than 200 m must be logically connected together by store and forward type devices such as bridges, routers or switches. However, this is not a bad thing, since it segregates the traffic within each collision domain, reducing the number of collisions on the network. The use of bridges and routers for traffic segregation, in this manner, is often done on industrial Ethernet networks to improve performance.

10.4.3 IEEE 802.3u 100BaseT standards arrangement

The IEEE 802.3u standard fits into the OSI model as shown in Figure 10.11. The unchanged IEEE 802.3 MAC layer sits beneath the LLC as the lower half of the data link layer of the OSI model.

Its Physical layer is divided into the following two sub layers and their associated interfaces:

- PHY physical medium independent layer
- MII medium independent interface
- PMD physical medium dependent layer
- MDI medium dependent interface

A convergence sub-layer is added for the 100BaseTX and FX systems, which use the ANSI X3T9.5 PMD layer. This was developed for the reliable transmission of 100 Mbps over the twisted pair version of FDDI. The FDDI PMD layer operates as a continuous full-dup1ex 125 Mbps transmission system, so a convergence layer is needed to translate this into the 100 Mbps half-duplex data bursts expected by the IEEE 802.3 MAC layer.

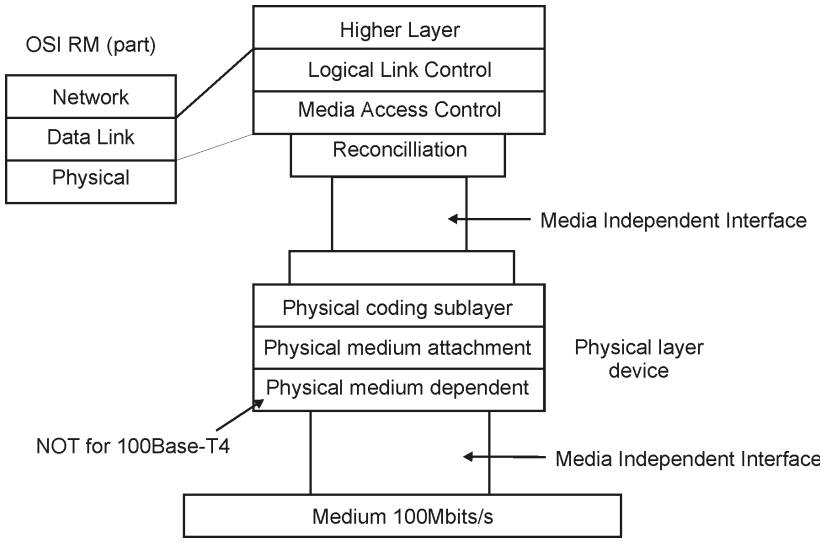


Figure 10.11
100BaseT standards architecture

10.4.4 Physical medium independent (PHY) sub layer

The PHY layer specifies the 4B/5B coding of the data, data scrambling and the 'non return to zero – inverted' (NRZI) data coding together with the clocking, data and clock extraction processes.

The 4B/5B technique selectively codes each group of four bits into a five-bit cell symbol. For example, the binary pattern 0110 is coded into the five-bit pattern 01110. In turn, this symbol is encoded using 'non return to zero – inverted' (NRZI) where a '1' is represented by a transition at the beginning of the cell, and a '0' by no transition at the beginning. This allows the carriage of 100 Mbps data by transmitting at 125 MHz, and gives a consequent reduction in component cost of some 80%.

With a five-bit pattern, there are 32 possible combinations. Obviously, there are only 16 of these that need to be used for the four bits of data, and of these, each is chosen so that there are no more than three consecutive zeros in each symbol. This ensures there will be sufficient signal transitions to maintain clock synchronization. The remaining 16 symbols are used for control purposes.

This selective coding is shown in Table 10.1.

This coding scheme is not self clocking so each of the receivers maintains a separate data receive clock which is kept in synchronization with the transmitting node, by the clock transitions in the data stream. Hence, the coding cannot allow more than three consecutive zeros in any symbol.

4-bit Data Pattern	5-bit Symbol
0000	11110
0001	01001
0010	10100
0011	10101
0100	01010
0101	01011
0110	01110
0111	01111
1000	10010
1001	10011
1010	10110
1011	10111
1100	11010
1101	11011
1110	11100
1111	11101

Table 10.1
4B/5B data coding

10.4.5 100BaseTX and FX physical media dependent (PMD) sub-layer

This uses the ANSI TP-X3T9.5 PMD layer and operates on two pairs of category 5 twisted pair. It uses stream cipher scrambling for data security and MLT-3 bit encoding. The multilevel threshold-3 (MLT-3) bit coding uses three voltage levels viz +1 V, 0 V and –1 V.

The level remains the same for consecutive sequences of the same bit, i.e. continuous '1s'. When a bit changes, the voltage level changes to the next state in the circular sequence 0 V, +1 V, 0 V, –1 V, 0 V etc. This results in a coded signal, which resembles a smooth sine wave of much lower frequency than the incoming bit stream.

Hence, for a 31.25 MHz baseband signal this allows for a 125 Mbps signaling bit stream providing a 100 Mbps throughput (4 B/5B encoder). The MAC outputs an NRZI code. This code is then passed to a scrambler, which ensures that there are no invalid groups in its NRZI output. The NRZI converted data is passed to the three level code block and the output is then sent to the transceiver. The code words are selectively chosen so the mean line signal line zero, in other words the line is DC balanced.

The three level code results in a lower frequency signal. Noise tolerance is not as high as in the case of 10BaseT because of the multilevel coding system; hence, category 5 cable is required.

Two pair wire, RJ-45 connectors and a hub are requirements for 100BaseTX. These factors and a maximum distance of 100 m between the nodes and hubs make for a very similar architecture to 10BaseT.

10.4.6 100BaseT4 physical media dependent (PMD) sub-layer

The 100BaseT4 systems use four pairs of category 3 UTP. It uses data encoded in an eight binary six ternary (8B/6T) coding scheme similar to the MLT-3 code. The data is

encoded using three voltage levels per bit time of +V, 0 volts and –V, these are usually written as simply +, 0 and –.

This coding scheme allows the eight bits of binary data to be coded into six ternary symbols, and reduces the required bandwidth to 25 MHz. The 256 code words are chosen so the line has a mean line signal of zero. This helps the receiver to discriminate the positive and negative signals relative to the average zero level. The coding utilizes only those code words that have a combined weight of 0 or +1, as well as at least two signal transitions for maintaining clock synchronization. For example, the code word for the data byte 20H is –++–00, which has a combined weight of 0 while 40H is –00+0+, which has a combined weight of +1.

If a continuous string of codeword of weight +1 is sent, then the mean signal will move away from zero known as DC wander. This causes the receiver to misinterpret the data since it is assuming the average voltage it is seeing, which is now tending to '+1', is its zero reference. To avoid this situation, a string of code words of weight +1 is always sent by inverting alternate code words before transmission.

Consider a string of consecutive data bytes 40H, the codeword is –00+0+, which has weight +1. This is sent as the sequence –00+0+, +00–0–, –00+0+, +00–0– etc, which results in a mean signal level of zero. The receiver consequently re-inverts every alternate codeword prior to decoding.

These signals are transmitted in half-duplex over three parallel pairs of Category 3, 4 or 5 UTP cable, while a fourth pair is used for reception of collision detection signals.

This is shown in Figure 10.12.

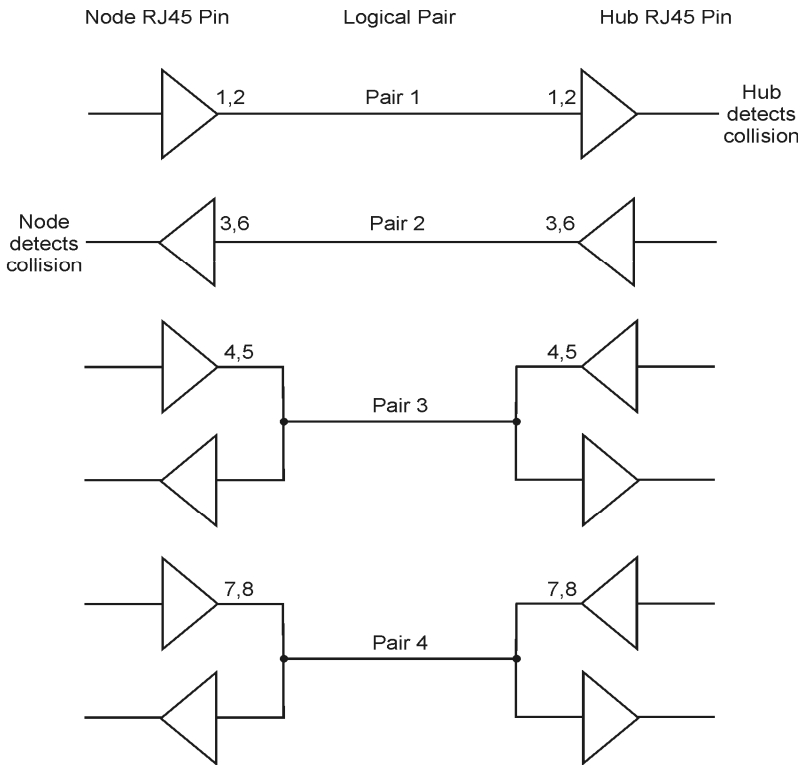


Figure 10.12
100BaseT4 wiring

100BaseTX and 100BaseT4 are designed to be interoperable at the transceivers using a media independent interface and compatible (Class 1) repeaters at the hub. Maximum node to hub distances of 100 m, and a maximum network diameter of 250 m are supported. The maximum hub-to-hub distance is 10 m.

10.4.7 100BaseT2

The IEEE published the 100BaseT2 system in 1996 as the IEEE 802.3y standard. It was designed to address the shortcomings of 100BaseT4, making full-duplex 100 Mbps accessible to installations with only two category 3 cable pairs available. The standard was completed two years after 100BaseTX, but never gained a significant market share. However it is mentioned here for reference only and because its underlying technology using digital signal processing (DSP) techniques and five-level coding (PAM-5) is used for the 1000BaseT systems on two category 5 pairs. These are discussed in detail under 1000BaseT systems.

The features of 100BaseT2 are:

- Uses two pairs of Category 3,4 or 5 UTP
- Uses both pairs for simultaneously transmitting and receiving – commonly known as dual-duplex transmission. This is achieved by using digital signal processing (DSP) techniques
- Uses a five-level coding scheme with five phase angles called pulse amplitude modulation (PAM 5) to transmit two bits per symbol

10.4.8 100BaseT hubs

The IEEE 802.3u specification defines two classes of 100BaseT hubs.

They are also called repeaters:

- Class I, or translational hubs, which can support both TX/FX and T4 systems
- Class II, or transparent hubs, which support only one signaling system

The Class I hubs have greater delays (0.7 microseconds maximum) in supporting both signaling standards and so only permit one hub in each collision domain. The Class I hub fully decodes each incoming TX or T4 packet into its digital form at the media independent interface (MII) and then sends the packet out as an analog signal from each of the other ports in the hub. Hubs are available with all T4 ports, all TX ports or combinations of TX and T4 ports, called Translational Hubs.

The Class II hubs operate like a 10BaseT hub connecting the ports (all of the same type) at the analog level. These then have lower inter-hub delays (0.46micro seconds maximum) and so two hubs are permitted in the same collision domain, but only 5 m apart. Alternatively, in an all fiber network, the total length of all the fiber segments is 228 meters. This allows two 100 m segments to the nodes with 28 m between the repeaters or any other combination. See Figures 10.13A and 10.13B show how class I and class II repeaters are connected.

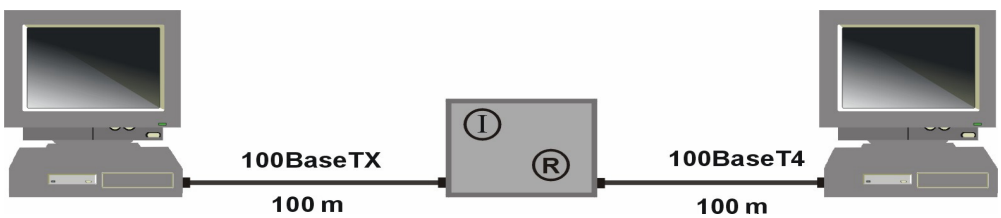


Figure 10.13A
100BaseTX and 100BaseT4 segments linked with a class I repeater

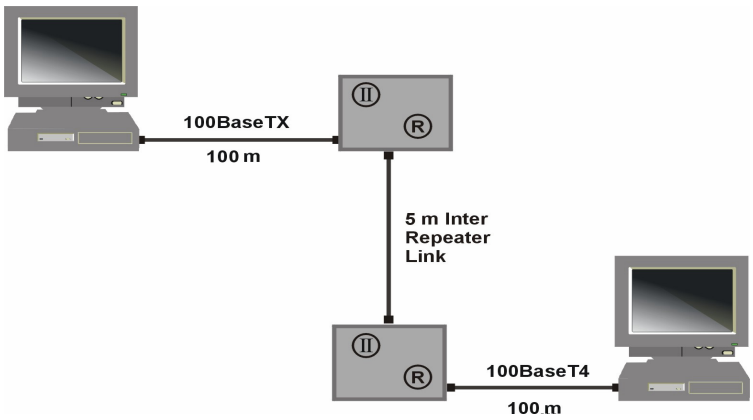


Figure 10.13B
Class II repeaters with an inter-repeater link

10.4.9 100BaseT adapters

Adapter cards are readily available as standard 100Mbps and as 10/100Mbps. The latter cards are interoperable at the hub on both speeds.

10.4.10 100 Mbps/fast Ethernet design considerations

UTP cabling distances 100BaseTX/T4

The maximum distance between a UTP hub and a desktop NIC is 100 meters, made up as follows:

- 5 meters from hub to patch panel
- 90 meters horizontal cabling from patch panel to office punch down block
- 5 meters from punch-down block to desktop NIC

Fiber optic cable distances 100BaseFX

The following maximum cable distances are in accordance with the 100BaseT bit budget.

Node to hub: maximum distance of multimode cable (62.5/125) is 160 meters (for connections using a single Class II hub).

Node to switch: maximum multimode cable distance is 210 meters.

Switch-to-switch: maximum distance of multimode cable for a backbone connection between two 100BaseFX switch ports is 412 meters.

Switch to switch full-duplex: maximum distance of multimode cable for a full-duplex connection between two 100BaseFX switch ports is 2000 meters.

Note: The IEEE has not included the use of single mode fiber in the IEEE 802.3u standard. However numerous vendors have products available enabling switch-to-switch distances of up to twenty kilometers using single mode fiber.

100BaseT hub (repeater) rules

The cable distance and the number of hubs that can be used in a 100BaseT collision domain depend on the delays in the cable, the time delay in the repeaters and NIC delays. The maximum round-trip delay for 100BaseT systems is the time to transmit 64 bytes or 512 bits and equals 5.12 microseconds. A frame has to go from the transmitter to the most remote node then back to the transmitter for collision detection within this round trip time. Therefore the one-way time delay will be half this.

The maximum sized collision domain can then be determined by the following calculation:

Repeater delays + Cable delays + NIC delays + Safety factor (5 bits Minimum) should be less than 2.56 microseconds.

The following Table 10.2 gives typical maximum one-way delays for various components. Repeater and NIC delays for specific components can be obtained from the manufacturer.

Components	Maximum Delay, in microseconds
Fast Ethernet NIC	0.25
Fast Ethernet Switch Port	0.25
Class I Repeater(Hub)	0.70 max.
Class II Repeater (Hub)	0.46 max.
UTP Cable (per 100 meters)	0.55
Multimode Fiber (per 100 meters)	0.50

Table 10.2
Maximum one-way fast Ethernet components delay

Notes

If the desired distance is too great, it is possible to create a new collision domain by using a switch instead of a hub.

Most 100BaseT hubs are stackable, which means multiple units can be placed on top of one another and connected together by means of a fast backplane bus. Such connections do not count as a repeater hop and make the ensemble function as a single repeater.

It should also be noted that these calculations assume CSMA/CD operations. They are irrelevant for full-duplex operations, and are also of no concern if switches are used instead of ordinary hubs.

Sample calculation

Can two Fast Ethernet nodes be connected together using two Class II hubs connected by 50 m fibers? One node is connected to the first repeater with 50 m UTP while the other has a 100 m fiber connection.

NIC	0.25 microseconds
50 m. UTP	0.275 microseconds
Hub Class II	0.46 microseconds
50 m. Fiber	0.25 microseconds
Hub Class II	0.46 microseconds
100 m. Fiber	0.50 microseconds
NIC	0.25 microseconds
Total Delay	2.445 microseconds

Table 10.3
Sample delay calculation

Calculation: Using the time delays in table 10.2:

The total one-way delay of 2.445 microseconds is within the required interval (2.56 microseconds) and allows at least 5 bits safety factor, so this connection is permissible.

10.5 Gigabit/1000 Mbps media systems

10.5.1 Gigabit Ethernet summary

Gigabit Ethernet uses the same IEEE 802.3 frame format as 10 Mbps and 100 Mbps Ethernet systems. It operates at ten times the clock speed of Fast Ethernet at 1Gbps. By retaining the same frame format as the earlier versions of Ethernet, backward compatibility is assured with earlier versions, increasing its attractiveness by offering a high bandwidth connectivity system to the Ethernet family of devices.

Gigabit Ethernet is defined by the IEEE 802.3z standard. This defines the Gigabit Ethernet media access control (MAC) layer functionality as well as three different physical layers: 1000BaseLX and 1000BaseSX using fiber and 1000BaseCX using copper.

These physical layers were originally developed by IBM for the ANSI fiber channel systems and used 8B/10B encoding to reduce the bandwidth required to send high-speed signals. The IEEE merged the fiber channel to the Ethernet MAC using a Gigabit media independent interface (GMII), which defines an electrical interface enabling existing fiber channel PHY chips to be used, and enabling future physical layers to be easily added. This development is defined by the IEEE 802.3ab standard.

These Gigabit Ethernet versions are summarized in Figure 10.14.

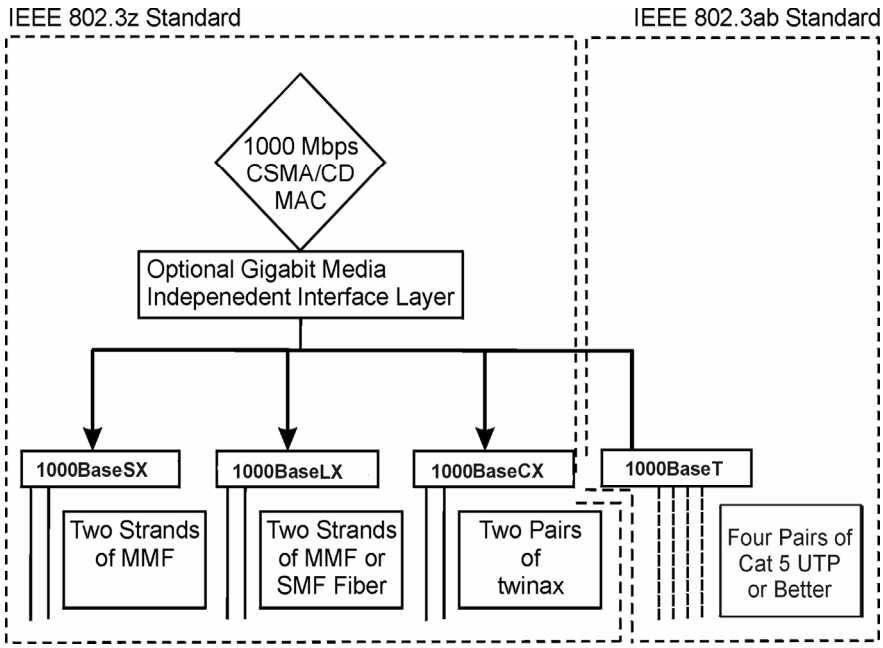


Figure 10.14
Gigabit Ethernet versions

10.5.2 Gigabit Ethernet MAC layer

Gigabit Ethernet retains the standard IEEE 802.3 frame format, however the CSMA/CD algorithm had to undergo a small change to enable it to function effectively at 1 Gbps. The slot time of 64 bytes used with both 10 Mbps and 100 Mbps systems have been increased to 512 bytes. Without this increased slot, time the network would have been impractically small at one tenth of the size of Fast Ethernet – only 25 meters.

The slot time defines the time during which the transmitting node retains control of the medium, and in particular is responsible for collision detection. With Gigabit Ethernet it was necessary to increase this time by a factor of eight to 4.096 microseconds to compensate for the tenfold speed increase. This then gives a collision domain of about 200 m.

If the transmitted frame is less than 512 bytes the transmitter continues transmitting to fill the 512-byte window. A carrier extension symbol is used to mark frames, which are shorter than 512 bytes, and to fill the remainder of the frame. This is shown in Figure 10.15.

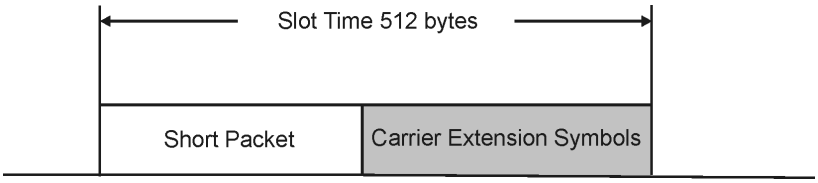


Figure 10.15
Carrier extension

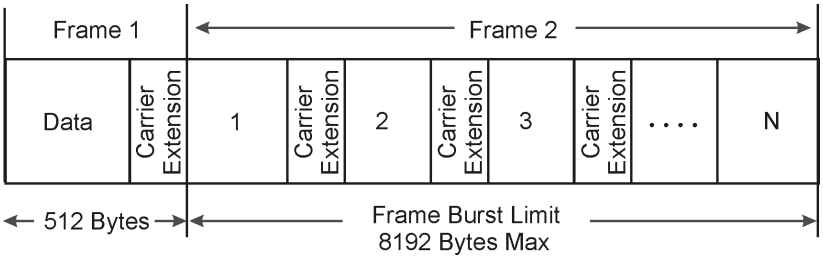


Figure 10.16
Packet bursting

While this is a simple technique to overcome the network size problem, it could cause problems with very low utilization if we send many short frames, typical of some industrial control systems. For example, a 64 byte frame would have 448 carrier extension symbols attached and result in a utilization of less than 10% This is unavoidable, but its effect can be minimized if we are sending a lot of small frames by a technique called packet bursting.

The first frame in a burst is transmitted in the normal way using carrier extension if necessary. Once the first frame is transmitted without a collision then the station can immediately send additional frames until the Frame Burst Limit of 65,536 bits has been reached. The transmitting station keeps the channel from becoming idle between frames by sending carrier extension symbols during the inter-frame gap. When the Frame Burst Limit is reached the last frame in the burst is started. This process averages the time wasted sending carrier extension symbols over a number of frames. The size of the burst varies depending on how many frames are being sent and their size. Frames are added to the burst in real-time with carrier extension symbols filling the inter-frame gap. The total number of bytes sent in the burst is totaled after each frame and transmission continues until at most 65,536 bits have been transmitted. This is shown in Figure 10.16.

10.5.3 Physical medium independent (PHY) sub layer

The IEEE 802.3z Gigabit Ethernet standard uses the three PHY sub-layers from the ANSI X3T11 fiber channel standard for the 1000BaseSX and 1000BaseLX versions using fiber optic cable and 1000BaseCX using shielded 150 ohm twinax copper cable.

The fiber channel PMD sub layer runs at 1Gbaud and specifies the 8B/10B coding of the data, data scrambling and the non return to zero – inverted (NRZI) data coding together with the clocking, data and clock extraction processes. This translated to a data rate of 800 Mbps. The IEEE then had to increase the speed of the fiber channel PHY layer to 1250 Mbaud to obtain the required throughput of 1Gbps.

The 8B/10B technique selectively codes each group of eight bits into a ten-bit symbol. Each symbol is chosen so that there are at least two transitions from ‘1’ to ‘0’ in each symbol. This ensures there will be sufficient signal transitions to allow the decoding device to maintain clock synchronization from the incoming data stream. The coding

scheme allows unique symbols to be defined for control purposes, such as denoting the start and end of packets and frames as well as instructions to devices.

The coding also balances the number of '1s' and '0s' in each symbol, called DC balancing. This is done so that the voltage swings in the data stream would always average to zero, and not develop any residual DC charge, which could result in any AC-coupled devices distorting the signal. This phenomenon is called 'baseline wander'.

10.5.4 1000BaseSX for horizontal fiber

This Gigabit Ethernet version was developed for the short backbone connections of the horizontal network wiring. The SX systems operate full duplex with multimode fiber only, using the cheaper 850 nm wavelength laser diodes.

The maximum distance supported varies between 200 and 550 meters depending on the bandwidth and attenuation of the fiber optic cable used. The standard 1000BaseSX NICs available today are full-duplex and incorporate SC fiber connectors.

10.5.5 1000BaseLX for vertical backbone cabling

This version was developed for use in the longer backbone connections of the vertical network wiring. The LX systems can use single mode or multimode fiber with the more expensive 1300 nm laser diodes.

The maximum distances recommended by the IEEE for these systems operating in full-duplex are 5 kilometer for single mode cable and 550 meters for multimode fiber cable. Many 1000BaseLX vendors guarantee their products over much greater distances; typically 10 km. Fiber extenders are available to give service over as much as 80 km. The standard 1000BaseLX NICs available today are full-duplex and incorporate SC fiber connectors.

10.5.6 1000BaseCX for copper cabling

This version of Gigabit Ethernet was developed for the short interconnection of switches, hubs or routers within a wiring closet. It is designed for 150-ohm shielded twisted pair cable similar to that used for IBM Token Ring systems.

The IEEE specified two types of connectors: The high-speed serial data connector (HSSDC) known as the fiber channel style 2 connector and the nine pin D-subminiature connector from the IBM token ring systems. The maximum cable length is 25 meters for both full- and half-duplex systems.

The preferred connection arrangements are to connect chassis-based products via the common back plane and stackable hubs via a regular fiber port.

10.5.7 1000BaseT for category 5 UTP

This version of the Gigabit Ethernet was developed under the IEEE 802.3ab standard for transmission over four pairs of category 5 or better cable. This is achieved by simultaneously sending and receiving over each of the four pairs. Compare this to the existing 100BaseTX system, which has individual pairs for transmitting and receiving. This is shown in Figure 10.17.

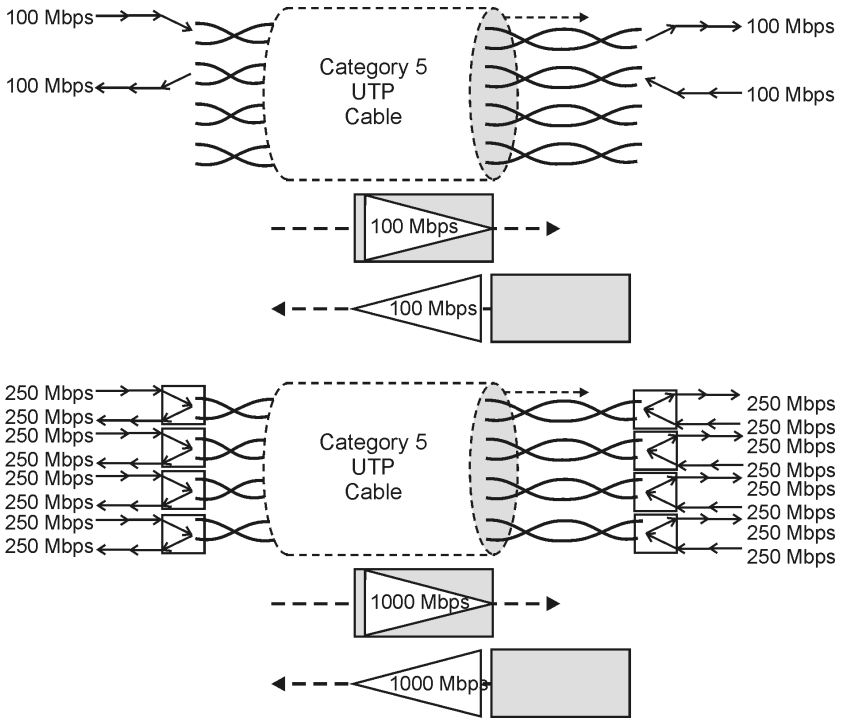


Figure 10.17
Comparison of 100BaseTX and 100BaseT

This system uses the same data-encoding scheme developed for 100BaseT2, which is PAM5. This utilizes five voltage levels so it has less noise immunity, however the digital signal processors (DSPs) associated with each pair overcome any problems in this area. The system achieves its tenfold speed improvement over 100BaseT2 by transmitting on twice as many pairs (4) and operating at five times the clock frequency (125 MHz).

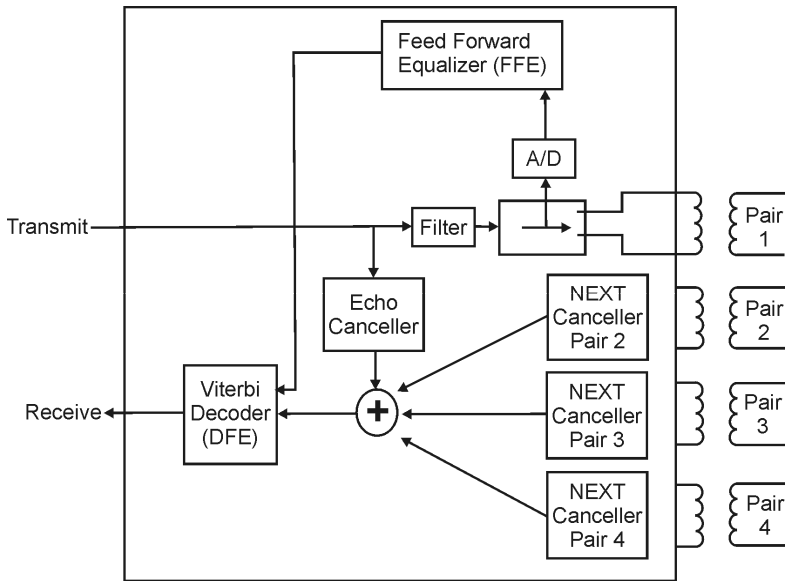


Figure 10.18
1000BaseT receiver uses DSP technology

10.5.8 Gigabit Ethernet full-duplex repeaters

Gigabit Ethernet nodes are connected to full-duplex repeaters also known as non-buffered switches or buffered distributors. As shown in Figure 10.19 these devices have a basic MAC function in each port, which enables them to verify that a complete frame is received and compute its frame check sequence (CRC) to verify the frame validity. Then the frame is buffered in the internal memory of the port before being forwarded to the other ports of the repeater. It is therefore combining the functions of a repeater with some features of a switch.

All ports on the repeater operate at the same speed of 1Gbps, and operate in full duplex so it can simultaneously send and receive from any port. The repeater uses IEEEE802.3x flow control to ensure the small internal buffers associated with each port do not overflow. When the buffers are filled to a critical level, the repeater tells the transmitting node to stop sending until the buffers have been sufficiently emptied.

The repeater does not analyze the packet address fields to determine where to send the packet, like a switch does, but simply sends out all valid packets to all the other ports on the repeater.

The IEEE does allow for half-duplex Gigabit repeaters.

10.5.9 Gigabit Ethernet design considerations

Fiber optic cable distances

The maximum cable distances that can be used between the node and a full duplex 1000BaseSX and LX repeater depend mainly on the chosen wavelength, the type of cable, and its bandwidth. The differential mode delay (DMD) limits the maximum transmission distances on multimode cable.

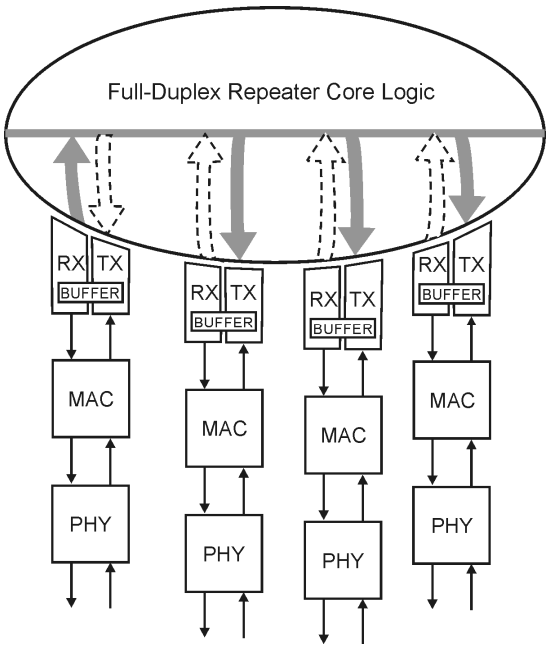


Figure 10.19
Gigabit Ethernet full duplex repeaters

The very narrow beam of laser light injected into the multimode fiber results in a relatively small number of rays going through the fiber core. These rays each have different propagation times because they are going through differing lengths of glass by

zigzagging through the core to a greater or lesser extent. These pulses of light can cause jitter and interference at the receiver. This is overcome by using a conditioned launch of the laser into the multimode fiber. This spreads the laser light evenly over the core of the multimode fiber so the laser source looks more like a Light Emitting Diode (LED) source. This spreads the light in a large number of rays across the fiber resulting in smoother spreading of the pulses, so less interference. This conditioned launch is done in the 1000BaseSX transceivers.

The following Table gives the maximum distances for full-duplex 1000BaseX fiber systems.

Wavelength (nm)	Cable Type	Bandwidth (Mhz. Km)	Attenuation (dB/km)	Maximum Distance (m)
850	50/125 Multimode	400	3.25	500
850	50/125 Multimode	500	3.43	550
850	62.5/125 Multimode	160	160	220
850	62.5/125 Multimode	200	200	275
1300	50/125 Multimode	500	2.32	550
1300	62.5/125 Multimode	500	1.0	550
1300	9/125 Single mode	Infinite	0.4	5000

Table 10.4
Maximum fiber distances for 1000BaseX (Full duplex)

Gigabit repeater rules

The cable distance and the number of repeaters, which can be used in a half-duplex 1000BaseT collision domain depends on the delay in the cable and the time delay in the repeaters and NIC delays. The maximum round-trip delay for 1000BaseT systems is the time to transmit 512 bytes or 4096 bits and equals 4.096 microseconds. A frame has to go from the transmitter to the most remote node then back to the transmitter for collision detection within this round trip time. Therefore the one-way time delay will be half this.

The maximum sized collision domain can then be determined by the following calculation:

Repeater Delays + Cable Delays + NIC Delays + Safety Factor (5 bits minimum) should be less than 2.048 microseconds.

It may be noted that all commercial systems are full-duplex systems, and, collision domain size calculations are not relevant in full-duplex mode. These calculations are relevant only if the backward compatibility with CSMA/CD mode is to be made use of.

Gigabit Ethernet network diameters

Table 10.5 gives the maximum collision diameters or in other words maximum network diameters for IEEE 802.3z half-duplex Gigabit Ethernet systems.

System	Maximum Collision Diameter Point-to Point Half-Duplex	Maximum Collision Diameter One Repeater Segment
1000BaseCX	25 m	50 m
1000BaseT	100 m	200 m
1000BaseSX or LX	31 m	220 m

Table 10.5
Maximum one-way gigabit Ethernet collision diameters

10.6 10 Gigabit Ethernet systems

Ethernet has continued to evolve and to become widely used because of its low implementation costs, reliability and simple installation and maintenance features. It is widely used; so much so, that nearly all traffic on the Internet originates or ends with an Ethernet network. Adaption to handle higher speeds has been concurrently occurring.

Gigabit Ethernets are already being used in large numbers and has begun transition from being only LANs to MANs and WANs as well.

An even faster 10 Gigabit Ethernet standard is now available, and the motivating force behind these developments were not only the exponential increase in data traffic, but also bandwidth-intensive applications such as video applications.

10 Gigabit Ethernet is significantly different in some aspects. It functions only over optical fibers and in full-duplex mode only. Packet formats are retained, and current installations are easily upgradeable to the new 10 Gigabit standard.

This section takes an overview of 10 gigabit standard (IEEE 802.3ae).

Information in the following pages is taken from a White Paper on 10 Gigabit Ethernet presented by '10 Gigabit Ethernet Alliance' and the complete text is available at http://www.10gea.org.

10.6.1 The 10 Gigabit Ethernet project and its objectives

The purpose of the 10 Gigabit Ethernet standard is to extend the IEEE 802.3 protocols to an operating speed of 10 Gbps and to expand the Ethernet application space to include WAN links.

This provides a significant increase in bandwidth while maintaining maximum compatibility with the installed base of IEEE 802.3 interfaces, previous investment in research and development, and principles of network operation and management.

In order to be adopted as a standard, the IEEE's 802.3ae Task Force has established five criteria for the new 10 Gigabit Ethernet standard:

- It needed to have broad market potential, supporting a broad set of applications, with multiple vendors supporting it, and multiple classes of customers
- It needed to be compatible with other existing IEEE 802.3 protocol standards, as well as with both open systems interconnection (OSI) and simple network management protocol (SNMP) management specifications
- It needed to be substantially different from the other IEEE 802.3 standards, making it a unique solution for problem rather than an alternative solution
- It needed to have demonstrated technical feasibility prior to final ratification
- It needed to be economically feasible for customers to deploy, providing reasonable cost, including all installation and management costs, for the expected performance increase

10.6.2 Architecture of 10-gigabit Ethernet standard

Under the International Standards Organization's open systems interconnection (OSI) model, Ethernet is fundamentally a layer 2 protocol. 10 Gigabit Ethernet uses the IEEE 802.3 Ethernet media access control (MAC) protocol, the IEEE 802.3 Ethernet frame format, and the minimum and maximum IEEE 802.3 frame size.

Just as 1000BaseX and 1000BaseT (Gigabit Ethernet) remained true to the Ethernet model, 10 Gigabit Ethernet continues the natural evolution of Ethernet in speed and distance. Since it is a full-duplex only and fiber-only technology, it does not need the

carrier-sensing multiple-access with collision detection (CSMA/CD) protocol that defines slower, half-duplex Ethernet technologies. In every other respect, 10 Gigabit Ethernet remains true to the original Ethernet model.

An Ethernet PHYsical layer device (PHY), which corresponds to layer 1 of the OSI model, connects the media (optical or copper) to the MAC layer, which corresponds to OSI layer 2. The Ethernet architecture further divides the PHY (Layer 1) into a physical media dependent (PMD) and a physical coding sublayer (PCS). Optical transceivers, for example, are PMDs. The PCS is made up of coding (e.g., 8B/10B) and a serializer or multiplexing functions.

The IEEE 802.3ae specification defines two PHY types: the LAN PHY and the WAN PHY (discussed below). The WAN PHY has an extended feature set added onto the functions of a LAN PHY. These PHYs are solely distinguished by the PCS. There is also be a number of PMD types.

10.6.3 Chip interface (XAUI)

Among the many technical innovations of the 10 Gigabit Ethernet task force is an interface called the XAUI (pronounced 'Zowie'). The 'AUI' portion is borrowed from the Ethernet attachment unit interface. The 'X' represents the Roman numeral for ten and implies ten gigabits per second. The XAUI is designed as an interface extender, and the interface, which it extends, is the XGMII – the 10 Gigabit media independent interface. The XGMII is a 74 signal wide interface (32-bit data paths for each of transmit and receive) that may be used to attach the Ethernet MAC to its PHY. The XAUI may be used in place of, or to extend, the XGMII in chip-to-chip applications typical of most Ethernet MAC to PHY interconnects.

The XAUI is a low pin count, self-clocked serial bus that is directly evolved from the Gigabit Ethernet 1000BaseX PHY. The XAUI interface speed is 2.5 times that of 1000BaseX. By arranging four serial lanes, the 4-bit XAUI interface supports the ten-times data throughput required by 10 Gigabit Ethernet.

The XAUI employs the same robust 8B/10B transmission code of 1000BaseX to provide a high level of signal integrity through the copper media typical of chip-to-chip printed circuit board traces. Additional benefits of XAUI technology include its inherently low EMI (electro-magnetic interference) due to its self-clocked nature, compensation for multibit bus skew – allowing significantly longer-distance chip-to-chip – error detection and fault isolation capabilities, low power consumption, and the ability to integrate the XAUI input/output within commonly available CMOS processes.

Multitudes of component vendors are delivering or have announced XAUI interface availability on standalone chips, custom ASICs (application-specific integrated circuits), and even FPGAs (field-programmable gate arrays). The 10 Gigabit Ethernet XAUI technology is identical or equivalent to the technology employed in other key industry standards such as InfiniBand(TM), 10 Gigabit fiber channel, and general purpose copper and optical back plane interconnects. This assures the lowest possible cost for 10 Gbps interconnects through healthy free market competition.

Specifically targeted XAUI applications include MAC to physical layer chip and direct MAC-to-optical transceiver module interconnects. The XAUI is the interface for the proposed 10 Gigabit plug-able optical module definition called the XGP. Integrated XAUI solutions together with the XGP enable efficient low-cost 10 Gigabit Ethernet direct multi-ports MAC to optical module interconnects with only PC board traces in between.

10.6.4 Physical media dependent (PMDs)

The IEEE 802.3ae task force has developed a standard that provides a physical layer that supports link distances for fiber optic media.

To meet these distance objectives, four PMDs were selected. The task force selected a 1310 nanometer serial PMD to meet its 2 km and 10 km single-mode fiber (SMF) objectives. It also selected a 1550 nm serial solution to meet (or exceed) its 40 km SMF objective. Support of the 40 km PMD is an acknowledgement that Gigabit Ethernet is already being successfully deployed in metropolitan and private, long distance applications. An 850-nanometer PMD was specified to achieve a 65-meter objective over multimode fiber using serial 850 nm transceivers.

Additionally, the task force selected two versions of the wide wave division multiplexing (WWDM) PMD, a 1310 nanometer version over single-mode fiber to travel a distance of 10km and a 1310 nanometer PMD to meet its 300-meter-over-installed-multimode-fiber objective.

10.6.5 Physical layer (PHYs)

The LAN PHY and the WAN PHY operate over common PMDs and, therefore, support the same distances. These PHYs are distinguished solely by the physical encoding sublayer (PCS).

The 10 Gigabit LAN PHY is intended to support existing Gigabit Ethernet applications at ten times the bandwidth with the most cost-effective solution. Over time, it is expected that the LAN PHY may be used in pure optical switching environments extending over all WAN distances. However, for compatibility with the existing WAN network, the 10 Gigabit Ethernet WAN PHY supports connections to existing and future installations of SONET/SDH (Synchronous Optical Network/Synchronous Digital Hierarchy) circuit-switched telephony access equipment.

The WAN PHY differs from the LAN PHY by including a simplified SONET/SDH framer in the WAN Interface sublayer (WIS). Because the line rate of SONET OC-192/SDH STM-64 is within a few percent of 10 Gbps, it is relatively simple to implement a MAC that can operate with a LAN PHY at 10 Gbps or with a WAN PHY payload rate of approximately 9.29 Gbps.

In order to enable low-cost WAN PHY implementations, the task force specifically rejected conformance to SONET/SDH jitter, stratum clock, and certain SONET/SDH optical specifications. The WAN PHY is basically a cost effective link that uses common Ethernet PMDs to provide access to the SONET infrastructure, thus enabling attachment of packet-based IP/Ethernet switches to the SONET/SDH and time division multiplexed (TDM) infrastructure. This feature enables Ethernet to use SONET/SDH for layer 1 transport across the WAN transport backbone.

It is also important to note that Ethernet remains an asynchronous link protocol where the timing of each message is independent. As in every Ethernet network, 10 Gigabit Ethernet's timing and synchronization must be synchronously maintained for each character in the bit stream of data, but the receiving hub, switch, or router may re-time and re-synchronize the data. In contrast, synchronous protocols, including SONET/SDH, require that each device share the same system clock to avoid timing drift between transmission and reception equipment and subsequent increases in network errors where timed delivery is critical.

The WAN PHY attaches data equipment such as switches or routers to a SONET/SDH or optical network. This allows simple extension of Ethernet links over those networks. Therefore, two routers will behave as though they are directly attached to each other over

a single Ethernet link. Since no bridges or store-and-forward buffer devices are required between them, all the IP traffic management systems for differentiated service operate over the extended 10 Gigabit Ethernet link connecting the two routers.

To simplify management of extended 10 Gigabit Ethernet links, the WAN PHY provides most of the SONET/SDH management information, allowing the network manager to view the Ethernet WAN PHY links as though they are SONET/SDH links. It is then possible to do performance monitoring and fault isolation on the entire network, including the 10 Gigabit Ethernet WAN link, from the SONET/SDH management station. The SONET/SDH management information is provided by the WAN interface sublayer (WIS), which also includes the SONET/SDH framer. The WIS operates between the 64B/66B PCS and serial PMD layers common to the LAN PHY.

10.6.6 10 Gigabit Ethernet applications in LANs

Ethernet technology is already the most deployed technology for high performance LAN environments. With the extension of 10 Gigabit Ethernet into the family of Ethernet technologies, the LAN now can reach farther and support up coming bandwidth hungry applications.

Similar to Gigabit Ethernet technology, the 10 Gigabit standard supports both single-mode and multi-mode fiber mediums. However in 10 Gigabit Ethernet, the distance for single-mode fiber has expanded from the 5 km that Gigabit Ethernet supports to 40 km in 10 Gigabit Ethernet.

The advantage for the support of longer distances is that it gives companies who manage their own LAN environments the option of extending their data centers to more cost-effective locations up to 40 km away from their campuses. This also allows them to support multiple campus locations within that 40 km range. Within data centers, switch-to-switch applications, as well as switch to server applications, can also be deployed over a more cost effective multi-mode fiber medium to create 10 Gigabit Ethernet backbones that support the continuous growth of bandwidth hungry applications.

With 10 Gigabit backbones installed, companies will have the capability to begin providing Gigabit Ethernet service to workstations and, eventually, to the desktop in order to support applications such as streaming video, medical imaging, centralized applications, and high-end graphics. 10 Gigabit Ethernet will also provide lower network latency due to the speed of the link and over-provisioning bandwidth to compensate for the bursty nature of data in enterprise applications. Additionally, the LAN environment must continue to change to keep up with the growth of the Internet.

10.6.7 10 Gigabit Ethernet metropolitan and storage area networks

Vendors and users generally agree that Ethernet is inexpensive, well understood, widely deployed and backwards compatible from Gigabit switched down to 10 Megabit shared. Today a packet can leave a server on a short-haul optic Gigabit Ethernet port, move cross-country via a DWDM (dense wave division multiplexing) network, and find its way down to a PC attached to a ‘thin coax’ BNC connector, all without any re-framing or protocol conversion. Ethernet is literally everywhere, and 10 Gigabit Ethernet maintains this seamless migration in functionality.

Gigabit Ethernet is already being deployed as a backbone technology for dark fiber metropolitan networks. With appropriate 10 Gigabit Ethernet interfaces, optical transceivers and single mode fiber, service providers will be able to build links reaching 40 km or more.

Additionally, 10 Gigabit Ethernet will provide infrastructure for both network-attached storage (NAS) and storage area networks (SAN).

Prior to the introduction of 10 Gigabit Ethernet, some industry observers maintained that Ethernet lacked sufficient horsepower to get the job done. Ethernet, they said, just doesn't have what it takes to move 'dump truck loads worth of data.' 10 Gigabit Ethernet, can now offer equivalent or superior data carrying capacity at similar latencies to many other storage networking technologies including 1 or 2 Gigabit fiber channel, Ultra160 or 320 SCSI, ATM OC-3, OC-12 & OC-192, and HIPPI (high performance parallel interface).

While Gigabit Ethernet storage servers, tape libraries and compute servers are already available; users should look for early availability of 10 Gigabit Ethernet end-point devices in the second half of 2001.

There are numerous applications for Gigabit Ethernet in storage networks today, which will seamlessly extend to 10 Gigabit Ethernet as it becomes available. These include:

- Business continuance/disaster recovery
- Remote backup
- Storage on demand
- Streaming media

10.6.8 10 Gigabit Ethernet in wide area networks

10 Gigabit Ethernet will enable Internet service providers (ISP) and network service providers (NSPs) to create very high speed links at a very low cost, between co-located, carrier-class switches and routers and optical equipment that is directly attached to the SONET/SDH cloud.

10 Gigabit Ethernet with the WAN PHY will also allow the construction of WANs that connect geographically dispersed LANs between campuses or POPs (points of presence) over existing SONET/SDH/TDM networks. 10 Gigabit Ethernet links between a service provider's switch and a DWDM (dense wave division multiplexing) device or LTE (line termination equipment) might in fact be very short – less than 300 meters.

10.6.9 Conclusion

As the Internet transforms long standing business models and global economies, Ethernet has withstood the test of time to become the most widely adopted networking technology in the world. Much of the world's data transfer begins and ends with an Ethernet connection. Today, we are in the midst of an Ethernet renaissance spurred on by surging e-business and the demand for low cost IP services that have opened the door to questioning traditional networking dogma. Service providers are looking for higher capacity solutions that simplify and reduce the total cost of network connectivity, thus permitting profitable service differentiation, while maintaining very high levels of reliability.

Ethernet is no longer designed only for the LAN. 10 Gigabit Ethernet is the natural evolution of the well-established IEEE 802.3 standard in speed and distance. It extends Ethernet's proven value set and economics to metropolitan and wide area networks by providing:

- Potentially lowest total cost of ownership (infrastructure/operational/human capital)
- Straightforward migration to higher performance levels
- Proven multi-vendor and installed base interoperability (plug and play)

- Familiar network management feature set

An Ethernet-optimized infrastructure build out is taking place. The metro area is currently the focus of intense network development to deliver optical Ethernet services. 10 Gigabit Ethernet is on the roadmaps of most switch, router and metro optical system vendors to enable:

- Cost effective Gigabit-level connections between customer access gear and service provider POPs (points of presence) in native Ethernet format
- Simple, very high speed, low-cost access to the metro optical infrastructure
- Metro-based campus interconnection over dark fiber targeting distances of 10/40 km and greater
- End to end optical networks with common management systems

11

Ethernet cabling and connectors

Objectives

When you have completed study of this chapter you will be able to:

- Describe the various types of physical transmission media used for local area networks
- Describe the structure of cables
- Examine factors affecting cable performance
- Describe factors affecting selection of cables
- Explain salient features of AUI cables, coaxial cables, twisted pair cables with their categories, and, fiber optic cables
- Discuss the advantages and disadvantages of each cable type
- Describe Ethernet cabling requirements for various Ethernet media systems
- Describe salient features of various cable connectors
- Describe the use of Ethernet cables and connectors in industrial environments

11.1 Cable types

Three main types of cable are used in networks:

- Coaxial cable, also called coax, which can be thin or thick
- Twisted pair cable, which can be shielded (STP) or unshielded (UTP)
- Fiber optic cables, which can be single-mode, multimode or graded-index multimode

There is also a fourth group of cables, known as IBM cable, which is essentially twisted pair cable, but designed to somewhat more stringent specifications by IBM. Several types are defined, and they are used primarily in IBM token ring networks.

11.2 Cable structure

All cable types have the following components in common:

- One or more conductors to provide a medium for the signal. The conductor might be a copper wire or glass
- Insulation of some sort around the conductors to help keep the signal in and interference out
- An outer sheath, or jacket, to encase the cable elements. The sheath keeps the cable components together, and may also help protect the cable components from water, pressure, or other types of damage

11.2.1 Conductor

For copper cable, the conductor is known as the signal, or carrier, wire, and it may consist of either solid or stranded wire. Solid wire is a single thick strand of conductive material, usually copper. Stranded wire consists of many thin strands of conductive material wound tightly together.

The signal wire is described in the following terms:

- The wire's conductive material (for example, copper)
- Whether the wire is stranded or solid
- The carrier wire's diameter, expressed directly in units of measurement (for example, in inches, centimeters, or millimeters), or in terms of the wire's gauge, as specified in the AWG (American Wire Gauge)
- The total diameter of the strand, which determines some of the wire's electrical properties, such as resistance and impedance. These properties, in turn, help determine the wire's performance
- For fiber optic cable, the conductor is known as the core. The core can be made from either glass or plastic, and is essentially a cylinder that runs through the cable. The diameter of this core is expressed in microns (millionths of a meter)

11.2.2 Insulation

The insulating layer keeps the transmission medium's signal from escaping and also helps to protect the signal from outside interference. For copper wires, the insulation is usually made of a dielectric such as polyethylene. Some types of coaxial cable have multiple protective layers around the signal wire. The size of the insulating layer determines the spacing between the conductors in a cable and therefore its capacitance and impedance.

For fiber optic cable, the 'insulation' is known as cladding and is made of material with a lower refractive index than the core's material. The refractive index is a measure that indicates the manner in which a material will reflect light rays. The lower refractive index ensures that light bounces back off the cladding and remains in the core.

Cable sheath

The outer casing, or sheath, of the cable provides a shell that keeps the cable's elements together. The sheath differs for indoor and outdoor exposure. Outdoor cable sheaths tend to be black, with appropriate resistance to UV light, and have enhanced water resistance. Two main indoor classes of sheath are plenum and non-plenum.

Plenum cable sheath

For certain environments, law requires plenum cable. It must be used when the cable is being run 'naked' (without being put in a conduit) inside walls, and should probably be used whenever possible. Plenum sheaths are made of non-flammable fluoro-polymers such as Teflon or Kynar. They are fire-resistant and do not give off toxic fumes when burning. They are also considerably more expensive (by a factor of 1.5 to 3) than cables with non-plenum sheaths. Studies have shown that cables with plenum sheaths have less signal loss than non-plenum cables. Plenum cable specified for networks installed in the United States should generally meet the National Electrical Code (NEC) CMP (communications plenum cable) or CL2P (class 2 plenum cable) specifications. Networks installed in other countries may have to meet equivalent safety standards, and these should be determined before installation. The cable should also be Underwriters Laboratories (UL) listed for UL-910, which subjects plenum cable to a flammability test.

Non-plenum cable sheath

Non-plenum cable uses less expensive material for sheaths, so it is consequently less expensive than cable with plenum sheaths, but it can often be used only under restricted conditions. Non-plenum cable sheaths are made of polyethylene (PE) or polyvinyl chloride (PVC), which will burn and give off toxic fumes. PVC cable used for networks should meet the NEC CMR (communications riser cable) or CL2R (class 2 riser cable) specifications. The cable should also be UL-listed for UL-1666, which subjects riser cable to a flammability test.

Cable packaging

Cables can be packaged in different ways, depending on what it is being used for and where it is located. For example, the IBM cable topology specifies a flat cable for use under carpets.

The following types of cable packaging are available:

- Simplex cable – one cable within one sheath, which is the default configuration. The term is used mainly for fiber optic cable to indicate that the sheath contains only a single fiber.
- Duplex cable – two cables, or fibers, within a single sheath. In fiber optic cable, this is a common arrangement. One fiber is used to transmit in each direction.
- Multi-fiber cable – multiple cables, or fibers, within a single sheath. For fiber optic cable, a single sheath may contain thousands of fibers. For electrical cable, the sheath will contain at most a few dozen cables.

11.3 Factors affecting cable performance

Copper cables are good media for signal transfer, but they are not perfect. Ideally, the signal at the end of a length of cable should be the same as at the beginning. Unfortunately, this will not be true in practical cables. All signals degrade when transmitted over a distance through any medium. This is because its amplitude decreases as the medium resists the flow of energy, and signals can become distorted because the shape of the electrical signal changes over distance. Any transmission also consists of signal and noise components. Signal quality degrades for several reasons, including attenuation, crosstalk, and impedance mismatches.

11.3.1 Attenuation

Attenuation is the decrease in signal strength, measured in decibels (dB) per unit length. Such loss happens as the signal travels along the wire. Attenuation occurs more quickly at higher frequencies and when the cable's resistance is higher. In networking environments, repeaters are responsible for regenerating a signal before passing it on. Many devices are repeaters without explicitly saying so. Since attenuation is sensitive to frequency, some situations require the use of equalizers to boost signals of different frequencies with the appropriate amount.

11.3.2 Characteristic impedance

The *impedance* of a cable is defined as the resistance offered to the flow of electrical current at a particular frequency. The characteristic impedance is the impedance of an infinitely long cable so that the signal never reaches end of the cable, and hence cannot bounce back. The same situation is replicated when a cable is terminated, so that the signal cannot bounce back. So the characteristic impedance is the impedance of a short cable when it is terminated as shown in Figure 11.1. Such a cable then appears electrically to be infinitely long and has no signal reflected from the termination. If one cable is connected to another of differing characteristic impedance, then signals are reflected at their interface. These reflections cause interference with the data signals and they must be avoided by using cables of the same characteristic impedance.

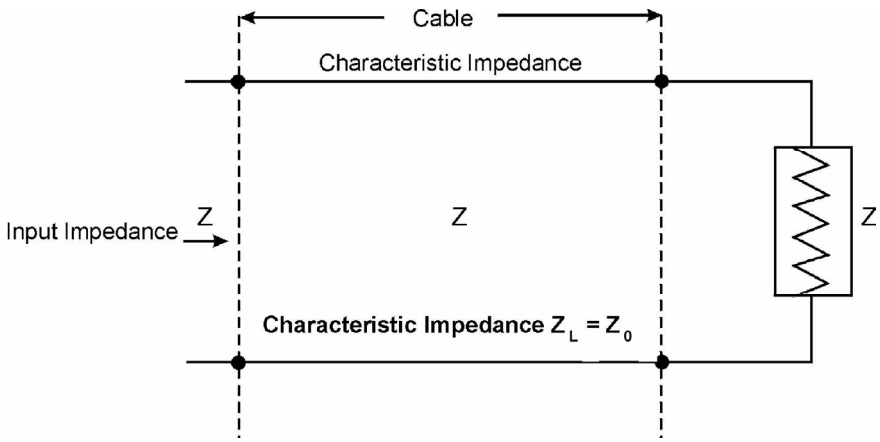


Figure 11.1
Characteristic impedance

11.3.3 Crosstalk

Crosstalk is electrical interference in the form of signals picked up from a neighboring cable or circuits; for example, signals on different wires in a multi-stranded twisted pair cable may interfere with each other. Crosstalk is non-existent in fiber optic cables.

The following forms of cross talk measurement are important for twisted pair cables:

- Near-end cross talk or NEXT: NEXT measurements (in dB) indicate the degree to which unwanted signals are coupled onto adjacent wire pairs. This unwanted 'bleeding over' of a signal from one wire pair to another can distort the desired signal. As the name implies, NEXT is measured at the 'near end' or the end closest to the transmitted signal. NEXT is a 'pair-to-pair' reading where each wire pair is tested for crosstalk relative to another pair. NEXT increases as the frequency of transmission increases. See Figure 11.2.

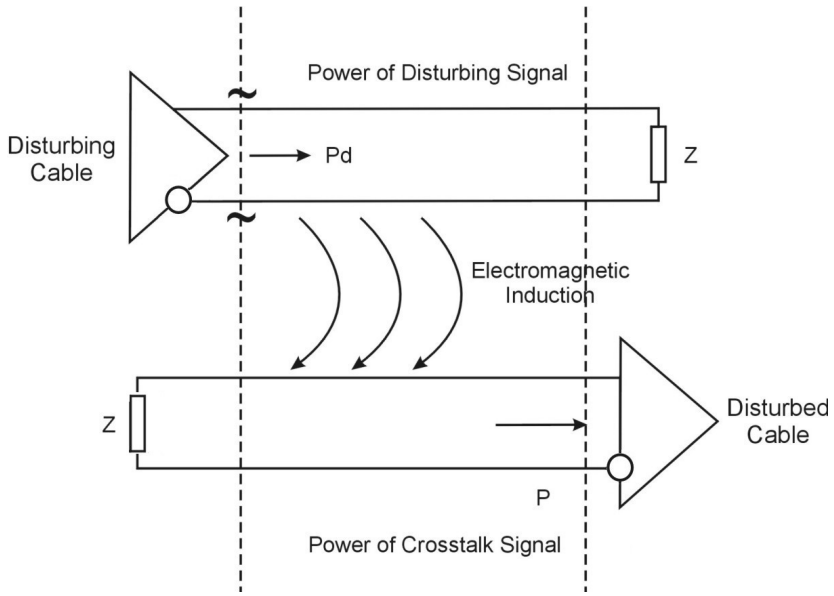


Figure 11.2
Near end crosstalk (NEXT)

- Far-end crosstalk or FEXT is similar in nature to NEXT, but crosstalk is measured at the opposite end from the transmitted signal. FEXT tests are affected by signal attenuation to a much greater degree than NEXT since FEXT is measured at the far end of the cabling link where signal attenuation is greatest. Therefore, FEXT measurements are a more significant indicator of cable performance if attenuation is accounted for
- Equal-level far-end crosstalk or ELFEXT: The comparative measurement of FEXT and attenuation is called equal level far end crosstalk or ELFEXT. ELFEXT is the arithmetic difference between FEXT and attenuation. Characterizing ELFEXT is important for cabling links intended to support 4 pair, full-duplex network transmissions
- Attenuation-to-cross talk ratio or ACR is not specifically a new test, but rather a relative comparison between NEXT and attenuation performance. Expressed in decibels (dB) the ratio is the arithmetic difference between NEXT and attenuation. ACR is significant because it is more indicative of cable performance than NEXT or attenuation alone. ACR is a measure of the strength of a signal compared to the crosstalk noise
- Power sum NEXT – power sum (in dB) is calculated from the six measured pair-to-pair cross talk results. Power Sum NEXT differs from pair-to-pair NEXT by determining the crosstalk induced on a given wire pair from 3 disturbing pairs. This methodology is critical for support of transmissions that utilize all four pairs in the cable such as Gigabit Ethernet. See Figure 11.3.

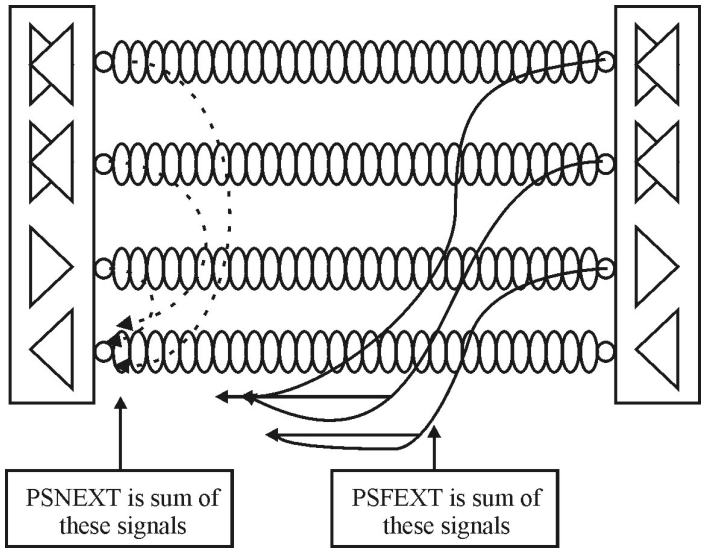


Figure 11.3
PSNEXT and PSFEXT

11.4 Selecting cables

Cables are used to meet all sorts of power and signaling requirements. The demands made on a cable depend on the location in which the cable is used and the function for which the cable is intended. These demands, in turn, determine the features a cable should have.

11.4.1 Function and location

Here are a few examples of considerations involving the cable's function and location:

- Cable designed to run over long distances, such as between floors or buildings, should be robust against environmental factors (moisture, temperature changes, and so on). This may require extra sheaths or sheaths made with a special material. Fiber optic cable performs well, even over distances much longer than a floor or even a building.
- Cable that must run around corners should bend easily, and the cable's properties and performance should not be affected by the bending. For several reasons, twisted pair cable is probably the best cable for such a situation (assuming it makes sense within the rest of the wiring scheme). Of course, another way to get around a corner is by using a connector. However, connectors may introduce signal-loss problems.
- Cables that must run through areas in which heavy current motors are operating (or worse, being turned on and off at random intervals) must be able to withstand magnetic interference. Large currents produce strong magnetic fields, which can interfere with and disrupt nearby signals. Because it is not affected by such electrical or magnetic fluctuations, fiber optic cable is the best choice in machinery-intensive environments.
- If you need to run many cables through a limited area, cable weight can become a factor, particularly if all cables will be running in the ceiling. In general, fiber optic and twisted pair cables tend to be the lightest.
- Cables being installed in barely accessible locations must be particularly reliable. It is worth considering installing a backup cable during the initial

installation. Because the installation costs in such locations are generally much more than the cable material cost, installation costs for the second cable add only marginally to the total cost. Generally, the suggestion is to make at least the second cable optical fiber

- Cables that need to interface with other worlds (for example, with a mainframe network or a different electrical or optical system) may need special properties or adapters. The kinds of cable required will depend on the details of the environments and the transition between them.

11.4.2 Main cable selection factors

Along with the function and location considerations, cable selections are determined by a combination of factors, including the following:

- The type of network being created (for example, Ethernet or token ring) – while it is possible to use just about any type of cable in any type of network, certain cable types have been more closely associated with particular network types.
- The amount of money available for the network –cable installation is a major part of the network costs.
- Cabling resources currently available (and useable) –available wiring that could conceivably be used for a network should be evaluated. It is almost certain, however, that at least some of that wire is defective or is not up to the requirements for the proposed network.
- Building or other safety codes and regulations.

11.5 AUI cable

Attachment unit interface cable (AUI) is a shielded multi-stranded cable used to connect Ethernet devices to Ethernet transceivers, and for no other purpose. AUI cable is made up of four individually shielded pairs of wire surrounded by a shielding double sheath. This shield makes the cable more resistant to signal interference, but increases attenuation over long distance.

Connection to other devices is made through DB15 connectors. Connectors at the end of the cable are male and female respectively. Any cable with male-male or female-female connectors at both ends are non-standard and should not be used.

AUI cable is used to connect transceivers to other Ethernet devices, and transceivers need power to operate. This power may be supplied to transceivers by an external power supply or by a pair of wires in the AUI cable dedicated to power supply.

AUI cable is available in two types, standard AUI and office AUI. Standard AUI cable is made up of 20 or AWG copper wire and can be used for distances up to 50 m. but is 0.420 inch thick and is somewhat inflexible. Office AUI cable is thinner (0.26 inch) and is made up of 28 AWG wire, and is relatively flexible. It can be used over distances of 16.5 m only.

Office AUI cable should be used only when standard version is found to be cumbersome due to its inflexibility.

11.6 Coaxial cables

In coaxial cables, two or more separate materials share a common central axis. Coaxial cables, often called coax, are used for radio frequency and data transmission. The cable is

remarkably stable in terms of its electrical properties at frequencies below 4 GHz, and this makes the cable popular for cable television (CATV) transmissions, as well as for creating local area networks (LANs).

11.6.1 Coaxial cable construction

A coaxial cable consists of the following layers (moving outward from the center) as shown in Figure 11.4.

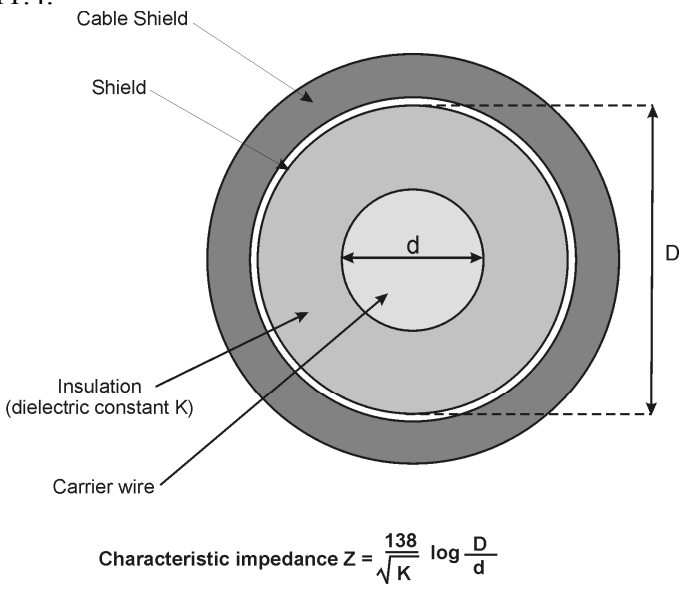


Figure 11.4
Cross-section of a coaxial cable

- **Carrier wire**
  A conductor wire or signal wire is in the center. This wire is usually made of copper and may be solid or stranded. There are restrictions regarding the wire composition for certain network configurations. The diameter of the signal wire is one factor in determining the attenuation of the signal over distance. The number of strands in a multi-strand conductor also affects the attenuation.
- **Insulation**
  An insulation layer consists of a dielectric around the carrier wire. This dielectric is usually made of some form of polyethylene or Teflon.
- **Foil shield**
  This thin foil shield around the dielectric usually consists of aluminum bonded to both sides of a tape. Not all coaxial cables have foil shielding. Some have two foil shield layers, interspersed with copper braid shield layers.
- **Braid shield**
  A braid, or mesh, conductor, made of copper or aluminum, that surrounds the insulation and foil shield. This conductor can serve as the ground for the carrier wire. Together with the insulation and any foil shield, the braid shield protects the carrier wire from electro magnetic interference (EMI) and radio frequency interference (RFI). It should be carefully note that the braid and foil shields provide good protection against electrostatic interference when earthed correctly, but little protection against magnetic interference.
- **Sheath**
  This is the outer cover that can be either plenum or non-plenum, depending on its composition. The layers surrounding the carrier wire also help prevent

signal loss due to radiation from the carrier wire. The signal and shield wires are concentric, or co-axial and hence the name.

11.6.2 Coaxial cable performance

The main features that affect the performance of coaxial cables are its composition, diameter, and impedance:

- The carrier wire's composition determines how good a conductor the cable will be. The IEEE specifies stranded copper carrier wire with tin coating for 'thin' coaxial cable, and solid copper carrier wire for 'thick' coaxial cable. (These terms will be defined shortly.)
- Cable diameter helps determine the electrical demands that can be made on the cable. In general, thick coaxial can support a much higher level of electrical activity than thin coaxial.
- Impedance is a measure of opposition to the flow of alternating current. The properties of the dielectric between the carrier wire and the braid help determine the cable's impedance.
- Impedance determines the cable's electrical properties and limits where the cable can be used. For example, Ethernet and ARCnet architectures can both use thin coaxial cable, but they have different characteristic impedances and so Ethernet and ARCnet cables are not compatible. Most LAN cables have an RG (recommended gauge) rating and cables with the same RG rating from different manufacturers can be safely mixed.

Recommended Gauge	Application	Characteristic impedance
RG-8	10Base5	50 ohms
RG-58	10Base2	50 ohms
RG-59	CATV	75 ohms
RG-2	ARCnet	93 ohms

Table 11.1
Common network coaxial cable impedances

In networks, the characteristic cable impedances range from 50 Ohms (for Ethernet) to 93 Ohms (for ARCnet). The impedance of the coaxial cable in Figure 11.4 is given by the formula:

$Z_0=(138/\sqrt{k})\log(D/d)$ in Ohms

Where k is the dielectric constant of the insulation.

11.6.3 Thick coaxial cable

Thick coaxial (RG-8) cable is 0.5 inch or 12.7 mm in diameter. It is used for 'Thick Ethernet' networks, also called 10Base5 or ThickNet networks. It can also be used for cable TV (CATV), and other connections. Thick coaxial cable is expensive and can be difficult to install and work with. Although 10Base5 is now obsolete, it remains in use in existing installations. The cable is a distinctive yellow, or orange, color, with black stripes every 2.5 meters (8 feet), indicating where node taps can be made.

This cable is constructed with a single solid copper core that carries the network signals, and a series of layers of shielding and insulator material.

Transceivers are connected to the cable at specified distances from one another, and standard transceiver cables connect these transceivers to the network devices.

Extensive shielding makes it highly resistant to electrical interference by outside sources such as lightning, machinery, etc. Bulkiness and limited flexibility of the cable limits its use to backbone media and is placed in cable runways or laid above ceiling tiles to keep it out of the way.

Thick coaxial cable is designed to access as a shared media. Multiple transceivers can be attached to the thick coaxial cable at multiple points on the cable itself. A properly installed length of thick coaxial cable can support up to 100 transceivers.

11.6.4 Thin coaxial cable

Thin coaxial cable (RG-58) is 3/16 inch or 4.76 mm in diameter. When used for IEEE802.3 networks, it is often known as Thin Ethernet. Such networks are also known as 10Base2, 'thinnet', or 'cheapernet'. When using this configuration, drop cables are not allowed. Instead, the T connector is connected directly to the Network Interface Card (NIC) at the node, since the NIC has an on-board transceiver.

It is smaller, lighter, and more flexible than thick coaxial cable. The cable itself resembles (but is not identical to) television coaxial cable.

Thin coaxial cable, due to its less extensive shielding capacity, can be run to a maximum length of 185 meters (606.7 ft).

50-ohm terminators are used on both cable ends.

11.6.5 Coaxial cable designations

Listed below are some of the available coaxial cable types.

- **RG-8:** Used for Thick Ethernet. It has 50 ohms impedance. The Thick Ethernet configuration requires an attachment unit interface (AUI) cable and a media access unit (MAU), or remote transceiver. The AUI cable required is a twisted pair cable that connects to the NIC. RG-8 is also known as N series Ethernet cable.
- **RG-58:** Used for Thin Ethernet. It has 50 ohms impedance and uses a BNC connector.

11.6.6 Advantages of a coaxial cable

A coaxial cable has the following general advantages over other types of cable that might be used for a network.

These advantages may change or disappear over time, as technology advances and products improve:

- The cable is relatively easy to install
- Coaxial cable is reasonably priced compared with other cable types

11.6.7 Disadvantages of coaxial cable

Coaxial cable has the following disadvantages when used for a network:

- It is easily damaged and sometimes difficult to work with, especially in the case of thick coaxial
- Coaxial is more difficult to work with than twisted pair cable
- Thick coaxial cable can be expensive to install, especially if it needs to be pulled through existing cable conduits
- Connectors can be expensive

11.6.8 Coaxial cable faults

The main problems encountered with coaxial cables are:

- Open or short circuited cables (and possible damage to the cable)
- Characteristic impedance mismatches
- Distance specifications being exceeded
- Missing or loose terminator

11.7 Twisted pair cable

Twisted pair cable is widely used, inexpensive, and easy to install. Twisted pair cable comes in two main varieties

- Shielded (STP)
- Unshielded (UTP)

It can transmit data at an acceptable rate – up to 1000 Mbps in some network architectures. The most common twisted pair wiring is telephone cable, which is unshielded and is usually voice-grade, rather than the higher-quality data-grade cable used for networks.

In a twisted pair cable, two conductor wires are wrapped around each other. Twisted pairs are made from two identical insulated conductors, which are twisted together along their length at a specified number of twists per meter, typically forty twists per meter (twelve twists per foot). The wires are twisted to reduce the effect of electromagnetic and electrostatic induction.

For full-duplex digital systems using balanced transmission, two sets of screened twisted pairs are required in one cable; each set with individual and overall screens. A protective PVC sheath then covers the entire cable. (Note: 10BaseT CSMA/CD is not full duplex but still needs 2 pairs)

Twisted pair cables are used with the following Ethernet physical layers

- 10BaseT
- 100BaseTX
- 100BaseT2
- 100BaseT4
- 1000BaseT

The capacitance of a twisted pair is fairly low at about 40 to 160 pF/m, allowing a reasonable bandwidth and an achievable slew rate. A signal is transmitted differentially between the two conductor wires. The current flows in opposite directions in each wire of the active circuit, as shown in Figure 11.5.

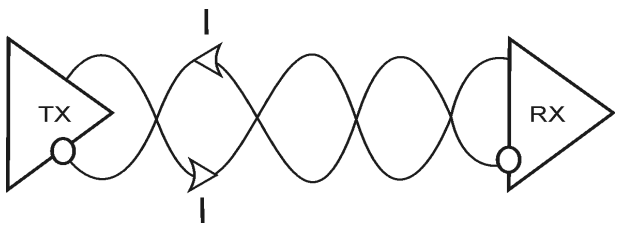


Figure 11.5
Current flow in a twisted pair

11.7.1 Elimination of noise by signal inversion

The twisting of associated pairs and method of transmission reduces the interference of the other strands of wire throughout the cable.

The network signals to transmitted in the form of changes of electrical state. Encoding turns ones and zeroes of network frames into these signals. In a twisted pair system, once a transceiver has been given an encoded signal to transmit, it will invert the polarity of that signal and transmit it on the other wire. The result of this a mirror image of the original signal.

Both the original and the inverted signal are then transmitted over the TX+ and TX– wires respectively. Since these wires are of the same length and have the same construction, the signal travels at the same rate through the cable. Since the pairs are twisted together, any outside electrical interference that affects one member of the pair will have the same effect on both signals.

The transmissions of the original signal and its mirror image reach a destination receiver. This receiver, operating in differential mode, inverts the signal on the TX- line before adding it to the signal on the TX- line. The signal on the TX- line is, however, already inverted so in reality an addition takes place. The noise component on the TX- line, however, is not inverted and as a result it is subtracted from the noise on the TX+ line, resulting in noise cancellation.

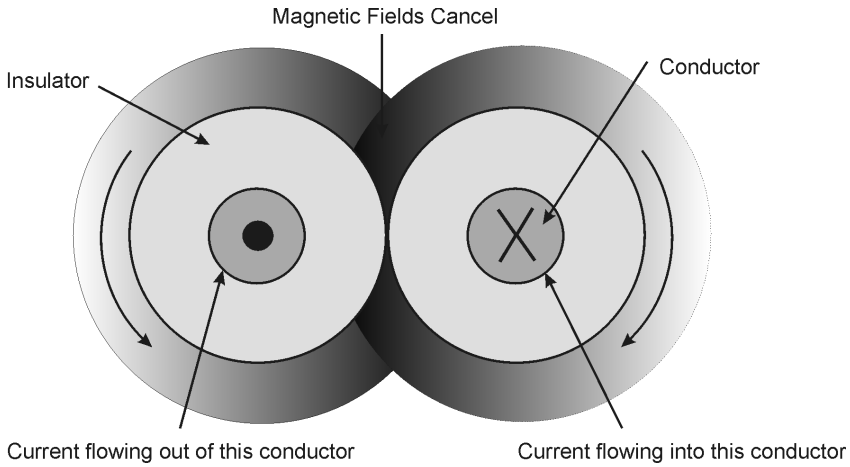


Figure 11.6
Magnetic shielding of twisted pair cables

Since the currents in the two conductors are equal and opposite, their induced magnetic fields also cancel each other. This type of cable is therefore self-shielding and is less prone to interference.

Twisting within a pair minimizes crosstalk between pairs. The twists also help deal with electro magnetic interference (EMI) and radio frequency interference (RFI), as well as balancing the mutual capacitance of the cable pair. The performance of a twisted pair cable can be influenced by changing the number of twists per meter in a wire pair. Each of the pairs in a 4-pair category 5 cable will have a different twist rate to reduce the crosstalk between them.

11.7.2 Components of twisted pair cable

A twisted pair cable has the following components:

- **Conductor wires**
  The signal wires for this cable come in pairs that are wrapped around each other. The conductor wires are usually made of copper. They may be solid (consisting of a single wire) or stranded (consisting of many thin wires wrapped tightly together). A twisted pair cable usually contains multiple twisted pairs; 2, 4, 6, 8, 25, 50, or 100 twisted pair bundles are common. For network applications, 2 and 4 pair cables are most commonly used
- **Shield**
  Some twisted pair cables have a shield in the form of a woven braid. This type of cable is referred to as shielded twisted pair or STP. **Sheath**
  The wire bundles are encased in a sheath made of polyvinyl chloride (PVC) or, in plenum cables, of a fire-resistant material, such as Teflon or Kynar. STP contains an extra shield or protective screen around each of the wire pairs to cut down on extraneous signals. This added protection also makes STP more expensive than UTP.

11.7.3 Shielded twisted pair (STP) cable

STP refers to the 150 ohm twisted pair cabling defined by the IBM cabling specifications for use with token ring networks. 150 ohm STP is not generally used with Ethernet. However, the Ethernet standard does describe how it can be adapted for use with 10BaseT, 100BaseTX, and 100BaseT2 Ethernet by installing special impedance matching transformers, or 'baluns', that convert the 100-ohm impedance of the Ethernet transceivers to the 150-ohm impedance of the STP cable. A balun (BALanced – UNbalanced) is an impedance matching transformer that converts the impedance of one interface to the impedance of the other interface. These are generally used to connect balanced twisted pair cabling with an unbalanced coaxial cabling. These are available from IBM, AMP, and Cambridge Connectors among others.

11.7.4 Unshielded twisted pair (UTP) cable

UTP cable does not include any extra shielding around the wire pairs. This type of cable is used in some slower speed Token Ring networks and can be used in Ethernet and ARCnet systems.

UTP is now the primary choice for many network architectures, with the IEEE approving standards for 10, 100 and 1000 Mbps Ethernet systems using UTP cabling. These are known as:

- 10BaseT for 10 Mbps
- 100BaseTX for 100 Mbps
- 1000 BaseT2 for 1000 Mbps on twisted pair cable

Because it lacks a conductive shield, UTP is not as good at blocking electrostatic noise and interference as STP or coaxial cable. Consequently, UTP cable segments must be shorter than when using other types of cable. For standard UTP, the length of a segment should never exceed 100 meters, or about 330 feet. Conversely, UTP is quite inexpensive, and is very easy to install and work with. The price and ease of installation make UTP tempting, but bear in mind that installation labor is generally the major part of the cabling expense and that other types of cable may be just as easy to install.

Four-pair UTP cable

UTP cabling most commonly includes 4 pairs of wires enclosed in a common sheath. 10BaseT, 100BaseTX, and 100BaseT2 use only two of the four pairs, while 100BaseT4 and 1000BaseT require all four pairs. Two-pair UTP is, however, available and is sometimes used in Industrial Ethernet installations.

The typical UTP cable is a polyvinyl chloride (PVC) or plenum-rated plastic jacket containing four pairs of wire. The majority of facility cabling in current and new installations is of this type. The dedicated (single) connections made using four-pair cable are easier to troubleshoot and replace than the alternative, bulk multi-pair cable such as 25-pair cable.

The insulation of each wire in a four-pair cable will have an overall color: brown, blue, orange, green, or white. In a four-pair UTP cable there is one wire each of brown, blue, green, and orange, and four wires of which the overall color is white. Periodically placed (usually within 1/2 inch of one another) rings of the other four colors distinguish the white wires from one another.

Wires with a unique base color are identified by that base color i.e. “blue”, “brown”, “green”, or “orange”. Those wires that are primarily white are identified as “white/<color>”, where “<color>” indicates the color of the rings.

The 10BaseT and 100BaseTX standards are concerned with the use of two pairs, pair 2 and pair 3 (of either EIA/TIA 568 specification). The A and B specifications are basically the same, except that pair 2 (orange) and pair 3 (green) are swapped. 10BaseT and 100BaseTX configure devices transmit over pair 3 of the EIA/TIA 568A specification (pair 2 of EIA/TIA 568B), and to receive from pair 2 of the EIA/TIA 568A specification (pair 3 of EIA/TIA 568B). The use of the wires of a UTP cable is shown in Table 11.2.

Wire Colour	EIA/TIA Pair	Ethernet Signal Use	
		568A	568Br
White/Blue (W-BL)	Pair 1	Not Used	
Blue (BL)			
White/Orange (W-OR)	Pair 2	RX+	TX+
Orange (OR)		RX-	TX-
White/Green (W-GR)	Pair 3	TX+	RX+
Green (GR)		TX-	RX-
White/Brown (W-BR)	Pair 4	Not Used	
Brown (BR)			

Table 11.2
Four-pair wire use for 10BaseT and 100BaseTX

Twenty-five pair cable

UTP cabling in large installations requiring several cable runs between two points is often 25-pair cable. This is a heavier, thicker form of UTP. The wires within the plastic jacket are of the same construction, and are twisted around associated wires to form pairs, but there are 50 individual wires twisted into 25 pairs in these larger cables. In most cases, 25-pair cable is used to connect wiring closets to one another, or to distribute large amounts of cable to intermediate distribution points, from which four-pair cable is run to the end stations.

Wires within a 25-pair cable are identified by color. The insulation of each wire in a 25-pair cable has an overall color: violet, green, brown, blue, red, orange, yellow, gray, black, and white.

In a 25-pair UTP, cable two colors identify all wires in the cable. The first color is the base color of the insulator; the second color is the color of narrow bands on the base color. These identifying bands are periodically placed on the wire, and repeated at regular intervals. A wire in a 25-pair cable is identified first by its base color, and then further specified by the color of the bands.

As a 25-pair cable can be used to make up to 12 connections between Ethernet stations (two wires in the cable are typically not used), the wire pairs need to be identified not only as transmit/receive pairs, but what other pair they are associated with.

There are two methods of identifying sets of pairs in a 25-pair cable.

The first is based on the connection of a 25-pair cable to a specific type of connector designed especially designed for it, the RJ-21 connector. The second is based on connection to a punch down block, a cable management device typically used to make the transition from a single 25-pair cable to a series of four-pair cables easier.

Crossover connections for 10BaseT and 100BaseTX

Crossing over is the reversal of transmit and receive pairs at opposite ends of a single cable. The 10BaseT and 100BaseTX specifications require that some UTP connections be crossed over.

Those cables that maintain the same pin numbers for transmit and receive pairs at both ends are called straight-through cables.

The 10BaseT and 100BaseTX specifications are designed around connections from the networking hardware to the end user stations being made through straight-through cabling. Thus, the transmit wires of a networking device such as a stand-alone hub or repeater connect to the receive pins of a 10BaseT or 100BaseTX end station.

If two similarly designed network devices, e.g. two hubs, are connected using a straight-through cable, the transmit pins of one device are connected to the transmit pins of the other device. In effect, the two devices will both attempt to transmit on the same pair.

A crossover must therefore be placed between two similar devices, so that transmit pins of one device to connect to the receive pins of the other device. When two similar devices are being connected using UTP cabling, an odd number of crossover cables, preferably only one, must be part of the cabling between them.

Screened twisted pair (ScTP) cables

Screened twisted pair cable, also referred to as foil twisted pair (FTP) is a 4-pair 100-ohm UTP, with a foil screen surrounding all four pairs in order to minimize EMI radiation and susceptibility to outside noise. This is simply a shielded version of the category 3, 4, and 5 UTP cable. It may be used in Ethernet applications in the same manner as equivalent category of UTP cable.

There are versions available where individual screens wrap around each pair.

11.7.5 EIA/TIA 568 cable categories

To distinguish varieties of UTP, the EIA/TIA has formulated several categories. The electrical specifications for these cables are detailed in EIA/TIA 568A, TSB-36, TSB-40 and their successor SP2840.

These categories are:

- **Category 1**
  Voice-grade, UTP telephone cable. This describes the cable that has been used for years in North America for telephone communications. Officially, such cable is not considered suitable for data-grade transmissions. In practice,

however, it works fine over short distances and under ordinary working conditions. It should be known that other national telecommunications providers have often used cable that does not even come up to this minimum standard, and as such, is unacceptable for data transmission.

- **Category 2**
  Voice-grade UTP, although capable of supporting transmission rates of up to 4 Mbps. IBM type 3 cable falls into this category.
- **Category 3**
  Data-grade UTP, used extensively for supporting data transmission rates of up to 10 Mbps. An Ethernet 10BaseT network requires at least this category of cable. Category 3 UTP cabling must not produce an attenuation of a 10 MHz signal greater than 98 dB/km at the control temperature of 20°C. Typically category 3 cable attenuation increases 1.5% per degree Celsius.
- **Category 4**
  Data-grade UTP, capable of supporting transmission rates of up to 16 Mbps. An IBM Token Ring network transmitting at 16 Mbps requires this type of cable. Category 4 UTP cabling must not produce an attenuation of a 10 MHz signal greater than 72 dB/km at the control temperature of 20°C.
- **Category 5**
  Data-grade UTP, capable of supporting transmission rates of up to 155 Mbps (but officially only up to 100 Mbps). Category 5 cable is constructed and insulated such that the maximum attenuation of a 10 MHz signal in a cable run at the control temperature of 20°C is 65 dB/km. TSB-67 contains specifications for the verification of installed UTP cabling links that consist of cables and connecting hardware specified in the TIA-568A standard.
- **Enhanced category 5 standard (category 5e)**
  "Enhanced Cat5" specifies transmission performance that exceeds that of Cat5, and it is used for 10BaseT, 100BaseTX, 155 Mbps ATM, etc. It has improved specifications for NEXT, PSELFEXT and attenuation.
  Category 5e directly supports the needs of Gigabit Ethernet.
  Its frequency range is measured from 1 through 100 MHz (not 100 Mbps).
- **Category 6**
  The specifications for Category 6 aim to deliver a 100 m (330 feet) channel of twisted pair cabling that provides a minimum ACR at 200-250 MHz that is approximately equal to the minimum ACR of a Category 5 channel at 100 MHz.
  Category 6 includes all of the CAT 5e parameters but sweeps the test frequency out to 200 MHz, greatly exceeding current Category 5 requirements The IEEE has proposed extending the test frequency to 250 MHz to characterize links that may be marginal at 200 MHz.
  Test parameters: All of the same performance parameters that have been specified for Category 5e.
  Frequency range for specifications: Category 6 components and links are to be tested to 250 MHz even though the ACR values for the installed links are negative at 250 MHz.
  Note: Several vendors are promoting 'proprietary category 6 solutions'. These 'proprietary category 6' cabling systems only deliver a comparable level of performance if every component of connecting hardware (socket and plug combination) is purchased from the same vendor and from the same product series.

Specially selected 8-pin modular connector jacks and plugs need to be matched because they are designed as a 'tuned pair' in order to achieve the high level of cross-talk performance for NEXT and FEXT. If the user mixes and matches connector components, the system will no longer deliver the promised 'category 6-like' performance.

- **Category 7**
  This is a proposed shielded twisted pair (STP) standard that aims to support transmission up to 600 MHz.

11.7.6 Category 3, 4 and 5 performance features

Twisted pair cable is categorized in terms of its electrical performance properties. The features that characterize the data grades of UTP cable are defined in EIA/TIA 568:

Attenuation

This value indicates how much power the signal loses and is dependant on the frequency of the transmission. The maximum attenuation per 1000 feet of UTP cable at 20º Celsius at various frequencies is specified as follows:

Since connectors are needed at each end of the cable, the standards specify a worst-case attenuation figure to be met by the connecting hardware assuming they have characteristic impedance of 100 ohms to match the UTP cable.

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
0.772	**6.8**	**5.7**	**5.5**
1.0	**7.8**	**6.5**	**6.3**
4.0	**17.0**	**13**	**13**
8.0	**26**	**19**	**18**
10	**30**	**22**	**20**
16	**40**	**27**	**25**
20	**--**	**31**	**28**
25	**--**	**--**	**32**
31.25	**--**	**--**	**30**
62.5	**--**	**--**	**52**
100	**--**	**--**	**67**

Table 11.4
Maximum attenuation per 1000 feet for Cat 3, 4, 5 cables

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
1.0	0.4	0.1	0.1
4.0	0.4	0.1	0.1
8.0	0.4	0.1	0.1
10	0.4	0.1	0.1
16	0.4	0.2	0.2
20	--	0.2	0.2
25	--	--	0.2
31.25	--	--	0.2
62.5	--	--	0.3
100	--	--	0.4

Table 11.5
Worst case of connecting hardware for 100 ohm UTP

Mutual capacitance

Cable capacitance is measured in capacitance per unit length e.g. pF/ft, and lower values indicate better performance. The standards equate to mutual capacitance (measured at 1 kHz and 20 ^{0}C) for Category 3 cable not exceeding 20 pF/ft and for categories 4 and 11 not exceeding cables 17 pF/ft.

Characteristic impedance

All UTP cable should have characteristic impedance of 100 ±15 Ohms over the frequency range from 1 MHz to the cables highest frequency rating. Note these measurements need to be made on a cable of length at least one-eighth of a wavelength.

NEXT

The near end crosstalk (NEXT) indicates the degree of interference from a transmitting pair to an adjacent passive pair in the same cable at the near (transmission) end. This is measured by applying a balanced signal to one pair of wires and measuring its disturbing effect on another pair, both of which are terminated in their nominal characteristic impedance of 100 Ohms. This was shown in Figure 11.2 earlier in the chapter.

NEXT is expressed in decibels, in accordance with the following formula:

NEXT = 10 log P_d / P_x

Where

P_d power of the disturbing signal

P_x power of the cross talk signal

NEXT depends on the signal frequency and cable category. Performance is better at lower frequencies and for cables in the higher categories. Higher NEXT values indicate small crosstalk interference.

The standard specifies minimum values for NEXT for the fixed 10BaseT cables, known as horizontal UTP cable and for the connecting hardware. The following tables show these values for the differing categories of cable at various frequencies.

Since each cable has a connector at each end, the contribution to the NEXT from these connectors can be significant as shown in the following figure:

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
0.15	54	68	74
0.772	43	58	64
1.0	41	56	62
4.0	32	47	53
8.0	28	42	48
10	26	41	47
16	23	38	44
20	--	36	42
25	--	--	41
31.25	--	--	40
62.5	--	--	35
100	--	--	32

Table 11.6
Minimum NEXT for horizontal UTP cable at 20ºC

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
1.0	58	65	65
4.0	46	58	65
8.0	40	52	62
10	38	50	60
16	34	46	56
20	--	44	54
25	--	--	52
31.25	--	--	50
62.5	--	--	44
100	--	--	40

Table 11.7
Minimum NEXT for connectors at 20ºC

Note that the twists in the UTP cable, which enhance its cross talk performance, need to be removed to align the conductors in the connector. To maintain adequate NEXT performance the amount of untwisted wire and the separation between the conductor pairs should be minimized. The amount of untwisting should not exceed 13 mm (0.5 inch) for category 5 cables and 25 mm (1 inch) for category 4 cables.

Structural return loss (SRL)

The structural return loss (SRL) is a measure of the degree of mismatch between the characteristic impedance of the cable and the connector. This is measured as the ratio of the input power to the reflected signal power.

SRL = 10 log (input power/reflected power) dB

Higher values are better implying less reflection. For example 23 dB SRL corresponds to a reflected signal of seven percent of the input signal.

Frequency Range	Cat. 4	Cat. 5	% Reflected
1 to 20	23 dB	23 dB	7 %
20 to 100	--	14 dB	20 %

Table 11.8
Minimum structural return loss (SRL) at 20ºC

Direct current resistance

The DC resistance is an indicator of the ability of the connectors to transmit DC and low frequency currents. The maximum resistance between the input and output connectors, excluding the cable, is specified as 0.3 Ohm for Category 3, 4 and 5 UTP cables.

Ground plane effects

Note that if cables are installed on a conductive ground plane, such as a metal cable tray or in a metal conduit, the transmission line properties of mutual capacitance, characteristic impedance, return loss and attenuation can become two or three percent worse. This is not normally a problem in practice.

11.7.7 Advantages of twisted pair cable

Twisted pair cable has the following advantages over other types of cables for networks:

- It is easy to connect devices to twisted pair cable
- STP and ScTP do a reasonably good job of blocking interference
- UTP is quite inexpensive
- UTP is very easy to install
- UTP may already be installed (but make sure it all works properly and that it meets the performance specifications a network requires)

11.7.8 Disadvantages of twisted pair cable

Twisted pair cable has the following disadvantages:

- STP is bulky and difficult to work with
- UTP is more susceptible to noise and interference than coaxial or fiber optic cable
- UTP signals cannot go as far as they can with other cable types before they need amplification
- Skin effect can increase attenuation. This occurs when transmitting data at a fast rate over twisted pair wire. Under these conditions, the current tends to flow mostly on the outside surface of the wire. This greatly decreases the cross-section of the wire being used, and thereby increases resistance. This, in turn, increases signal attenuation

11.7.9 Selecting and installing twisted pair cable

When deciding on a category of cable, take future developments in the network and in technology into account. It is better to install Cat5e if Cat5 currently suited to the needs. Do not, however, install Cat6 if Cat5e is sufficient, as the bandwidth of Cat6 opens the door to unwanted interference, especially in industrial environments.

Check the wiring sequence before purchasing the cable. Different wiring sequences can hide behind the same modular plug in a twisted pair cable. (A wiring sequence, or wiring scheme, describes how wires are paired up and which locations each wire occupies in the plug.) If a plug that terminates one wiring scheme into a jack that continues with a different sequence is connected, the connection may not provide reliable transmission. If existing cable uses an incompatible wiring scheme, then a ‘cross wye’ as an adapter between the two schemes can be used.

If any of the cable purchases include patch cables (for example, to connect a computer to a wall plate), be aware that these cables come in straight through or reversed varieties. For networking applications, use the straight through cable, which means that wire 1 coming in connects to wire 1 going out. In a reversed cable; wire 2 connects to wire 2 rather than to wire 7, and so on.

11.8 Fiber optic cable

Fiber optic communication uses light signals and so transmissions are not subject to electromagnetic interference. Since a light signal encounters little resistance on its path (compared to an electrical signal traveling along a copper wire), this means that fiber optic cable can be used for much longer distances before the signal must be amplified, or

repeated. Some fiber optic segments can be several kilometers long before a repeater is needed.

In principle, data transmission using a fiber optic cable is many times faster than with copper and speeds of over 10 Gbps are possible. In reality, however, this advantage is nebulous because we are still waiting for the transmission and reception technology to catch up. Nevertheless, fiber optic connections deliver transmissions that are more reliable over greater distances, although at a somewhat greater cost. Cables of this type differ in their physical dimensions and composition and in the wavelength(s) of light with which the cable transmits.

Fiber optic cables are generally cheaper than coaxial cables, especially when comparing data capacity per unit cost. However, the transmission and receiving equipment, together with more complicated methods of terminating and joining these cables, makes fiber optic cable the most expensive medium for data communications.

The main benefits of fiber optic cables are:

- Enormous bandwidth (greater information carrying capacity)
- Low signal attenuation (greater speed and distance characteristics)
- Inherent signal security
- Low error rates
- Noise immunity (impervious to EMI and RFI)
- Logistical considerations (light in weight, smaller in size)
- Total galvanic isolation between ends (no conductive path)
- Safe for use in hazardous areas
- No crosstalk

11.8.1 Theory of operation

A fiber optic system has three components – light source, transmission medium, and detector. A pulse of light indicates one bit and absence of light indicates zero bits. The transmission medium is an ultra-thin fiber of glass. The detector generates an electrical impulse when light falls on it. A light source at one end of the optical fiber and a detector at the other end then results in a unidirectional data transmission system that accepts an electrical signal, converts it into light pulses, and then re-converts the light output to an electrical signal at receiving end.

A property of light, called refraction (bending of a ray of light when it passes from one medium into other medium) is used to ‘guide’ the ray through the whole length of fiber. The amount of refraction depends on media, and above a certain critical angle, all rays of light are refracted back into the fiber, and can propagate in this way for many kilometers with little loss.

There will be many rays bouncing around inside the optical conductor, each ray is said to have a different mode. Such a conductor is called multi-mode fiber. If the conductor diameter is reduced to within a few wavelengths of light, then the light will propagate only in a single ray. Such a fiber is called single-mode fiber.

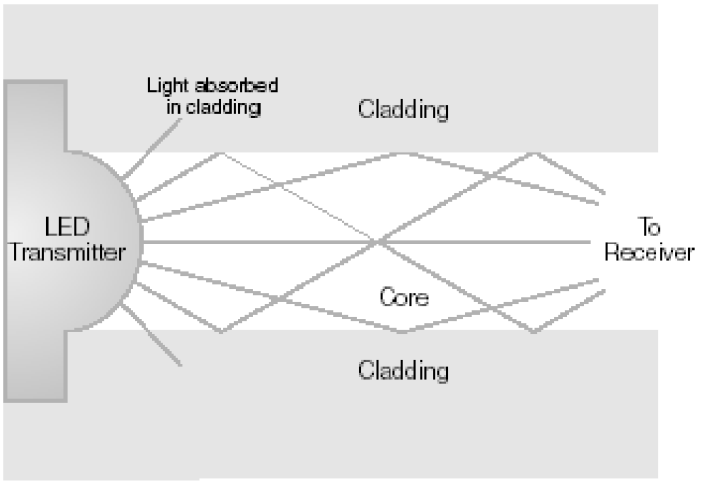


Figure 11.7
LED source coupled to a multi-mode fiber

11.8.2 Multimode fibers

The light takes many paths between the two ends as it reflects from the sides of the fiber core. This causes the light rays to arrive both out of phase and at different times resulting in a spreading of the original pulse shape. As a result, the original sharp pulses sent from one end become distorted by the time they reach the receiving end.

The problem becomes worse as data rates increase. Multimode fibers, therefore, have a limited maximum data rate (bandwidth) as the receiver can only differentiate between the pulsed signals at a low data rate. The effect is known as 'modal dispersion' and its result referred to as 'inter-symbol interference'. For slower data rates over short distances, multimode fibers are quite adequate and speeds of up to 300 Mbps are readily available.

A further consideration with multimode fibers is the 'index' of the fiber (how the impurities are applied in the core). The cable can be either 'graded index' (more expensive but better performance) or 'step index' (less expensive). The type of index affects the way in which the light waves reflect or refract off the walls of the fiber. Graded index cores focus the modes as they arrive at the receiver, and consequently improve the permissible data rate of the fiber.

The core diameters of multimode fibers typically range between 50–100µm. The two most common core diameters are 50 and 62.5µm.

Multimode fibers are easier and cheaper to manufacture than mono-mode fibers. Multimode cores are typically 50 times greater than the wavelength of the light signal they will propagate. With this type of fiber, an LED transmitter light source is normally used because it can be coupled with less precision than a laser diode. With the wide aperture and LED transmitter, the multimode fiber will send light in multiple paths (modes) toward the receiver.

One measure of signal distortion is modal dispersion, which is represented in nanoseconds of signal spread per kilometer (ns/km). This value represents the difference in arrival time between the fastest and slowest of the alternate light paths. The value also imposes an upper limit on the bandwidth. With step-index fiber, expect between 15 and 30 ns/km. Note that a modal dispersion of 20 ns/km yields a bandwidth of less than 50 Mbps.

11.8.3 Monomode/single mode fibers

'Monomode' or 'single mode' fibers are less expensive to manufacture but more difficult to interface. They allow only a single path or mode for the light to travel down the fiber with minimal reflections. Monomode fibers typically use lasers as light sources.

Monomode fibers do not suffer from major dispersion or overlap problems and permit a very high rate of data transfer over much longer distances.

The core of the fibers is much thinner than multimode fibers at approximately 8.5µm. The cladding diameter is 125µm, the same as for multimode fibers.

Optical sources must be powerful and aimed precisely into the fiber to overcome any misalignment (hence the use of laser diodes). The thin monomode fibers are difficult to work with when splicing, terminating, and are consequently expensive to install.

Single-mode fiber has the least signal attenuation, usually less than 0.25dB per kilometer. Transmission speeds of 50 Gbps and higher are possible.

Even though the core of single-mode cable is shrunk to very small sizes, the cladding is not reduced accordingly. For single-mode fiber, the cladding is made the same size as for the popular multimode fiber optic cable. This both helps create a *de facto* size standard and also makes the fiber and cable easier to handle and more resistant to damage.

Specification of optical fiber cables

Optical fibers are specified based on diameter. A fiber specified as 50/150 has a core of 50 µm and a cladding diameter of 150 µm. The most popular sizes of multimode fibers are 50/125, used mainly in Europe, and 62.5/125, used mainly in Australia and the USA.

Another outer layer provides an external protection against abrasion and shock. Outer coatings can range from 250 - 900 µm in diameter, and very often cable specifications include this diameter, for example: 50/150/250.

To provide additional mechanical protection, the fiber is often placed inside a loose, but stiffer, outer jacket which adds thickness and weight to the cable. Cables made with several fibers are most commonly used. The final sheath and protective coating on the outside of the cable depends on the application and where the cable will be used. A strengthening member is normally placed down the center of the cable to give it longitudinal strength. This allows the cable to be pulled through a conduit or hung between poles without causing damage to the fibers. The tensile members are made from steel or Kevlar, the latter being more common. In industrial and mining applications, fiber cores are often placed inside cables used for other purposes, such as trailing power cables for large mining, stacking or reclaiming equipment.

Experience has shown that optic fibers are likely to break during a 25-year provision period. In general, the incremental cost of extra fiber cores in cables is not high when compared to overall costs (including installation and termination costs). So it is usually worthwhile specifying extra cores as spares, for future use.

Joining optical fibers

In the early days of optic fibers, connections and terminations were a major problem. Largely, this has improved but connections still require a great deal of care to avoid signal losses that will affect the overall performance of the communications system.

There are three main methods of splicing optic fibers:

- **Mechanical:** Where the fibers are fitted into mechanical alignment structures
- **Chemical:** Where the two fibers are fitted into a barrel arrangement with epoxy glue in it. They are then heated in an oven to set the glue
- **Fusion splicing:** Where the two fibers are heat-welded together

To overcome the difficulties of termination, fiber optic cables can be provided by a supplier in standard lengths such as 10 m, 100 m or 1000 m with the ends cut and finished with a mechanical termination ferrule that allows the end of the cable to slip into a closely matching female socket. This enables the optical fiber to be connected and disconnected as required.

The mechanical design of the connector forces the fiber into a very accurate alignment with the socket and results in a relatively low loss. Similar connectors can be used for in-line splicing using a double-sided female connector. Although the loss through this type of connector can be an order of magnitude greater than the loss of a fused splice, it is much quicker and requires less special tools and training. Unfortunately, mechanical damage or an unplanned break in a fiber requires special tools and training to repair and re-splice.

One way around this problem is to keep spare standard lengths of pre-terminated fibers that can quickly and easily be plugged into the damaged section. The techniques for terminating fiber optic cables are constantly being improved to simplify these activities.

Limitations of fiber optic cables

On the negative side, the limitations of fiber optic cables are as follows:

- Cost of source and receiving equipment is relatively high
- It is difficult to 'switch' or 'tee-off' a fiber optic cable so fiber optic systems are most suitable for point-to-point communication links
- Techniques for joining or terminating fibers (mechanical and chemical) are difficult and require precise physical alignment. Special equipment and specialized training are required
- Equipment for testing fiber optic cables is different and more expensive than traditional methods used for electronic signals
- Fiber optic systems are used almost exclusively for binary digital signals and are not really suitable for long distance analog signals

Uses of optical fiber cables

Optical fiber cables are used less often to create a network than to connect together two networks or network segments. For example, cable that must run between floors is often fiber optic cable, most commonly of the 62.5/125 varieties with an LED (light-emitting diode) as the light source.

Being impervious to electromagnetic interference, fiber is ideal for such uses because the cable is often run through the lift, or elevator, shaft, and the drive motor puts out strong interference when the cage is running.

A disadvantage of fiber optic networks has been price. Network interface cards (NICs) for fiber optic nodes can cost many times the cost of some Ethernet and ARCnet cards. It is not always necessary, however, to use the most expensive fiber optic connections. For short distances and smaller bandwidth, inexpensive cable is adequate. Generally, a fiber optic cable will always allow a longer transmission than a copper cable segment.

11.8.4 Fiber optic cable components

The major components of a fiber optic cable are the core, cladding, buffer, strength members, and jacket. Some types of fiber optic cable even include a conductive copper wire that can be used to provide power for example, to a repeater.

Fiber optic core and cladding

The core of fiber optic cable consists of one or more glass or plastic fibers through which the light signal moves. Plastic is easier to manufacture and use but works over shorter distances than glass.

In networking contexts, the most popular core sizes are 50, 62.5 and 100 microns. Most of the fiber optic cable used in networking has two core fibers: one for communicating in each direction.

The core and cladding are actually manufactured as a single unit. The cladding is a protective layer of glass or plastic with a lower index of refraction than the core. The lower index means that light that hits the core walls will be redirected back to continue on its path. The cladding will be anywhere between a hundred microns and a millimeter (1000 microns) or so.

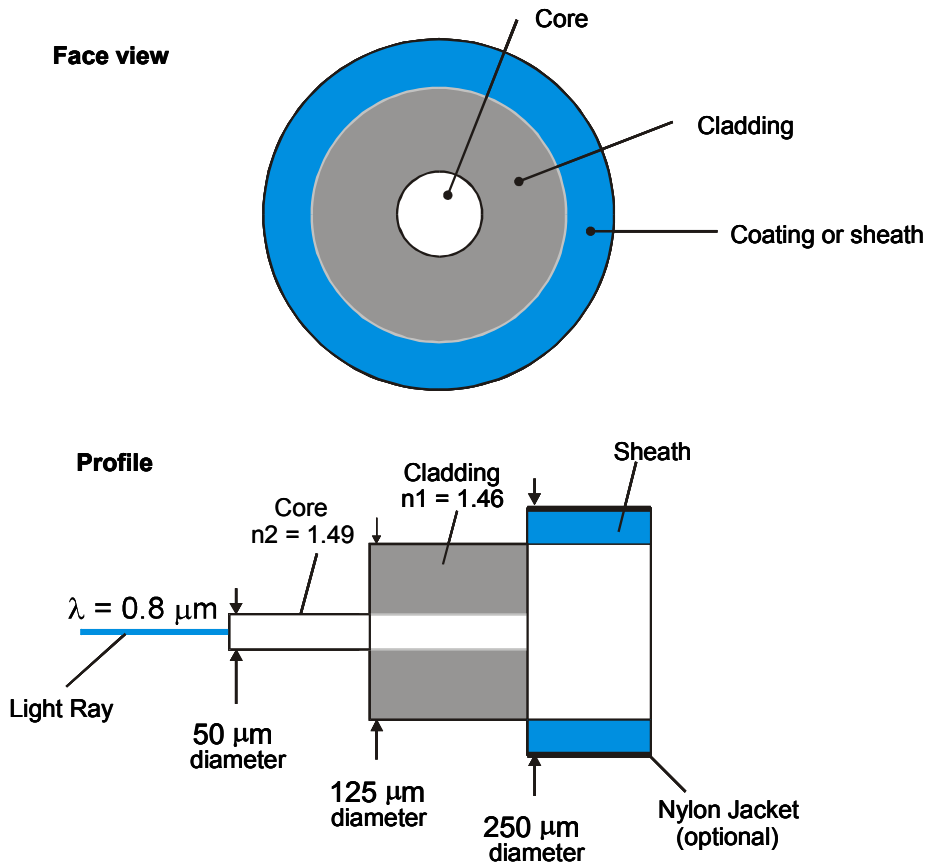


Figure 11.8
Fiber optic cable components

Fiber optic buffer

The buffer of a fiber optic cable is a one or more layer of plastic surrounding the cladding. The buffer helps strengthen the cable, thereby decreasing the likelihood of micro cracks, which can eventually grow into larger breaks in the cable. The buffer also protects the core and cladding from potential corrosion by water or other materials in the operating environment. The buffers can double the diameter of some cable.

A buffer can be loose or tight. A loose buffer is a rigid tube of plastic with one or more fibers (consisting of core and cladding) running loosely through it. The tube takes on all the stresses applied to the cable, buffering the fiber from these stresses. A tight buffer fits

snugly around the fiber(s). A tight buffer can protect the fibers from stress due to pressure and impact, but not from changes in temperature. Loose-buffered cables are normally used for external applications while tight-buffered fibers are usually restricted to internal cables.

Strength members

Fiber optic cable also has strength members, which are strands of very tough material (such as steel, fiberglass or Kevlar) that provide extra strength for the cable. Each of the substances has advantages and drawbacks. For example, steel attracts lightning, which will not disrupt an optical signal but may seriously damage the equipment or the operator sending or receiving such a signal.

Fiber optic jacket

The jacket of a fiber optic cable is an outer casing that can be plenum or non-plenum, as with electrical cable. In cable used for networking, the jacket usually houses at least two fiber/cladding pairs: one for each direction.

11.8.5 Fiber core refractive index changes

One reason why optical fiber makes such a good transmission medium is that the different indexes of refraction for the cladding and core help to contain the light signal within the core, producing a wave-guide for the light. Cable can be constructed by changing abruptly or gradually from the core refractive index to that of the cladding. The two major types of multimode fiber differ in this feature.

Step-index cable

Cable with an abrupt change in refraction index is called step-index cable. In step-index cable, the change is made in a single step. Single-step multimode cable uses this method, and it is the simplest, least expensive type of fiber optic cable. It is also the easiest to install. The core is usually between 50 and 62.5 microns in diameter; the cladding is at least 125 microns. The core width gives light quite a bit of room to bounce around in, and the attenuation is high (at least for fiber optic cable): between 10 and 50 dB/km. Transmission speeds between 200 Mbps and 3 Gbps are possible, but actual speeds are much lower.

Graded-index cable

Cable with a gradual change in refraction index is called graded-index cable, or graded-index multimode. This fiber optic cable type has a relatively wide core, like single-step multimode cable. The change occurs gradually and involves several layers, each with a slightly lower index of refraction. A gradation of refraction indexes controls the light signal better than the step-index method. As a result, the attenuation is lower, usually less than 5 dB/km. Similarly, the modal dispersion can be 1 ns/km and lower, which allows more than ten times the bandwidth of step-index cable. Graded-index multimode cable is the most commonly used type for network wiring.

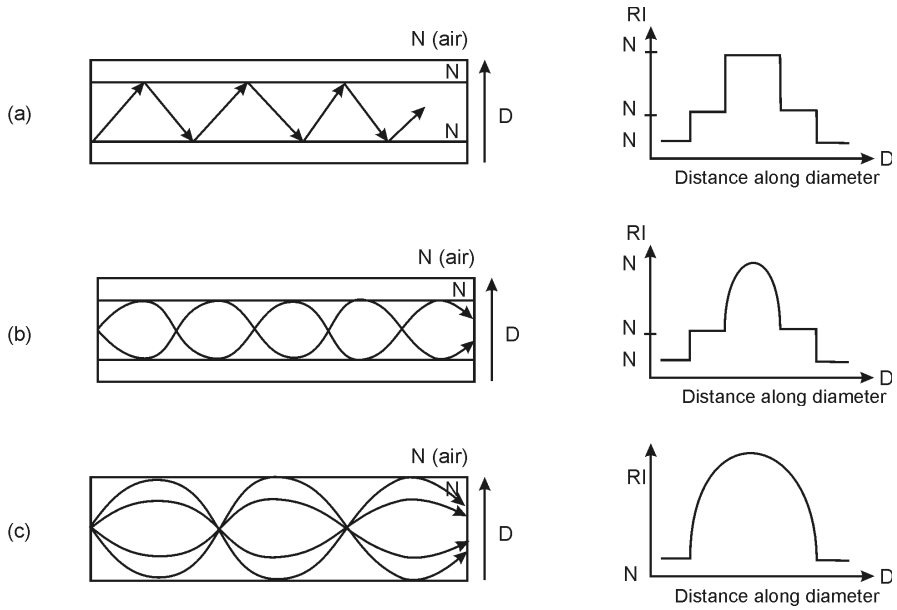


Figure 11.9
Fiber refractive index profiles

Fiber composition

Fiber core and cladding may be made of plastic or glass. The following list summarizes the composition combinations, going from highest quality to lowest:

- **Single-mode glass**: has a narrow core, so only one signal can travel through
- **Graded-index glass**: this is a multi-mode fiber, and the gradual change in refractive index helps give more control over the light signal and significantly reduces modal dispersion
- **Step-index glass**: this is also multi-mode. The abrupt change from the refractive index of the core to that of the cladding means the signal is less controllable, producing low bandwidth fibers
- **Plastic-coated silica (PCS):** has a relatively wide core (200 microns) and a relatively low bandwidth (20 MHz)
- **Plastic:** this should be used only for low speed (e.g. 56k bps) over short distances (15 m)

To summarize, fiber optic cables may consist of a glass core and glass cladding (the best available). Glass yields much higher performance, in the form of higher bandwidth over greater distances. Single-mode glass with a small core is the highest quality. Cables may also consist of glass core and plastic cladding. Finally, the lowest grade fiber composition is plastic core and plastic cladding. Step-index plastic has the lowest performance.

Fiber optic cable quality

Some points about fiber optic cable quality:

- The smaller the core, the better the signal
- Fiber made of glass is better than fiber made of plastic
- The purer and cleaner the light, the better the signal. (Pure, clean light is a single color, with minimal spread around the primary wavelength of the color)
- Certain wavelengths of light behave better than others

Fiber optic cable designations

Fiber optic cables are specified in terms of their core and cladding diameters. For example, a 62.5/125 cable has a core with a 62.5-micron diameter and cladding with twice that diameter.

Following are some commonly used fiber optic cable configurations:

- **8/125:** A single-mode cable with an 8-micron core and a 125-micron cladding. Systems using this type of cable are expensive and currently used only in contexts where extremely large bandwidths are needed (such as in some real-time applications) and/or where large distances are involved. An 8/125-cable configuration is likely to use a light wavelength of 1300 or 1550 nm
- **62.5/125:** The most popular fiber optic cable configuration, used in most network applications. Both 850 and 1300 nm wavelengths can be used with this type of cable
- **100/140:** The configuration that IBM first specified for fiber optic wiring for a Token Ring network. Because of the tremendous popularity of the 62.5/125 configurations, IBM now supports both configurations

If purchasing fiber optic cable, it is important that the correct core size is bought. If the type of desired network is determined, constrains of a particular core size will arise.

Advantages of fiber optic cables

Fiber optic connections offer the following advantages over other types of cabling systems:

- Light signals are impervious to interference from EMI or electrical cross talk.
- Light signals do not interfere with other signals. As a result, fiber optic connections can be used in extremely adverse environments, such as in lift shafts or assembly plants, where powerful motors and engines produce lots of electrical noise.
- Fiber optic lines are much harder to tap into, so they are more secure for private lines.
- Light has a much higher bandwidth, or maximum data transfer rate, than electrical connections. This speed advantage is not always achieved in practice, however.
- The signal has a much lower loss rate, so it can be transmitted much further than it could be with coaxial or twisted pair cable before amplification is necessary.
- Optical fiber is much safer, because there is no electricity and so no danger of electrical shock or other electrical accidents. However, if a laser source is used, there is danger of eye damage.
- Fiber optic cable is generally much thinner and lighter than electrical cable, and so it can be installed more unobtrusively. (Fiber optic cable weighs about 30 grams per meter; coaxial cable weighs nearly ten times that much).
- The installation and connection of the cables is nowadays much easier than it was at first.

Disadvantages of fiber optic cable

The disadvantages of fiber optic connections include the following:

- Fiber optic cable is more expensive than other types of cable.

- Other components, particularly 'fiber' NICs are expensive.
- Certain components, particularly couplers, are subject to optical cross talk.
- Fiber connectors are not designed to use incessantly. Generally, they are designed for fewer than a thousand mating. After that, the connection may become loose, unstable, or misaligned. The resulting signal loss may be unacceptably high.
- Many more parts can break in a fiber optic connection than in an electrical one.

11.9 The IBM cable system

IBM designed the IBM cable system for use in its token ring networks and also for general-purpose premises wiring. IBM has specified nine types of cable, mainly twisted pair, but with more stringent specifications than for the generic twisted pair cabling. The types also include fiber optic cable, but exclude coaxial cable.

The twisted pair versions differ in the following ways:

- Whether the type is shielded or unshielded
- Whether the carrier wire is solid or stranded
- The gauge (diameter) of the carrier wire
- The number of twisted pairs

11.9.1 IBM type 1 cable specifications

Specifications have been created for seven of the nine types with types 4 and 7 undefined. However, the only type relevant to Ethernet users is type 1, as it may be used instead of the usual EIA/TIA-type UTP cable, provided the appropriate impedance-matching baluns are employed.

Type 1 cable is shielded twisted pair, with two pairs of 22-gauge solid wire. Its impedance is 150 ohms and the maximum frequency allowed is 16 MHz. Although not required by the specifications, a plenum version is also available, at about twice the cost of the non-plenum cable. IBM specification numbers are 4716748 for non-plenum data cable, 4716749 for plenum data cable, 4716734 for outdoor data cable, and 6339585 for riser cable.

Type 1A cable

This consists of two 'data grade' shielded twisted pairs and uses 22 AWG solid conductors. Its impedance is 150 ohms and the maximum allowed frequency is 300 MHz. IBM specification numbers are 33G2772 for non-plenum data cable, 33G8220 for plenum data cable, and 33G2774 for riser cable.

11.10 Ethernet cabling requirement overview

This section now lists in brief cabling requirements and minimum specification followed in industry.

10BaseT

Type of cable	Cat. 3,4 or 5 UTP
Maximum length	100 m
Max. impedance allowed	75–165 ohms
Max. attenuation allowed	11.5 db at 5–10 MHz
Max. jitter allowed	5 ns
Delay	1 microsecond
Crosstalk	60 dB for a 10 MHz link
Other considerations	Use plenum cable for ambients higher than 20ºC

Table 11.9
10BaseT requirements

10Base2

Type of cable	Thin coaxial
Maximum length	185 m
Max. number of stations	30
Max. impedance allowed	50 ohms
Other considerations	Termination at both ends required using 50 ohm terminators

Table 11.10
10Base 2 Cable requirements

10Base5

Type of cable	Thick coaxial
Maximum length	500 m
Max. number of stations	100
Max. impedance allowed	50 ohms
Other considerations	Termination at both ends required using 50 ohm terminators

Table 11.11
10Base5 cable requirements

10BaseT full-duplex

Type of cable	Cat. 3,4 or 5 UTP
Maximum length	100 m
Max. impedance allowed	75–165 ohms
Max. attenuation allowed	11.5 db at 5–10 MHz
Max. jitter allowed	5 ns
Delay	Delay is not a factor
Crosstalk	60 dB for a 10 MHz link
Other considerations	Use plenum cable for ambients higher than 20ºC

Table 11.12
10BaseT Full duplex cabling requirements

10BaseF multimode

Type of cable	50/125 or 62.5/125 or 100/140 micron multimode fiber
Maximum length	2 km
Max. attenuation allowed	<13 dB for 50/125 micron <16 dB for 62.5/125 micron <19 dB for 100/140 micron
Max. delay	Total 25.6 microseconds one way

Table 11.13
10BaseF Multi-mode cabling requirements

100BaseFX multimode cabling requirements

Type of cable	50/125 or 62.5/125 or 100/140 micron multimode fiber
Maximum length	2 km for simplex, 412 m for duplex
Max. attenuation allowed @ 850 nm	<13 dB for 50/125 micron <16 dB for 62.5/125 micron <19 dB for 100/140 micron
Max. delay	Total 25.6 microseconds one way

Table 11.14
100BaseFX Cabling requirements

100Base TX

Type of cable	Cat. 5 UTP cable
Maximum length	100 m
Max. attenuation allowed @ 100 MHz	240 dB
Max. delay	Total 1 microseconds
Max. impedance allowed	75–165 ohms
Max. allowed crosstalk	27 dB

Table 11.15
00BaseTX cabling requirements

11.11 Cable connectors

11.11.1 AUI cable connectors

AUI cable is uses DB15 connectors.

The DB15 connector (male or female) provides 15 pins or channels depending in gender. The mapping of these 15 pins of the AUI cable has been dealt with in section 4.1.

11.11.2 Coaxial cable connectors

A segment of coaxial cable has a connector at each end. The cable is attached through these end connectors to a T connector, a barrel connector, another end connector, or to a terminator. Through these connectors, another cable or a hardware device is attached to the coaxial cable. In addition to their function, connectors differ in their attachment mechanism and components. For example, BNC connectors join two components by

plugging them together and then turning the components to click the connection into place. Different size coaxial cables require different sized connectors, matched to the characteristic impedance of the cable, so the introduction of connectors causes minimal reflection of the transmitted signals.

For coaxial cable, the following types of connectors are available:

- N series connectors are used for thick coaxial cable
- BNC is used for thin coaxial cable
- TNC (threaded nut connector) may be used in the same situations as a BNC, provided that the other connector is also using TNC

N Type

These connectors are used for the termination of thick coaxial cables and for the connection of transceivers to the cable. When used to provide a transceiver tap, the coaxial cable is broken at an annular ring and two N type connectors are attached to the resulting bare ends. These N type connectors, once in place, are screwed onto a barrel housing.

The barrel housing contains a center channel across which the signals are passed and a pin or cable that contacts this center channel, providing access to and from the core of the coaxial cable. The pin that contacts the center channel is connected to the transceiver assembly and provides the path for signal transmission and reception.

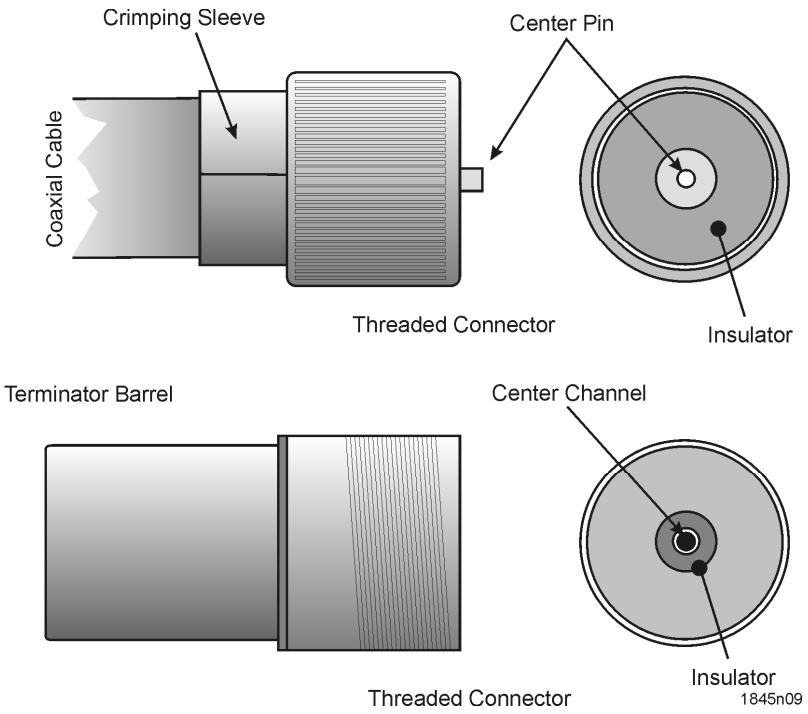


Figure 11.10
N Type connector and terminator

Thick coaxial cables require termination with N type connectors. As the coaxial cable carries network transmissions as voltage, both ends of the thick coaxial cable must be terminated to keep the signal from reflecting throughout the cable, which would disrupt network operation. The terminators used for thick coaxial cable are 50-Ohm. These terminators are screwed into an N type connector placed at the end of a run.

BNC

The BNC connector, used for 10Base2, is an intrusive connector much like the N type connector used with thick coaxial cable. The BNC connector (shown in Figure 11.11) requires that the coaxial cable be broken to make the connection. Two BNC connectors are either screwed onto or crimped to the resulting bare ends.

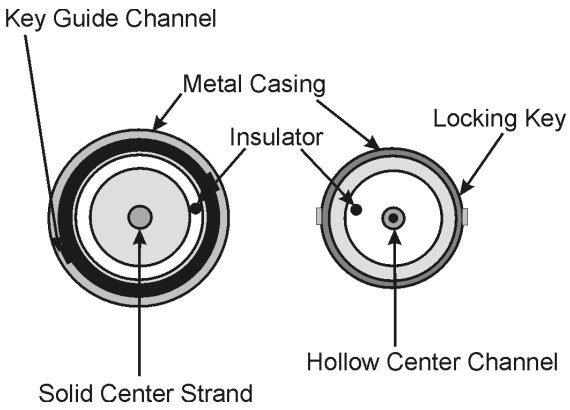


Figure 11.11
BNC connector

BNC male connectors are attached to BNC female terminators or T connectors (Figure 11.12). The outside metal housing of the BNC male connector has two guide channels that slip over corresponding locking key posts on the female BNC connector. When the outer housing is placed over the T connector and turned, the connectors will snap securely into place.

Tapping of coax cable

Tapping a thick coaxial cable is done without breaking the cable itself by use of a non-intrusive, or 'vampire' tap (Figure 11.12). This tap inserts a solid pin through the thick insulating material and shielding of the coaxial cable. The solid pin reaches in through the insulator to the core wire where signals pass through the cable. The signals travel through the pin to and from the core.

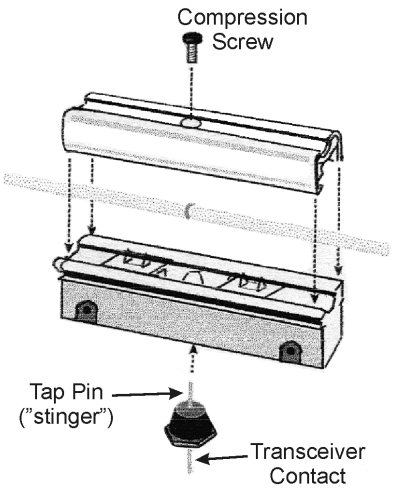


Figure 11.12
'Vampire' non-intrusive tap and cable saddle

Non-intrusive taps are made up of saddles, which bind the connector assembly to the cable, and tap pins, which burrow through the insulator to the core wire.

Non-intrusive connector saddles are clamped to the cable to hold the assembly in place, and usually are either part of, or are easily connected to, an Ethernet transceiver assembly. (See figure 11.13) The non-intrusive tap's cable saddle is then inserted into a transceiver assembly. The contact pin, that carries the signal from the tap pin's connection to the coaxial cable core, makes a contact with a channel in the transceiver housing. The transceiver breaks the signal up and carries it to a DB15 connector, to which an AUI cable may be connected.

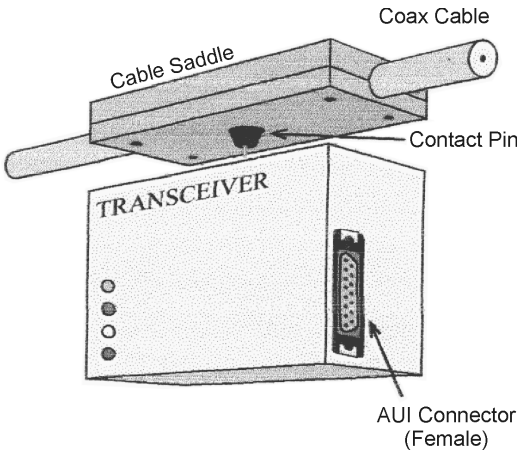


Figure 11.13
Cable saddle and transceiver assembly

Thin coax T connectors

Connections from the cable to network nodes are made using T connectors, which provide taps for additional runs of coaxial cable to workstations or network devices. T connectors, as shown in figure 11.14 below, provide three BNC connections, two of which attach to male BNC connectors on the cable itself and one of which is used for connection to the female BNC connection of a transceiver or network interface card on a workstation.

T connectors should be attached directly to the BNC connectors of network interface cards or other Ethernet devices.

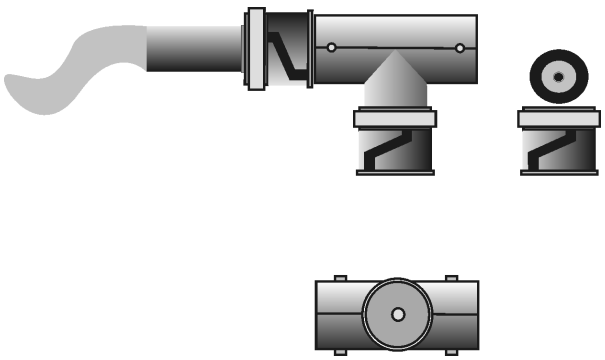


Figure 11.14
Thin coax T connector

The use of the crimp-on BNC connectors is recommended for more stable and consistent connections. BNC connectors use the same pin-and-channel system to provide a contact that is used in the thick coaxial N type connector.

The so-called crimpless connectors should be avoided at all costs. A good quality-crimping tool is very important for BNC connectors. The handle and jaws of the tool should have a ratchet action, to ensure the crimp is made to the required compression. Ensure that the crimping tool has jaws wide enough to cover the entire crimped sleeve at one time. Anything less is asking for problems.

The typical crimping sequence is normally indicated on the packaging with the connector. Ensure that the center contact is crimped onto the conductor before inserting into the body of the connector.

Typical dimensions are shown in the Figure 11.15 below.

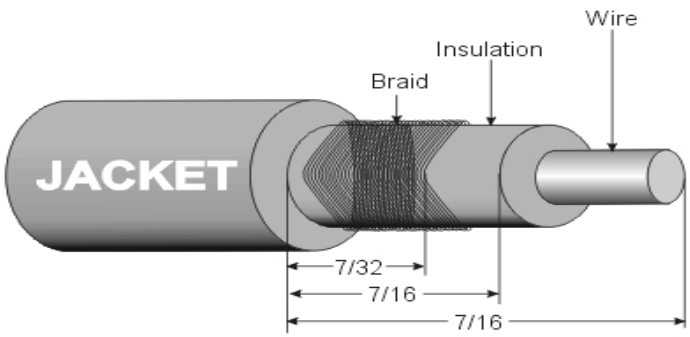


Figure 11.15
BNC coaxial cable termination

11.11.3 UTP cable connectors

RJ-45 cable connectors

The RJ-45 connector is a modular, plastic connector that is often used in UTP cable installations. The RJ-45 is a keyed connector, designed to plug into an RJ-45 port only in the correct alignment. The connector is a plastic housing that is crimped onto a length of UTP cable using a custom RJ-45 die tool. The connector housing is often transparent, and consists of a main body, the contact blades or pins, the raised key, and a locking clip and arm.

The 8-wire RJ-45 connector is small, inexpensive and popular. As a matter of interest, the RJ-45 is different to the RJ-11, although they look the same. RJ-45 is an eight-position plug or jack as opposed to the RJ-11, which is a six-position jack. ‘RJ’ stands for registered jack and is supposed to refer to a specific wiring sequence.

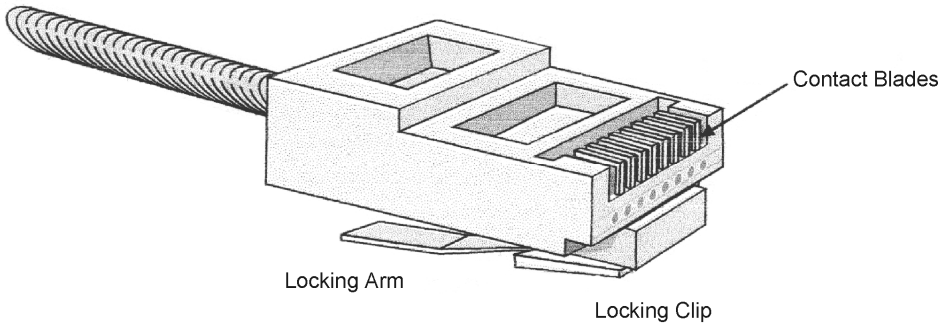


Figure 11.16
RJ-45 connector

The locking clip, part of the raised key assembly, secures the connector in place after a connection is made. When the RJ-45 connector is inserted into a port, the locking clip is pressed down and snaps up into place. A thin arm, attached to the locking clip, allows the clip to be lowered to release the connector from the port.

Stranded or solid conductors

RJ-45 connectors for UTP cabling are available in two basic configurations; stranded and solid. These names refer to the type of UTP cabling that they are designed to connect to. The blades of the RJ-45 connector end in a series of points that pierce the jacket of the wires and make the connection to the core. Different types of connections are required for each type of core composition.

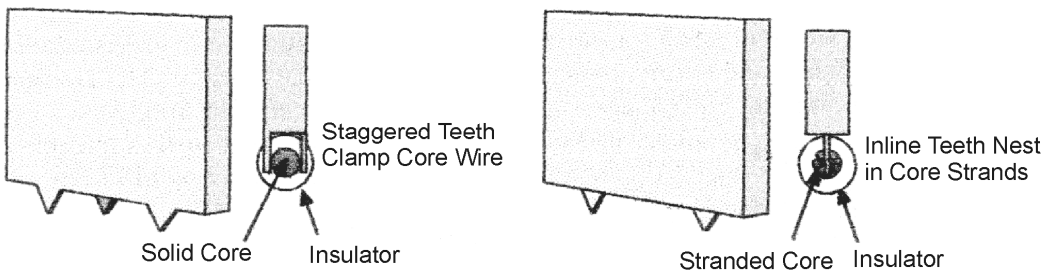


Figure 11.17
Solid and stranded RJ-45 blades

A UTP cable that uses stranded core wires will allow the contact points to nest among the individual strands. The contact blades in a stranded RJ-45 connector, therefore, are laid out with their contact points in a straight line. The contact points cut through the insulating material of the jacket and make contact with several strands of the core.

The solid UTP connector arranges the contact points of the blades in a staggered fashion. The purpose of this arrangement is to pierce the insulator on either side of the core wire and make contacts on either side. As the contact points cannot burrow into the solid core, they clamp the wire in the middle of the blade, providing three opportunities for a viable connection.

There are two terms often used with connectors:

- **Polarization** means the physical form and configuration of connectors
- **Sequence** refers to the order of the wire pairs

An RJ-45 crimping tool shown in Figure 11.18 is often referred to as a 'plug presser' because of its pressing action. When a cable is connected to the cable, the plastic connector is placed in a die in the jaw of the crimping tool. The wires are carefully dressed and inserted into the open connector and then the handles of the crimping tool are closed to force the connector together. The following section discusses the various pin-outs required for the RJ-45 connector.

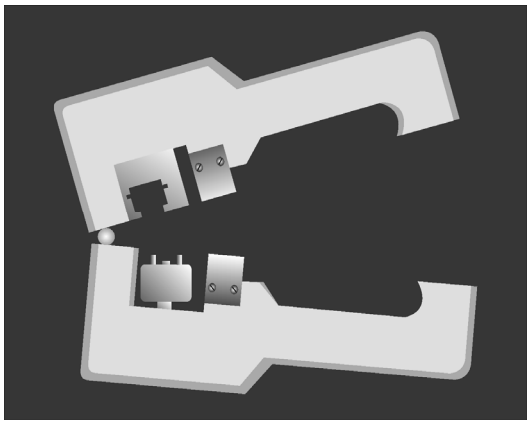


Figure 11.18
Crimping tool

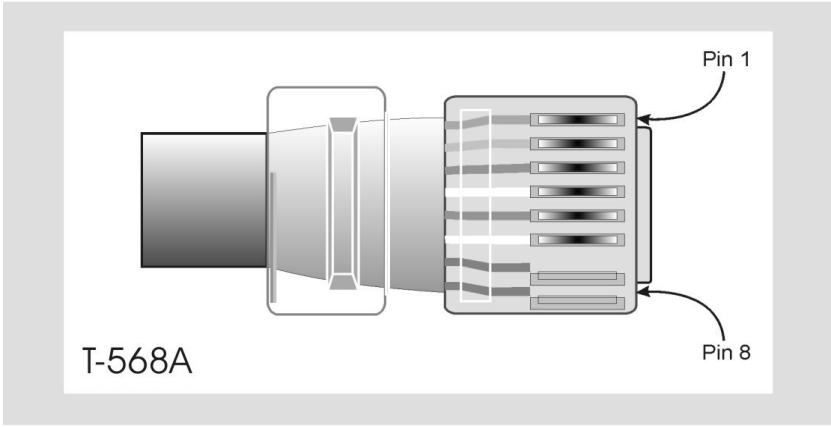
With UTP horizontal wiring, there is general agreement on how to color code each of the wires in a cable. There is however, no general agreement on the physical connections for mating UTP wires and connectors. There are various connector configurations in use, but the main ones are EIA/TIA 568A and EIA/TIA 568B. The difference between them is that the green and orange pairs are interchanged.

RJ-45 pin assignments

Table 11.17 shows pin assignments for each of the Ethernet twisted pair cabling systems:

Contact	10BaseT signal	100BaseTX signal	100BaseT4 signal	100BaseT2 signal	1000BaseT signal
1	TX+	TX+	TX D1+	B1 DA+	B1 DA+
2	TX–	TX–	TX D1–	B1 DA–	B1 DA–
3	RX+	RX+	RX D2+	B1 DB+	B1 DB+
4	Not used	Not used	B1 D3+	Not used	B1 DC+
5	Not used	Not used	B1 D3-	Not used	B1 DC–
6	RX–	RX–	RX D2-	B1 DB–	B1 DB–
7	Not used	Not used	B1 D4+	Not used	B1 DD+
8	Not used	Not used	B1 D4–	Not used	B1 DD–
TX= Transmit data B= Bidirectional data RX= Receive data					

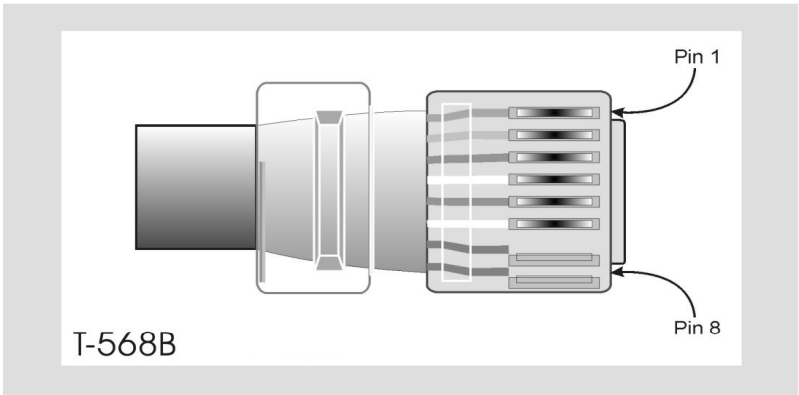
Table 11.17
RJ-45 pin assignments for various Ethernet twisted pair cables



Green/White	1
Green	2
Orange/White	3
Blue	4
Blue/White	5
Orange	6
Brown/White	7
Brown	8

Figure 11.19
T-568A pin assignments

The sequence is green-orange-blue-brown. Wires with white are always on the odd-numbered pins, and pins 3/6 "straddle" pins 4/5.



Orange/White	1
Orange	2
Green/White	3
Blue	4
Blue/White	5
Green	6
Brown/White	7
Brown	8

Figure 11.20
T-568 pin assignments

For T-568B, the orange and green pairs are swapped, so the sequence is now: orange-green-blue-brown, still with the white wires on odd-numbered pins.

If a crossover cable is required, the transmit and receive pairs (1/2 and 3/6) on the one end of the cable have too be swapped. The cable will therefore look like T-568A on one end and T-568B on the other end.

Medium independent interface (MII) connector

MII interface were discussed in Chapter 4 (see Figure 4.2) along with its pin assignments and pin functions.

RJ-21 connector for 25-pair UTP cable

25-pair UTP cable was briefly discussed earlier in this chapter. This cable is connected by RJ-21 connectors. The RJ-21 connector, also known as a 'Telco connector', is a D-shaped metal or plastic housing that is wired and crimped to a UTP cable made up of 50 wires, a 25-pair cable. The RJ-21 connector can only be plugged into an RJ-21 port. The connector itself is large, and the cables that it connects to are often quite heavy, so the RJ-21 relies on a tight fit and good cable management practices to keep itself in the port. Some devices may also incorporate a securing strap that wraps over the back of the connector and holds it tight to the port.

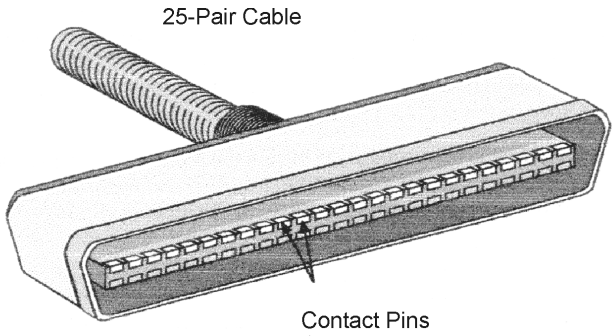


Figure 11.21
RJ-21 connector for 25-pair UTP cables

The RJ-21 is used in locations where 25-pair cable is being run either to stations or to an intermediary cable management device such as a patch panel or punch down block. Due to the bulk of the 25-pair cable and the desirability of keeping the wires within the insulating jacket, as much as possible, 25-pair cable is rarely run directly to Ethernet stations.

The RJ-21 connector, when used in a 10BaseT environment, must use the EIA/TIA 568A pin out scheme. The numbers of the RJ-21 connector's pins are detailed in Figure 11.24 below.

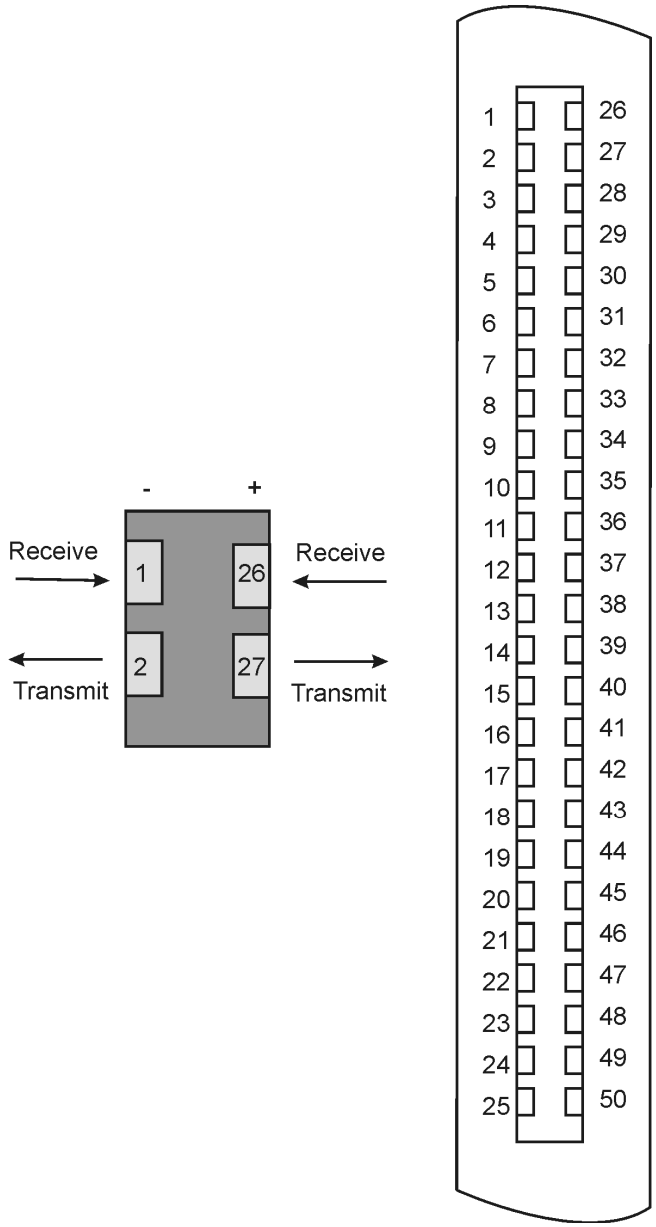


Figure 11.22
RJ-21 pin mapping for 10BaseT

Punch down blocks

While not strictly a connector type, the punch down block is a fairly common component in many Ethernet 10BaseT installations that use 25-pair cable. The punch downs are bayonet pins to which UTP wire strands are connected. The bayonet pins are arranged in 50 rows of four columns each. The pins that make up the punch down block are identified by the row and column they are members of.

Each of the four columns is lettered A, B, C, or D, from leftmost to rightmost. The rows are numbered from top to bottom, one to 50. Thus, the upper left hand pin is identified as A1, while the lower right hand pin is identified as D50.

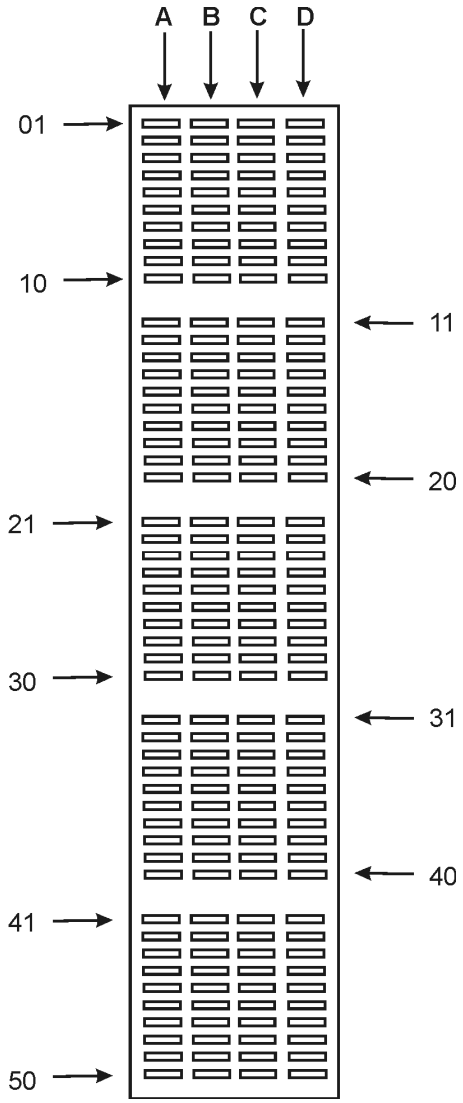


Figure 11.23
Punch down block map for UTP cabling

11.11.4 Connectors for fiber optic cables

Both multimode and single mode fiber optics use the same standard connector in the Ethernet 10BaseFL and FOIRL specifications.

Straight-tip (ST) connector

The 10BaseFL standard and FOIRL specification for Ethernet networks define one style of connector as being acceptable for both multimode and single mode fiber optic cabling – the straight-tip or ST connector (note that ST connectors for single mode and multimode fiber optics are different in construction and are not to be used interchangeably). Designed by AT&T, the ST connector replaces the earlier sub-miniature assembly or SMA connector. The ST connector is a keyed, locking connector that automatically aligns the center strands of the fiber optic cabling with the transmission or reception points of the network or cable management device it is connecting to.

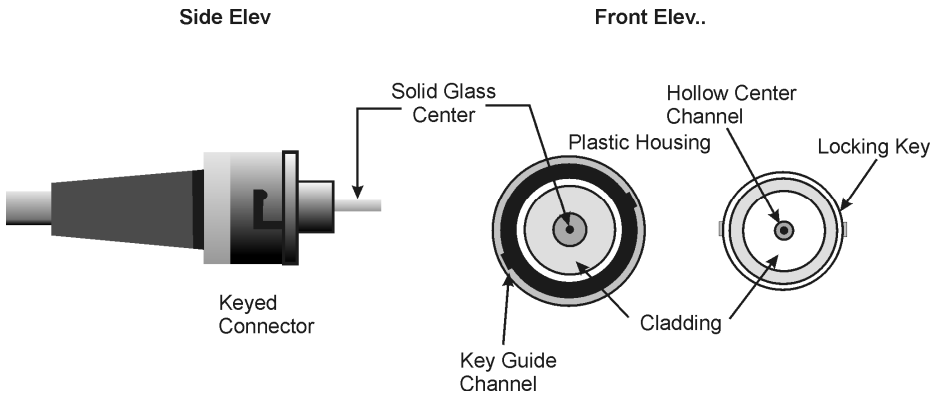


Figure 11.24
ST connector

The key guide channels of the male ST connector allow the ST connector to only be connected to a female ST connector in the proper alignment. The alignment keys of the female ST connector ensure the proper rotation of the connector and, at the end of the channel, lock the male ST connector into place at the correct attitude. An integral spring holds the ST connectors together and provides strain relief on the cables.

SC connector

The SC connector is a gendered connector that is recommended for use in Fast Ethernet networks that incorporate multimode fiber optics adhering to the 100BaseFX specification. It consists of two plastic housings, the outer and inner. The inner housing fits loosely into the outer, and slides back and forth with a travel of approximately 2 mm (0.08 in). The fiber is terminated inside a spring-loaded ceramic ferrule inside the inner housing. These ferrules are very similar to the floating ferrules used in the FDDI MIC connector.

The 100BaseFX specification requires very precise alignment of the fiber optic strands in order to make an acceptable connection. In order to accomplish this, SC connectors and ports each incorporate 'floating' ferrules that make the final connection between fibers. These floating ferrules are spring loaded to provide the correct mating tension. This arrangement allows the ferrules to move correctly when making a connection. This small amount of movement manages to accommodate the subtle differences in construction found from connector to connector and from port to port. The sides of the outer housing are open, allowing the inner housing to act as a latching mechanism when the connector is inserted properly in an SC port.

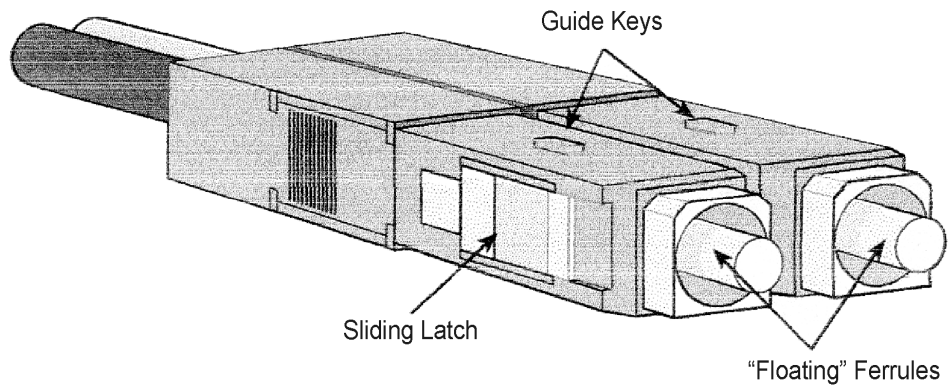


Figure 11.25
SC connectors (showing 2) for Fast Ethernet

12

LAN system components

Objectives

When you have completed this chapter you should be able to:

- Explain the basic function of each of the devices listed under 12.1
- Explain the fundamental differences between the operation and application of switches (layer 2 and 3), bridges and routers

12.1 Introduction

In the design of an Ethernet system there are a number of different components that can be used. These include:

- Repeaters
- Media converters
- Bridges
- Hubs
- Switches
- Routers
- Gateways
- Print servers
- Terminal servers
- Remote access servers
- Time servers
- Thin servers

The lengths of LAN segments are limited due to physical and collision domain constraints and there is often a need to increase this range. This can be achieved by means of a number of interconnecting devices, ranging from repeaters to gateways. It may also be necessary to partition an existing network into separate networks for reasons of security or traffic overload.

In modern network devices the functions mentioned above are often mixed. Here are a few examples:

- A shared 10BaseT hub is, in fact, a multi-port repeater
- A Layer 2 switch is essentially a multi-port bridge
- Segmentable and dual-speed shared hubs make use of internal bridges
- Switches can function as bridges, a two-port switch being none other than a bridge
- Layer 3 switches function as routers

These examples are not meant to confuse the reader, but serve to emphasize the fact that the functions should be understood, rather than the 'boxes' in which they are packaged.

12.2 Repeaters

A repeater operates at the physical layer of the OSI model (layer 1) and simply retransmits incoming electrical signals. This involves amplifying and re-timing the signals received on one segment onto all other segments, without considering any possible collisions. All segments need to operate with the same media access mechanism and the repeater is unconcerned with the meaning of the individual bits in the packets. Collisions, truncated packets or electrical noise on one segment are transmitted onto all other segments.

12.2.1 Packaging

Repeaters are packaged either as stand-alone units (i.e. desktop models or small cigarette package-sized units) or 19" rack-mount units. Some of these can link two segments only, while larger rack-mount modular units (called concentrators) are used for linking multiple segments. Regardless of packaging, repeaters can be classified either as local repeaters (for linking network segments that are physically in close proximity), or as remote repeaters for linking segments that are some distance apart.

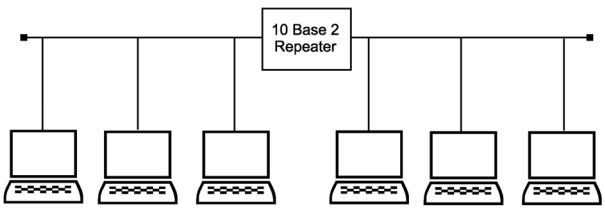


Figure 12.1
Repeater application

12.2.2 Local Ethernet repeaters

Several options are available:

- Two-port local repeaters offer most combinations of 10Base5, 10Base2, 10BaseT and 10BaseFL such as 10Base5/10Base5, 10Base2/10Base2, 10Base5/10Base2, 10Base2/10BaseT, 10BaseT/10BaseT and 10BaseFL/10BaseFL. By using such devices (often called boosters or extenders) one can, for example, extend the distance between a computer and

a 10BaseT Hub by up to 100m, or extend a 10BaseFL link between two devices (such as bridges) by up 2-3 km

- Multi-port local repeaters offer several ports of the same type (e.g. 4x 10Base2 or 8x 10Base5) in one unit, often with one additional connector of a different type (e.g. 10Base2 for a 10Base5 repeater.) In the case of 10BaseT the cheapest solution is to use an off-the-shelf 10BaseT shared hub, which is effectively a multi-port repeater
- Multi-port local repeaters are also available as chassis-type units; i.e. as frames with common back planes and removable units. An advantage of this approach is that 10Base2, 10Base5, 10BaseT and 10BaseFL can be mixed in one unit, with an option of SNMP management for the overall unit. These are also referred to as concentrators

12.2.3 Remote repeaters

Remote repeaters, on the other hand, have to be used in pairs with one repeater connected to each network segment and a fiber-optic link between the repeaters. On the network side they typically offer 10Base5, 10Base2 and 10BaseT. On the interconnecting side the choices include 'single pair Ethernet', using telephone cable up to 457m in length, or single mode/multi-mode optic fiber, with various connector options. With 10BaseFL (backwards compatible with the old FOIRL standard), this distance can be up to 1.6 km.

In conclusion it must be emphasized that although repeaters are probable the cheapest way to extend a network, they do so without separating the collision domains, or network traffic. They simply extend the physical size of the network. All segments joined by repeaters therefore share the same bandwidth and collision domain.

12.3 Media converters

Media converters are essentially repeaters, but interconnect mixed media viz. copper and fiber. An example would be 10BaseT/10BaseFL. As in the case of repeaters, they are available in single and multi-port options, and in stand-alone or chassis type configurations. The latter option often features remote management via SNMP.

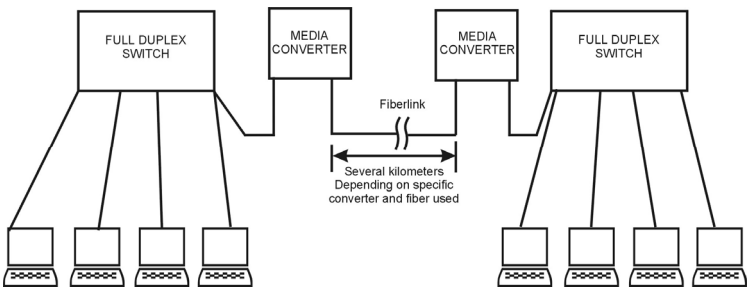


Figure 12.2
Media converter application

Models may vary between manufacturers, but generally Ethernet media converters support:

- 10 Mbps (10Base2, 10BaseT, 10BaseFL- single and multi-mode)
- 100 Mbps (Fast) Ethernet (100BaseTX, 100BaseFX- single and multimode)
- 1000 Mbps (Gigabit) Ethernet (single and multimode)

An added advantage of the fast and Gigabit Ethernet media converters is that they support full duplex operation that effectively doubles the available bandwidth.

12.4 Bridges

Bridges operate at the data link layer of the OSI model (layer 2) and are used to connect two separate networks to form a single large continuous LAN. The overall network, however, still remains one network with a single network ID (NetID). The bridge only divides the network up into two segments, each with its own collision domain and each retaining its full (say, 10 Mbps) bandwidth. Broadcast transmissions are seen by all nodes, on both sides of the bridge.

The bridge exists as a node on each network and passes only valid messages across to destination addresses on the other network. The decision as to whether or not a frame should be passed across the bridge is based on the layer 2 address, i.e. the media (MAC) address. The bridge stores the frame from one network and examines its destination MAC address to determine whether it should be forwarded across the bridge.

Bridges can be classified as either MAC or LLC bridges, the MAC sub-layer being the lower half of the data link layer and the LLC sub-layer being the upper half. For MAC bridges the media access control mechanism on both sides must be identical; thus it can bridge only Ethernet to Ethernet, token ring to token ring and so on. For LLC bridges, the data link protocol must be identical on both sides of the bridge (e.g. IEEE 802.2 LLC); however, the physical layers or MAC sub-layers do not necessarily have to be the same. Thus the bridge isolates the media access mechanisms of the networks. Data can therefore be transferred, for example, between Ethernet and token ring LANs. In this case, collisions on the Ethernet system do not cross the bridge nor do the tokens.

Bridges can be used to extend the length of a network (as with repeaters) but in addition they improve network performance. For example, if a network is demonstrating fairly slow response times, the nodes that mainly communicate with each other can be grouped together on one segment and the remaining nodes can be grouped together in another segment. The busy segment may not see much improvement in response rates (as it is already quite busy) but the lower activity segment may see quite an improvement in response times. Bridges should be designed so that 80% or more of the traffic is within the LAN and only 20% cross the bridge. Stations generating excessive traffic should be identified by a protocol analyzer and relocated to another LAN.

12.4.1 Intelligent bridges

Intelligent bridges (also referred to as transparent or spanning-tree bridges) are the most commonly used bridges because they are very efficient in operation and do not need to be taught the network topology. A transparent bridge learns and maintains two address lists corresponding to each network it is connected to. When a frame arrives from the one Ethernet network, its source address is added to the list of source addresses for that network. The destination address is then compared to that of the two lists of addresses for each network and a decision made whether to transmit the frame onto the other network. If no corresponding address to the destination node is recorded in either of these two lists the message is retransmitted to all other bridge outputs (flooding), to ensure the message is delivered to the correct network. Over a period of time, the bridge learns all the addresses on each network and thus avoids unnecessary traffic on the other network. The bridge also maintains time out data for each entry to ensure the table is kept up to date and old entries purged.

Transparent bridges cannot have loops that could cause endless circulation of packets. If the network contains bridges that could form a loop as shown in Figure 12.3, one of the bridges (C) needs to be made redundant and deactivated.

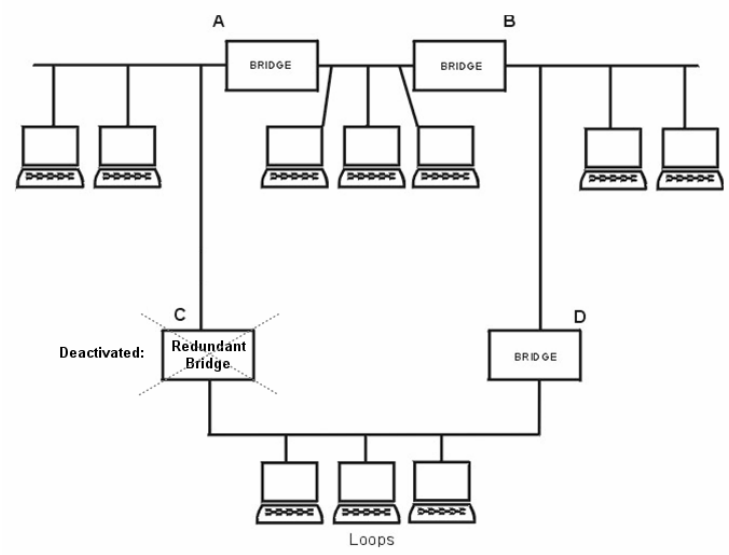


Figure 12.3
Avoidance of loops in bridge networks

The spanning tree algorithm (IEEE 802.1d) is used to manage paths between segments having redundant bridges. This algorithm designates one bridge in the spanning tree as the root and all other bridges transmit frames towards the root using a least cost metric. Redundant bridges can be reactivated if the network topology changes.

12.4.2 Source-routing bridges

Source routing (SR) Bridges are popular for IBM token ring networks. In these networks, the sender must determine the best path to the destination. This is done by sending a discovery frame that circulates the network and arrives at the destination with a record of the path token. These frames are returned to the sender who can then select the best path. Once the path has been discovered, the source updates its routing table and includes the path details in the routing information field in the transmitted frame.

12.4.3 SRT and translational bridges

When connecting Ethernet networks to token ring networks, either **source routing transparent** (SRT) bridges or translational bridges are used. SRT bridges are a combination of a transparent and source routing bridge, and are used to interconnect Ethernet (IEEE802.3) and token ring (IEE802.5) networks. It uses source routing of the data frame if it contains routing information; otherwise it reverts to transparent bridging. Translational bridges, on the other hand, translate the routing information to allow source routing networks to bridge to transparent networks. The IBM 8209 is an example of this type of bridge.

12.4.4 Local vs. remote bridges

Local bridges are devices that have two network ports and hence interconnect two adjacent networks at one point. This function is currently often performed by switches, being essentially intelligent multi-port bridges.

A very useful type of local bridge is a 10/100 Mbps Ethernet bridge, which allows interconnection of 10BaseT, 100BaseTX and 100BaseFX networks, thereby performing the required speed translation. These bridges typically provide full duplex operation on 100BastTX and 100BaseFX, and employ internal buffers to prevent saturation of the 10BaseT port.

Remote bridges, on the other hand, operate in pairs with some form of interconnection between them. This interconnection can be with or without modems, and include

RS-232/V.24, V.35, RS-422, RS-530, X.21, 4-wire, or fiber (both single and multi-mode). The distance between bridges can typically be up to 1.6 Km.

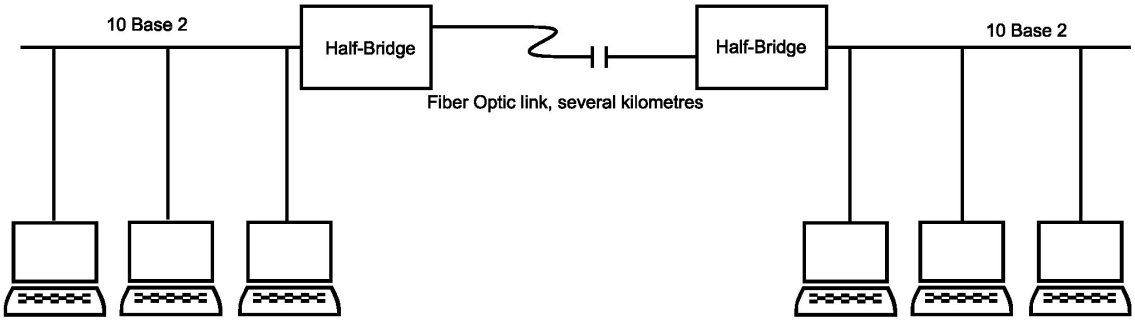


Figure 12.4
Remote bridge application

12.5 Hubs

Hubs are used to interconnect hosts in a physical star configuration. This section will deal with Ethernet hubs, which are of the 10/100/100BaseT variety. They are available in many configurations, some of which will be discussed below.

12.5.1 Desktop vs stackable hubs

Smaller desktop units are intended for stand-alone applications, and typically have 5 to 8 ports. Some 10BaseT desktop models have an additional 10Base2 port. These devices are often called workgroup hubs.

Stackable hubs, on the other hand, typically have up to 24 ports and can be physically stacked and interconnected to act as one large hub without any repeater count restrictions. These stacks are often mounted in 19-inch cabinets.

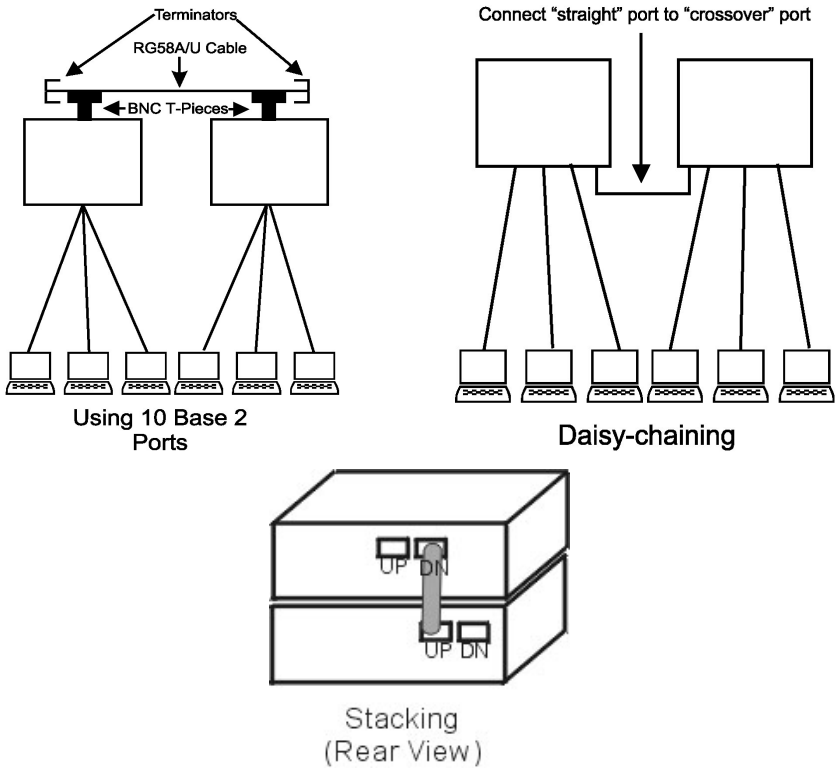


Figure 12.5
10BaseT hub interconnection

12.5.2 Shared vs. switched hubs

Shared hubs interconnect all ports on the hub in order to form a logical bus. This is typical of the cheaper workgroup hubs. All hosts connected to the hub share the available bandwidth since they all form part of the same collision domain.

Although they physically look alike, switched hubs (better known as switches) allow each port to retain and share its full bandwidth only with the hosts connected to that port. Each port (and the segment connected to that port) functions as a separate collision domain. This attribute will be discussed in more detail in the section on switches.

12.5.3 Managed hubs

Managed hubs have an on-board processor with its own MAC and IP address. Once the hub has been set up via a PC on the hub's serial (COM) port, it can be monitored and controlled via the network using SNMP or Telnet. The user can perform activities such as enabling/ disabling individual ports, performing segmentation (see next section), monitoring the traffic on a given port, or setting alarm conditions for a given port.

12.5.4 Segmentable hubs

On a non-segmentable (i.e. shared) hub, all hosts share the same bandwidth. On a segmentable hub, however, the ports can be grouped, under software control, into several shared groups. All hosts on each segment then share the full bandwidth on that segment, which means that a 24-port 10BaseT hub segmented into 4 groups effectively supports 40

Mbps. The configured segments are internally connected via bridges, so that all ports can still communicate with each other if needed.

12.5.5 Dual-speed hubs

Some hubs offer dual-speed ports, e.g. 10BaseT/100BaseT. These ports are auto-configured, i.e. each port senses the speed of the NIC connected to it, and adjusts its own speed accordingly. All the 10BaseT ports connect to a common low-speed internal segment, while all the 100BaseT ports connect to a common high-speed internal segment. The two internal segments are interconnected via a speed-matching bridge.

12.5.6 Modular hubs

Some stackable hubs are modular, allowing the user to configure the hub by plugging in a separate module for each port. Ethernet options typically include both 10 and 100 Mbps, with either copper or fiber. These hubs are sometimes referred to as chassis hubs.

12.5.7 Hub interconnection

Stackable hubs are best interconnected by means of special stacking cables attached to the appropriate connectors on the back of the chassis.

An alternative method for non-stackable hubs is by 'daisy-chaining' an interconnecting port on each hub by means of a UTP patch cord. Care has to be taken not to connect the 'transmit' pins on the ports together (and, for that matter, the 'receive' pins) – it simply will not work. This is similar to interconnecting two COM ports with a 'straight' cable i.e. without a null modem. Connect transmit to receive and vice versa by (a) using a crossover cable and interconnecting two 'normal' ports, or (b) using a normal ('straight') cable and utilizing a crossover port on one of the hubs. Some hubs have a dedicated uplink (crossover) port while others have a port that can be manually switched into crossover mode.

A third method that can be used on hubs with a 10Base2 port is to create a backbone. Attach a BNC T-piece to each hub, and interconnect the T-pieces with RG-58 coax cable. The open connections on the extreme ends of the backbone obviously have to be terminated.

Fast Ethernet hubs need to be deployed with caution because the inherent propagation delay of the hub is significant in terms of the 5.12 microsecond collision domain size. Fast Ethernet hubs are classified as Class I, II or II+, and the class dictates the number of hubs that can be interconnected. For example, Class II dictates that there may be no more than two hubs between any given pair of nodes, that the maximum distance between the two hubs shall not exceed 5 m, and that the maximum distance between any two nodes shall not exceed 205 m. The safest approach, however, is to follow the guidelines of each manufacturer.

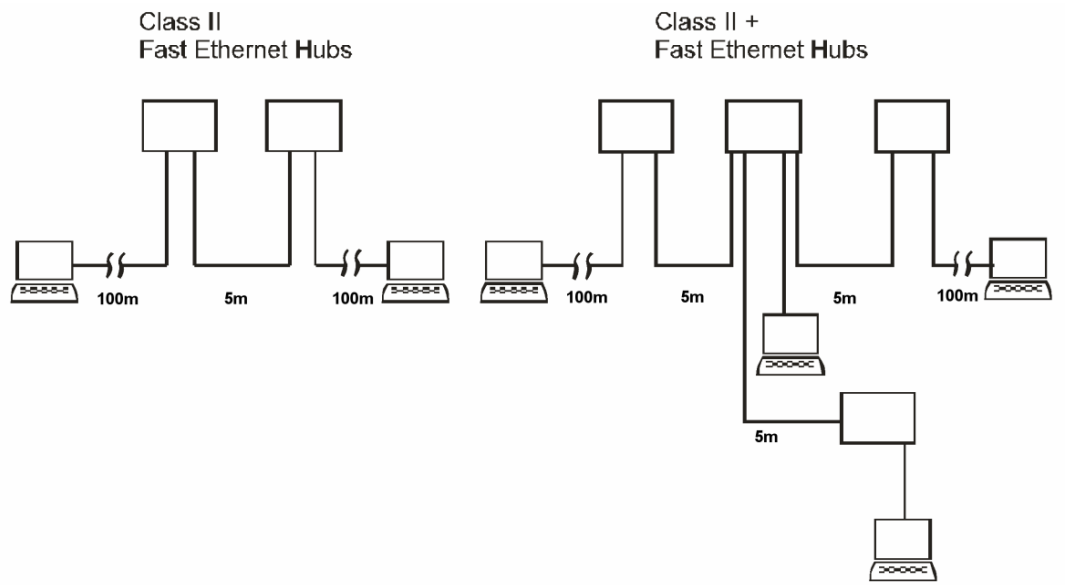


Figure 12.6
Fast Ethernet hub interconnection

12.6 Switches

Ethernet switches are an expansion of the concept of bridging and are, in fact, intelligent (self-learning) multi-port bridges. They enable frame transfers to be accomplished between any pair of devices on a network, on a per-frame basis. Only the two ports involved 'see' the specific frame. Illustrated below is an example of an 8 port switch, with 8 hosts attached. This comprises a physical star configuration, but it does not operate as a logical bus as an ordinary hub does. Since each port on the switch represents a separate segment with its own collision domain, it means that there are only 2 devices on each segment, namely the host and the switch port. Hence, in this particular case, there can be no collisions on any segment!

In the sketch below hosts 1 & 7, 3 & 5 and 4 & 8 need to communicate at a given moment, and are connected directly for the duration of the frame transfer. For example, host 7 sends a packet to the switch, which determines the destination address, and directs the package to port 1 at 10 Mbps.

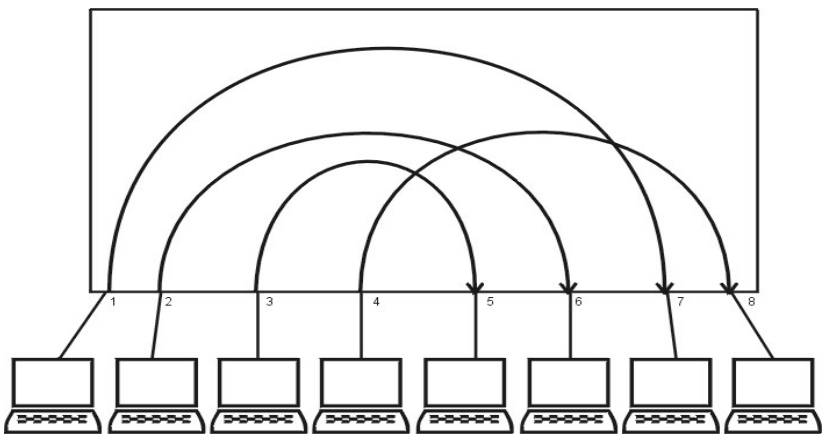


Figure 12.7
8-Port Ethernet switch

If host 3 wishes to communicate with host 5, the same procedure is repeated. Provided that there are no conflicting destinations, a 16-port switch could allow 8 concurrent frame exchanges at 10 Mbps, rendering an effective bandwidth of 80 Mbps. On top of this, the switch could allow full duplex operation, which would double this figure.

12.6.1 Cut-through vs store-and-forward

Switches have two basic architectures, cut-through and store-and-forward. In the past, cut-through switches were faster because they examined the packet destination address only before forwarding the frame to the destination segment. A store-and-forward switch, on the other hand, accepts and analyses the entire packet before forwarding it to its destination. It takes more time to examine the entire packet, but it allows the switch to catch certain packet errors and keep them from propagating through the network. The speed of modern store-and-forward switches has caught up with cut-through switches so that the speed difference between the two is minimal. There are also a number of hybrid designs that mix the two architectures.

Since a store-and-forward switch buffers the frame, it can delay forwarding the frame if there is traffic on the destination segment, thereby adhering to the CSMA/CD protocol. In the case of a cut-through switch this is a problem, since a busy destination segment means that the frame cannot be forwarded, yet it cannot be stored either. The solution is to force a collision on the source segment, thereby enticing the source host to retransmit the frame.

12.6.2 Layer 2 switches vs. layer 3 switches

Layer 2 switches operate at the data link layer of the OSI model and derive their addressing information from the destination MAC address in the Ethernet header. Layer 3 switches, on the other hand, obtain addressing information from the Network Layer, namely from the destination IP address in the IP header. Layer 3 switches are used to replace routers in LANs as they can do basic IP routing (supporting protocols such as RIP and RIPv2) at almost ‘wire-speed’; hence they are significantly faster than routers.

12.6.3 Full duplex switches

An additional advancement is full duplex Ethernet where a device can simultaneously transmit AND receive data over one Ethernet connection. This requires a different Ethernet NIC in the host, as well as a switch that supports full duplex. This enables two devices two transmit and receive simultaneously via a switch. The node automatically negotiates with the switch and uses full duplex if both devices can support it.

Full duplex is useful in situations where large amounts of data are to be moved around quickly, for example between graphics workstations and file servers.

12.6.4 Switch applications

High-speed aggregation

Switches are very efficient in providing a high-speed aggregated connection to a server or backbone. Apart from the normal lower-speed (say, 10BaseT) ports, switches have a high-speed uplink port (100BaseTX). This port is simply another port on the switch, accessible by all the other ports, but features a speed conversion from 10 Mbps to 100 Mbps.

Assume that the uplink port was connected to a file server. If all the other ports (say, eight times 10BaseT) wanted to access the server concurrently, this would necessitate a bandwidth of 80 Mbps in order to avoid a bottleneck and subsequent delays. With a 10BaseT uplink port this would create a serious problem. However, with a 100BaseTX uplink there is still 20 Mbps of bandwidth to spare.

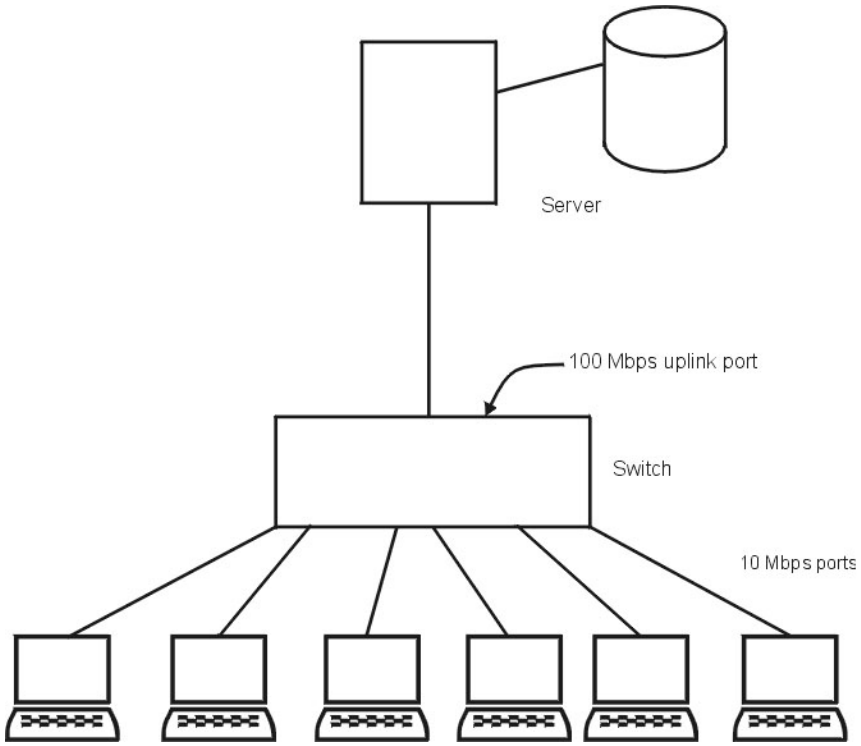


Figure 12.8
Using a Switch to connect users to a Server

Backbones

Switches are very effective in backbone applications, linking several LANs together as one, yet segregating the collision domains. An example could be a switch located in the basement of a building, linking the networks on different floors of the building. Since the actual 'backbone' is contained within the switch, it is known in this application as a 'collapsed backbone'.

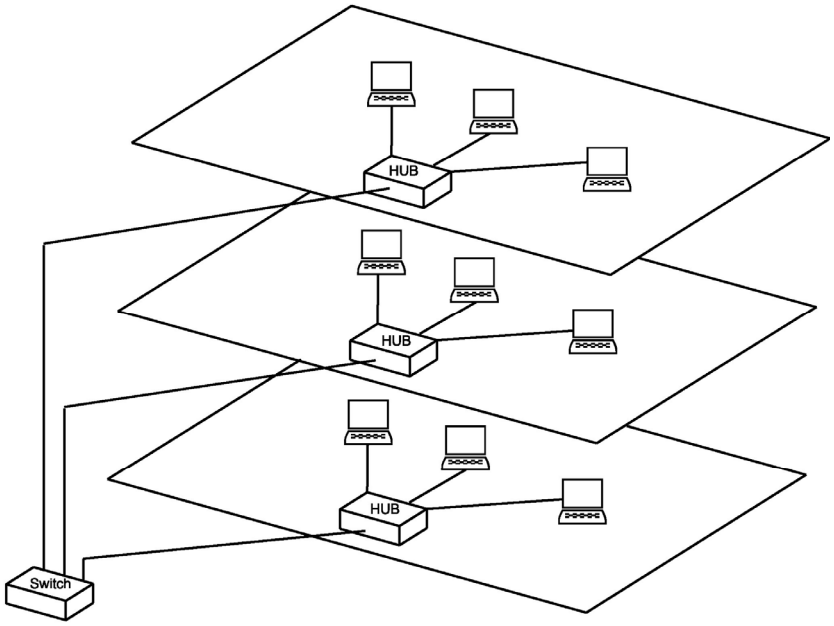


Figure 12.9
Using a switch as a backbone

VLANs and deterministic Ethernet

Provided that a LAN is constructed around switches that support VLANs, individual hosts on the physical LAN can be grouped into smaller Virtual LANs (VLANs), totally invisible to their fellow hosts. Unfortunately, neither the 'standard' Ethernet nor the IEEE802.3 header contains sufficient information to identify members of each VLAN; hence, the frame had to be modified by the insertion of a 'tag', between the Source MAC address and the type/length fields. This modified frame is known as an Ethernet 802.1Q tagged frame and is used for communication between the switches.

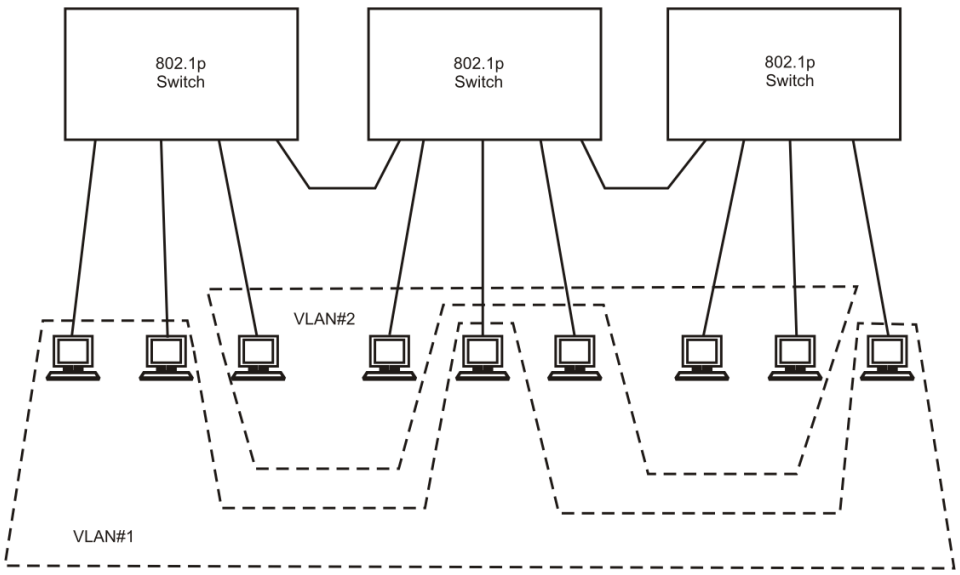


Figure 12.10
Virtual LAN's using switches

The IEEE802.1p committee has defined a standard for packet-based LAN's that supports Layer 2 traffic prioritization in a switched LAN environment. IEEE802.1p is

part of a larger initiative (IEEE802.1p/Q) that adds more information to the Ethernet header (as shown in Fig 12.11) to allow networks to support VLANs and traffic prioritization.

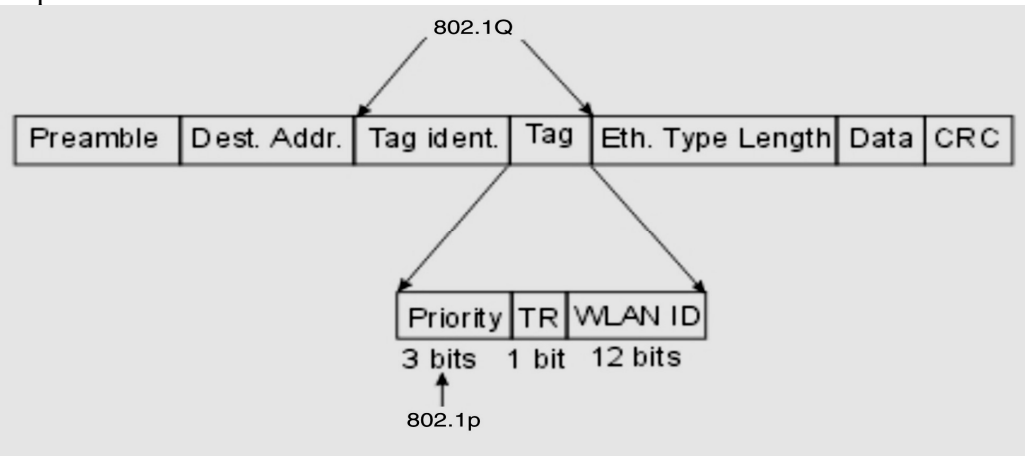


Figure 12.11
IEEE802.1p/Q modified Ethernet header

802.1p/Q adds 16 bits to the header, of which three are for a priority tag and twelve for a VLAN ID number. This allows for eight discrete priority Layers from 0 (high) to 7 (low) that supports different kinds of traffic in terms of their delay-sensitivity. Since IEEE802.1p/Q operates at Layer II, it supports prioritization for all traffic on the VLAN, both IP and non-IP. This introduction of priority layers enables so-called deterministic Ethernet where, instead of contending for access to a bus, a source node can pass a frame directly to a destination node on the basis of its priority, and without risk of any collisions.

The TR bit simply indicates whether the MSB of the VLAN ID is on the left or the right.

12.7 Routers

Unlike bridges and layer 2 switches, routers operate at layer 3 of the OSI model, namely at the network layer (or, the Internet layer of the DOD model). They therefore ignore address information contained within the data link layer (the MAC addresses) and rather delve deeper into each frame and extract the address information contained in the network layer. For TCP/IP this is the IP address.

Like bridges or switches, routers appear as hosts on each network that it is connected to. They are connected to each participating network through an NIC, each with an MAC address as well as an IP address. Each NIC has to be assigned an IP address with the same NetID as the network it is connected to. This IP address allocated to each network is known as the default gateway for that network and each host on the internetwork requires at least one default gateway (but could have more). The default gateway is the IP address to which any host must forward a packet if it finds that the NetID of the destination and the local NetID does not match, which implies remote delivery of the packet.

A second major difference between routers and bridges or switches is that routers will not act autonomously but rather have to be GIVEN the frames that need to be forwarded. A host to the designated Default Gateway forwards such frames.

Protocol dependency

Because routers operate at the network layer, they are used to transfer data between two networks that have the same Internet layer protocols (such as IP) but not necessarily the same physical or data link protocols. Routers are therefore said to be protocol dependent, and have to be able to handle all the Internet layer protocols present on a particular network. A network utilizing Novell Netware therefore requires routers that can accommodate IPX (Internet packet exchange) – the network layer component of SPX/IPX. If this network has to handle Internet access as well, it can only do this via IP, and hence the routers will need to be upgraded to models that can handle both IPX and IP.

Routers maintain tables of the networks that they are connected to and of the optimum path to reach a particular network. They then redirect the message to the next router along that path.

12.7.1 Two-port vs. multi-port routers

Multi-port routers are chassis-based devices with modular construction. They can interconnect several networks. The most common type of router is, however, a 2-port router. Since these are invariably used to implement WANs, they connect LANs to a 'communications cloud'; the one port will be a local LAN port e.g. 10BaseT, but the second port will be a WAN port such as X.25.

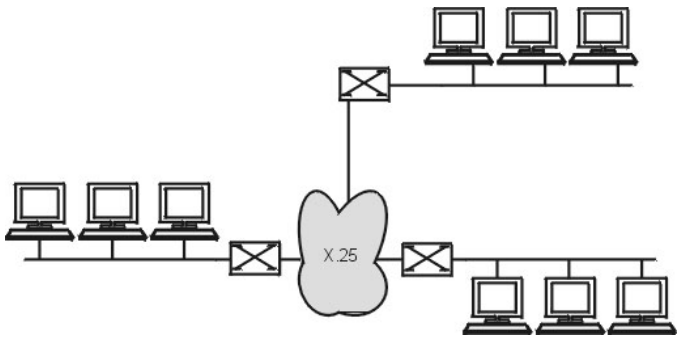


Figure 12.12
Implementing a WAN with 2-port routers (gateways)

12.7.2 Access routers

Access routers are 2-port routers that use dial-up access rather than a permanent (e.g. X.25) connection to connect a LAN to an ISP and hence to the 'communications cloud' of the Internet. Typical options are ISDN or dial-up over telephone lines, using either the V.34 (ITU 33.6kbps) or V.90 (ITU 56kbps) standard. Some models allow multiple phone lines to be used, using multi-link PPP, and will automatically dial up a line when needed or redial when a line is dropped, thereby creating a 'virtual leased line'.

12.7.3 Border routers

Routers within an autonomous system normally communicate with each other using an interior gateway protocol such as RIP. However, routers within an autonomous system that also communicate with remote autonomous systems need to do that via an exterior gateway protocol such as BGP-4. Whilst doing this, they still have to communicate with

other routers within their own autonomous system, e.g. via RIP. These routers are referred to as border routers.

12.7.4 Routing vs. bridging

It sometimes happens that a router is confronted with a layer 3 (network layer) address it does not understand. In the case of an IP router, this may be a Novell IPX address. A similar situation will arise in the case of NetBIOS/NetBEUI, which is non-routable. A 'brouter' (bridging router) will revert to a bridge and try to deal with the situation at layer 2 if it cannot understand the layer 3 protocol, and in this way forward the packet towards its destination. Most modern routers have this function built in.

12.8 Gateways

Gateways are network interconnection devices, not to be confused with 'default gateways' which are the ROUTER IP addresses to which packets are forwarded for subsequent routing (indirect delivery).

A gateway is designed to connect dissimilar networks and could operate anywhere from layer 4 to layer 7 of the OSI model. In a worst case scenario, a gateway may be required to decode and re-encode all seven layers of two dissimilar networks connected to either side, for example when connecting an Ethernet network to an IBM SNA network. Gateways thus have the highest overhead and the lowest performance of all the internetworking devices. The gateway translates from one protocol to the other and handles differences in physical signals, data format, and speed.

Since gateways are, per definition, protocol converters, it so happens that a 2-port (WAN) router could also be classified as a 'gateway' since it has to convert both layer 1 and layer 2 on the LAN side (say, Ethernet) to layer 1 and Layer 2 on the WAN side (say, X.25). This leads to the confusing practice of referring to (WAN) routers as gateways.

12.9 Print servers

Print servers are devices, attached to the network, through which printers can be made available to all users. Typical print servers cater for both serial and parallel printers. Some also provide concurrent multi-protocol support, which means that they support multiple protocols and will execute print jobs on a first-come first-served basis regardless of the protocol used. Protocols supported could include SPX/IPX, TCP/IP, AppleTalk/EtherTalk, NetBIOS/NetBEUI, or DEC LAT.

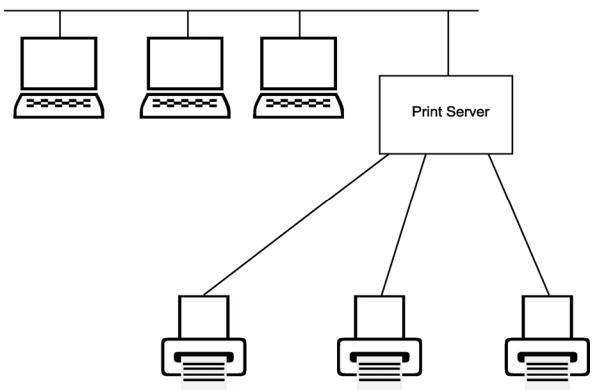


Figure 12.13
Print server applications

12.10 Terminal servers

Terminal servers connect multiple (typically up to 32) serial (RS-232) devices such as system consoles, data entry terminals, bar code readers, scanners, and serial printers to a network. They support multiple protocols such as TCP/IP, SPX/IPX, NetBIOS/NetBEUI, AppleTalk and DEC LAT, which means that they not only can handle devices which support different protocols, but that they can also provide protocol translation between ports.

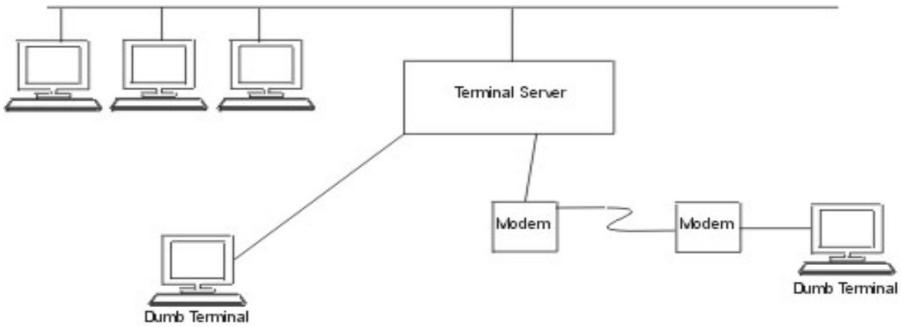


Figure 12.14
Terminal server applications

12.11 Thin servers

Thin servers are essentially single-channel terminal servers. They provide connectivity between Ethernet (10BaseT/100BaseTX) and any serial devices with RS-232 or RS-485 ports. They implement the bottom 4 layers of the OSI model with Ethernet and layer 3/4 protocols such as TCP/IP, SPX/IPX and DEC LAT.

A special version, the industrial thin server, is mounted in a rugged DIN rail package. It can be configured over one of its serial ports, and managed via Telnet or SNMP. A software redirector package enables a user to remove a serial device such as a weigh-bridge from its controlling computer, locate it elsewhere, then connect it via a thin server to an Ethernet network through the nearest available hub. All this is done without modifying any software. A software package called a port redirector makes the computer 'think' that it is still communicating via the weighbridge via the COM port while, in fact, the data and control messages to the device are routed via the network.

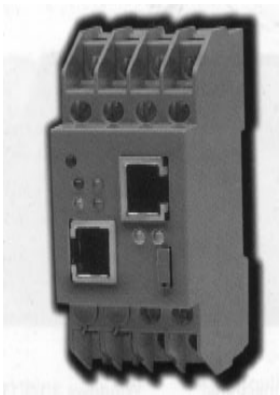


Figure 12.15
Industrial thin server (courtesy of Lantronix)

12.12 Remote access servers

Remote access servers are devices that allow users to dial into a network via analog telephone or ISDN. Typical remote access servers support between 1 and 32 dial-in users via PPP or SLIP. User authentication can be done via Radius, Kerberos or SecurID. Some offer dial-back facilities whereby the user authenticates to the server's internal table, after which the server dials back the user so that the cost of the connection is carried by the network and not the remote user.

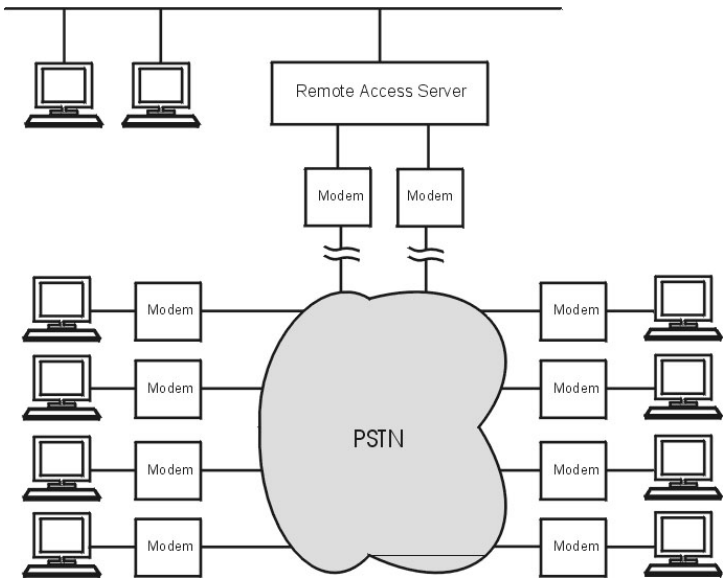


Figure 12.16
Remote access server application

12.13 Network timeservers

Network time servers are standalone devices that compute the correct local time my means of a global positioning system (GPS) receiver, and then distribute it across the network by means of the network time protocol (NTP).

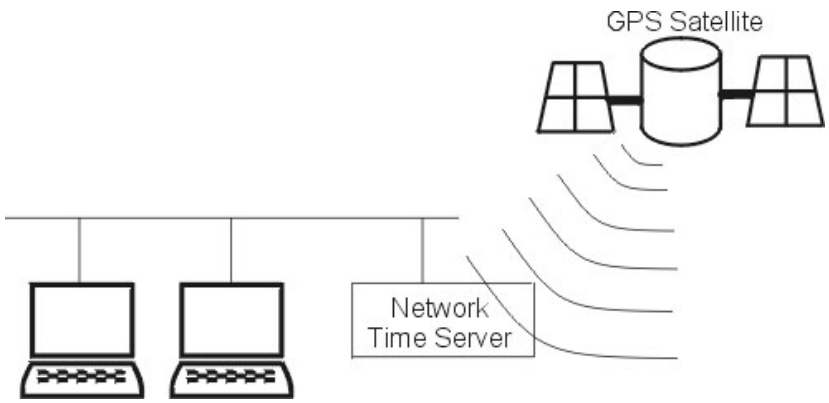


Figure 12.17
Network timeserver application

13

Structured cabling

Objectives

This chapter deals with structured cabling. This chapter will familiarize you with:

- Concept, objectives, and, advantages of structured cabling
- TIA/EIA standards relevant for structured cabling
- Components of structured cabling
- Concept of horizontal cabling
- Cables, outlets/connectors, patch cables etc
- Need for proper documentation and identification systems
- Likely role of fiber optic cables in coming years
- Possibility of overcoming of 100 m limits by use of collapsed and/or centralized cabling concepts
- Next generation fiber-optic cabling products

13.1 Introduction

A computer network is said to be as good as its cabling. Cabling is to a computer network what veins and arteries are to the human body.

Small networks for a few stations can be cabled easily with use of a high quality hub and a few patch cables. The majority of networks of today, however, support a large number of stations spread over several floors of a building or even a few buildings. Cabling for such networks is a different matter all together, far more complex than that required for a small network. A systematic and planned approach is called for in such cases. Structured cabling is the name given to a planned and systematic way of execution of such cabling projects.

Structured cabling, also called structured wiring system (SWS), refers to all of the cabling and related hardware selected and installed in a logical and hierarchical way.

A well-planned and well-executed structured cabling should be able to accommodate constant growth while maintaining order. Computer networking, indeed all of Information Technology, is growing at exponential rates. Newer technologies, faster speeds, more and diverse applications are becoming order of the year, if not order of the day.

It is therefore essential that a growth plan that scales well in terms of higher speeds as well as more and more connections be considered. Accommodating new technology, adding users, and moving connections/stations around is referred to as 'moves, adds, and changes' (MAC).

A good cabling system will make MAC cycles easy because it:

- Is designed to be relatively independent of the computers or telephone network that uses it; so that either can be upgraded/updated with minimum rework on the cabling
- Is consistent, meaning there is the same cabling system for everything
- Encompasses all communication services, including voice, video, and data.
- Is vendor independent
- Will, as far as possible, look into the future in terms of technology, if not be future-proof
- Will take into account ease of maintenance, environmental issues, and security considerations
- Will comply with relevant building and municipal codes

13.2 TIA/EIA cabling standards

Standards lay down compliance norms and bring about uniformity in implementations. Several vendor-independent structured cabling standards have been developed.

The Telecommunications Industry Association (TIA), Electronic Industries Association (EIA), American National Standards Institute (ANSI) and the ISO are the professional bodies involved in laying down these standards. Both the TIA and EIA are members of ANSI, which is a coordinating body for voluntary standards within United States of America. ANSI, in turn, is a member of the ISO, the international standards body.

The important standards relevant here are:

- **ANSI/TIA/EIA-568-A:** This standard lays down specifications for a Structured Cabling System. It includes specifications for horizontal cables, backbone cables, cable spaces, interconnection equipment, etc
- **ANSI/TIA/EIA-569-A:** This lays down specifications for the design and construction practices to be used for supporting Structured Cabling Systems. These include specifications for Telecommunication Closets, Equipment Rooms, and Cable Pathways etc
- **ANSI/TIA/EIA-570:** This standard specifies norms for premises wiring systems for homes or small offices
- **ANSI/TIA/EIA-606:** This standard lays down norms for cabling systems
- **ANSI/TIA/EIA–607:** This standard lays down norms for grounding practices needed to support the equipment used in cabling systems
- **ISO/IEC 11801:** This is an international standard on 'Generic Cabling for Customer Premises. Topics covered are same as those covered by the TIA/EIA-568 standard. It also includes category-rating system for cables. It lists four classes of performance for a link from class A to class D. The classes C and D are similar to Category 3 and Category 5 links of the TIA/EIA 568A standard
- **CENELEC EN 50173:** This is a European cabling standard while the British equivalent is BS EN 50173

13.3 Components of structured cabling

The TIA/EIA 568 standard lists six basic elements of a structured cabling system. They are:

Building entrance facilities: Equipment (such as cables, surge protection equipment, connecting hardware) used to connect a campus data network or public telephone network to the cabling inside the building.

Equipment room: Equipment rooms are used for major cable terminations and for any grounding equipment needed to make a connection to the campus data network and public telephone network

Backbone cabling: Building backbone cabling based on a star topology is used to provide connections between telecommunication closets, equipment rooms, and the entrance facilities.

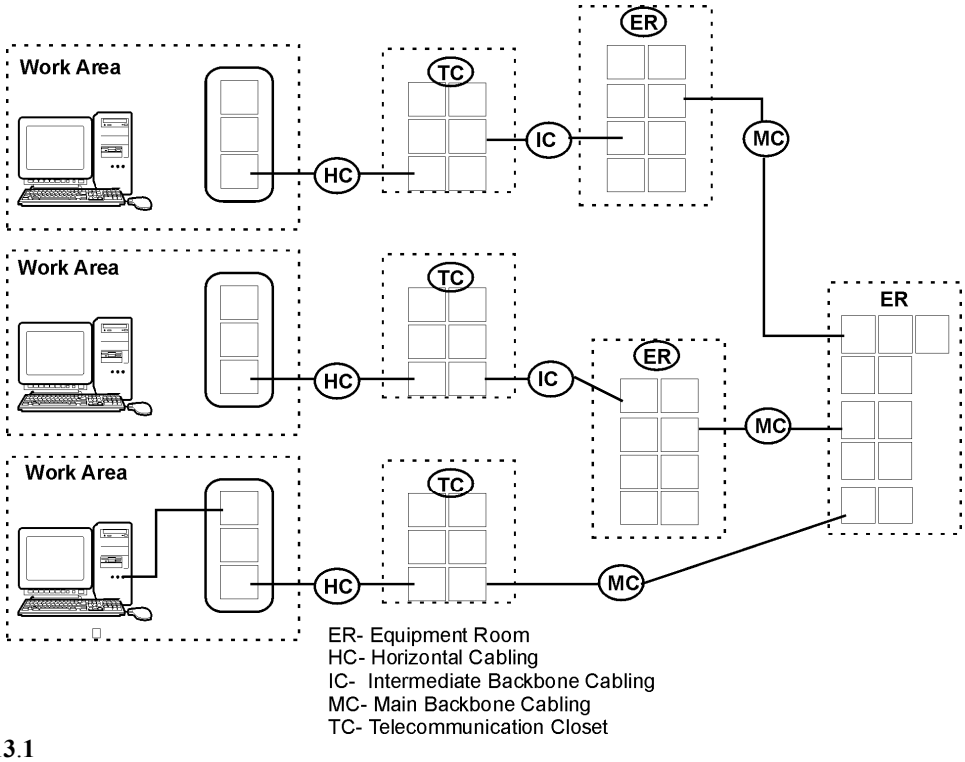


Figure 13.1
Typical elements of a structured cabling system

Telecommunication closet: A telecommunication closet, also called a wiring closet is primarily used to provide a location for the termination of the horizontal cable on a building floor. This closet houses the mechanical cable terminations and cross-connects, if any, between the horizontal and the backbone cabling system. It may also house hubs and switches.

Work area: This is an office space or any other area where computers are placed. The work area equipment includes patch cables used to connect computers, telephones, or other devices, to outlets on the wall.

Horizontal cabling: This includes cables extending from telecommunication closets to communication outlets located in work area. It also includes any patch cables needed for cross-connects between hub equipment in the closet and the horizontal cabling.

13.4 Star topology for structured cabling

The TIA/EIA 568 standard specifies a star topology for the structured cabling backbone system. It also specifies that there be no more than two levels of hierarchy within a building. This means that a cable should not go through more than one intermediate cross-connect device between the main cross connect (MC) located in an equipment room and the horizontal cross connect (HC) located in a wiring closet.

The star topology has been chosen because of its obvious advantages:

- It is easier to manage 'moves-adds-changes'
- It is faster to troubleshoot
- Independent point-to-point links prevent cable problems on any link from affecting other links
- Network speed can be increased by upgrading hubs without a need for recabling the whole building

13.5 Horizontal cabling

13.5.1 Cables used in horizontal cabling

Both twisted-pair cables and fiber-optic cables can be used in structured cabling systems.

The TIA/EIA stipulates holds that twisted-pair cables of a category better than Category 2 be used. Category 5 or better is recommended for new horizontal cable installations if twisted-pair cabling is the choice.

Specifically, TIA/EIA 568 gives the following options for use in horizontal links:

- Four-pair, 100 ohm impedance UTP cable of Category 5 or better using 24 AWG solid conductors is recommended. The connector recommended is the eight position RJ-45 modular jack
- Two-pair 150 ohm shielded twisted pair (STP) using an IEEE802.5 four-position shielded token ring connector is recommended
- Two-fiber, 62.5/125 multimode fiber optical cables are recommended. The recommended connector is the SCFOC/2.5 duplex connector 9, also known as the SC connector
- Coaxial cables are still recognized in TIA/EIA standards but are not recommended for new installations. Coaxial cabling is in fact being phased out from future versions of the standards

13.5.2 Telecommunication outlet/connector

The TIA/EIA specifies a minimum of two work area outlets (WAOs) for each work area with each area being connected directly to a telecommunication closet. One outlet should connect to one four-pair UTP cable. The other outlet may connect to another four-pair UTP cable, an STP cable, or to a fiber-optic cable as needed. Any active or passive adapters needed at the work area should be external to the outlet.

13.5.3 Cross-connect patch cables

Patch cables should not exceed 6 meters in length. A total allowance of 10 meters is provided for all patch cables in the entire length from the closet to the computer. This, combined with the maximum of 90 meters of horizontal link cable distance, makes a total

of 100 meters for the maximum horizontal channel distance from the network equipment in the closet to the computer.

13.5.4 Horizontal channel and basic link as per TIA/EIA telecommunication system bulletin 67 (TSB- 67)

TSB-67 specifies requirements for testing and certifying installed UTP horizontal cabling. It defines a basic link and a channel for testing purposes as shown in figure 13.2. The basic length consists of the fixed cable that travels between the wall plate in the work area and the wire termination point in the wiring closet. This basic link is limited to a length of 90 meters. This link is to be tested and certified according to guidelines in TSB-67 before hooking up any network equipment or telephone equipment.

Only Category 5e cables and components should be used for horizontal cabling. Components of lower category will not give the best results and may not accommodate high-speed data transfer.

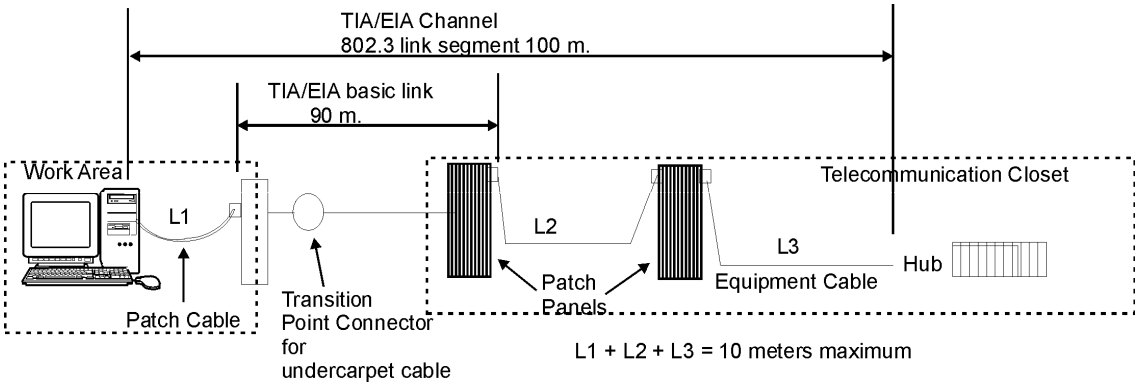


Figure13.2
Basic link and link segment

13.5.5 Documentation and identification

Even a small network cannot be organized and managed without proper documentation. A comprehensive listing of all cable installations with floor plans, locations, and, identification numbers is necessary. Time spent in preparing this will save time and trouble in 'moves, adds and changes' cycles, as well as in times of trouble shooting. Software packages for cable management should be used if the network is large and complex.

Cable identification is the basis for any cabling documentation. An identification convention suitable for the network can be set up easily. It is critical to be consistent at the time of installation as well as at times of 'moves, adds, changes' cycles.

Labels specifically designed to stick to cables should be used.

13.6 Fiber-optics in structured cabling

Future computer networks will certainly be faster, support a greater number of applications, and provide service to a number of geographically diverse users.

Fiber-optic cables are most likely to play a major role in this growth because of their unique advantages.

This sub-section is based on extracts from an article written by Paul Kopera, Anixter Director of Fiber Optics. The original article was written in 1996-97 and then revised in 2001.

'With network requirements changing constantly, it is important to employ a cabling system that can keep up with the demand.

Cabling systems, the backbone of any data communications system, must become utilities. That is, they must adapt to network requirements on demand. When a network needs more speed, the media should deliver it. The days of recabling to adopt new networking technologies should be phased out. Today's Structured Cabling System should provide seamless migration to tomorrow's network services.

One media that provides utility-like service is optical fiber. Fiber optic cabling has been used in telecommunication networks for over 20 years, bringing unsurpassed reliability and expandability to that industry. Over the past decade, optical fibers have found their way into cable television networks—increasing reliability, providing expanded service, and reducing costs. In the Local Area Network (LAN), fiber cabling has been deployed as the primary media for campus and building backbones, offering high-speed connections between diverse LAN segments.

Today, with increasingly sophisticated applications like high-speed ISPs and e-commerce becoming standard, it's time to consider optical fiber as the primary media to provide data services to the desktop.'

13.6.1 Advantages of fiber-optic technology

Fiber-optic cable has the following advantages:

- It has the largest bandwidth compared to any media available. It can transmit signals over the longest distance at the lowest cost, without errors and the least amount of maintenance
- Fiber is immune to EMI and RFI
- It cannot be tapped, so it's very secure
- Fiber transmission systems are highly reliable. Network downtime is limited to catastrophic failures such as a cable cut, not soft failures such as loading problems
- Interference does not affect fiber traffic and as a result, the number of retransmissions is reduced and network efficiency is increased. There are no cross talk issues with optical fiber
- It is impervious to lightning strikes and does not conduct electricity or support ground loops
- Fiber-based network segments can be extended 20 times farther than copper segments. Since the current structured cabling standard allows 100 meter lengths of horizontal fiber cabling from the telecom closet (this length is based on assumption use of twisted-pair cable), each length can support several GHz of optical bandwidth
- Recent developments in multimode fiber optics include enhanced glass designed to accommodate even higher-speed transmissions. With capabilities well above today's 10/100 Mbps Ethernet systems, fiber enables the migration to tomorrow's 10 Gigabit Ethernet ATM and SONET networking schemes without recabling
- Optical fiber is independent of the transmission frequency on a network. There are no cross talk or attenuation mechanisms that can degrade or limit the performance of fiber as network speeds increase. Further, the bandwidth

of an optical fiber channel cannot be altered in any manner by installation practices

Once a fiber is installed, tested, and certified as good, then that channel will work at 1 Mbps, 10 Mbps, 100 Mbps, 500 Mbps, 1 Gbps or 10 Gbps. This guarantees that a fiber cable plant installed today will be capable of handling any networking technology that may come along in the next 15 to 20 years. Thus, the installed fiber-optic cable need not be changed for the next 15 years. So rather than terming it 'upgradable', it may be termed 'future-proof'.

13.6.2 The 100 meter limit

Optical fiber specifications laid down by the TIA/EIA for structured cabling are summarized in table 13.1 below:

Optical Fiber Type	Multimode		Single Mode
Dimensions	6.25/125 microns	50/125 microns	8.3/125 microns
Bandwidth			
Low Speed (850 nm)	160 MHz-km	500 MHz.km	--
High Speed (1300 nm)	500 MHz*km	500 MHZ8km	>100 Ghz
Attenuation at			
Low Speed	3.5 dB/km	--	--
High Speed	1.5 dB/km	--	--
Backbone Cable Length	2000 m.	2000 m.	3000 m.
Horizontal Cable Length	100 m.	100 m.	Not Recommended
Applications	E-net, TR, FDDI 155 Mbps ATM Baseband Video Security Systems	E-net, TR FDDI 1.2 Gbps ATM Fiber Chanel Broadband Video	Channel Extension
Connector Type (T568-SC)	Beige Shroud	Beige Shroud	Blue Shroud

Table 13.1
TIA/EIA optical fiber specifications for structured cabling

The horizontal distance limitations of 100 m. laid down by TIA/EIA 568-A is based on the performance characteristics of copper cabling. The TIA/EIA 568-B.3 Committee is currently evaluating the extended performance capabilities of optical fiber. The objective is to take advantage of the fiber's bandwidth and operating distance characteristics to create structured cabling systems that are more robust.

13.6.3 Present structured cabling topology as per TIA/EIA

TIA/EIA 568-A recommends multiple wiring closets distributed throughout a building irrespective of whether backbone and horizontal cabling is copper or fiber based.

A network can be vertical with multiple wiring closets on each floor, or horizontal with multiple satellite closets located throughout a plant. The basic cabling structure is a star-cabled plant with the highest functionality networking components residing in the main distribution center (MDC).

The MDC is interconnected via fiber backbone cable to intermediate distribution centers (IDCs) in the case of a campus backbone, or to telecommunications closets (TCs) in case of a network occupying a single building.

From the TC to the desktop, up to 100 meters of Cat 5 UTP cable or optical fiber cable can be deployed. Typically, lower level network electronics are located in a TC and provide floor-level management and segmentation of a network. A TC also provides a point of presence for structured cabling support components, namely cable interconnect or cross-connect centers, cable storage and splices to backbone cabling.

13.6.4 Collapsed cabling design alternative

The fiber's superior performance can be used to revise the 100 m limit so that a horizontal distribution system can be redesigned to more efficiently use networking components, increase reliability and reduce maintenance and cost.

One method is to collapse all horizontal networking products into one closet and run fiber cables from this central TC to each user.

Since optical fiber systems have sufficient transmission bandwidth to support most horizontal distances, it is not necessary to have multiple wiring closets throughout each floor. With this network design, management is centralized and the number of maintenance sites or troubleshooting points is reduced. Cutting the number of wiring closets saves money and space. It reduces the number of locations that must be fitted with additional power, heating, ventilating, and air-conditioning facilities in a horizontal space. Testing, troubleshooting and documentation become easier. Moves, adds and changes are facilitated through network management software rather than patch cord manipulation. With this architecture, newly developed open-office cabling schemes (TIA/EIA TSB 75) can also be easily integrated into a network.

13.6.5 Centralized cabling design alternative

While collapsed cabling is one alternative, centralized cabling is a second alternative.
In a centralized cabling system, all network electronics reside in either the MDC or IDC. The idea is to connect the user directly from the desktop or workgroup to the centralized network electronics.

There are no active components at floor level. Connections are made between horizontal and riser cables through splice points or interconnect centers located in a TC. For short runs, a technique called fiber home run is used. It connects a workstation directly to the MDC. Low count (2 or 4 fibers) horizontal cable can be run to each workstation or office. In addition, multi-fiber cables (12 or more fibers) can support multiple users, providing backbone connections to workgroups in a modular office environment.

A centralized cabling network design provides the same benefits as a collapsed network – condensed electronics and more efficient use of chassis and rack spaces. By providing one central location for all network electronics, maintenance is simplified, troubleshooting time reduced and security enhanced. Moves, adds and changes are again addressed by software.

Centralized cabling is described by the Technical Service Bulletin, TIA/EIA TSB 72, which recommends a maximum distance of 300 meters to allow Gigabit applications to be supported.

13.6.6 Fiber zone cabling – mix of collapsed and centralized cabling approaches

This design concept is an interesting mix between a collapsed backbone and a centralized cabling scheme. Fiber zone cabling is a very effective way to bring fiber to a work area.

It utilizes low-cost, copper-based electronics for Ethernet data communications, while providing a clear migration path to higher-speed technologies.

Like centralized cabling, a fiber zone cabling scheme has one central MDC. Multifiber cables are deployed from the MDC through a TC to the user group. A typical cable might contain 12 or 24 fibers.

At the workgroup, the fiber cable is terminated in a multi-user telecommunications outlet (MUTO) and two of the fibers are connected to a workgroup hub. This local hub, supporting six to twelve users, has a fiber backbone connection and UTP user ports. Connections are made between the hub and workstation with simple UTP patch cords. The station network interface card (NIC) is also UTP-based. The remaining optical fibers are unused or left 'dark' in the MUTO for future needs.

Dark fibers provide a simple mechanism for adding user channels to the workgroup or for upgrading the workgroup to more advanced high-speed network architectures like ATM, SONET, or Gigabit Ethernet. Upgrades are accomplished by removing the hub and installing fiber jumper cables from the multi-user outlets to the workstation.

Network electronics also need to be upgraded. This process converts the network segment to a fiber home run or centralized cabling scheme. It is a very flexible and cost-effective way to deploy fiber today while providing a future migration strategy for a network. Further, an investment made in UTP-based Ethernet connectivity products is not wasted; it is, in effect, extended.

Two new cabling products have entered the marketplace, offering zone-cabling enclosures. One style mounts above a suspended ceiling, holding fiber and copper UTP cross connects, between hubs, switches, and workstations. The other style, a much larger unit, replaces a 2'×4' ceiling tile and has enough room to house a hub or other active electronics, as well as cross connects.

13.6.7 New next generation products

Over the past year, several new products have been developed that will aid in the deployment of optical fiber-to-the-desk.

To date, the standards committees are evaluating new, higher performance optical components that offer increased performance, ease of installation and lower costs. Among some of these exciting developments are small-form-factor connectors (SFFC), vertical cavity surface-emitting lasers (VCSEL) and next-generation fiber.

Advancements in fiber connectors are continuing to make fiber as viable an answer as copper.

Traditionally, fiber systems required twice as many connectors as copper cabling – crowding telecommunication closets with additional patch panels and electronics. Recently, manufacturers have introduced small-form-factor connectors that provide twice as much density as previous fiber connectors. These mini-fiber connectors hold the send and receive fibers in one housing. This reduces the space required for a fiber connection. More importantly, it decreases the footprint required on the hubs and switches for fiber transceivers. The net result is a cost reduction nearly four times to that of a conventional fiber system.

Complimenting the SFFC components are new vertical cavity surface-emitting lasers. This fiber optic transmission source combines the power and bandwidth of a laser at the

lower cost of a light-emitting diode (LED). VCSELs, when integrated into SFFC transceivers, allow for the development of higher-speed, higher-bandwidth optical systems, further extending the reach and capability of the FTTD cable system.

Next-generation fiber is 50/125 micron with a laser bandwidth greater than 2000 MHz/km at 850 nm. This fiber allows serial transmission at 10 Gigabits up to 300 meters. This next-generation fiber coupled with a 10 Gigabit, 850 nm VCSEL allows the lowest cost 10 Gigabit solutions.

Recent developments in fiber optics include:

- Enhanced glass design to accommodate high-speed transmission
- Smaller-size connectors that save space and lower cost
- Vertical cavity surface emitting laser technology for high-speed transmission over longer distances at low cost
- A vast array of new support hardware designed for Fiber Zone Cabling
- Fiber-to-the-desk is a cost-effective design that utilizes fiber in today's low-speed network while providing a simple migration strategy for tomorrow's high-speed connections. Fiber-to-the-desk combines the best attributes of a copper-based network (low-cost electronics) with the best of fiber (superior physical characteristics and upgradability) to provide unequalled network service and reliability

14

Multi-segment configuration guidelines for half-duplex Ethernet systems

Objectives

This chapter provides some design and evaluation insights into multi-segment configuration guidelines for half-duplex Ethernet systems. Study of this chapter will provide:

- An understanding of approaches for verifying the configuration of a half-duplex shared Ethernet channels
- Information on Model I and Model II guidelines laid down in the IEEE 802.3 standard
- Rules for combining multiple segments with repeater hubs
- Detailed analysis of methods of building complex half-duplex Ethernet systems operating at 10, 100, and, 1000 Mbps
- Examples of sample network configurations

Note: This topic is only relevant for CSMA/CD systems. Most modern Ethernet systems are full duplex switched systems, with no timing constraints.

14.1 Introduction

The basic configuration of simple half-duplex Ethernet systems using a single medium is easily done by studying the properties of the medium to be used and studying the standard rules. But, when it comes to more complex half-duplex systems based on repeater hubs, multi-segment configuration guidelines need to be studied and applied.

The official configuration guidelines lay down two methods for configuring these systems, the methods being called Model I and Model II.

Model I comprises of a set of ready-to-use rules. Model II lays down calculation aids for evaluation of more complex topologies that cannot be easily configured by the application of Model I rules.

The guidelines are applicable to equipment described in, or made to confirm to, the IEEE 802.3 standard. The segments must be built as per recommendations of the standard. If this is not adhered to, verification and evaluation of operations in terms of signal timing is not possible.

Proper documentation of each network link must be prepared when it is installed. Information about the cable length of each segment, cable types, cable ID numbers etc, should be recorded in the documentation. The IEEE standard recommends creation of documentation formats based on Table 14.1 shown below:

	Horrizontal Cabling	Transceiver Cabling	Wiring Closet Patch Cords	Station Patch Cords
Length				
Type				
Cable Manufacturer				
Cable Code/ID				

Table 14.1
Cable segment documentation form

14.2 Defining collision domains

The multi-segment configuration guidelines apply to the MAC protocol and half-duplex Ethernet collision domains described earlier in this manual.

A collision domain is defined as a network within which there will be a collision if two computers attached to the system transmit at the same time. An Ethernet system made up of a single segment or of a multiple segments linked together with repeater hubs make a single collision domain. A typical collision domain is shown in Figure 14.1.

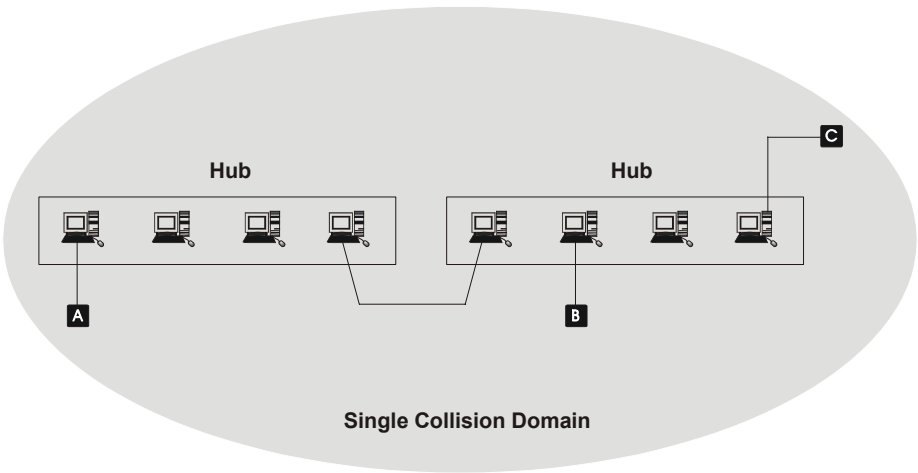


Figure 14.1
Repeaters create a single collision domain

All segments within a collision domain must operate at the same speed. For this reason, there are separate configuration guidelines for Ethernet segments with of different speeds.

IEEE 802.3 lays down guidelines for the operation of a single half-duplex LAN, and does not say anything about combining multiple single collision domains. Switching hubs, however, enable the creation of a new collision domain on each port of a switching hub, thereby linking many networks together. Segments with different speeds can also be linked this way.

14.3 Model I configuration guidelines for 10 Mbps systems

Model I in the IEEE 802.3 standard describes a set of multi-segment configuration rules for combining various 10 Mbps Ethernet segments. Most of the terms and phrases used below have been taken directly from the IEEE standard.

The guidelines are as follows:

Repeater sets are required for all segment interconnections. A 'repeater set' is a repeater and its associated transceivers if any. Repeaters must comply with all specifications in the IEEE 802.3 standard.

MAUs that are a part of repeater sets do not count towards the maximum number of MAUs on a segment. Twisted-pair, fiber optic and thin coax repeater hubs typically use internal MAUs located inside each port of the repeater. Thick Ethernet repeaters use an outboard MAU to connect to the thick coax.

The transmission path permitted between any two DTEs may consist of up to five segments, four repeater sets (including optional AUIs), two MAUs, and two AUIs. The repeater sets are assumed to have their own MAUs, which are not counted in this rule.

AUI cables for 10BaseFP and 10BaseFL shall not exceed 25 m. Since two MAUs per segment are required, 25 m per MAU results in a total AUI cable length of 50 m per segment.

When a transmission path consists of four repeaters and five segments, up to three of the segments may be mixing segments and the remainder must be link segments. When five segments are present, each fiber optic link segment (FOIRL, 10BaseFB, or 10BaseFL) shall not exceed 500 m, and each 10BaseFP segment shall not exceed 300 m. A mixing segment is one that may have more than two medium dependent interfaces attached to it, e.g. a coaxial cable segment. A link segment is a point-to-point full-duplex medium that connects two and only two MAUs.

When a transmission path consists of three repeater sets and four segments, the following restrictions apply:

The maximum allowable length of any inter-repeater fiber segment shall not exceed 1000 m for FOIRL, 10BaseFB, and 10BaseFL segments and shall not exceed 700 m for 10BaseFP segments.

The maximum allowable length of any repeater to DTE fiber segment shall not exceed 400 m for 10BaseFL segments and shall not exceed 300 m for 10BaseFP segments and 400 m for segments terminated in a 10BaseFL MAU.

There is no restriction on the number of mixing segments in this case. When using three repeater sets and four segments, all segments may be mixing segments if so desired.

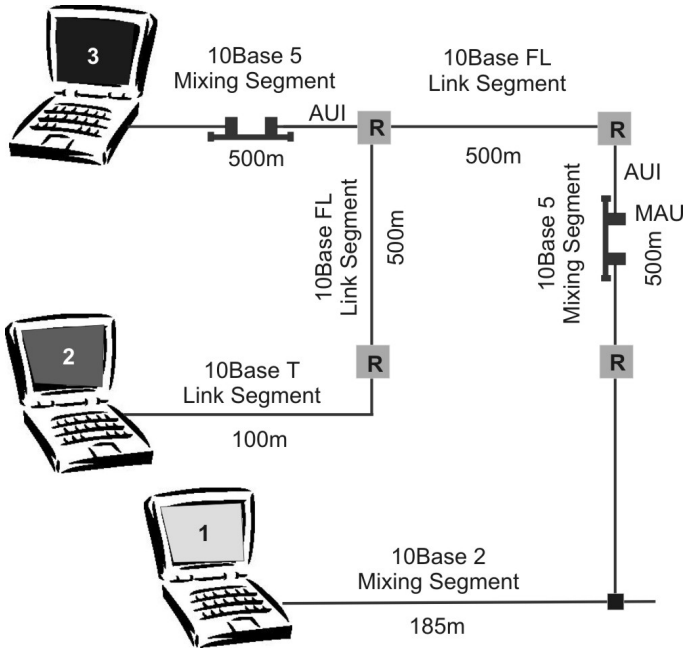


Figure 14.2
Model I 10 Mbps configuration

Figure 14.2 shows an example of one possible maximum Ethernet configuration that meets the Model I configuration rules. The maximum packet transmission path in this system is between station 1 and station 2, since there are four repeaters and five media segments in that particular path. Two of the segments in the path are mixing segments, and the other three are link segments.

The Model I configuration rules are based on conservative timing calculations. That however should not be taken to mean that these rules could be relaxed. In spite of the allowances made in the standards for manufacturing tolerances and equipment variances, there isn't much margin left in maximum-sized Ethernet networks. For maximum performance and reliability, it is better to conform to the published guidelines.

14.4 Model II configuration guidelines for 10 Mbps

The Model II configuration guidelines provide a set of calculation aids that make it possible to check the validity of more complex Ethernet systems. This is a simple process based on multiplication and addition.

There are two sets of calculations provided in the standard that must be performed for each Ethernet system. The first set of calculations verifies the round-trip signal delay time, while the second set verifies that the amount of inter-frame gap shrinkage is within normal limits. Both calculations are based on network models that evaluate the worst-case path through the network.

14.4.1 Models of networks and transmission delay values

The network models and transmission delay values provided in the Model 2 guidelines deliberately hide a lot of complexity while still making it possible to calculate the timing values for any Ethernet system. Each component in an Ethernet system provides a certain amount of transmission delay and all of these are listed in the 802.3 standard in detail.

The choice of equipment used influences the transmission and delay of Ethernet signals through the system.

Complex delay calculations and delay considerations covered earlier are found in the Model II guidelines. The standard also provides a set of network models and segment delay values. The worst-case path of a system is defined as the path through a network that has the longest segments and the most number of repeaters between any two stations. A standard network model used in calculating the round-trip timing of a system's worst-case path is shown in Figure 14.3 This calculation model, which is perhaps the most commonly used one, includes a left and right end segment, and many middle segments. The number of middle segments used in the calculation is dependent on individual systems, although a minimum number is shown in the Figure 14.3.

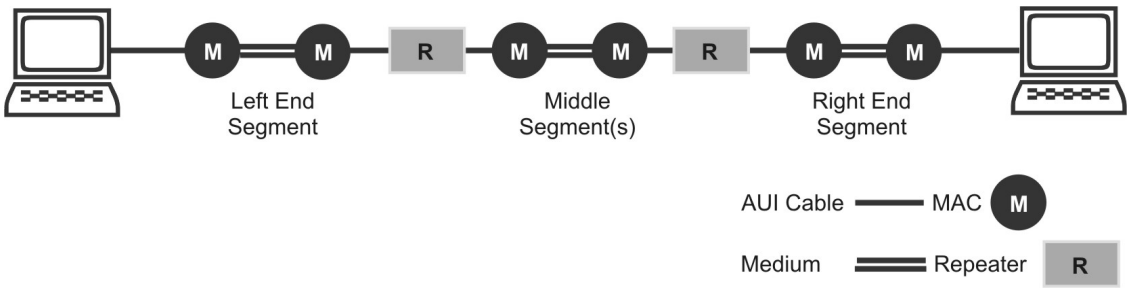


Figure 14.3
Network model for round-trip timing

A similar model is used to check the worst-case path's round-trip timing on any network under evaluation. Later we will use this model to evaluate the round-trip timing of two sample networks Interframe gap shrinkage is also calculated by using a similar model, as will be demonstrated later.

14.4.2 The worst-case path

The first step is to locate the path with the maximum delay in a network. As defined earlier, this is the path with the longest round-trip time and the largest number of repeaters between two stations. In some cases, there may be more than one worst-case path in the system. If such a problem is encountered then it is prudent to identify all the paths through the given network and classify them using the definition of a worst-case path. Once this is done, calculate the round-trip timing or interframe gap for each path. Whenever the results exceed the limits prescribed by the standard classify the network as 'failed to pass the test'.

A complete and up-to-date map of the network should be available to find the worst-case path between two stations. The information needed in such a map must include:

- Segment types used (twisted pair, fiber optic, coax)
- Segment length
- Repeater locations for the entire system
- Segment and repeater layouts for the system

Once the worst-case path is found, the next thing needed is to draw your path based on the standard model shown in Figure 14.3. This is done by assigning the segment at one end of the worst-case path to be a left end segment, leaving a right end segment with one or more middle segments.

For doing this, draw a sketch of your worst-case path, noting the segment types and lengths. Then arbitrarily assign one of the end segments to be the left end; the type of

segment assigned doesn't really matter. This leaves a right end segment. Consequently all other segments in the worst-case path are classified as middle segments.

14.4.3 Calculating the round-trip delay time

If and when any two stations on a half-duplex Ethernet channel transmit at the same time, they must have fair access to the system. This is one of the issues that the configuration guidelines try to address. When this is achieved, each station attempting to transmit must be notified of channel contention (a possible collision). Each station receiving a collision signal within the correct collision-timing window achieves this.

Calculating the total path delay, or *round-trip timing*, of the worst-case path determines whether an Ethernet system meets the limits or not. This is done using segment delay values. Each Ethernet media type has a value provided in terms of bit time that eventually determines the delay value. A bit time is the amount of time required to send one data bit on the network. For a 10 Mbps Ethernet system the value is 100 nanoseconds (ns). Table 14.2 gives the segment delay values provided in the standard. These are used in calculating the total worst-case path delay.

Segment Type	Max Length (meters)	Left End		Middle Segment		Right End		RT Delay/meter
		Base	Max	Base	Max	Base	Max	
10Base5	500	11.75	55.05	46.5	89.8	169.5	212.8	0.0866
10Base2	185	11.75	30.73	146.5	65.48	169.5	188.48	0.1026
10BaseT	100	15.25	26.55	42.0	53.3	165	176.3	0.113
10BaseFL	2000	12.25	212.25	33.5	233.5	156.5	356.5	0.1
Excess AUI	48	0	4.88	0	4.88	0	4.88	0.1026

Table 14.2
Round trip delay value in bit times

The total round-trip delay is found by adding up the delay values found on the worst-case path in network. Once calculations have been done on the segment delay values for each segment in the worst-case path, add the segment delay values together to find the total path delay. The standard recommends addition of a margin of 5 bit times to this total. If the result is less than or equal to 575 bit times, the path passes the test.

This value ensures that a station at the end of a worst-case path will be notified of a collision and stop transmitting within 575 bit times. This includes 511 bits of the frame plus the 64 bits of frame preamble and start frame delimiter (511 + 64 = 575). Once it is known that the round-trip timing for the worst-case path is okay, and then one can be sure that all other paths must be okay as well.

There is need to check one more item in the calculation for total path delay. If the path being checked has left and right end segments of different segment types, then check the path twice. The first time through, use the left end path delay values of one of the segment types, and the second time through use the left end path delay values of the other segment type. The total path delay must pass the delay calculations no matter which set of path delay values are used.

14.4.4 The inter-frame gap shrinkage

The inter-frame gap is a 96-bit time delay provided between frame transmissions to allow the network interfaces and other components some recovery time between frames. As

frames travel through an Ethernet system, the variable timing delays in network components combined with the effects of signal reconstruction circuits in the repeaters can result in an apparent shrinkage of the inter-frame gap.

Too small a gap between frames can overrun the frame reception capability of network interfaces, leading to lost frames. Therefore, it's important to ensure that a minimum inter-frame gap is maintained at all receivers (stations).

The network model for checking the inter-frame gap shrinkage is shown in figure 14.4.

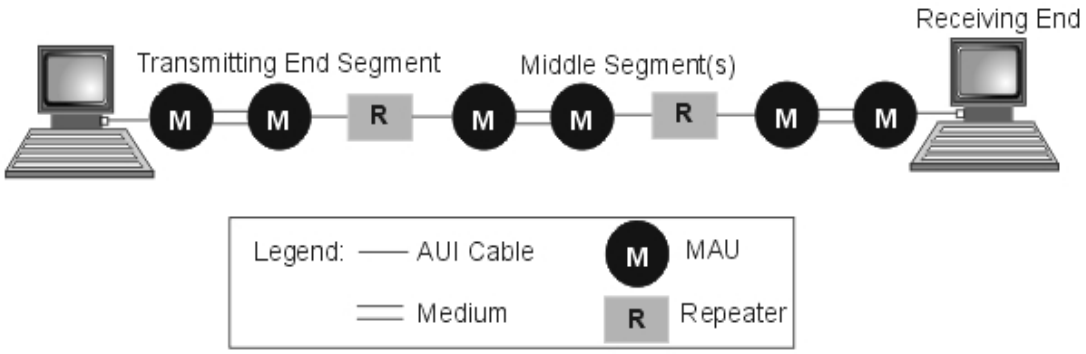


Figure 14.4
Network model for interframe gap shrinkage

Figure 14.4 looks a lot like the round-trip path delay model (Figure 14.3), except that it includes a *transmitting end* segment. When one is doing the calculations for inter-frame gap shrinkage, only the transmitting end and the middle segments are of interest, since only signals on these segments must travel through a repeater to reach the receiving end station. The final segment connected to the receiving end station does not contribute any gap shrinkage and is therefore not included in the interframe gap calculations. Table 14.3 gives the values used for calculating inter-frame gap shrinkage.

Segment Type	Transmitting End	Mid-Segment
Coax	16	11
Link Segment	10.5	8

Table 14.3
Interframe gap shrinkage in bit times

When the receive and transmit end segments are not of the same media type, the standard lays down use of the end segment with the largest number of shrinkage bit times as the transmitting end for the purposes of this calculation. This will provide the worst-case value for interframe gap shrinkage. If the total is less than or equal to 49 bit times, then the worst-case path passes the shrinkage test.

14.5 Model 1-configuration guidelines for Fast Ethernet

Transmission system Model 1 of the Fast Ethernet standard ensures that the important Fast Ethernet timing requirements are met, so that the medium access control (MAC) protocol will function correctly.

The basic rules for Fast Ethernet configuration include:

- All copper (twisted-pair) segments must be less than or equal to 100 meters in length
- Fiber segments must be less than or equal to 412 meters in length

- If Medium Independent Interface (MII) cables are used, they must not exceed 0.5 meters each

When it comes to evaluating network timing, delays attributable to the MII do not need to be accounted for separately, since these delays are incorporated into station and repeater delays.

Table 14.4 shows the maximum collision domain diameter for segments using Class I and Class II repeaters. The maximum collision domain diameter in a given Fast Ethernet system is the longest distance between any two stations (DTEs) in the collision domain.

Repeater Type	All Copper	All Fiber	Copper+Fiber eg.T4+FX	Copeer+Fiber eg. TX+FX
DTE-DTE Single Segment	100	412	N/A	N/A
One Class I Repeater	200	272	231(1)	260.8 (1)
One Class II Repeater	200	320	N/A (2)	308.8 (1)
Two Class II Repeaters	205	228	N/A (2)	216.2 (3)

Notes:(1) Assumes 100 m. Of twisted-pair cabling and one fiber link
(2) FX Fiber and T4 copper cannot be linked in a class II repeater
(3) Assumes 105 m. Of twisted-pair cable and one fiber link

Table 14.4
Maximum Fast Ethernet collision domain in meters as per Model I guidelines

The first row in the above table shows that a DTE-to-DTE (station-to-station) link with no intervening repeater may be made up of a maximum of 100 meters of copper, or 412 meters of fiber optic cable. The next row provides the maximum collision domain diameter when using a Class I repeater, including the case of all-twisted-pair and all-fiber optic cables, or a network with a mix of twisted-pair and fiber cables. The third row shows the maximum collision domain length with a single Class II repeater in the link.

The last row of the shows the maximum collision domain allowed when two Class II repeaters are used in a link. In this last configuration, the total twisted-pair segment length is assumed 105 meters on the mixed fiber and twisted-pair segment. This includes 100 meters for the segment length from the repeater port to the station, and five meters for a short segment that links the two repeaters together in a wiring closet.

Figure 14.5 shows an example of a maximum configuration based on the 100 Mbps simplified guidelines we've just seen. Note that the maximum collision domain diameter includes the distance:

A (100 m) + B (5 m) + C (100 m)

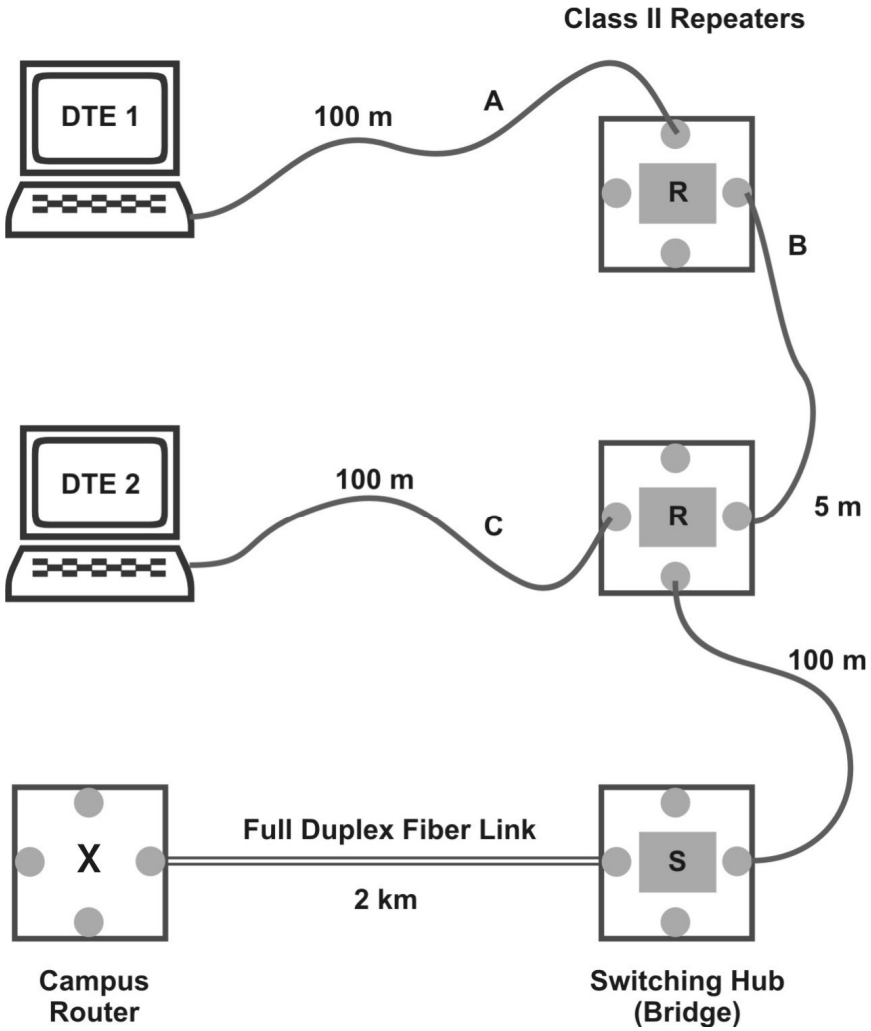


Figure 14.5
One possible maximum 100 Mbps configuration

The inter-repeater segment length can be longer than 5 m as long as the maximum diameter of the collision domain does not exceed the guidelines for the segment types and repeaters being used. Segment *B* in above figure could be 10 meters in length, for instance, as long as other segment lengths are adjusted to keep the maximum collision diameter to 205 meters. While it's possible to vary the length of the inter-repeater segment in this fashion, you should be wary of doing so and carefully consider the consequences.

14.5.1 Longer inter-repeater links

Using longer inter-repeater links has some shortcomings. Their use makes network timing rely on the use of shorter than standard segments from the repeater ports to the stations, which could cause confusion and problems later on. These days it assumed that twisted-pair segment lengths could be up to 100 meters long. Because of that, a new segment that's 100 meters long could be attached to a system with a long inter-repeater link later. In this case, the maximum diameter between some stations could then become 210 meters. If the signal delay on this long path exceeds 512 bit times, then the network may experience problems, such as late collisions. This can be avoided by keeping the length of inter-repeater segments to five meters or less.

A switching hub is just another station (DTE) as far as the guidelines for a collision domain are concerned. The switching hub shown in figure 14.5 provides a way to link separate network technologies – in this case, a standard 100BaseT segment and a full-duplex Ethernet link. The switching hub is shown linked to a campus router with a full-duplex fiber link that spans up to two kilometers. This makes it possible to provide a 100 Mbps Ethernet connection to the rest of a campus network using a router port located in a central section of the network.

Figure 14.6 shows an example of a maximum configuration based on a mixture of fiber optic and copper segments. Note that there are two paths representing the maximum collision domain diameter. This includes the distance A (100 m) + C (208.8 m), or the distance B (100 m) + C (208.8 m), for a total of 308.8 meters in both cases.

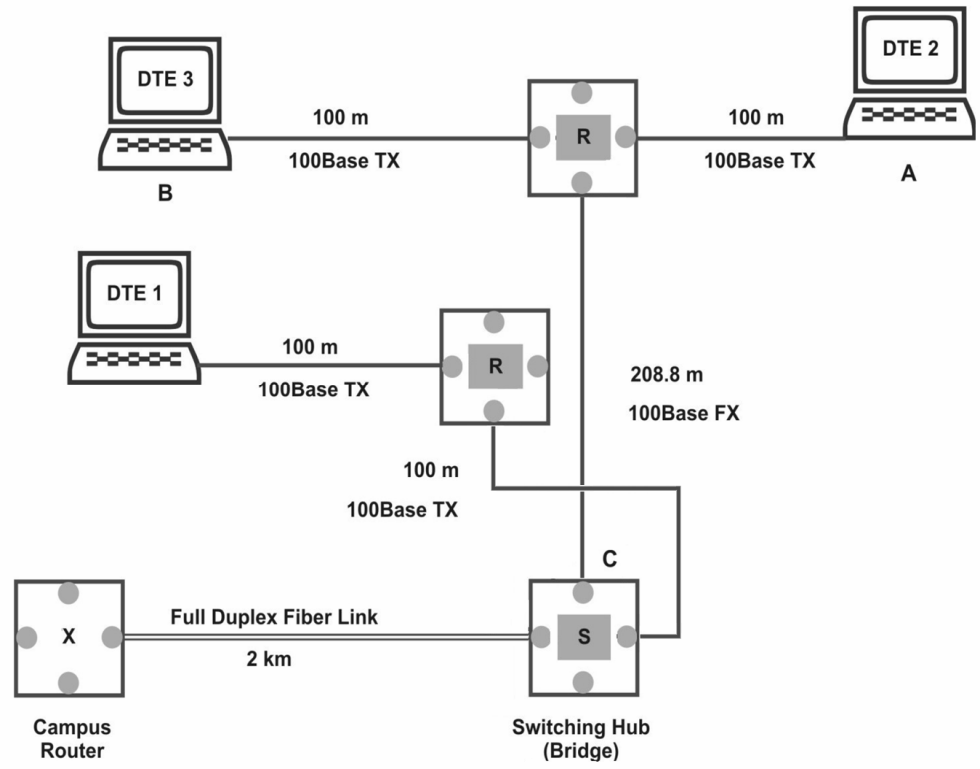


Figure 14.6
Mixed fiber and copper 100 Mbps configuration

A Class II repeater can be used to link the copper (TX) and fiber (FX) segments, since these segments both use the same encoding scheme.

14.6 Model 2 configuration guidelines for Fast Ethernet

Transmission system Model 2 for Fast Ethernet segments provides a set of calculations for verifying the signal-timing budget of more complex half-duplex Fast Ethernet LANs. These calculations are much simpler than the model 2 calculations used in the original 10 Mbps system, since the Fast Ethernet system uses only link segments.

The maximum diameter and the number of segments and repeaters in a half-duplex 100BaseT systems are limited by the round-trip signal timing required to ensure that the collision detect mechanism will work correctly. The Model 2 configuration calculations provide the information needed to verify the timing budget of a set of standard 100BaseT

segments and repeaters. This ensures that their combined signal delays fit within the timing budget required by the standard.

It may be noticed that these calculations appear to have a different round-trip timing budget than the timing budget provided in the 10 Mbps media system. This is because media segments in the Fast Ethernet system are based on different signaling systems than 10 Mbps Ethernet, and because the conversion of signals between the Ethernet interface and the media segments consumes a number of bit times.

It may also be noted that that there is no calculation for inter-frame gap shrinkage, unlike the one found in the 10 Mbps Model 2 calculations. That's because the maximum number of repeaters allowed in a Fast Ethernet system is limited, thus eliminating the risk of excessive inter-frame gap shrinkage.

14.6.1 Calculating round-trip delay time

Once the worst-case path has been found, the next step is to calculate the total round-trip delay. This can be accomplished by taking the sum of all the delay values for the individual segment in the path, plus the station delays and repeater delays. The calculation model in the standard provides a set of delay values measured in bit times, as shown in Table 14.5.

Component	Round-Trip Delay in Bit Times per m.	Maximum Round-Trip Delay in Bit Times per m.
Two TX/FX DTEs	N/A	100
Two T4 DTEs	N/A	138
One T4 & one TX/FX DTE (1)	N/A	127
Category 3 Cable	1.14	114 (100 m.)
Category 4 Cable	1.14	114 (100 m.)
Category 5 Cable	1.112	111.2 (100 m.)
Shielded Twisted Pair Cable	1.112	111.2 (100 m.)
Fiber Optic Cable	1.0	412 (412 m.)
Class I Repeater	N/A	140
Class II Repeater with all ports TX/FX	N/A	92
Class II Repeater with any T4 Port	N/A	67

Note (1): Worst Case values are used

Table 14.5
100BaseT component delays

It may be noted that the Round-Trip Delay in Bit Times per Meter only applies to the cable types in the table. The device types in the table (DTE, repeater) have only a maximum round-trip delay through each device listed.

To arrive at the round-trip delay value, multiply the length of the segment (in meters) times the round-trip delay in bit times per meter listed in the table for the segment type. This results in the round-trip delay in bit times for that segment. If the segment is at the maximum length one can use the maximum round-trip delay in bit times value listed in the table for that segment type. If not sure, of the segment length, one can also use the maximum length in your calculations just to be safe.

Once the segment delay values for each segment in the worst-case path are calculated, add the segment delay values together. One should also add the delay values for two stations (DTEs), and the delay for any repeaters in the path, to find the total path delay. The vendor may provide values for cable, station, and repeater timing, which one can use instead of the ones in the table.

To this total path delay value, add a safety margin of zero to four bit times, with four bit times of margin recommended in the standard. This helps account for unexpected delays, such as those caused by long patch cables between a wall jack in the office and the computer. If the result is less than or equal to 512 bit times, the path passes the test.

14.6.2 Calculating segment delay values

The segment delay value varies depending on the kind of segment used, and on the quality of cable in the segment if it is a copper segment. More accurate cable delay values may be provided by the manufacturer of the cable. If propagation delay of the cable being used is known, one can also look up the delay for that cable in Table 14.6 given below.

Speed relative to speed of light	Bit Time Per Meter		
	Nanoseconds per meter	100 Mbps Fast Ethernet	1000 Mbps Gigabit Ethernet
0.4	8.34	0.834	8.34
0.5	6.67	0.667	6.67
0.51	6.54	0.654	6.54
0.52	6.41	0.641	6.41
0.53	6.29	0.629	6.29
0.54	6.18	0.618	6.18
0.55	6.06	0.606	6.06
0.56	5.96	0.596	5.96
0.57	5.85	0.585	5.85
0.58	5.75	0.575	5.75
0.5852	5.70	0.57	5.7
0.59	5.65	0.565	5.65
0.60	5.56	0.556	5.56
0.61	5.47	0.547	5.47
0.62	5.38	0.538	5.38
0.63	5.29	0.529	5.29
0.64	5.21	0.521	5.21
0.65	5.13	0.513	5.13
0.654	5.10	0.51	5.10
0.66	5.05	0.505	5.05
0.666	5.01	0.501	5.01
0.67	4.98	0.498	4.98
0.68	4.91	0.491	4.91
0.69	4.83	0.483	4.83
0.70	4.77	0.477	4.77
0.80	4.17	0.417	4.17
0.90	3.71	0.371	3.71

Table 14.6
Conversion table for cable propagation times

Table 14.6 values are taken from the standard and provide a set of delay values in bit times per meter. These are listed in terms of the speed of signal propagation on the cable. The speed (propagation time) is provided as a percentage of the speed of light. This is also called the nominal velocity of propagation, or NVP, in vendor literature.

If one knows the NVP of the cable being used, then this table can provide the delay value in bit times per meter for that cable. A cable's total delay value can be calculated by multiplying the bit time/meter value by the length of the segment. The result of this calculation must be multiplied by two to get the total round-trip delay value for the segment. The only difference between 100 Mbps Fast Ethernet and 1000 Mbps Gigabit Ethernet in the above table is that the bit time in Fast Ethernet is ten times longer than the bit time in Gigabit Ethernet. For example, since the bit time is one nanosecond in Gigabit Ethernet, a propagation time of 8.34 nanoseconds per meter translates to 8.34 bit times.

14.6.3 Typical propagation values for cables

Typical propagation rates for Category 5 cable provided by two major vendors are given below. These values apply to both 100 Mbps Fast Ethernet and 1000 Mbps Gigabit Ethernet systems.

AT&T: Part No. 1061, Jacket: Non-Plenum, NVP= 70%
AT&T: Part No. 2061, Jacket: Plenum, NVP= 75%
Belden: Part No. 1583A, Jacket: Non-Plenum, NVP= 72%
Belden: Part No. 1585A, Jacket: Plenum, NVP= 75%

14.7 Model 1 configuration guidelines for Gigabit Ethernet

Transmission system Model 1 rules for half-duplex Gigabit Ethernet configuration are:

- The system is limited to a single repeater
- Segment lengths are limited to the lesser of 316 meters (1,036.7 feet) or the maximum transmission distance of the segment media type

The maximum length in terms of bit times for a single segment is 316 meters. However, any media signaling limitations, which reduce the maximum transmission distance of the link to below 316 meters, take precedence. Table 14.7 shows the maximum collision domain diameter for a Gigabit Ethernet system for the segment types shown. The maximum diameter of the collision domain is the longest distance between any two stations (DTEs) in the collision domain.

Configuration Configuration	Cat. 5 UTP	1000BASE CX	Fiber Optic 1000BASE -SX/LX	Cat.5 + Fiber Optic	1000BASE -CX and 1000BASE SX/LX
DTE-DTE Single Segment	100	25	316 (1)	N/A	N/A
One Repeater	200	50	220	210 (2)	220 (3)

Notes: (1) May be limited by the max. Transmission distance of the link
(2) Assumes 100 m. Of Cat. 5 unshielded twisted-pair (UTP) Cable and one fiber-optic link of 110 m.
(3) Assumes 25 m. Of 1000BASE-CX and one optical fiber link of 195 m.

Table 14.7
Model I maximum gigabit Ethernet collision domain in meters

The first row in table 14.7 shows the maximum lengths for a DTE-to-DTE (station-to-station) link. With no intervening repeater, the link may be up of a maximum of 100 m of copper, 25 m of 1000BaseCX cable, or 316 m of fiber optic cable. Some of the Gigabit Ethernet fiber optic links are limited to quite a bit less than 316 m due to signal transmission considerations. In those cases, one will not be able to reach the 316 m maximum allowed by the bit-timing budget of the system.

The row labeled *one repeater* provides the maximum collision domain diameter when using the single repeater allowed in a half-duplex Gigabit Ethernet system. This includes the case of all twisted-pair cable (200 m), all fiber optic cable (220 m) or a mix of fiber optic and copper cables.

14.8 Model 2 configuration guidelines for Gigabit Ethernet

Transmission system Model 2 for Gigabit Ethernet segments provides a set of calculations for verifying the signal-timing budget of more complex half-duplex Gigabit Ethernet LANs. These calculations are much simpler than the Model 2 calculations for either the 10 Mbps or 100 Mbps Ethernet systems, since Gigabit Ethernet only uses link segments and only allows one repeater. Therefore, the only calculation needed is the worst-case path delay value (PDV).

14.8.1 Calculating the path delay value

Once worst-case path has been determined, calculate the total round-trip delay value for the path, or PDV. The PDV is made up of the sum of segment delay values, repeater delay, DTE delays, and a safety margin.

The calculation model in the standard provides a set of delay values measured in bit times, as shown in Table 14.8. To calculate the round-trip delay value, multiply the length of the segment (in meters) times the round-trip delay in bit times per meter listed in the table for the segment type. This results in the round-trip delay in bit times for that segment.

Component	Round-trip Delay in Bit times per meter	Max. Round-trip in Bit Times
Two DTEs	N/A	864
Cat. 5 UTP Cable Segment	11.12	1112 (100 m)
Shielded Jumper Cable (CX)	10.10	253 (25 m)
Fiber Optic Cable Segment	10.10	1111 (110 m)
Repeater	N/A	976

Table 14.8
1000BaseT components delays

The result of this calculation is the round-trip delay in bit times for that segment. One can use the maximum round-trip delay in bit times value listed in the table for that segment type if the segment is at the maximum length. The max delay values can also be used if one is not sure of the segment length and want to use the maximum length in calculations just to be safe. To calculate cable delays, one can use the conversion values provided in right-hand column of Table 14.6.

To complete the PDV calculation, add the entire set of segment delay values together, along with the delay values for two stations (DTEs), and the delay for any repeaters in the path. Vendors may provide values for cable, station and repeater timing, which one can use instead of the ones in the tables provide here.

To this total path delay value, add a safety margin of from zero to 40 bit times, with 32 bit times of margin recommended in the standard. This helps account for any unexpected delays, such as those caused by extra long patch cords between a wall jack in the office and the computer. If the result is less than or equal to 4,096 bit times, the path passes the test.

14.9 Sample network configurations

A few sample network configurations will be undertaken to show how the configuration rules work in the real world. The 10 Mbps examples will be the most complex, since the 10 Mbps, system has the most complex set of segments and timing rules. After that, a single example for the 100 Mbps system will be discussed, since the configuration rules

are much simpler for Fast Ethernet. There is no need for a Gigabit Ethernet example, as the configuration rules are extremely simple, allowing for only a single repeater hub. In addition, all Gigabit Ethernet equipment being sold today only supports full-duplex mode, which means there are no half-duplex Gigabit Ethernet systems.

14.9.1 Simple 10 Mbps model 2 configurations

Figure 14.7 shows a network with three 10BaseFL segments connected to a fiber optic repeater. Two of the segments are 2 km (2,000 m) in length, and one is 1.5 km in length.

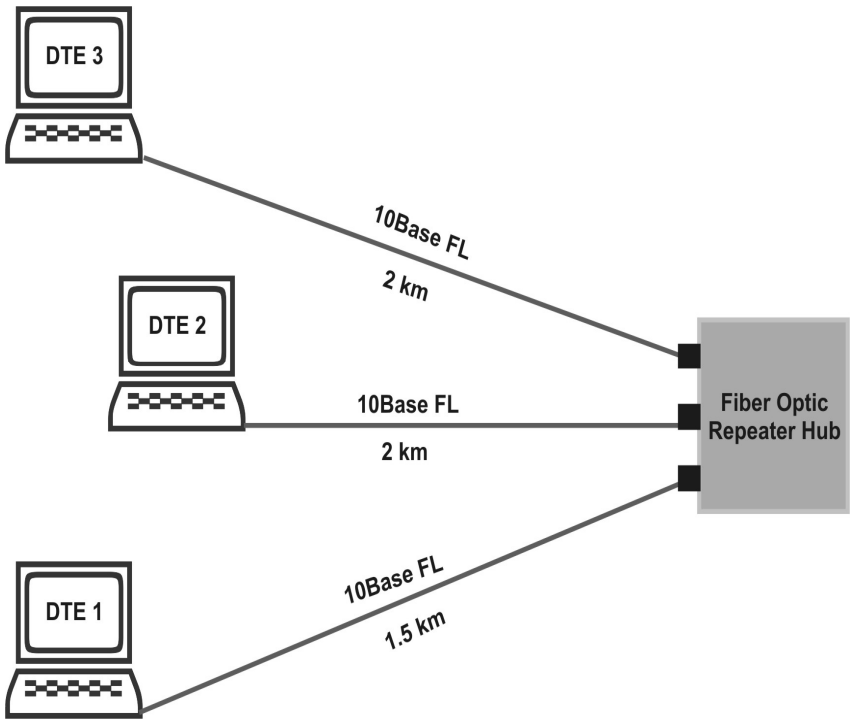


Figure 14.7
Simple 10 Mbps configuration example

This is a simple network, but its configuration is not described in the Model 1 configuration rules. The only way to verify its operation is to perform the Model 2 calculations. Figure 14.7 shows that that the worst-case delay path is between station 3 and station 2.

14.9.2 Round-trip delay

There are only two media segments in the worst-case path, and hence the network model for round-trip delay only has a left and right end segment. There are no middle segments to deal with. For the purposes of this example it shall be assumed that the fiber optic transceivers are connected directly to the stations and repeater. This eliminates the need to add extra bit times for transceiver cable length. Both segments in the worst-case path are the maximum allowable length. This means that using the 'max' values from Table 14.2 is the simplest way of calculating this.

According to the table, the max. left end segment delay value for a 2 km 10BaseFL link is 212.25 bit times. For the 2 km right end segment, the max. delay value is 356.5 bit times. Add them together, plus the five bit times margin recommended in the standard, and the total is: 573.75 bit times. This is less than the 575 maximum bit time budget

allowed for a 10 Mbps network, which means that the worst-case path is okay. All shorter paths will have smaller delay values; so all paths in this Ethernet system meet the requirements of the standard as far as round-trip timing is concerned. To complete the interframe gap calculation, one will need to compute the gap shrinkage in this network system.

14.9.3 Inter-frame gap shrinkage

Since there are only two segments, one only looks at a single transmitting end segment when calculating the inter-frame gap shrinkage. There are no middle segments to deal with, and the receive end segment does not count in the calculations for inter-frame gap. Since both segments are of the same media type, finding the worst-case value is easy. According to Table 14.3, the inter-frame gap value for the link segments is 10.5 bit times, and that becomes our total shrinkage value for this worst-case path. This is well under the 49 bit times of inter-frame shrinkage allowed for a 10 Mbps network.

As one can see, the example network meets both the round-trip delay requirements and the inter-frame shrinkage requirements, thus it qualifies as a valid network according to the Model 2 configuration method.

14.9.4 Complex 10 Mbps Model 2 configurations

The next example is more difficult, comprised of many different segment types, extra transceiver cables, etc. All these extra bits and pieces also make the example more complicated to explain, although the basic process of looking up the bit times and adding them together is still quite simple.

For this complex configuration example, refer back to figure 14.2 earlier in the chapter. This figure shows one possible maximum-length system using four repeaters and five segments. According to the Model 1 rule-based configuration method, it has already been seen that this network complies with the standards. To check that, one has to check this network again, this time using the calculation method provided for model 2.

First step is finding the worst-case path in the sample network. By examination, one can see that the path between stations 1 and 2 in figure 14.2 is the maximum delay path. It contains the largest number of segments and repeaters in the path between any two stations in the network. Next, one makes a network model out of the worst-case path. Start the process by arbitrarily designating the thin Ethernet end segment as the left end segment. That leaves three middle segments composed of a 10Base5 segment and two fiber optic link segments, and a right end segment comprised of a 10BaseT link segment.

Next, one has to calculate the segment delay value for the 10Base2 left end segment. This can be accomplished by adding the left end base value for 10Base2 coax (11.75) to the product of the round-trip delay times the length in meters ($185 \times 0.1026 = 18.981$) results in a total segment delay value of 30.731 for the thin coax segment. However, since 185 m is the maximum segment length allowed for 10Base2 segments, one can simply look up the max left hand segment value from table 14.2, which, not surprisingly, is 30.731. The 10Base2 thin Ethernet segment is shown attached directly to the DTE and the repeater, and there is no transceiver cable in use. Therefore, one does not have to add any excess AUI cable length timing to the value for this segment.

14.9.5 Calculating separate left end values

Since the left and right end segments in worst-case path are different media types, the standard notes that one needs to do the path delay calculations twice. First, calculate the total path delay using the 10Base2 segment as the left end segment and the 10BaseT segment as the right end. Then swap their places and make the calculation again, using the 10Base-T segment as the left end segment this time and the 10Base2 segment as the right end segment. The largest value that results from the two calculations is the value one must use in verifying the network.

14.9.6 AUI delay value

The segment delay values provided in the table include allowances for a transceiver cable (AUI) of up to two meters in length at each end of the segment. This allowance helps takes care of any timing delays that may occur due to wires inside the ports of a repeater.

Media systems with external transceivers connected with transceiver cables require that we account for the timing delay in these transceiver cables. One can find out how long the transceiver cables are, and use that length multiplied by the round-trip delay per meter to develop an extra transceiver cable delay time, which is then added to the total path delay calculation. If one is not sure how long, the transceiver cables are in your network, one can use the maximum delay shown for a transceiver cable, which is 4.88 for all segment locations, left end, middle, or right end.

14.9.7 Calculating middle segment values

In the worst-case path for the network in figure 14.2, there are three middle segments composed of a maximum length 10Base5 segment, and two 500 m long 10BaseFL fiber optic segments. By looking in table 14.2 under the Middle Segments column, one finds that the 10Base5 segment has a Max delay value of 89.8.

Note that the repeaters are connected to the 10Base5 segment with transceiver cables and outboard MAUs. That means one needs to add the delay for two transceiver cables. Let's assume that one does not know how long the transceiver cables are. Therefore, one has to use the value for two maximum-length transceiver cables in the segment, one at each connection to a repeater. That gives a transceiver cable delay of 9.76 to add to the total path delay.

One can calculate the segment delay value for the 10BaseFL middle segments by multiplying the 500-meter length of each segment by the RT Delay/meter value, which is 0.1, giving a result of 50. Add 50 to the middle segment base value for a 10Base-FL segment, which is 33.5, for a total segment delay of 83.5.

Although it's not shown in Figure 14.2, fiber optic links often use outboard fiber optic transceivers and transceiver cables to make a connection to a station. Just to make things a little harder, let it be assumed that one uses two transceiver cables, each being 25 m in length, to make a connection from the repeaters to outboard fiber optic transceivers on the 10Base-FL segments. That gives a total of 50 m of transceiver cable on each 10BaseFL segment. Since now there are two such middle segments, one can represent the total transceiver cable length for both segments by adding 9.76 extra bit times to the total path delay.

14.9.8 Completing the round-trip timing calculation

Here our calculations started with the 10Base2 segment assigned to the left end segment, which leaves us with a 10BaseT right end segment. This segment is 100 m long, which is the length provided in the 'Max' column for a 10Base-T segment. Depending on the cable

quality, a 10BaseT segment can be longer than 100 m, but we'll assume that the link in our example is 100 m. That makes the Max value for the 10BaseT right end segment 176.3. Adding all the segment delay values together, one gets the result shown in table 14.9.

Link	Media	Bit-Time Delay
Left End	10BASE-T	30.731
Mid-Segment	10BASE5	89.8
Mid-Segment	10BASE-FL	83.5
Mid-Segment	10BASE-FL	83.5
Right End	10Base-T	176.3
Excess Length		
AUI	Quan. Four	19.52
Total Path Delay		483.351

Table 14.9
Round-trip path delay with 10Base2 left end segments

To complete the process, one needs to perform a second set of calculations with the left and right segments swapped. In this case, the left end becomes a maximum length 10BaseT segment, with a value of 26.55, and the right end becomes a maximum length 10Base2 segments with a value of 188.48. Note that the excess length AUI values do not change. As shown in Table 14.2, the bit time values for AUI cables are the same no matter where the cables are used. Adding the bit time values again, one gets the following result in Table 14.10.

Link	Media	Bit-Time Delay
Left End	10BASE-T	26.55
Mid-Segment	10BASE5	89.8
Mid-Segment	10BASE-FL	83.5
Mid-Segment	10BASE-FL	83.5
Right End	10Base-2	188.48
Excess Length		
AUI	Quan. Four	19.52
Total Path Delay		491.35

Table 14.10
Round-trip path delay with 10BaseT left end segments

The second set of calculations shown in table 14.10 produced a larger value than the total from Table 14.9. According to the standard, one must use this value for the worst-case round-trip delay for this Ethernet. The standard also recommends adding a margin of five bit times to form the total path delay value. One is allowed to add anywhere from zero to five bits margin, but five bit times is recommended.

Adding five bit times for margin brings us up to a total delay value of 496.35 bit times, which is less than the maximum of 575 bit times allowed by the standard. Therefore, the complex network is qualified in terms of the worst-case round-trip timing delay. All shorter paths will have smaller delay values, which means that all paths in the Ethernet system shown in figure 14.2 meet the requirements of the standard as far as round-trip timing is concerned.

14.9.9 Inter-frame gap shrinkage

Evaluation of the complex network example shown in figure 14.2 by calculating the worst-case interframe gap shrinkage for that network is now considered. This is done by evaluating the same worst-case path we used in the path delay calculations. However, for the purposes of calculating gap shrinkage evaluate only the transmitting and mid-segments.

Once again one starts by applying a network model to the worst-case path, in this case the network model for inter-frame gap shrinkage. To calculate inter-frame gap shrinkage, the transmitting segment should be assigned the end segment in the worst-case path of your network system that has the largest shrinkage value. As shown in table 14.3, the coax media segment has the largest value, so for the purposes of evaluating our sample network we will assign the 10Base2 thin coax segment to the role of transmitting end segment. That leaves us with middle segments consisting of one coax and two link segments, and a 10Base-T receive end segment which is simply ignored. The totals are shown below:

PVV for transmitting End Coax	=	16
PVV for Mid-Segment Coax	=	11
PVV for Mid-Segment Link	=	8
PVV for Mid-Segment Link	=	8
Total PVV	**=**	**43**

It can be seen that, the total path variability value for our sample network equals 43. This is less than the 49-bit time maximum allowed in the standard, which means that this network meets the requirements for inter-frame gap shrinkage.

14.9.10 100 Mbps model 2 configuration

For this example, refer back to figure 14.5, which shows one possible maximum length network. As we've seen, the Model 1 rule-based configuration method shows that this system is okay. To check that, we'll evaluate the same system using the calculation method provided in Model 2.

14.9.11 Worst-case path

In the sample network, the two longest paths are between Station 1 and Station 2, and between Station 1 and the switching hub. Signals from Station 1 must go through two repeaters and two 100 m segments, as well as a 5 m inter-repeater segment to reach either Station 2 or the switching hub. As far as the configuration guidelines are concerned, the switching hub is considered as another station.

Both of these paths in the network include the same segment lengths and number of repeaters, so we will evaluate one of them as the worst-case path. Let's assume that all three segments are 100Base-TX segments, based on Category 5 cables. By looking up the Max Delay value in Table 14.5 for a Category 5 segment, we find 111.2 bit times.

The delay of a 5 m inter-repeater segment can be found by multiplying the round-trip Delay per Meter for Category 5 cable (1.112) times the length of the segment in meters (5). This results in 5.56 bit times for the round-trip delay on that segment. Now that we know the segment round-trip delay values, we can complete the evaluation by following the steps for calculating the total round-trip delay for the worst-case path.

To calculate the total round-trip delay, we use the delay times for stations and repeaters found in table 14.5. As shown in below, the total round-trip path delay value for the sample network is 511.96 bit times when using Category 5 cable. This is less than the maximum of 512 bit times, which means that the network passes the test for round-trip delay.

Delay for two TX DTEs	=	100
Delay for 100 m. Cat. 5 segment	=	111.2
Delay for 100 m. Cat. 5 segment	=	111.2
Delay for 5 meter Cat. 5 segment	=	5.56
Delay for Class II repeater	=	92
Delay for Class II repeater	=	92
Total Round-Trip Delay	**=**	**511.96 bits**

It may be noted that there is no margin of up to 4 bit times provided in this calculation. There are no spare bit times to use for margin, because the bit time values shown in Table 14.5 are all worst-case maximums. This table provides worst-case values that you can use if you don't know what the actual cable bit times, repeater timing, or station-timing values are.

For a more realistic look, let's see what happens if we work this example again, using actual cable specifications provided by a vendor. Assume that the Category 5 cable is AT&T type 1061 cable, a non-plenum cable that has an NVP of 70 percent as shown below. If we look up that speed in table 14.6, we find that a cable with a speed of 0.7 is rated at 0.477 bit times per meter. The round-trip bit time will be twice that, or 0.954 bit times. Therefore, timing for 100 m will be 95.4 bit times, and for 5m it will be 4.77 bit times. How things add up using these different cable values is shown below:

Delay for Two TX DTEs	=	100
Delay for 100 m. Cat. 5 Segment	=	95.4
Delay for 100 m. Cat. 5 segment	=	95.4
Delay for 5 m. Cat. 5 Segment	=	4.77
Delay for Class II repeater	=	92
Delay for Class II repeater	=	92
Total Delay	**=**	**483.57 bits**

When real-world cable values are used instead of the worst-case default values in table 7.5, there is enough timing left to provide for 4 bit times of margin. This meets the goal of 512 bit times, with bit times to spare.

14.9.12 Working with bit time values

Some vendors note that their repeater delay values are smaller than the values listed in Table 14.5, which will make it easier to meet the 512-bit time maximum. While these extra bit times could theoretically be used to provide an inter-repeater segment longer than five meters, this approach could lead to problems.

While providing a longer inter-repeater link might appear to be a useful feature, one should consider what would happen if that vendor's repeater failed and had to be replaced with another vendor's repeater whose delay time was larger. If that were to occur, then the worst-case path in your network might end up with excessive delays due to the bit times consumed by the longer inter-repeater segment you had implemented. One can

avoid this problem by designing the network conservatively and not pushing things to the edge of the timing budget.

One can use more than one Class I or two Class II repeaters in a given collision domain. This can be done if the segment lengths are kept short enough to provide the extra bit time budget required by the repeaters. However, the majority of network installations are based on building cabling systems with 100 m segment lengths (typically implemented as 90 m 'in the walls' and 10 m for patch cables). A network design with so many repeaters that the segments must be kept very short to meet the timing specifications is not going to be useful in most situations.

15

Industrial Ethernet

Objectives

When you have completed study of this chapter, you will be able to:

- Describe the concept of Industrial Ethernet with specific reference to:
  - Connectors and cabling
  - Packaging
  - Determinism
  - Power on the bus
  - Redundancy

15.1 Introduction

Early Ethernet was not entirely suitable for control functions as it was primarily developed for office–type environments. The Ethernet technology has, however, made rapid advances over the past few years. It has gained such widespread acceptance in Industry that it is becoming the *de facto* field bus technology. An indication of this trend is the inclusion of Ethernet as the levels 1 and 2 infrastructure for Modbus/TCP (Schneider), Ethernet/IP (Rockwell Automation and ODVA), ProfiNet (Profibus) and Foundation Fieldbus HSE.

The following sections will deal with problems related to early Ethernet, and how they have been addressed in subsequent upgrades.

15.2 Connectors and cabling

Earlier industrial Ethernet systems such as the first–generation Siemens SimaticNet (Sinec–H1) were based on the 10Base5 configuration, and thus the connectors involved include the screw–type N–connectors and the D–type connectors, which are both fairly rugged. The heavy–gauge twin–screen (braided) RG–8 coaxial cable is also quite impervious to electrostatic interference.

Most modern industrial Ethernet systems are, however, based on a 10BaseT/100BaseTX configuration and thus have to contend with RJ–45 connectors and Cat5-type UTP cable. The RJ-45 connectors could be problematic. They are everything

but rugged and are suspect when subjected to great temperature extremes, contact with oils and other fluids, dirt, UV radiation, EMI as well as shock, vibration and mechanical loading.

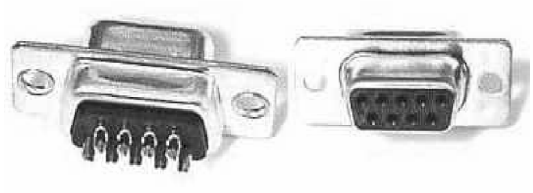


Figure 15.1
D-type connectors

As in interim measure, some manufacturers have started using D–type (known also as DB or D–Subminiature) connectors. These are mechanically quite rugged, but are neither waterproof nor dustproof either. They can therefore be used only in IP20 rated environments, i.e. within enclosures in a plant.

Ethernet I/O devices have become part of modern control systems. Ethernet makes it possible to use a variety of TCP/IP protocols to communicate with decentralized components virtually to sensor level. As a result, Ethernet is now installed in areas that were always the domain of traditional Fieldbus systems. These areas demand IP67 class protection against dirt, dust and fluids. This requires that suitable connector technology meeting IP67 standards have to be defined for transmission speeds up to 100Mbps. Two solutions to this problem are emerging. The one is a modified RJ-45 connector while the other is an M12 (micro-style) connector.

Figure 15.2
Modified RJ-45 connector(RJ-LNxx)
Courtesy: AMC Inc

Standardization groups are addressing the problem both nationally and internationally. User organizations such as IAONA (Industrial Automation Open Networking Alliance), Profibus user organization and ODVA (Open DeviceNet Vendor Association) are also trying to define standards within their organizations.

Network connectors for IP67 are not easy to implement. Three different approaches can be found. First, there is the RJ-45 connector sealed within an IP67 housing. Then there is an M12 (micro-style) connector with either four or eight pins. A third option is a hybrid connector based on RJ-45 technology with additional contacts for power distribution. Two of the so-called sealed RJ-45 connectors are in the process of standardization. Initially the 4-pin M12 version will be standardized in Europe. Connectors will be tested against existing standards (e.g., VDE 0110) and provide the corresponding IP67 class

protection at 100 Mbps. In the US, the ODVA has standardized a sealed version of the RJ-45 connector for use with Ethernet/IP.

For the use of the standard M12 in Ethernet systems is covered in standard EN 61076-2-101. The transmission performance of 4-pin M12 connector for Ethernet up to 100Mbps is comparable, if not better than standardized office grade Ethernet products. In office environments, four-pair UTP cabling is common. For industrial applications two-pair cables are less expensive and easier to handle. Apart from installation difficulties, 8-pin M12 connectors may not meet all the electrical requirements described in EN 50173 or EIA/TIA-568B.

Figure 15.3
M12 connector (EtherMate)
Courtesy: Lumberg Inc.

Typical M12 connectors for Ethernet are of the 8–pole variety, with threaded connectors. They can accept various types of Cat5/5e twisted pair wiring such as braided or shielded wire (solid or stranded), and offer excellent protection against moisture, dust, corrosion, EMI, RFI, mechanical vibration and shock, UV radiation, and extreme temperatures (–40ºC to 75ºC).

As far as the media is concerned, several manufacturers are producing Cat5 and Cat5e wiring systems using braided or shielded twisted pairs. An example an integrated approach to industrial Ethernet cabling is Lumberg's etherMATE™ system, which includes both the cabling and an M12 connector system.

Some vendors also use 2-pair Cat5+ cable, which has an outer diameter similar to Cat5 cable, but have wire with a thicker cross-section. This, together with special punch-down connectors, greatly simplifies installation.

15.3 Packaging

Commercial Ethernet equipment (hubs, switches, etc) are only rated to IP20, in other words, they have to be deployed in enclosures for industrial applications. They are also typically rated to 40 ^{o}C. Additional issues relate to vibration and power supplies.

Some manufacturers are now offering industrially-hardened switches with DIN-rail mounts, IP67 (waterproof and dustproof) rating, industrial temperature rating (60^{o}C), DIN-rail mounts and redundant power supplies

Figure 15.4
Industrial grade switch
Courtesy: Siemens

15.4 Deterministic versus stochastic operation

One of the most common complaints with early Ethernet was that it uses CSMA/CD (a probabilistic or stochastic method) as opposed to other automation technologies such as Fieldbus that use deterministic access methods such as token passing or the publisher–subscriber model. CSMA/CD essentially means that it is impossible to guarantee delivery of a possibly critical message within a certain time. This could be due either to congestion on the network (often due to other less critical traffic) or to collisions with other frames. In office applications there is not much difference between 5 seconds and 500 milliseconds, but in industrial applications a millisecond counts. Industrial processes often require scans in a 5 to 20–millisecond range, and some demanding processes could even require 2 to 5 milliseconds. On 10BaseT Ethernet, for example, the access time on a moderately loaded 100–station network could range from 10 to 100mS, which is acceptable for office applications but not for processes.

There is a myth doing the rounds that Ethernet will experience an exponential growth in collisions and traffic delays- resulting ultimately in a collapse of the network- if loaded above 40%. In fact, Ethernet delays for 10 Mbps Ethernet are linear and can be consistently maintained under 2 ms for a lightly loaded network (<10%) and 30 ms for a heavily loaded network (<50%).

It is therefore important for the loading or traffic needs to be carefully analyzed to ensure that the network is not overwhelmed at critical or peak operational times. While a typical utilization factor on a commercial Ethernet LAN of 30% is acceptable, figures of less than 10% utilization on an industrial Ethernet LAN are required. Most industrial networks run at 3 or 4% utilization with fairly large number of I/O points being transferred across the system.

The advent of Fast and Gigabit Ethernet, switching hubs, IEEE 802.3Q VLAN technology, IEEE 802.3p traffic prioritization and full duplex operation has resulted in

very deterministic Ethernet operation and has effectively put this concern to rest for most applications.

15.5 Size and overhead of Ethernet frame

Data link encoding efficiency is another problem, with the Ethernet frames taking up far more space than for an equivalent fieldbus frame. If the TCP/IP protocol is used in addition to the Ethernet frame, the overhead increases dramatically. The efficiency of the overall system is, however, more complex than simply the number of bytes on the transmitting cable and issues such as raw speed on the cable and the overall traffic need to be examined carefully. For example, if 2 bytes of data from an instrument had to be packaged in a 60 byte message (because of TCP/IP and Ethernet protocols being used) this would result in an enormous overhead compared to a fieldbus protocol. However, if the communications link were running at 100 Mbps or 1 Gbps with full duplex communications, this would put a different light on the problem and make the overhead issue almost irrelevant.

15.6 Noise and interference

Due to higher electrical noise near to the industrial LANs some form of electrical shielding and protection is useful to minimize errors in the communication. A good choice of cable is fiber–optic (or sometimes coaxial cable). Twisted pair can be used but care should be taken to route the cables far away from any potential sources of noise. If twisted pair cable is selected; a good decision is to use screened twisted pair cable (ScTP) rather than the standard UTP.

It should be noted here that Ethernet–based networks are installed in a wide variety of systems and rarely have problems reported due to reliability issues. The use of fiber ensures that there are minimal problems due to earth loops or electrical noise and interference.

15.7 Partitioning of the network

It is very important that the industrial network operates separately from that of the commercial network, as speed of response and real time operation are often critical attributes of an industrial network. An office type network may not have the same response requirements. In addition, security is another concern where the industrial network is split off from the commercial networks so any problems in the commercial network will not affect the industrial side.

Industrial networks are also often partitioned into individual sub–networks for reasons of security and speed of response, by means of bridges and switches.

In order to reduce network traffic, some PLC manufacturers use exception reporting. This requires only changes in the various digital and analog parameters to be transmitted on the network. For example, if a digital point changes state (from ‘on’ to ‘off’ or ‘off’ to ‘on’), this change would be reported. Similarly, an analog value could have associated with it a specified change of span before reporting the new analog value to the master station.

15.8 Switching technology

Both the repeating hub and bridge technologies are being superseded by switching technology. This allows traffic between two nodes on the network to be directly

connected in a full duplex fashion. The nodes are connected through a switch with extremely low levels of latency. Furthermore, the switch is capable of handling all the ports communicating simultaneously with one another without any collisions. This means that the overall speed of the switch backplane is considerably greater than the sum of the speeds of the individual Ethernet ports.

Most switches operate at the data link layer and are also referred to as switching hubs and layer 2 switches. Some switches can interpret the network layer addresses (e.g. the IP address) and make routing decisions on that. These are known as layer 3 switches.

Advanced switches can be configured to support virtual LANs. This allows the user to configure a switch so that all the ports on the switch are subdivided into predefined groups. These groups of ports are referred to as virtual LANs (VLANs)– a concept that is very useful for industrial networks. Only switch ports allocated to the same VLAN can communicate with each other.

Switches do have performance limitations that could affect a critical industrial application. If there is traffic on a switch from multiple input ports aimed at one particular output port, the switch may drop some of the packets. Depending on the vendor implementation, it may force a collision back to the transmitting device so that the transmitting node backs off long enough for the congestion to clear. This means that the transmitting node does not have a guarantee on the transmission between two nodes – something that could impact a critical industrial application.

In addition, although switches do not create separate broadcast domains, each virtual LAN effectively forms one (if this is enabled on the switch). An Ethernet broadcast message received on one port is retransmitted onto all ports in the VLAN. Hence a switch will not eliminate the problem of excessive broadcast traffic that can cause severe performance degradation in the operation of the network. TCP/IP uses the Ethernet broadcast frame to obtain MAC addresses and hence broadcasts are fairly prevalent here.

A problem with a switched network is that duplicate paths between two given nodes could cause a frame to be passed around and around the 'ring' caused by the two alternative paths. This possibility is eliminated by the 'Spanning Tree' algorithm, the IEEE802.1d standard for layer 2 recovery. However, this method is quite slow and could take from 2 to 5 seconds to detect and bypass a path failure and could leave all networked devices isolated during the process. This is obviously unacceptable for industrial applications.

A solution to the problem is to connect the switches in a dual redundant ring topology, using copper or fiber. This poses a new problem, as an Ethernet broadcast message will be sent around the loop indefinitely. Several vendors now market switches with added redundancy management capabilities. One of the switches in the system acts as a redundancy manager and allows a physical 200 Mbps ring to be created, by terminating both ends of the traditional Ethernet bus in itself. Although the bus is now looped back to itself, the redundancy manager logically breaks the loop, preventing broadcast messages from wreaking havoc. Logically, the redundancy manager behaves like two nodes, sitting back to back, transmitting and receiving messages to the other around the ring using 802.1p/Q frames. This creates a deterministic path through any 802.1p/Q compliant switches (up to 50) in the ring, which results in a real time 'awareness' of the state of the ring. Up to 50 switches can be interconnected in this way, using 3 km fiber connections. As a result, a dual fiber-optic ring with a circumference of 150 km can be created!

When a network failure is detected (i.e. the loop is broken), the redundancy manager interconnects the two segments attached to it, thereby restoring the loop. This takes place in between 20 and 500 milliseconds, depending on the size of the ring.

15.9 Power on the bus

Industry often expects device power to be delivered over the same wires as those used for communicating with the devices. Examples of such systems are DeviceNet and Foundation Fieldbus. This is, however, not an absolute necessity as the power can be delivered separately. Profibus DP, for example, does not provide this feature yet it is one of the leading Fieldbuses.

Ethernet does, however, provide the ability to deliver some power. The IEEE 802.3af standard was ratified by the IEEE Standards Board in June 2003 and allows a source device (a hub or a switch) to supply a minimum of 300 mA at 48 Volts DC to the field device. This is in the same range as FF and DeviceNet.

The standard allows for two alternatives, namely the transmission of power over the signal pairs (1/2 and 3/6) or the transmission of power over the unused pairs (4/5 and 7/8). Intrinsic safety issues still need addressing.

15.10 Fast and Gigabit Ethernet

The recent developments in Ethernet technology are making it even more relevant in the industrial market. Fast Ethernet as defined in the IEEE specification 802.3u is essentially Ethernet running at 100 Mbps. The same frame structure, addressing scheme and CSMA/CD access method are used as with the 10 Mbps standard. In addition, Fast Ethernet can also operate in full duplex mode as opposed to CSMA/CD, which means that there are no collisions. Fast Ethernet operates at ten times the speed of that of standard IEEE 802.3 Ethernet. Video and audio applications can enjoy substantial improvements in performance using Fast Ethernet. Smart instruments that require far smaller frame sizes will not see such an improvement in performance. One area however, where there may be significant throughput improvements, is in the area of collision recovery. The back-off times for 100 Mbps Ethernet is a tenth that of standard Ethernet. Hence a heavily loaded network with considerable individual messages and nodes would see performance improvements. If loading and collisions are not really an issue on the slower 10 Mbps network, then there will not be many tangible improvements in the higher LAN speed of 100 Mbps.

Note that with the auto–negotiation feature built into standard switches and many Ethernet cards, the device can operate at either the 10 or 100 Mbps speeds. In addition, the Cat5 wiring installed for 10BaseT Ethernet is adequate for the 100 Mbps standard as well.

Gigabit Ethernet is another technology that could be used to connect instruments and PLCs. However its speed would probably not be fully exploited by the instruments for the reasons indicated above.

15.11 TCP/IP and industrial systems

The TCP/IP suite of protocols provides for a common open protocol. In combination with Ethernet this can be considered to be a truly open standard available to all users and vendors. However, there are some problems at the application layer area. Although TCP/IP implements four layers which are all open (network interface, internet, transport and application layers), most industrial vendors still implement their own specific application layer. Hence equipment from different vendors can coexists on the factory shop floor but cannot inter–operate. Protocols such as MMS (manufacturing messaging service) have been promoted as truly ‘open’ automation application layer protocols but with limited acceptance to date.

15.12 Industrial Ethernet architectures for high availability

There are several key technology areas involved in the design of Ethernet based industrial automation architecture. These include available switching technologies, quality of service (QoS) issues, the integration of existing (legacy) field buses, sensor bus integration, high availability and resiliency, security issues, long distance communication and network management– to name but a few.

For high availability systems a single network interface represents a single point of failure (SPOF) that can bring the system down. There are several approaches that can be used on their own or in combination, depending on the amount of resilience required (and hence the cost). The cost of the additional investment in the system has to be weighed against the costs of any downtime.

For a start, the network topology could be changed to represent a switched ring. Since the Ethernet architecture allows an array of switches but not a ring, this setup necessitates the use of a special controlling switch (redundancy manager) which controls the ring and protects the system against a single failure on the network. It does not, however, guard against a failure of the network interface on one of the network devices. The redundancy manager is basically a switch that "divides" itself in two internally, resulting in two halves that periodically check each other by sending high-priority messages around the loop to each other.

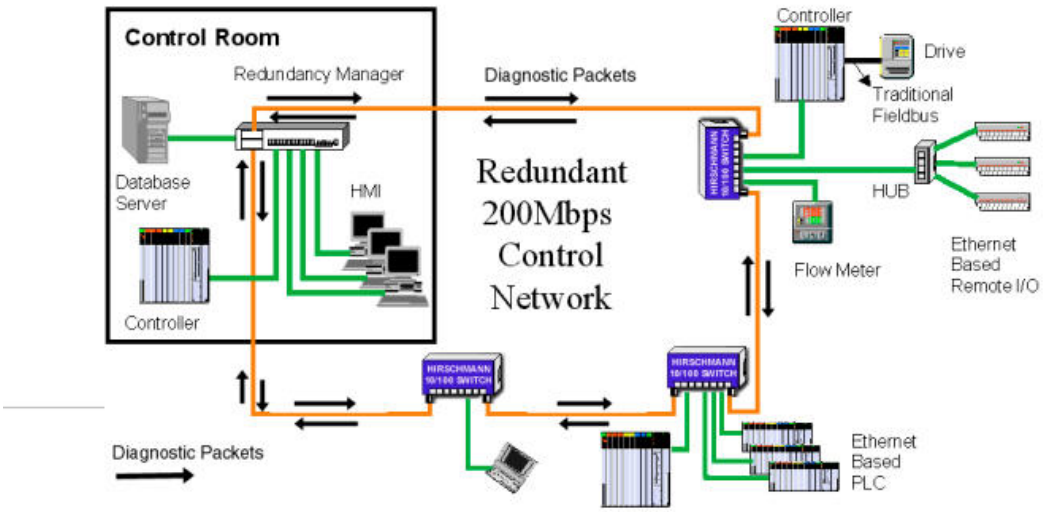


Figure 15.5
Redundant switch ring
Courtesy: Hirschmann

If a failure occurs anywhere on the ring, the redundancy manager becomes aware of it through a failure to communicate with itself, and "heals" itself.

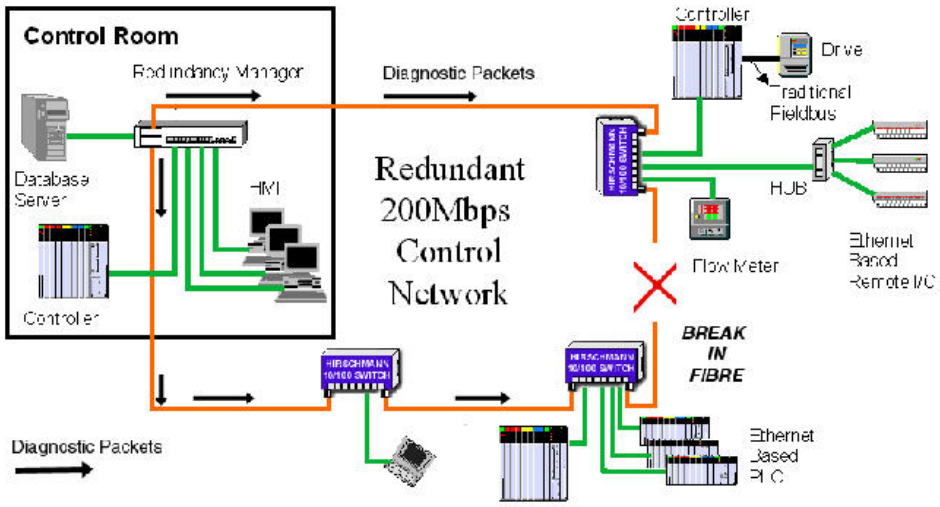


Figure 15.6
Redundant switch ring with failure
Courtesy: Hirschmann

The next level of resiliency would necessitate two network interfaces on the controller (that is, changing it to a dual–homed system), each one connected to a different switch. This setup would be able to tolerate both a single network failure and a network interface failure.

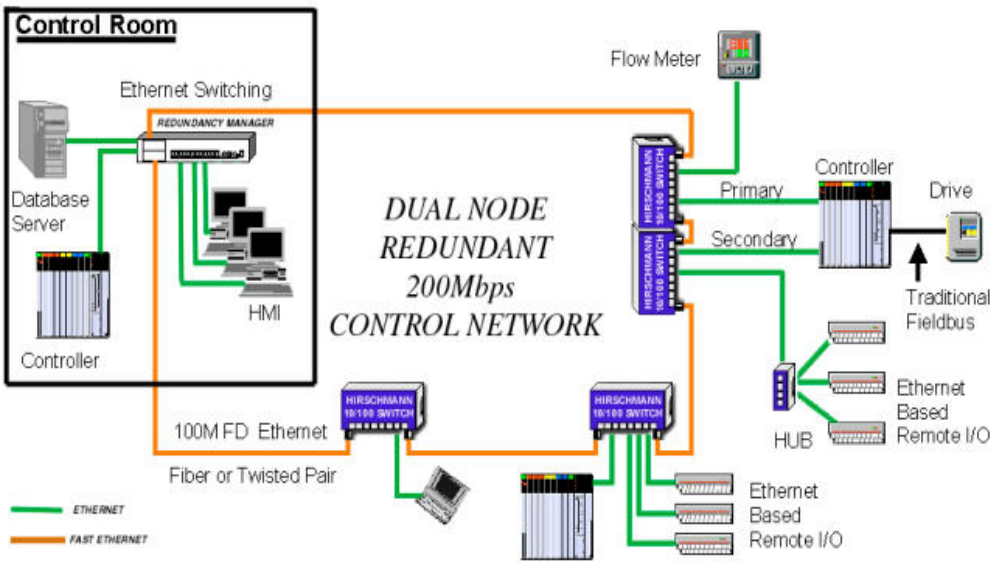


Figure 15.7
Redundant switch ring with dual access to controller
Courtesy: Hirschmann

Ultimately one could protect the system against a total failure by duplicating the switched ring, connecting each port of the dual–homed system to a different ring.

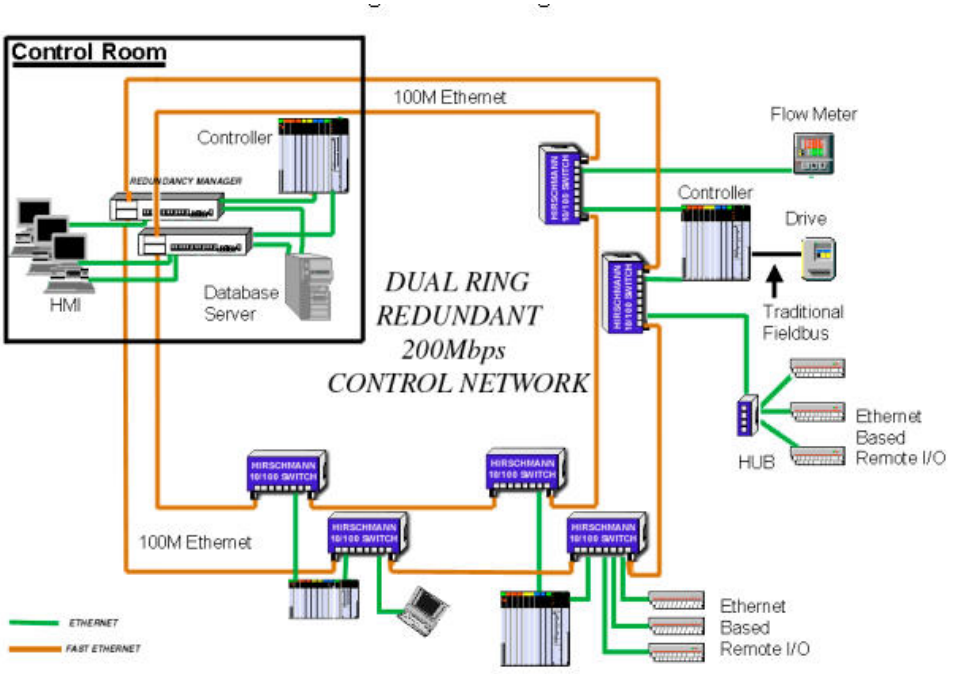


Figure 15.8
Dual redundant switch ring
Courtesy: Hirschmann

Other factors supporting a high degree of resilience would include hot swappable switches and NICs, dual redundant power supplies and on–line diagnostic software.

16

Troubleshooting Ethernet

Objectives

When you have completed study of this chapter, you will be able to identify, troubleshoot and fix problems such as:

- Thin and thick coax cable and connectors
- UTP cabling
- Incorrect media selection
- Jabber
- Too many nodes
- Excessive broadcasting
- Bad frames
- Faulty auto–negotiation
- 10/100 Mbps mismatch
- Full-/half-duplex mismatch
- Faulty hubs
- Switched networks
- Loading

16.1 Introduction

This section deals with addressing common faults on Ethernet networks. Ethernet encompasses layers 1 and 2, namely the physical and data Link layers of the OSI model. This is equivalent to the bottom layer (the network interface layer) in the ARPA model. This section will focus on those layers only, as well as on the actual medium over which the communication takes place.

16.2 Common problems and faults

Ethernet hardware is fairly simple and robust, and once a network has been commissioned, providing the cabling has been done professionally and certified, the network should be fairly trouble–free.

Most problems will be experienced at the commissioning phase, and could theoretically be attributed to the cabling, the LAN devices (such as hubs and switches), the network interface cards (NICs) or the protocol stack configuration on the hosts.

The wiring system should be installed and commissioned by a certified installer. The suppliers of high–speed Ethernet cabling systems, such as ITT, will in any case not guarantee their wiring if not installed by an installer certified by them. This effectively rules out wiring problems for new installations, although old installations could be suspect.

If the LAN devices such as hubs and switches are from reputable vendors, it is highly unlikely that they will malfunction in the beginning. Care should nevertheless be taken to ensure that intelligent (managed) hubs and switches are correctly set up.

This applies to NICs also. NICs rarely fail and nine times out of ten the problem lies with a faulty setup or incorrect driver installation or an incorrect configuration of the higher level protocols such as IP.

16.3 Tools of the trade

In addition to fundamental understanding of the technologies involved, spending sufficient time, employing a pair of eyes and patience, one can be successful in isolating Ethernet related problems with the help of the following tools:

16.3.1 Multimeters

A simple multimeter can be used to check for continuity and cable resistance, as will be explained in this section.

16.3.2 Handheld cable testers

There are many versions available in the market, ranging from simple devices that basically check for wiring continuity to sophisticated devices that comply with all the prerequisites for 1000BaseT wiring infrastructure tests. Testers are available from several vendors such as MicroTest, Fluke, and Scope.

Figure 16.1
Cable tester

16.3.3 Fiber optic cable testers

Fiber optic testers are simpler than UDP testers, since they basically only have to measure continuity and attenuation loss. Some UDP testers can be turned into fiber optic testers by purchasing an attachment that fits onto the existing tester. For more complex problems such as finding the location of a damaged section on a fiber optic cable, an alternative is to use a proper optical time domain reflectometer (OTDR) but these are expensive instruments and it is often cheaper to employ the services of a professional wire installer (with his own OTDR) if this is required.

16.3.4 Traffic generators

A traffic generator is a device that can generate a pre–programmed data pattern on the network. Although they are not used for fault finding strictly speaking, they can be used to predict network behavior due to increased traffic, for example, when planning network changes or upgrades. Traffic generators can be stand–alone devices or they can be integrated into hardware LAN analyzers such as the Hewlett Packard 3217.

16.3.5 RMON probes

An RMON (Remote MONitoring) probe is a device that can examine a network at a given point and keep track of captured information at a detailed level. The advantage of a RMON probe is that it can monitor a network at a remote location. The data captured by the RMON probe can then be uploaded and remotely displayed by the appropriate RMON management software. RMON probes and the associated management software are available from several vendors such as 3COM, Bay Networks and NetScout. It is also possible to create an RMON probe by running commercially available RMON software on a normal PC, although the data collection capability will not be as good as that of a dedicated RMON probe.

16.3.6 Handheld frame analyzers

Handheld frame analyzers are manufactured by several vendors, for up to Gigabit Ethernet speeds. These little devices can perform link testing, traffic statistics gathering etc. and can even break down frames by protocol type. The drawback of these testers is the small display and the lack of memory, which results in a lack of historical or logging functions on these devices.

Some frame testers are non–intrusive, i.e. they are clamp–style meters that simply clamp on to the wire and do not have to be attached to a hub port.

Figure 16.2
Psibernet Gigabit Ethernet probe

16.3.7 Software protocol analyzers

Software protocol analyzers are software packages running on PCs and using either a general purpose or a specialized NIC to capture frames from the network. The NIC is controlled by a so–called promiscuous driver, which enables it to capture all packets on the medium and not only those addressed to it in broadcast or unicast mode.

On the lower end of the scale, simple analyzers are available for download from the Internet as freeware or shareware. The free packages work well but rely heavily on the user for interpreting the captured information. Top of the range software products such as Network Associates' 'Sniffer' or WaveTek Wandel Goltemann's 'Domino' suite have sophisticated expert systems that can aid in the analysis of the captured software. Unfortunately, this comes at a price.

16.3.8 Hardware based protocol analyzers

Several manufacturers such as Hewlett Packard, Network Associates and WaveTek also supply hardware based protocol analyzers using their protocol analysis software running on a proprietary hardware infrastructure. This makes them very expensive but dramatically increases the power of the analyzer. For fast and gigabit Ethernet, this is probably the better approach.

16.4 Problems and solutions

16.4.1 Noise

If excessive noise is suspected on a coax or UTP cable, an oscilloscope can be connected between the signal conductor(s) and ground. This method will show up noise on the conductor, but will not necessarily give a true indication of the amount of power in the noise. A simple and cheap method to pick up noise on the wire is to connect a small loudspeaker between the conductor and ground. A small operational amplifier can be used as an input buffer, so as not to 'load' the wire under observation. The noise will be heard as an audible signal.

The quickest way to get rid of a noise problem, apart from using screened UTP (ScTP), is to change to a fiber–based instead of a wire–based network, for example, by using 100BaseFX instead of 100BaseTX.

Noise can to some extent be counteracted on a coax–based network by earthing the screen AT ONE END ONLY. Earthing it on both sides will create an earth loop. This is normally accomplished by means of an earthing chain or an earthing screw on one of the terminators. Care should also be taken not to allow contact between any of the other connectors on the segment and ground.

16.4.2 Thin coax problems

Incorrect cable type

The correct cable for thin Ethernet is RG58A/U or RG58C/U. This is a 5–mm diameter coaxial cable with 50–ohm characteristic impedance and a stranded center conductor. Incorrect cable used in a thin Ethernet system can cause reflections, resulting in CRC errors, and hence many retransmitted frames.

The characteristic impedance of coaxial cable is a function of the ratio between the center conductor diameter and the screen diameter. Hence other types of coax may closely resemble RG58, but may have different characteristic impedance.

Loose connectors

The BNC coaxial connectors used on RG58 should be of the correct diameter and should be properly crimped onto the cable. An incorrect size connector or a poor crimp could lead to intermittent contact problems, which are very hard to locate. Even worse is the 'Radio Shack' hobbyist type screw–on BNC connector that can be used to quickly make up a cable without the use of a crimping tool. These more often than not lead to very poor connections. A good test is to grip the cable in one hand, and the connector in another, and pull very hard. If the connector comes off, the connector mounting procedures need to be seriously reviewed.

Excessive number of connectors

The total length of a thin Ethernet segment is 185 m and the total number of stations on the segment should not exceed 30. However, each station involves a BNC T–piece plus two coax connectors and there could be additional BNC barrel connectors joining the cable. Although the resistance of each BNC connector is small, they are still finite and can add up. The total resistance of the segment (cable plus connectors) should not exceed 10 ohms otherwise problems can surface.

An easy method of checking the loop resistance (the resistance to the other end of the cable and back) is to remove the terminator on one end of the cable and measure the resistance between the connector body and the center contact. The total resistance equals the resistance of the cable plus connectors plus the terminator on the far side. This should be between 50 and 60 ohms. Anything more than this is indicative of a problem.

Overlong cable segments

The maximum length of a thin net segment is 185 m. This constraint is not imposed by collision domain considerations but rather by the attenuation characteristics of the cable. If it is suspected that the cable is too long, its length should be confirmed. Usually, the cable is within a cable trench and hence it cannot be visually measured. In this case, a time domain reflectometer (TDR) can be used to confirm its length.

Stub cables

For thin Ethernet (10Base2), the maximum distance between the bus and the transceiver electronics is 4 cm. In practice, this is taken up by the physical connector plus the PC board tracks leading to the transceiver, which means that there is no scope for a drop cable or 'stub' between the NIC and the bus. The BNC T–piece has to be mounted directly on to the NIC.

Users might occasionally get away with putting a short stub between the T–piece and the NIC but this invariably leads to problems in the long run.

Incorrect terminations

10Base2 is designed around 50 ohm coax and hence requires a 50 ohm terminator at each end. Without the terminators in place, there would be so many reflections from each end that the network would collapse. A slightly incorrect terminator is better than no terminator, yet may still create reflections of such magnitude that it affects the operation of the network.

A 93 ohm terminator looks no different than a 50 ohm terminator; therefore it should not be automatically assumed that a terminator is of the correct value.

If two 10Base2 segments are joined with a repeater, the internal termination on the repeater can be mistakenly left enabled. This leads to three terminators on the segment, creating reflections and hence affecting the network performance.

The easiest way to check for proper termination is by alternatively removing the terminators at each end, and measuring the resistance between connector body and center pin. In each case, the result should be 50 to 60 ohms. Alternatively, one of the T–pieces in the middle of the segment can be removed from its NIC and the resistance between the connector body and the center pin measured. The result should be the value of the two half cable segments (including terminators) in parallel, that is, 25 to 30 ohms.

Invisible insulation damage

If the internal insulation of coax is inadvertently damaged, for example, by placing a heavy point load on the cable, the outer cover could return to its original shape whilst leaving the internal dielectric deformed. This leads to a change of characteristic impedance at the damaged point resulting, in reflections. This, in turn, could lead to standing waves being formed on the cable.

An indication of this problem is when a work station experiences problems when attached to a specific point on a cable, yet functions normally when moved a few meters to either side. The only solution is to remove the offending section of the cable. It cannot be seen by the naked eye and the position of the damage has to be located with a TDR because of the nature of the damage. Alternatively, the whole cable segment has to be replaced.

Invisible cable break

This problem is similar to the previous one, with the difference that the conductor has been completely severed at a specific point. Despite the terminators at both ends of the cable, the cable break effectively creates two half segments, each with an un–terminated end, and hence nothing will work.

The only method to discover the location of the break is by using a TDR.

Thick coax problems

Thick coax (RG8), as used for 10Base5 or thick Ethernet, will basically exhibit the same problems as thin coax yet there are a few additional complications.

Loose connectors

10Base5 use N–type male screw on connectors on the cable. As with BNC connectors, incorrect procedures or a wrong sized crimping tool can cause sloppy joints. This can lead to intermittent problems that are difficult to locate.

Again, a good test is to grab hold of the connector and to try and rip it off the cable with brute force. If the connector comes off, it was not properly installed in the first place.

Dirty taps

The MAU transceiver is often installed on a thick coax by using a vampire tap, which necessitates pre-drilling into the cable in order to allow the center pin of the tap to contact the center conductor of the coax. The hole has to go through two layers of braided screen and two layers of foil. If the hole is not properly cleaned pieces of the foil and braid can remain and cause short circuits between the signal conductor and ground.

Open tap holes

When a transceiver is removed from a location on the cable, the abandoned hole should be sealed. If not, dirt or water could enter the hole and create problems in the long run.

Tight cable bends

The bend radius on a thick coax cable may not exceed 10 inches. If it does, the insulation can deform to such an extent that reflections are created leading to CRC errors. Excessive cable bends can be detected with a TDR.

Excessive loop resistance

The resistance of a cable segment may not exceed 5 ohms. As in the case of thin coax, the easiest way to do this is to remove a terminator at one end and measure the loop resistance. It should be in a range of 50 – 55 ohms.

UTP problems

The most commonly used tool for UTP troubleshooting is a cable meter or pair scanner. At the bottom end of the scale, a cable tester can be an inexpensive tool, only able to check for the presence of wire on the appropriate pins of a RJ–45 connector. High–end cable testers can also test for noise on the cable, cable length, and crosstalk (such as near end signal crosstalk or NEXT) at various frequencies. It can check the cable against CAT5/5e specifications and can download cable test reports to a PC for subsequent evaluation.

The following is a description of some wiring practices that can lead to problems.

Incorrect wire type (solid/stranded)

Patch cords must be made with stranded wire. Solid wire will eventually suffer from metal fatigue and crack right at the RJ–45 connector, leading to permanent or intermittent open connection/s. Some RJ–45 plugs, designed for stranded wire, will actually cut through the solid conductor during installation, leading to an immediate open connection. This can lead to CRC errors resulting in slow network performance, or can even disable a workstation permanently.

The permanently installed cable between hub and workstation, on the other hand, should not exceed 90 m and must be of the solid variety. Not only is stranded wire more expensive for this application, but the capacitance is higher, which may lead to a degradation of performance.

Incorrect wire system components

The performance of the wire link between a hub and a workstation is not only dependent on the grade of wire used, but also on the associated components such as patch panels, surface mount units (SMUs) and RJ–45 type connectors. A single substandard connector on a wire link is sufficient to degrade the performance of the entire link.

High quality fast and Gigabit Ethernet wiring systems use high–grade RJ–45 connectors that are visibly different from standard RJ–45 type connectors.

Incorrect cable type

Care must be taken to ensure that the existing UTP wiring is of the correct category for the type of Ethernet being used. For 10BaseT, Cat3 UTP is sufficient, while Fast Ethernet

(100BaseT) requires Cat5 and Gigabit Ethernet requires Cat5e or better. This applies to patch cords as well as the permanently installed ('infrastructure') wiring.

Most industrial Ethernet systems nowadays are 100BaseX based and hence use Cat5 wiring. For such applications, it might be prudent to install screened Cat5 wiring (ScTP) for better noise immunity. ScTP is available with a common foil screen around 4 pairs or with an individual foil screen around each pair.

A common mistake is to use telephone grade patch ('silver satin') cable for the connection between an RJ–45 wall socket (SMU) and the network interface card in a computer. Telephone patch cables use very thin wires that are untwisted, leading to high signal loss and large amounts of crosstalk. This will lead to signal errors causing retransmission of lost packets, which will eventually slow the network down.

'Straight' vs. crossover cable

A 10BaseT/100BaseTX patch cable consists of 4 wires (two pairs) with an RJ–45 connector at each end. The pins used for the TX and RX signals are 1, 2 and 3, 6. Although a typical patch cord has 8 wires (4 pairs), the 4 unused wires are nevertheless crimped into the connector for mechanical strength. In order to facilitate communication between computer and hub, the TX and RX ports on the hub are reversed, so that the TX on the computer and the RX on the hub are interconnected whilst the TX on the hub is connected to the RX on the hub. This requires a 'straight' interconnection cable with pin 1 wired to pin 1, pin 2 wired to pin 2 etc.

If the NICs on two computers are to be interconnected without the benefit of a hub, a normal straight cable cannot be used since it will connect TX to TX and RX to RX. For this purpose, a crossover cable has to be used in the same way as a 'null' modem cable. Crossover cables are normally color coded (for example, green or black) in order to differentiate them from straight cables.

A crossover cable can create problems when it looks like a normal straight cable and the unsuspecting person uses it to connect a NIC to a hub or a wall outlet. A quick way to identify a crossover cable is to hold the two RJ–45 connectors side by side and observe the colors of the 8 wires in the cable through the clear plastic of the connector body. The sequence of the colors should be the same for both connectors.

Hydra cables

Some 10BaseT hubs feature 50 pin connectors to conserve space on the hub. Alternatively, some building wire systems use 50 pin connectors on the wiring panels but the hub equipment has RJ–45 connectors. In both cases, hydra or octopus cable has to be used. This consists of a 50 pin connector connected to a length of 25 pair cable, which is then broken out as a set of 12 small cables, each with a RJ–45 connector. Depending on the vendor the 50–pin connector can be attached through locking clips, Velcro strips or screws. It does not always lock down properly, although at a glance it may seem so. This can cause a permanent or intermittent break of contact on some ports.

For 10BaseT systems, near end crosstalk (NEXT), which occurs when a signal is coupled from a transmitting wire pair to a receiving wire pair close to the transmitter, (where the signal is strongest) causes most problems. This is not a serious problem on a single pair cable, as only two pairs are used but on the 25 pair cable, with many signals in close proximity, this can create problems. It can be very difficult to troubleshoot since it will require test equipment that can transmit on all pairs simultaneously.

Excessive untwists

On Cat5 cable, crosstalk is minimized by twisting each cable pair. However, in order to attach a connector at the end the cable has to be untwisted slightly. Great care has to be taken since excessive untwists (more than 1 cm) is enough to create excessive crosstalk, which can lead to signal errors. This problem can be detected with a high quality cable tester.

Stubs

A stub cable is an abandoned telephone cable leading from a punch–down block to some other point. This does not create a problem for telephone systems but if the same Cat3 telephone cabling is used to support 10BaseT, then the stub cables may cause signal reflections that result in bit errors. Again, a high quality cable tester can only detect this problem.

Damaged RJ–45 connectors

On RJ–45 connectors without protective boots, the retaining clip can easily break off especially on cheaper connectors made of brittle plastic. The connector will still mate with the receptacle but will retract with the least amount of pull on the cable, thereby breaking contact. This problem can be checked by alternatively pushing and pulling on the connector and observing the LED on the hub, media coupler or NIC– wherever the suspect connector is inserted. Because of the mechanical deficiencies of RJ–45 connectors, they are not commonly used on industrial Ethernet systems.

T4 on 2 pairs

100BaseTX is a direct replacement for 10BaseT in that it uses the same 2 wire pairs and the same pin allocations. The only prerequisite is that the wiring must be Cat5.

100BaseT4, however, was developed for installations where all the wiring is Cat3, and cannot be replaced. It achieves its high speed over the inferior wire by using all 4 pairs instead of just 2. In the event of deploying 100BaseT4 on a Cat3 wiring infrastructure, a cable tester has to be used to ensure that in fact, all 4 pairs are available for each link and have acceptable crosstalk.

100BaseT4 required the development of a new physical layer technology, as opposed to 100BaseTX/FX that used existing FDDI technology. Therefore, it became commercially available only a year after 100BaseX and never gained real market acceptance. As a result, very few users will actually be faced with this problem.

Fiber optic problems

Since fiber does not suffer from noise, interference and crosstalk problems there are basically only two issues to contend with, namely, attenuation and continuity.

The simplest way of checking a link is to plug each end of the cable into a fiber hub, NIC or fiber optic transceiver. If the cable is all right, the LEDs at each end will light up. Another way of checking continuity is by using an inexpensive fiber optic cable tester consisting of a light source and a light meter to test the segment.

More sophisticated tests can be done with an optical time domain reflectometer (OTDR). OTDRs can not only measure losses across a fiber link, but can also determine the nature and location of the losses. Unfortunately, they are very expensive but most professional cable installers will own one.

10BaseFX and 100BaseFX use LED transmitters that are not harmful to the eyes, but Gigabit Ethernet uses laser devices that can damage the retina of the eye. It is therefore

dangerous to try and stare into the fiber (all systems are infrared and therefore invisible anyway).

Incorrect connector installation

Fiber optic connectors can propagate light even if the two connector ends are not touching each other. Eventually, the gap between fiber ends may be so far apart that the link stops working. It is therefore imperative to ensure that the connectors are properly latched.

Dirty cable ends

A speck of dust or some finger oil deposited by touching the connector end is sufficient to affect communication because of the small diameter of the fiber (8-62 microns) and the low light intensity. Dust caps must be left in place when the cable is not in use and a fiber optic cleaning pad must be used to remove dirt and oils from the connector point before installation to avoid this problem.

Component ageing

The amount of power that a fiber optic transmitter can radiate diminishes during the working lifetime of the transmitter. This is taken into account during the design of the link but in the case of a marginal design, the link could start failing intermittently towards the end of the design life of the equipment. A fiber optic power meter can be used to confirm the actual amount of loss across the link but an easy way to trouble shoot the link is to replace the transceivers at both ends of the link with new ones.

16.4.3 AUI problems

Excessive cable length

The maximum length of the AUI cable is 50 m assuming that it is a proper IEEE 802.3 cable. Some installations use lightweight office grade cables that are limited to 12 m in length. If these cables are too long, the excessive attenuation can lead to intermittent problems.

DIX latches

The DIX version of the 15 pin D–connector uses a sliding latch. Unfortunately, not all vendors adhere to the IEEE 802 specifications and some use lightweight latch hardware, which results in a connector that can very easily become unstuck. There are basically two solutions to the problem. The first solution is to use a lightweight (office grade) AUI cable, provided the distance would not be a problem. This places less stress on the connector. The second solution is to use a special plastic retainer such as the ‘ET Lock’ made specifically for this purpose.

SQE test

The signal quality error (SQE) test signal is used on all AUI based equipment to test the collision circuitry. This method is only used on the old 15 pin AUI based external

transceivers (MAUs) and sends a short signal burst (about 10 bit times in length) to the NIC just after each frame transmission. This tests both the collision detection circuitry and the signal paths. The SQE operation can be observed by means of an LED on the MAU.

The SQE signal is only sent from the transceiver to the NIC and not on to the network itself. It does not delay frame transmissions but occurs during the inter–frame gap and is not interpreted as a collision.

The SQE test signal must, however, be disabled if an external transceiver (MAU) is attached to a repeater hub. If this is not done, the hub will detect the SQE signal as a collision and will issue a jam signal. As this happens after each packet, it can seriously delay transmissions over the network. The problem is that it is not possible to detect this with a protocol analyzer.

16.4.4 NIC problems

Basic card diagnostics

The easiest way to check if a particular NIC is faulty is to replace it with another (working) NIC. Modern NICs for desktop PCs usually have auto–diagnostics included and these can be accessed, for example, from the device manager in MS Windows. Some cards can even participate in a card to card diagnostic. Provided there are two identical cards, one can be set up as an initiator and one as a responder. Since the two cards will communicate at the data link level, the packets exchanged will, to some extent, contribute to the network traffic but will not affect any other devices or protocols present on the network.

The drivers used for card auto–diagnostics will usually conflict with the NDIS and ODI drivers present on the host, and a message is usually generated, advising the user that the Windows drivers will be shut down, or that the user should re–boot in DOS.

With PCMCIA cards, there is an additional complication in that the card diagnostics will only run under DOS, but under DOS the IRQ (interrupt address) of the NIC typically defaults to 5, which happens to be the IRQ for the sound card. Therefore, the diagnostics will usually pass every test, but fail on the IRQ test. This result can then be ignored safely if the card passes the other diagnostics. If the card works, it works!

Incorrect media selection

Some cards support more than one medium, for example, 10Base2/10Base5, or 10Base5/10BaseT, or even all three. It may then happen that the card fails to operate since it fails to ‘see’ the attached medium.

It is imperative to know how the selection is done. Modern cards usually have an auto–detect function but this only takes place when the machine is booted up. It does NOT re–detect the medium if it is changed afterwards. Therefore, if the connection to a machine is changed from 10BaseT to 10Base2, for example, the machine has to be re–booted.

Some older cards need to have the medium set via a setup program, whilst even older cards have DIP switches on which the medium has to be selected.

Wire hogging

Older interface cards find it difficult to maintain the minimum 9.6 micro second inter-frame spacing (IFS) and as a result of this, nodes tend to return to and compete for access

to the bus in a random fashion. Modern interface cards are so fast that they can sustain the minimum 9.6 microsecond IFS rate. As a result, it becomes possible for a single card to gain repetitive sequential access to the bus in the face of slower competition and hence 'hogging' the bus.

With a protocol analyzer, this can be detected by displaying a chart of network utilization versus time and looking for broad spikes above 50 percent. The solution to this problem is to replace shared hubs with switched hubs and increase the bandwidth of the system by migrating from 10 – 100 megabits per second, for example.

Jabbers

A jabber is a faulty NIC that transmits continuously. NICs have a built–in jabber control that is supposed to detect a situation whereby the card transmits frames longer than the allowed 1518 bytes and shut the card down. However, if this does not happen, the defective card can bring the network down. This situation is indicated by a very high collision rate coupled with a very low or nonexistent data transfer rate. A protocol analyzer might not show any packets since the jabbering card is not transmitting any sensible data. The easiest way to detect the offending card is by removing the cables from the NICs or the hub one–by–one until the problem disappears in which case the offending card is located.

Faulty CSMA/CD mechanism

A card with a faulty CSMA/CD mechanism will create a large number of collisions since it transmits legitimate frames but does not wait for the bus to be quiet before transmitting. As in the previous case, the easiest way to detect this problem is to isolate the cards one by one until the culprit is detected.

Too many nodes

A problem with CSMA/CD networks is that the network efficiency decreases as the network traffic increases. Although Ethernet networks can theoretically utilize well over 90% of the available bandwidth, the access time of individual nodes increase dramatically as network loading increases. The problem is similar to that encountered on many urban roads during peak hours. During rush hours, the traffic approaches the design limit of the road. This does not mean that the road stops functioning. In fact, it carries a very large number of vehicles, but to get into the main traffic from a side road becomes problematic.

For office type applications, an average loading of around 30% is deemed acceptable while for industrial applications, 3% is considered maximum. Should the loading of the network be a problem, the network can be segmented using switches instead of shared hubs. In many applications, it will be found that the improvement created by changing from shared to switched hubs, is larger than the improvement to be gained by upgrading from 10 Mbps to Fast Ethernet.

Improper packet distribution

Improper packet distribution takes place when one or more nodes dominate most of the bandwidth. This can be monitored by using a protocol analyzer and checking the source address of individual packets. Another way of checking this easily is by using the NDG software Web Boy facility and checking the contribution of the most active transmitters.

Nodes like this are typically performing tasks such as video conferencing or database access, which require a large bandwidth. The solution to the problem is to give these

nodes separate switch connections or to group them together on a faster 100BaseT or 1000BaseT segment.

Excessive broadcasting

A broadcast packet is intended to reach all the nodes in the network and is sent to a MAC address of FF-FF-FF-FF-FF-FF. Unlike routers, bridges and switches forward broadcast packets throughout the network and therefore cannot contain the broadcast traffic. Too many simultaneous broadcast packets can degrade network performance.

In general, it is considered that if broadcast packets exceed 5% of the total traffic on the network, it would indicate a broadcast overload problem. Broadcasting is a particular problem with Netware servers and networks using NetBIOS/NetBEUI. Again, it is fairly easy to observe the amount of broadcast traffic using the WebBoy utility.

A broadcast overload problem can be addressed by adding routers, layer 3 switches or VLAN switches with broadcast filtering capabilities.

Bad packets

Bad packets can be caused by poor cabling infrastructure, defective NICs, external noise, or faulty devices such as hubs, devices or repeaters. The problem with bad packets is that they cannot be analyzed by software protocol analyzers.

Software protocol analyzers obtain packets that have already been successfully received by the NIC. That means they are one level removed from the actual medium on which the frames exist and hence cannot capture frames that are rejected by the NIC. The only solution to this problem is to use a software protocol analyzer that has a special custom NIC, capable of capturing information regarding packet deformities or by using a more expensive hardware protocol analyzer.

16.4.5 Faulty packets

Runts

Runt packets are shorter than the minimum 64 bytes and are typically created by a collision–taking place during the slot time.

As a solution, try to determine whether the frames are collisions or under–runs. If they are collisions, the problem can be addressed by segmentation through bridges and switches. If the frames are genuine under–runs, the packet has to be traced back to the generating node that is obviously faulty.

CRC errors

CRC errors occur when the CRC check at the receiving end does not match the CRC checksum calculated by the transmitter.

As a solution, trace the frame back to the transmitting node. The problem is either caused by excessive noise induced into the wire, corrupting some of the bits in the frames, or by a faulty CRC generator in the transmitting node.

Late collisions

Late collisions are typically caused when the network diameter exceeds the maximum permissible size. This problem can be eliminated by ensuring that the collision domains

are within specified values, i.e. 2500 meters for 10 Mbps Ethernet, 250 m for Fast Ethernet and 200 m for Gigabit Ethernet.

Check the network diameter as outlined above by physical inspection or by using a TDR. If that is found to be a problem, segment the network by using bridges or switches.

Misaligned frames

Misaligned frames are frames that get out of sync by a bit or two, due to excessive delays somewhere along the path or frames that have several bits appended after the CRC checksum.

As a solution, try and trace the signal back to its source. The problem could have been introduced anywhere along the path.

Faulty auto–negotiation

Auto–negotiation is specified for

- 10BaseT,
- 100BaseTX,
- 100BaseT2,
- 100BaseT4 and
- 1000BaseT.

It allows two stations on a link segment (a segment with only two devices on it) e.g. an NIC in a computer and a port on a switching hub to negotiate a speed (10/100/1000Mbps) and an operating mode (full/half duplex). If auto–negotiation is faulty or switched off on one device, the two devices might be set for different operating modes and as a result, they will not be able to communicate.

On the NIC side the solution might be to run the card diagnostics and to confirm that auto–negotiation is, in fact, enabled.

On the switch side, this depends on the diagnostics available for that particular switch. It might also be an idea to select another port, or to plug the cable into another switch.

10/100 Mbps mismatch

This issue is related to the previous one since auto–negotiation normally takes care of the speed issue.

Some system managers prefer to set the speeds on all NICs manually, for example, to 10 Mbps. If such an NIC is connected to a dual–speed switch port, the switch port will automatically sense the NIC speed and revert to 10 Mbps. If, however, the switch port is only capable of 100 Mbps, then the two devices will not be able to communicate.

This problem can only be resolved by knowing the speed (s) at which the devices are supposed to operate, and then by checking the settings via the setup software.

Full/half duplex mismatch

This problem is related to the previous two.

A 10BaseT device can only operate in half–duplex (CSMA/CD) whilst a 100BaseTX can operate in full duplex OR half–duplex.

If, for example, a 100BaseTX device is connected to a 10BaseT hub, its auto–negotiation circuitry will detect the absence of a similar facility on the hub. It will therefore know, by default, that it is ‘talking’ to 10BaseT and it will set its mode to half–duplex. If, however, the NIC has been set to operate in full duplex only, communications will be impossible.

16.4.6 Host related problems

Incorrect host setup

Ethernet only supplies the bottom layer of the DOD model. It is therefore able to convey data from one node to another by placing it in the data field of an Ethernet frame, but nothing more. The additional protocols to implement the protocol stack have to be installed above it, in order to make networked communications possible.

In industrial Ethernet networks, this will typically be the TCP/IP suite, implementing the remaining layers of the ARPA model as follows.

The second layer of the DOD model (the internet layer) is implemented with IP (as well as its associated protocols such as ARP and ICMP).

The next layer (the host–to host layer) is implemented with TCP and UDP.

The upper layer (the application layer) is implemented with the various application layer protocols such as FTP, Telnet etc. The host might also require a suitable application layer protocol to support its operating system in communicating with the operating system on other hosts, on Windows, that is NetBIOS by default.

As if this is not enough, each host needs a network 'client' in order to access resources on other hosts, and a network 'service' to allow other hosts to access its own resources in turn. The network client and network service on each host do not form part of the communications stack but reside above it and communicate with each other across the stack.

Finally, the driver software for the specific NIC needs to be installed, in order to create a binding ('link') between the lower layer software (firmware) on the NIC and the next layer software (for example, IP) on the host. The presence of the bindings can be observed, for example, on a Windows 95/98 host by clicking 'settings' –> 'control panel' –> 'networks'–>'configuration,' then selecting the appropriate NIC and clicking 'Properties' –> 'Bindings.'

Without these, regardless of the Ethernet NIC installed, networking is not possible.

Failure to log in

When booting a PC, the Windows dialogue will prompt the user to log on to the server, or to log on to his/her own machine. Failure to log in will not prevent Windows from completing its boot–up sequence but the network card will not be enabled. This is clearly visible as the LED's on the NIC and hub will not light up.

16.4.7 Hub related problems

Faulty individual port

A port on a hub may simply be 'dead.' Everybody else on the hub can 'see' each other, except the user on the suspect port. Closer inspection will show that the LED for that particular channel does not light up. The quickest way to verify this is to remove the UTP cable from the suspect hub port and plugging it into another port. If the LEDs light up on the alternative port, it means that the original port is not operational.

On managed hubs, the configuration of the hub has to be checked by using the hub's management software to verify that the particular port has not, in fact, been disabled by the network supervisor.

Faulty hub

This will be indicated by the fact that none of the LEDs on the hub are illuminated and that none of the users on that particular hub are able to access the network. The easiest to check this is by temporarily replacing the hub with a similar one and checking if the problem disappears.

Incorrect hub interconnection

If hubs are interconnected in a daisy chain fashion by means of interconnecting ports with a UTP cable, care must be taken to ensure that either a crossover cable is used or that the crossover/uplink port on one hub ONLY is used. Failure to comply with this precaution will prevent the interconnected hubs from communicating with each other although it will not damage any electronics.

A symptom of this problem will be that all users on either side of the faulty link will be able to see each other but nobody will be able to see anything across the faulty link. This problem can be rectified by ensuring that a proper crossover cable is being used or, if a straight cable is being used, that it is plugged into the crossover/uplink port on one hub only. On the other hub, it must be plugged into a normal port.

16.5 Troubleshooting switched networks

Troubleshooting in a shared network is fairly easy since all packets are visible everywhere in the segment and as a result, the protocol analysis software can run on any host within that segment. In a switched network, the situation changes radically since each switch port effectively resides in its own segment and packets transferred through the switch are not seen by ports for whom they are not intended.

In order to address the problem, many vendors have built traffic monitoring modules into their switches. These modules use either RMON or SNMP to built up statistics on each port and report switch statistics to switched management software.

Capturing the packets on a particular switched port is also a problem, since packets are not forwarded to all ports in a switch hence there is no place to plug in a LAN analyzer and view the packets.

One solution implemented by vendors is port liaising, also known as port mirroring or port spanning. The liaising has to be set up by the user and the switch copies the packets from the port under observation to a designated spare port. This allows the LAN user to plug in a LAN analyzer onto the spare port in order to observe the original port.

Another solution is to insert a shared hub in the segment under observation that is between the host and the switch port to which it was originally connected. The LAN analyzer can then be connected to the hub in order to observe the passing traffic.

16.6 Troubleshooting Fast Ethernet

The most diagnostic software is PC based and it uses a NIC with a promiscuous mode driver. This makes it easy to upgrade the system by simply adding a new NIC and driver. However, most PCs are not powerful enough to receive, store and analyze incoming data rates. It might therefore be necessary to rather consider the purchase of a dedicated hardware analyzer.

Most of the typical problems experienced with fast Ethernet, have already been discussed. These include a physical network diameter that is too large, the presence of Cat3 wiring in the system, trying to run 100BaseT4 on 2 pairs, mismatched 10BaseT/100BaseTX ports, and noise.

16.7 Troubleshooting Gigabit Ethernet

Although Gigabit Ethernet is very similar to its predecessors, the packets arrive so fast that they cannot be analyzed by normal means. A Gigabit Ethernet link is capable of transporting around 125 MB of data per second and few analyzers have the memory capability to handle this. Gigabit Ethernet analyzers such as those made by Hewlett Packard (LAN Internet Advisor), Network Associates (Gigabit Sniffer Pro) and WaveTech Wandel Goltemann (Domino Gigabit Analyzer) are highly specialized Gigabit Ethernet analyzers. They minimize storage requirements by filtering and analyzing capture packets in real time, looking for a problem. Unfortunately, they come at a price tag of around US$ 50 000.

17

Network protocols, part one – Internet Protocol (IP)

Objectives

This chapter deals with the first part of network protocols. The Internet Protocol (IP) is the most important protocol used with Ethernet systems. This protocol is dealt with in this chapter. When you study this chapter, you will:

- Be introduced to network protocols
- Read about origins of TCP/IP, the most important of the protocols
- Learn about the Internet Protocol (IP) suite of protocols
- Study IPv4, i.e. version 4 of IP
- Study the basics of IPv6, i.e., version 6 of IP
- Learn about address resolution protocol (ARP) and reverse address resolution protocol (RARP)
- Learn about the Internet control message protocol (ICMP)
- Learn about routing protocols
- Learn about interior and exterior gateway protocols

17.1 Introduction

Network protocols were very briefly dealt with in chapter one of this manual. A more detailed look will now be taken in the following section.

A protocol is defined as a set of rules for exchanging data in a manner that is understandable to both the transmitter and the receiver. There must be a formal and agreed set of rules if the communication is to be successful. The rules for a data link protocol relate to such responsibilities as error detection and correction methods as well as flow control methods. A physical layer standard such as RS-232 covers voltage and current standards. In addition, other properties are such as size of data packets is important in LAN protocols.

An important responsibility of network layer protocols is the method of routing the packet, once it has been assembled. In a self-contained LAN, i.e. intranetwork, this is not

a problem, since all packets will eventually reach their destinations by virtue of design. However, if the packet is to be switched across networks, i.e. on an internetwork- such as a WAN- then routing decisions must be made.

Network hardware/software is conceptually organized as a series of levels (or layers), one above each other. The number, names and functions of layers can vary from network to network. In any type of network, however, the purpose of any layer is to provide certain services to higher layers, hiding from them the details of how these services are implemented.

Layer N1 of a network protocol stack on a computer carries on communication with layer N1 of another computer; the communication being carried out as per the rules of the layer N1 protocol.

The communication being carried out between the two N1 layers of the two computers is at a logical level. Actually, the data is not directly transferred between these N1 layers. Each layer passes data and control information to the layer immediately below it, until the lowest layer is reached. Below the bottom layer is the physical medium through which actual communication occurs.

Between each pair of layers is an interface that specifies the services that the lower layer offers to the upper layer. A clean and unambiguous interface simplifies layer replacement, substituting implementation of one layer with a completely different implementation (if the need arises), because all that is required of the new implementation is that it offers exactly the same services to its upper layer as was done in the previous case.

The actual data to be transmitted between two computers is carried in the data field of the Ethernet frame. The high-level protocol information carried inside each Ethernet frame is what actually establishes communication between applications running on the computers attached to the network.

It must be understood that the high-level protocols are independent of the Ethernet system. An Ethernet LAN with its hardware carrying an Ethernet frame is simply a kind of courier service for data being sent by applications. The Ethernet LAN itself does not know, nor is required to know about high-level protocol data being carried in the Ethernet frame.

Since the Ethernet system does not concern itself with the contents of the data field in the frame, different computers running different high-level protocols can share the same Ethernet network.

The most widely used system of high-level network protocols is called the transmission control protocol/internet protocol (TCP/IP) suite.

17.1.1 The origins of TCP/IP

In the early 1960s, the American Department of Defense (DOD) indicated the need for a wide-area communication system, covering the United States and allowing the interconnection of heterogeneous hardware and software systems.

In 1967, the Stanford Research Institute was contracted to develop the suite of protocols for this network, initially to be known as ARPANet. Other participants in the project included the University of Berkeley (California) and the private company BBN (Bolt, Barenek and Newman). Development work commenced in 1970 and by 1972, approximately 40 sites were connected via TCP/IP. In 1973, the first international connection was made and in 1974, TCP/IP was released to the public. Initially the network was used to interconnect government, military and educational sites together. As

time progressed, commercial companies were allowed access and by 1990 the backbone of the Internet, as it was now known, was being extended into one country after another.

One of the major reasons why TCP/IP has become the *de facto* standard worldwide for industrial and telecommunications applications is the fact that the Internet was designed around it and that without it, no Internet access is possible.

17.1.2 The ARPA model vs the OSI model

Whereas the OSI model was developed in Europe by the International Standards Organization (ISO), the ARPA model (also known as the DOD or Department of Defense model) was developed in the USA by the Advanced Projects Research Agency. Although they were developed by different bodies and at different points in time, both serve as models for a communications infrastructure and hence provide 'abstractions' of the same reality. The remarkable degree of similarity is therefore not surprising.

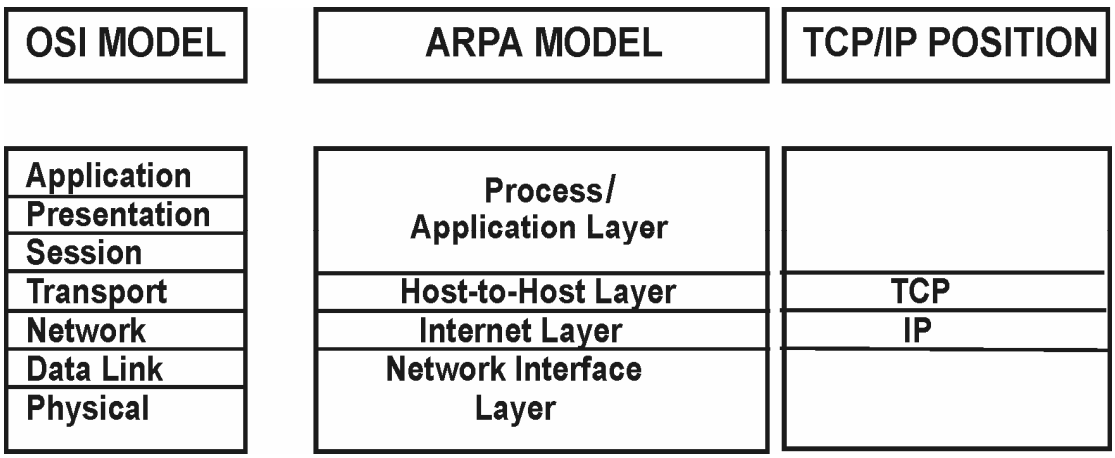


Figure 17.1
Comparison of OSI and ARPA models

Whereas the OSI model has 7 layers, the ARPA model has 4 layers. The OSI layers map onto the ARPA model as follows:

- The OSI session, presentation and applications layers are contained in the ARPA process/application Layer (nowadays simply referred to by the Internet community as the application level)
- The OSI transport layer maps onto the ARPA host-to-host layer (nowadays referred to by the Internet community as the host level)
- The OSI network layer maps onto the ARPA internet layer (nowadays referred to by the Internet community as the network level)
- The OSI physical and data link layers map onto the ARPA network interface layer

The relationship between the two models is depicted in Figure 17.1

17.1.3 The TCP/IP protocol suite vs the ARPA model

TCP/IP, or rather the TCP/IP protocol suite, is not limited to the TCP and IP protocols, but consists of a multitude of interrelated protocols that occupy the upper three layers of

the ARPA model. TCP/IP does NOT include the bottom network access layer, but depends on it for access to the medium.

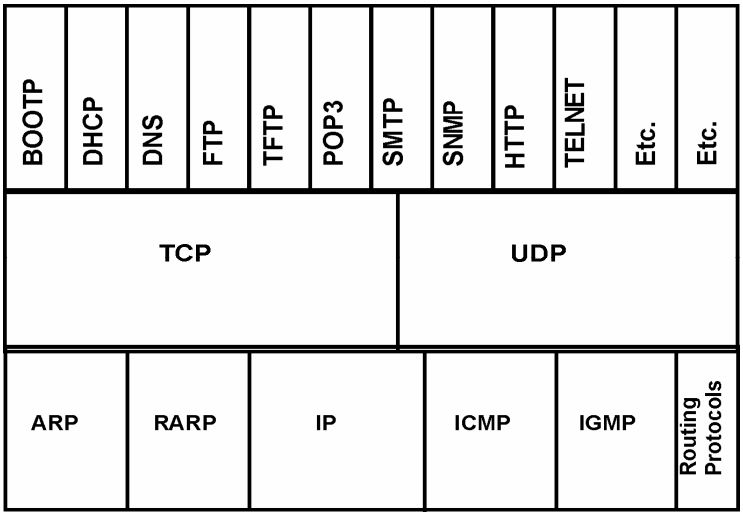


Figure 17.2
Some protocols in the TCP/IP protocol suite

The network interface layer

The network interface layer is responsible for transporting data (frames) between hosts on the same physical network. It is implemented in the network interface card or NIC, using both hardware and firmware (i.e. software resident in ROM).

The NIC employs the appropriate medium access control methodology, such as CSMA/CA, CSMA/CD, token passing, or polling, and is responsible for placing the data received from the upper layers within a frame before transmitting it. The frame format is dependent on the system being used (example Ethernet or frame relay), and holds the hardware address of the source and destination hosts as well as a checksum for data integrity.

RFCs (requests for comments) that apply to the network interface layer include:

- Asynchronous transfer mode (ATM), described in RFC 1438
- Switched multi-megabit data service (SMDS), described in RFC 1209
- Ethernet, described in RFC 894
- ARCNET, described in RFC 1201
- Serial line internet protocol (SLIP), described in RFC 1055
- Frame relay, described in RFC 1490
- Fiber distributed data interface (FDDI), described in RFC 1103

Note: Any Internet-related specification is originally submitted as a 'request for comments' or RFC. As time progresses an RFC may become a standard, or a recommended practice, and so on. Regardless of the status of an RFC, it can be obtained from various sources on the Internet such as *www.rfc-editor.org*.

The Internet layer

This layer is primarily responsible for the routing of packets from one host to another. The emphasis is on 'packets' as opposed to frames, since at this level the data exists in software only. Each packet contains the address information needed for its routing through the internetwork to the receiving host.

The dominant protocol at this level is IP (as in TCP/IP), namely the Internet protocol. There are, however, several other additional protocols required at this level.

These protocols include:

- **Address resolution protocol (ARP), RFC 826:** This is a protocol used for the translation of an IP address to a hardware (MAC) address, such as required by Ethernet
- **Reverse address resolution protocol (RARP), RFC 903:** This is the complement of ARP and translates a hardware address to an IP address
- **Internet control message protocol (ICMP), RFC 792:** This is a protocol used for sending control or error messages between routers or hosts. One of the best-known applications here is the *ping* or echo request that is used to test a communications link

The host-to-host layer

This layer is primarily responsible for data integrity between the sender host and receiver host regardless of the path or distance used to convey the message.

Communications errors are detected and corrected at this level.

It has two protocols associated with it, these being:

- **User data protocol (UDP):** This is a connectionless (unreliable) protocol used for higher layer port addressing. It offers minimal protocol overhead and is described in RFC 768
- **Transmission control protocol (TCP):** This connection-oriented protocol offers vastly improved protection and error control. This protocol, the TCP component of TCP/IP, is the heart of the TCP/IP suite of applications. It provides a very reliable method of transferring data in byte (octet) format, between applications. This is described in RFC 793

The application layer

This layer provides the user or application programs (clients and servers) with interfaces to the TCP/IP stack.

At this level there are many protocols used, some of the more common ones being:

- **File transfer protocol (FTP),** which as the name implies, is used for the transfer of files between two hosts using TCP. It is described in RFC 959
- **Trivial file transfer protocol (TFTP),** which is an 'economic' version of FTP and uses UDP instead of TCP for, reduced overhead. It is described in RFC 783
- **Simple mail transfer protocol (SMTP),** which is an example of an application that provides access to TCP and IP for programs sending e-mail. It is described in RFC 821
- **TELNET (TELecommunications NETwork),** which is used to emulate terminals and for remote access to servers. It can, for example, emulate a VT100 terminal across a network

Other application layer protocols include POP3, RPC, RLOGIN, IMAP, HTTP, and NTP, to name but a few. Users can also develop their own application layer protocols by means of a developer's kit such as Winsock.

17.2 Internet Protocol (IP)

It was seen earlier that the Internet layer is not populated by a single protocol, but rather by a collection of protocols.

They include:

- The Internet protocol (IP)
- The Internet control message protocol (ICMP)
- The address resolution protocol (ARP)
- The reverse address resolution protocol (RARP)
- Routing protocols (such as RIP, OSPF, BGP-4, etc.)

17.3 Internet Protocol version 4 (IPv4)

IP is at the core of the TCP/IP suite. It is primarily responsible for routing packets to their destination, from router to router. This routing is performed based on the IP addresses, embedded in the header attached to each packet forwarded by IP.

The only version of IP in use until 1999 was IPv4, which uses a 32-bit address. However, IPv4 is now being superseded by IPv6, which uses a 128-bit address. IPv4 is sill widely used and is likely to remain so for stand-alone industrial systems. This chapter will focus on version 4 as a vehicle for explaining the fundamental processes involved.

17.3.1 Source of IP addresses

The ultimate responsibility for the issuing of IP addresses is vested in the Internet Assigned Numbers Authority (IANA). This responsibility is then delegated to the three Regional Internet Registries (RIRs). They are:

- **APNIC –** Asia-Pacific Network Information Center (*http://www.apnic.net*)
- **ARIN –** American Registry for Internet Numbers (*http://www.arin.net*)
- **RIPE NCC –** Reseau IP Europeans (*http://www.ripe.net*)

The RIRs allocate blocks of IP addresses to Internet Service Providers (ISPs) under their jurisdiction, for subsequent issuing to users or sub-ISPs.

The use of ‘legitimate’ IP addresses is a pre-requisite for connecting to the Internet. For systems NOT connected to the Internet, any IP addressing scheme may be used. It is recommended that so-called ‘private’ Internet addresses be used for this purpose, as outlined in this chapter.

17.3.2 The purpose of the IP address

The MAC or hardware address (also called the media address or Ethernet address) discussed earlier is unique for each node, and is usually allocated to that particular node e.g. network interface card at the time of its manufacture. The equivalent for a human being would be its ID or social security number. As with a human ID number, the MAC address belongs to that node and follows it wherever it goes. This number works fine for identifying hosts on a LAN where all nodes can ‘see’ (or rather, ‘hear’) each other.

With human beings the problem arises when the intended recipient is living in another city, or worse, in another country. In this case, the ID number is still relevant for final identification, but the message (e.g. a letter) first has to be routed to the destination by the postal system. For the postal system, a name on the envelope has little meaning. It requires a postal address.

The TCP/IP equivalent of this postal address is the IP address. As with the human postal address, this IP address does not belong to the node, but rather indicates its place of residence. For example, if an employee has a fixed IP address at work and he resigns, he will leave his IP address behind and his successor will 'inherit' it. Since each host (which already has a MAC or hardware address) needs an IP address in order to communicate across the Internet, resolving host MAC addresses versus IP.

Addressing resolution is a mandatory function. This is performed by the address resolution protocol (ARP), which is to be discussed later on in this chapter.

17.3.3 IPv4 address notation

The IPv4 address consists of 32 bits, e.g.

11000000011001000110010000000001

Since this number is fine for computers but a little difficult for human beings, it is divided into four octets, which for ease of reference could be called w, x, y and z. Each octet is converted to its decimal equivalent.

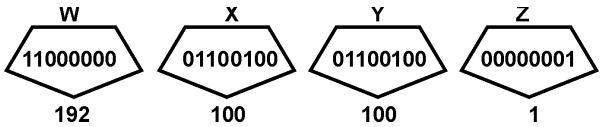


Figure 17.3
IP address structure for address 192.100.100.1

The result of the conversion is written as 192.100.100.1. This is known as the 'dotted decimal' or 'dotted quad' notation.

17.3.4 Network ID and host ID

Refer to the following postal address.

4 Kingsville Street
Claremont 6010
Perth WA
Australia.

The first part, viz. 4 Kingsville Street, enables the local postal deliveryman at the Australian post office in Claremont, Perth (zip code 6010) to deliver a letter to that specific residence. This assumes that the latter has already found its way to the local post office.

The second part (lines 2–4) enables the international postal system to route the letter towards its destination post office anywhere in the world. In similar fashion, an IP address has two distinct parts. The first part, the network ID ('NetID') is a unique number identifying a specific network and allows the Internet routers to forward a packet towards its destination network from anywhere in the world.

The second part, the host ID ('HostID') is a number allocated to a specific machine (host) on the destination network and allows the router servicing that host to deliver the packet directly to the host.

For example, in IP address 192.100.100.5, the computer or HostID would be 5, and it would be connected to network or NetID number 192.100.100.0.

17.3.5 Address classes

Originally, the intention was to allocate IP addresses in so-called 'address classes'. Although the system proved to be problematic, and IP addresses are currently issued 'classless', the legacy of IP address classes remains and has to be understood.

To provide for flexibility in assigning addresses to networks, the interpretation of the address field was coded to specify either:

- A small number of networks with a large number of hosts (class A)
- A moderate number of networks with a moderate number of hosts (class B)
- A large number of networks with a small number of hosts (class C)

Class D was intended for multicasting whilst E was reserved for possible future use. For class A, the first bit is fixed as '0', for class B the first 2 bits are fixed as '10', and, for class C the first three bits are fixed as '110'.

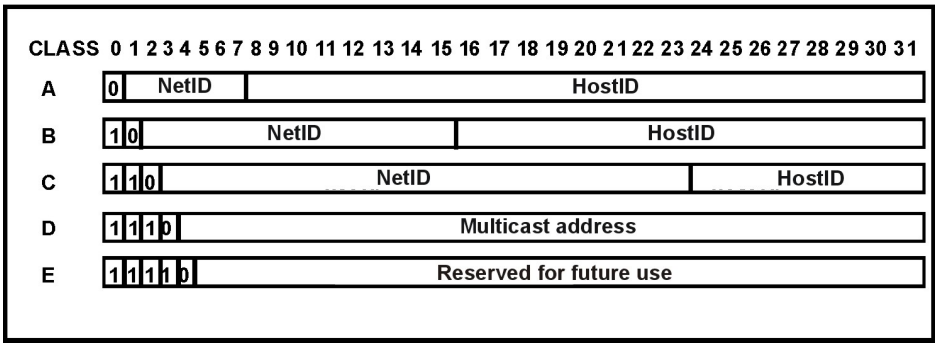


Figure 17.4
Address structure for IPv4

17.3.6 Determining the address class by inspection

The NetID should normally not be all 0s as this indicates a local network. With this in mind, analyze the first octet ('w').

For class A, the first bit is fixed at zero. The binary values for 'w' can therefore only vary between 00000000_2 (0_{10}) and 01111111_2 (127_{10}). Zero is not allowed. However, 127 is also a reserved number, with 127.x.y.z reserved for loop-back testing. In particular, 127.0.0.1 is used to test that the TCP/IP protocol is properly configured by sending information in a loop back to the computer that originally sent the packet, without it travelling over the network.

The values for "w" can therefore only vary between 1 and 126, which allows for 126 possible class A NetID's.

For class B, the first two bits are fixed at 10. The binary values for 'w' can therefore only vary between 10000000_2 (128_{10}) and 10111111_2 (191_{10}). For class C, the first three bits are fixed at 110. The binary values for 'w' can therefore only vary between 11000000_2 (192_{10}) and 11011111_2 (223_{10}).

The relationship between 'w' and the address class can therefore be summarized as follows.

Class	Range
A	1.x.y.z to 126.x.y.z
B	128.x.y.z to 191.x.y.z
C	192.x.y.z to 223.x.y.z

Figure 17.5
IPv4 Classes and their address ranges

17.3.7 Number of networks and hosts per address class

Note that there are two reserved host numbers, irrespective of class. These are 'all zeros' or 'all ones' for HostID. An IP address with a host number of zero is used as the address of the whole network. For example, on a class C network with the NetID = 200.100.100, the IP address 200.100.100.0 indicates the whole network. A hostID of 255 (as in 200.100.100.255) means 'all the hosts on the network'.

To summarize:

HostID = 'all zeros' means 'this network'.

HostID = 'all ones' means 'all hosts on this network'

For class A, the number of NetIDs is determined by octet 'w'. Unfortunately, the first bit (fixed at 0) is used to indicate class A and hence cannot be used. This leaves seven usable bits. Seven bits allow 2^7 = 128 combinations, from 0 to 127. 0 and 127 are reserved; hence, only 126 netIDs are possible. The number of hostIDs, on the other hand, is determined by octets 'x', 'y', and 'z'. From these 24 bits, 2^{24} = 16,777,218 combinations are available. All zeros and all ones are not permissible, which leaves 16,777,216 usable combinations.

For class B, the number of netIDs is determined by octets 'w' and 'x'. The first bits (10) are used to indicate class B and hence cannot be used. This leaves fourteen usable bits. Fourteen bits allow 2^{14} = 16384 combinations. The number of hostIDs is determined by octet 'y' and 'z'. From these 16 bits, 2^{16} = 65536 combinations are available. All zeros and all ones are not permissible, which leaves 65534 usable combinations.

For class C, the number of netIDs is determined by octets 'w', 'x' and 'y'. The first three bits (110) are used to indicate class C and hence cannot be used. This leaves twenty-two usable bits. Twenty-two bits allow 2^{21} = 2097152 combinations. The number of hostIDs is determined by octet 'z'. From these 8 bits, 2^8 = 256 combinations are available. Once again, all zeros and all ones are not permissible which leaves 254 usable combinations. The number of networks and number of hosts per network for the three classes are shown in the table in the next section.

17.3.8 Subnet masks

Class	Number of Networks	Number of Hosts per Network
A	126	16,777,216
B	16,384	65,534
C	2,097,152	254

Figure 17.6
Number of networks and hosts per class

Strictly speaking, one should be referring to 'netmasks' in general, or to 'subnet masks' in the case of defining network masks for the purposes of subnetting. Unfortunately, most

people (including Microsoft) have confused the two issues and are referring to subnet masks in all cases.

For routing purposes it is necessary for a device to strip the hostID off an IP address, in order to ascertain whether or not the remaining NetID portion of the IP address matches the network address of that particular network.

Whilst it is easy for human beings, it is not the case for a computer and the latter has to be 'shown' which portion is the NetID, and which is the HostID. This is done by defining a netmask in which a '1' is entered for each bit which is a part of the netID, and a '0' for each bit which is a part of the HostID. The computer takes care of the rest. The '1s' start from the left and run in a contiguous block.

For example: A conventional class C IP address, 192.100.100.5, written in binary, would be represented in digital as 11000000 01100100 01100100 00000101. Since it is a class C address, the first 24 bits represent the NetID and would therefore be masked by 1s. The subnet mask would therefore be:

11111111 11111111 1111111 00000000.

To summarize:

IP Address:	01100100	01100100	01100100	00000101
Subnet Mask:	11111111	11111111	11111111	00000000
	|<——————— Net ID———————>|<Host ID>|			

The mask, written in decimal dotted notation, becomes 255.255.255.0. This is the so-called 'default netmask' for class C. Default netmasks for classes A and B can be configured in the same manner.

Currently IP addresses are issued classless, which means that it is not possible to determine the boundary between NetID and HostID by analyzing the IP address itself. This makes the use of a subnet mask even more necessary.

IP address class	Default netmask
A	255.0.0.0
B	255.255.0.0
C	255.255.255.0

Figure 17.7
Default netmasks

17.3.9 Subnetting

Although it is theoretically possible, one would never place all the hosts (for example, all 65534 hosts on a class B address) on a single segment – the sheer volume of traffic would render the network useless. For this reason one would have to revert to subnetting.

Assume that a class C address of 192.100.100.0 has been allocated to a network. As shown earlier, 254 hosts are possible. Now assume further that the company has four networks, connected by a router (or routers).

Creating subnetworks under the 192.100.100.0 network address and assigning a different subnetwork number to each LAN segment could solve the problem.

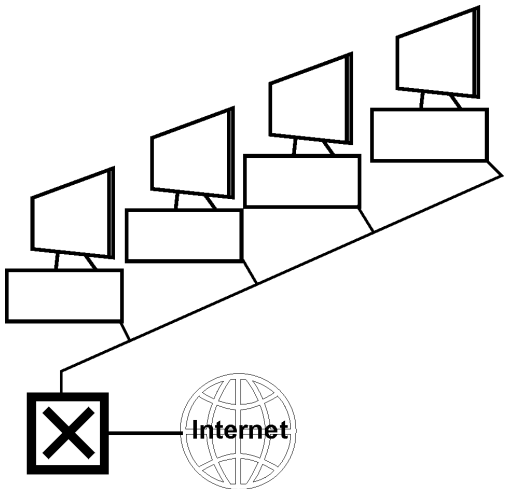


Figure 17.8
Network 192.100.100.0 Class C network before subnetting

To create a subnetwork, 'steal' some of the bits assigned to the HostID and use them for a subnetwork number, leaving fewer bits for HostID. Instead of NetID + HostID, the IP address will now represent NetID + SubnetID + HostID.

To calculate the number of bits to be reassigned to the SubnetID, choose a number of bits 'n' so that (2n)-2 is bigger than or equal to the number of subnets required. This is because two of the possible bit combinations of the new SubnetID, namely all 0s and all 1s are not allowed. In this case, 4 subnets are required so 3 bits have to be 'stolen' from the HostID since $(2^3)–2 = 6$, which is sufficient in view of the 4 subnets we require.

Since only 5 bits are now available for HostID (3 of the 8 'stolen'), each subnetwork can now only have 30 HostIDs numbered 00001 (1_{10}) through 11110 (30_{10}), since neither 00000 nor 11111 is allowed. To be technically correct, each subnetwork will only have 29 computers (not 30) since one HostID will be allocated to the router on that subnetwork.

The 'z' of the IP address is calculated by concatenating the SubnetID and the HostID. For example, for HostID = 1 (00001) on SubnetID = 3 (011), z would be 011 appended to 00001 which gives 01100001 in binary, or 97_{10}.

Note that the total available number of HostIDs has dropped from 254 to 180. In the preceding example, the first 3 bits of the HostID have been allocated as SubnetID, and have therefore effectively become part of the NetID. A default Class C subnet mask would unfortunately obliterate these 3 bits, with the result that the routers would not be able to route messages between the subnets.

Subnet ID	Use For	Host ID	From IP	To IP
000	N/A			
001	Subnet 1	00001-11110	192.100.100.33	192.100.100.62
010	Subnet 2	00001-11110	192.100.100.65	192.100.100.94
011	Subnet 3	00001-11110	192.100.100.97	192.100.100.126
100	Subnet 4	00001-11110	192.100.100.129	192.100.100.158
101	Spare	00001-11110	192.100.100.161	192.100.100.190
110	Spare	00001-11110	192.100.100.193	192.100.100.222
111	N/A			

Figure 17.9
Ipv4 Address allocation for 6 subnets on Class C address

For this reason the subnet mask has to be EXTENDED another 3 bits to the right, so that it becomes 11111111 11111111 11111111 *111*00000. The extra bits have been typed in Italics, for clarity.

The subnet mask is now 255.255.255.224 or /27. The /27 is the so called 'prefix' indicating that there are 27 '1s' in the mask.

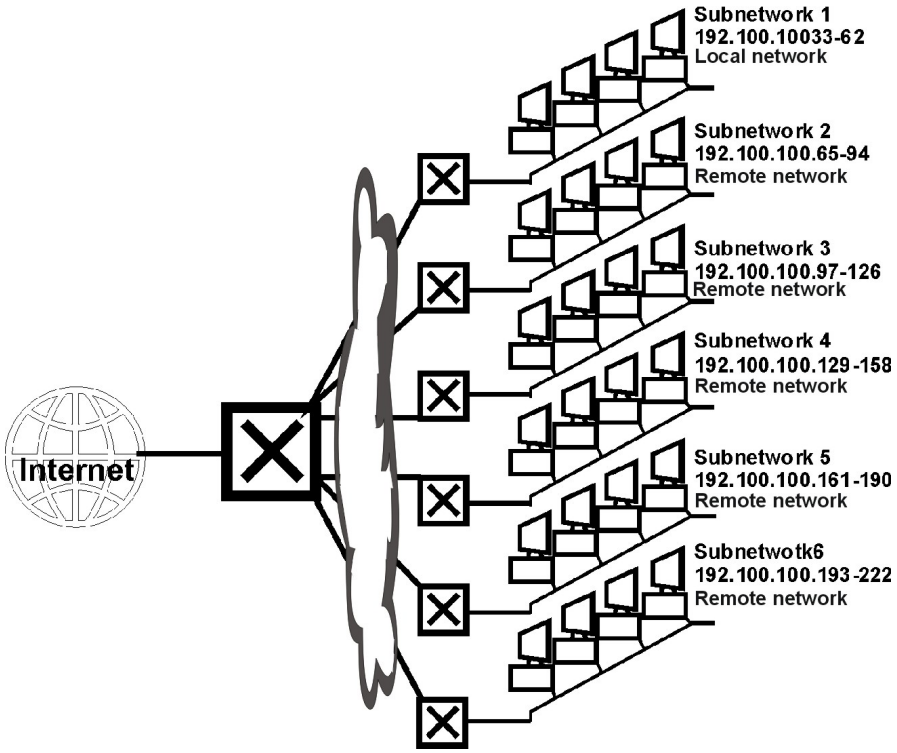


Figure 17.10
Network 192.100.100.0 after subnetting

17.3.10 Private vs Internet-unique IP addresses

If it is certain that a network will never be connected to the Internet, any IP address can be used as long as the IP addressing rules are followed. To keep things simple, it is advisable to use class C addresses.

Assign each LAN segment its own class C network number. Then it is possible to assign each host a complete IP address simply by appending the decimal host number to the decimal network number. With a unique class C network number for each LAN segment, there can be 254 hosts per segment.

If there is a possibility of connecting a network to the Internet, one should not use IP addresses that might result in address conflicts. In order to prevent such conflicts, either obtain Internet-unique IP addresses from an ISP, or use private IP addresses with address translation. The first method is the 'safest' one since none of the IP addresses will be used anywhere else on the Internet. The ISP may charge a fee for this privilege.

The second method of preventing IP address conflicts on the Internet is using addresses reserved for private networks. IANA has reserved several blocks of IP addresses for this purpose as shown in Figure 17.11

Class	From IP	To IP	Prefix
A	10.0.0.0	10.255.255.255	/8
B	172.16.0.0	172.31.255.255	/12
C	192.168.0.0	192.168.255.255	/16

Figure 17.11
Private IP addresses

Hosts on the Internet are not supposed to be assigned private IP addresses. Thus, if the network is eventually connected to the Internet, even if traffic from one of the hosts on the network somehow gets to the Internet, there should be no address conflicts. Furthermore, reserved IP addresses are not routed on the Internet because Internet routers are programmed not to forward messages sent to or from reserved IP addresses.

The disadvantage of using IP addresses reserved for private networks is that when a network is eventually connected to the Internet, all the hosts on that network might need to be re-configured. Each host will need to reconfigure with an Internet-unique IP address, or one will have to configure the connecting router to translate the reserved IP addresses into Internet-unique IP addresses that have been assigned by an ISP. For more information about IP addresses reserved for private networks, refer to RFC1918.

17.3.11 Classless addressing

Initially, the IPv4 Internet addresses were only assigned in classes A, B, and C. This approach turned out to be extremely wasteful, as large amounts of allocated addresses were not being used. Not only were the Class D and E address spaces under-utilized, but a company with 500 employees that was assigned a class B address would have 65,034 addresses that no-one else could use.

Presently, IPv4 addresses are considered classless. The issuing authorities simply hand down a block of contiguous addresses to ISPs, who can then issue them one by one, or break the large block up into smaller blocks for distribution to sub-ISPs, who will then repeat the process. Because of the fact that the 32 bit IPv4 addresses are no longer considered 'classful', the traditional distinction between classes A, B and C addresses and the implied boundaries between the NetID and HostID can be ignored. Instead, whenever an IPv4 network address is assigned to an organization, it is done in the form of a 32-bit network address and a corresponding 32-bit mask. The 'ones' in the mask cover the NetID, and the 'zeros' cover the HostID. The 'ones' always run contiguously from the left and are called the **prefix.**

An address of 202.13.3.12 with a mask of 11111111111111111111111111000000 ('ones' in the first 26 positions) would therefore be said to have a prefix of 26 and would be written as **202.13.13.12/26.**

The subnet mask in this case would be 255.255.255.192

Note that this address, in terms of the conventional classification, would have been regarded as a class C address and hence would have been assigned a prefix of /24 (subnet mask with 'ones' in the first 24 positions) by default.

17.3.12 Classless Inter-Domain Routing (CIDR)

A second problem with the fashion in which the IP addresses were allocated by the Network Information Center (NIC), was the fact that it was done more or less at random and that each address had to be advertised individually in the Internet routing tables.

Consider, for example, the case of the following four private ('traditional' class C) networks, each one with its own contiguous block of 256 (254 useable) addresses.

Network A: 200.100.0.0 (IP addresses 200.100.0.1 –200.100.0.254)
Network B: 192.33.87.0 (IP addresses 192.33.87.1 – 192.33.87.254)
Network C: 194.27.11.0 (IP addresses 194.27.11.1 – 194.27.11.254)
Network D: 202.15.16.0 (IP addresses 202.15.16.1 – 202.15.16.254)

If there were no reserved addresses, then the concentrating router at the ISP would have to advertise 4 × 256 = 1024 separate network addresses. In a real life situation, the ISP's router would have to advertise tens of thousands of addresses. It would also be seeing hundreds of thousands, if not millions, of addresses advertised by the routers of other ISPs across the globe. In the early Nineties, the situation was so serious it was expected that, by 1994, the routers on the Internet would no longer be able to cope with the multitude of routing table entries.

To alleviate this problem, the concept of classless inter-domain routing (CIDR) was introduced. CIDR removes the imposition of the class A, B and C address masks and allows the owner of a network to 'super-net' multiple addresses together. It then allows the concentrating router to aggregate (or 'combine') these multiple contiguous network addresses into a single route advertisement on the Internet.

Take the same example as before, but this time allocate contiguous addresses. Note that 'w' can have any value between 1 and 255 since the address classes are no longer relevant.

	w	**x**	**y**	**z**
Network A:	220.100.0.			0
Network B:	220.100.1.			0
Network C:	220.100.2.			0
Network D:	220.100.3.			0

CIDR now allows the router to advertise all 1000 computers under one advertisement, using the starting address of the block (220.100.0.0) and a CIDR (Supernet Mask) of 255.255.252.0. This is achieved as follows.

As with subnet masking, CIDR uses a mask, but it is less (shorter) than the network mask. Whereas the '1s' in the network mask indicate the bits that comprise the network ID, the '1s' in the CIDR (supernet) mask indicates the bits in the IP address that do not change. The total number of computers in this 'supernet' can be calculated as follows:

Number of '1s' in network (subnet) mask = 22
Number of hosts per network = $2^{(32-24)} = 2^8 = 256$ (minus 2 of course)
Number of '1s' in CIDR mask = 14
X= (Number of '1s' in network mask – number of '1s' in CIDR mask) = 2
Number of networks aggregated = $2^X = 2^2 = 4$
Total number of hosts = 4 × 256 = 1024

The route advertisement of 220.100.0.0 255.255.252.0 implies a supernet comprising 4 networks, each with 254 possible hosts. The lowest IP address is 220.100.0.1 and the highest is 220.100.3.254.

CIDR and the concept of classless addressing go hand in hand since it is obvious that the concept can only work if the ISPs are allowed to exercise strict control over this issue and on the allocation of IP addresses. Before the advent of CIDR, clients could obtain IP addresses and regard it as their 'property'. Under the new dispensation, the ISP needs to keep control over its allocated block(s) of IP addresses. A client can therefore only 'rent' IP addresses from an ISP and the latter may insist on its return, should the client decide to change to another ISP.

17.3.13 IPv4 header structure

The IP header is appended to the data that IP accepts from higher-level protocols, before routing it around the network. The IP header consists of five or six 32-bit "long words" and is made up as follows:

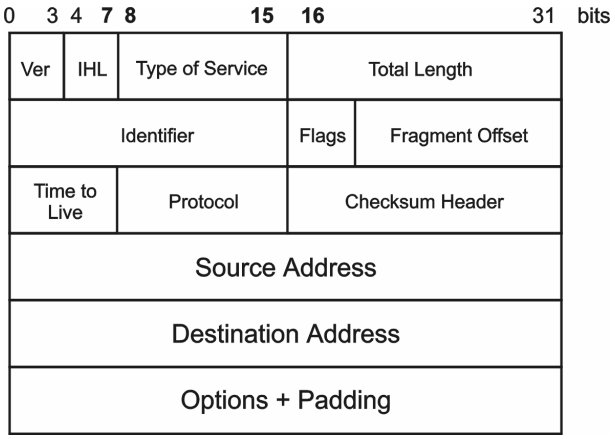


Figure 17.12
Structure of IPv4 header

Ver (4 bits)

The version field indicates the version of the IP protocol in use, and hence the format of the header. In this case, it is 4.

IHL (4 bits)

The Internet header length (IHL) is the length of the IP header in 32 bit 'long words', and thus points to the beginning of the data. This is necessary since the IP header can contain options and therefore has a variable length. The minimum value is 5, representing 5×4 = 20 bytes.

Type of service (8 bits)

The Type of Service (TOS) field is intended to provide an indication of the parameters of the quality of service desired. These parameters are used to guide the selection of the actual service parameters when transmitting a datagram through a particular network.

Some networks offer service precedence, which treats high precedence traffic as more important than other traffic (generally by accepting only traffic above a certain precedence at time of high load). The choice involved is a three-way trade-off between low delay, high reliability, and high throughput.

The TOS field is composed of a 3-bit precedence field (which is often ignored) and an unused (LSB) bit that must be 0.

The remaining 4 bits may only be turned on one at a time, and are allocated as follows:

- Bit 3: minimize delay
- Bit 4: maximize throughput
- Bit 5: maximize reliability
- Bit 6: minimize monetary cost

RFC 1340 (corrected by RFC 1349) specifies how all these bits should be set for standard applications. Applications such as TELNET and RLOGIN need minimum delay

since they transfer small amounts of data. FTP needs maximum throughput since it transfers large amounts of data. Network Management (SNMP) requires maximum reliability and Usenet News (NNTP) needs to minimize monetary cost.

Most TCP/IP implementations do not support the TOS feature, although some newer implementations of BSD and routing protocols such as OSPF and IS-IS can make routing decisions on it.

Total length (16 bits)

Total length is the length of the datagram, measured in bytes, including the header and data. Using this field and the header length, it can be determined where the data starts and ends. This field allows the length of a Datagram to be up to 2^{16} = 65,536 bytes, the maximum size of the segment handed down to IP from the protocol above it.

Such long datagrams are, however, impractical for most hosts and networks. All hosts must at least be prepared to accept datagrams of up to 576 octets (whether they arrive whole or in fragments). It is recommended that hosts only send datagrams larger than 576 octets if they have the assurance that the destination is prepared to accept the larger datagrams.

The number 576 is selected to allow a reasonable sized data block to be transmitted in addition to the required header information. For example, this size allows a data block of 512 octets plus 64 header octets to fit in a Datagram, which is the maximum size permitted by X.25. A typical IP header is 20 octets, allowing some space for headers of higher-level protocols.

Identification (16 bits)

This number uniquely identifies each datagram sent by a host. It is normally incremented by one for each datagram sent. In the case of fragmentation, it is appended to all fragments of the same datagram for the sake of reconstructing the original datagram at the receiving end. It can be compared to the 'tracking' number of an item delivered by registered mail or UPS.

Flags (3 bits)

There are two flags, the third bit is not used and remains '0':

- The DF (Don't Fragment) flag is set (=1) by the higher-level protocol (e.g. TCP) if IP is NOT allowed to fragment a datagram. If such a situation occurs, IP will not fragment and forward the datagram, but simply return an appropriate ICMP message to the sending host
- The MF (More Flag) is used as follows. If fragmentation DOES occur, MF=1 will indicate that there are more fragments to follow, whilst MF=0 indicates that it is the last fragment

Fragment offset (13 bits)

This field indicates where in the original datagram this fragment belongs. The fragment offset is measured in units of 8 bytes (64 bits). The first fragment has offset zero. In other words, the transmitted offset value is equal to the actual offset divided by eight. This constraint necessitates fragmentation in such a way that the offset is always exactly divisible by eight. The 13 bit offset also limits the maximum sized Datagram that can be fragmented to 64kB.

Time to live (8 bits)

The purpose of this field is to cause undeliverable datagrams to be discarded. Every router that processes a datagram must decrease the TTL by one and if this field contains the value zero, then the datagram must be destroyed.

The original design called for TTL to be decremented one for every second on the Internet (hence the 'time' to live). Currently all routers simply decrement it every time they pass a datagram.

Protocol (8 bits)

This field indicates the next (higher) level protocol used in the data portion of the Internet datagram, in other words the protocol that resides above IP in the protocol stack and which has passed the datagram on to IP. Typical values are 0x0806 for ARP and 0x8035 for RARP. (0x meaning 'hex'.)

Header checksum (16 bits)

This is a checksum on the header only, referred to as a 'standard Internet checksum'. Since some header fields change (e.g. TTL), this is recomputed and verified at each point that the IP header is processed. It is not necessary to cover the data portion of the datagram, as the protocols making use of IP, such as ICMP, IGMP, UDP and TCP, all have a checksum in their headers to cover their own header and data. To calculate it, the header is divided into 16 bit words. These words are then added together (normal binary addition with carry) one by one, and the interim sum stored in a 32-bit accumulator. When done, the upper 16 bits of the result is stripped off and added to the lower 16 bits. If, after this, there is a carry out to the 17th bit, it is carried back and added to bit 0. The result is then truncated to 16 bits.

Source and destination addresses (32 bit each)

These are the 32 bit IP addresses of both the origin and the destination of the datagram.

17.3.14 Packet fragmentation

It should be clear by now that IP might often have difficulty in sending packets across a network since, for example, Ethernet can only accommodate 1500 octets at a time and X.25 is limited to 576. This is where the fragmentation process comes into play. The relevant field here is 'fragment offset' (13 bits) while the relevant flags are DF and MF.

Consider a datagram consisting of an IP header followed by 3500 bytes of data. This cannot be transported over an Ethernet network, so it has to be fragmented in order to 'fit'.

The datagram will be broken up into three separate datagrams; each with their own IP header with the first two frames around 1500 bytes and the last fragment around 500 bytes. The three frames will travel to their destination independently, and will be recognized as fragments of the original datagram by virtue of the number in the identifier field. However, there is no guarantee that they will arrive in the correct order, and the receiver needs to reassemble them.

For this reason the fragment offset field indicates the distance or offset between the start of this particular fragment of data, and the starting point of the original frame. There is one problem that occurs, since only 13 bits are available in the header for the fragment offset (instead of 16). This offset is divided by 8 before transmission, and again multiplied by 8 after reception, requiring the data size (i.e. the offset) to be a multiple of 8 – so an offset of 1500 won't do. 1480 will be OK since it is divisible by 8. The data will be transmitted as fragments of 1480, and finally the remainder of 540 bytes. The fragment

offsets will be 0, 1480 and 2960 bytes respectively, or 0, 185 and 370 – after division by 8.

Incidentally, another reason why the data per fragment cannot exceed 1480 bytes for Ethernet is that the IP header has to be included for each datagram (otherwise individual datagrams will not be routable). Hence, 20 of the 1500 bytes have to be forfeited to the IP header.

The first frame will be transmitted with 1480 bytes of data, fragment offset = 0, and MF = 1

The second frame will be transmitted with the next 1480 bytes of data, fragment offset = 185, and MF = 1

The last third frame will be transmitted with 540 bytes of data, fragment offset = 370, MF = 0.

Some protocol analyzers will indicate the offset in hexadecimal; hence, it will be displayed as 0xb9 and 0x172, respectively.

For any given type of network, the packet size cannot exceed the so-called MTU (maximum transmission unit) for that type of network.

The following are some default values:

- 16 Mbps (IBM) Token Ring: 17914 (bytes)
- 4 Mbps (IEEE802.5) Token Ring: 4464
- FDDI: 4352
- Ethernet/ IEEE802.3: 1500
- X.25: 576
- PPP (low delay): 296

The fragmentation mechanism can be checked by doing a 'ping' across a network, and setting the data (–l) parameter to exceed the MTU value for the network.

17.4 Internet Protocol version 6 (IPv6/ IPng)

17.4.1 Introduction

IPng ('IP new generation'), as documented in RFC 1752, was approved by the Internet Engineering Steering Group in November 1994 and made a proposed standard. The formal name of this protocol is IPv6. After extensive testing, IANA gave permission for its deployment in mid-1999.

IPv6 is an update of IPv4, to be installed as a 'backwards compatible' software upgrade, with no scheduled implementation dates. It runs well on high performance networks such as ATM, and at the same time remains efficient, enough for low bandwidth networks such as wireless LANs. It also makes provision for Internet functions such as audio broadcasting and encryption.

Upgrading to and deployment of IPv6 can be achieved in stages. Individual IPv4 hosts and routers may be upgraded to IPv6 one at a time without affecting any other hosts or routers. New IPv6 hosts and routers can be installed one by one. There are no prerequisites to upgrading routers, but in the case of upgrading hosts to IPv6, the DNS server must first be upgraded to handle IPv6 address records.

When existing IPv4 hosts or routers are upgraded to IPv6, they may continue to use their existing address. They do not need to be assigned new IPv6 addresses; neither do administrators have to draft new addressing plans.

The simplicity of the upgrade to IPv6 is brought about through the transition mechanisms built into IPv6. They include the following:

- The IPv6 addressing structure embeds IPv4 addresses within IPv6 addresses, and encodes other information used by the transition mechanisms
- All hosts and routers upgraded to IPv6 in the early transition phase will be "dual" capable (i.e. implement complete IPv4 and IPv6 protocol stacks)
- Encapsulation of IPv6 packets within IPv4 headers will be used to carry them over segments of the end-to-end path where the routers have not yet been upgraded to IPv6

The IPv6 transition mechanisms ensure that IPv6 hosts can inter-operate with IPv4 hosts anywhere in the Internet up until the time when IPv4 addresses run out, and allows IPv6 and IPv4 hosts within a limited scope to inter-operate indefinitely after that. This feature protects the huge investment users have made in IPv4 and ensures that IPv6 does not render IPv4 obsolete. Hosts that need only a limited connectivity range (e.g., printers) need never be upgraded to IPv6.

17.4.2 IPv6 Header format

The header contains the following fields:

Ver (4 bits)

The IP version number, viz. 6.

Class (8 bits)

Class value. This replaces the 4-bit *priority* value envisaged during the early stages of the design and is used in conjunction with the Flow label.

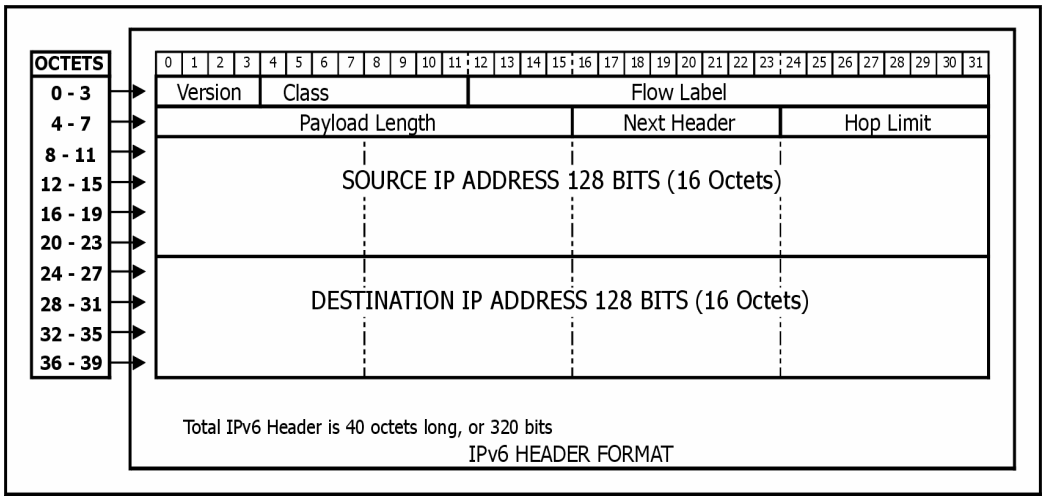


Figure 17.13
Structure of IPv6 header

Flow label (20 bits)

A flow is a sequence of packets sent from a particular source to a particular (unicast or multicast) destination for which the source desires special handling by the intervening routers. This is an optional field to be used if specific non-standard ('non-default') handling. It is required to support applications that require some degree of consistent throughput in order to minimize delay and/or jitter. These types of applications are commonly described as 'multi-media' or 'real-time' applications.

The flow label will affect the way the packets are handled but will not influence the routing decisions.

Payload length (16 bits)

The payload is the rest of the packet following the IPv6 header, in octets. The maximum payload that can be carried behind a standard IPv6 header cannot exceed 65,536 bytes. With an extension header, this is possible. The datagram is then referred to as a jumbo datagram. Payload length differs slightly from the IPv4 in that the 'total length' field does not include the header.

Next HDR (8 bits)

This identifies the type of header immediately following the IPv6 header, using the same values as the IPv4 protocol field. Unlike IPv4, where this would typically point to TCP or UDP, this field could either point to the next protocol header (TCP) or to the next IPv6 extension header.

Hop limit (8 bits)

This is an unsigned integer, similar to TTL in IPv4. It is decremented by 1 by each node that forwards the packet. The packet is discarded if the hop limit is decremented to zero.

Source address (128 bits)

This is the address of the initial sender of the packet.

Destination address (128 bits)

This is the address of the intended recipient of the packet, which is not necessarily the ultimate recipient, if an optional routing header is present.

17.4.3 IPv6 extensions

IPv6 includes an improved option mechanism over IPv4. Instead of placing extra options bytes within the main header, IPv6 options are placed in separate extension headers that are located between the IPv6 header and the transport layer header in a packet.

Most IPv6 extension headers are not examined or processed by routers along a packet's path until it arrives at its final destination. This leads to a major improvement in router performance for packets containing options. In IPv4, the presence of any options requires the router to examine all options.

IPv6 extension headers can be of arbitrary length and the total amount of options carried in a packet is not limited to 40 bytes as with IPv4. They are also not carried within the main header, as with IPv4, but are only used when needed, and are carried behind the main header. This feature plus the manner in which they are processed, permits IPv6 options to be used for functions, which were not practical in IPv4.

A good example of this is the IPv6 authentication and security encapsulation options. In order to improve the performance when handling subsequent option headers and the transport protocol which follows, IPv6 options are always an integer multiple of 8 octets long, in order to retain this alignment for subsequent headers.

The IPv6 extension headers currently defined are:

- Routing header (for extended routing, similar to the IPv4 loose source route)
- Fragment header (for fragmentation and re-assembly)
- Authentication header (for integrity and authentication)
- Encrypted security payload (for confidentiality)

- Hop-by-hop options header (for special options that require hop by hop processing)
- Destination options header (for optional information to be examined by the destination node)

17.4.4 IPv6 addresses

IPv6 addresses are 128-bits long and are identifiers for individual interfaces or sets of interfaces. IPv6 addresses of all types are assigned to *interfaces* (i.e. network interface cards) and NOT to nodes i.e. hosts. Since each interface belongs to a single node, any of that node's interfaces' unicast addresses may be used as an identifier for the node. A single interface may be assigned multiple IPv6 addresses of any type.

There are three types of IPv6 addresses. These are unicast, anycast, and multicast.

- **Unicast** addresses identify a single interface
- **Anycast addresses** identify a set of interfaces such that a packet sent to an anycast address will be delivered to one member of the set
- **Multicast** addresses identify a group of interfaces, such that a packet sent to a multicast address is delivered to all of the interfaces in the group. There are no broadcast addresses in IPv6, their function being superseded by multicast addresses

The IPv6 address is four times the length of IPv4 addresses (128 bit vs 32 bit). This is 4 billion times 4 billion (2^{96}) times the size of the IPv4 address space (2^{32}). This works out to be 340,282,366,920,938,463,463,374,607,431,768,211,456. Theoretically, this is approximately 665,570,793,348,866,943,898,599 addresses per square meter of the surface of the planet Earth (assuming the earth surface is 511,263,971,197,990 square meters).

In more practical terms, considering that the creation of addressing hierarchies, which reduces the efficiency of the usage of the address space, IPv6 is still expected to support between 8×10^{17} to 2×10^{33} nodes. Even the most pessimistic estimate provides around 1,500 addresses per square meter of the surface of the Earth.

The leading bits in the address indicate the specific type of IPv6 address. The variable length field comprising these leading bits is called the Format Prefix (FP). The initial allocation of these prefixes is as follows:

Allocation	Prefix (Binary)	Fraction of Address Space
Reserved	0000 0000	1/256
Unassigned	0000 0001	1/256
Reserved for NSAP Allocation	0000 001	1/128
Reserved for IPX Allocation	0000 010	1/128
Unassigned	0000 011	1/128
Unassigned	0000 1	1/32
Unassigned	0001	1/16
Unassigned	001	1/8
Provider-Based Unicast Address	010	1/8
Unassigned	011	1/8
Reserved for Neutral-Interconnect-Based Unicast Addresses	100	1/8
Unassigned	101	1/8
Unassigned	110	1/8
Unassigned	1110	1/16
Unassigned	1111 0	1/32
Unassigned	1111 10	1/64
Unassigned	1111 110	1/128
Unassigned	1111 1110 0	1/512
Link Local Addresses	1111 1111 10	1/1024
Site Local Use Addresses	1111 1110 11	1/1024
Multicast Addresses	1111 1111	1/256

Figure 17.14
Allocation of format prefixes in IPv6

Approximately fifteen percent of the address space is initially allocated. The remaining 85% is reserved for future use.

Unicast addresses

There are several forms of unicast address assignment in IPv6. These are:

- Aggragatable global unicast addresses
- Unspecified addresses
- Loopback addresses
- IPv4-based addresses
- Site Local addresses
- Link Local addresses

Global unicast addresses

These addresses are used for global communication. They are similar in function to IPv4 addresses under CIDR. Their format is:

3 bits	13 bits	32 bits	16 bits	64 bits
001	TLA	NLA	SLA	Interface ID

Figure 17.15
Global unicast addresses

The first 3 bits identify the address as a global unicast address. The next, 13-bit, field (TLA) identifies the top level aggregator. This number will be used to identify the relevant Internet ‘exchange point’, or long-haul (‘backbone’) provider. These numbers

(8,192 of them) will be issued by IANA, to be further distributed via the three regional registries (ARIN, RIPE and APNIC), who could possibly further delegate the allocation of sub-ranges to national or regional registries such as the French NIC managed by INRIA for French networks.

The third, 32-bit, field (NLA) identifies the next level aggregator. This will be structured by long-haul providers to identify a second-tier provider by means of the first n bits, and to identify a subscriber to that second-tier provider by means of the remaining 32-n bits.

The fourth, 16-bit, field is the SLA or site local aggregator. This will be allocated to a link within a site, and is not associated with a registry or service provider. In other words, it will remain unchanged despite a change of service provider. Its closest equivalent in IPv4 would be the 'NetID'.

The last field is the 64-bit interface ID. This is the equivalent of the 'HostID' in IPv4. However, instead of an arbitrary number it would consist of the hardware address of the interface, e.g. the Ethernet MAC address.

- All identifiers will be 64-bits long even if there are only a few devices on the network, and
- Where possible these identifiers will be based o the IEEE EUI-64 format. Existing 48-bit MAC addresses are converted to EUI-64 format by splitting them in the middle and inserting the string FF-FE in between the two halves

Unspecified address

This can be written as 0:0:0:0:0:0:0:0, or simply "::" (double colon). This address can be used as a source address by a station that has not yet been configured with an IP address. It can never be used as a destination address. This is similar to 0.0.0.0 in IPv4

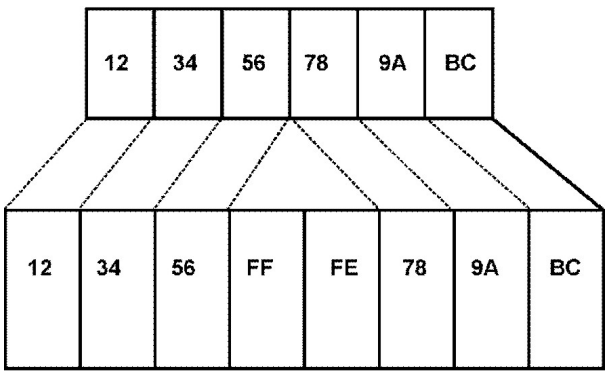


Figure 17.16
Converting a 48-bit MAC address to EUI-64 format

Loopback addresses

The loopback address 0:0:0:0:0:0:0:1 can be used by a node to send a datagram to itself. It is similar to the 127.0.0.1 of IPv4.

IPv4-based addresses

It is possible to construct an IPv6 address out of an existing IPv4 address. This is done by prepending 96 zero bits to a 32-bit IPv4 address. The result is written as 0:0:0:0:0:0:192.100.100.3, or simply::192.100.100.3.

Site local unicast addresses

Site local unicast addresses are equivalent to the IPv4 private addresses. The site local addressing prefix 1111 1110 11 has been reserved for this purpose. A typical site local address will consist of this prefix, a set of 38 zeros, a subnet ID, and the interface identifier. Site local addresses cannot be routed in the Internet, but only between two stations on a single site.

The last 80 bits of a site local unicast address is identical to the last 80 bits of an aggregatable address. This allows for easy renumbering where a site has to be connected to the Internet.

Link local unicast addresses

Stations that are not yet configured with either a provider-based address or a site local address may use link local unicast addresses. Theses are composed of the link local prefix, 1111 1110 10, a set of 0s, and an interface identifier. These addresses can only be used by stations connected to the same local network and packets addressed in this way cannot traverse a router.

Anycast addresses

An IPv6 anycast address is an address that is assigned to more than one interface (typically belonging to different nodes), with the property that a packet sent to an anycast address is routed to the 'nearest' interface having that address, according to the routing protocols' measure of distance. Anycast addresses, when used as part of a route sequence, permit a node to select which of several Internet Service Providers it wants to carry its traffic. This capability is sometimes called 'source selected policies'.

This would be implemented by configuring anycast addresses to identify the set of routers belonging to Internet service providers (e.g. one anycast address per Internet service provider). These anycast addresses can be used as intermediate addresses in an IPv6 routing header, to cause a packet to be delivered via a particular provider or sequence of providers.

Other possible uses of anycast addresses are to identify the set of routers attached to a particular subnet, or the set of routers providing entry into a particular routing domain.

Anycast addresses are allocated from the unicast address space, using any of the defined unicast address formats. Thus, anycast addresses are syntactically indistinguishable from unicast addresses. When a unicast address is assigned to more than one interface, thus turning it into an anycast address, the nodes to which the address is assigned must be explicitly configured to know that it is an anycast address.

Multicast addresses

An IPv6 multicast address is an identifier for a group of interfaces. An interface may belong to any number of multicast groups. Multicast addresses have the following format:

The **11111111** (0xFF) at the start of the address identify the address as being a multicast.

- **FLAGS**. Four bits are reserved for Flags. The first 3 bits are currently reserved, and set to 0. The last bit (the one on the right) is called T for 'transient'. T = 0 indicates a permanently assigned ('well-known') multicast address, assigned by IANA, while T = 1 indicates a non-permanently assigned ('transient') multicast address
- **SCOP** is a 4-bit multicast scope value used to limit the scope of the multicast group, for example to ensure that packets intended for a local videoconference are not spread across the Internet

  The values are:

0	Reserved	8	Organization local scope
1	Interface local scope	9	(unassigned)
2	Link local scope	A	(unassigned)
3	Subnet local scope	B	(unassigned)
4	Admin. local scope	C	(unassigned)
5	Site local scope	D	(unassigned)
6	(unassigned)	E	Global scope
7	(unassigned)	F	Reserved

- **GROUP ID** identifies the multicast group, either permanent or transient, within the given scope. Permanent Group IDs are assigned by IANA

The following example shows how it all fits together. In the multicast address FF:08::43 points to all NTP servers in a given organization, in the following way:

- FF indicates that this is a multicast address
- 0 indicates that the T flag is set to 0, i.e. this is a permanently assigned multicast address
- 8 points to all interfaces in the same organization as the sender (see SCOP options above)
- Group ID = 43 has been permanently assigned to Network Time Protocol (NTP) servers

17.5 Address resolution protocol (ARP) and reverse address resolution protocol (RARP)

17.5.1 Introduction

ARP is used with IPv4. Initially the designers of IPv6 assumed that it would use ARP as well, but subsequent work by the SIP, SIPP and IPv6 working groups led to the development of the IPv6 'neighbor discovery' procedures that encompass ARP, as well as those of router discovery.

Some network technologies make address resolution difficult. Ethernet interface boards, for example, come with built-in 48-bit hardware addresses.

This creates several difficulties:

- No simple correlation, applicable to the whole network, can be created between physical (MAC) addresses and Internet protocol (IP) addresses
- When the interface board fails and has to be replaced the IP address then has to be remapped to a different MAC address
- The MAC address is too long to be encoded into the 32-bit IP address

To overcome these problems in an efficient manner, and to eliminate the need for applications to know about MAC addresses, the address resolution protocol (ARP) (RFC 826) resolves addresses dynamically.

When a host wishes to communicate with another host on the same physical network, it needs the destination MAC address in order to compose the basic level 2 frame. If it does not know what the destination MAC address is, but has its IP address, it broadcasts a special type of datagram in order to resolve the problem. This is called an address resolution protocol (ARP) request. This datagram requests the owner of the unresolved IP

address to reply with its MAC address. All hosts on the network will receive the broadcast, but only the one that recognizes its own IP address will respond.

While the sender could just broadcast the original datagram to all hosts on the network, this would impose an unnecessary load on the network, especially if the datagram was large. A small ARP request, followed by a small address resolution protocol (ARP) reply, followed by a direct transmission of the original datagram, is a much more efficient way of resolving the problem.

17.5.2 Address resolution cache

Because communication between two computers usually involves transfer of a succession of datagrams, it is prudent for the sender to 'remember' the MAC information it receives – at least for a while. Thus, when the sender receives an ARP reply, it stores the MAC address it receives as well as the corresponding IP address in its ARP cache. Before sending any message to a specific IP address it checks first to see if the relevant address binding is in the cache. This saves it from repeatedly broadcasting identical ARP requests.

The ARP cache holds 4 fields of information for each device:

- **IF index:** the number of the entry in the table
- **Physical address:** the MAC address of the device
- **Internet protocol (IP) address:** the corresponding IP address
- **Type:** the type of entry in the ARP cache

There are 4 possible types:

- 4 = static – the entry will not change
- 3 = dynamic – the entry can change
- 2 = the entry is invalid
- 1 = none of the above

ARP header

Fields of an ARP Header are listed below:

Hardware type (16 bits)

Specifies the hardware interface type of the target, e.g.:

- 1 = Ethernet
- 3 = X.25
- 4 = Token Ring
- 6 = IEEE 802.x
- 7 = ARCnet

Protocol type (16 bits)

Specifies the type of high-level protocol address the sending device is using. For example,

204810 (0x800): IP
205410 (0x806): ARP
328210 (0xcd2): RARP

HA length (8 bits)

This is the length, in bytes, of the hardware (MAC) address. For Ethernet it is 6.

PA length (8 bits)
This is the length, in bytes, of the internetwork protocol address.
For IP it is 4.

Operation (16 bits)
This indicates the type of ARP datagram:

- 1 = ARP request
- 2 = ARP reply
- 3 = RARP request
- 4 = RARP reply

Sender HA (48 bits)
This is the hardware (MAC) address of the sender.

Sender PA (48 bits)
This is the (internetwork) protocol address of the sender.

Target HA (48 bits)
This is the hardware (MAC) address of the target host

Target PA (48 bits)
This is the (internetwork) protocol address of the target host. Because of the use of fields to indicate the lengths of the hardware and protocol addresses, the address fields can be used to carry a variety of address types, making ARP applicable to a number of different types of network.

The broadcasting of ARP requests presents some potential problems. Networks such as Ethernet employ connectionless delivery systems i.e. the sender does not receive any feedback as to whether datagrams it has transmitted were received by the target device.

If the target is not available, the ARP request destined for it will be lost without trace and no ARP response will be generated. Thus, the sender must be programmed to retransmit its ARP request after a certain time, and must be able to store the Data gram it is attempting to transmit in the interim. It must also remember what requests it has sent out so that it does not send out multiple ARP requests for the same address. If it does not receive an ARP reply, it will eventually have to discard the outgoing datagrams.

Because it is possible for a machine's hardware address to change, as happens when an Ethernet interface fails and has to be replaced, entries in an ARP cache have a limited life span after which they are deleted. Every time a machine with an ARP cache receives an ARP message, it uses the information to update its own ARP cache. If the incoming address binding already exists, it overwrites the existing entry with the fresh information and resets the timer for that entry.

The host trying to determine another machine's MAC address will send out an ARP request to that machine. In the datagram it will set Operation = 1 (ARP request), and insert its own IP and MAC addresses as well as the destination machine's IP address in the header. The field for the destination machine's MAC address will be left at zero. It will then broadcast this message using all 'ones' in the destination address of the LLC frame so that all hosts on that subnet will 'see' the request.

If a machine is the target of an incoming ARP request, its own ARP software will reply. It swaps the target and sender address pairs in the ARP datagram (both HA and PA), inserts its own MAC address into the relevant field, changes the operation code to 2 (ARP reply), and sends it back to the requesting host.

17.5.3 Proxy ARP

Proxy ARP enables a router to answer ARP requests made to a destination node that is not on the same subnet as the requesting node. Assume that a router connects two subnets, A and B. If host A1 on subnet A tries to send an ARP request to host B1 on subnet B, this would normally not work as an ARP can only be performed between hosts on the same subnet (where all hosts can 'see' and respond to the FF:FF:FF:FF:FF:FF broadcast MAC address). The requesting host, A1, would therefore not get a response.

If proxy ARP has been enabled on the router, it will recognize this request and issue its own ARP request, on behalf of A1, to B1. Upon obtaining a response from B1, it would report to A1 on behalf of B1. It must be understood that the MAC address returned to A1 will not be that of B1, but rather that of the router NIC connected to subnet A, as this is the physical address where A1 will send data destined for B1.

17.5.4 Gratuitous ARP

Gratuitous ARP occurs when a host sends out an ARP request looking for its own address. This is normally done at the time of boot-up. This can be used for two purposes.

Firstly, a host would not expect a response to the request. If a response does appear, it means that another host with a duplicate IP address exists on the network. Secondly, any host observing an ARP request broadcast will automatically update its own ARP cache if the information pertaining to the destination node already exists in its cache. If a specific host is therefore powered down and the NIC replaced, all other hosts with the powered down host's IP address in their caches will update when the host in question is re-booted.

17.5.5 Reverse address resolution protocol (RARP)

As its name suggests, the reverse address resolution protocol (RARP) (RFC 903) does the opposite to ARP. It is used to obtain an IP address when the physical address is known.

Usually, a machine holds its own IP address on its hard drive, where the operating system can find it on start-up. However, a diskless workstation is only aware of its own hardware address and has to recover its IP address from an address file on a remote server at start-up. It uses RARP to retrieve its IP address.

A diskless workstation broadcasts an RARP request on the local network using the same datagram format as an ARP request. It has, however, an Opcode of 3 (RARP request), and identifies itself as both the sender and the target by placing its own physical address in both the sender hardware address field and the target hardware address field.

Although the RARP request is broadcast, only a RARP server (i.e. a machine holding a table of addresses and programmed to provide RARP services) can generate a reply.

There should be at least one RARP server on a network, often there are more. The RARP server changes the Opcode to 4 (RARP reply). It then inserts the missing address in the target IP address field, and sends the reply directly back to the requesting machine. The requesting machine then stores it in memory until the next time it reboots.

All RARP servers on a network will reply to an RARP request, even though only one reply is required. The RARP software on the requesting machine sets a timer when sending a request and retransmits the request if the timer expires before a reply has been received.

On a best-effort local area network, such as Ethernet, the provision of more than one RARP server reduces the likelihood of RARP replies being lost or dropped because the server is down or overloaded. This is important because a diskless workstation often requires its own IP address before it can complete its bootstrap procedure.

To avoid multiple and unnecessary RARP responses on a broadcast-type network such as Ethernet, each machine on the network is assigned a particular server, called its primary RARP server. When a machine broadcasts an RARP request, all servers will receive it and record its time of arrival, but only the primary server for that machine will reply. If the primary server is unable to reply for any reason, the sender's timer will expire, it will rebroadcast its request and all non-primary servers receiving the rebroadcast so soon after the initial broadcast will respond.

Alternatively, all RARP servers can be programmed to respond to the initial broadcast, with the primary server set to reply immediately, and all other servers set to respond after a random time delay. The retransmission of a request should be delayed long enough for these delayed RARP replies to arrive.

RARP has several drawbacks. It has to be implemented as a server process. It is also prudent to have more than one server, since no diskless workstation can boot up if the single RARP server goes down. In addition to this, very little information (only an IP address) is returned. Finally, RARP uses a MAC address to obtain an IP address, hence it cannot be routed.

17.6 Internet control message protocol (ICMP)

Errors occur in all networks. These arise when destination nodes fail, or become temporarily unavailable, or when certain routes become overloaded with traffic. A message mechanism called the **Internet control message protocol** (ICMP) is incorporated into the TCP/IP protocol suite to report errors and other useful information about the performance and operation of the network.

17.6.1 ICMP Message structure

ICMP communicates between the Internet layers on two nodes and is used by both gateways (routers) and individual hosts. Although ICMP is viewed as residing within the Internet layer, its messages travel across the network encapsulated in IP datagrams in the same way as higher layer protocol (such as TCP or UDP) datagrams. This is done with the Protocol field in the IP header set to 0x1, indicating that an ICMP datagram is being carried.

The reason for this approach is that, due to its simplicity, the ICMP header does not include any IP address information and is therefore in itself not routable. It therefore has little choice but to rely in IP for delivery. The ICMP message, consisting of an ICMP header and ICMP data, is encapsulated as 'data' within an IP datagram with the resultant structure indicated in the figure below.

The complete IP datagram, in turn, has to depend on the lower network interface layer (for example, Ethernet) and is thus contained as a payload within the Ethernet data area.

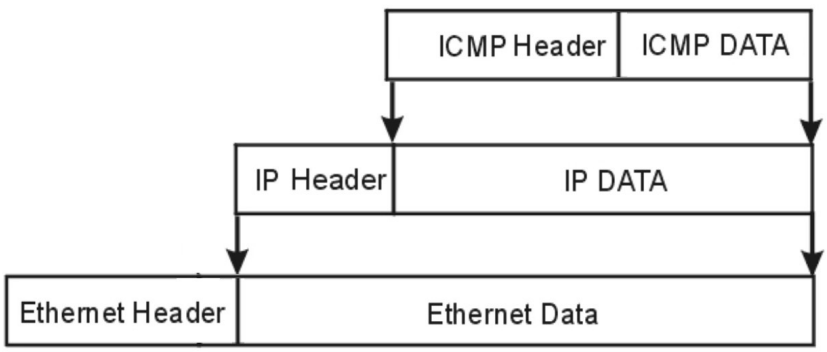


Figure 17.17
Encapsulation of ICMP message

17.6.2 ICMP applications

The various uses for ICMP include:

- Exchanging messages between hosts to synchronize clocks
- Exchanging subnet mask information
- Informing a sending node that its message will be terminated due to an expired TTL
- Determining whether a node (either host or router) is reachable
- Advising routers of better routes
- Informing a sending host that its messages are arriving too fast and that it should back off

There is a variety of ICMP messages, each with a different format, yet the first 3 fields as contained in the first 4 bytes or 'long word' is the same for all.

The three common fields are:

- ICMP message type (8 bits)
- Code (8 bits)
- Checksum (16 bits)

ICMP message type (8 bits)

This is a code that identifies the type of ICMP message, and, **a** code in which interpretation depends on the type of ICMP message.

The various codes for type fields and their descriptions are:

- 0 Echo, reply
- 3 Destination unreachable
- 4 Source quench
- 5 Redirect (change a route
- 8 Echo request
- 11 time exceeded (datagram)
- 12 Parameter problem (datagram)
- 13 Time stamp request
- 14 Time stamp reply
- 17 Address mark request
- 18 Address mark reply

Code (8 bits)

This is a code in which the interpretation depends on the type of ICMP message.

Checksum (16 bits)

This is a 16-bit checksum that is calculated on the entire ICMP datagram.

ICMP messages can be further subdivided into two broad groups viz. ICMP error messages and ICMP query messages as follows.

ICMP error messages:

- Destination Unreachable

- Time Exceeded
- Invalid Parameters
- Source Quench
- Redirect

ICMP query messages:

- Echo Request and Reply Messages
- Timestamp Request and Reply Messages
- Subnet Mask Request and Reply Messages

Too many ICMP error messages in the case of a network experiencing errors due to heavy traffic can exacerbate the problem, hence the following conditions apply:

- No ICMP messages are generated in response to ICMP messages
- No ICMP error messages are generated for multicast frames

ICMP error messages are only generated for the first frame in a series of segments

17.7 Routing protocols

17.7.1 Routing basics

Unlike the host-to-host layer protocols (e.g. TCP), which control end-to-end communications, the Internet layer protocol (IP) is rather 'short-sighted'. Any given IP node (host or router) is only concerned with routing (switching) the datagram to the *next* node, where the process is repeated. Very few routers have knowledge about the entire internetwork, and often the datagrams are forwarded based on default information without any knowledge of where the destination actually is.

Before discussing the individual routing protocols in any depth, the basic concepts of IP routing have to be clarified. This section will discuss the concepts and protocols involved in routing.

17.7.2 Direct vs indirect delivery

When the source host prepares to send a message to another host, a fundamental decision has to be made, namely: is the destination host also resident on the local network or not? If the NetID portions of the IP address match, the source host will assume that the destination host is resident on the same network, and will attempt to forward it locally. This is called direct delivery.

If not, the message will be forwarded to the local default gateway (i.e. the local router), which will forward it. This is called indirect delivery. The process will now be repeated. If the router can deliver it directly i.e. the host resides on a network directly connected to the router, it will. If not, it will consult its routing tables and forward it to the next appropriate router.

This process will repeat itself until the packet is delivered to its final estimation.

17.7.3 Static versus dynamic routing

Each router has a table with the following format:

Active Routes for 207.194.66.100:

Network Address	Netmask	Gateway Address	Interface	Metric
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1

207.194.66.0	255.255.255.224	207.194.66.100	207.194.66.100	1
207.194.66.100	255.255.255.255	127.0.0.1	127.0.0.1	1
207.194.66.255	255.255.255.255	207.194.66.100	207.194.66.100	1
224.0.0.0	224.0.0.0	207.194.66.100	207.194.66.100	1
255.255.255.255	255.255.255.255	207.194.66.100	0.0.0.0	1

It reads as follows: 'If a packet is destined for network 207.194.66.0, with a Netmask of 255.255.255.224, then forward it to the router port: 207.194.66.100', etc. It is logical that a given router cannot contain the whereabouts of each network in the world in its routing tables; hence, it will contain default routes as well. If a packet cannot be specifically routed, it will be forwarded on a default route, which should (hopefully) move it closer to its intended destination.

These routing tables can be maintained in two ways. In most cases, the routing protocols will do this automatically. The routing protocols are implemented in software that runs on the routers, enabling them to communicate on a regular basis and allowing them to share their 'knowledge' about the network with each other. In this way, they continuously 'learn' about the topology of the system, and upgrade their routing tables accordingly. This process is called *dynamic* routing.

If, for example, a particular router is removed from the system, the routing tables of all routers containing a reference to that router will change. However, because of the interdependence of the routing tables, a change in any given table will initiate a change in many other routers and it will be a while before the tables stabilize. This process is known as *convergence.*

Dynamic routing can be further sub-classified as Distance Vector, Link-State, or Hybrid, depending on the method by which the routers calculate the optimum path.

In distance vector dynamic routing, the 'metric' or yardstick used for calculating the optimum routes is simply based on distance, i.e. which route results in the least number of 'hops' to the destination. Each router constructs a table, which indicates the number of hops to each known network. It then periodically passes copies of its tables to its immediate neighbours. Each recipient of the message then simply adjusts its own tables based on the information received from its neighbour.

The major problem with the distance vector algorithm is that it takes some time to converge to a new understanding of the network. The bandwidth and traffic requirements of this algorithm can also affect the performance of the network. The major advantage of the distance vector algorithm is that it is simple to configure and maintain as it only uses the distance to calculate the optimum route.

Link-state routing protocols are also known as shortest path first protocols. This is based on the routers exchanging link-state advertisements to the other routers. Link-state advertisement messages contain information about error rates and traffic densities and are triggered by events rather than running periodically as with the distance routing algorithms.

Hybridized routing protocols use both the methods described above and are more accurate than the conventional distance vector protocols. They converge more rapidly to an understanding of the network than distance vector protocols and avoid the overheads of the link-state updates. The best example of this one is the enhanced interior routing protocol (EIGRP).

It is also possible for a network administrator to make *static* entries into routing tables. These entries will not change, even if a router that they point to is not operational.

17.7.4 Autonomous systems

For routing a TCP/IP-based Internet, work can be divided into several autonomous systems (ASs) or domains. An autonomous system consists of hosts, routers and data links that form several physical networks that are administered by a single authority such as a service provider, university, corporation, or government agency.

Autonomous systems can be classified under one of three categories:

- **Stub AS**: This is an AS that has only one connection to the 'outside world' and therefore does not carry any third-party traffic. This is typical of a smaller corporate network
- **Multi-homed non-transit AS**: This is an AS that has two or more connections to the 'outside world' but is not set up to carry any third party traffic. This is typical of a larger corporate network
- **Transit AS**: This is an AS with two or more connections to the outside world, and is set up to carry third party traffic. This is typical of an ISP network

Routing decisions that are made within an AS are totally under the control of the administering organization. Any routing protocol, using any type of routing algorithm, can be used within an AS since the routing between two hosts in the system is completely isolated from any routing that occurs in other ASs. Only if a host within one AS communicates with a host outside the system, will another AS (or ASs) and possibly the Internet backbone be involved.

17.7.5 Interior, exterior and gateway-to-gateway protocols

There are three categories of TCP/IP routing protocols, namely interior gateway protocols (IGPs), exterior gateway protocols (EGPs), and gateway-to-gateway protocols (GGPs).

Two routers that communicate directly with one another and are both part of the same AS are said to be interior neighbours and are called interior gateways. They communicate with each other using interior gateway protocols (IGPs).

In a simple AS consisting of only a few physical networks, the routing function provided by IP may be sufficient. In larger ASs, however, sophisticated routers using adaptive routing algorithms may be needed. These routers will communicate with each other using IGPs such as RIP, Hello, IS-IS or OSPF.

Routers in different ASs, however, cannot use IGPs for communication for more than one reason. Firstly, IGPs are not optimized for long-distance path determination.

Secondly, the owners of ASs (particularly Internet service providers) would find it unacceptable for their routing metrics (which include sensitive information such as error rates and network traffic) to be visible to their competitors. For this reason routers that communicate with each other and are resident in different ASs communicate with each other using EGPs.

The routers on the periphery, connected to other ASs, must be capable of handling both the appropriate IGPs and EGPs.

The most common EGP currently used in the TCP/IP environment is the Border Gateway Patrol (BGP), the current version being BGP-4. A third type of routing protocol is used by the core routers (gateways) that connect users to the Internet backbone. They use Gateway-to-Gateway protocols (GGPs) to communicate with each other.

17.8 Interior gateway protocols

The protocols that will be discussed are RIPv2 (Routing Information Protocol version 2), EIGRP (Enhanced Interior Gateway Routing Protocol), and OSPF (Open Shortest Path First).

RIPv2

RIPv2 originally saw the light as RIP (RFC 1058, 1388) and is one of the oldest routing protocols. The original RIP had a shortcoming in that it could not handle variable length subnet masks, and hence could not support CIDR. This capability has been included with RIPv2.

RIPv2 is a distance vector routing protocol where each router, using a special packet to collect and share information about distances, keeps a routing table of its perspective of the network showing the number of hops required to reach each network. RIP uses the hop counts as a metric (i.e. form of measurement).

In order to maintain their individual perspective of the network, routers periodically pass copies of their routing tables to their immediate neighbors. Each recipient adds a distance vector to the table and forwards the table to its immediate neighbors. The hop count is incremented by one every time the packet passes through a router. RIP only records one route per destination (even if there are more).

The RIP routers have fixed update intervals and each router broadcasts its entire routing table to other routers at 30-second intervals (60 seconds for Netware RIP).

Each router takes the routing information from its neighbor, adds or subtracts one hop to the various routes to account for itself and then broadcasts its updated table. Every time a router entry is updated, the timeout value for the entry is reset. If an entry has not been updated within 180 seconds, it is assumed suspect, the hop field set to 16 to mark the route as unreachable, and it is later removed from the routing table.

One of the major problems with distance vector protocols like RIP is the convergence time, which is the time it takes for the routing information on all routers to settle in response to some change to the network. For a large network, the convergence time can be long and there is a greater chance of frames being misrouted.

RIPv2 (RFC1723) also supports:

- **Authentication**: This prevents a routing table from being corrupted with incorrect data from a bad source.
- **Subnet masks:** The IP address and its subnet mask enable the RIPv2 to identify the type of destination that the route leads to. This enables it to discern the network subnet from the host address.
- **IP identification:** This makes RIPv2 more effective than RIP as it prevents unnecessary hops. This is useful where multiple routing protocols are used simultaneously and some routes may never be identified. The IP address of the next hop router would be passed to neighboring routers via routing table updates. These routers would then force datagrams to use a specific route whether or not that route had been calculated to be the optimum route or not using least-hop-count.
- **Multicasting of RIPv2 messages:** This is a method of simultaneously advertising routing data to multiple RIP or RIPv2 devices. This is useful when multiple destinations must receive identical information

EIGRP

EIGRP is an enhancement of the original IGRP, a proprietary routing protocol developed by Cisco Systems for use on the Internet. IGRP is outdated since it cannot handle CIDR and variable-length subnet masks.

EIGRP is a link-state routing protocol that uses a composite metric for route calculations. It allows for multi-path routing, load balancing across 2, 3 or 4 links, and automatic recovery from a failed link. Since it does not only take hop count into consideration, it has better real time appreciation of the link status between routers and is more flexible than RIP. Like RIP it completely broadcasts routing table updates, but at 90 second intervals.

Each of the metrics used in the calculation of the distance vectors has a weighting factor. The metrics used in the calculation are as follows:

- Hop count – unlike RIP, EIGRP does not stop at 16 hops and can operate up to a maximum of 255
- Packet size (MTU)
- Link bandwidth
- Delay
- Loading
- Reliability

The metric used is:

Metric = K1 * bandwidth + (K2 * bandwidth)/(256 – Load) + K3 * Delay.

(K1, K2 and K3 are weighting factors.)

Reliability is also added in using the metric:

Metric modified = Metric * K5/(reliability + K4). This modifies the existing metric calculated in the first equation above.

One of the key design parameters of EIGRP is complete independence from routed protocols. Hence, EIGRP has implemented a modular approach to supporting routed protocols and can easily be retrofitted to support any other routed protocol.

OSPF

This was designed specifically as an IP routing protocol; hence, it cannot transport IPX or Appletalk protocols. It is encapsulated directly in the IP protocol. OSPF can quickly detect topological changes by flooding link state advertisements to all the other neighbors with reasonably quick convergence.

OSPF is a link-state routing or shortest path first (SPF) protocol detailed in RFC's 1131, 1247 and 1583. Here each router periodically uses a broadcast mechanism to transmit information to all other routers about its own directly connected routers and the status of the data links to them. Based on the information received from all the other routers each router then constructs its own network routing tree using the shortest path algorithm.

These routers continually monitor the status of their links by sending packets to neighboring routers. When the status of a router or link changes, this information is broadcast to the other routers that then update their routing tables. This process is known as flooding and the packets sent are very small representing only the link-state changes.

Using cost as the metric, OSPF can support a much larger network than RIP, which is limited to 15 routers. A problem area can be in mixed RIP and OSPF environments if routers go from RIP to OSPF and back when hop counts are not incremented correctly.

17.9 Exterior gateway protocols (EGP)

One of the earlier EGPs was, in fact called EGP! The current *de facto* Internet standard for inter-domain (AS) routing is Border Gateway Protocol version 4, or simply BGP-4.

BGP- 4

BGP-4, as detailed in RFC 1771, performs intelligent route selection based on the shortest AS path. In other words, whereas IGPs such as RIP make decisions on the number of ROUTERS to a specific destination, BGP-4 bases its decisions on the number of AUTONOMOUS SYSTEMS to a specific destination. It is a so-called Path Vector protocol, and runs over TCP (port 179).

BGP routers in one AS speak BGP to routers in other ASs, where the 'other' AS might be that of an Internet service provider, or another corporation. Companies with an international presence and a large, global WAN, may also opt to have a separate AS on each continent (for example running OSPF internally) and run BGP between them in order to create a clean separation.

BGP comes in two variations namely internal BGP (iBGP) and external BGP (eBGP). iBGP is used within an AS and eBGP between ASs. In order to ascertain which one is used between two adjacent routers, one should look at the AS number for each router. BGP uses a formally registered AS number for entities that will advertise their presence in the Internet. Therefore, if two routers share the same AS number, they are probably using iBGP and if they differ, the routers speak eBGP. Incidentally, BGP routers are referred to as 'BGP speakers', all BGP routers are 'peers', and two adjacent BGP speakers are 'neighbors'.

The range of non-registered (i.e. private) AS numbers is 64512–65535 and these are typically issued by ISPs to stub ASs i.e. those that do not carry third-party traffic.

18

Network protocols part two – TCP, UDP

Objectives

This is the second of two chapters on Ethernet related network protocols. On studying of this chapter, you will:

- Learn about the transmission control protocol (TCP) and user datagram protocol (UDP), both of which are important transport layer protocols
- Become familiar with Internet packet exchange (IPX) and sequential packet exchange (SPX), which are Novell's protocols for the network layer and transport layer respectively
- Learn about the network basic input/output system (NetBIOS) which is a high level interface, and NetBIOS extended user interface (NetBEUI) which is a transport protocol used by NetBIOS
- Become familiar with the concept of Modbus/TCP where a Modbus frame is embedded in a TCP frame for carrying Modbus messages

18.1 Transmission control protocol (TCP)

18.1.1 Basic functions

The transport layer, or host-to-host communication layer, is primarily responsible for ensuring delivery of packets transmitted by the Internet protocols. This additional reliability is needed to compensate for the lack of reliability in IP.

There are only two relevant protocols in the transport layer, namely TCP and UDP. TCP will be discussed in following pages.

TCP is a connection-oriented protocol and is therefore reliable, although the word 'reliable' is used in a data communications context and not in an everyday sense. TCP establishes a connection between two hosts before any data is transmitted. Because a connection is set up beforehand, it is possible to verify that all packets are received on the other end and to arrange re-transmission in the case of lost packets. Because of all these built-in functions, TCP involves significant additional overhead in terms of processing time and header size.

TCP includes the following functions:

- Segmentation of large chunks of data into smaller segments that can be accommodated by IP. The word 'segmentation' is used here to differentiate it from the 'fragmentation' performed by IP
- Data stream reconstruction from packets received
- Receipt acknowledgement
- Socket services for providing multiple connections to ports on remote hosts
- Packet verification and error control
- Flow control
- Packet sequencing and reordering

In order to achieve its intended goals, TCP makes use of ports and sockets, connection oriented communication, sliding windows, and sequence numbers/acknowledgements.

18.1.2 Ports

Whereas IP can route the message to a particular machine based on its IP address, TCP has to know for which process (i.e. software program) on that particular machine it is destined. This is done by means of port numbers ranging from 1 to 65535.

Port numbers are controlled by IANA (the Internet assigned numbers authority) and can be divided into three groups:

- Well-known ports, ranging from 1 to 1023, have been assigned by IANA and are globally known to all TCP users. For example, HTTP uses port 80.
- Registered ports are registered by IANA in cases where the port number cannot be classified as 'well-known', yet it is used by a significant number of users. Examples are port numbers registered for Microsoft Windows or for specific types of PLCs. These numbers range from 1024 to 49151, the latter being 75% of 65536.
- A third class of port numbers is known as ephemeral ports. These range from 49152 to 65535 and can be used on an *ad-hoc* basis.

18.1.3 Sockets

In order to identify both the location and application to which a particular packet is to be sent, the IP address (location) and port number (process) is combined into a functional address called a socket. The IP address is contained in the IP header and the port number is contained in the TCP or UDP header.

In order for any data to be transferred under TCP, a socket must exist both at the source and at the destination. TCP is also capable of creating multiple sockets to the same port.

18.1.4 Sequence numbers

A fundamental notion in the TCP design is that every BYTE of data sent over the TCP connection has a unique 32-bit sequence number. Of course, this number cannot be sent along with every byte, yet it is nevertheless implied. However, the sequence number of the FIRST byte in each segment is included in the accompanying TCP header, for each subsequent byte that number is simply incremented by the receiver in order to keep track of the bytes.

Before any data transmission takes place, both sender and receiver (e.g. client and server) have to agree on the initial sequence numbers (ISNs) to be used. This process is described under 'establishing a connection'.

Since TCP supports full-duplex operation, both client and server will decide on their initial sequence numbers for the connection, even though data may only flow in one direction for that specific connection.

The sequence number, for obvious reasons, cannot start at 0 every time, as it will create serious problems in the case of short-lived multiple sequential connections between two machines. A packet with a sequence number from an earlier connection could easily arrive late, during a subsequent connection. The receiver will have difficulty in deciding whether the packet belongs to a former or to the current connection. It is easy to visualize a similar problem in real life. Imagine tracking a parcel carried by UPS if all UPS agents started issuing tracking numbers beginning with 0 every morning.

The sequence number is generated by means of a 32-bit software counter that starts at 0 during boot-up and increments at a rate of about once every 4 microseconds (although this varies depending on the operating system being used). When TCP establishes a connection, the value of the counter is read and used as the initial sequence number. This creates an apparently random choice of the initial sequence number.

At some point during a connection, the counter could rollover from 2^{32}–1 and start counting from 0 again. The TCP software takes care of this.

18.1.5 Acknowledgement numbers

TCP acknowledges data received on a PER SEGMENT basis, although several consecutive segments may be acknowledged at the same time. In practice, segments are made to fit in one frame i.e. if Ethernet is used at layers 1 and 2, TCP makes the segments smaller or equal to 1500 bytes.

The acknowledgement number returned to the sender to indicate successful delivery equals the number of the last byte received plus one, hence it points to the next expected sequence number. For example: 10 bytes are sent, with sequence number 33. This means that the first byte is numbered 33 and the last byte is numbered 42. If received successfully, an acknowledgement number (ACK) of 43 will be returned. The sender now knows that the data has been received properly, as it agrees with that number.

TCP does not issue selective acknowledgements, so if a specific segment contains errors, the acknowledgement number returned to the sender will point to the first byte in the defective segment. This implies that the segment starting with that sequence number, and all subsequent segments (even though they may have been transmitted successfully) have to be retransmitted.

From the previous paragraph, it should be clear that a duplicate acknowledgement received by the sender means that there was an error in the transmission of one or more bytes following that particular sequence number.

Please note that the sequence number and the acknowledgement number in one header are NOT related at all. The former relates to outgoing data, the latter refers to incoming data. During the connection establishment phase the sequence numbers for both hosts are set up independently, hence these two numbers will never bear any resemblance to each other.

18.1.6 Sliding windows

Obviously there is a need to get some sort of acknowledgment back to ensure that there is guaranteed delivery. This technique, called positive acknowledgment with retransmission, requires the receiver to send back an acknowledgment message within a given time. The transmitter starts a timer so that if no response is received from the destination node

within a given time, another copy of the message will be transmitted. An example of this situation is given in Figure 18.1.

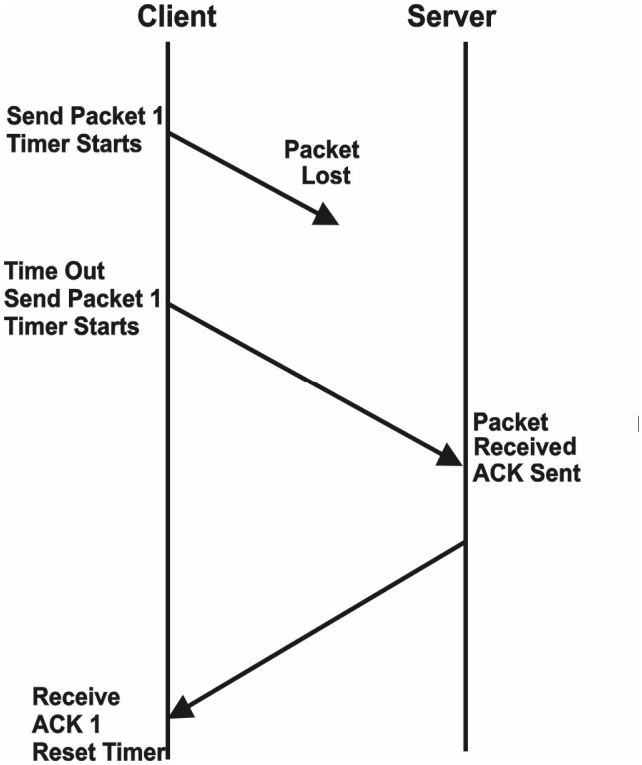


Figure 18.1
Positive acknowledgement philosophy

The sliding window form of positive acknowledgment is used by TCP, as it is very time consuming waiting for each individual acknowledgment to be returned for each packet transmitted. Hence, the idea is that a number of packets (with the cumulative number of bytes not exceeding the window size) are transmitted before the source may receive an acknowledgment to the first message (due to time delays, etc). As long as acknowledgments are received, the window slides along and the next packet is transmitted.

During the TCP connection phase, each host will inform the other side of its permissible window size. For example, for Windows this is typically 8k or around 8192 bytes. This means that, using Ethernet, 5 full data frames comprising 5 × 1460 = 7300 bytes can be sent without acknowledgement. At this stage, the window size has shrunk to less than 1000 bytes, which means that unless an ACK is generated, the sender will have to pause its transmission.

18.1.7 Establishing a connection

A three-way SYN/ SYN_ACK/ACK handshake (as indicated in *Figure 18.2*) is used to establish a TCP connection. As this is a full-duplex protocol, it is possible (and necessary) for a connection to be established in both directions at the same time.

As mentioned before, TCP generates pseudo-random sequence numbers by means of a 32-bit software counter that resets at boot-up and then increments every four microseconds. The host establishing the connection reads a value ‘x’ from the counter

where x can vary between 0 and 2^{32}–1) and inserts it in the sequence number field. It then sets the SYN flag = 1 and transmits the header (no data yet) to the appropriate IP address and Port number. If the chosen sequence number were 132, this action would then be abbreviated as SYN 132.

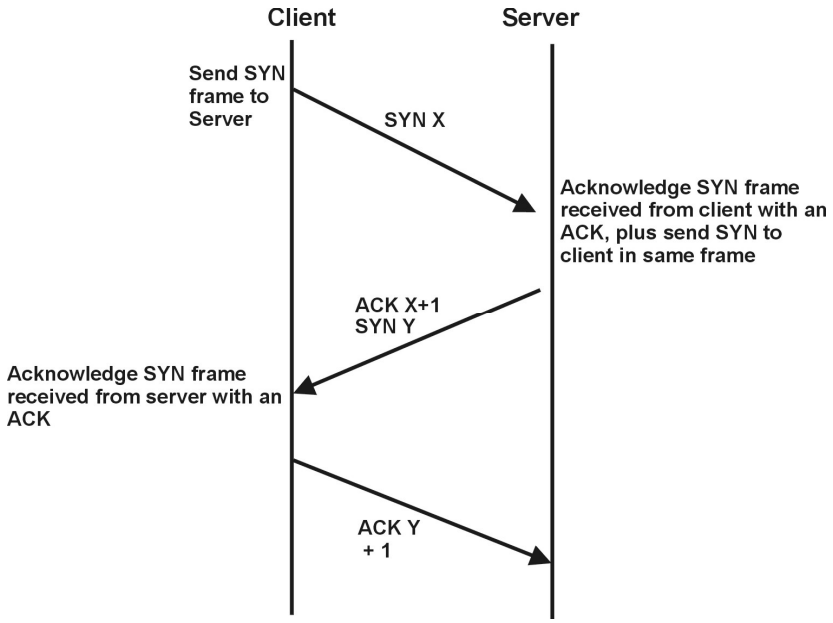


Figure 18.2
TCP connection establishment

The receiving host (e.g. the server) acknowledges this by incrementing the received sequence number by one, and sending it back to the originator as an acknowledgement number. It also sets the ACK flag = 1 to indicate that this is an acknowledgement. This results in an ACK 133. The first byte expected would therefore be numbered 133. At the same time, the Server obtains its own sequence number (y), inserts it in the header, and sets the SYN flag in order to establish a connection in the opposite direction. The header is then sent off to the originator (the client), conveying the message e.g. SYN 567. The composite 'message' contained within the header would thus be ACK 133, SYN 567.

The originator receives this, notes that its own request for a connection has been complied with, and acknowledges the other node's request with an ACK 568. Two-way communication is now established.

18.1.8 Closing a connection

An existing connection can be terminated in several ways.

Firstly, one of the hosts can request to close the connection by setting the FIN flag. The other host can acknowledge this with an ACK, but does not have to close immediately as it may need to transmit more data. This is known as a half-close. When the second host is also ready to close, it will send a FIN that is acknowledged with an ACK. The resulting situation is known as a full close.

Secondly, either of the nodes can terminate its connection with the issue of RST, resulting in the other node also relinquishing its connection and (although not necessarily) responding with an ACK.

Both situations are depicted in Figure 18.3.

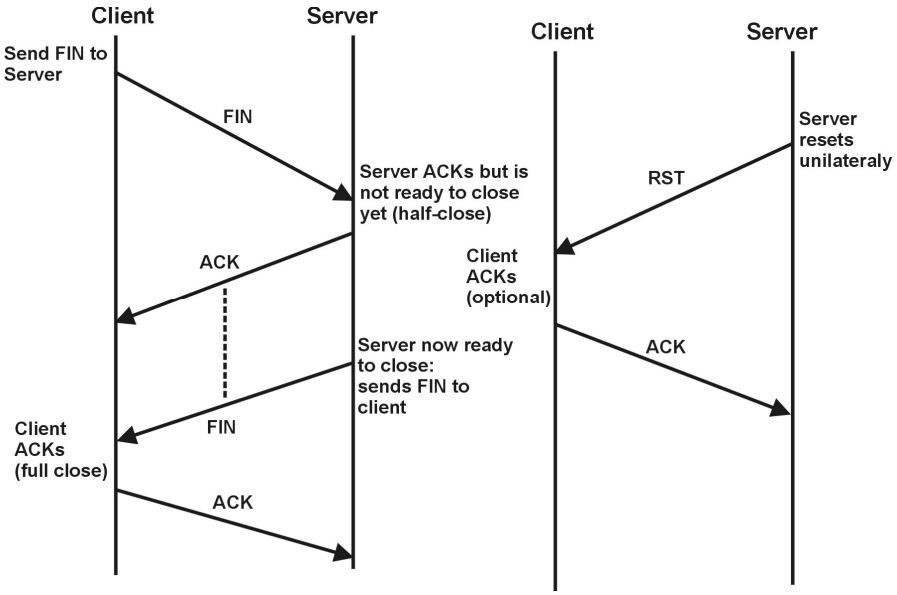


Figure 18.3
Closing a connection

18.1.9 The push operation

TCP normally breaks the data stream into what it regards are appropriately sized segments, based on some definition of efficiency. However, this may not be swift enough for an interactive keyboard application. Hence the push instruction (PSH bit in the code field) used by the application program forces delivery of bytes currently in the stream and the data will be immediately delivered to the process at the receiving end.

18.1.10 Maximum segment size

Both the transmitting and receiving nodes need to agree on the maximum size segments they will transfer. This is specified in the options field. On the one hand, TCP 'prefers' IP not to perform any fragmentation as this leads to a reduction in transmission speed due to the fragmentation process, and a higher probability of loss of a packet and the resultant retransmission of the entire packet.

On the other hand, there is an improvement in overall efficiency if the data packets are not too small and a maximum segment size is selected that fills the physical packets that are transmitted across the network. The current specification recommends a maximum segment size of 536 (this is the 576 byte default size of an X.25 frame minus 20 bytes each for the IP and TCP headers). If the size is not correctly specified, for example too small, the framing bytes (headers etc.) consume most of the packet size resulting in considerable overhead. Refer to RFC 879 for a detailed discussion on this issue.

18.1.11 The TCP frame

The TCP frame consists of a header plus data and is structured as follows:

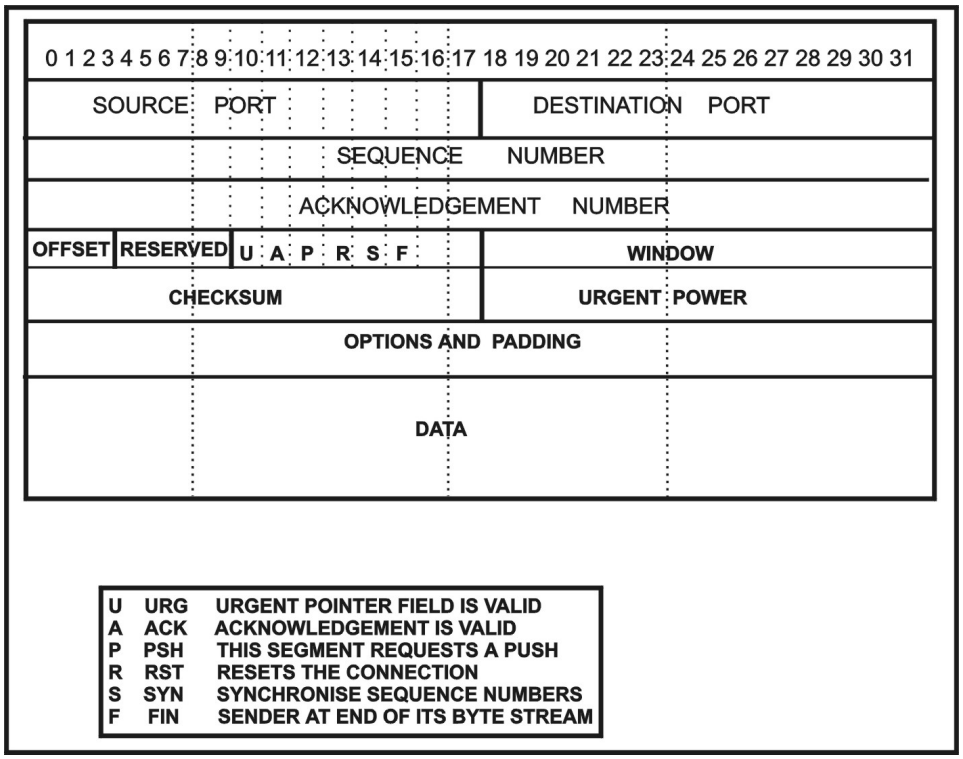


Figure 18.4
TCP frame format

The various fields within the header are as follows:

Source port: 16 bits
The source port number

Destination port: 16 bits
The destination port number

Sequence number: 32 bits
The sequence number of the first data byte in the current segment, except when the SYN flag is set. If the SYN flag is set, a connection is still being established and the sequence number in the header is the initial sequence number (ISN). The first subsequent data byte is ISN+1

Acknowledgement number: 32 bits
If the ACK flag is set, this field contains the value of the next sequence number the sender of this message is expecting to receive. Once a connection is established this is always sent

Offset: 4 bits
The number of 32 bit words in the TCP header. (Similar to IHL in the IP header). This indicates where the data begins. The TCP header (even one including options) is always an integral number of 32 bits long

Reserved: 6 bits

Reserved for future use. Must be zero

Control bits (flags): 6 bits

(From left to right)
URG: Urgent pointer field significant
ACK: Acknowledgement field significant
PSH: Push Function
RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: No more data from sender

Checksum: 16 bits

This is known as the standard Internet checksum, and is the same as the one used for the IP header. The checksum field is the 16-bit one's complement of the one's complement sum of all 16-bit words in the header and text. If a segment contains an odd number of header and text octets to be check-summed, the last octet is padded on the right with zeros to form a 16-bit word for checksum purposes. The pad is not transmitted as part of the segment.

While computing the checksum, the checksum field itself is replaced with zeros.

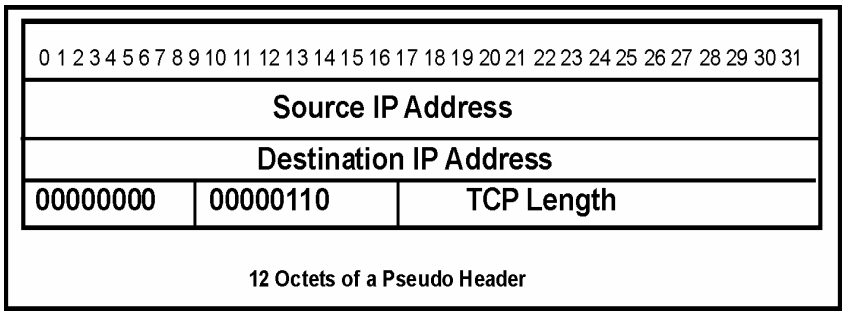


Figure 18.5
Pseudo TCP header format

The checksum also covers a 96-bit 'pseudo header' conceptually appended to the TCP header. This pseudo header contains the source IP address, the destination IP address, the protocol number (06), and TCP length. It must be emphasized that this pseudo header is only used for computation purposes and is NOT transmitted. This gives TCP protection against misrouted segments

Window: 16 bits

The number of data octets beginning with the one indicated in the acknowledgement field, which the sender of this segment is willing or able to accept

Urgent pointer

Urgent data is placed in the beginning of a frame, and the urgent pointer points at the last byte of urgent data (relative to the sequence number i.e. the number of the first byte in the frame). This field is only being interpreted in segments with the URG control bit set

Options

Options may occupy space at the end of the TCP header and are a multiple of 8 bits in length. All options are included in the checksum

18.2 User datagram protocol (UDP)

18.2.1 Basic functions

The second protocol that occupies the host-to-host layer is the UDP. As in the case of TCP, it makes use of the underlying IP protocol to deliver its datagrams.

UDP is a 'connectionless' or non-connection-oriented protocol and does not require a connection to be established between two machines prior to data transmission. It is therefore said to be an 'unreliable' protocol – the word 'unreliable' used here as opposed to 'reliable' in the case of TCP.

As in the case of TCP, packets are still delivered to sockets or ports. However, no connection is established beforehand and therefore UDP cannot guarantee that packets are retransmitted if faulty, received in the correct sequence, or even received at all. In view of this, one might doubt the desirability of such an unreliable protocol.

There are, however, some good reasons for its existence. Sending a UDP datagram involves very little overhead in that there are no synchronization parameters, no priority options, no sequence numbers, no retransmit timers, no delayed acknowledgement timers, and no retransmission of packets. The header is small; the protocol is quick, and functionally streamlined. The only major drawback is that delivery is not guaranteed. UDP is therefore used for communications that involve broadcasts, for general network announcements, or for real-time data.

A particularly good application is with streaming video and streaming audio where low transmission overheads are a pre-requisite, and where retransmission of lost packets is not only unnecessary but also definitely undesirable.

The UDP frame

The format of the UDP frame and the interpretation of its fields are described in RFC-768. The frame consists of a header plus data and contains the following fields:

Source port: 16 bits

This is an optional field. When meaningful, it indicates the port of the sending process, and may be assumed to be the port to which a reply must be addressed in the absence of any other information. If not used, a value of zero is inserted.

Destination port: 16 bits

Same as for source port

Message length: 16 bits

This is the length in bytes of this datagram including the header and the data. (This means the minimum value of the length is eight.)

Checksum: 16 bits

This is the 16-bit one's complement of the one's complement sum of a pseudo header of information from the IP header, the UDP header, and the data, padded with '0' bytes at the end (if necessary) to make a multiple of two bytes. The pseudo header conceptually prefixed to the UDP header contains the source address, the destination address, the

protocol, and the UDP length. As in the case of TCP, this header is used for computational purposes only, and is NOT transmitted.

This information gives protection against misrouted datagrams. This checksum procedure is the same as is used in TCP. If the computed checksum is zero, it is transmitted as all ones (the equivalent in one's complements arithmetic). An all zero transmitted checksum value means that the transmitter generated no checksum (for debugging or for higher level protocols that don't care).

UDP is numbered protocol 17 (21 octal) when used with the Internet protocol.

19

Ethernet based plant automation solutions

19.1 MODBUS TCP/IP

19.1.1 MODBUS messaging

MODBUS is an application layer (OSI layer 7) messaging protocol that provides client/server communication between devices connected to different types of buses or networks. The MODBUS protocol implements a client/server architecture and operates essentially in a "request/ response" mode, irrespective of the media access control used at layer 2. This client/server model is based on four types of messages namely:

- MODBUS Requests, the messages sent on the network by the clients to initiate transactions,
- MODBUS Confirmations, the response messages received on the client side,
- MODBUS Indications, the request messages received on the server side, and
- MODBUS Responses, the response messages sent by the servers

These messaging services of the client/server model are used to exchange real-time information between two device applications, between device applications and devices, or between devices and HMI/SCADA applications.

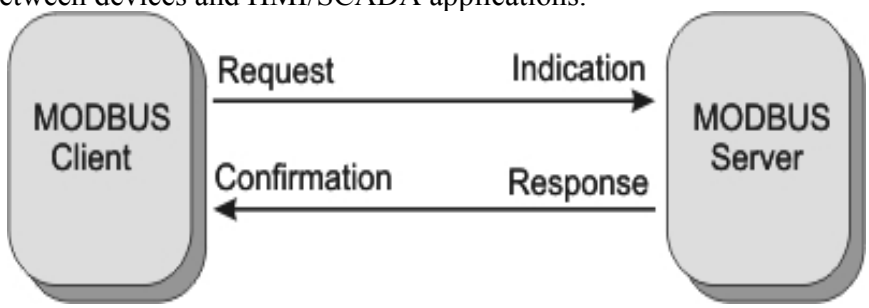


Figure 19.1
MODBUS client/server interaction

In an error-free scenario, the exchange of information between client and server can be illustrated as follows. The client (on the master device) initiates a request. The MODBUS messaging protocol (layer 7) then generates a protocol data unit or PDU, consisting of a function code and a data request. At layer 2, this PDU is converted to an application data unit (ADU) by the addition of some bus or network related fields, such as a slave address and a checksum for error detection purposes. This process is depicted in Figure 19.2

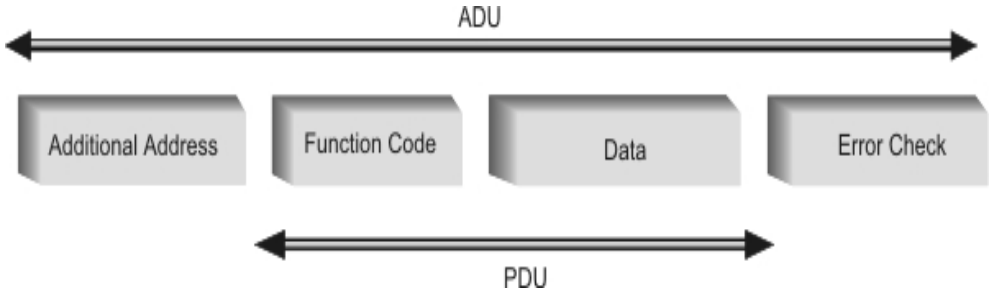


Figure 19.2
General MODBUS frame

The server (on the slave device) then performs the required action and initiates a response. The interaction between client and server is shown in Figure 19.3

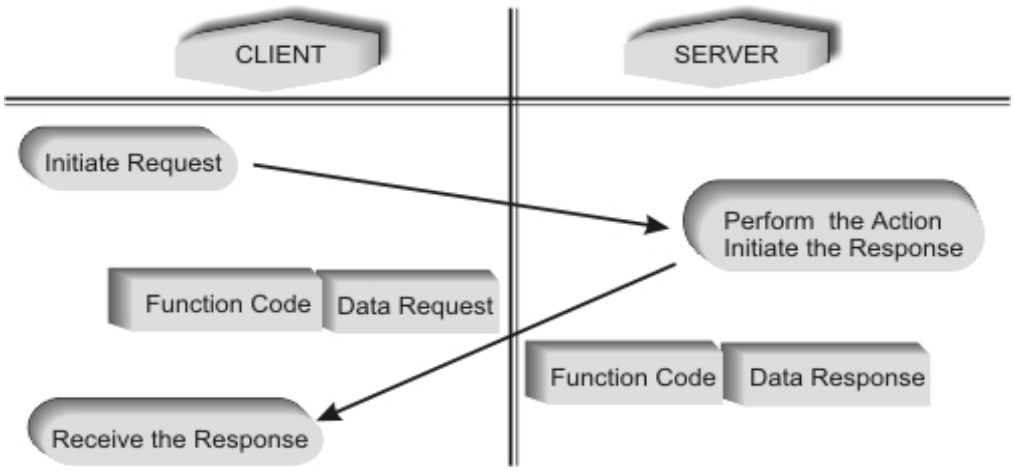


Figure 19.3
MODBUS transaction

The various types of function codes, with their associated requests and responses, have already been described in detail in chapter 1.

The MODBUS Messaging Protocol (layer 7) needs additional support at the lower layers in order to get the message across. A popular method is the use of a master/slave (half-duplex) layer 2 protocol, transmitting the data in serial format over RS-232, RS-485 or Bell 202 type modem links. Other methods include MODBUS+ (half-duplex over RS-485), or MAP. A recent addition is the use of TCP/IP and Ethernet to convey data from client to server. The TCP/IP approach enables client/server interaction over routed networks, albeit at the cost of additional overheads (processing time, headers, etc). An additional sub-layer is required to map the MODBUS application layer on to TCP. The

function of this sub-layer is to encapsulate the MODBUS PDU so that it can be transported as a packet of data by TCP/IP.

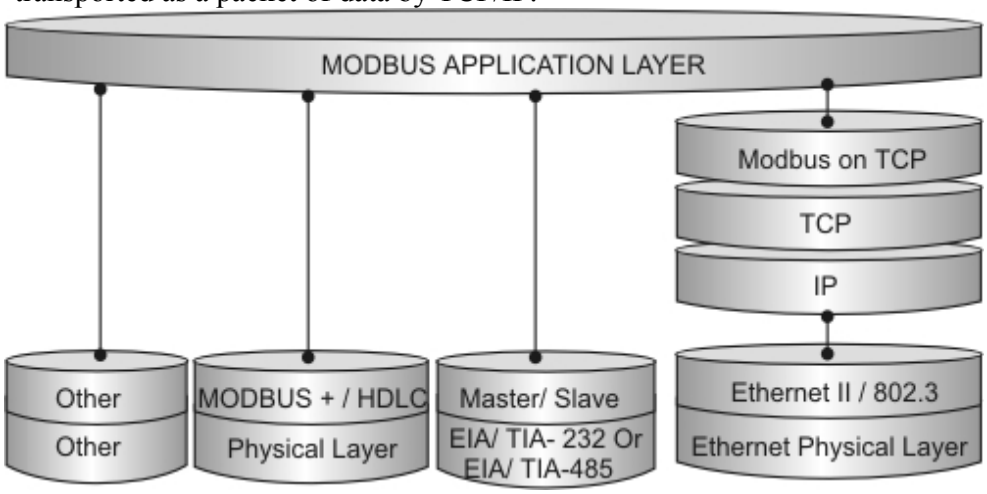


Figure 19.4
MODBUS communication stack

19.1.2 MODBUS encapsulation

System developers familiar with both TCP/IP and the MODBUS protocol might well ask why connection-oriented TCP is used, rather than the datagram-oriented UDP. TCP has more overheads, and as a result it is slower than UDP. The main reason for this choice is to keep control of an individual 'transaction' by enclosing it in a connection which can be identified, supervised, and canceled without requiring specific action on the part of the client or server applications. This gives the mechanism a wide tolerance to network performance changes, and allows security features such as firewalls and proxies to be easily added.

The PDU consisting of data and function code is encapsulated by adding a "*MODBUS* on TCP/IP A*p*plication" (MBAP) header in front of the PDU. The resulting MODBUS TCP/IP ADU, consisting of the PDU plus MBAP header, is then transported as a chunk of data via TCP/IP and Ethernet.

This header differs from a conventional MODBUS RTU header in the following respects:

- The "slave address" is replaced by a 1-byte "unit identifier" that is used to communicate with serial (non-IP) devices via IP devices such as routers.
- All MODBUS requests and responses are designed in such a way that the recipient knows when the message has ended. Therefore, if the message has a variable length data field in it, an additional byte count is included.
- This byte count is also useful in the case of long messages being split up by TCP, ensuring that the recipient is kept informed of the exact number of bytes transmitted.

Fields	Length	Description -	Client	Server
Transaction Identifier	2 Bytes	Identification of a MODBUS Request / Response transaction.	Initialized by the client	Recopied by the server from the received request
Protocol Identifier	2 Bytes	0 = MODBUS protocol	Initialized by the client	Recopied by the server from the received request
Length	2 Bytes	Number of following bytes	Initialized by the client (request)	Initialized by the server (Response)
Unit Identifier	1 Byte	Identification of a remote slave connected on a serial line or on other buses.	Initialized by the client	Recopied by the server from the received request

Table 19.1
MBAP Fields

The MBAP header is 7 bytes long and comprises the following 4 fields.

- The *transaction identifier* is a pseudo-random number used for pairing requests and responses. The MODBUS server copies this number received from the client in its response to the client.
- The *protocol identifier* is used for multiplexing between systems. The MODBUS protocol is defined as value 0.
- The *length* field is a byte count of all the fields following it, including the unit identifier and data fields.
- The *unit identifier* is used for routing between systems, typically to a MODBUS or MODBUS+ serial line slave through a gateway between the serial line and a TCP/IP network. It is set by the client and the same value must be returned by the server.

All MODBUS/TCP ADUs are sent via registered port 502 and the fields are encoded big-endian, which means that if a number is represented by more than one byte, the most significant byte is sent first.

The entire MODBUS ADU is transported by TCP/IP as data.

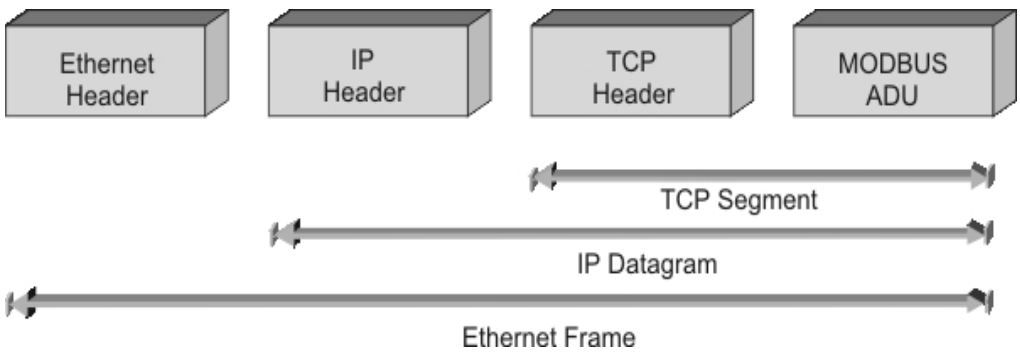


Figure 19.5
Transportation of MODBUS ADU

19.1.3 MODBUS component architecture model

Figure 19.6 shows a system with both client and server devices (masters and slaves). Some are connected via Ethernet, while others are serial (RS-232 or RS-485) devices connected to the Ethernet network via gateways.

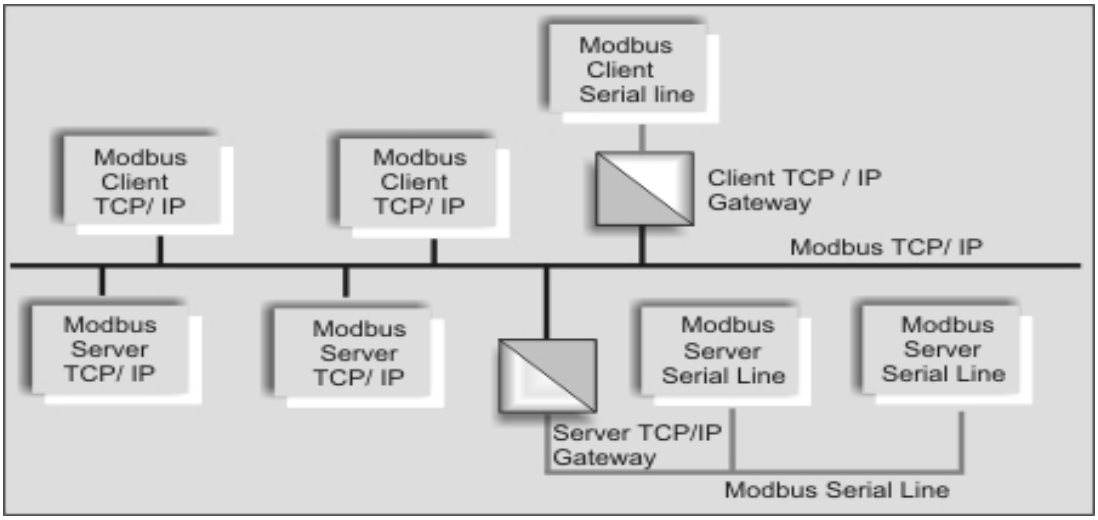


Figure 19.6
MODBUS TCP communications architecture

Each of the TCP/IP enabled devices in Figure 19.6 supports the MODBUS messaging service architecture. The following is a graphical representation of this architecture, and the way it relates to the TCP/IP stack. The communication application layer corresponds to layer 7 of the OSI model, while the TCP/IP stack corresponds to OSI layers 3 and 4. The TCP management layer acts as an interface between the two. Somewhat unusual is the location of the client and the server, as these are usually implemented as part of the user application (above the stack) and not as part of the application layer (within the stack).

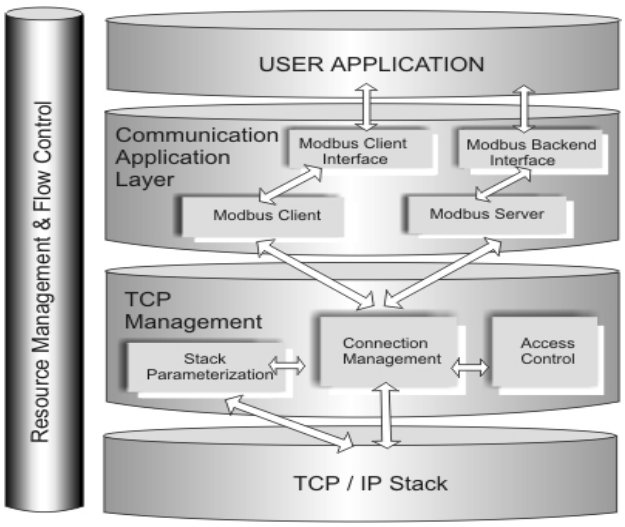


Figure 19.7
MODBUS messaging service architecture

A MODBUS device can be a client device (master) or a server device (slave) and as such it can provide a client and/or a server interface to the user application. The server interface is called a "backend interface" as it allows indirect access to user application objects such as discrete inputs, coils, input registers and holding registers. The section on MODBUS in Chapter 1 explains how MODBUS requests are mapped onto the device's application memory.

The application program on the client device (master) sends explicit instructions to a remote server device (slave) by exchanging control information with the MODBUS client. The MODBUS client, in turn, builds a MODBUS request with parameters obtained from the application program and passes this message on to the server. The processing of this request involves waiting for a reply, and the generation of a MODBUS confirmation.

The MODBUS client interface allows the application to exchange information with the MODBUS client through an applications programming interface (API).

The MODBUS server maintains a constant listening watch on port 502. When it receives a request from the client, it actions the appropriate read, write or other function to the application program via the backend interface. It then returns the appropriate response to the client.

In order to control the equilibrium in the flow of inbound and outbound messages, flow control is implemented at various levels. It is primarily based on TCP flow control, with some additional control at the data link and application layers.

19.1.4 MODBUS TCP operation

Communication between a MODBUS client and MODBUS server requires a TCP connection. This connection can be established explicitly by the user application module, or it can be taken care of automatically by the TCP connection management module. The number of concurrent TCP connections is not dictated by the MODBUS specification, but is dependent on the capabilities of the device.

The following implementation rules are also prescribed by the specification:

- The TCP connection should be kept open and not closed and re-opened for every transaction.
- The number of concurrent connections between a client and a server should be kept to a minimum.
- Several MODBUS transactions can be activated on the same TCP connection (albeit with different transaction identifiers)
- For a bidirectional client/server link, a TCP connection needs to be established in each direction
- A TCP frame may only carry one MODBUS ADU

In order to establish a connection, a client and a server must negotiate a TCP connection with a reserved port number bigger than 1024 on the client, and a well-known port number (502) on the server. On the server side only port 502 is used, but on the client side each subsequent connection will require a different port number. The triple handshake procedure (SYN X, ACK X+1 SYN Y, ACK Y+1) is explained in the chapter on TCP.

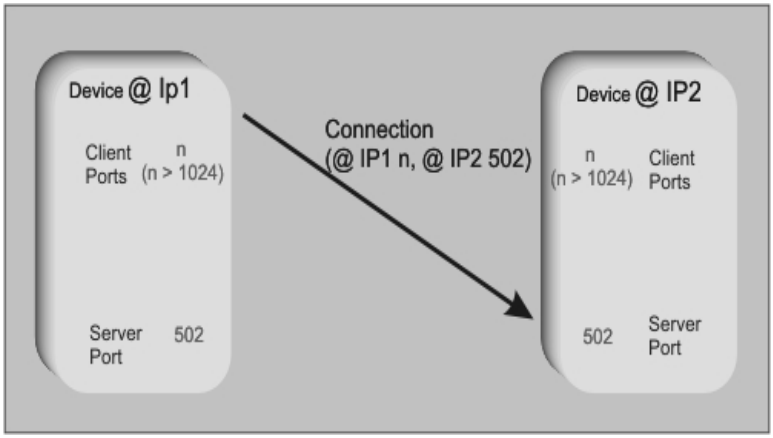


Figure 19.8
MODBUS TCP connection establishment

Once the connection is established, the client and server will exchange requests and responses. This will continue until the client is done, at which point it will attempt to close the connection with a FIN. The server may respond in kind (FIN) or simply acknowledge with an ACK, because it is not ready to close the connection yet, resulting in a half-close. When the server is ready to close as well, it will issue a FIN to which the client will respond with an ACK. The connection is then closed.

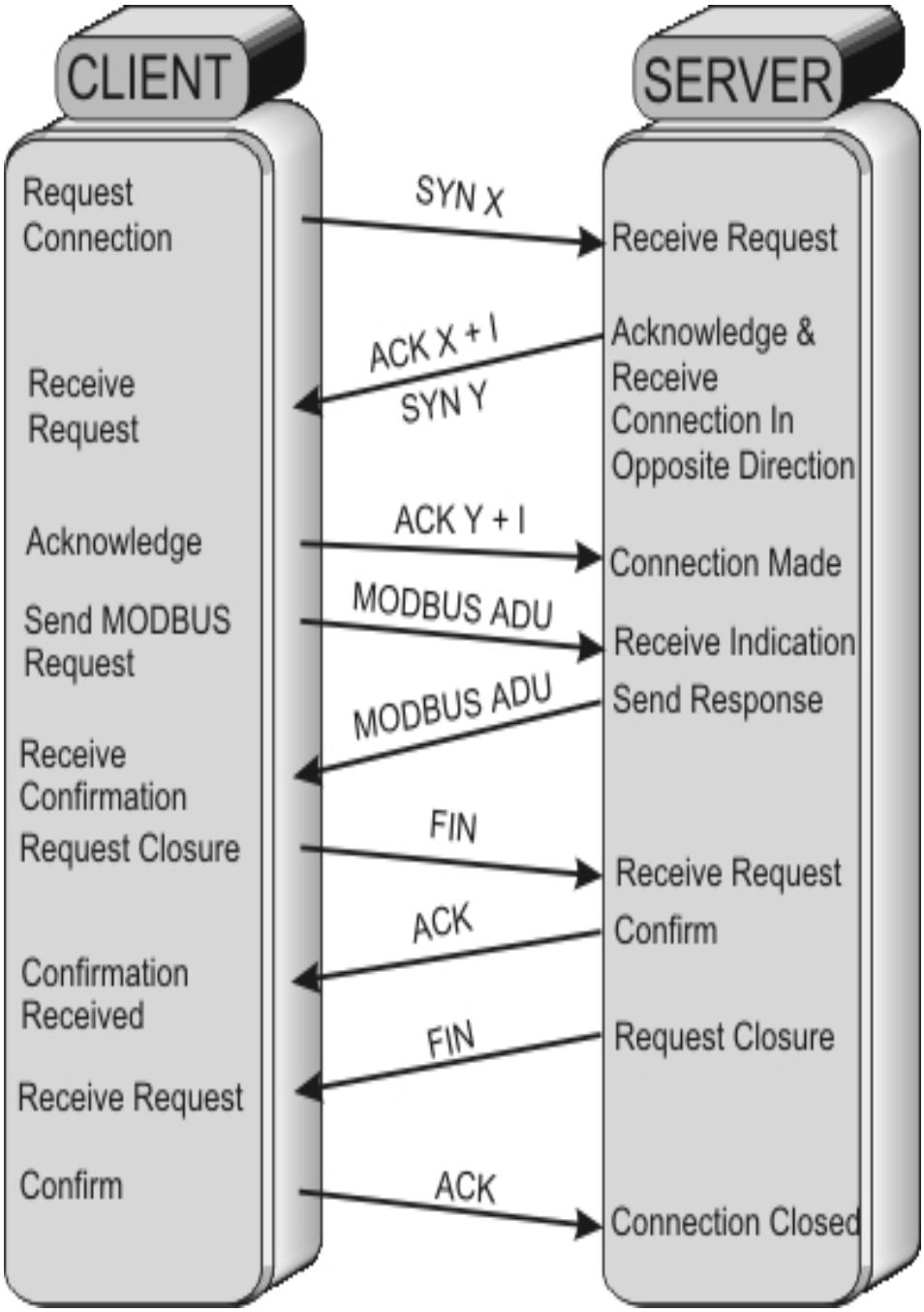


Figure 19.9
Client/server interaction

19.1.5 MODBUS and IDA

IDA (Interface for Distributed Automation) is a new approach to plant automation currently (2004) under development in the IDA group with strong support from Schneider Automation and Jetter. IDA supplies the infrastructure for modular, distributed and reusable automation solutions. It is an object oriented communication system that defines (a) methods for real time communication and (b) methods for management communication among the nodes. The methodology is based on the architecture introduced in the evolving draft Function Block standard IEC 61499 and will result in a

system along the same lines as PROFInet. The scope of IDA further includes web-based device management via standard Internet browsers, plug-and-work methods based on XML device descriptions as well as synchronization methods to permit clock synchronization of devices as required for axis coordination of drives. Safety will be another integral part of IDA which is achieved by definition of a Safety layer allowing users to combine safe and non-safe devices and tools in one application over one Ethernet-TCP/IP based network simultaneously.

The IDA real-time communication is based exclusively on the use of the Real-Time Publish/Subscribe protocol (RTPS). RTPS is implemented by "middleware" (OSI layers 4 -7) and is common to all IDA devices. The RTPS protocol and the middleware are built on top of the UDP protocol. Real-time services in general have the highest priority of all IDA communication services. Depending on the type of application, real-time communication relationships include e.g. preconfigured or dynamic, cyclic or on-demand, point to point or group oriented, single-source or redundant.

Another important feature of the IDA technology is the web-based device management. All field devices have their own built-in web page which contains their configuration, operation and diagnostic parameters. Users have access to this information via a standard Internet browser, such as Microsoft's Internet Explorer. XML-based device descriptions will simplify system configuration and support device interchangeability. IDA is not only the "missing" application layer in Industrial Ethernet. It goes much further and defines all communication features required for new automation concepts with distributed intelligence.

Recently there had been a merger between the MODBUS and IDA working groups, (see www.MODBUS-ida.org) which means that MODBUS will feature strongly in the IDA concept.

19.2 Ethernet/IP (Ethernet/Industrial Protocol)

19.2.1 Introduction

DeviceNet™ and ControlNet™ are two well-known industrial networks based on CIP, the Control and Information Protocol. Both networks have been developed by Rockwell Automation, but are now owned and maintained by the two manufacturers' organizations ODVA (Open DeviceNet Vendors Association) and CI (ControlNet International). ODVA and CI have recently introduced the newest member of this family; viz. EtherNet/IP. This chapter describes the techniques and mechanisms that are used to implement Ethernet/IP. The full Ethernet/IP specification can be downloaded from the ODVA website. It specifies issues such as object modeling, explicit and implicit messaging, communication objects, a general object library, device profiles, electronic data sheets (EDSs), explicit messaging services and data. This section will attempt to give an overall view of the system, taking into account the fact that layers 1 thru 4 of the OSI model (the bottom three layers of the TCP model) have already been dealt with in another chapter.

19.2.2 Plant automation hierarchies

Automation systems should ideally provide users with three primary services namely control, configuration and data collection. The first, control, involves the exchange of time-critical data between controlling devices such as programmable logic controllers (PLCs) and input/output (I/O) devices such as actuators and sensors. Networks that are involved in the transmission of this data must provide some level of priority setting

and/or interrupt capabilities, and should behave in a fairly deterministic fashion.

The second type of functionality, namely configuration, typically involves a personal computer (PC) or a similar device in order for users to set up and maintain their systems. This activity is typically performed during commissioning or maintenance operations, but can also take place during runtime, e.g. recipe management in batch operations.

The third involves the collection of data for the purposes of display (e.g. in HMI stations), data analysis, trending, troubleshooting or maintenance.

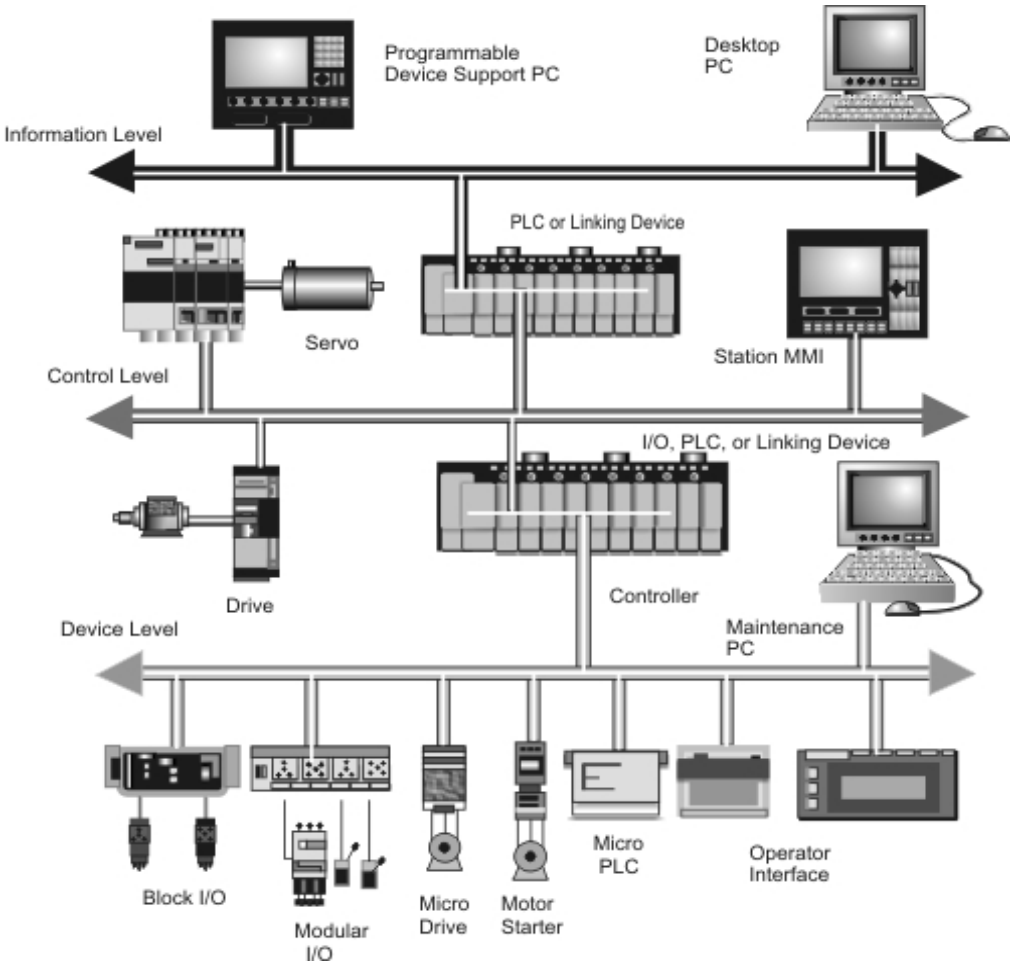


Figure 19.10
Hierarchy of plant levels

Figure 19.10 shows a generic view of an automation system architecture. At the device level, information is exchanged primarily between devices and networks deployed on the plant floor. Fast cycle times are required, networks at this level are bit-or byte oriented, and data packets are fairly small. Examples are ASi, DeviceNet, PROFIBUS DP and Foundation Fieldbus H1.

At the control level data is primarily exchanged between MMIs (or HMIs, to be politically more correct) and PLCs . At this level speed is less critical and the amount of data exchanged in a packet is, generally speaking, bigger. These are systems at this level is said to be message oriented, and examples are ControlNet, PROFIBUS FMS and Foundation Fieldbus HSE.

19.2.3 Ethernet/IP vs. DeviceNet and ControlNet

There is a world-wide trend to develop plant automation systems that use Ethernet and TCP/IP. This, in conjunction with appropriate software at layers 4-8 (layer 8, the "user" layer, is not defined in the OSI model) makes it possible to easily exchange data right across the three plant hierarchies and, in fact, across a WAN or VPN. Efforts this regard include IDA (the Interface for Distributed Automation), ProfiNet and Ethernet/IP.

The "IP" in Ethernet/IP stands for Industrial Protocol (and not for "Internet Protocol" as in TCP/IP). Ethernet/IP is an open industrial network standard based on Ethernet , using commercial off-the-shelf (COTS) technology and the well-established military-standard TCP/IP protocol suite, on which the Internet is based. It allows users to collect, configure and control data, and provides interoperability between equipment from various vendors, of which there are several hundred already.

The system is defined in terms of several open standards, which have a wide level of acceptance. They are Ethernet (IEEE802.3), TCP/IP, and CIP (Control Information Protocol, EN50170 and IEC 61158). The latter is already in use in DeviceNet and ControlNet.

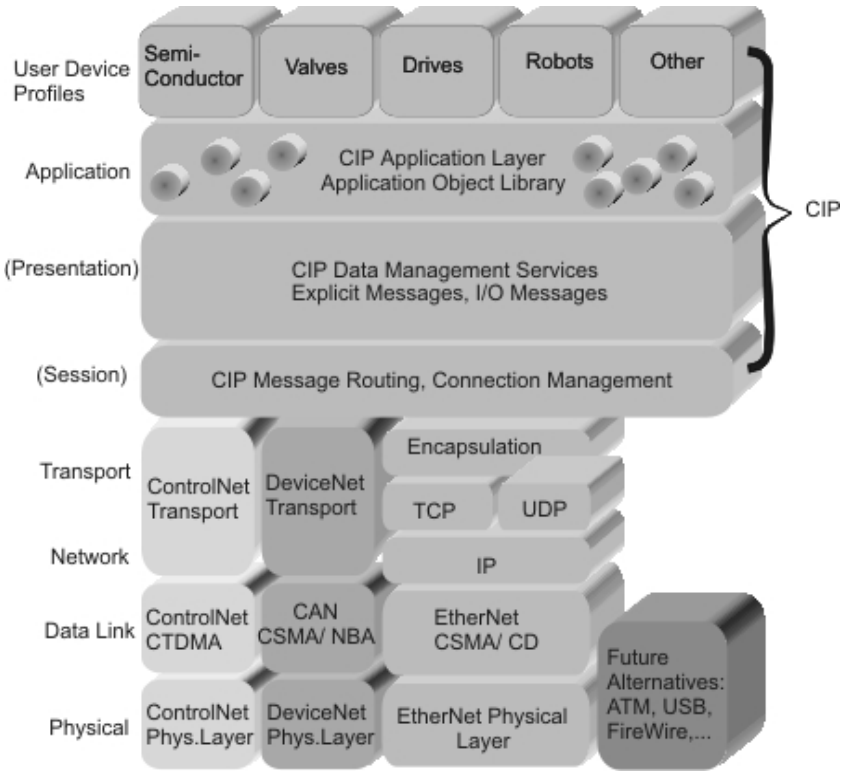


Figure 19.11
Ethernet/IP, DeviceNet and ControlNet stacks

As Figure 19.11 shows, CIP has already been in use with DeviceNet and ControlNet, the only difference between those two systems being the implementation of the four bottom layers. Now TCP/IP has been added as an alternative network layer/transport layer, but CIP remains intact.

The operation of Ethernet/IP will now be discussed, using the OSI model as a framework.

19.2.4 The medium

"Layer 0", the medium, is implemented with the media prescribed in the IEEE802.3 standards.

Note that the OSI model has no layer 0 as such, but layer 1, the physical layer, dictates the type of medium to be used. The medium is formally specified in a separate specification, for example TIA/EIA 568 in the case of Cat5 wiring. The preferred topology for industrial Ethernet is a star (hub) configuration, hence the wiring for short runs (less than 100m) will be TIA/EIA Cat5, Cat5e or Cat5i, with the screened/shielded variety preferred. For longer runs (up to 3000m) fiber is required. The Ethernet/IP Media Planning and Installation Manual (from Rockwell Automation; downloadable on the Internet) gives detailed recommendations in this regard. Although there is no prescribed Industrial connector for Cat5 (yet), most Industrial Ethernet vendors use watertight/dustproof connectors (IP67) such as the modified RJ-45 connectors or M-12 style connectors.

Figure 19.12
Industrial RJ-45 connectors (Courtesy: Siemon)

19.2.5 The network interface layer

The lowest layer of the TCP/IP model, the network interface layer, corresponds with layers 1 (physical) and 2 (data link) of the OSI model. Ethernet provides a set of physical media definitions, a scheme for sharing that physical media (CSMA/CD or full duplex), and a simple frame format and hardware source/destination addressing scheme (MAC addresses) for moving packets of data (frames) between devices on a LAN. On its own, however, Ethernet lacks the more complex features required of a fully functional industrial network. For that reason, all installed Ethernet networks support one or more communication protocols that run on top of Ethernet and provide more sophisticated data transfer and network management functionality. It is the higher layer protocols that determine what level of functionality is supported by the network, what types of devices may be connected to the network, and how devices interoperate on the network.

The network interface layer is implemented with IEEE802.3 Ethernet. It must be understood that Ethernet is not a replacement for any "field bus" as it only implements OSI layers 1 and 2, i.e. it only serves as a vehicle to get the data packets (frames) from point A to point B. It lacks all the other layers from 3 upwards.

For many years users have steered away from the use of Ethernet in industrial applications, mainly because of its perceived lack of determinism. This was due to its CSMA/CD medium access method, which is essentially stochastic in nature. Other issues that affected its industrial application included connectors and cabling, packaging, power supplies, switching requirements, speed, power over the cable requirements and provision for redundancy.

Modern Ethernet systems, however, differ from the old cable-based legacy systems like chalk and cheese. Switched Ethernet systems can now operate in full duplex mode, which, for all practical purposes, eliminate collisions. There are now many vendors that offer industrial devices, with features such as IP67 environmental rating, rail mounting, redundant DC power supplies, VLAN capability, prioritized switching (IEEE802.1p/Q) and redundant ring operation.

Industry often expects device power to be delivered over the same wires as those used for communicating with the devices. Examples of such systems are DeviceNet and Foundation Fieldbus H1. This is, however, not an absolute necessity as the power can be delivered separately. PROFIBUS DP, for example, does not provide this feature yet it is one of the leading field buses. Ethernet does, however, now provide the ability to deliver power over the cable. The IEEE 802.3af standard was ratified by the IEEE Standards Board in June 2003 and allows a source device (a hub or a switch) to supply a minimum of 300 mA at 48 Volts DC to the field device. This is in the same range as Foundation Fieldbus and DeviceNet. The standard allows for two alternatives, namely the transmission of power over the signal pairs (1/2 and 3/6) or the transmission of power over the unused pairs (4/5 and 7/8). Intrinsic safety issues still need addressing.

The issues of switch interconnection, VLANs (802.1Q), prioritized switching (IEEE802.1p) and redundant switched rings have already been described elsewhere.

To summarize: In theory any type of Ethernet (even 10Base5) could be used, but these older systems should be seen as legacy issues. The current trend is to use 10/100BaseT networks, with switches in place of hubs wherever possible. Although the idea is to use COTS components, that does not mean that users can install inexpensive commercial grade equipment in the field. Some locations will require industrial grade switches, which come at a price. Issues such as system resilience in the case of power or device failure should also be carefully considered.

19.2.6 The internet layer

This layer corresponds to layer 3, the network layer, in the OSI model. Some of the network layer protocols that have been implemented over Ethernet are DECnet, Novell IPX, MAP, TOP, AppleTalk and IP. Of these, IP is receiving the most attention due to the emergence of the Internet as well as the popularity of corporate Intranets. TCP/IP is the protocol of the Internet and, although TCP/IP will run on physical media other than Ethernet, and Ethernet supports other communication protocols, the two have become increasingly linked due to the desire of organizations to integrate their internal Intranets with the global Internet. Therefore, TCP and IP have become the dominant "middle layer" protocols, not only in the corporate environment but on the factory floor as well.

Here, Ethernet/IP uses the “standard” suite of protocols as discussed elsewhere in this manual. These include IP, ARP, ICMP, etc. Of particular importance here is IGMP, the Internet Group Management Protocol, as IP multicasting is very a prominent feature in Ethernet/IP.

19.2.7 The host-to-host layer

This corresponds to layer 4, the transport layer, of the OSI model. Here, both TCP and UDP are used, depending on whether connection-oriented or connectionless operation is required.

TCP runs on top of IP and operates only in unicast (point-to-point) mode. It is also used by applications such as Telnet, FTP and HTTP. In an industrial automation application, TCP is typically used to download ladder programs between a workstation and a PLC, for

MMI software that reads or writes PLC data tables, or for peer-to-peer messaging between two PLCs. This type of communication is referred to as "explicit" communication.

Through TCP, for example, Ethernet/IP is able to send explicit (connection oriented) messages, in which the data field carries both protocol information and instructions. Here, nodes must interpret each message, execute the requested task, and generate responses. These types of messages are used for device configuration and various diagnostics.

UDP is a much simpler transport protocol. It is connectionless and provides a very simple capability to send "datagrams" between two devices. It does not guarantee that the data will get from one device to another, does not perform retries, and does not even know if the target device has received the data successfully. Application layers that implement their own handshaking or connection management between two devices and, therefore, only need a minimal transport service, will use UDP. For instance, UDP is used by applications such as SNMP. UDP is smaller, simpler and faster than TCP due to its minimal capabilities and use of resources. In an industrial automation application, UDP is typically used for network management functions, applications that do not require reliable data transmission, applications that are willing to implement their own reliability scheme, such a flash memory programming of network devices, and for input/output (I/O) operations.

For connectionless real-time messaging, multicasting, and for sending implicit messages, Ethernet/IP uses UDP. Implicit messages contain no protocol information in the data field, only real-time I/O data. The meaning of the data is predefined in advance processing time in the node during runtime is reduced. Because these messages are low on overhead and short, they can pass quickly enough to be useful for certain time-critical control applications.

Between TCP/UDP and CIP is an encapsulating protocol, which appends its own encapsulating header and checksum to the CIP data to be sent before passing it on to TCP or UDP. In this way the CIP information is simply passed on by TCP/IP or UDP/IP as if it is a chunk of data**.**

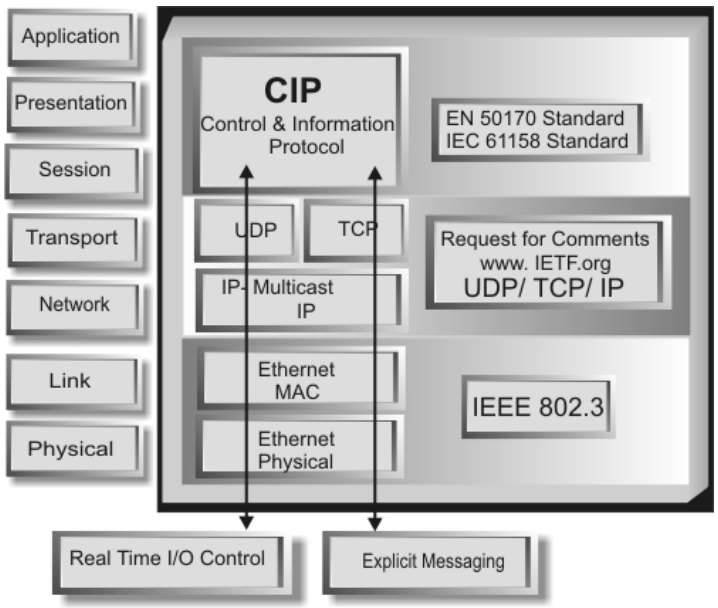


Figure 19.13
Explicit vs. implicit messaging

19.2.8 The process/application layer

As shown in figure 19.12, CIP covers not only layers 5, 6 and 7 of the OSI model, but also the "user layer" (layer 8), which includes the user device profiles. Apart from the common device profiles, CIP also includes the common object library, the common control services and the common routing services. The OSI model has no layer 8, but items such as device profiles do not fit within the conceptual structure of the OSI model, hence vendors add a "layer 8" above layer 7 for this purpose.

CIP gives generic Ethernet an "industrial functionality" and allows the control, collection and configuration of data. It also allows a producer-consumer model to run on Ethernet.

For users that already have Ethernet/IP, DeviceNet and ControlNet systems in place, CIP provides the following common features:

- It provides the user with a standard set of messaging services for all three networks
- It lets the user connect to any network and configure and collect data from any network
- It saves time and effort during system configuration because no routing tables or added logic are necessary to move data between networks
- It reduces the amount of training needed when moving to different networks (DeviceNet, ControlNet, Ethernet/IP) by providing similar configuration tools and features

CIP co-exists with the other TCP/IP application layer protocols as shown in the following figure.

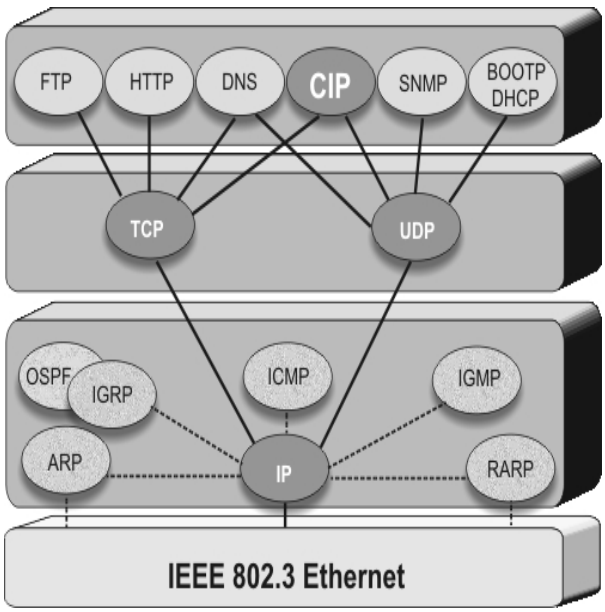


Figure 19.14
CIP vs. other application layer protocols

19.2.9 Description of the CIP protocol

The following description of the CIP protocol is an edited version of a white paper describing the operation of CIP, and we wish to give full recognition to the ODVA in this regard.

CIP is a versatile protocol that has been designed with the automation industry in mind, but because of its very open nature, it can be applied to many areas. The general CIP protocol specification is available for download from the ODVA website. It is beyond the scope of the following paragraphs to fully describe each and every detail of the specification, but the key features will be addressed. The specification is subdivided into several chapters and appendices which describe the following features:

- Object modeling
- Messaging protocol
- Communication objects
- General object library
- Device profiles
- Electronic data sheets
- Services
- Data management

Object modeling

CIP makes use of abstract object modeling to describe items such as the suite of communication services available, the externally visible behavior of a CIP node, and a common means by which information within CIP products is accessed and exchanged.

Every CIP node is modeled as a collection of objects. An object provides an abstract representation of a particular component within a product. Anything not described in object form is not visible through the CIP protocol. CIP objects are structured into classes, instances, and attributes.

A class is a set of objects that all represent the same kind of system component. An object instance is the actual representation of a particular object within a class. Each instance of a class has the same attributes, but it has its own particular set of attribute values. As figure 9.15 illustrates, multiple object instances within a particular class can reside within a CIP node. An object instance and/or object class has attributes, provides services and implements a behavior.

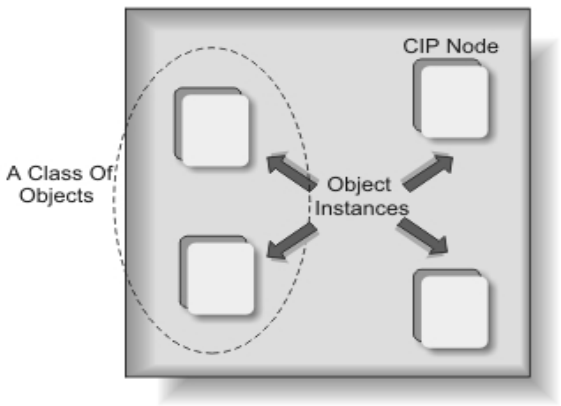


Figure 19.15
A class of objects

The objects and their components are addressed by a uniform addressing scheme consisting of a:

- Media access control identifier (MAC ID), an integer identification value assigned to each node on a CIP network

- Class identifier (Class ID), an integer identification value assigned to each object class accessible from the network
- Instance identifier (Instance ID), an integer identification value assigned to an object instance that identifies it among all instances of the same class
- Attribute identifier (Attribute ID), an integer identification value assigned to a class and/or instance attribute
- Service code, an integer identification value which denotes a particular object instance and/or object class function.

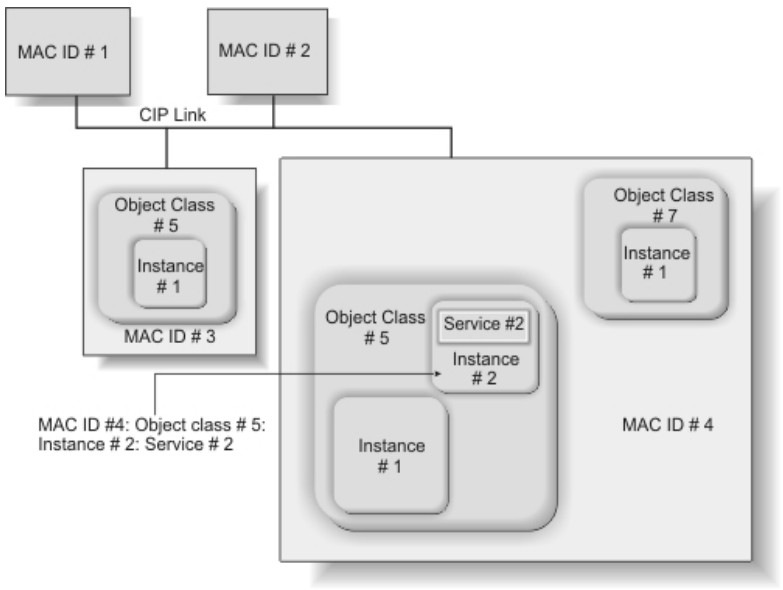


Figure 19.16
Object addressing example

Figure 19.16 shows an example of this object addressing scheme. More details of the object modeling can be found in chapters 1 and 4 of the CIP specification.

Messaging protocol

CIP is layered on top of a connection-based network. A CIP connection provides a path between multiple applications. When a connection is established, the transmissions associated with that connection are assigned a connection ID (CID). If the connection involves a bi-directional exchange, then two connection ID values are assigned. See Figure 19.17.

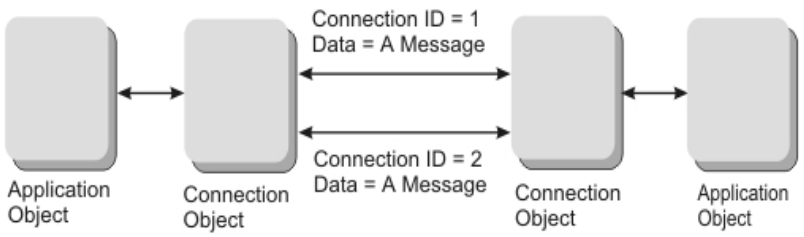


Figure 19.17
Connections and connection IDs

The definition and format of the connection ID is network dependent. For example, the connection ID for CIP connections over DeviceNet is based on the CAN identifier field.

Since most messaging on a CIP network is done through connections, a process has been defined to establish such connections between devices that are not "connected" yet. This is done through the unconnected message manager (UCMM), which is responsible for processing.

All connections in a CIP network can be divided into I/O connections and explicit messaging connections.

- I/O connections provide dedicated, special purpose communication paths between a producing application and one or more consuming applications. Application-specific I/O data moves through these ports and is often referred to as implicit messaging.
- Explicit messaging connections provide generic, multi-purpose communication paths between two devices. These connections are often referred to as just "messaging connections." Explicit messages provide the typical request/response-oriented network communication.

More details of the messaging protocol can be found in Chapter 2 of the CIP specification.

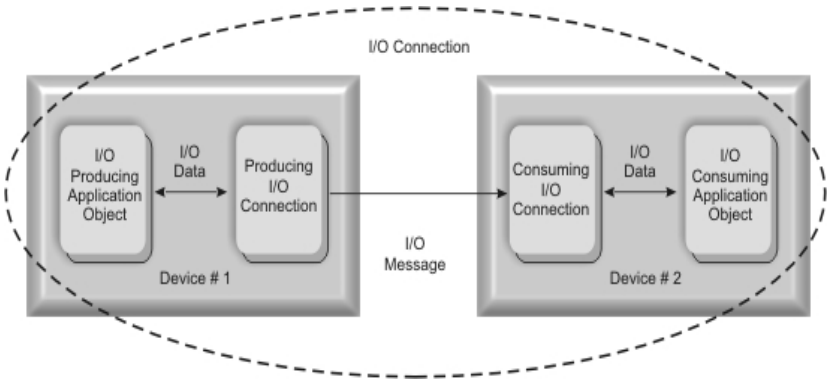


Figure 19.18
CIP I/O Connection

Communication objects

The CIP communication objects manage and provide the runtime exchange of messages. While these objects follow the overall principles and guidelines for CIP objects, the communication objects are unique in a way since they are the focal point for all CIP communication. It therefore makes sense to have a look at them in more detail.

Every instance of a communication object contains a link producer part or a link consumer part or both. I/O connections may be producing only, consuming only or producing and consuming, while explicit messaging connections always are producing and consuming.

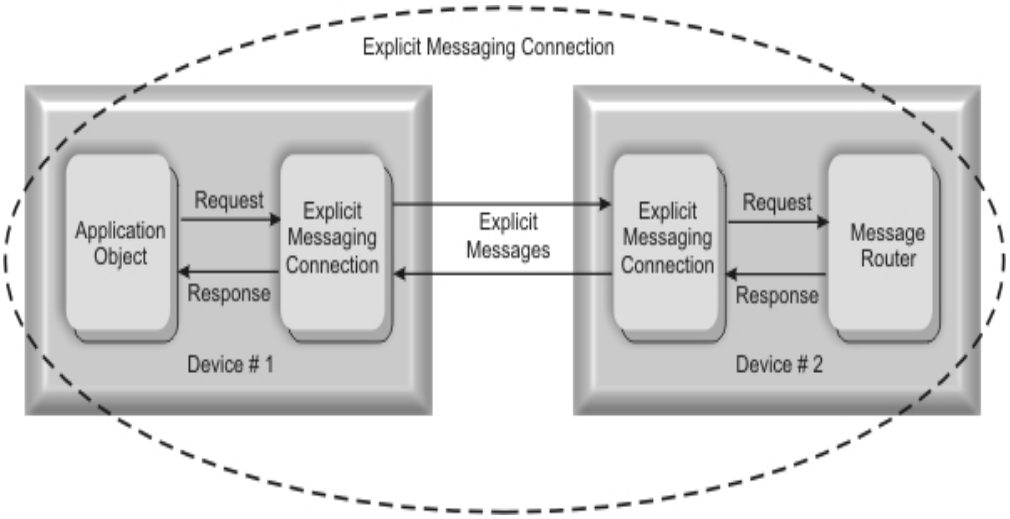


Figure 19.19
CIP explicit messaging connection

Figures 19.18 and 19.19 show the typical connection arrangement for CIP I/O messaging and CIP explicit messaging.

The attribute values in the connection objects define a set of attributes that describe vital parameters of this connection.

First of all, they state what kind of connection this is. They specify whether this is an I/O connection or an explicit messaging connection, but also the maximum size of the data to be exchanged across this connection and the source and sink of this data.

Further attributes define the state of this connection and what kind of behavior this connection is to show. Of particular importance is how messages are triggered (from the application, through change of state or change of data, through cyclic events, or by network events), and the timing of the connections (time-out associated with this connection and pre-defined action if a time-out occurs). CIP allows multiple connections to coexist in a device, although simple devices, e.g. simple DeviceNet slaves, will only have very few connections alive at any given point in time.

More details of the communication objects can be found in chapter 3 of the CIP specification.

General object library

The CIP family of protocols contains a fairly large collection of commonly defined objects. The overall set of object classes can be subdivided into three types:

- General use objects
- Application specific objects
- Network specific objects

Apart from the objects that are network specific, all other objects are common objects that can and will be used in all three protocols.

The following are examples of objects for general use:

- Identity object
- Message router object
- Assembly object
- Connection object
- Parameter object

A further group consists of application specific objects:

- Discrete input point
- Discrete input point object
- Discrete output point object
- Analog input point object
- Analog output point object
- Presence sensing object
- Discrete input group object
- Discrete output group object
- Analog input group object
- Analog output group object
- Position controller supervisor object
- Position controller object block
- Sequencer object
- Command block object
- Motor data object
- Control supervisor object
- AC/DC drive object
- Overload object
- Softstart object
- Selection object
- S-Device supervisor object
- S-Analog sensor object
- S-Analog actor object
- S-Single stage controller object
- S-Gas calibration object
- Trip Point Object

The last group consists of network specific objects:

- DeviceNet object (specific to DeviceNet only)
- ControlNet object (specific to ControlNet only)
- ControlNet keeper object (specific to ControlNet only)
- ControlNet scheduling object (specific to ControlNet only)
- TCP/IP interface object (specific to Ethernet/IP only)
- Ethernet link object (specific to Ethernet/IP only)

The general use objects can be found in many different devices, while the application specific objects are typically only found in devices hosting such applications.

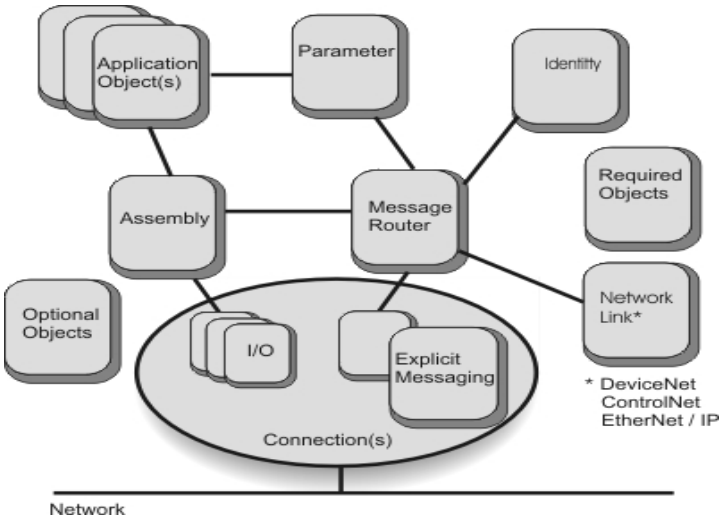


Figure 19.20
Typical device object model

This looks like a huge number of object types, but typical devices only implement a subset of these objects. Figure 19.20 shows the object model of a such a typical device.

The objects required in a typical device are:

- At least one connection object
- An identity object
- One or several network link related object (depends on network)
- A message router object (or at least its function)

Further objects are added according to the functionality of the device. This allows very good scalability of devices so that small devices (e.g. a proximity sensor on DeviceNet) are not burdened with unnecessary overhead. A developer typically uses publicly defined object (see above list), but can also create his own objects in the vendor specific areas, e.g. class ID 100 – 199. However, it is strongly encouraged to work in the special interest groups (SIGs) of ODVA and ControlNet International to create common definitions for further objects instead of inventing private ones.

As an example of the many existing objects, the identity object (class code: 1) is described below. Since the vast majority of devices only support one instance of the identity object, there is typically no requirement for any class attributes. Thus only instance attributes are required in most cases. These are mandatory attributes, such as vendor ID, device type, product code, revision, status, serial number and product name, as well as optional attributes such as state and heartbeat interval.

Typically, devices do not change their identity, so all attributes (with the exception of the heartbeat interval attribute) are read-only.

More details of the CIP objects in general and the identity object in particular can be found in chapter 5 of the CIP specification.

Device profiles

With the definitions of communication links and objects, it would very well be possible to design products. However, similar products could easily have quite different structures inside and could also show quite different behavior. To overcome this situation and to make the application of CIP devices much easier, devices of similar functionality have

been grouped into device types with associated profiles. Such a CIP profile contains the full description of the object structure and their behavior. The following device types and associated profiles have been fully defined by August 2001:

- Generic device
- AC drives
- Motor overload
- Limit switch
- Inductive proximity switch
- Photoelectric sensor
- General purpose discrete I/O
- Resolver
- Communication adapter
- ControlNet programmable logic controller – position controller
- DC drives
- Contactor
- Motor starter
- Soft start
- Human machine interface
- Mass flow controller
- Pneumatic valves
- Vacuum pressure gauge
- ControlNet physical layer

Device developers must use a profile. Any device that does not fall into the scope of one of the specialized profiles must use the generic device profile or a vendor specific profile. What profile is used and which parts of it are implemented must be described in the user documentation of the device.

Every profile consists of a set of objects, some required, some optional, and a behavior associated with that particular type of device. Most profiles also define one or several I/O data formats that define the meaning of the individual bits and bytes of the I/O data. In addition to the publicly defined object set and I/O data assemblies, manufacturers can add objects and assemblies of their own if they have additional data not described in the public profile. Again, it is encouraged to coordinate additional data used by many manufacturers through discussion in the SIGs and eventually create additions to the public profiles for everybody's use and for the benefit of the device users.

More details of the CIP profiles can be found in chapter 6 of the CIP specification.

Electronic data sheets

CIP has made provisions for several options to configure devices:

- A printed data sheet
- Parameter objects and parameter object stubs
- An electronic data sheet (EDS)
- A combination of an EDS and parameter object stubs
- A configuration assembly and any of the above methods

When using configuration information collected on a printed data sheet, configuration tools can only provide prompts for service, class instance, and attribute data, and relay

this information to a device. While this procedure can do the job, it is the least desirable solution since it does not determine the context, content, or format of the data.

Parameter objects, on the other hand, provide a full description of all configurable data of a device. This allows a configuration tool to gain access to all parameters and maintain a user-friendly interface since the device itself provides all the necessary information. However, this method burdens a device with full parameter information and this may be too much for a small device, e.g. a simple DeviceNet slave. Therefore, an abbreviated version of the parameter object, called parameter object stub may be used. This still allows access to the parameter data, but it does not describe any meaning of this data. This is where an EDS is very handy. An EDS supplies all the information that a full parameter object contains on top of what the parameter object stub provides. The combination of EDS and parameter object stub thus provides the full functionality and ease of use of the parameter object without burdening the individual devices.

Finally, a configuration assembly allows the bulk upload and download of a full block of parameters. More details of electronic data sheets used in CIP can be found in chapter 7 of the CIP specification.

Services

Service codes are used to define the action that is to take place when an object of parts of an object are addressed using the addressing scheme described in appendix A of the CIP specification. Apart from the simple read and write functions, a set of CIP Common Services has been defined. These CIP common services are common in nature which means that they are universal in use and a large number of objects typically support these services. Furthermore, there are object specific service codes which may have a different meaning for the same code depending on the object. Finally, there is a possibility to define vendor specific services according to the requirements of the developer. While this gives a lot of flexibility, the disadvantage of vendor specific services is that they may not be understood universally. More details of the CIP service codes can be found in appendix A of the CIP specification.

Data management

The data management part of the CIP specification describes addressing models for CIP entities and the data structure of the entities themselves.

The entity addressing is done by so-called segments, a method that allows to be used in a very flexible way so that many different types of addressing methods can be accommodated. In its simplest form, the segment addressing provides the class/instance/attribute addressing required for CIP objects.

The data types in CIP follow the requirements of IEC 61131-3. There is a set of elementary data types and data types that are derived from these elementary types.

More details of the CIP data management can be found in appendix B of the CIP specification.

19.3 PROFInet

19.3.1 Introduction

PROFInet is an open standard intended as a basis for plant-wide industrial automation, based on Industrial Ethernet. It supports implementation of distributed automation, integration of existing field devices and the implementation of time-critical real-time

applications such as motion control. It is now integrated in IEC 61158 and also exceeds the scope of IEC 61499.

PROFInet tries to define only the minimum number of common points required for an open manufacturer-independent automation system, and can be integrated with existing PROFIBUS DP, PA and FMS systems, as well as systems from other vendors such as ASi and Foundation Fieldbus.

PROFInet's openness is based on the use of several existing open standards such as:

- Industrial Ethernet
- TCP/IP
- Application layer protocols such as RPC and HTTP
- Microsoft COM/DCOM
- ActiveX
- Web technologies such as HTML and XML
- Existing PROFIBUS bus segments and PROFIBUS DP field services

Unlike the relationship between DeviceNet and Ethernet/IP, PROFInet is not merely "PROFIBUS over Ethernet". Although it is compatible with PROFIBUS, it embodies a radically different automation concept and should rather be compared with IDA.

The following considerations were taken into account during the development of PROFInet:

- Openness (using universally accepted standards)
- Consistency (communication and cooperation of devices on all levels via similar methods)
- Integration of existing PROFIBUS systems with a homogenous engineering perspective
- Easy of use
- Upward compatibility to PROFIBUS and existing engineering tools (e.g. PLC programming)
- Component and object oriented. Applications are created by interconnecting objects via graphics, text or scripts.

The fact that PROFInet uses Microsoft's DCOM does not mean that it is confined to Microsoft operating systems. Independent DCOM and RPC implementations are widely available, even on embedded controllers.

Distributed field devices are integrated through PROFInet IO. This is the "usual" IO view of PROFIBUS DP, where the inputs and outputs of the field devices are cyclically transmitted to the process image of the PLC. PROFINET IO uses a device model based on that of PROFIBUS DP, and comprises slots and channels. The characteristics of the field devices are described in a GSD (General station data) file by means of XML.

PROFINET, does, however, also cater for distributed automation scenarios. Here, the autonomous modules of machines or plants ate described in a reusable "component model". This component model is described via an XML based PCD (PROFInet component description), using either the PROFInet component editor or the component generator of the manufacturer.

19.3.2 Distributed field devices

The distributed field devices are directly interconnected with Ethernet. Physically the PROFInet architecture appears similar to that of PROFIBUS, except for the RS-485 that is now replaced with Ethernet.

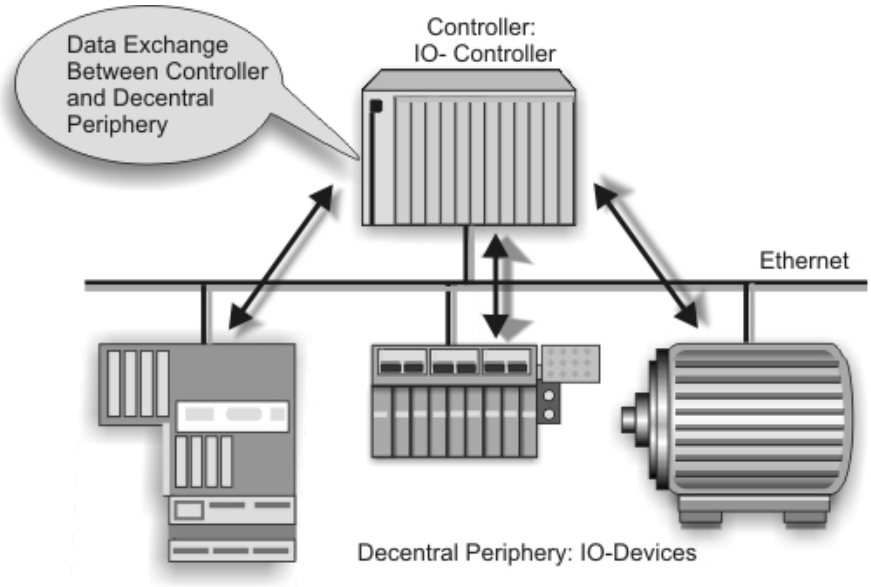


Figure 19.21
PROFINET physical architecture

However, because of the peer-to-peer relationship between Ethernet devices, the master/slave relationship of PROFIBUS can now be replaced with a producer-consumer model. PROFInet distinguishes between 3 types of devices. They are:

- IO controllers. These are the PLCs on which the automation programs are run.
- IO Devices. These are remotely assigned field devices, which are assigned to IO controllers.
- IO Supervisors. These are programming units or PCs with commissioning and diagnostic functions.

Data can now be transferred between IO controller and IO devices over various channels. These are logical channels, all data packets are physically transported by the same Ethernet network. The difference between the real time channel and the standard channel will be described in the following paragraphs. Data is allocated to the various channels as follows.

- Cyclic IO data over the real-time channel
- Event-controlled interrupts over the real-time channel
- Parameter assignment, configuration and diagnostics over the standard channel using UDP/IP

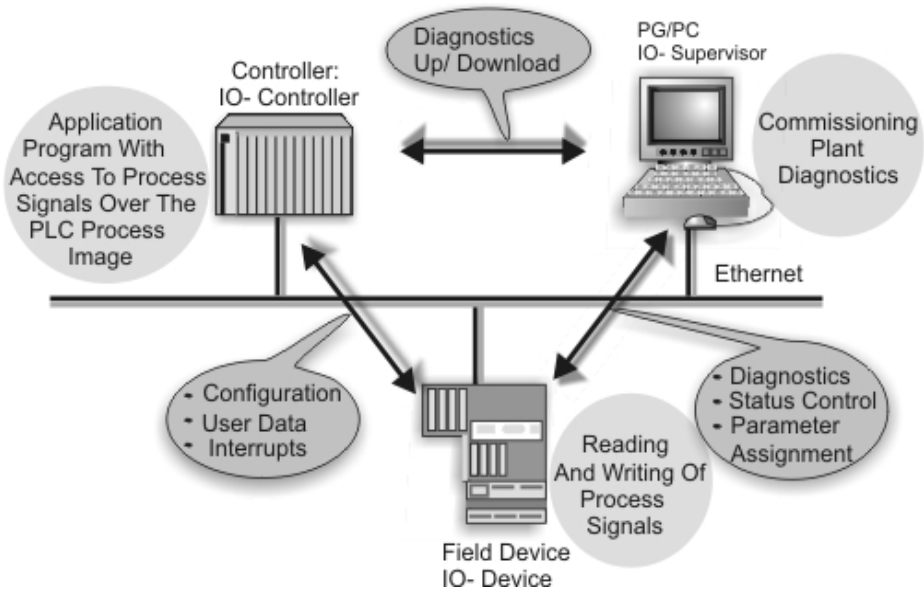


Figure 19.22
Functional scope of PROFInet IO

Initially, application relations (IO-AR) are established between the controller and the device, using the standard (UDP/IP) channel. The user then transfers configuration data to the device. Once it is running, the device may use the real-time channel to exchange IO information or interrupts.

19.3.3 Device models

A uniform device model is specified for PROFInet IO devices, which enables the configuration of modular and compact field devices. This is orientated towards the characteristics of PROFIBUS DP and, for a modular field device, comprises slots for the insertion of modules. These modules are fitted with IO channels which serve the input and output of process signals.

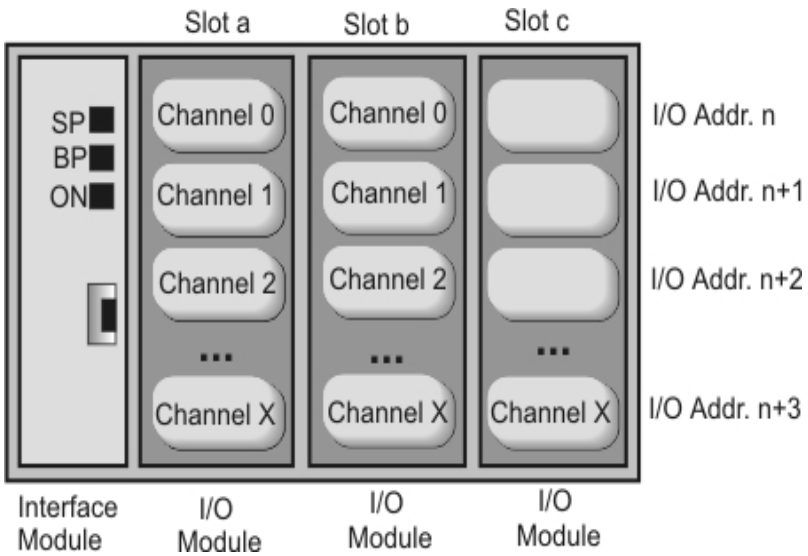


Figure 19.23
PROFInet IO device model

This modular design ensures that the existing PROFIBUS DP range of IO modules can also be incorporated in PROFInet without requiring any modification. This ensures investment protection for device manufacturers and operators or owners (e.g. spare parts inventory).

Each IO device is assigned a globally unique device ID within the framework of PROFInet IO. This 32-bit device-ident-number is broken down into a 16-bit manufacturer ID and a 16-bit device ID. The manufacturer identification is assigned by PROFINET International. The device ID can be individually assigned by manufacturers to suit their own product development.

A PROFInet IO device is integrated in the configuration tool in the same manner as a PROFIBUS DP device, i.e. via a device description. The characteristics of an IO device are described in a GSD (General Station Description), which contains all the information that the field device requires:

- Properties of the IO device (e.g. communication parameters)
- Plug-in modules (quantity and type)
- Configuration data of the individual modules (e.g. analog input modules)
- Parameters of the modules (e.g. 4…20mA)
- Error texts for diagnostics (e.g. wire break, short-circuit)

The GSD is XML-based and corresponds to ISO 15745. The fact that XML is an open, widespread and accepted standard for describing data means availability of powerful tools and characteristics such as the creation and validation through implementation of standard tools, the integration of foreign languages and hierarchical structuring.

19.3.4 Communication

PROFInet's Ethernet communication is scalable and can be run at three different levels, depending on the requirements of a specific subsystem. Using the same Ethernet infrastructure, communications can be scaled from demanding real-time applications to relatively time-independent IT-type communication. All three types of communication can share the network simultaneously.

The transmission time between devices over a 100 Mbps Ethernet is negligible when compared to the processing delays in the stack. For this reason, any improvement in the real-time response of a system has to be achieved primarily by optimizing the stack for both the producer and the consumer.

TCP/IP

Non-time-critical data, such as configuration data, parameter assignment and other interconnection information is transported over a "standard" channel which utilizes TCP/IP. Despite the overheads imposed by TCP/IP, this approach enables communication not only between control and device level entities, but also with information level entities such as MES and ERP systems- within the plant, Intranet or even the Internet. Ethernet plus TCP, UDP and IP, however, implement only the lower 4 layers in the OSI stack, so additional higher layer protocols such as SNMP, FTP and HTTP are needed.

SRT

Within the production plant, time-critical process data is transported over a "real time" channel, called *soft real time* (SRT). This also utilizes Ethernet, but bypasses the higher

layer protocols such as IP and above. Minimizing the stack not only reduces the time needed to send a message, but also reduces the time needed by the recipient to receive the message and produce a response. Not only is the stack minimized to increase performance, but packets are prioritized in accordance with IEEE 802.1Q. Data packets flowing through the switches on the network are then handled on the basis of this prioritization. Priority 6 is the standard for SRT data, giving these packets precedence over Voice over IP applications with priority 5. SRT is implemented in software on certain controllers. The total protocol overhead for real-time data is only 28 bytes.

SRT can be cyclic, as in the case of produce-consume operations, or acyclic as in the case of alarms.

IRT

For truly real-time operations requiring synchronous communications, such as motion control applications, isochronous real-time communication (IRT) is available with a jitter of less than 1 microsecond for consecutive cycles at a clock rate of 1 kHz, for up to 100 nodes. To implement this, PROFInet has defined a time-slot transmission called IRT on top of layer 2 for Fast Ethernet. Participating devices are synchronized via the bus, and then the communication is broken up into repetitive cycles. The first part of the cycle is allocated to the IRT channel and used for critical real-time communication. The remainder of the cycle is the so-called "open channel", used by TCP/IP and SRT. In this way, congestion on the open channel cannot affect the flow of critical real-time information.

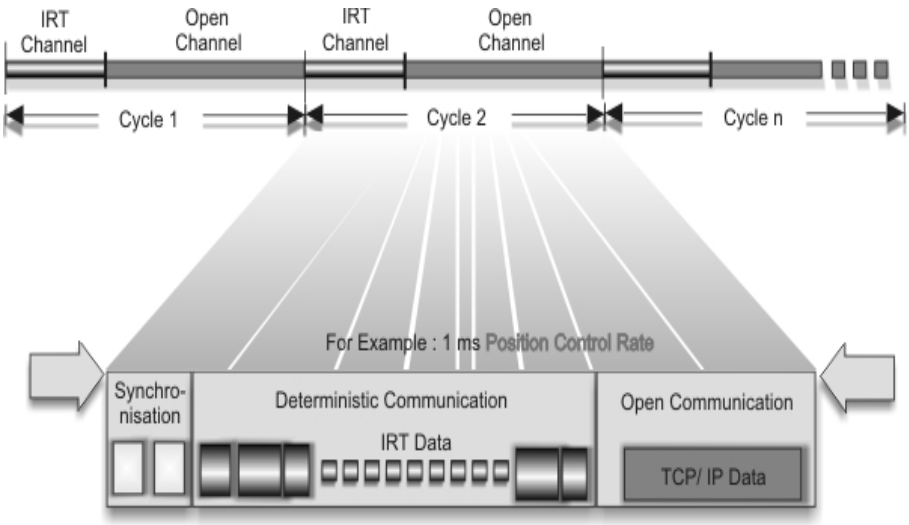


Figure 19.24
Scheduling of IRT

The communication stack

The following figure shows the communication stack for PROFInet devices, especially for the integration of SRT.

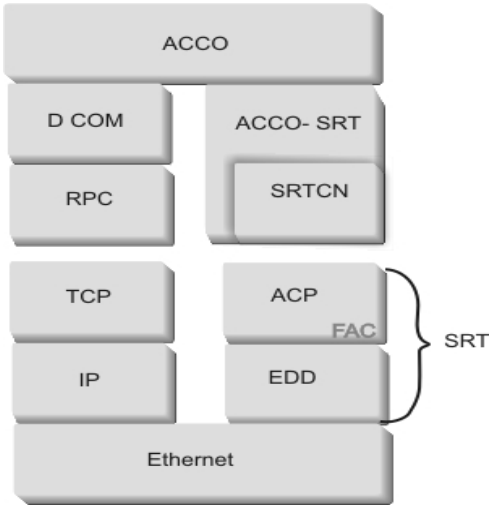


Figure 19.25
PROFINET communication stack

Notice the two "legs" of the stack. In the case of non-time-critical communication, ACCO (the active control connection), communicates via DCOM (distributed COM) and RPC (remote procedure call) over TCP, IP and Ethernet.

In the case of time-critical communications, the right hand side of the stack is used. The following initialisms need some explanation.

- EDD stands for Ethernet device driver.
- ACP stands for alarms (acyclic SRT), consumer (the receive side of cyclic SRT) and producer (the send side of cyclic SRT)
- EAC (Ethernet access control) is a port that the ACP needs to interface with the Ethernet device driver. The performance of the stack is very dependent on how effectively the EAC is integrated into the actual EDD.
- SRT stands for soft real time and covers EDD, ACP and EAC.
- ACCO-SRT is a feature which the PROFInet runtime provides, and which integrates SRT into the ACCO.
- SRTCN handles the application triggered copy of provider items and consumer items in and out of SRT frames.

The following figure shows the internal structure of the ACCO.

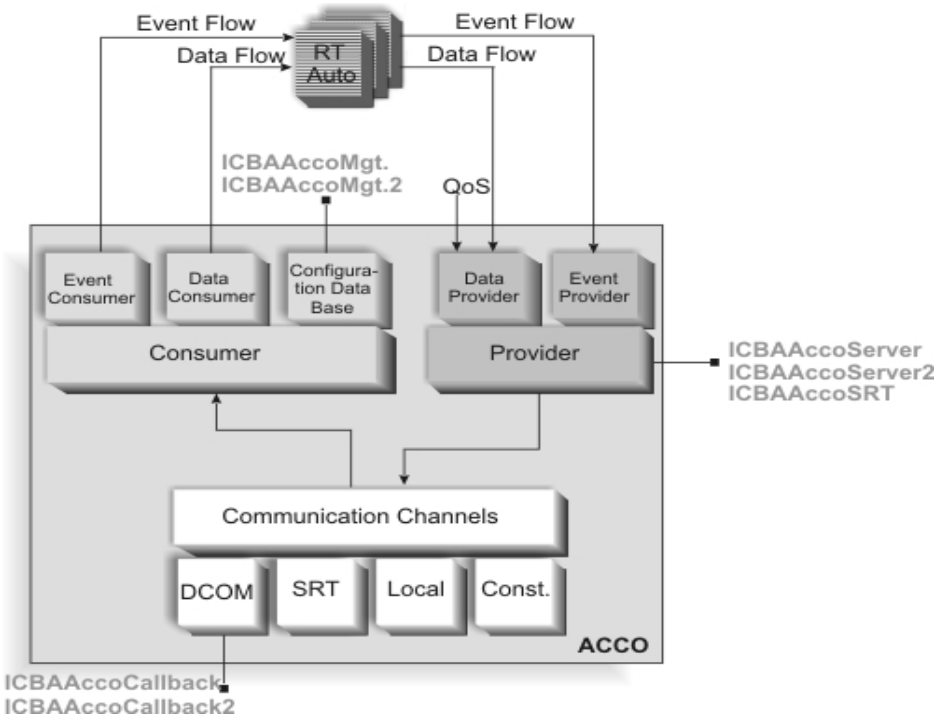


Figure 19.26
ACCO component

The ACCO is, strictly speaking, just a software "black box " (object) with defined interfaces viz. ICBAAccoMgt, ICBAAccoServer etc. Unfortunately a discussion of object-oriented programming as well as DCOM concepts is well beyond the scope of this discussion, so some readers may have to take certain things for granted.

The ACCO consists of the following components:

- The *consumer* implements the functionality required for configuring connections and connection sinks
- The *provider* implements the functionality required for connection sources, and
- *Communication* abstracts the employed communication path for producer and consumer

The QoS (quality of service) is defined as the maximum time duration for transmitting an item change on the provider to the consumer. The QoS is affected by communication times and communication cycles. The number of communication cycles, in turn, depends on whether the provider and/or consumer reside on Ethernet (or some field bus), the communication channel used (DCOM, SRT etc), the TCP stack delays, Ethernet medium delays, switch jams, controller jams and Field bus cycles. Unfortunately not all of these are within the control of PROFInet.

The following paragraphs provide a short overview of the interaction between the ACCO components. The process starts when a user draws "lines" in some PROFInet configuration tool to configure his connections and applies information like substitute value and QoS. On "download" the configuration tool transfers the connection information to the ACCO that carries the consumer of some connections to establish (connection sinks). The consumer then negotiates with the provider the productive

operation of the connections involved. During the productive operation, the provider transmits the connection data via the negotiated communication channel to the consumer.

For *data connections*, this works according to the following block diagram:

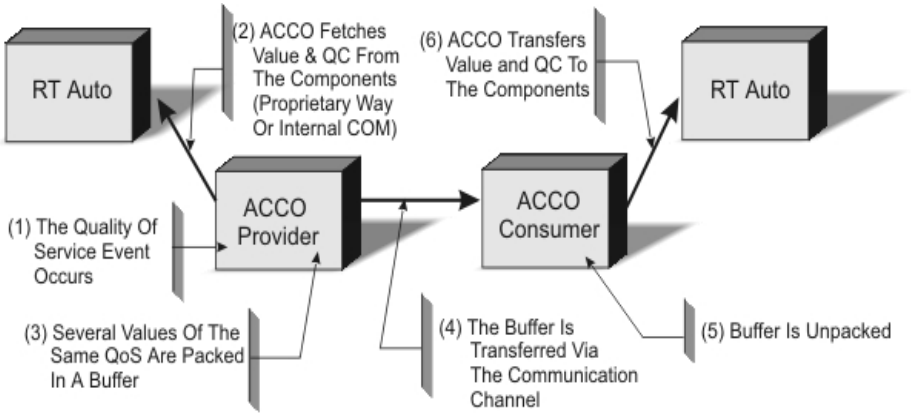


Figure 19.27
Productive operation of data connections

For data connections, productive operation is performed in the following steps:

- An event for the configured QoS (e.g. a time interval of 100 ms expires) occurs at the provider. The way in which that is signaled to the provider depends on the employed QoS and on the implementation of the operating system integration. The signal can be a timer interrupt, for example.
- The connection data is read via the access methods at the RTAutos (real time autos). Note that the time to read the connection data is dependent on the communication channel used. For the DCOM channel the connection data is read when the QoS event triggers the provider. For local and SRT connection the read of connection data is triggered by the application.
- The connection data is filtered for changes (this applies only to the DCOM channel) and packed in a buffer.
- The data item is transferred to the consumer via the communication channel.
- The buffer is unpacked.
- The connection data is transferred to the RTAutos. Note that the write of connection data is done differently according to the communication channel used. For the DCOM channel the connection data is written when the buffer is unpacked. For local and SRT connections the connection data is written at an application specific time.

19.3.5 Cabling and connectors (Layer 0)

Since most stations are supplied with 24V, a hybrid wiring system is ideal. Hybrid cables are available as Cu/FOC cable (2 optical fibers for data, 4 wires for power) as well as Cu/Cu (4 wires for data and 4 for power).

For 100BASE-TX, Cat5 cable is required in accordance with IEC 11801. The entire path has to meet the requirements of class D of IEC11801. PROFInet cables also have a cross-section of AWG 22 instead of the usual AWG 24. For fiber, ISO/IEC 9314-3 (multimode) and ISO/IEC 9314-4 (single mode) is recommended.

PROFInet prescribes the use of ruggedized RJ-45 connectors (IP20) in switchgear cabinets. However, in very harsh environments the use of a RJ-45 with IP65, IP67 or

even IP68 is recommended. The RJ-45 connectors for PROFInet are versions 4 and 5 specified in draft IEC 61076-3-106.

Figure 19.28
RJ-45, rated IP 20 (Courtesy: PI)

Another connector that can be used in harsh environments is the M12 connector. The one used for PROFInet is the shielded D-coded version specified in draft IEC 61076-2-101.

Figure 19.29
RJ-45, rated IP 67 (Courtesy: PI)

For fiber optic systems, the duplex DC connector system in accordance with ISO/IEC 11801 is used.

A hybrid connector is used where field modules are connected with a cable carrying data and power. These connectors are shock-hazard-protected and permit the use of identical connectors at both ends die to the pin-socket changeover is no longer necessary due to the integrated protection.

Figure19.30
Hybrid connector rated IP 67(Courtesy: PI)

19.3.6 Network interface layer (OSI layers 1 and 2)

10BASE-T is supported, but the recommended technology for PROFIBUS is Fast Ethernet (100 Mbps, IEEE 802.u). This includes 100BASE-TX (copper) and 100BASE-FX (fiber). Switches suitable for PROFInet should be rated for Industrial use and should support IEEE 802.1Q as well as auto polarity exchange, auto-negotiation, and auto crossover. Port mirroring for diagnostic purposes is considered optional.

20

Interconnecting, Fieldbuses

20.1 Introduction

Interconnecting bus-based process automation systems can present quite a challenge, especially if they are different in most respects. Often there is little choice but to connect them. For example, ASi systems may be deployed on the plant floor, but the limitation on the bus length may require a connection to another system, such as PROFIBUS, in order to get their information to the control room. Fortunately this problem affects almost everybody in the industry with the result that a multitude of vendors are developing solutions. The elegance of the solution, however, depends on the openness of the specific system and whether it supports certain technologies such as OPC, DCOM etc or not. This section will explore some of the methods used to interconnect industrial networks.

20.2 DeviceNet, ControlNet, Ethernet/IP

It was shown in the previous chapter that these three systems share a common application layer protocol in the form of CIP. They only differ in the implementation of the lower three OSI layers. As a result, it is fairly easy to interconnect them with routers, although not the common IT variety. The reason is that despite their similarities the three systems use different network layer protocols.

The solution is to interconnect them with custom designed routers available from several vendors. There are three types of routers available, depending on what is to be interconnected. There are three options: an Ethernet/IP-DeviceNet router, an Ethernet/IP-ControlNet router and a ControlNet-DeviceNet router. These routers are available as stand-alone units, or as plug-in modules to be used in existing chassis-based equipment. With CIP, every network device presents as a series of objects. Each object is simply a grouping of the related data values in device. Since these CIP message packets are identical for Ethernet/IP, DeviceNet and ControlNet, they can be sent and received to any device at any layer of the system architecture.

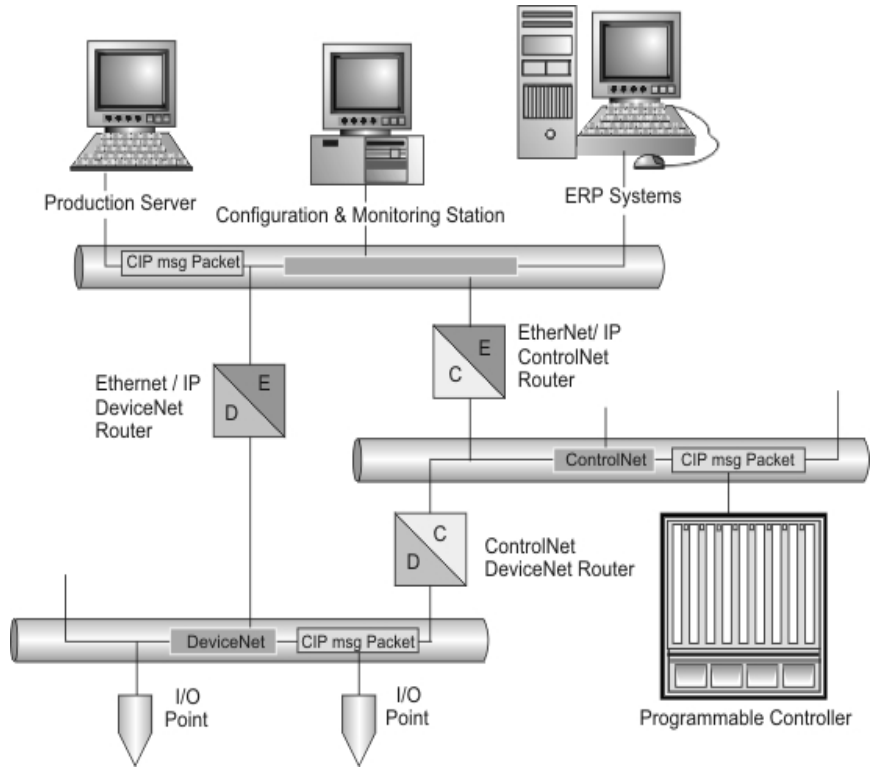


Figure 20.1
Routing between Ethernet/IP, ControNet and DeviceNet

20.3 Gateways

Older ("legacy") systems pose a problem. These are the older systems (non-Ethernet and non-TCP/IP) from different vendors, such as DeviceNet, Foundation Fieldbus H1, PROFIBUS, and ASi. Since these systems belong to different "families" and are quite different in their implementation, the only solution might be a Gateway from a third part vendor.

Gateways between most of the popular systems are readily available, but they might only provide a partial solution as the gateway does not guarantee interoperability of the application software.

20.4 Proxies

Some of the more modern systems such as PROFInet allows communication between field buses by means of a proxy. This means that a user can construct a system consisting of a random mixture of fieldbus and Ethernet based subsystems. On the Ethernet side, the proxy is the "representative" for one or more fieldbus devices. This "representative" ensures transparent communication between networks. For example, it ensures transparent forwarding of cyclic data to the fieldbus device. (Transparency in this case means here that the protocols are not tunneled).

From the perspective of PROFIBUS DP, the proxy would be the PROFIBUS master. The PROFIBUS master coordinates the data exchange between the PROFIBUS nodes and is also the device that communicates with PROFInet (via Ethernet) on the other side. These proxies can be implemented as PLCs, on PCs, or as simple gateways. The fact that the PROFIBUS nodes are connected via a proxy, is not visible to the user on the Ethernet

side. From the user on the Ethernet side, the DP slaves on PROFIBUS are treated like any other IO device. As an alternative solution, an entire fieldbus application can be mapped as a PROFInet component within the framework of the PROFInet component model.

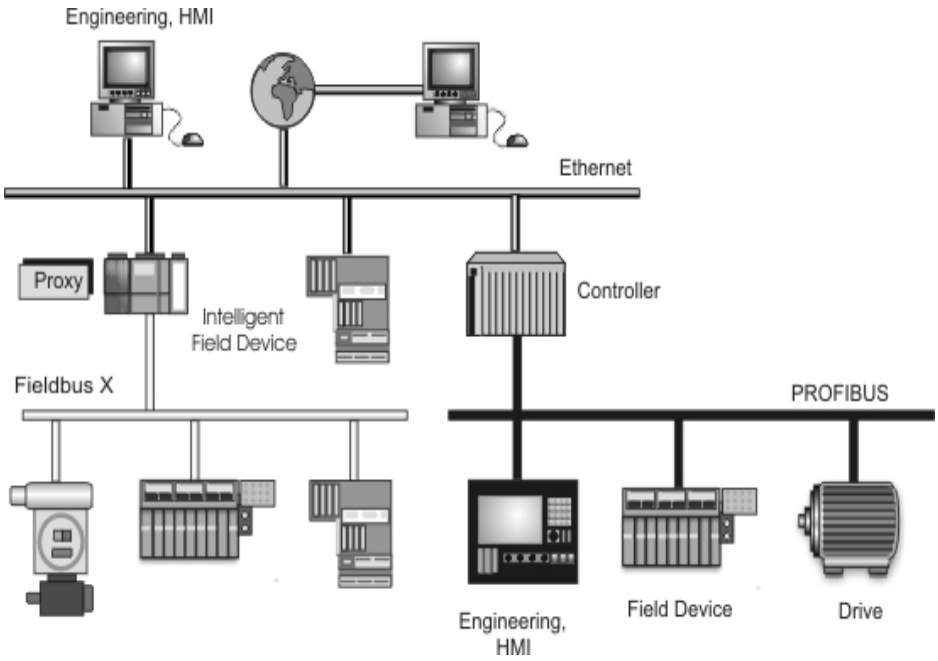


Figure 20.2
Interconnection by proxy

20.4 OPC

Before we discuss OPC, just a few words about COM/DCOM. COM is Microsoft's Component Object Model and DCOM is the distributed version thereof, i.e. DCOM objects can reside on different processors, even on a LAN or a WAN. This is a component software architecture that can be used to build software components (black boxes) that can interoperate regardless of where they are situated. DCOM also provides the mechanisms for the communication between these components, shared memory management between them, and the dynamic loading of components if and when required. These objects are written in an object oriented language such as C++ or VB and the (software) interfaces that can be used to access their functions (or *methods* as they are called) are generally given names starting with an uppercase I such as IOPCShutdown. Each object and each interface is considered unique and is given a 128-bit "GUID" (Globally Unique ID) number that uniquely identifies it. The interfaces are software interfaces, and defined in a way that makes sense (only!) to object oriented programmers.

Unfortunately DCOM is a complicated issue and a more detailed discussion thereof is beyond the scope of this chapter.

OPC, on the other hand, initially stood for OLE (Object Linking and Embedding) for Process Control. Nowadays we refer to ActiveX rather than OLE, but the initialism "OPC" remains. It is an open, standard infrastructure for the exchange of process data, built on DCOM. It specifies a set of software interfaces and logical objects as well as the functions (methods) of those objects. It is a software standard supported by all major software control system vendors and users can develop their own clients and servers if they wish, with or without programming knowledge. The main reason for the existence

of OPC is to access process control information regardless of the operating system or hardware involved. In other words, OPC data is accessed in the same way regardless of who the vendor of the system is. OPC is built around the client-server concept where the server collects plant data and passes it on to a client for display.

Since OPC was designed around Microsoft DCOM it was initially limited to Windows operating systems. Despite its legacy, DCOM is currently implemented on all major operating systems as well as several imbedded real-time operating systems and is therefore effectively vendor neutral. In fact, much of the non-real-time PROFInet communication is built on DCOM.

OPC effectively creates a "software bus" allowing multiple clients to access data from multiple servers without the need for any special drivers.

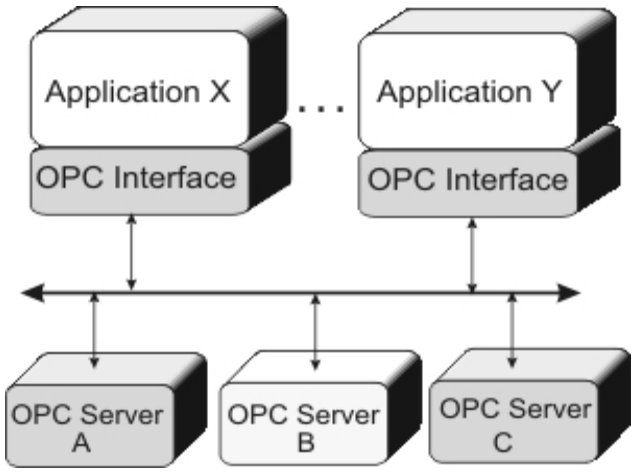


Figure 20.3
The concept of a "software bus"

The following figure shows an application (a client, for example) obtaining plant data from an OPC server. The OPC server, in turn, obtains its information from some physical IO or by virtue of a bridging process from a SCADA system which, in turn, gathers physical plant information through some physical IO.

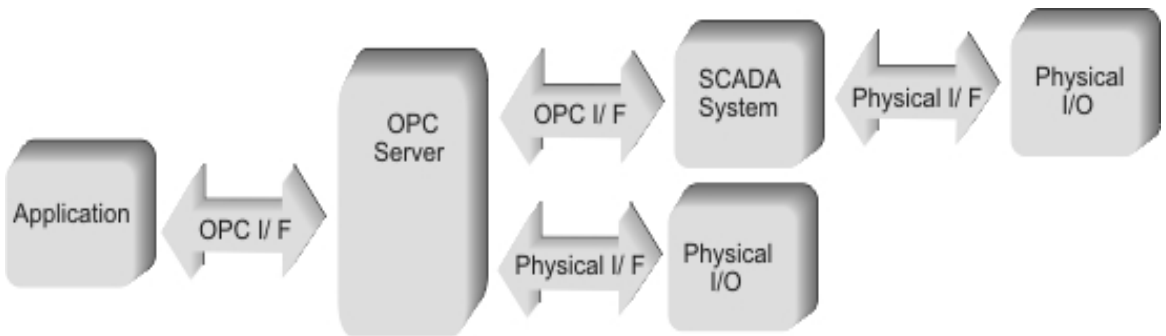


Figure 20.4
Relationship between OPC client and OPC server

There are several OP specifications. The most common one is OPC DA (Data Access). This standard defines a set of application interfaces, allowing software developers to develop clients that will retrieve data from the server. Through the client, the user can

locate individual OPC servers and perform simple browsing in the name spaces (i.e. all the available tags) of the OPC server. Although OPC DA allows a client to write back data to the server, it is primarily intended to read data from the server and hence to create a "window" into the plant.

Since PROFInet is also built on DCOM and, in fact, defines a functionality that goes beyond OPC, each PROFInet node can, in principle, also be accessed as an OPC server. The basic functionality required for this is already implemented by the PROFInet run time implementation (in particular ACCO). There are two options here. As a first option, individual PROFInet nodes can act as OPC servers. As a second option, the OPC server could be implemented on a PC, and the plant data mapped on to the server by the OPC Objectizer.

Another emerging OPC standard is OPC DX (Data Exchange). OPC DX is an extension of the OPC DA specification and defines a communication standard for the higher-level exchange of non-time-critical user data at system levels between different makes of control systems, e.g. between Foundation Fieldbus HSE, Ethernet/IP and PROFInet. However, OPC DX does not permit direct access to the field level of the operating system.

OPC DX allows system integrators to configure controllers from different vendors through a standard browser interface without any regard for the manufacturer of a specific node. In other words, it allows a common view of all devices.

The functionality and performance of PROFINet is greater than that of OPC. However, OPC provides a higher degree of interoperability with other systems.

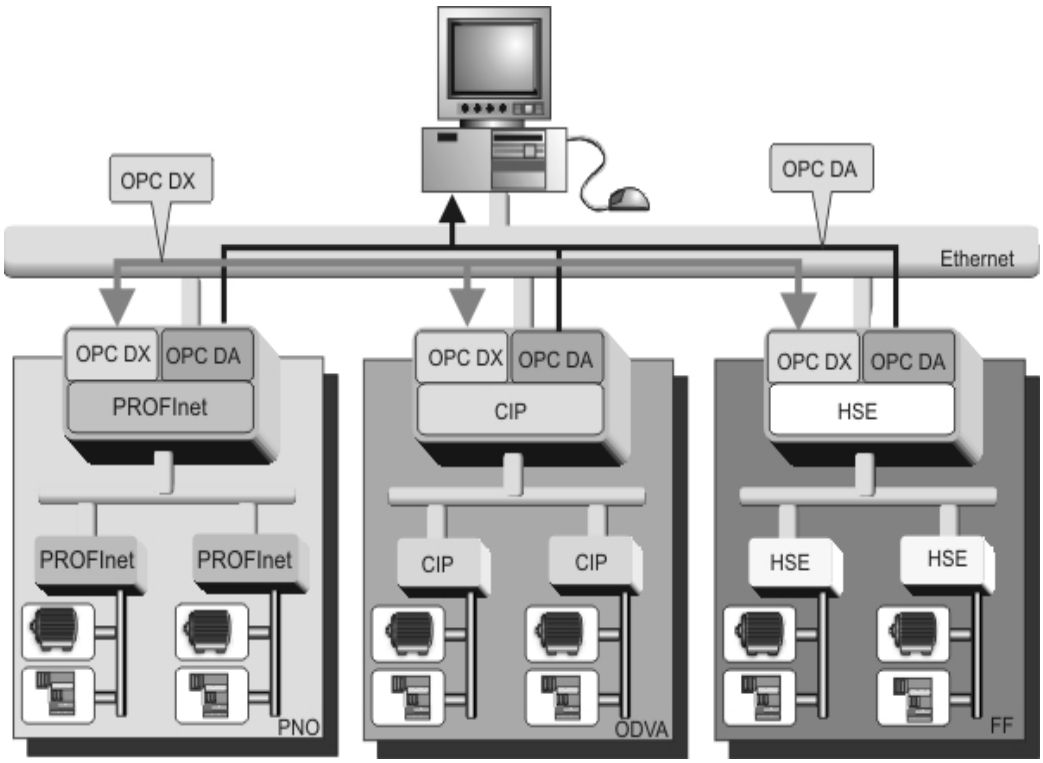


Figure 20.5
Using OPC DX between various systems

21

Virtual LAN

Virtual LAN or VLAN as it is called, is primarily a performance enhancement method and aims at limiting broadcast traffic being limited to segments to which they are meant. This also results in improved security though security is not the primary purpose of a VLAN implementation. VLAN cannot be the sole method of Network security but has to be implemented in combination with other methods discussed in the previous modules.

Objectives

On completing the study of this module, you will learn about:

- Need for Virtual LAN (VLAN), benefits and constraints
- Operating principle of a VLAN
- Methods of implementing a VLAN
- Possible methods of connection
- Filtering table
- Tagging

21.1 Introduction

We have learnt in earlier modules about *Local Area Networks* (LANs) and about the different types of networks such as Ethernet, Token Ring and FDDI. We have also had a look at various network connectivity devices such as repeaters and hubs. Ethernet is by far the most popular networking topology adopted the world over for interconnecting computers. Switched Ethernet has become almost a *de facto* standard with the widespread availability of rugged and low cost switches. Since Ethernet uses a collision sensing type of communication protocol, large networks face problems of network throughput. Segmenting the network somewhat mitigates this problem.

When a LAN is divided into segments using a switch, with each port serving a smaller number of network nodes, the chances of collision reduce. Moreover, the devices that normally communicate with one another are placed in one segment so that the need for forwarding the packets to other ports also gets reduced. In some cases, machines that

require very high bandwidth (for example, a server or a high performance workstation) are connected directly to a switch port, thus enabling them to have almost the entire bandwidth of one segment dedicated to them.

We will learn in this chapter how Virtual LAN (VLAN) technology improves the LAN performance and also helps in making the network more secure. At the same time, it should be noted that security is not the main consideration in setting up VLANs nor they can be the sole approach to network security. When used in combination with other security implementations, a VLAN can result in improved network security.

21.2 Need for VLAN

To begin with, LANs were just that: networks that served users in a small area, usually a small department or a workgroup with one server and a few clients. Very often, different floors in an office each had a LAN. Sometimes these LANs were connected by routers to obtain a set of interconnected LANs (Figure 21.1).

The routers, which connected the LANs, enabled communication between LANs but they were often complicated, slow and expensive. With the advent of Ethernet switches a need for a better solution was felt. Routers were therefore pushed to the periphery, say, for example, communication links leading to the external world (a WAN or the Internet). See Figure 21.2.

A switched LAN also has some disadvantages. While switches are able to segment unicast traffic (one node to another), multicast or broadcast traffic is allowed to pass through, unlike routers. In other words, while each segment is a collision domain, the entire LAN is a broadcast domain. This can become a bottleneck restricting LAN throughput.

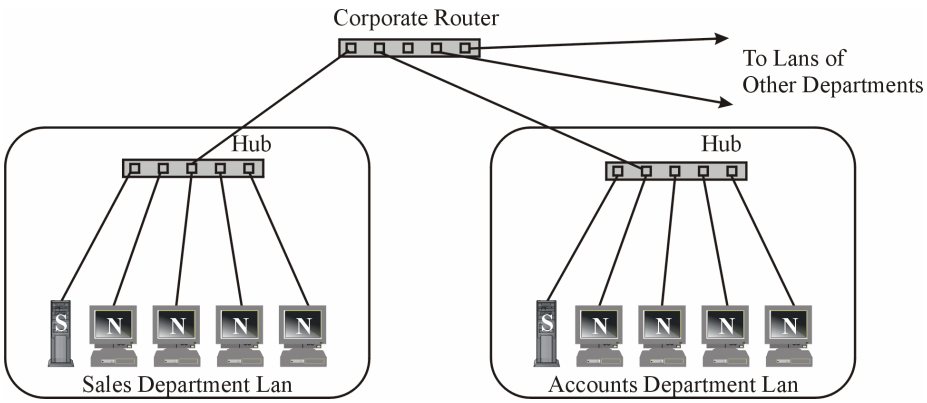


Figure 21.1
Interconnected LANs

The other factor, which affects performance, is due to changes in the present business environment. Very often the personnel involved in a particular project or those belonging to a particular department are not confined to a given area and are spread throughout a building or campus. Product design teams may be cross functional groups and usually exist for short periods of time. In such cases, grouping the users into one physical segment is not feasible. Refer to figure 21.3. In these cases, more packets have to travel from one physical segment (or switch port) to another, thus increasing the network loading. VLANs offer a way to overcome these problems.

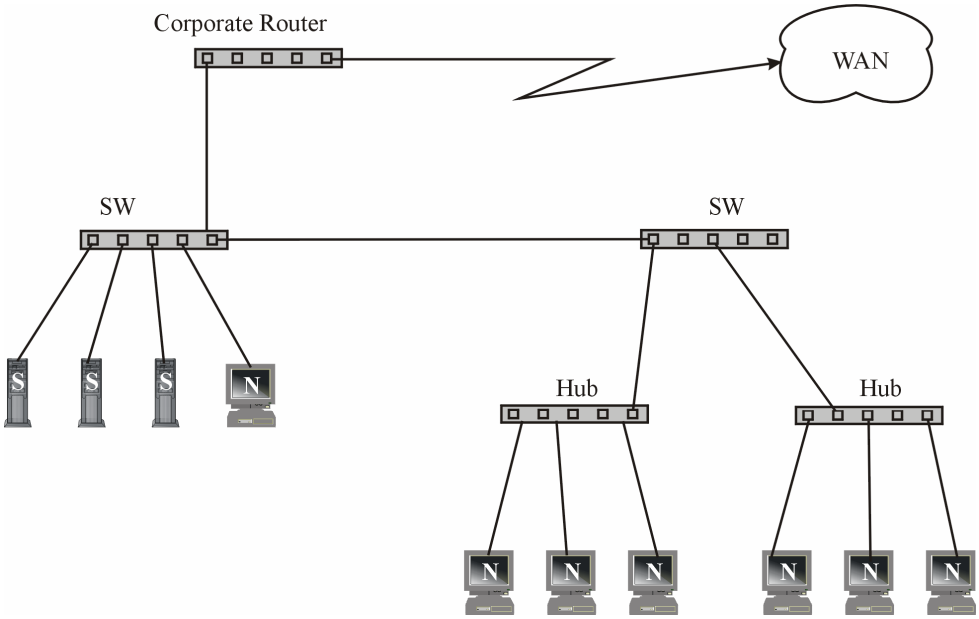


Figure 21.2
A switched LAN

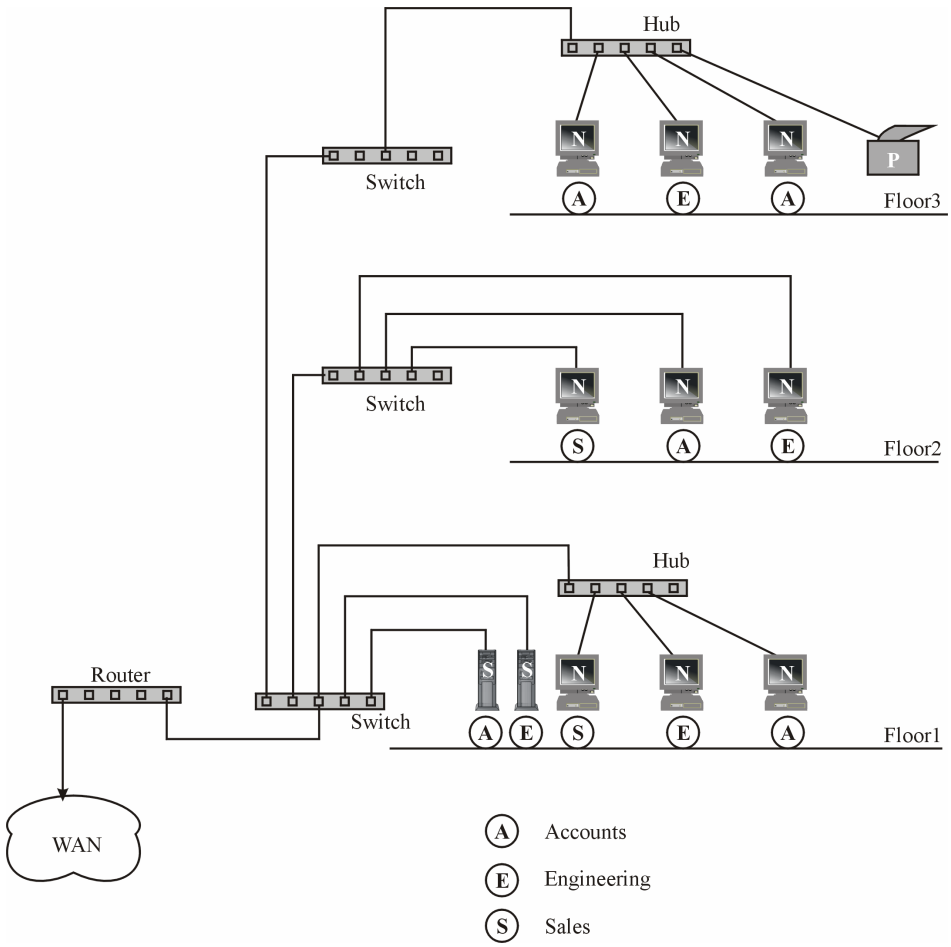


Figure 21.3
Workgroup distribution within a building

A VLAN logically groups switch ports into workgroups. Since broadcasts and multicasts between the users of a workgroup are likely to be high, a VLAN that includes members of a given workgroup limits the broadcast traffic to within the particular virtual network. Thus a VLAN performs like a virtual broadcast domain. Figure 21.4 shows the logically defined VLAN network.

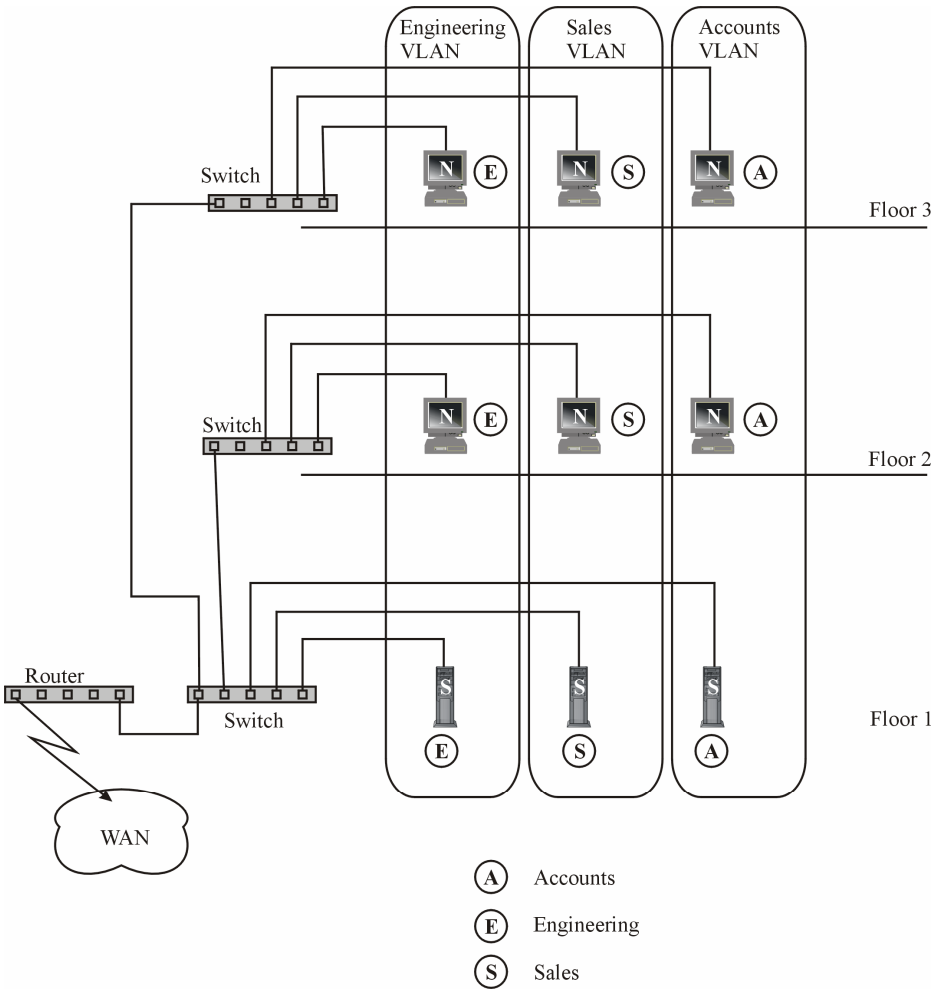


Figure 21.4
A typical network of VLANs

21.3 Benefits of a VLAN

VLANs offer a number of advantages over the traditional LAN implementation. We will discuss these advantages below:

Performance improvement

In networks having a high proportion of broadcast traffic, a VLAN can improve network performance by limiting the broadcast from going to destinations for which the broadcast is not intended. No doubt a router could perform this function, but the greater amount of processing required for a router to route the frames or packets increases latency and therefore reduces the performance of the network. VLAN is simply more efficient.

Improved security

By the ability to contain the broadcast activity within a workgroup, access to sensitive data broadcast on a network is limited only to those for whom such broadcast is intended. This improves data security of the network.

Ability to set up virtual workgroups

As discussed earlier in this chapter, VLAN makes it possible for the network administrator to define a set of users as belonging to a virtual broadcast domain irrespective of their physical location. The only alternative to VLAN is to physically move them. Given that present day organizations are more dynamic, frequent changes in workgroups have also become necessary. This can be achieved by VLAN without the need of physical relocation every time a change takes place.

Reduced administration

A considerable part of the administration efforts go towards additions, movements, and changes, all of which involve reconfiguration of hubs, routers, and station addressing and sometimes re-cabling. A rough estimate is that 70% or more of administration time is spent on such activities. A VLAN reduces the need for such changes and with good management tools; all that has to be done is a simple drag and drop action to change a user from one VLAN to another.

Reduced cost

The ease of administration, the avoidance of physical movement/cabling changes and elimination of the need for expensive routers to contain broadcast activity keep the costs lower.

21.4 VLAN constraints

VLANs are not without problems. When resources such as a printer have to be shared within a logical group, it may cause problems for some who will have the printers assigned to them on another floor of the office far from where they work.

Also, the present trend is to group different servers in a common server farm, with increased physical security, environment control, fire prevention measures etc. These servers may have to be accessed by members of more than one VLAN. If it is not possible to assign the server to more than one VLAN, such access is not possible, unless the servers are put in a separate VLAN of their own and connected to the rest of the network by a router. This can slow down network throughput.

Thus the ability for a node to be assigned to more than one VLAN is a critical success factor in a VLAN implementation. The constraints pointed out in the previous factor cannot be otherwise overcome.

Also, VLAN implementations are by and large proprietary, which makes inter-operation between switches difficult. This means that anyone wanting to establish a VLAN has to procure all products from the same vendor. Standards for VLAN such as IEEE 802.1Q are still under evolution and in the absence of such standardization VLANs remain somewhat restrictive.

21.5 Operating principle of a VLAN

We will see how a VLAN functions as a virtual broadcast domain. The switch is the core component of any VLAN implementation. It serves as the point of communication to data destined for a node or data received from a node connected to it. The network

administrator decides how VLANs will be grouped. The switch implements this decision. The switch provides the intelligence required to identify the group to which a packet belongs. This can be carried out either by packet filtering or by packet identification (also called tagging).

Packet filtering is similar to the technique used by routers. In this method, each packet received by the switch is examined and compared with a filtering table (also called a filtering database). The filtering table that is developed for each switch contains information on user groups based either on port address, MAC (Media Access Control) station address, protocol type or application type (we will learn about these in greater detail later in this chapter). The switch takes appropriate action based on the comparison. Packet filtering thus provides an additional layer of processing for deciding as to how the packet is to be handled. It thus increases the switch latency. It is also necessary that all the switches in the network must maintain and synchronize their filtering tables, which involves administrative overheads.

Packet identification is a relatively new method and consists of placing a unique identifier in the header of each packet as it travels through the switch fabric. The identifier is examined by each switch prior to any broadcast or transmissions to other switches/routers or workstations in the network. When a packet exits the switch fabric (routed to the target end station) the switch removes the identifier from the header. Packet identification functions in layer 2 and does not involve administrative overheads.

The latter method of adding an identifier to the header for identification is called explicit tagging. On the other hand, the method of packet filtering is said to use implicit tagging.

In explicit tagging, the switch needs to know whether the device to which a packet is being sent is a VLAN aware device or not; in other words, whether the device is provided with the intelligence to interpret the tag information. The identifier is added only if the device is VLAN aware.

21.6 VLAN-Implementation methods

The membership of an end user device in a VLAN can be implemented in several ways.

In general, they will fall under one of the following five categories:

- Grouping by switch port
- Grouping by MAC address
- Grouping by network layer information
- Multicast group
- Higher layer grouping

We will review each of the approaches in detail.

Grouping by switch port

In this method, the membership to a VLAN depends on the switch port to which it is connected. Two types of grouping are possible.

In the first, grouping can be within a single switch, say ports 1, 3, 5 and 7 make up one group and 2, 4, 6 and 8 make up a second group. (Refer to figure 21.5).

In the other method, which will involve more than one switch, a VLAN can include specific ports of different switches. As an example: ports 1, 2, 3 and 4 of switch-1 and ports 1,2, 3 and 4 of switch-2 may together form VLAN-A and ports 5, 6, 7 and 8 of switch-1 and ports 5, 6, 7 and 8 of switch-2 form another VLAN-B. (Refer to figure 21.6).

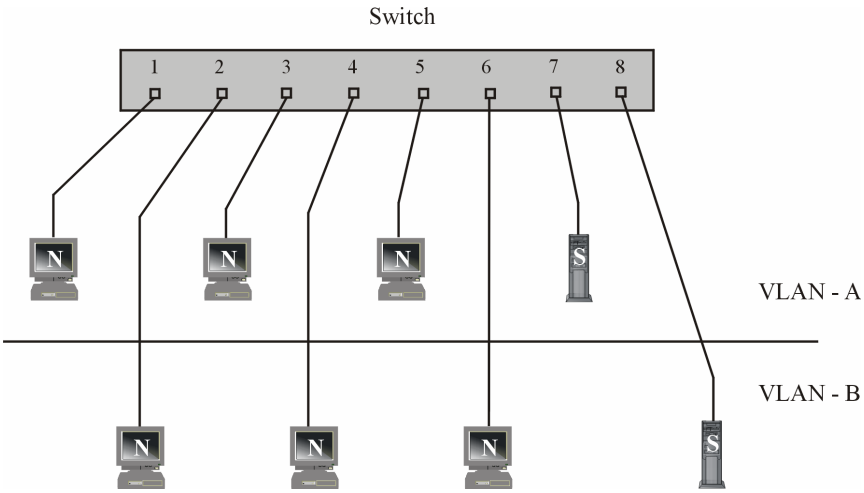


Figure 21.5
Single switch with ports grouped

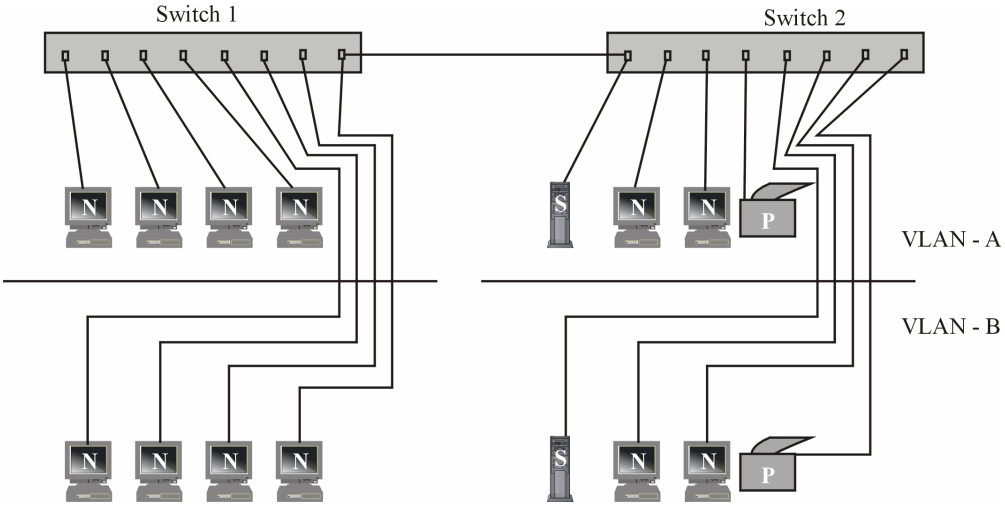


Figure 21.6
Two switches with ports grouped

The second method is the one that is more commonly adopted. The limitation of the method of using the switch port to indicate grouping is that whenever a particular user moves to a new location involving a different switch/port, the network administrator has to reconfigure the VLAN membership information.

Also, this grouping works best for nodes that are directly connected to a switch port. If a machine is connected to a port through a hub, all the machines connected to the hub have to be necessarily members of the same VLAN as one shared media or physical segment cannot be assigned to multiple VLANs. However, hubs with a multi-backplane connection feature overcome this limitation to some extent, since each backplane of the hub can be allocated to a particular VLAN.

The switch port grouped implementation works well in an environment where network moves are done in a controlled fashion and where robust VLAN management software is available to configure the ports.

Grouping by MAC address

The problem of manual reconfiguration is overcome by grouping different machines based on their MAC Addresses. These addresses are machine specific and are decided based on the hardwired information unique to each individual network interface card (NIC). This ensures that whenever a user moves from one switch port to another, the grouping is still undisturbed and whatever is the move, the user's VLAN membership is retained without change. For this reason, this method can be thought of as a user based VLAN implementation.

This method has some drawbacks too. The MAC address based implementation requires that initially each MAC address be assigned to at least one VLAN. This means that to begin with, there will be one large VLAN containing a very large number of users and this may lead to some performance degradation. Some vendors who provide automated tools to create groups based on the initial network status overcome this. In other words, a MAC address based VLAN is created automatically for each subnet of the LAN.

Serious performance issues can arise if members of different VLANs coexist in a shared media environment (within a single segment). Also, in the case of large networks, the method of communicating the VLAN membership information between different switches to keep them always synchronized can cause performance degradation too.

The use of MAC address to identify VLAN membership can be problematic in a network where a large number of laptop computers are used and are connected to the network by means of docking stations. The NIC and therefore the MAC address is a part of the docking station, which usually remains on a particular desk. In a situation where the user and his/her laptop move around to different desks/docking stations their MAC address gets changed when they move to a different desk. This makes tracking groups based on MAC addresses difficult since reconfiguration will be needed whenever a user moves to a different docking station.

Grouping by network layer information

This method is also called as Layer 3-based VLAN implementation. Here, grouping is based on layer 3 information such as protocol type, or in many cases network-layer address (such as subnet address in IP networks). The switches examine the information regarding the subnet address contained in each packet which decides to which VLAN the user belongs and make the routing decision on this basis. It should be noted that even though this method uses layer 3 information, this does not constitute routing. This is because no route calculation is done, frames are usually bridged according to the spanning tree algorithm and connectivity within any VLAN is seen as a flat-bridged topology. See figure 21.7 for an example of this type of implementation.

Some vendors, however, incorporate varying amounts of intelligence in their switches so that they carry out functions normally associated with layer 3 devices. Some of these layer 3 aware devices have the packet forwarding or routing function built into ASIC chips on the switch, which make them faster than CPU, based routers. However, the process of examining layer 3 addresses in a packet is definitely slower than looking at MAC addresses in frames. These implementations are therefore slower than those, which use layer 2 information.

VLANs grouped using layer 3 information are particularly effective in dealing with networks using TCP/IP than those, which use protocols such as IPX or DECnet, which do not involve manual configuration at the desktop. Particularly, end stations using un-routable protocols such as NetBIOS cannot be differentiated and therefore cannot be included in these VLANs.

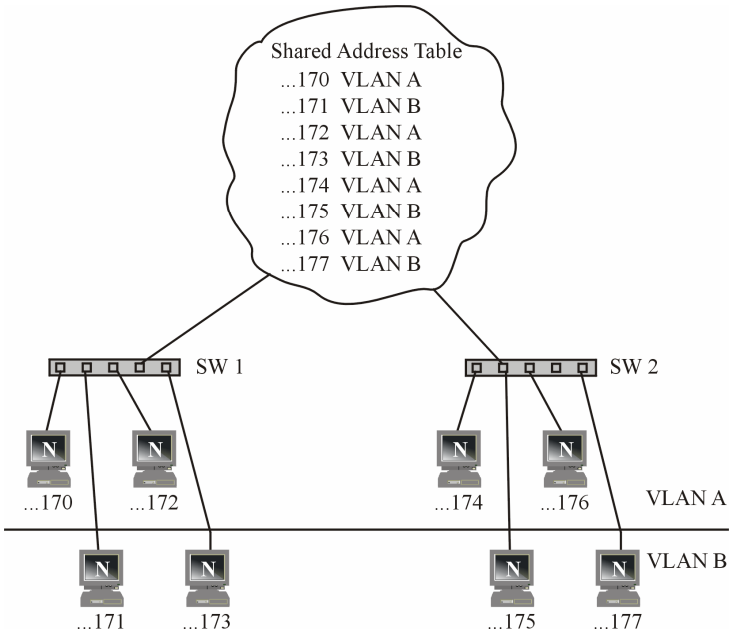


Figure 21.7
VLAN grouping by network-layer address

VLANs using multicast groups

This represents a different approach to VLANs although the concept of a VLAN as a broadcast domain is still applicable. When an IP packet is sent via multicast, it is sent to an address that is a proxy for a group of IP addresses dynamically defined. Each workstation is given a chance to join this group for a certain period of time by sending an affirmative response to a broadcast. They are however members of the multicast group only for a certain period of time. This approach is thus highly flexible and exhibits a high degree of application sensitivity. Such groups can also span routers and therefore WAN connections.

VLANs using higher layer grouping

VLANs implemented using this method of grouping are defined using applications or services or a combination. For example File Transfer Protocol (FTP) applications can belong to one VLAN and TELNET applications can belong to another. Such complex VLAN implementations need a high degree of automation in configuration and management.

Combination of definitions

Considering that each of the above categories have their unique advantages and drawbacks, several vendors are attempting to use multiple methods of VLAN grouping in their products. This will give greater flexibility in configuring VLANs by network administrators depending on the specific features of their networks. For example a network may use both IP and NetBIOS protocols. The ability to define grouping based on IP Subnet addresses as well as by MAC addresses will enable separate VLANs to be set up in the same environment for IP subnets and NetBIOS end stations.

21.7 Method of connections

Devices in a VLAN can be connected in different ways depending on whether the device is ‘VLAN aware’ or ‘VLAN unaware’. As we saw earlier in this chapter, a VLAN aware

device can understand explicit tagging in frames and interpret to which VLAN the frame is to be directed.

These are called:

- Trunk link
- Access link and
- Hybrid link

We will discuss these in a little detail.

Trunk link

All devices connected to a trunk link should be VLAN aware. This includes workstations. All frames on a trunk line should be explicitly tagged. These special frames are known as 'tagged frames'. Figure 21.8 below shows an example.

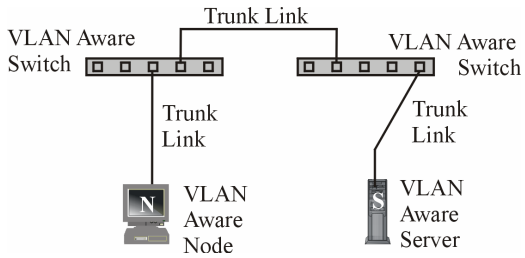


Figure 21.8
An example of trunk link

Access link

An access link connects a VLAN unaware device to a VLAN aware bridge. It is usual to find VLAN unaware LAN segments in legacy LANs. Since VLAN unaware devices cannot handle explicitly tagged frames, the frames must be implicitly tagged, or in other words, they must be untagged. Refer to figure 21.9 for an illustration.

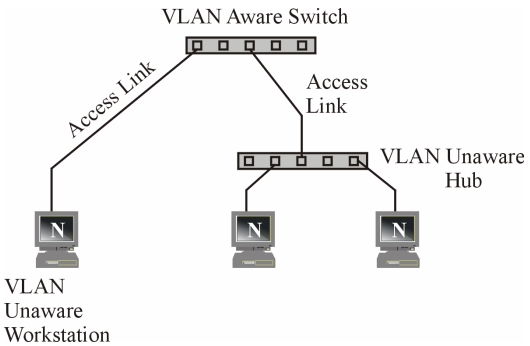


Figure 21.9
Access link

Hybrid link

As the name indicates, Hybrid link is one that connects both VLAN aware and VLAN unaware devices together. A hybrid link can have tagged as well as untagged frames, but all the frames for a specific VLAN should be either tagged or untagged. Refer to figure 21.10 below.

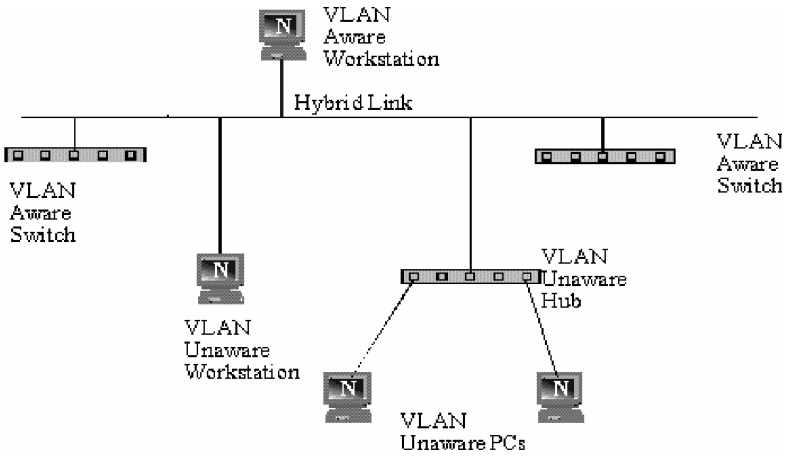


Figure 21.10
Hybrid link

All three types of links can be present in a single network.

21.8 Filtering table

As we saw earlier in this chapter, the filtering table in each switch is a database that stores the VLAN membership information that permits a switch to decide about how a packet is to be handled. There are essentially two types of entries, viz. static and dynamic.

Static entries

The static entries are those that are created, modified or deleted by the network administrator. These are not automatically deleted by ageing, unlike dynamic entries.

Static entries can either be filtering entries or registration entries. A filtering entry is made for each port whether the frames to be specific address in a VLAN should be forwarded, discarded or should be decided by a dynamic entry. Static registration entries specify whether frames to be sent to a specific VLAN should be explicitly tagged and which ports are members of a VLAN.

Dynamic entries

Dynamic entries are automatically created by the switch and cannot be added, or changed by the administrator. Unlike static entries they are automatically removed from the table after a certain time decided by he administrator and can vary between ten seconds to a million seconds. The learning process is by observing the packets sent by a port identifying the address and VLAN membership and the entries are made accordingly.

The entries can be of the dynamic filtering type, dynamic registration type or group registration type. Dynamic filtration entries specify whether a packet to be sent to a specific address in a particular VLAN should be forwarded or discarded. Dynamic registration entries specify as to which ports are registered in a specific VLAN. Addition or deletion of entries is done using a protocol called GVRP. Group registration entries indicate for each port whether frames to be sent to a group MAC address should be filtered or discarded. These entries are added or deleted using *Group Multicast Registration Protocol* (GMRP).

GVRP has another function too, that of communicating information to other VLAN aware switches. Such switches register and propagate VLAN membership to all the ports that are part of the currently active topology of the VLAN.

21.9 Tagging

Tagging is the process of adding information to the frame in the form of a tag header to enable switches to forward frames only to specific ports that belong to a particular VLAN instead of to all ports as done otherwise. Frames containing tag headers are called tagged frames.

The tag header also includes the following features:

- It allows user priority information to be specified
- It allows source routing information to be specified
- It includes MAC address format

The formats of tag frames are different for Ethernet and for Token Ring/FDDI networks. There are two parts to a tag header. They are: *Tag Protocol IDentifier* (TPID) and *Tag Control Information* (TCI). TPID is 2 bytes long for Ethernet and 8 bytes for Token Ring/FDDI, and contains a predefined number that identifies the frame as a modified (i.e. tagged) frame. TCI is 2 bytes long and contains a three-bit priority number as well as a twelve-bit VLAN ID number.

21.10 Summary

VLANs are useful when a network suffers from heavy broadcast traffic that needs to be contained or where the cost of frequent moves of network end user devices both due to recabling and due to consequent administrative overheads has become too high. They improve network security by preventing network traffic from going beyond the group for whom they are meant. It should be remembered that VLANs are proprietary solutions, which may need all hardware and management tools to be purchased from a single vendor. Also VLANs will NOT be effective if the network has a large presence of computers with shared media access. The actual implementation strategy will depend on the needs and practices of the user organization.

Appendix A

FIELDBUS COMPARISON SPREADSHEET- Note: whilst every effort is made to ensure the accuracy of this spreadsheet, please report any errors or blanks to the author

FIELDBUS	Standard	Continuous Or Digital	Topology	max Segment Length	max Speed	Wire Type	max devices	max data/ PDU	Inter-Operability	Device Plug and Play	Loop Powered	Control in field	On-Line Config	Time Stamp and synch	Obsolescence robustness	Predictive Maint
ASI	EN50295	Continuous and Digital	bus/tree	100m 300m total	167kb/s	2	32	**4 bits**			Optional					
CAN	ISO11898 ISO11519	DIGITAL	bus	500m/125kb/s 40m/1Mb/s	1Mb/s	2	64	8 bytes								
ControlNet	EN50170 IEC61158	DIGITAL	bus/star/ tree	5km 250m/48 nodes	5Mb/s	**coax**	99	510 bytes								
DeviceNet (ODVA)	EN50325	DIGITAL	bus	500m/125kb/s 100m/500kb/s	500kb/s	4	64	8 bytes								
Foundation Fieldbus	IEC61158 EN50170	Continuous and Digital	bus	2000m 9.5km total	31.25kb/s-H1 100Mb/s –HSE	2	240	246 bytes				patented				
FIP	EN50170 EN50254 IEC61158	Continuous and Digital	bus	1000m	2.5Mb/s	2	256	256 bytes			Optional					
INTERBUS-S	EN50254 IEC61158	Continuous and Digital	ring	12.8km	500kb/s	2/8	255	64 bytes								
LON	ANSI 709-1-A-1999	DIGITAL	bus/tree	6.1km/5kb/s	1.2Mb/s	2	2	228 bytes								

Modbus plus	Proprietary	Continuous and Digital	bus	1.8km	1Mb/s	2	32	32 bytes								
P-Net	EN50170 IEC61158	Continuous and Digital	bus/tree	1200m 9.3 km total	76.8kb/s	2	32 masters 125 slav	56 bytes								
PROFIBUS FMS	EN50170	DIGITAL	bus	19.2km/9.6kb/s 200m/500kb/s	500kb/s	2	127	246 bytes								
PROFIBUS DP	IEC61158	DIGITAL	bus	1km/12Mb/s (4 repeater)	12Mb/s	2	127	246 bytes								
PROFIBUS PA	IEC61158		bus	1.9km	93.75kb/s	2	32	246 bytes								
SERCOS	IEC61491	DIGITAL	ring	250m	16Mb/s	2/fiber	245	16 bytes								
Seriplex	Proprietary	DIGITAL	bus	1000 feet	~250kb/s	4	510	32 bytes								
SwiftNet	Proprietary IEC61158	DIGITAL	bus	360m	5Mb/s	2	>1024	896 bytes								

Courtesy of Jim Russell of www.iceweb.com.au

Technology Training that Works

Pre-Workshop Questionnaire

PRACTICAL FIELDBUS, DeviceNet and Ethernet for Industry

Full Name _____

City/Country _____ Date _____

Would you kindly answer the questions below.

Please answer all questions to the best of your ability.

1. What are the two main reasons for you attending this training workshop?

   _____

   _____

   _____

2. Briefly describe your main responsibilities in your current job?

   _____

   _____

   _____

3. Where did you hear about this workshop?

   ☐ IDC Technologies Brochure ☐ Web Site

   ☐ Colleague ☐ Other _____

4. Have you been on a previous IDC Technologies workshop? ☐ Yes ☐ No

5. In which area do you work?

   ☐ Trades ☐ Manager

   ☐ Technician ☐ IT

   ☐ Technologist ☐ Engineer

   ☐ Other _____

6. What size is your organisation?

   ☐ less than 50 people ☐ More than 100 People

   ☐ Between 50 and 100 people

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Technology Training that Works

Technical Questions

1. What is the difference between RS-232 and RS-485?

2. What are the basic characteristics of ASi Bus?

3. What is the maximum length you can run DeviceNet over?

4. What are the main features of Industrial Ethernet ?

5. How far can you run 100BaseT Ethernet ?

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Post Course Questionnaire

Workshop Name: ……..

Name: ……Company: ……. Date: ……..

To help us improve the quality of future technical workshops your honest and frank comments will provide us with valuable feedback.
Please complete the following:

How would you rate the following?

Please place a cross (x) in the appropriate column and make any comments below.		Poor 0	1	2	3	4	Average 5	6	7	8	9	Excellent 10
1.	Subject matter presented											
2.	Practical demonstrations											
3.	Materials provided (training manual, software)											
4.	Overhead slides											
5.	Venue											
6.	Instructor											
7.	How well did the workshop meet your expectations?											
8.	Other, please specify…….											

Miscellaneous

Which section/s of the workshop did you feel was the most valuable? .....

Was there a section of the workshop that you would like us to remove/modify? .....

Based on your experience today, would you attend another IDC Technologies workshop? Yes ☐ No ☐

If no, please give your reason .....

.....

.....

Do you have any comments either about the workshop or the instructor that you would like to share with us? .....

.....

.....

.....

Do we have your permission to use your comments in our marketing? Yes ☐ No ☐

Would you like to receive updates on new IDC Technologies workshops and technical forums? Yes ☐ No ☐

If yes, please supply us with your email address **(please print clearly)**: .....

and postal address: .....

Do you think your place of work may be interested in IDC Technologies presenting a customised in-house training workshop? Yes ☐ No ☐
If yes, please fill in the following information:

Contact Person: ..... Position: .....

E-mail: ..... Phone Number: .....

New IDC Workshops

We need your help. We are constantly researching and producing new technology training workshops for Engineers and Technicians to help you in your work. We would appreciate it if you would indicate which workshops below are of interest to you.
Please cross (x) the appropriate boxes.

	INSTRUMENTATION AUTOMATION & PROCESS CONTROL		INFORMATION TECHNOLOGY
	Practical Automation and Process Control using PLCs		Practical Web-Site Development & E-Commerce Systems for Industry
	Practical Data Acquisition using Personal Computers & Standalone Systems		Industrial Network Security for SCADA, Automation, Process Control and PLC Systems
	Practical On-line Analytical Instrumentation for Engineers and Technicians		
	Practical Flow Measurement for Engineers and Technicians		**ELECTRICAL**
	Practical Intrinsic Safety for Engineers and Technicians		Safe Operation and Maintenance of Circuit Breakers and Switchgear
	Practical Safety Instrumentation and Shut-down Systems for Industry		Practical Power Systems Protection for Engineers and Technicians
	Practical Process Control for Engineers and Technicians		Practical High Voltage Safety Operating Procedures
	Practical Industrial Programming using 61131-3 for PLCs		Practical Solutions to Power Quality Problems for Engineers and Technicians
	Practical SCADA Systems for Industry		Wind & Solar Power – Renewable Energy Technologies
	Fundamentals of OPC (OLE for Process Control)		Practical Power Distribution
	Practical Instrumentation for Automation and Process Control		Practical Variable Speed Drives for Instrumentation and Control Systems
	Practical Motion Control for Engineers and Technicians		
	Practical HAZOPS, Trips and Alarms		**ELECTRONICS**
			Practical Digital Signal Processing Systems for Engineers and Technicians
	DATA COMMUNICATIONS & NETWORKING		Shielding, EMC/EMI, Noise Reduction, Earthing and Circuit Board Layout
	Practical Data Communications for Engineers and Technicians		Practical EMC and EMI Control for Engineers and Technicians
	Practical DNP3, 60870.5 & Modern SCADA Communication Systems		
	Practical FieldBus and Device Networks for Engineers and Technicians		**MECHANICAL ENGINEERING**
	Troubleshooting & Problem Solving of Industrial Data Communications		Fundamentals of Heating, Ventilation & Airconditioning (HVAC) for Engineers and Technicians
	Practical Fibre Optics for Engineers and Technicians		Practical Boiler Plant Operation and Management
	Practical Industrial Networking for Engineers and Technicians		Practical Centrifugal Pumps – Efficient use for Safety & Reliability
	Practical TCP/IP & Ethernet Networking for Industry		
	Practical Telecommunications for Engineers and Technicians		**PROJECT & FINANCIAL MANAGEMENT**
	Best Practice in Industrial Data Communications		Practical Project Management for Engineers and Technicians
	Practical Routers & Switches (including TCP/IP and Ethernet) for Engineers and Technicians		Practical Financial Management and Project Investment Analysis
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Position: .....

Company Name: .....

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Email: .....

Name: .....

Position: .....

Company Name: .....

Address: .....

Email: .....

Thank you for completing this questionnaire,
your opinion is important to us.